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(54) **FOCUSED STERILIZATION AND
STERILIZED SUB-ASSEMBLIES FOR
ANALYTE MONITORING SYSTEMS**

(71) Applicant: **ABBOTT DIABETES CARE INC.**,
Alameda, CA (US)

(72) Inventors: **Christopher A. Thomas**, Alameda, CA
(US); **Louis Pace**, San Carlos, CA
(US); **Vincent M. Dipalma**, Oakland,
CA (US); **Vivek S. Rao**, Alameda, CA
(US); **Steven T. Mitchell**, Pleasant Hill,
CA (US); **Peter G. Robinson**, Alamo,
CA (US); **Matthew Simmons**,
Pleasanton, CA (US); **Jörn Meissner**,
Neubiberg (DE)

(73) Assignee: **ABBOTT DIABETES CARE INC.**,
Alameda, CA (US)

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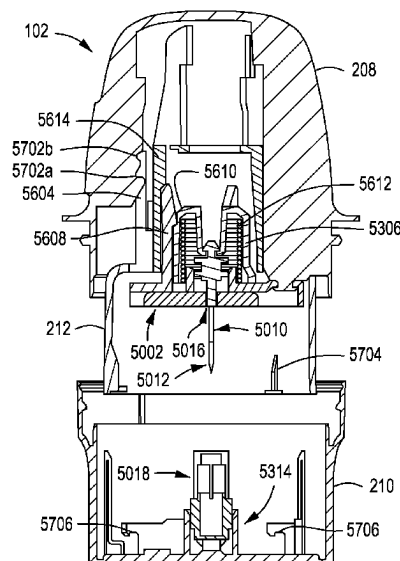
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Primary Examiner — Monzer R Chorbaji
(74) *Attorney, Agent, or Firm* — Baker Botts L.L.P.

(57) **ABSTRACT**

A system includes a sensor applicator, a sensor control device arranged within the sensor applicator and including an electronics housing and a sensor extending from a bottom of the electronics housing, and a cap coupled to one of the sensor applicator and the sensor control device, wherein the cap is removable prior to deploying the sensor control device from the sensor applicator.

20 Claims, 109 Drawing Sheets



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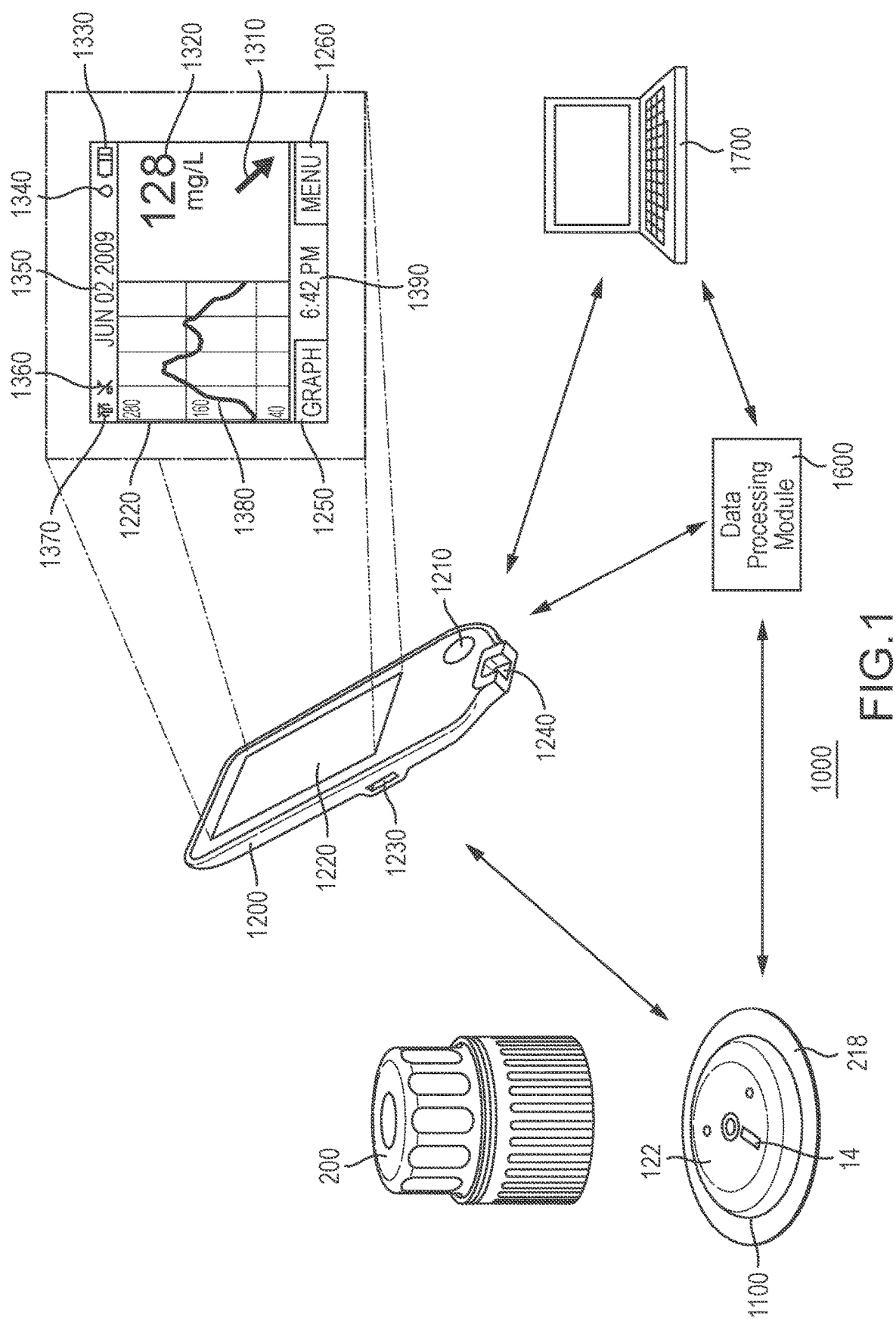
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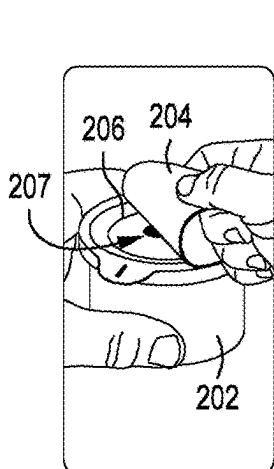


FIG. 2A

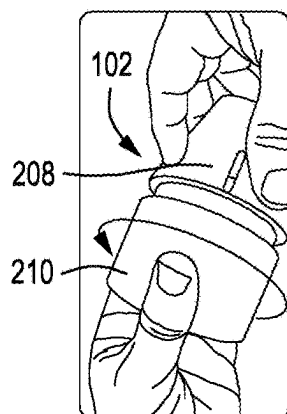


FIG. 2B

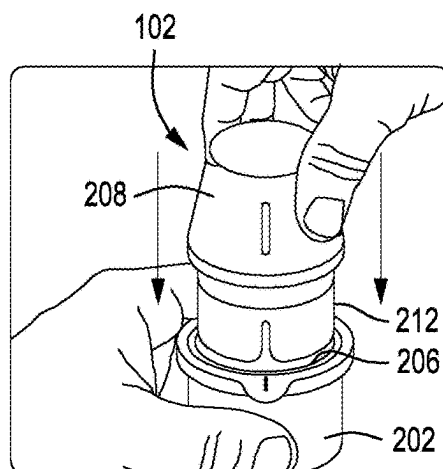


FIG. 2C

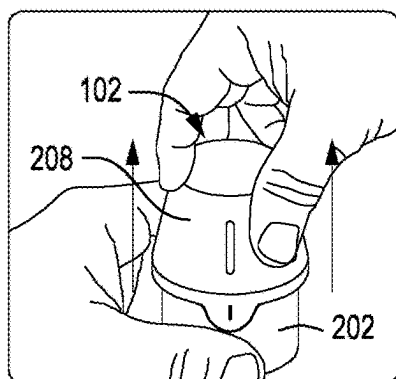


FIG. 2D

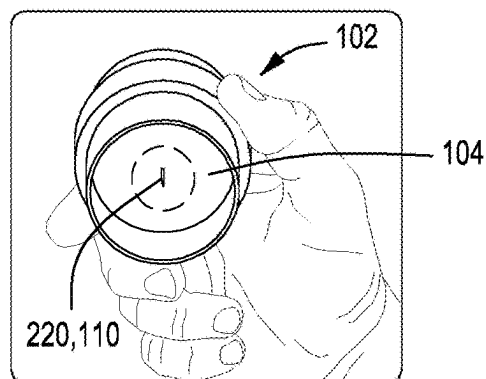


FIG. 2E

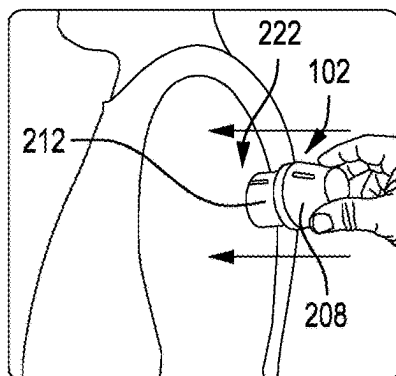


FIG. 2F

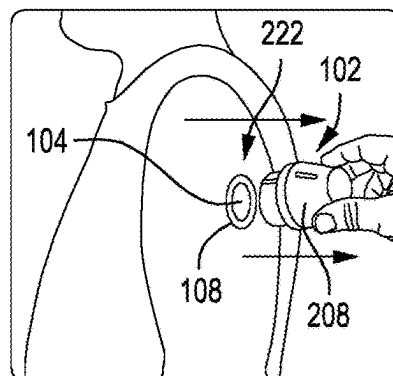


FIG. 2G

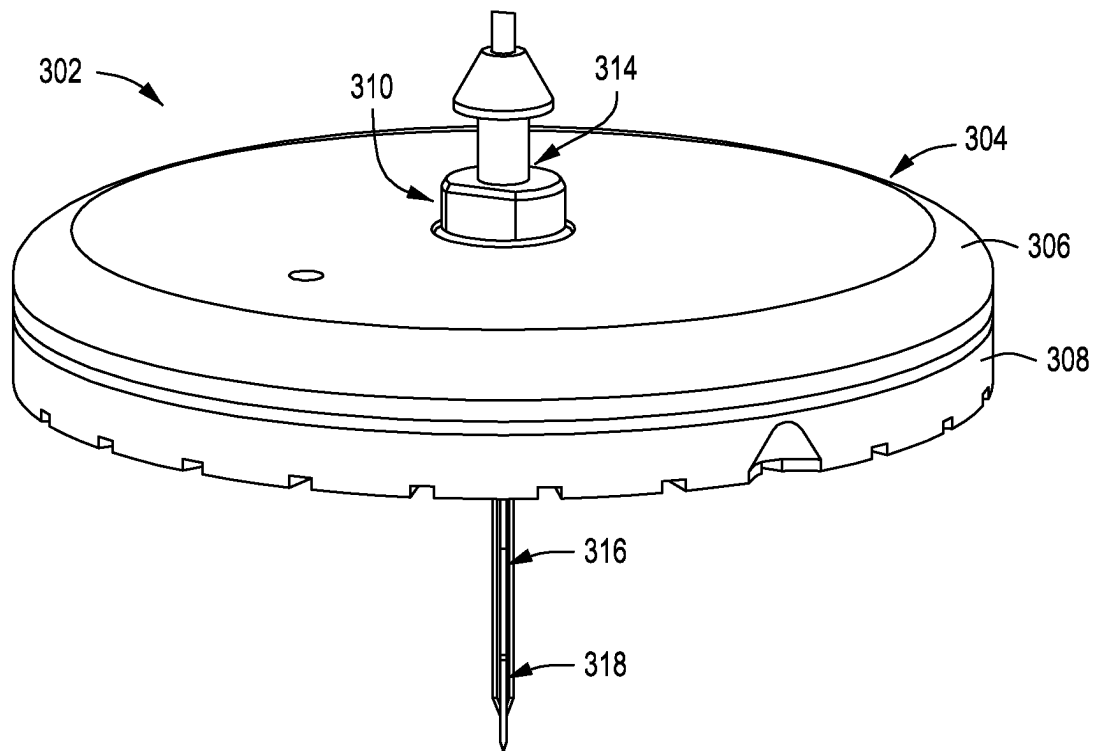


FIG. 3A

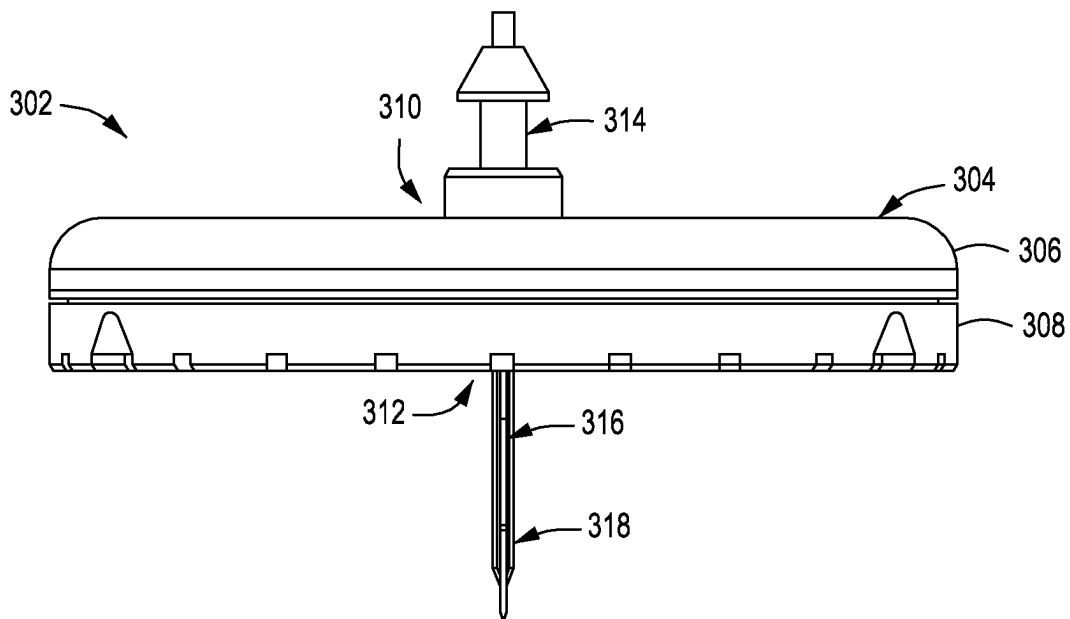


FIG. 3B

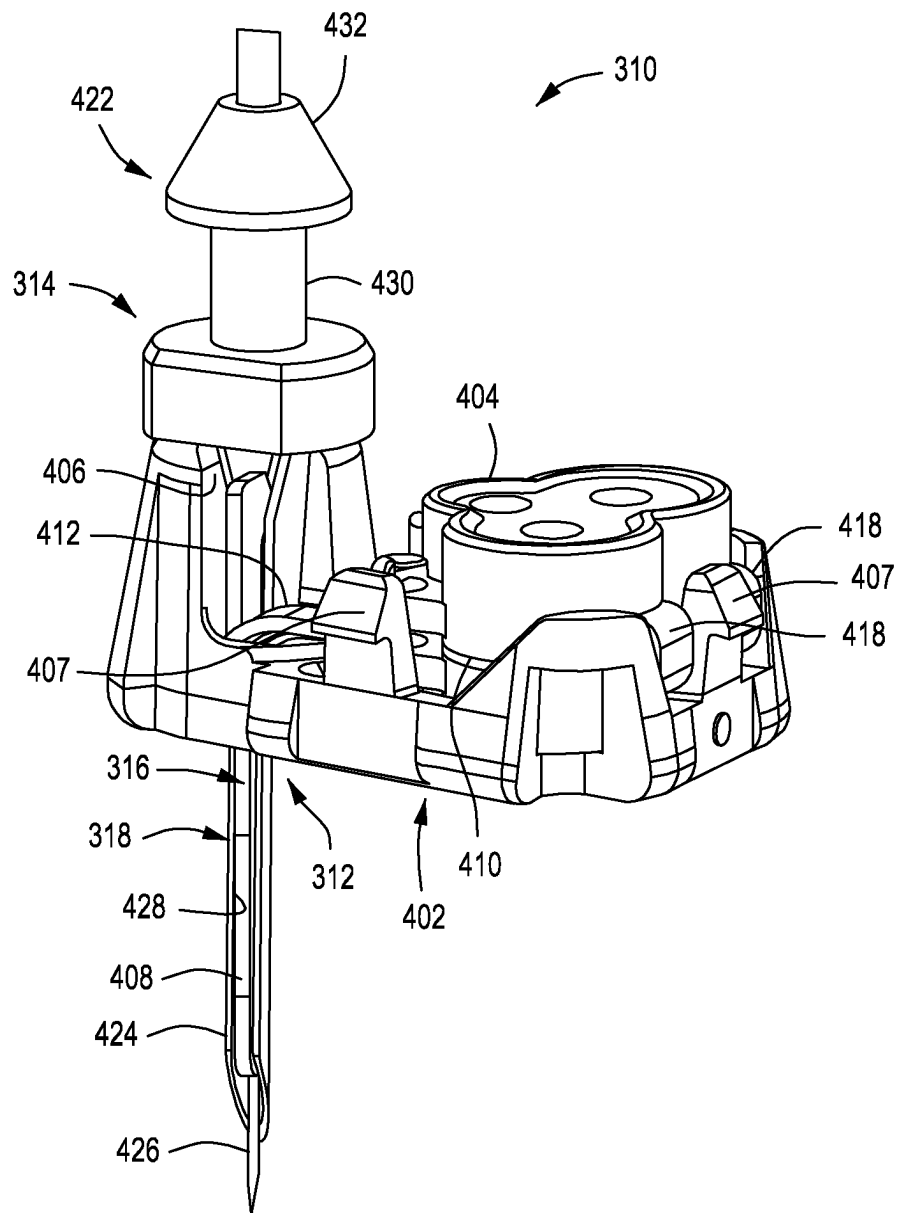


FIG. 4A

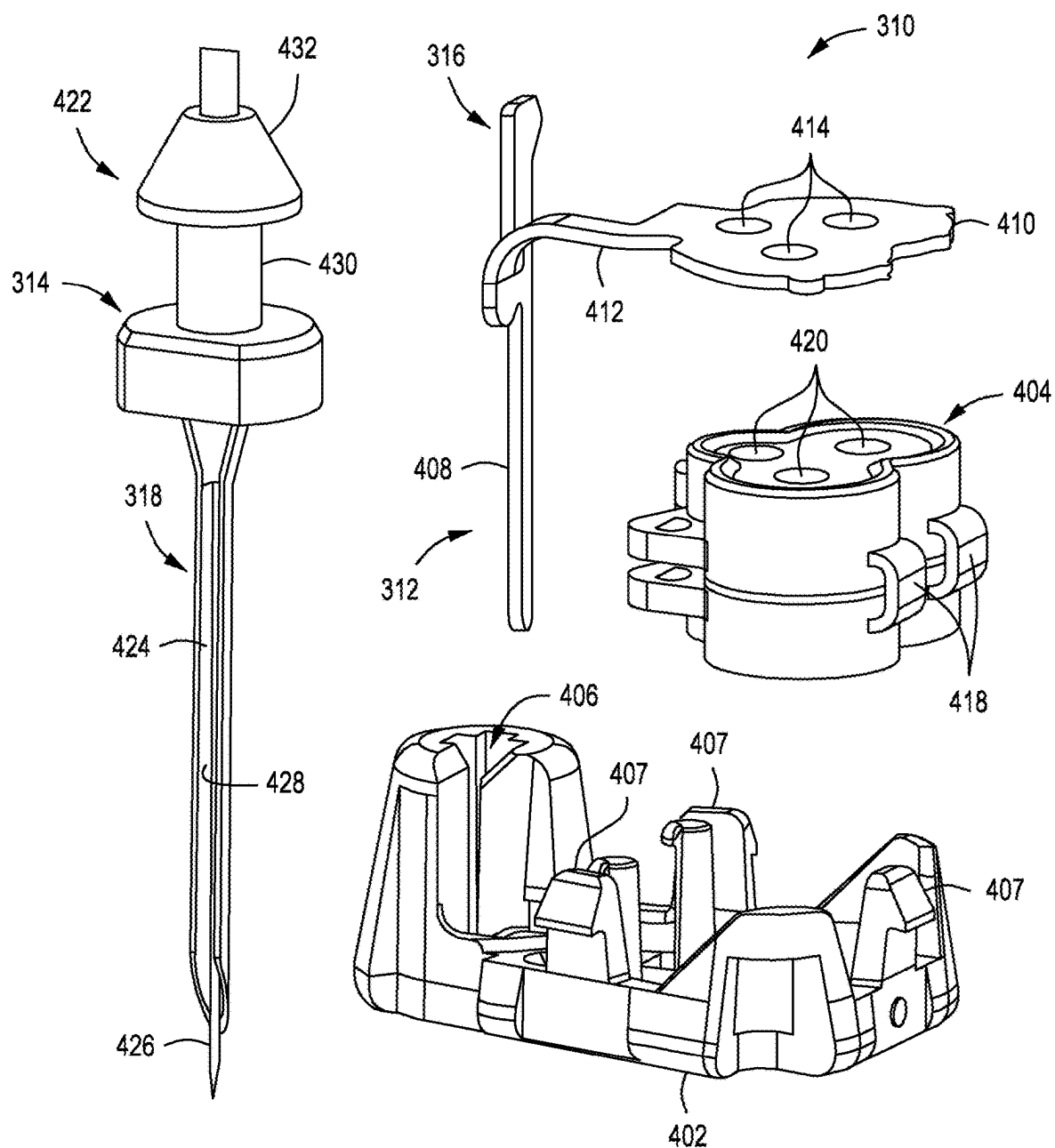


FIG. 4B

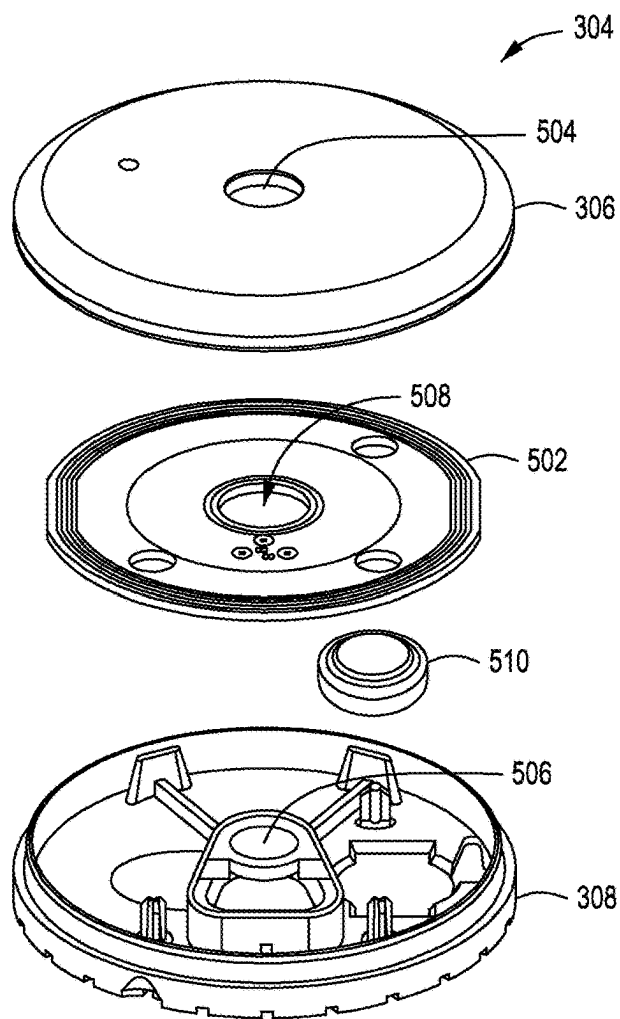


FIG. 5A

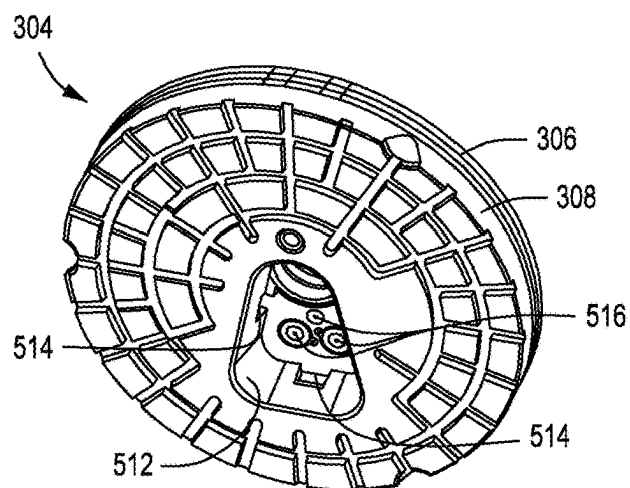


FIG. 5B

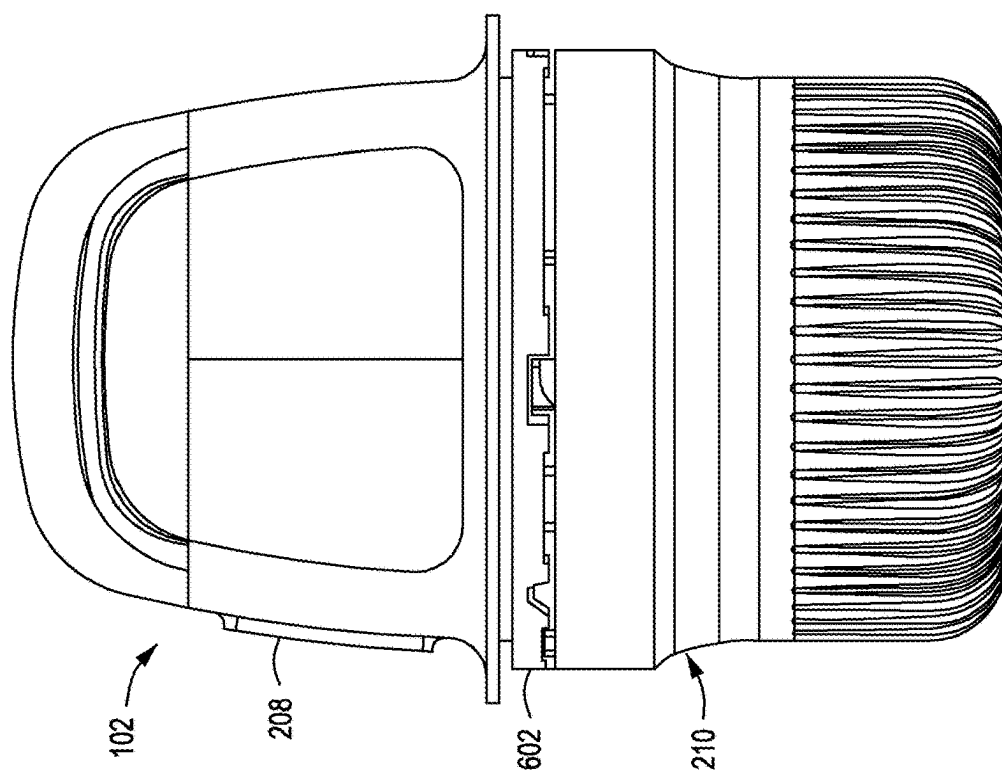


FIG. 6A

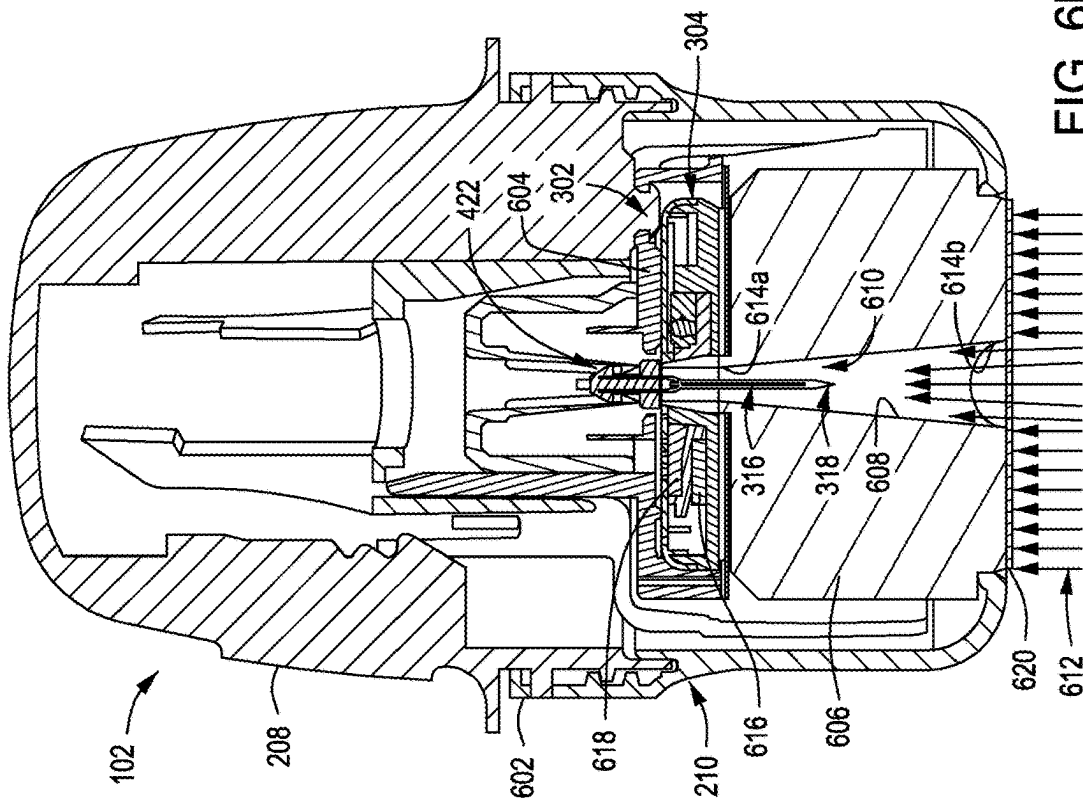
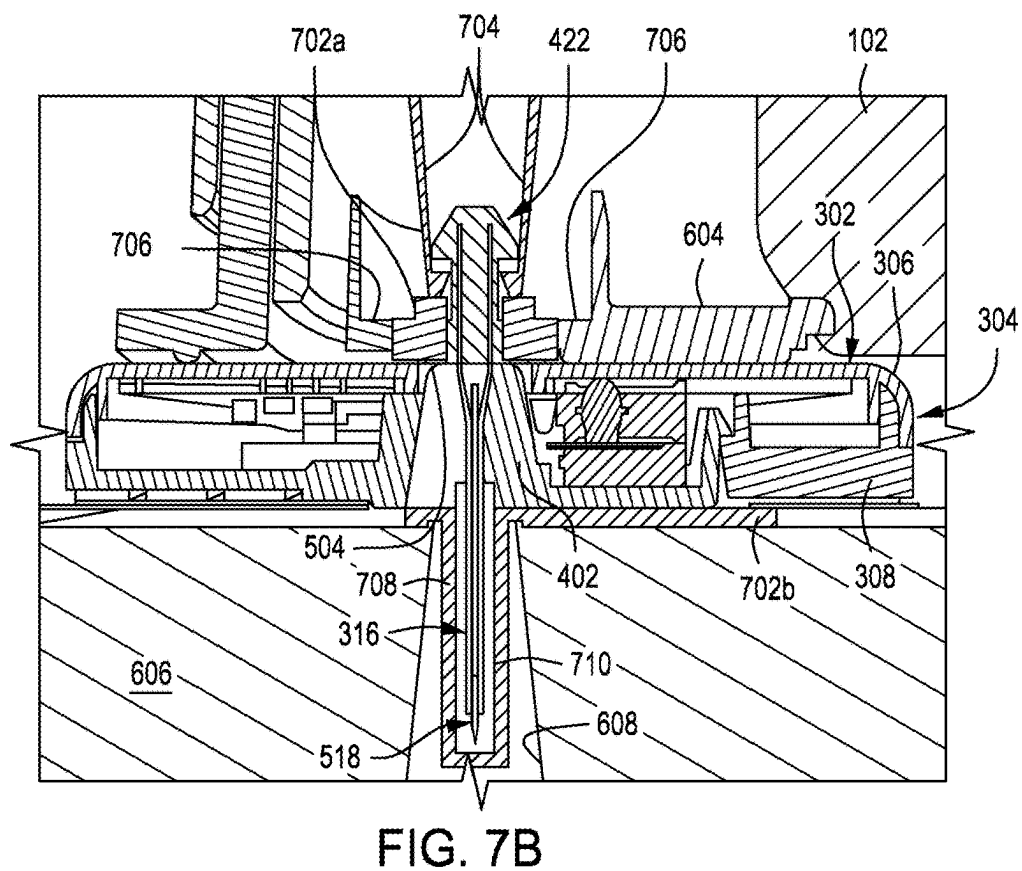
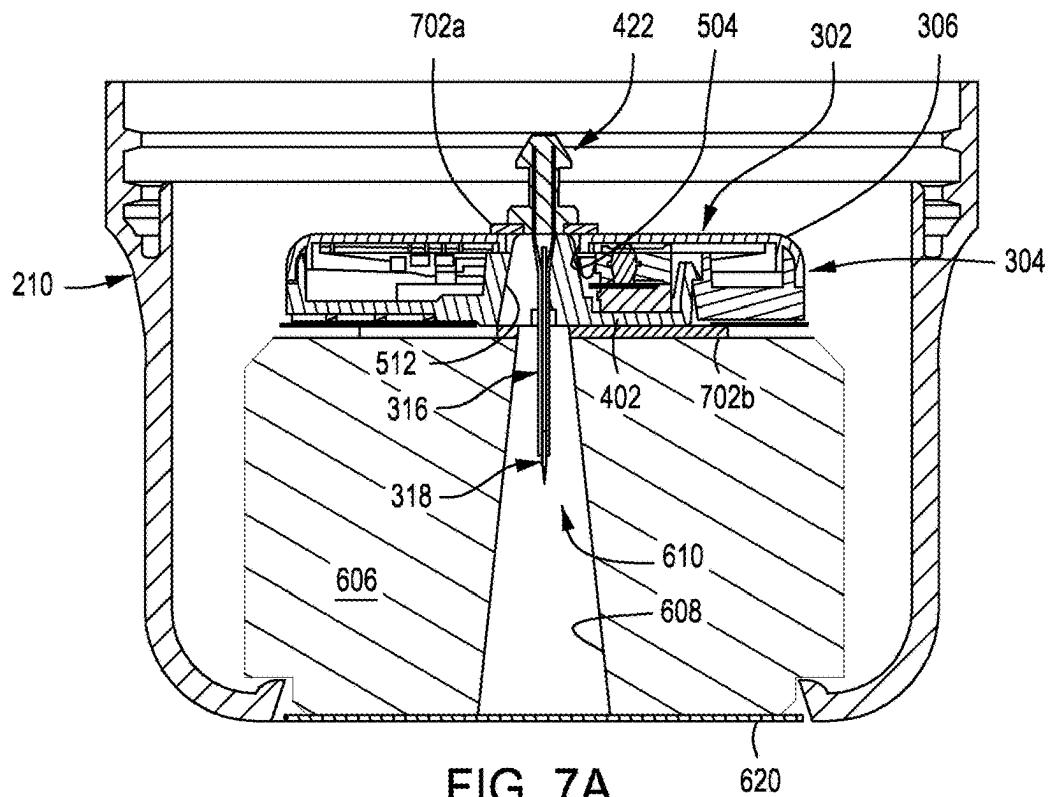


FIG. 6B



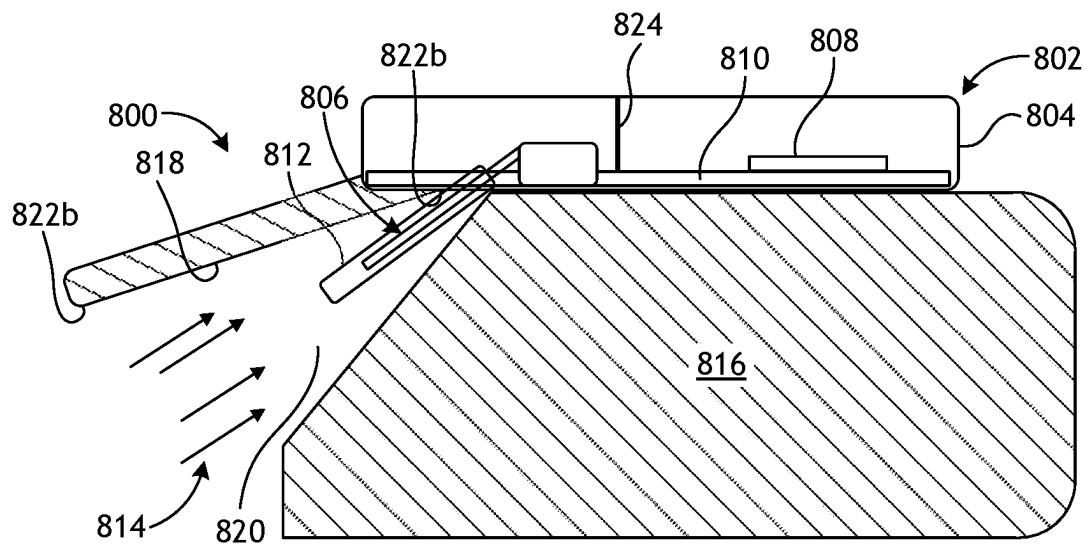


FIG. 8

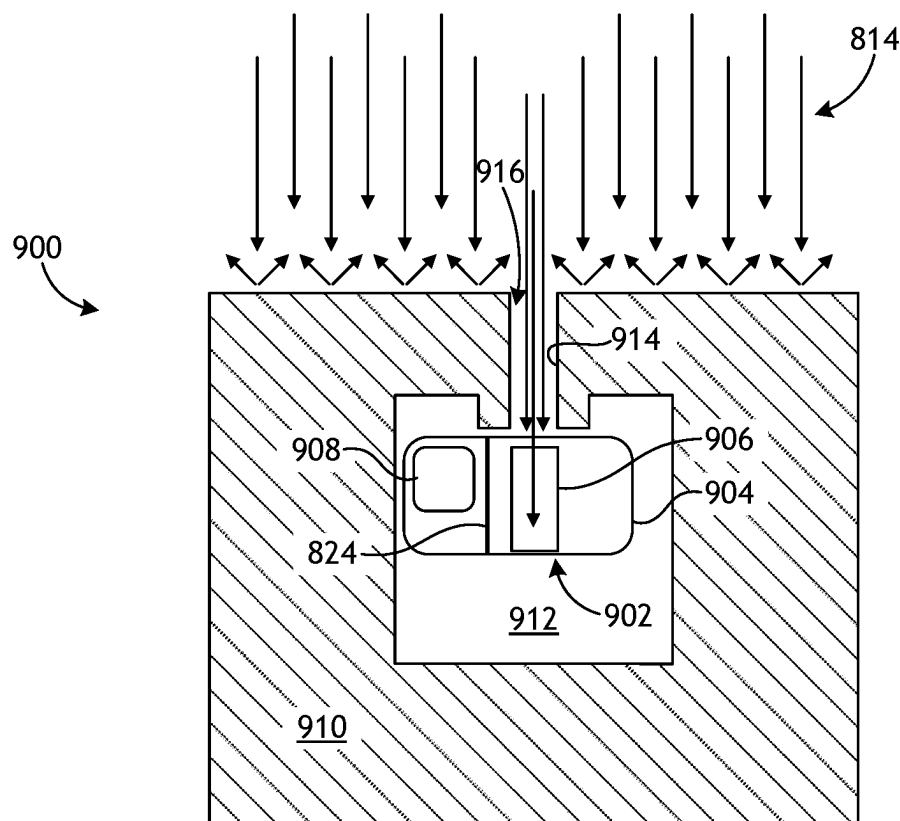


FIG. 9

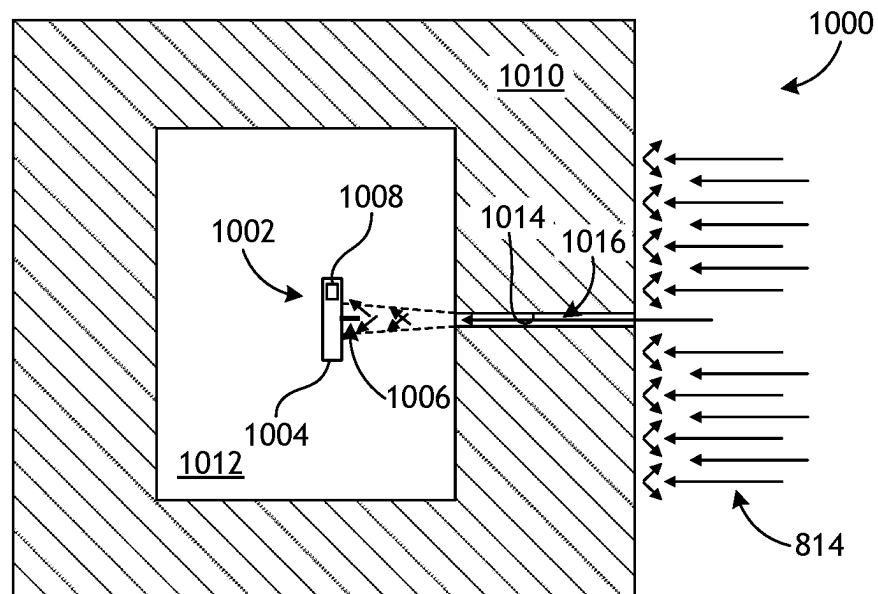


FIG. 10

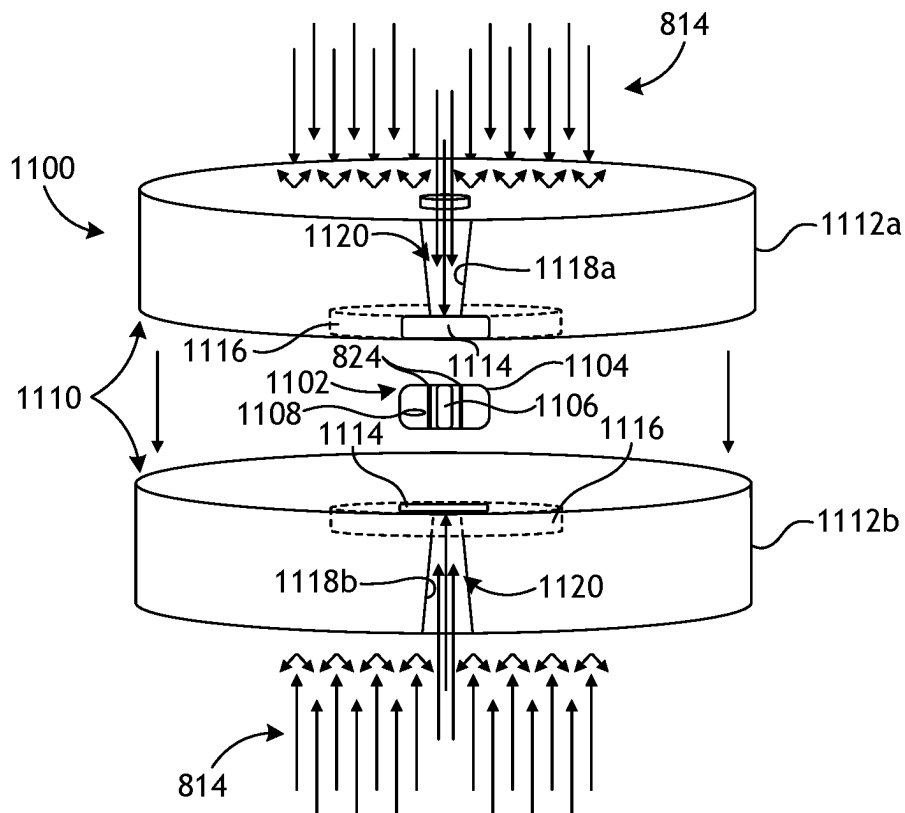


FIG. 11

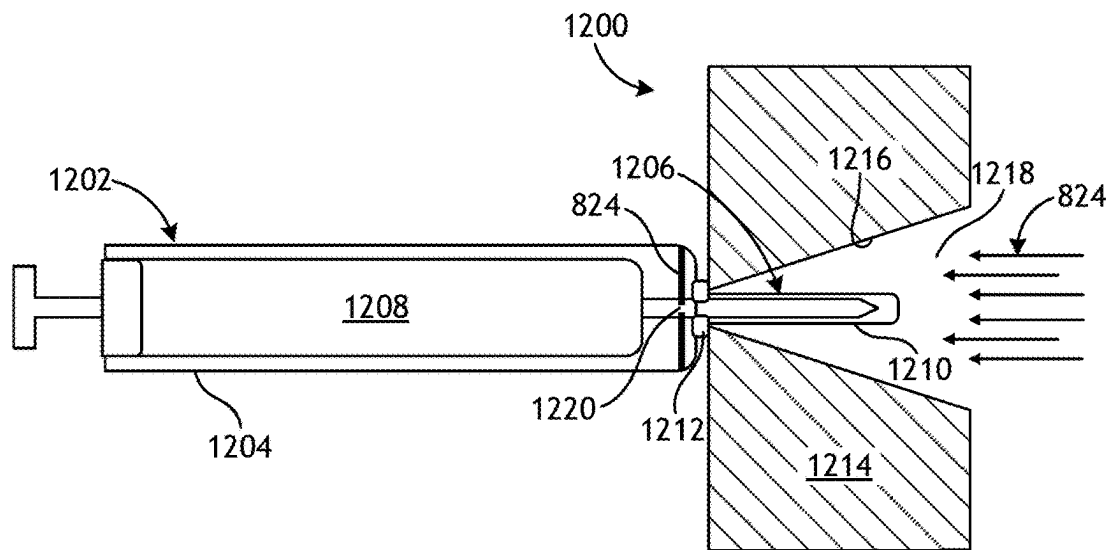


FIG. 12

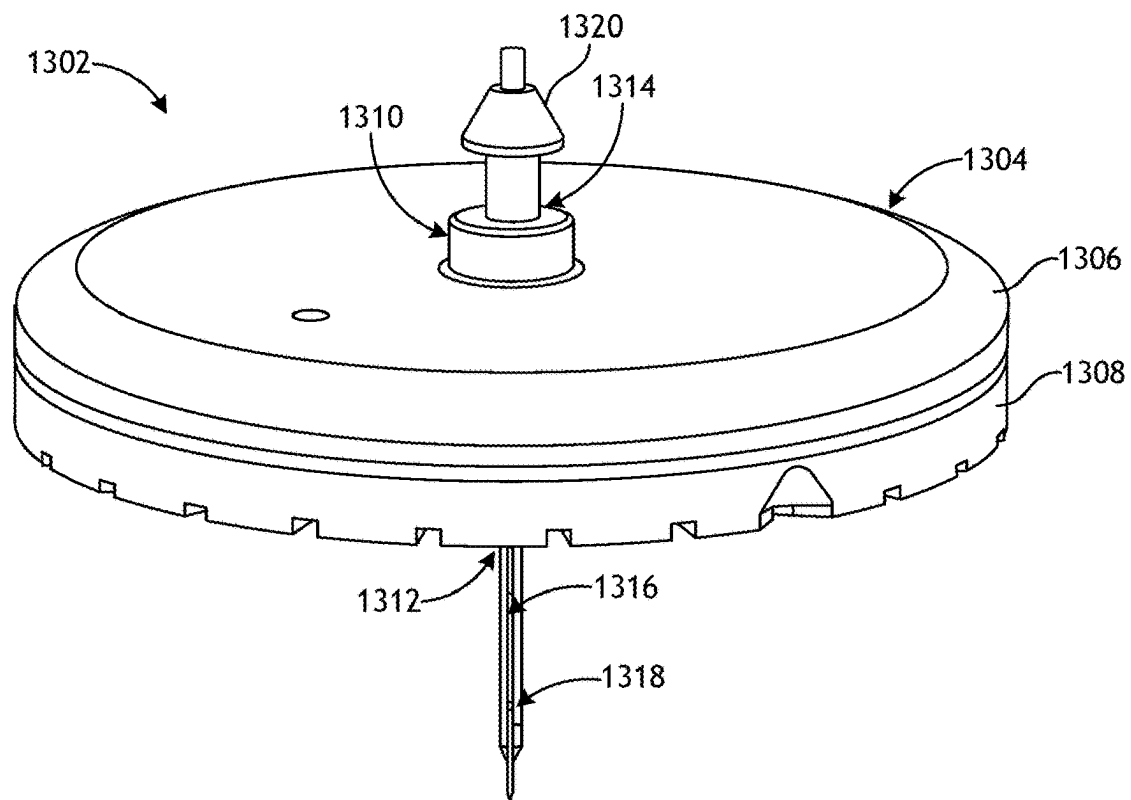


FIG. 13

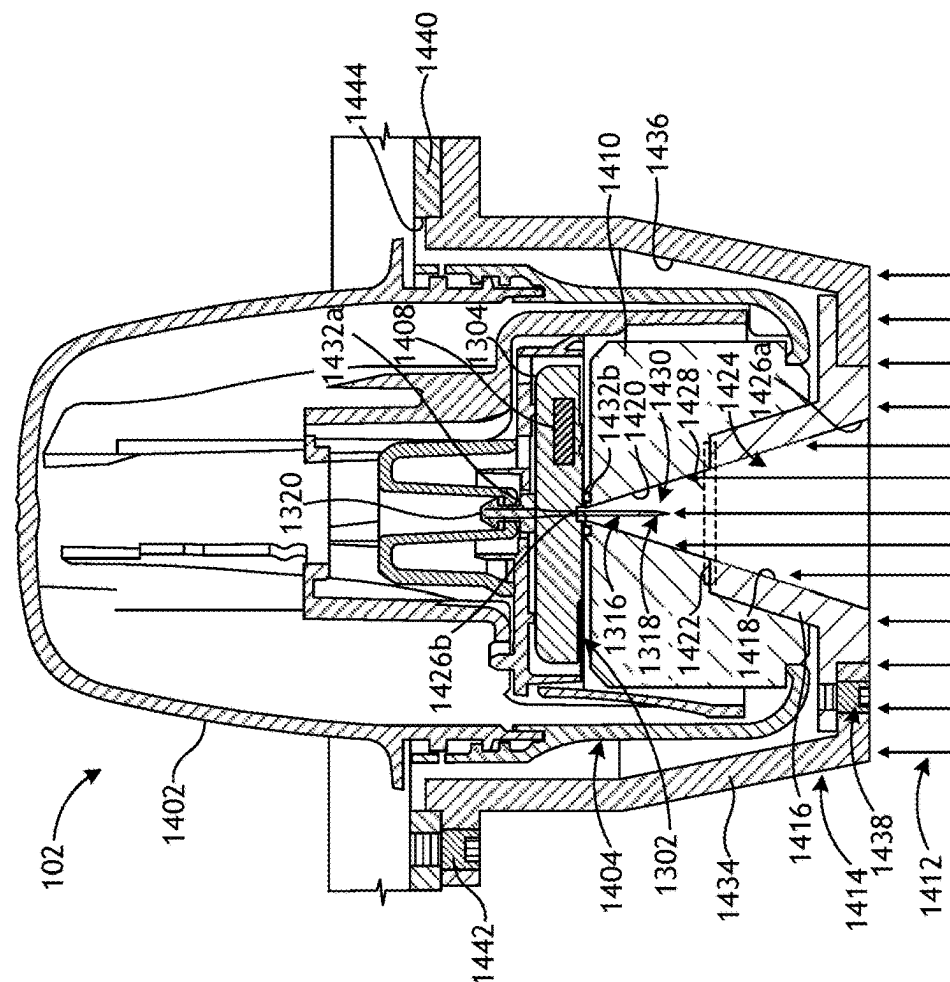


FIG. 14A

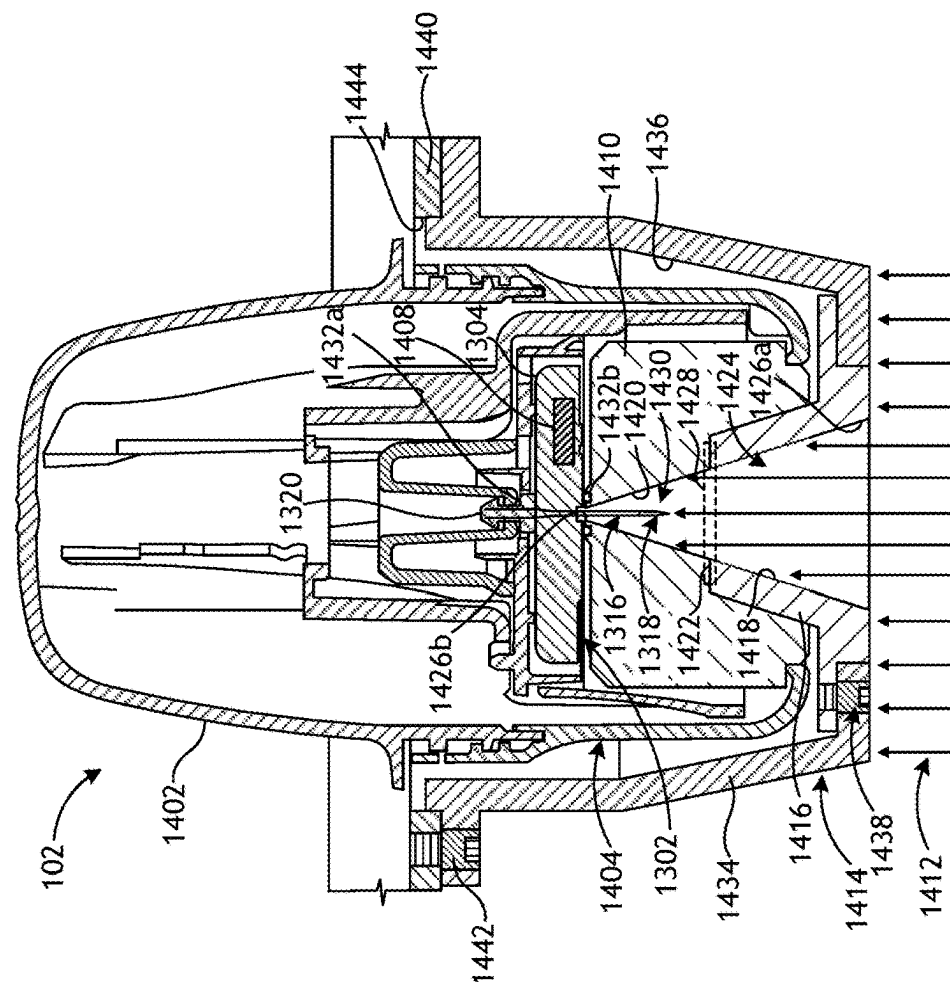


FIG. 14B

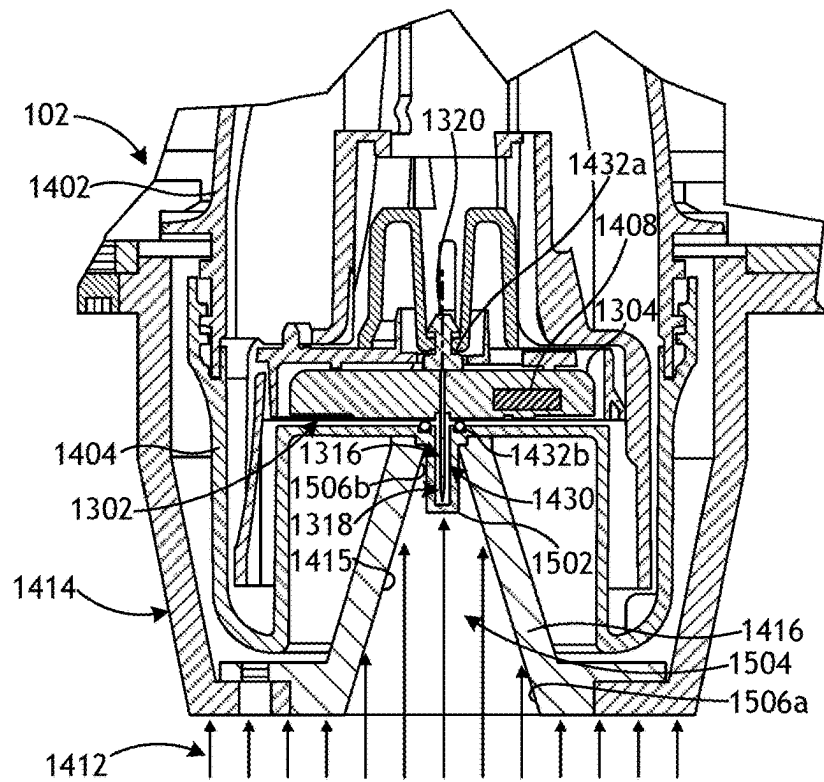


FIG. 15

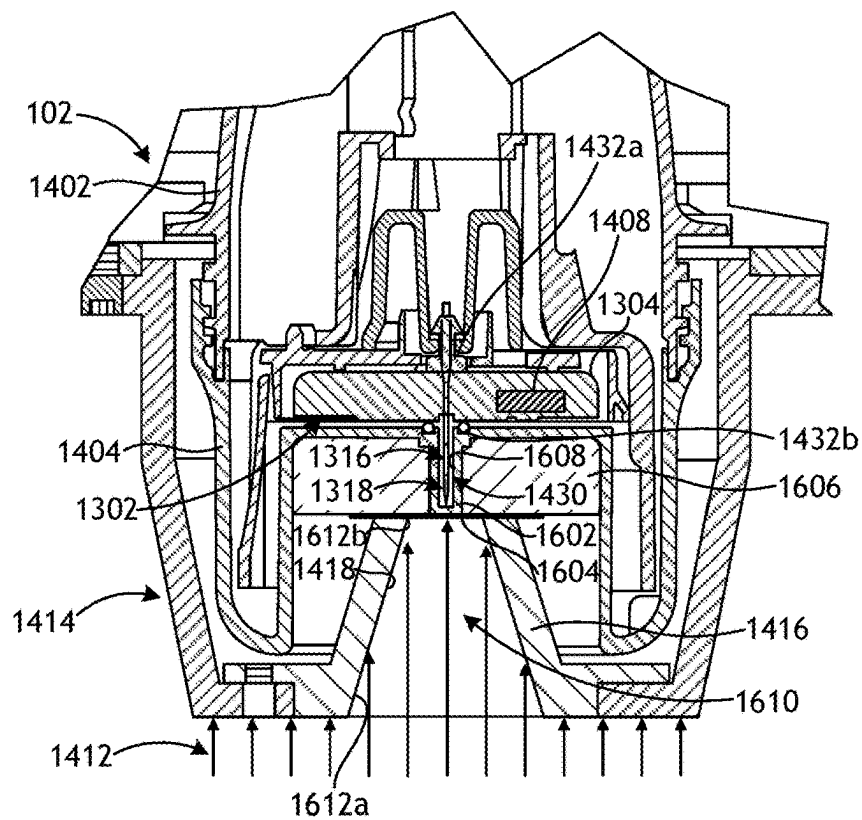


FIG. 16

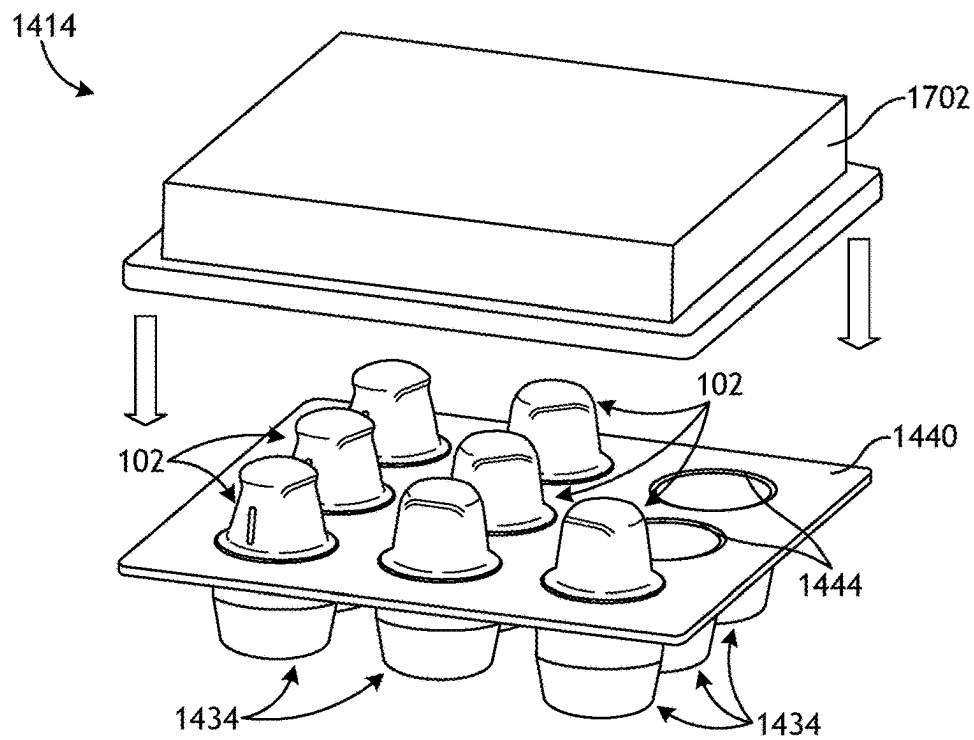


FIG. 17A

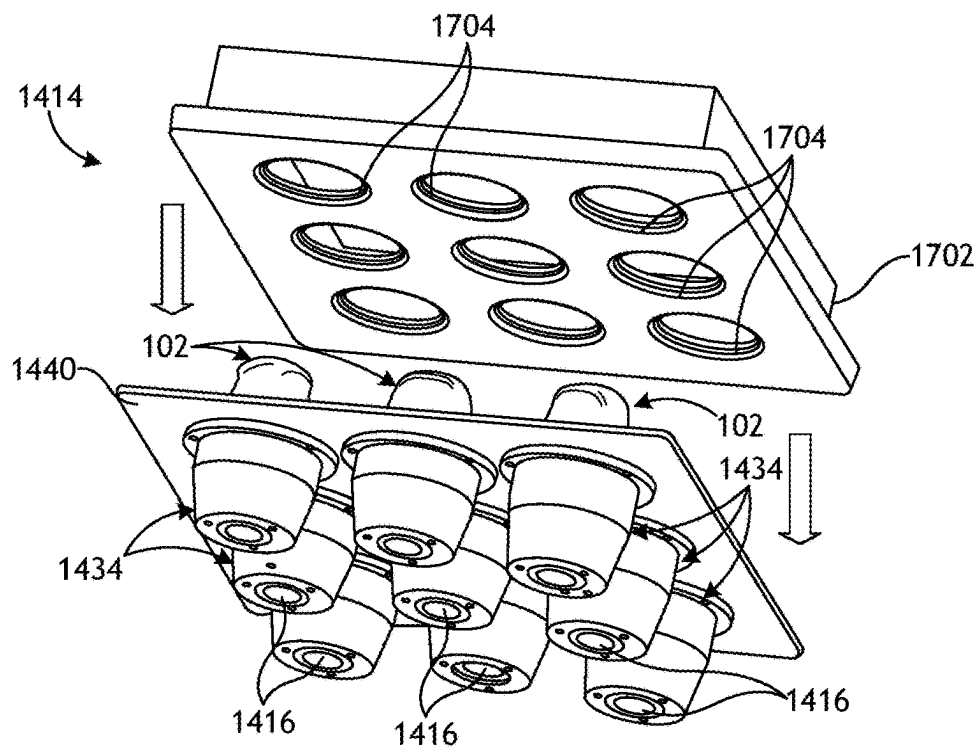


FIG. 17B

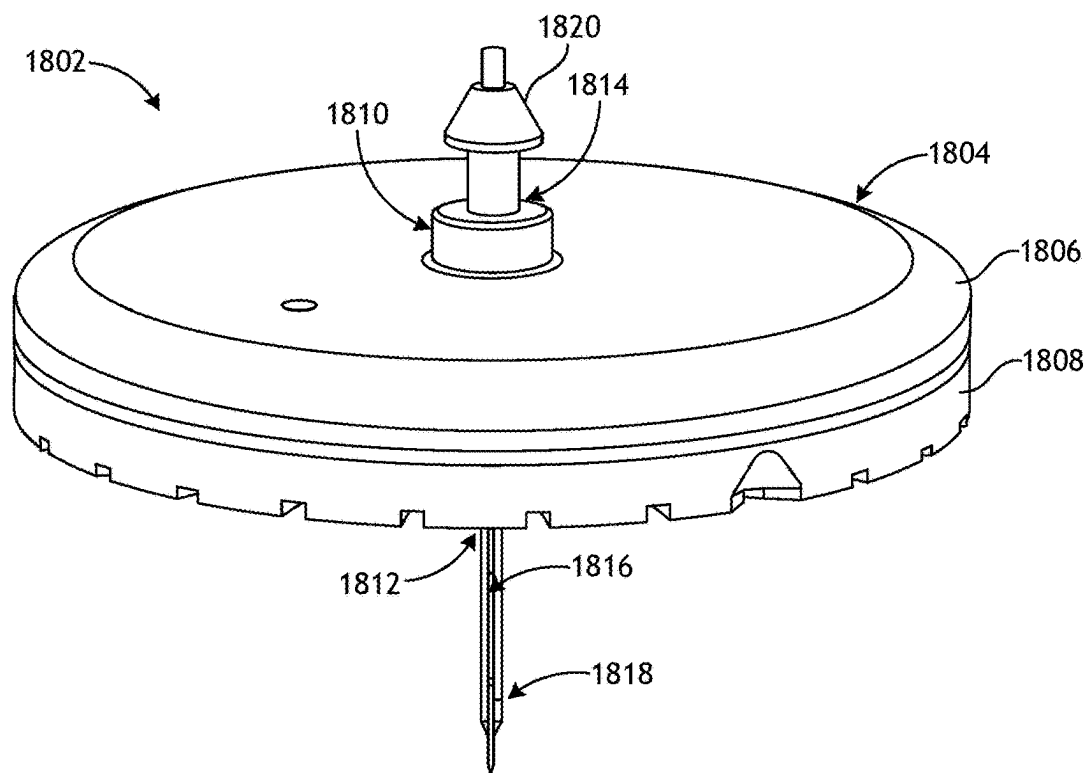


FIG. 18

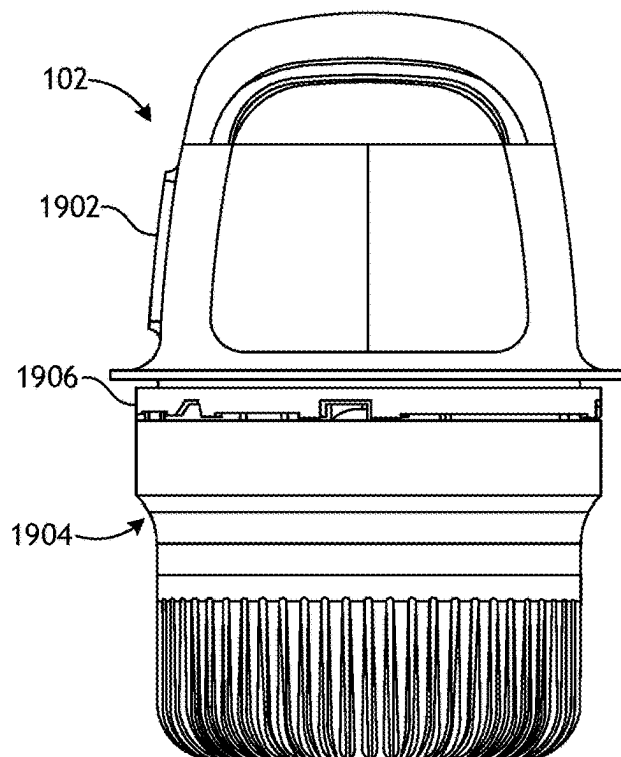


FIG. 19A

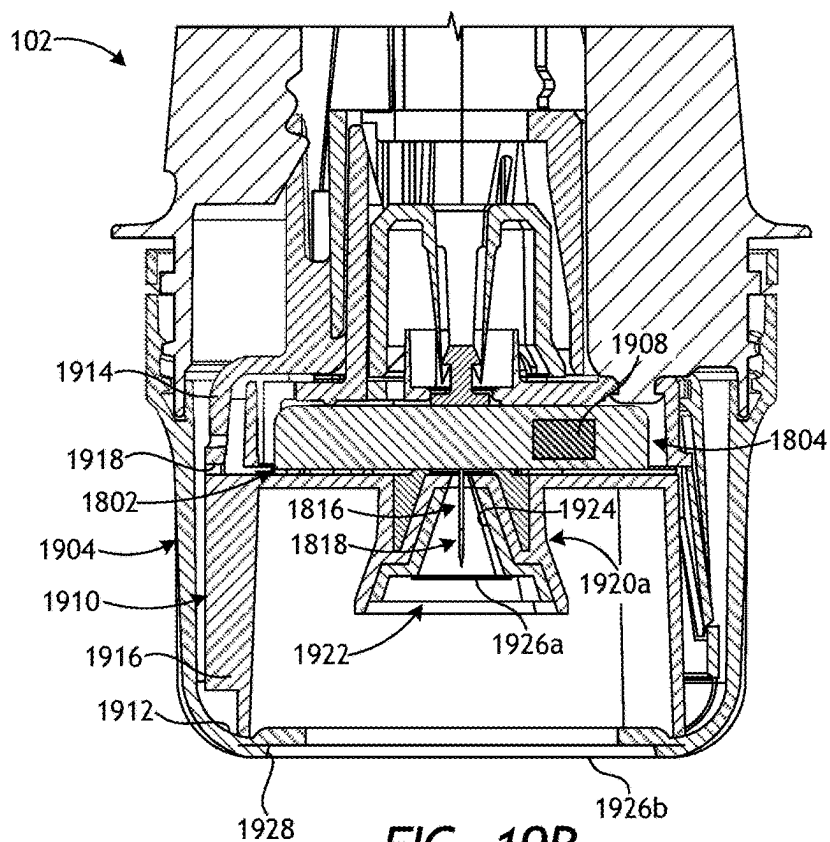


FIG. 19B

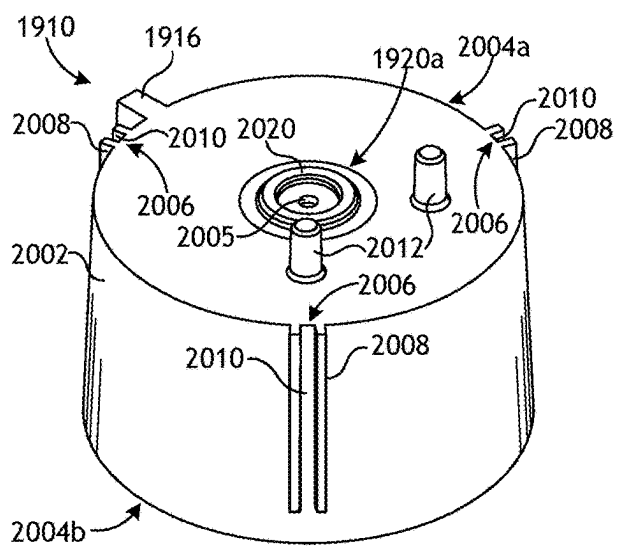


FIG. 20A

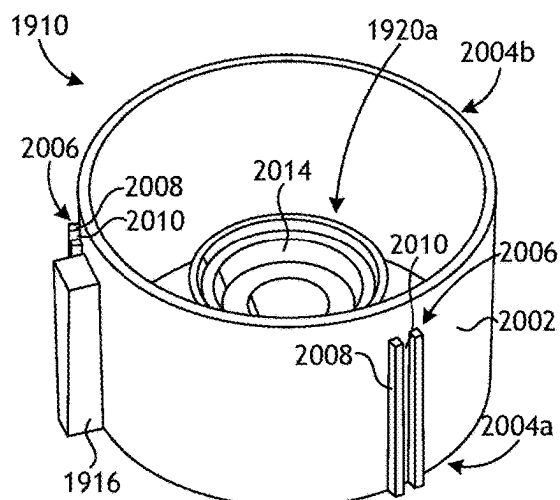


FIG. 20B

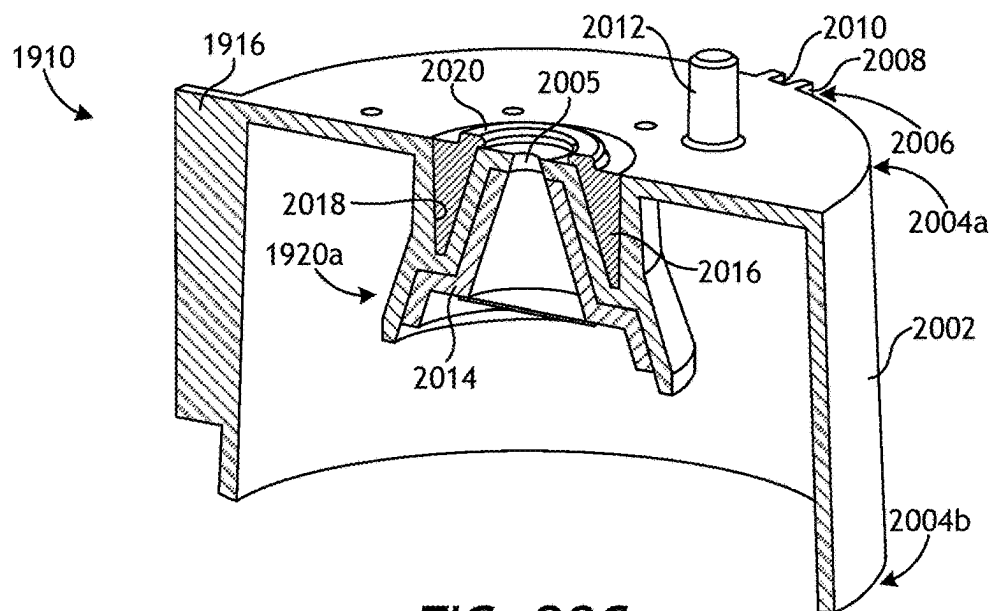


FIG. 20C

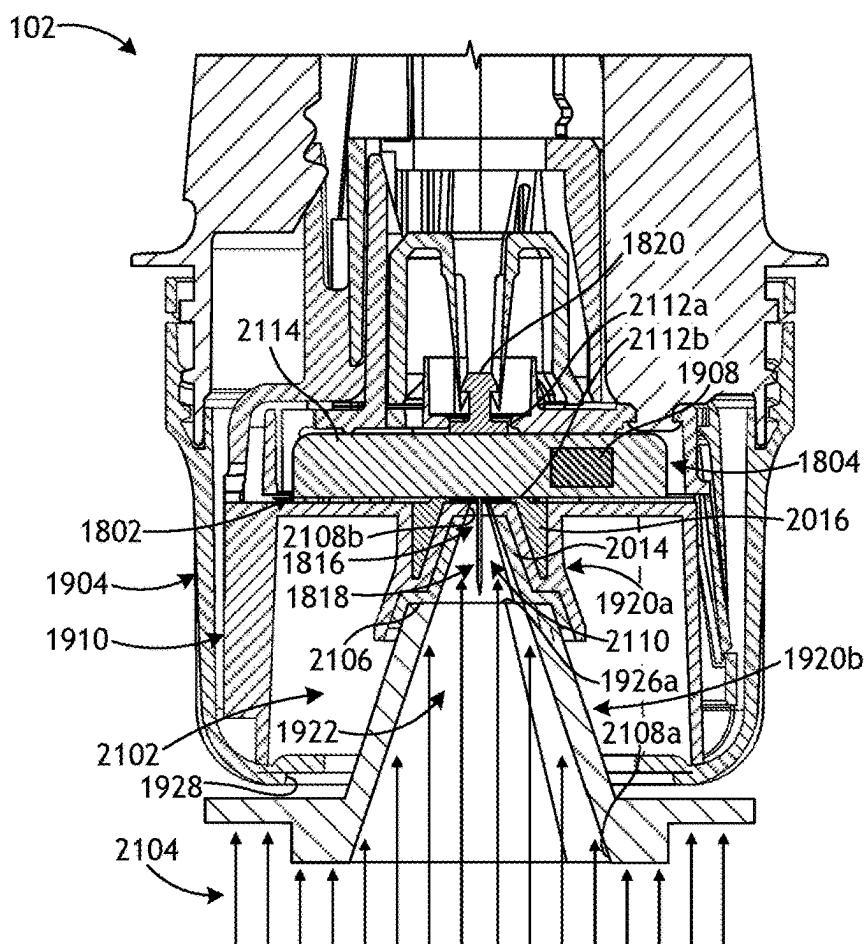


FIG. 21

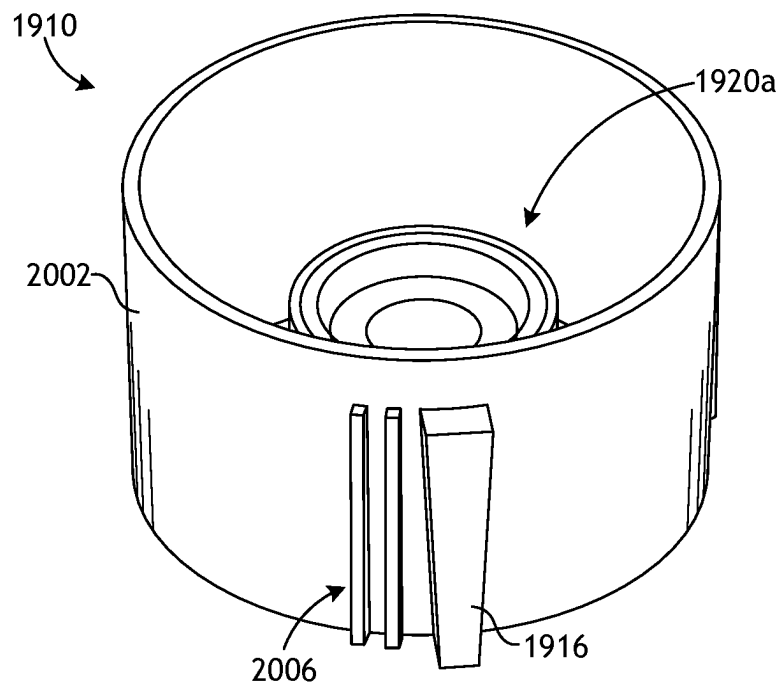


FIG. 22A

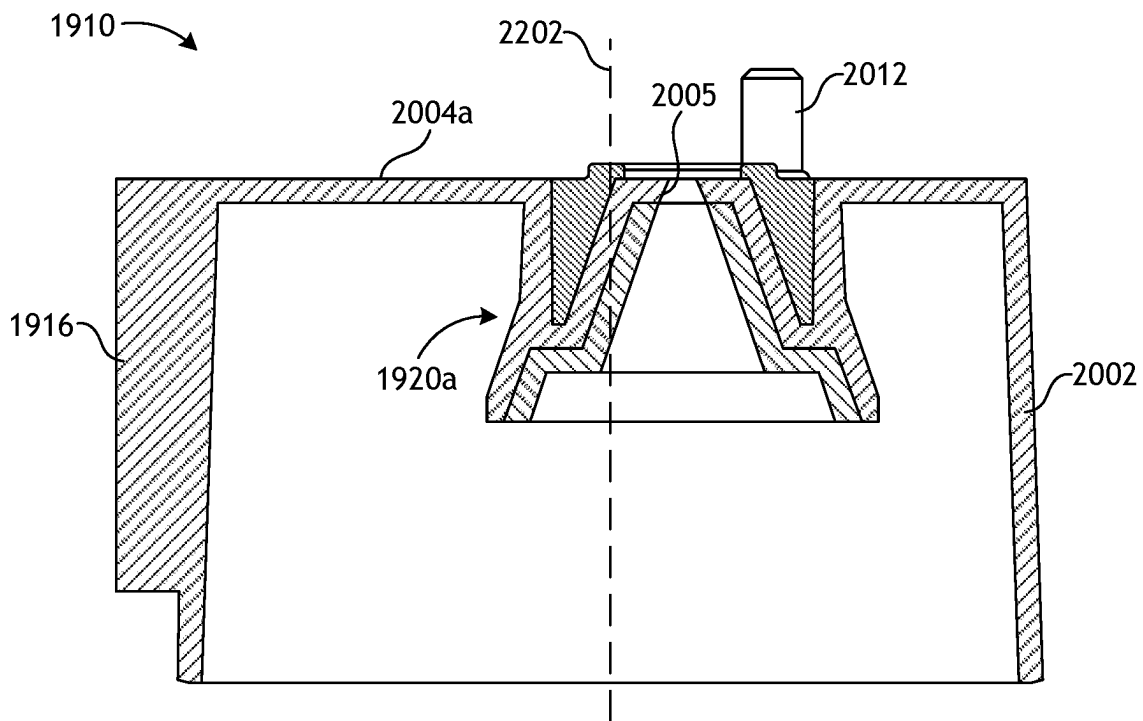


FIG. 22B

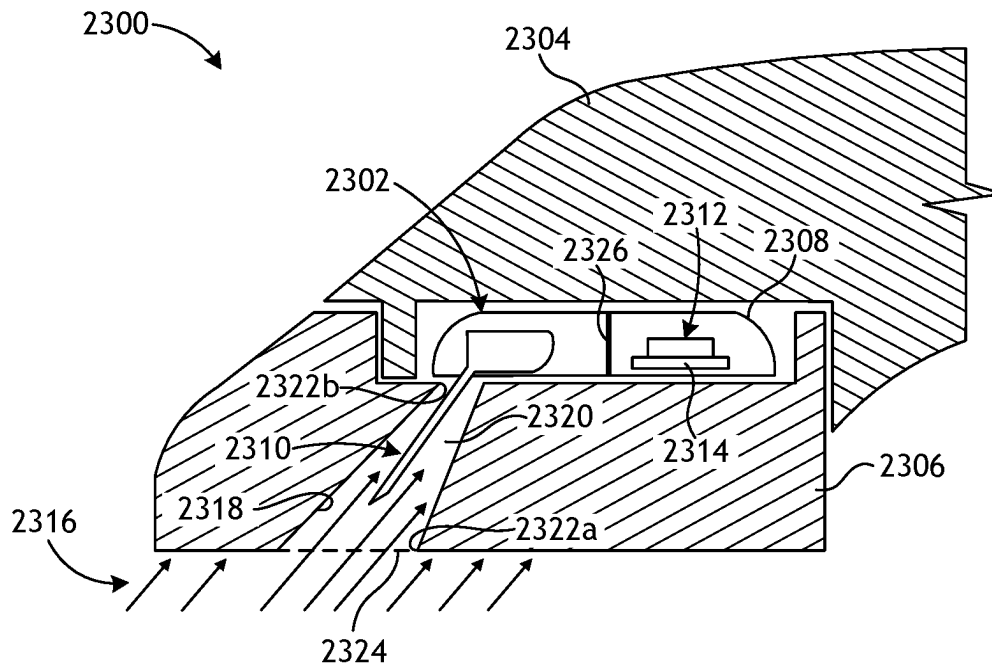


FIG. 23

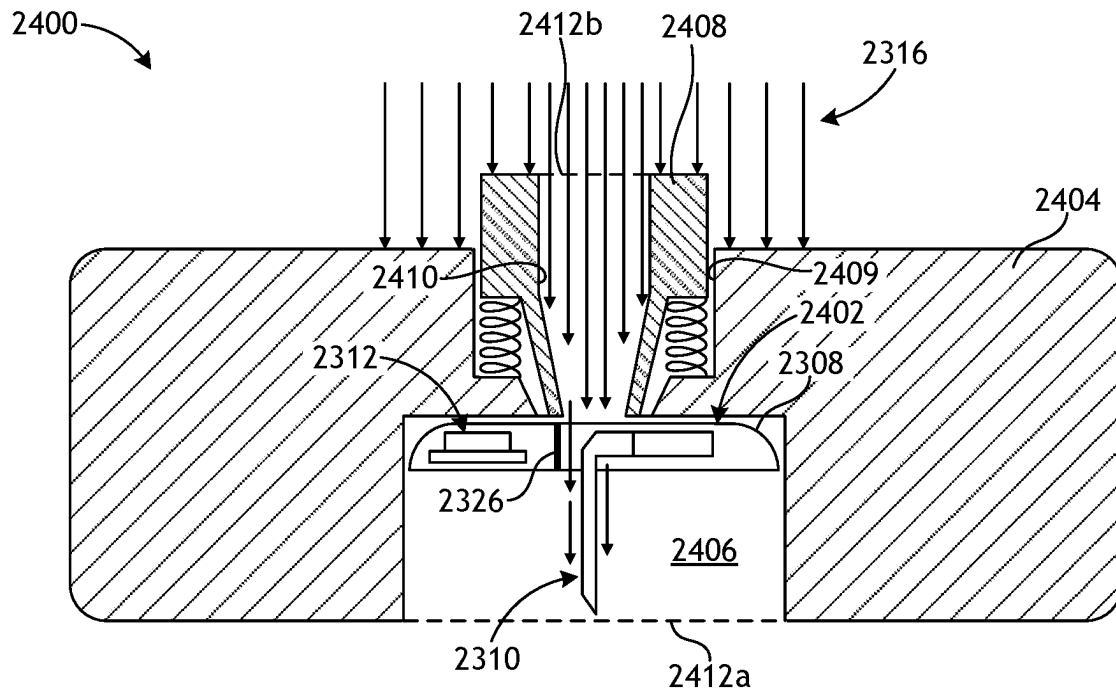


FIG. 24

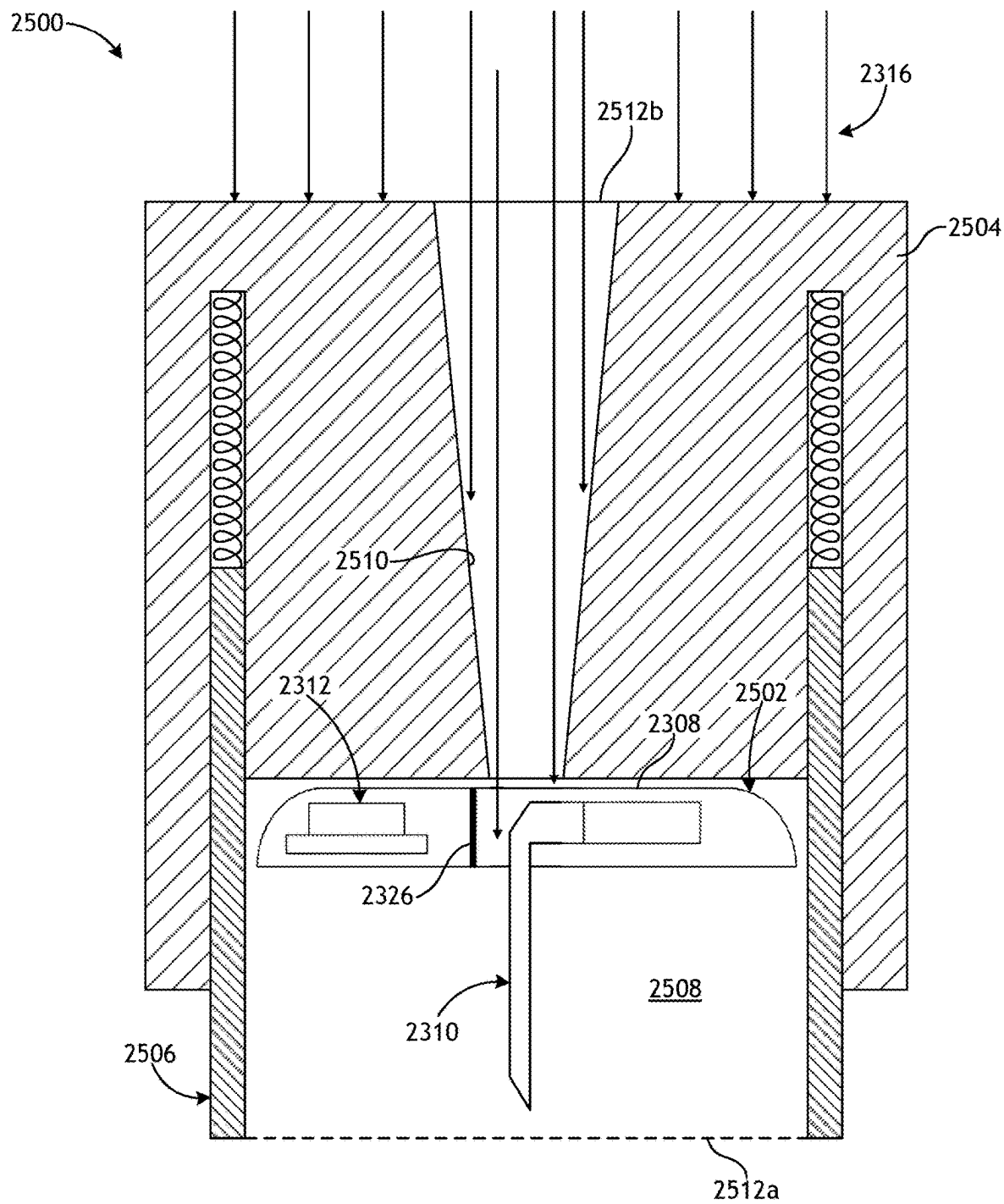


FIG. 25

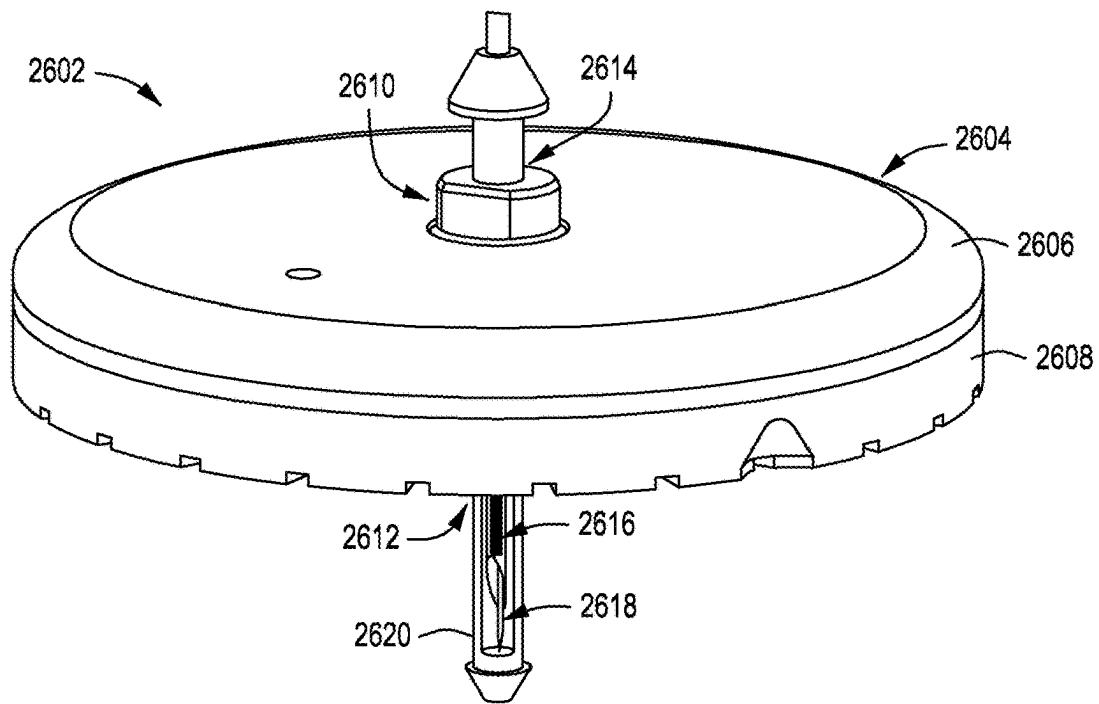


FIG. 26A

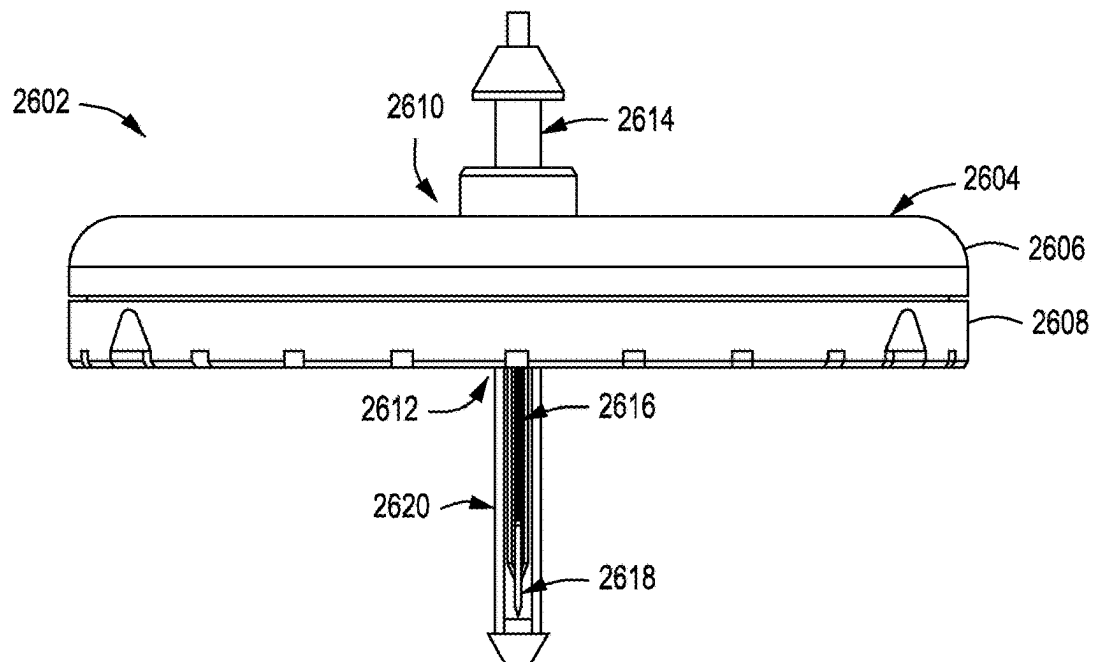


FIG. 26B

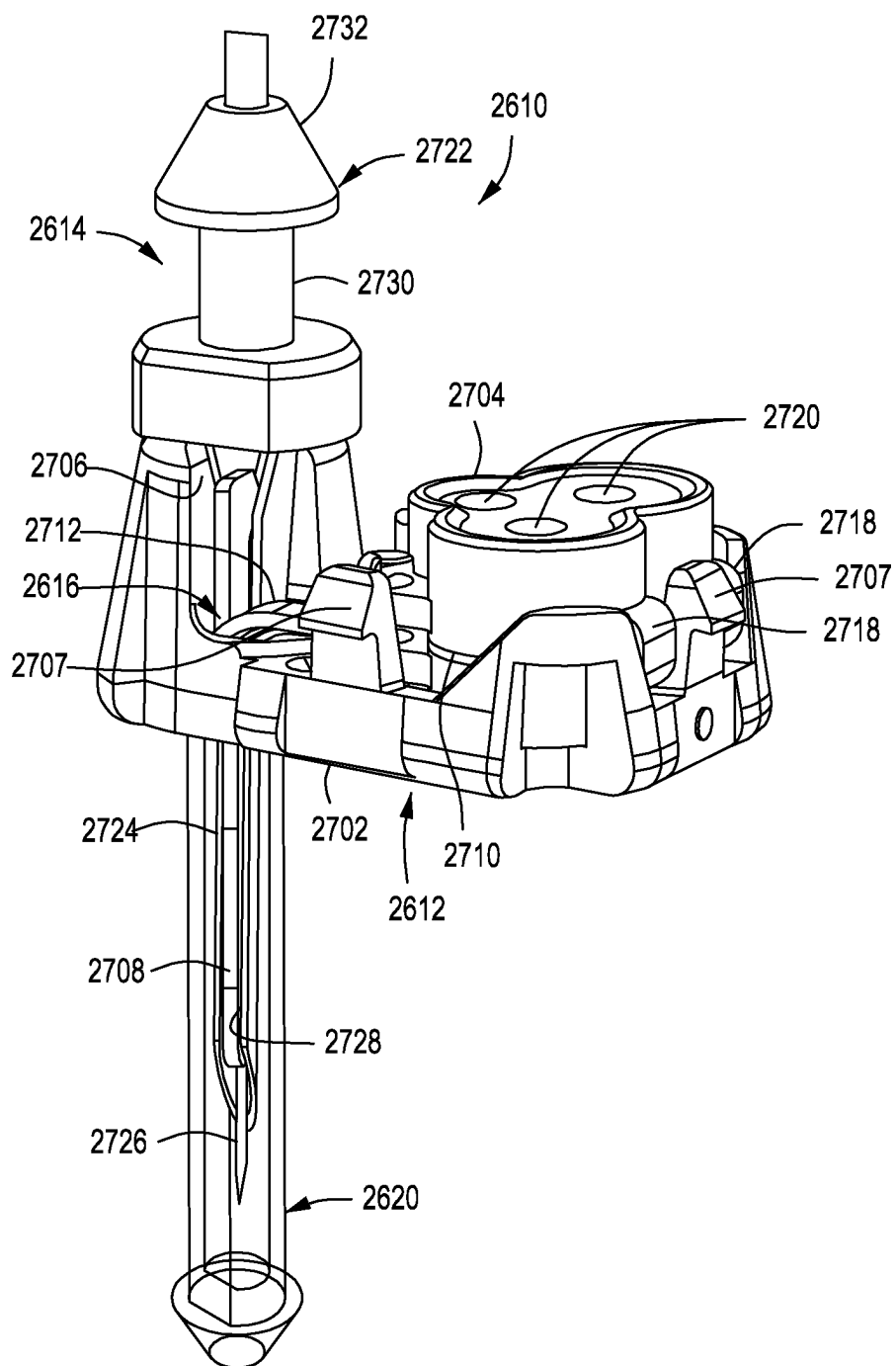
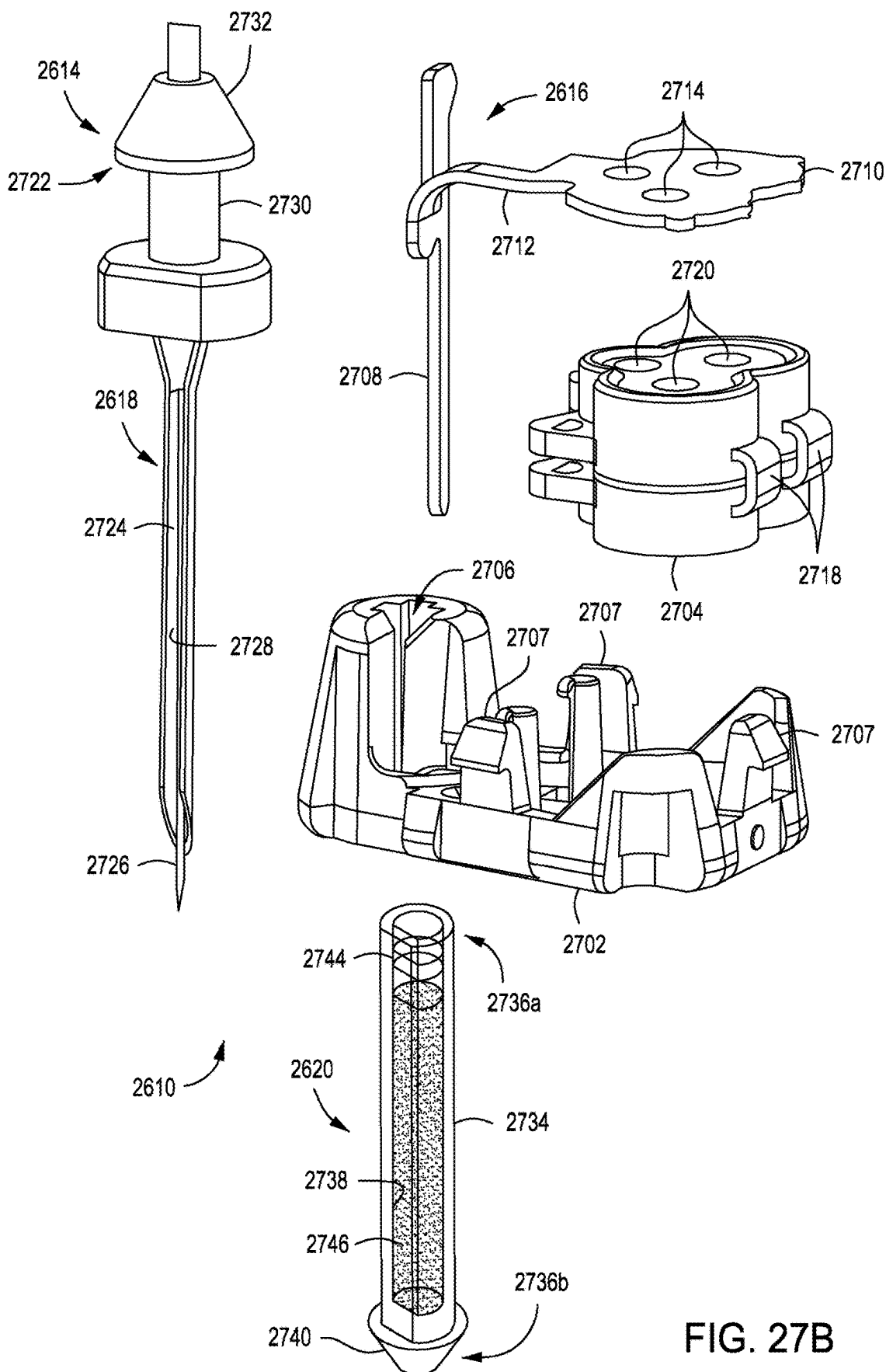


FIG. 27A



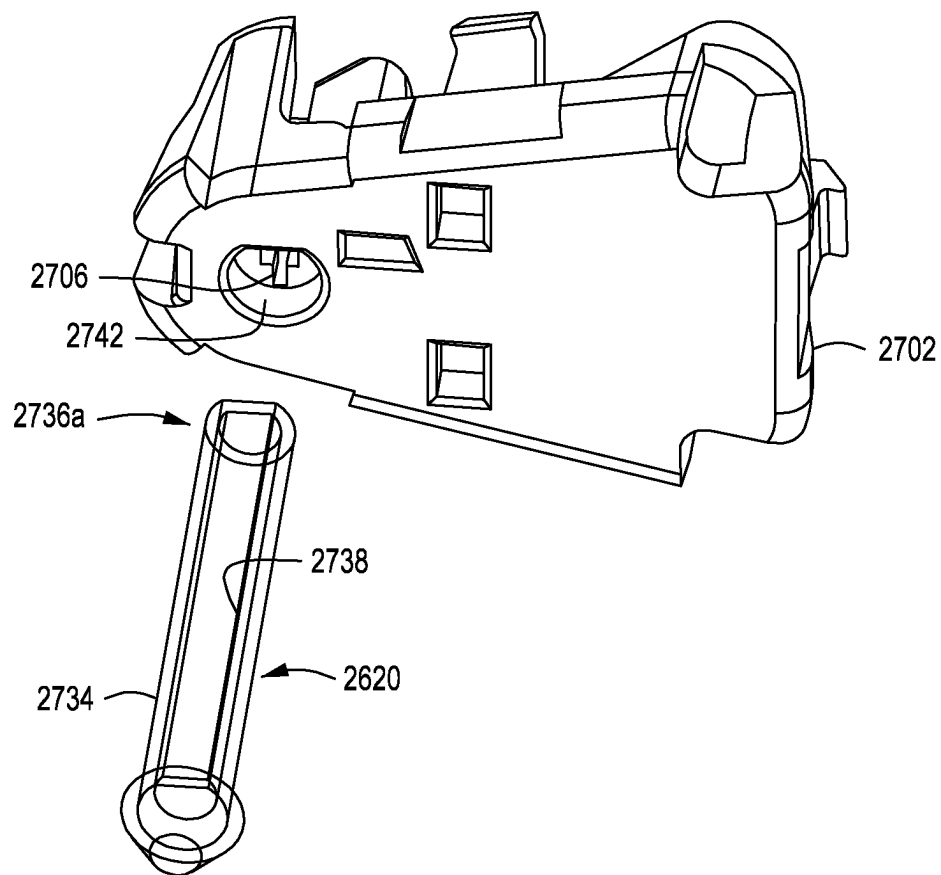


FIG. 27C

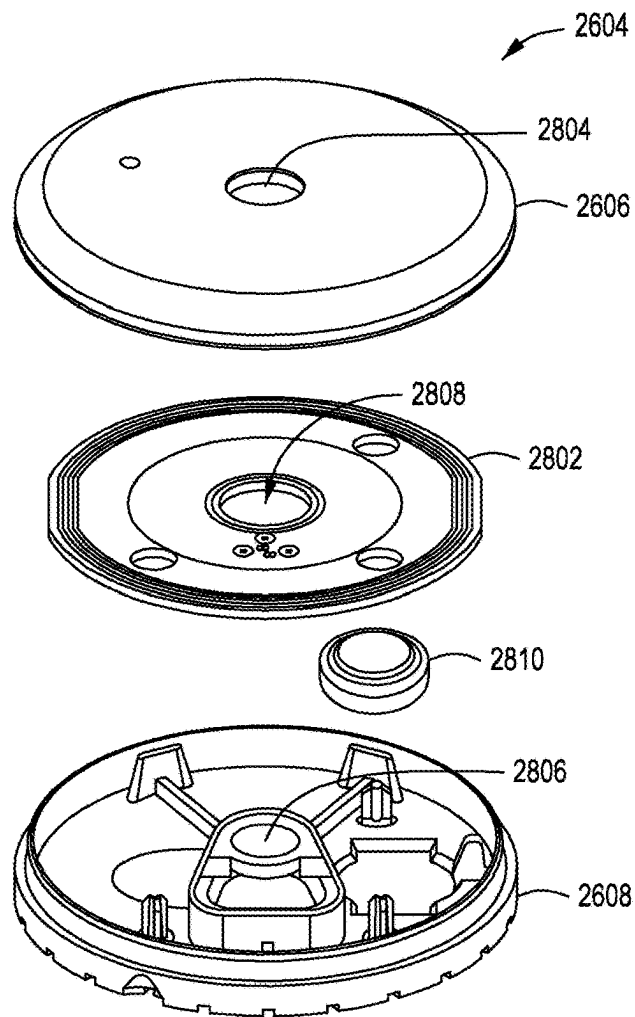


FIG. 28A

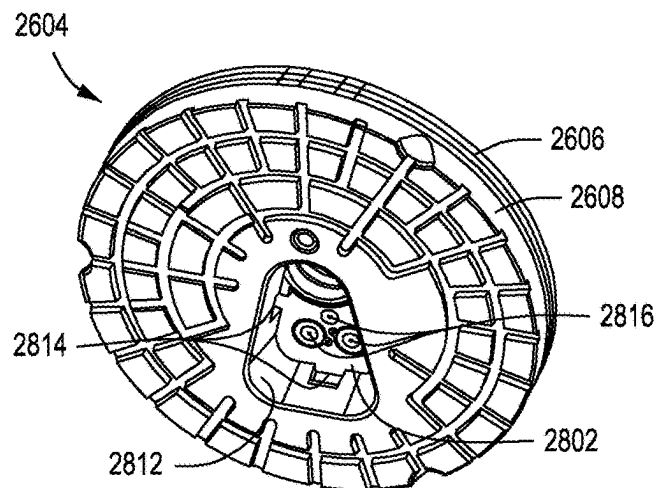


FIG. 28B

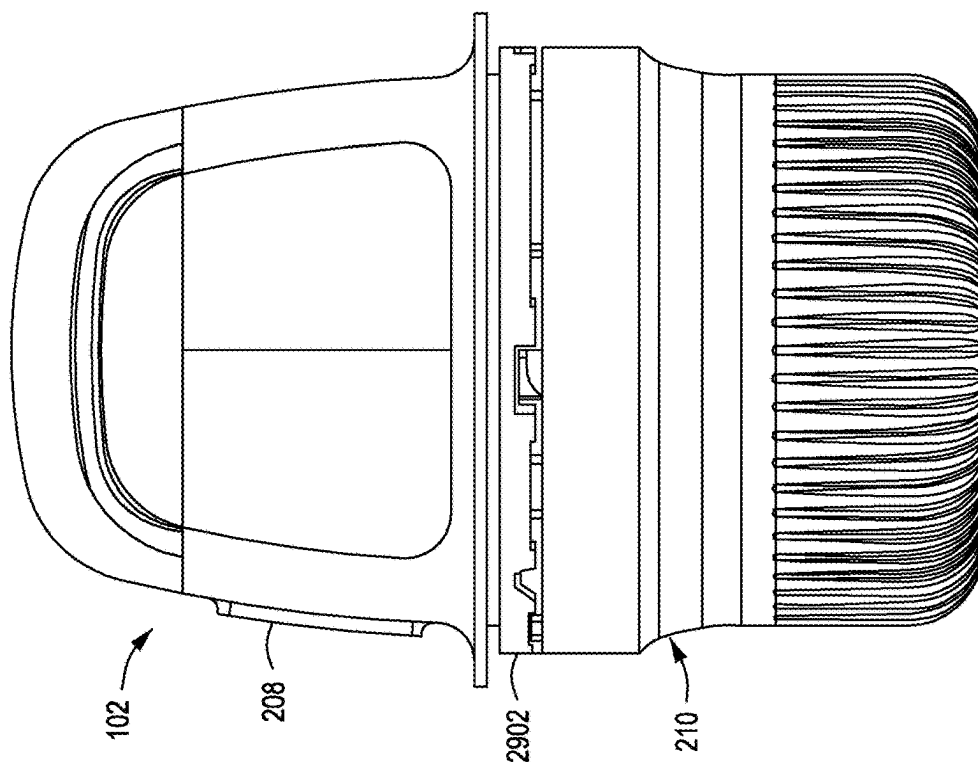


FIG. 29A

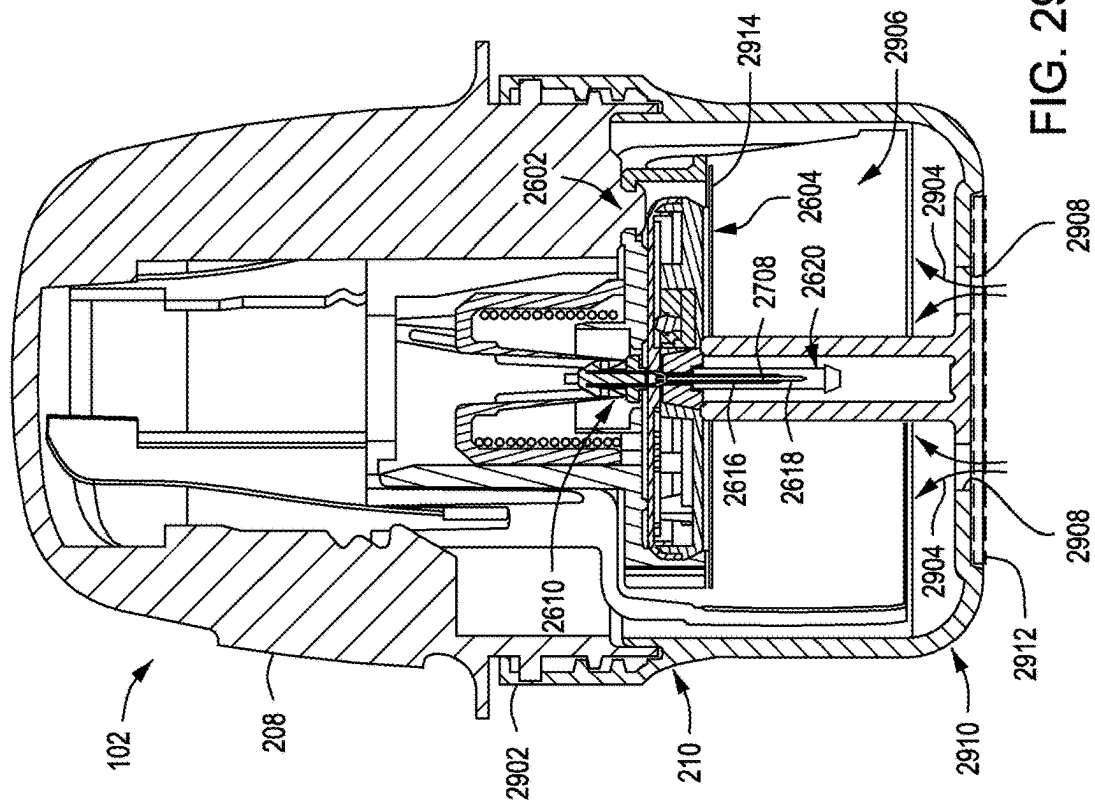


FIG. 29B

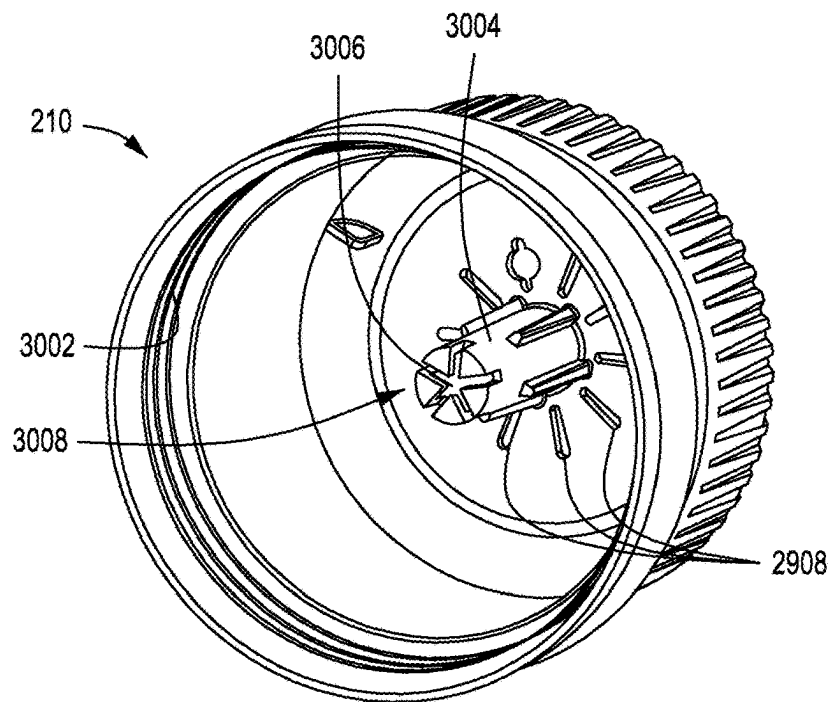


FIG. 30

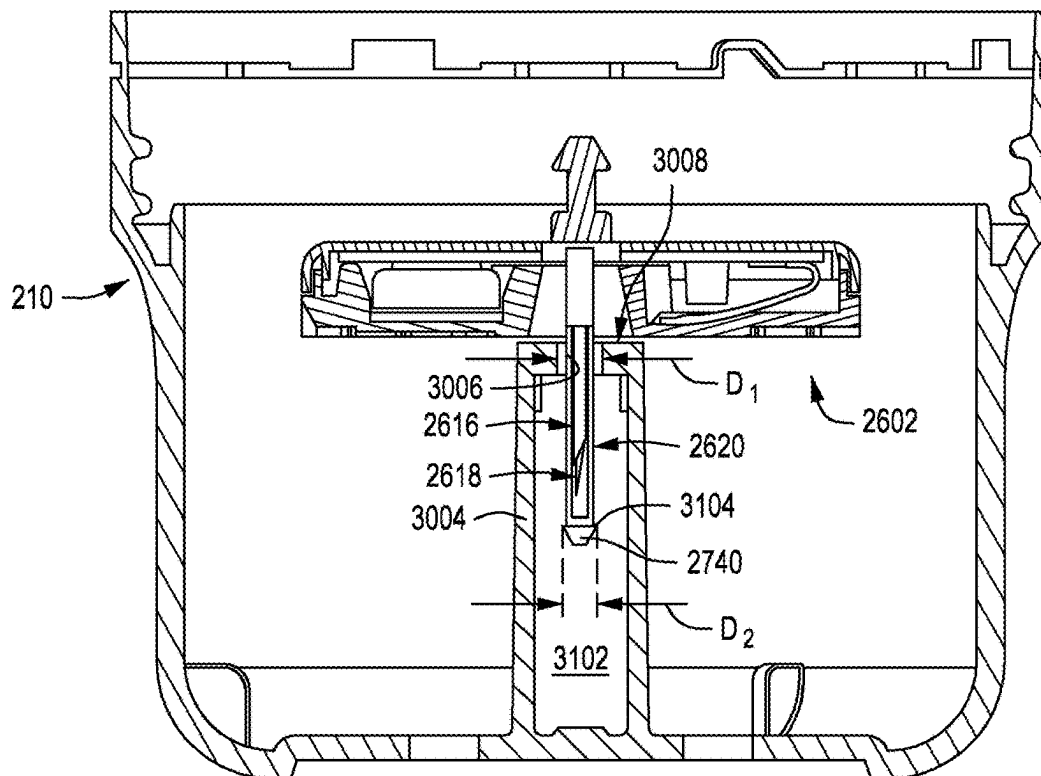


FIG. 31

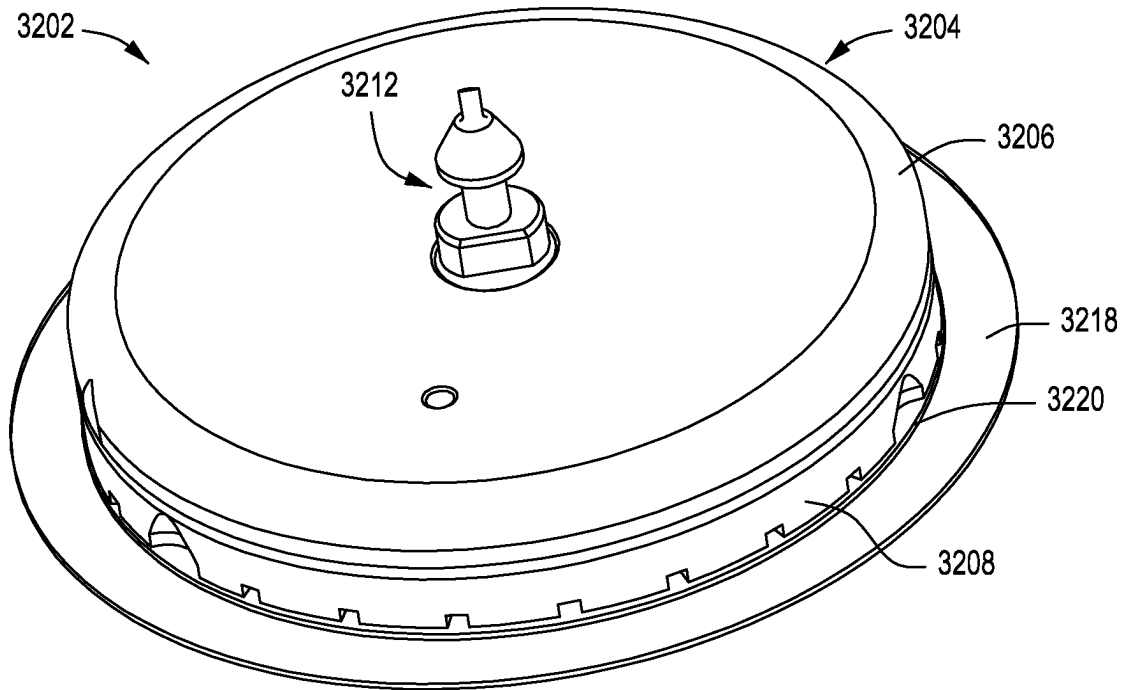


FIG. 32A

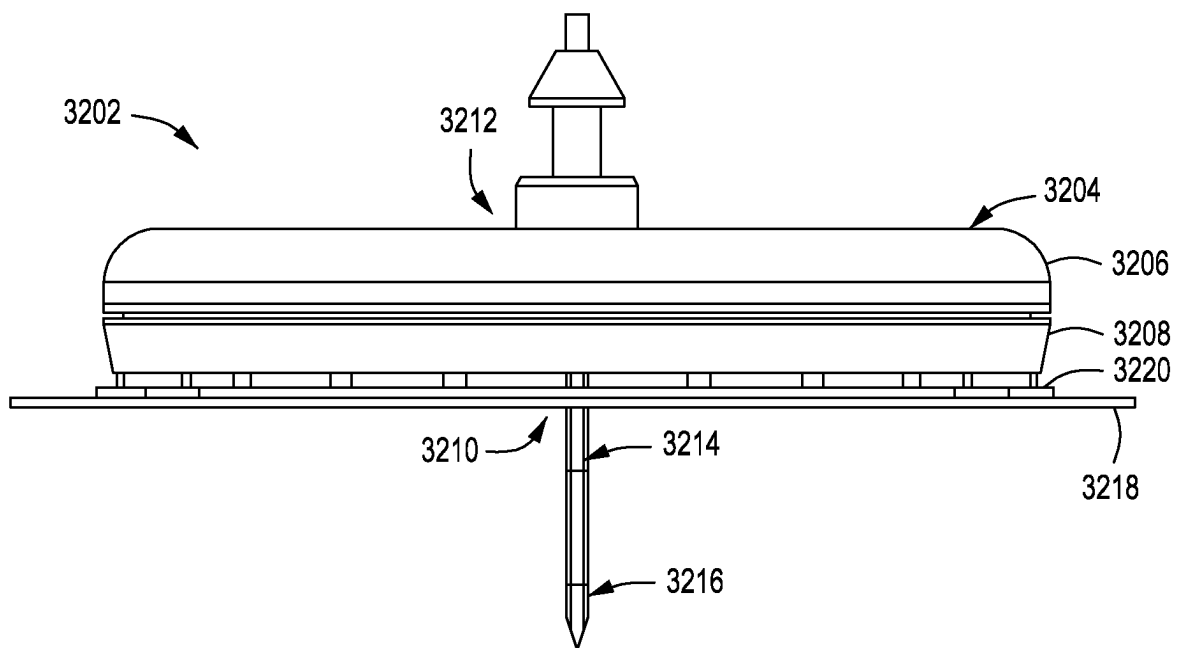


FIG. 32B

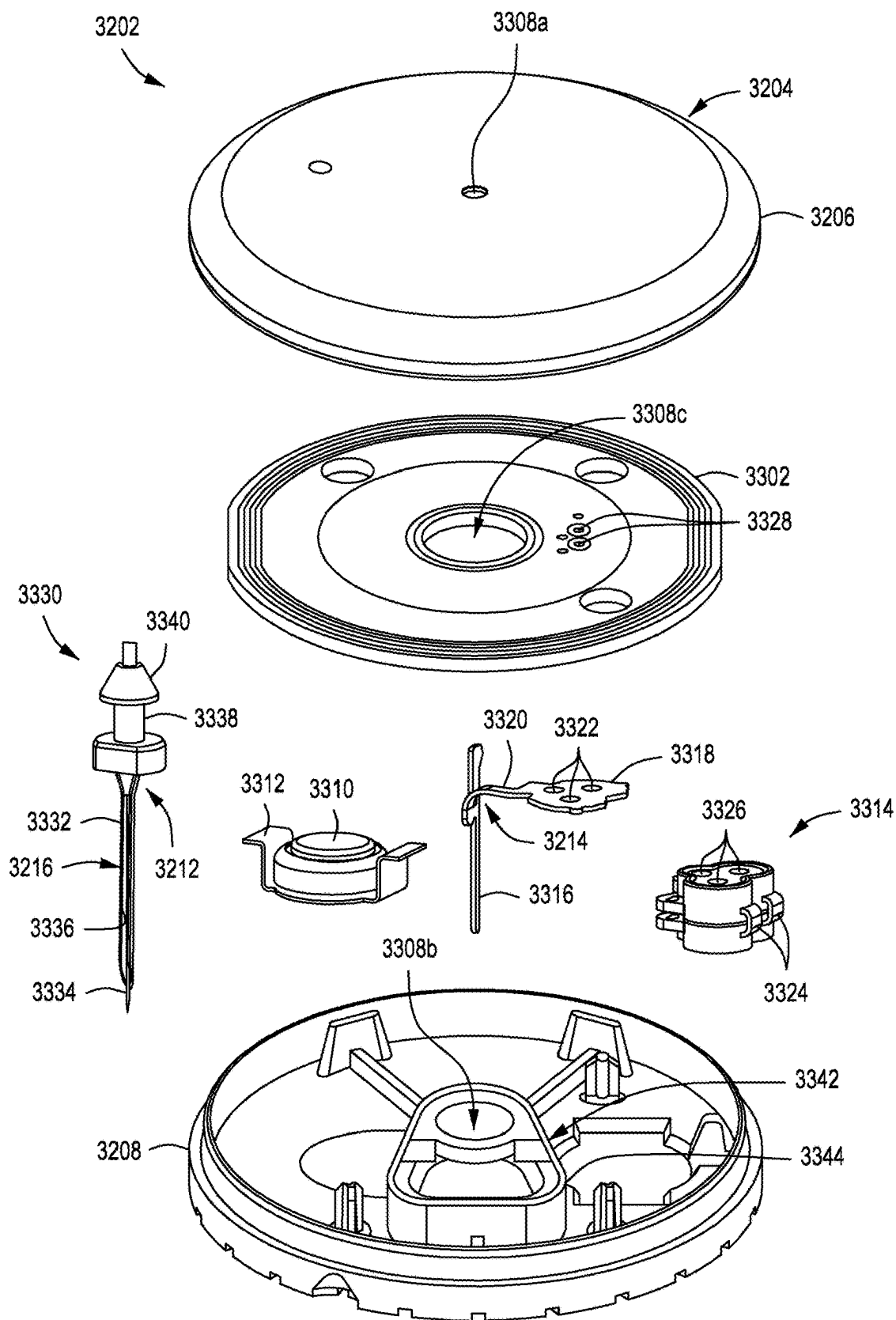


FIG. 33A

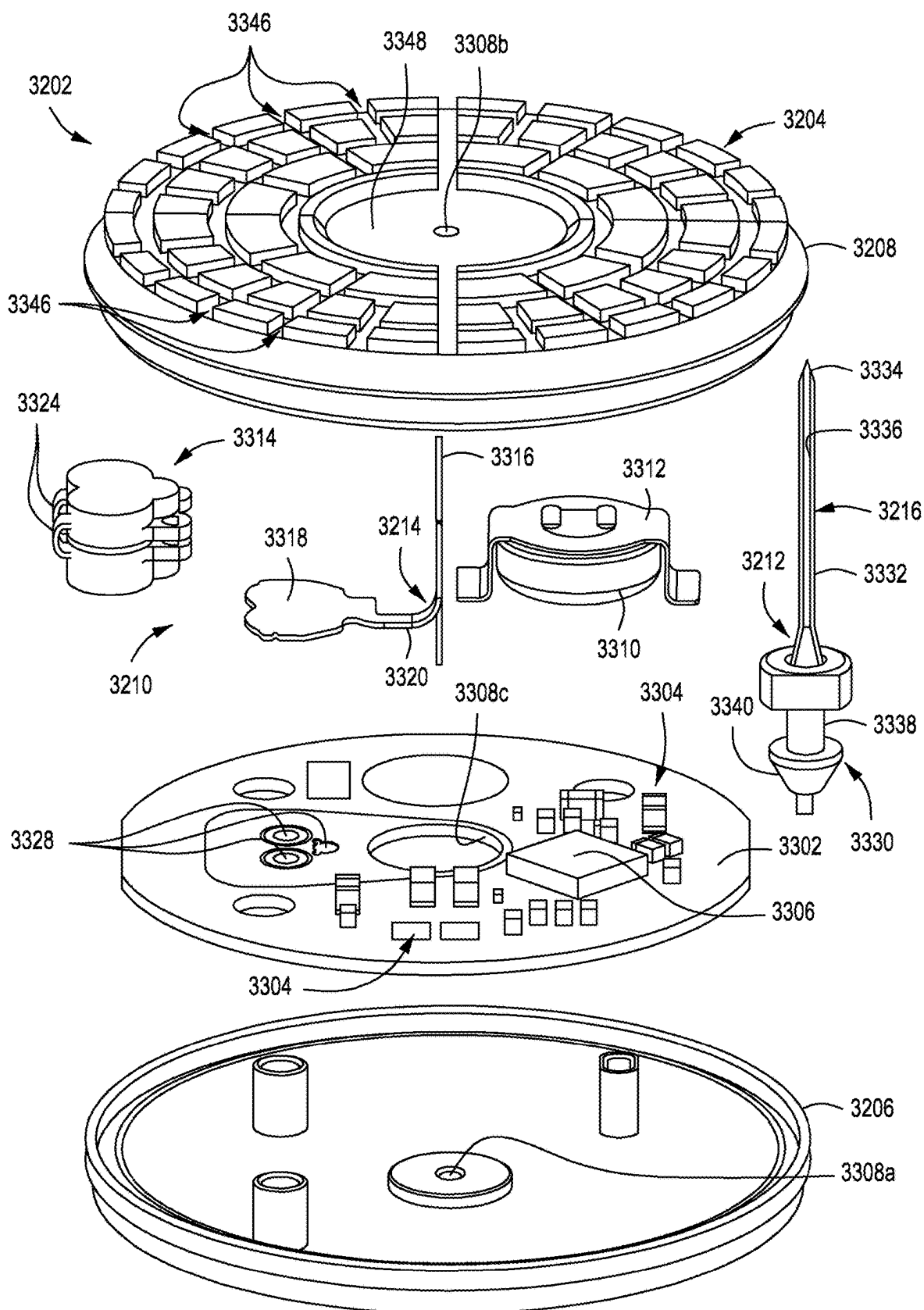


FIG. 33B

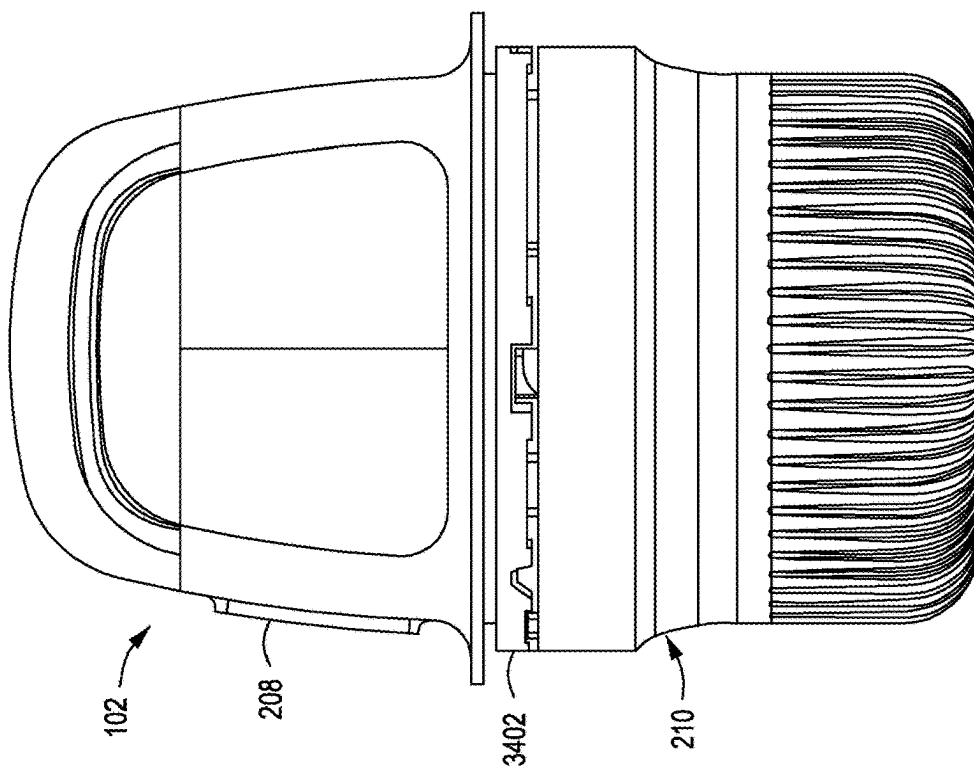


FIG. 34A

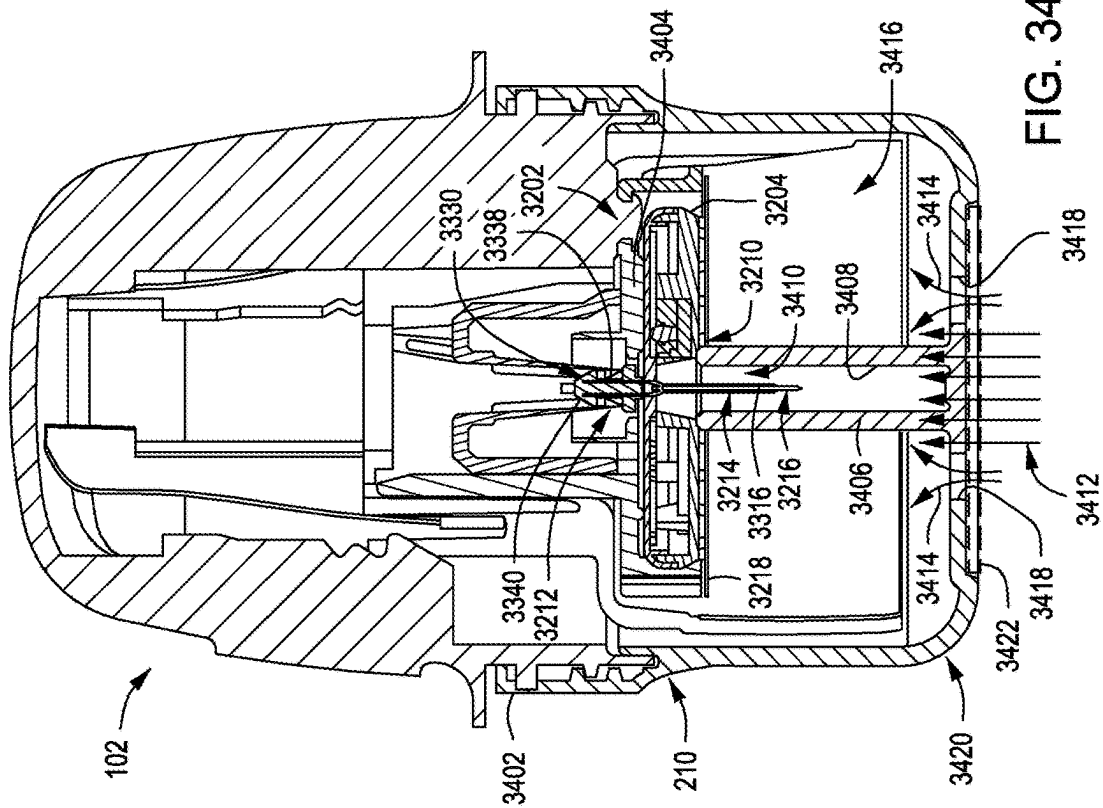


FIG. 34B

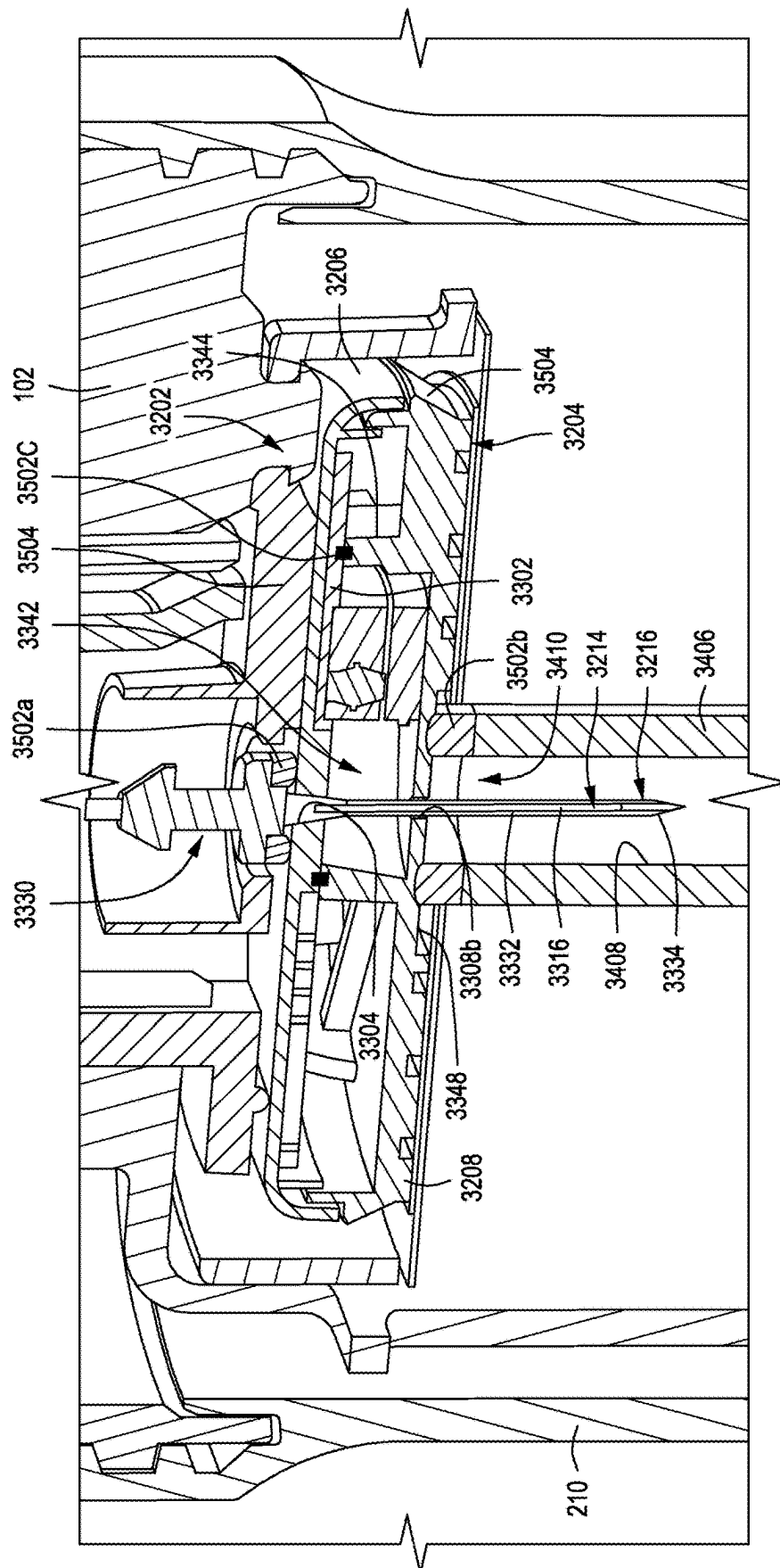


FIG. 35

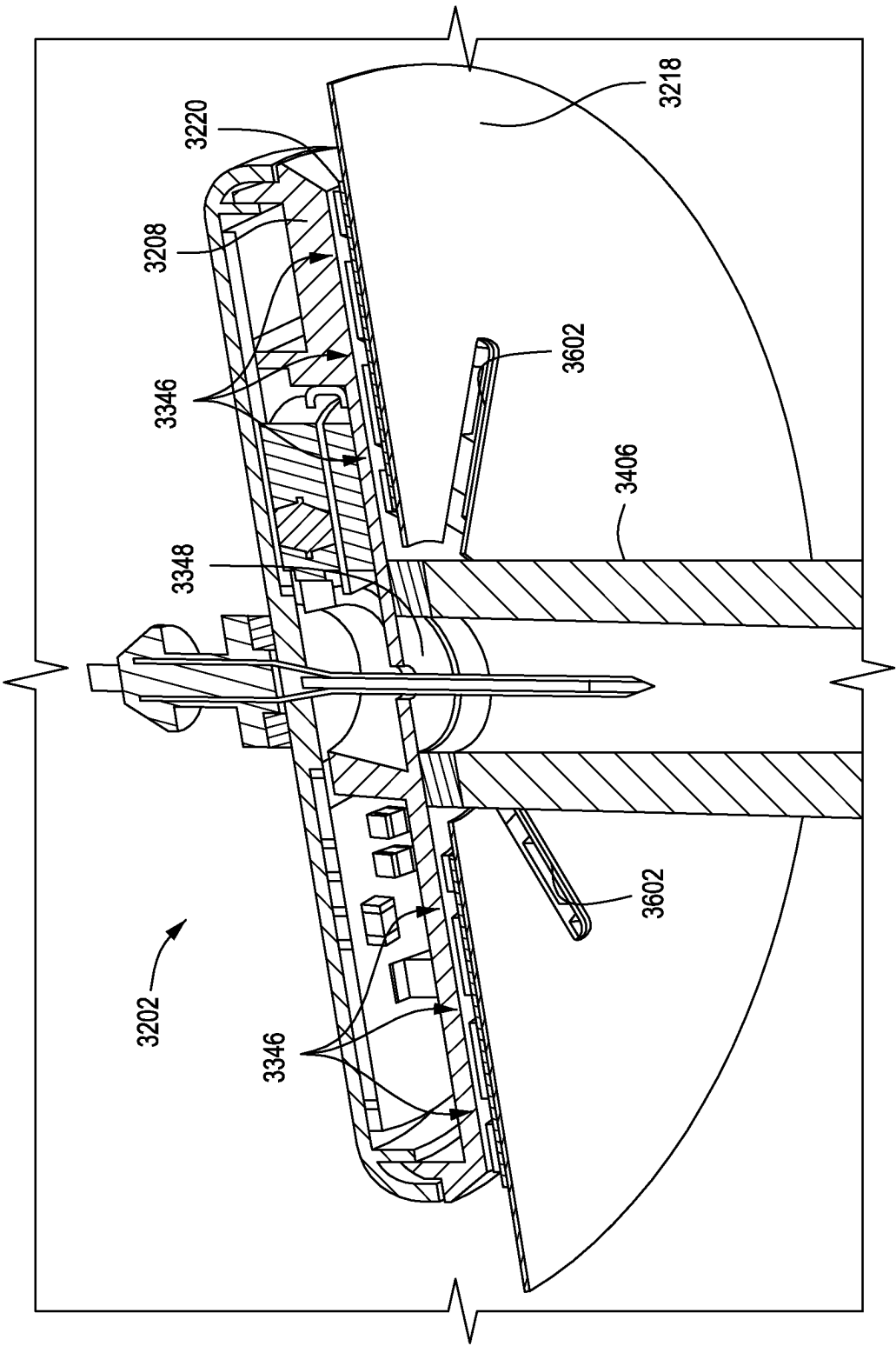


FIG. 36

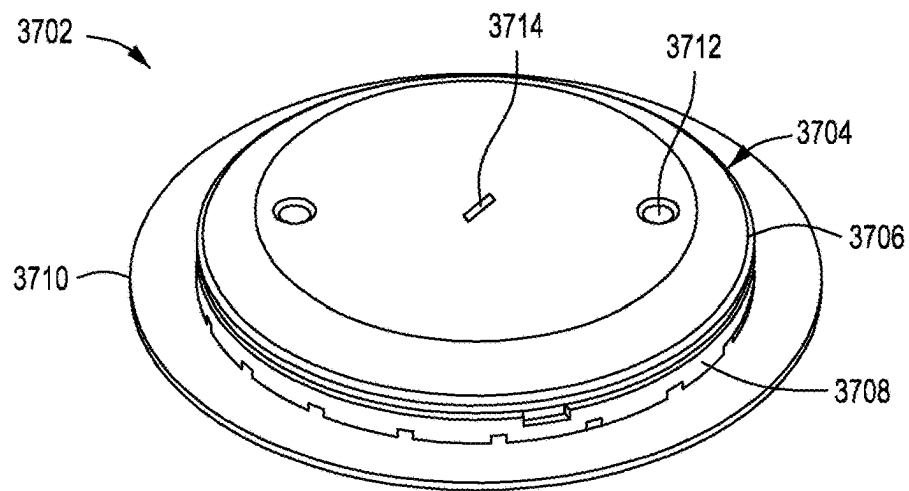


FIG. 37A

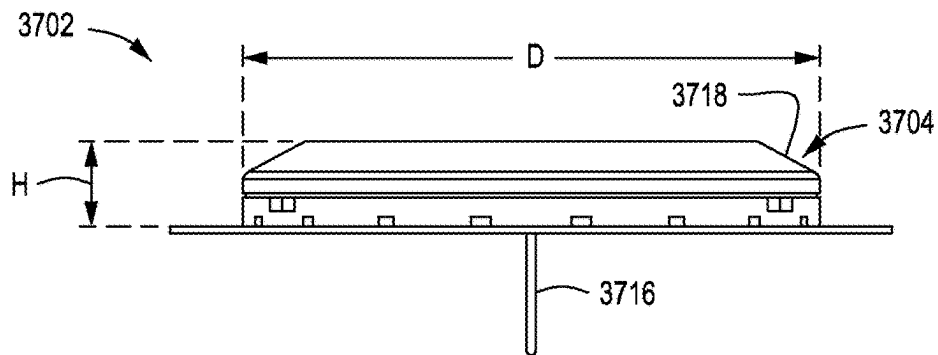


FIG. 37B

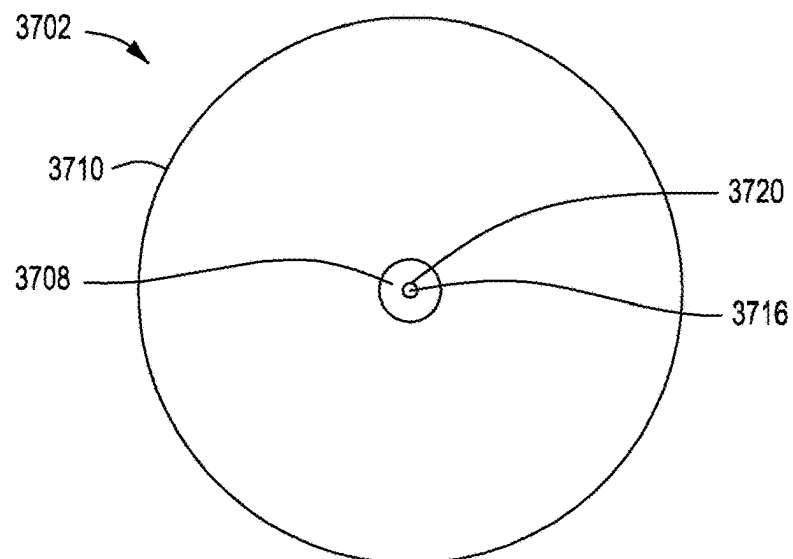


FIG. 37C

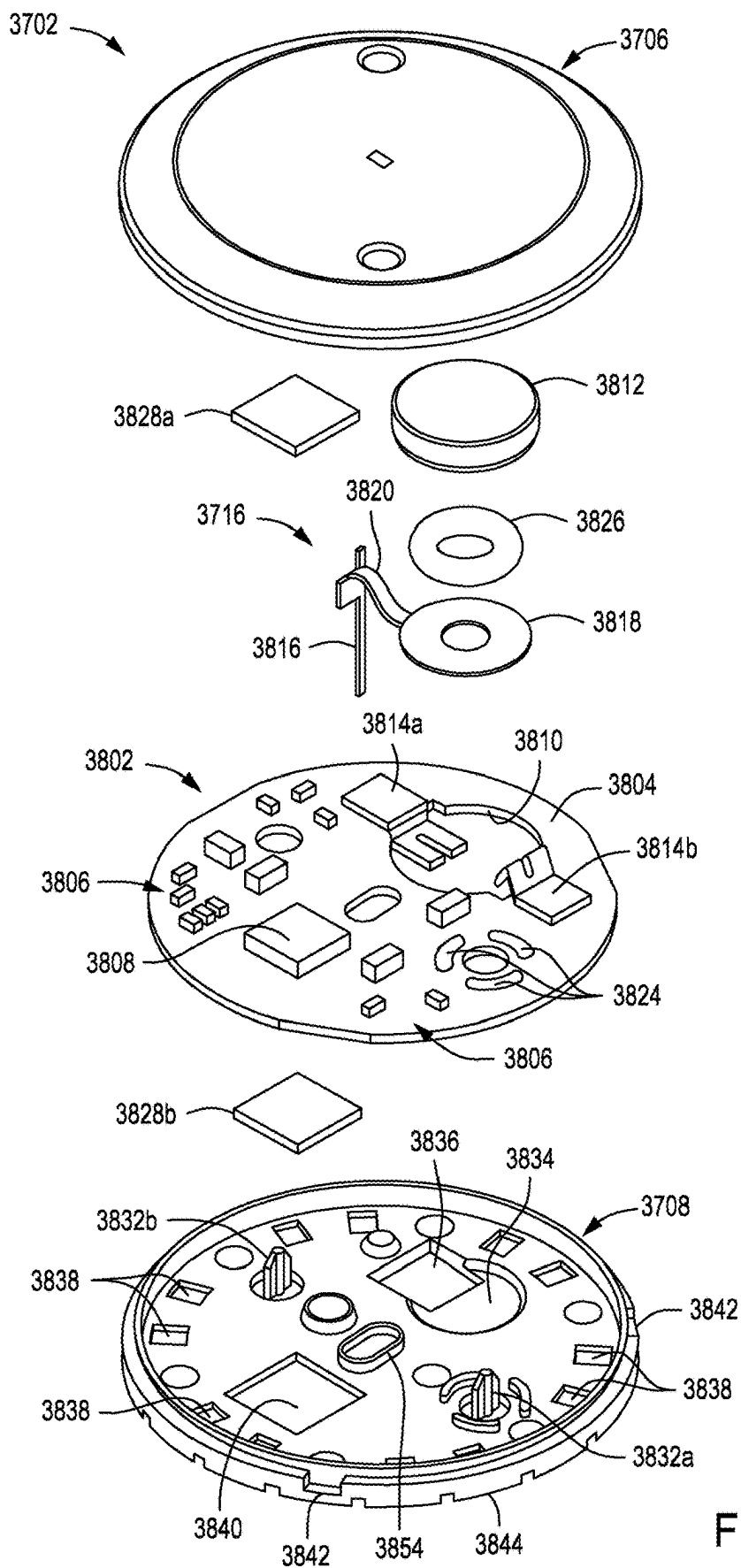


FIG. 38A

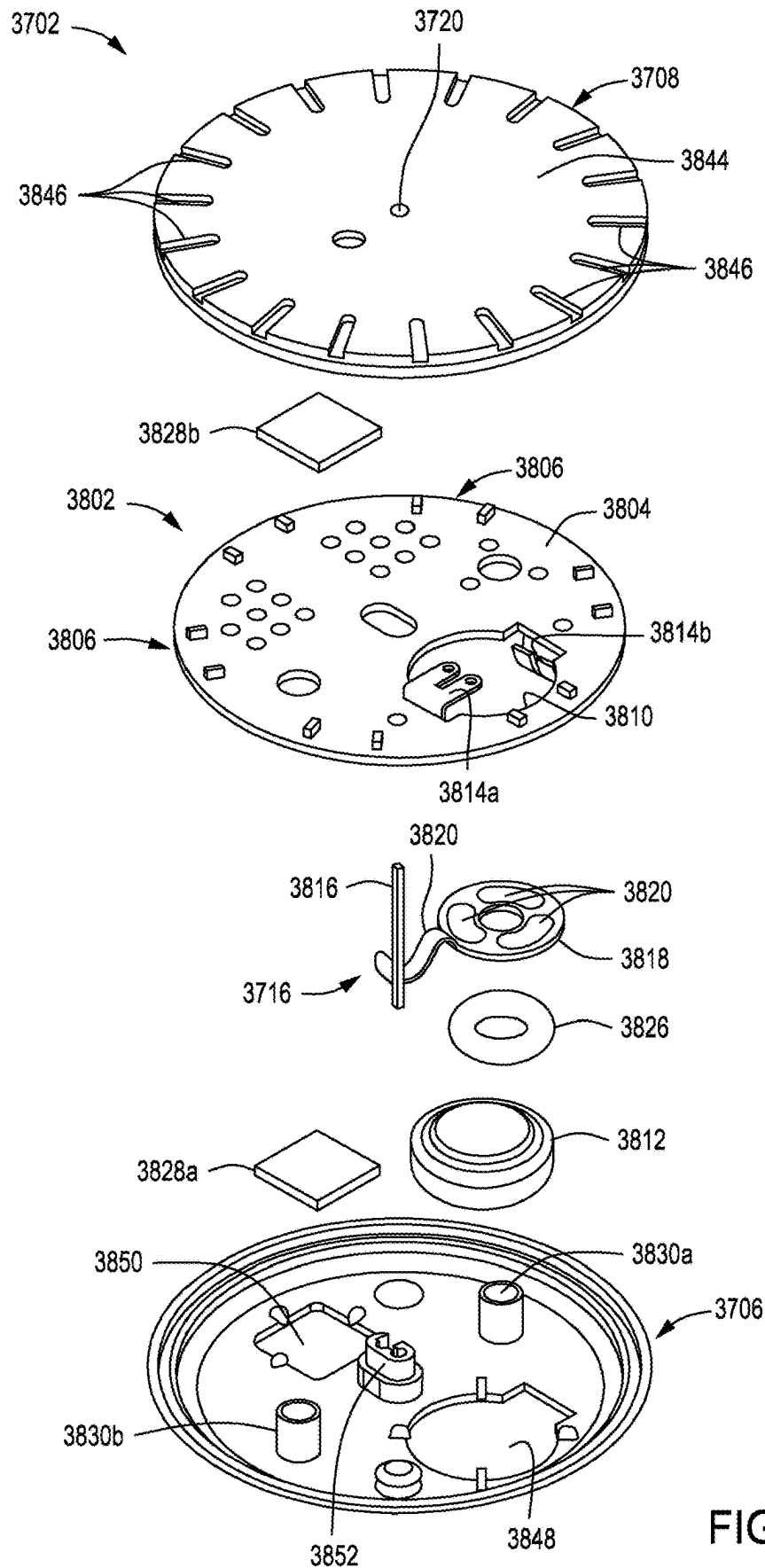


FIG. 38B

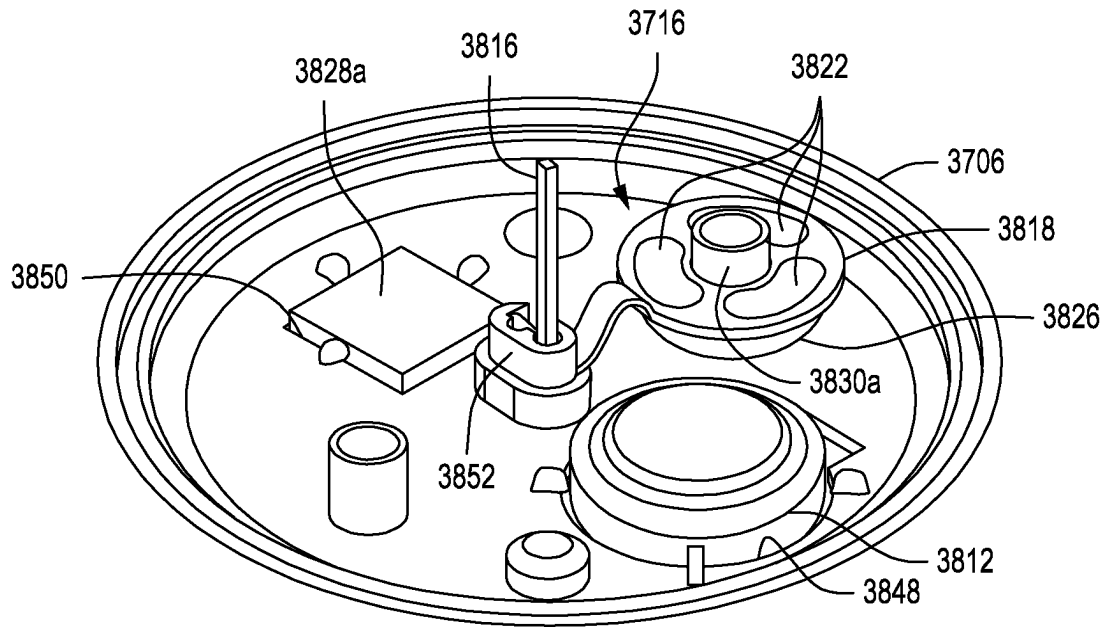


FIG. 39A

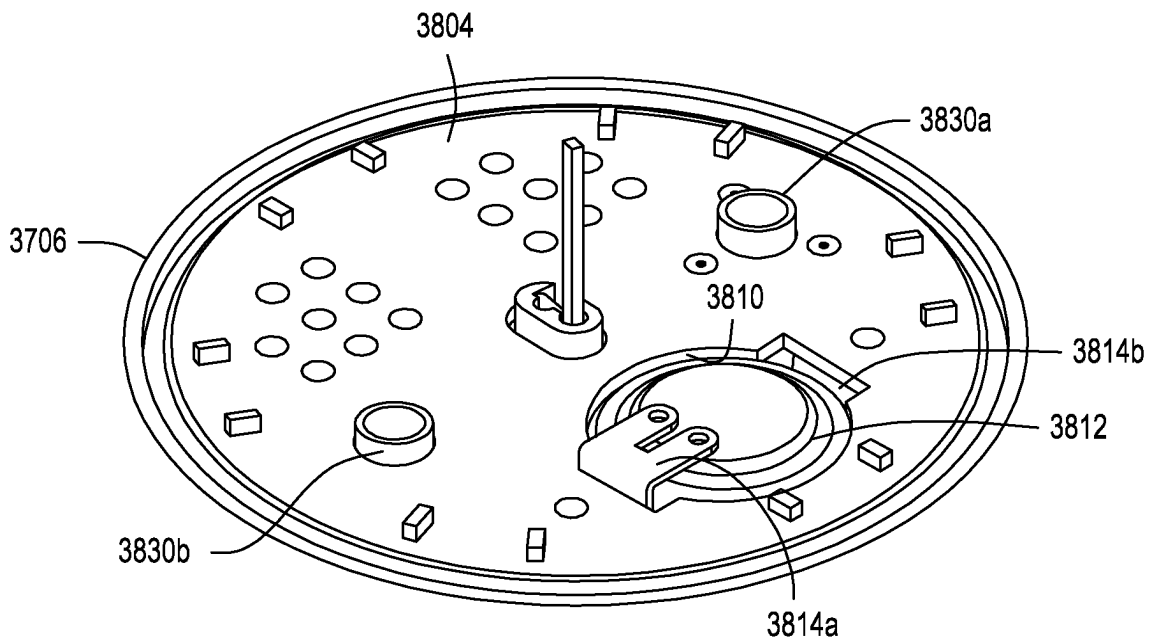


FIG. 39B

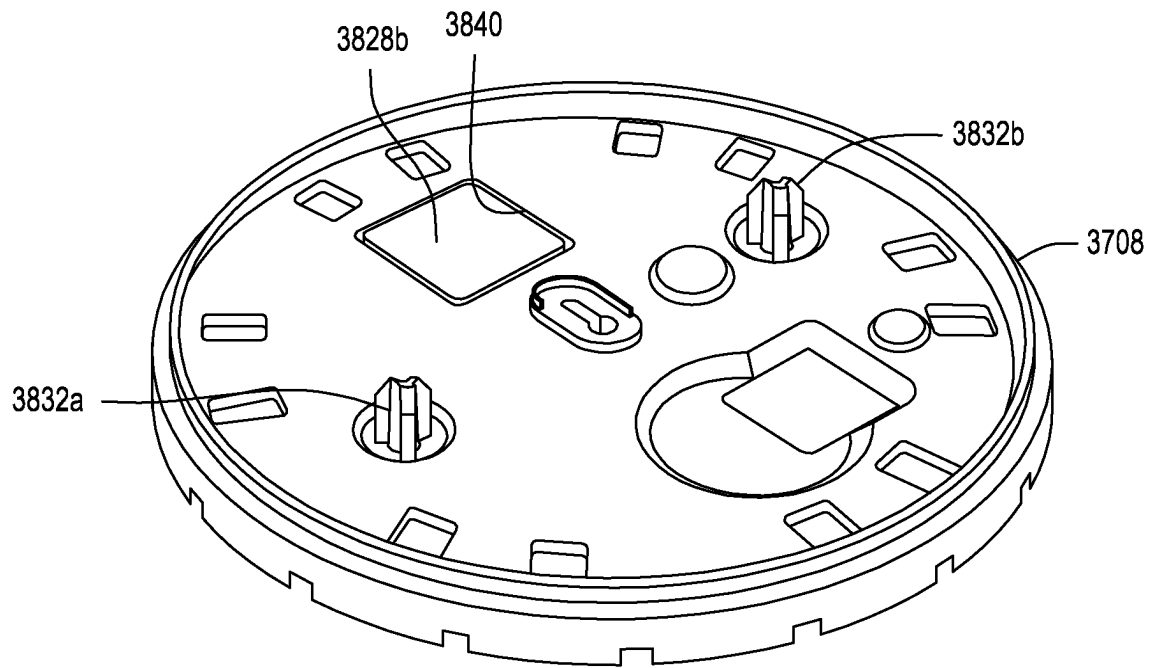


FIG. 39C

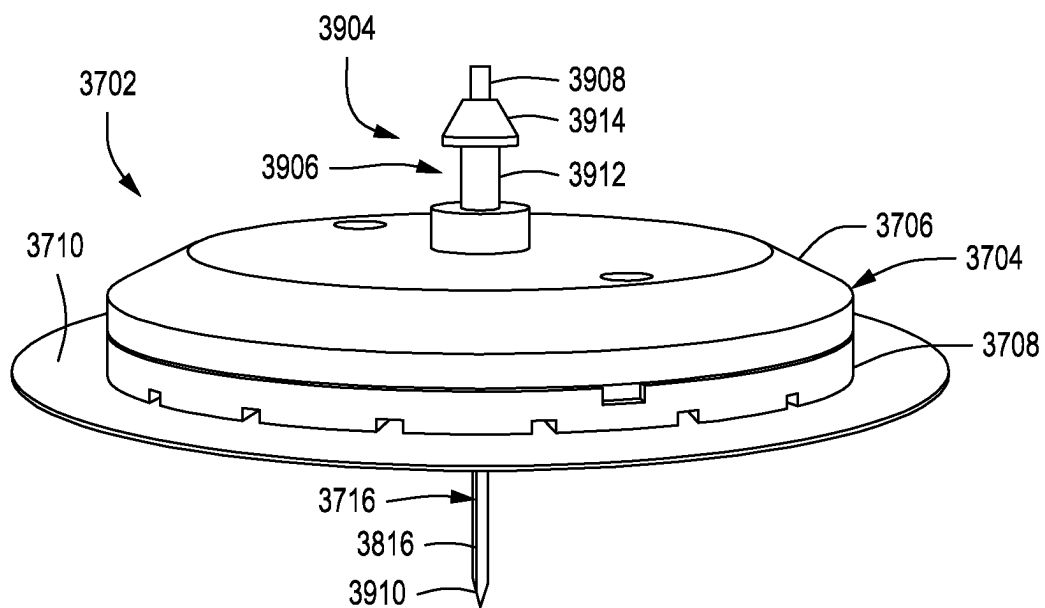


FIG. 39D

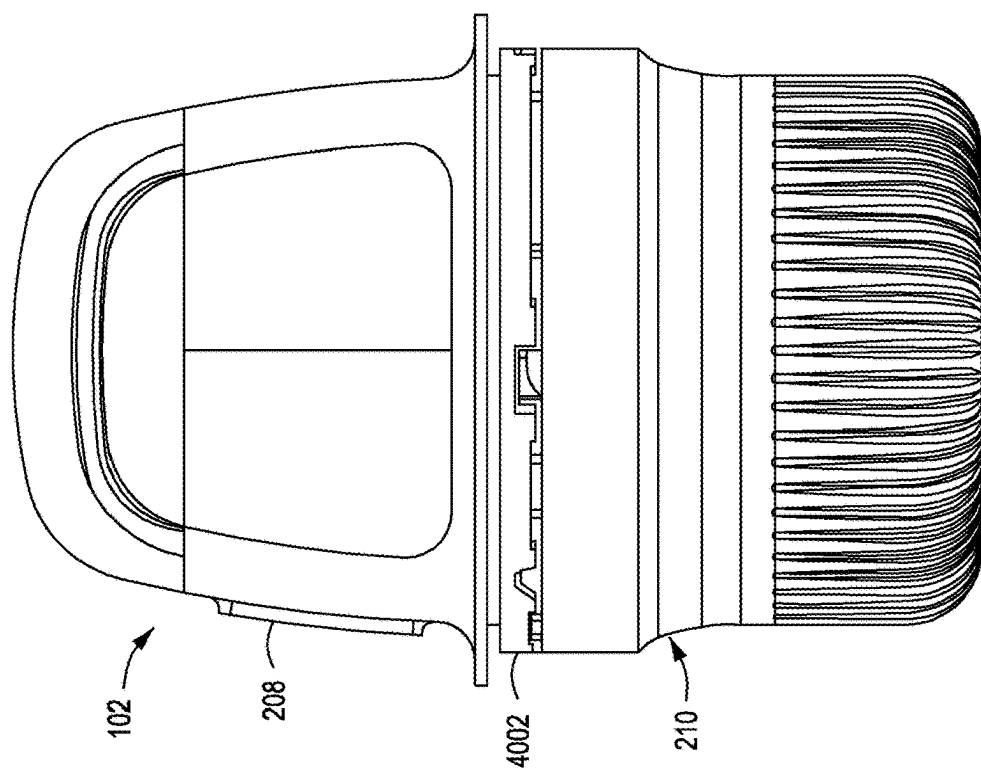


FIG. 40A

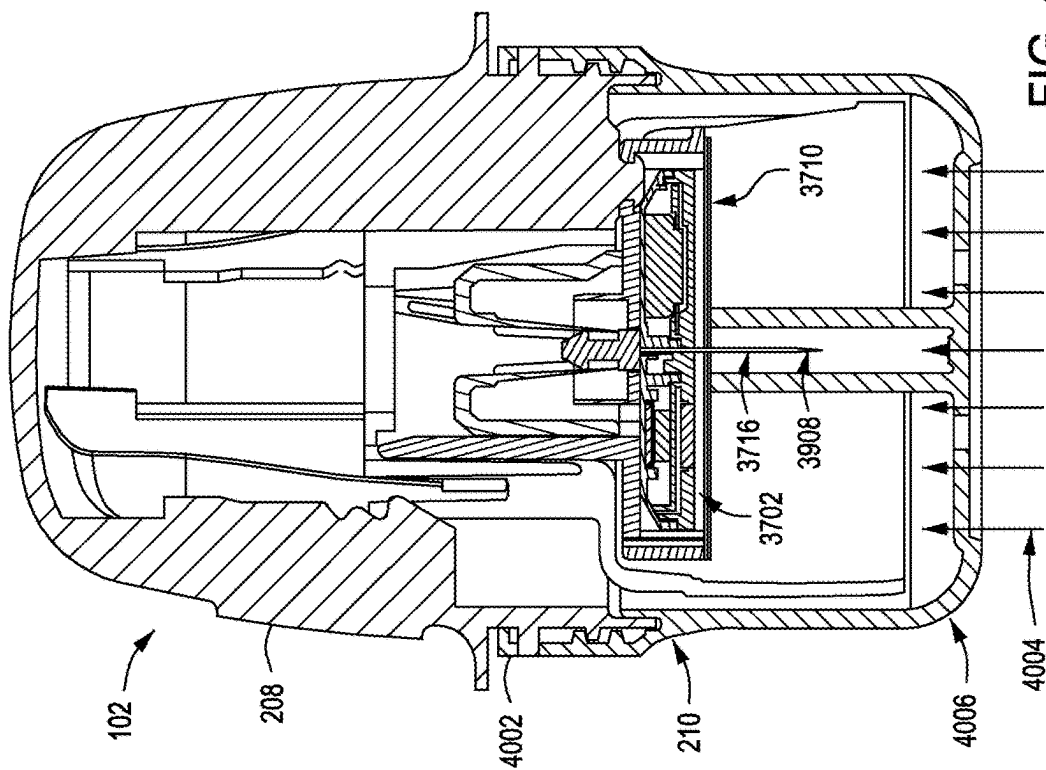


FIG. 40B

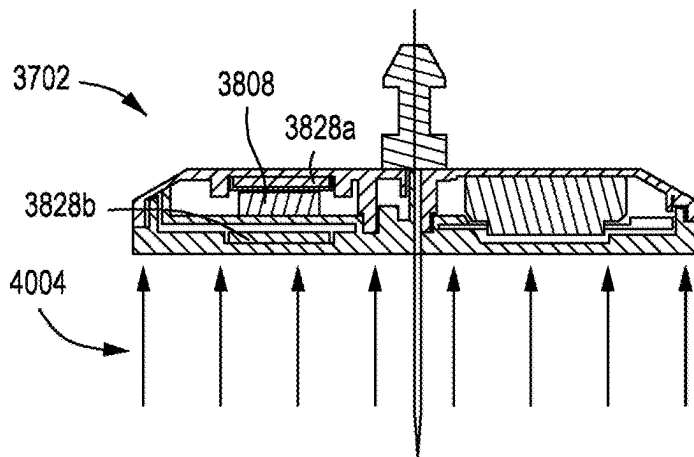


FIG. 41A

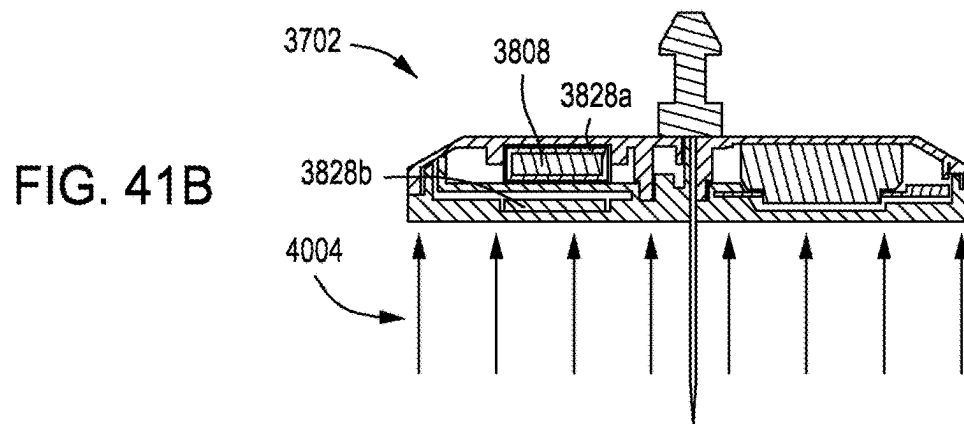


FIG. 41B

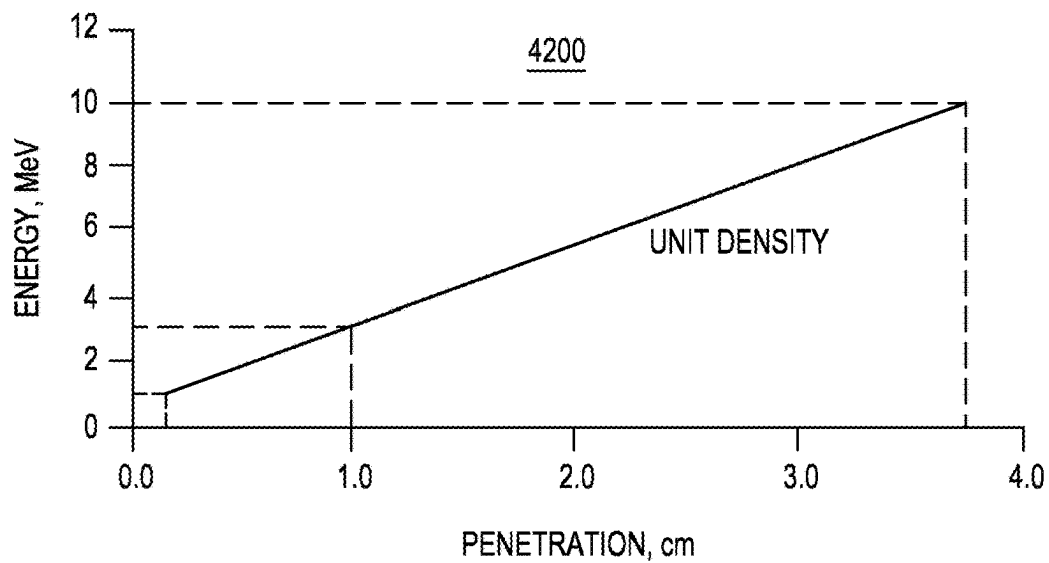


FIG. 42

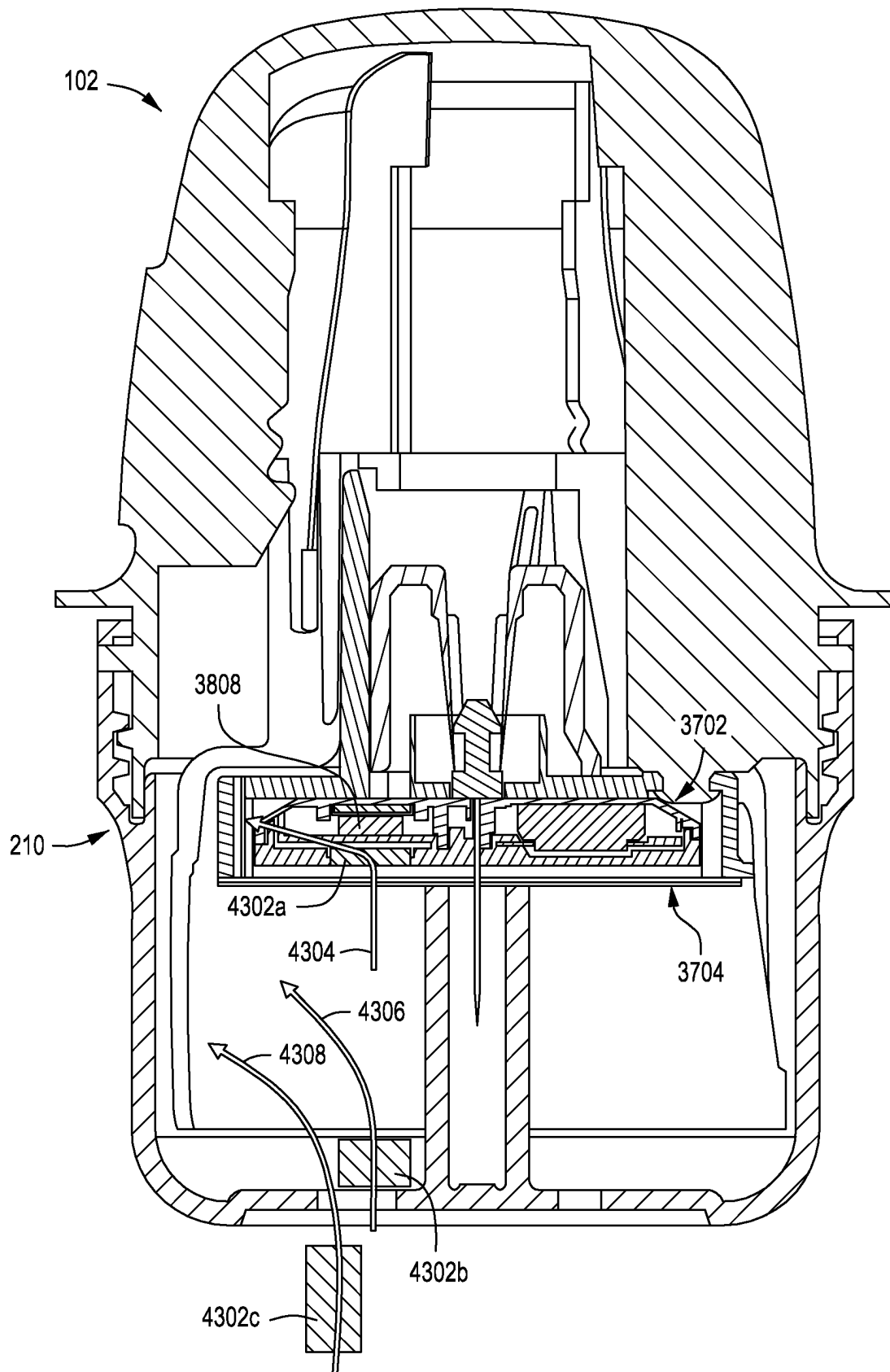


FIG. 43

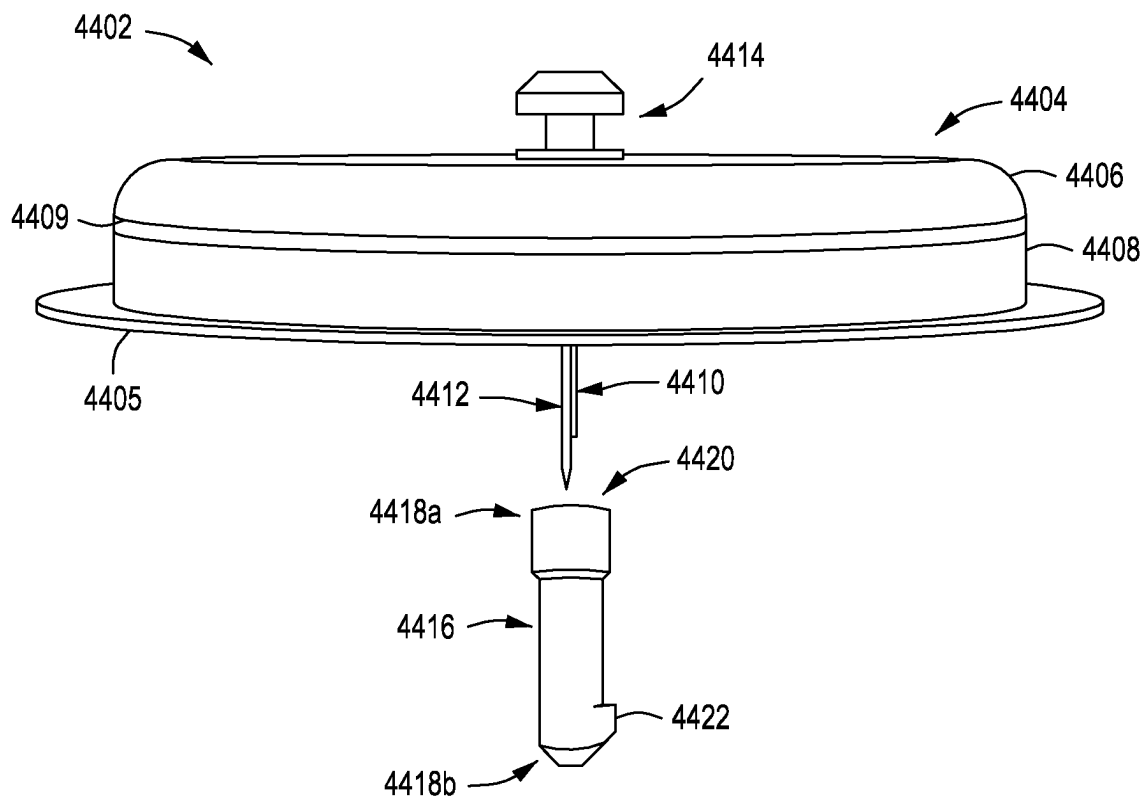


FIG. 44

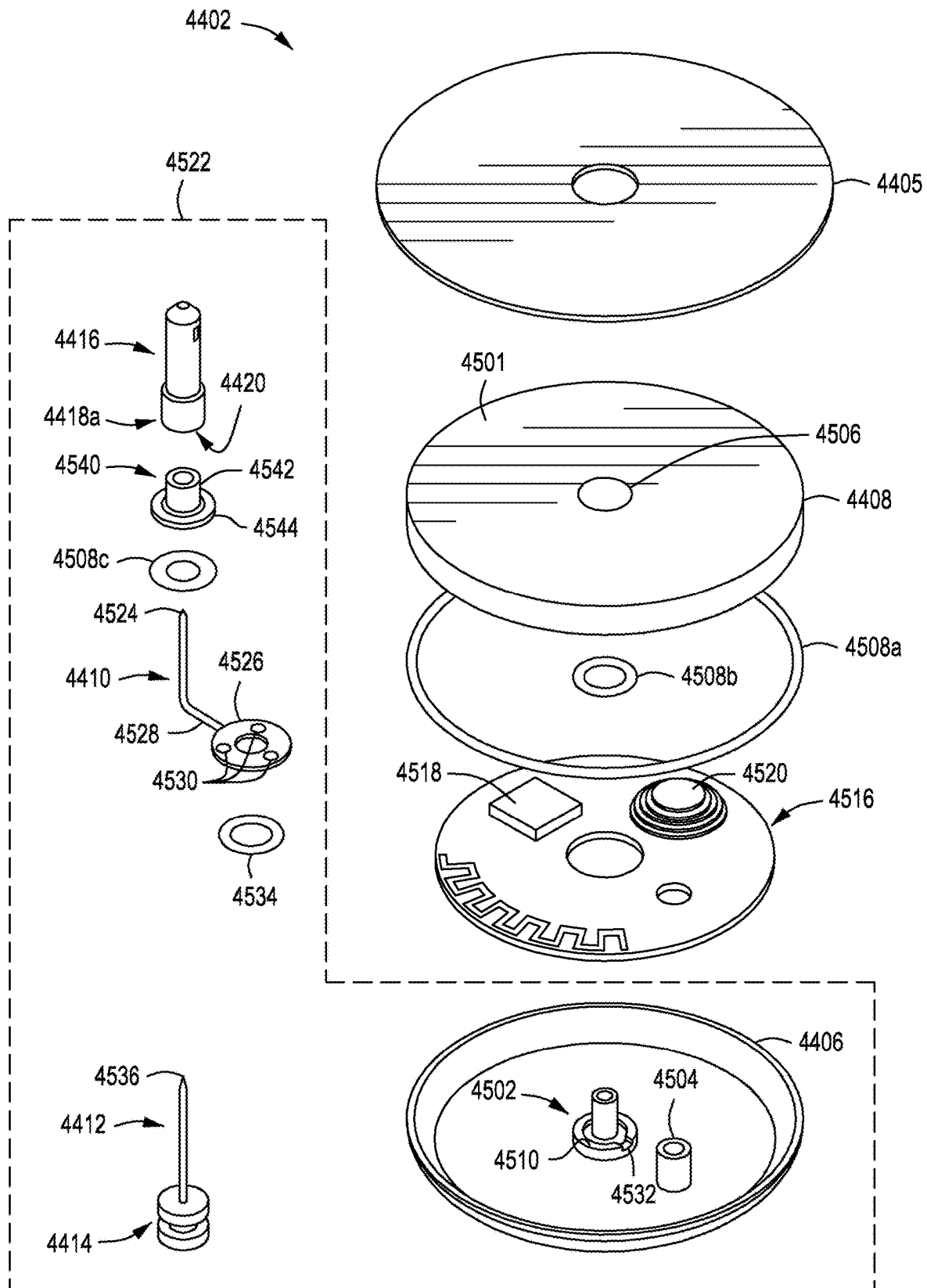


FIG. 45

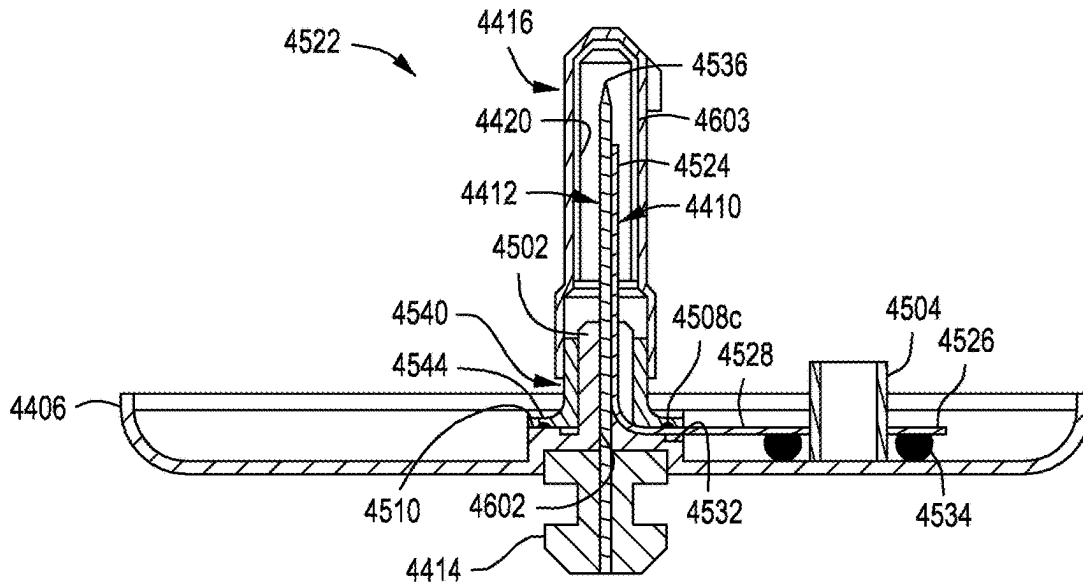


FIG. 46A

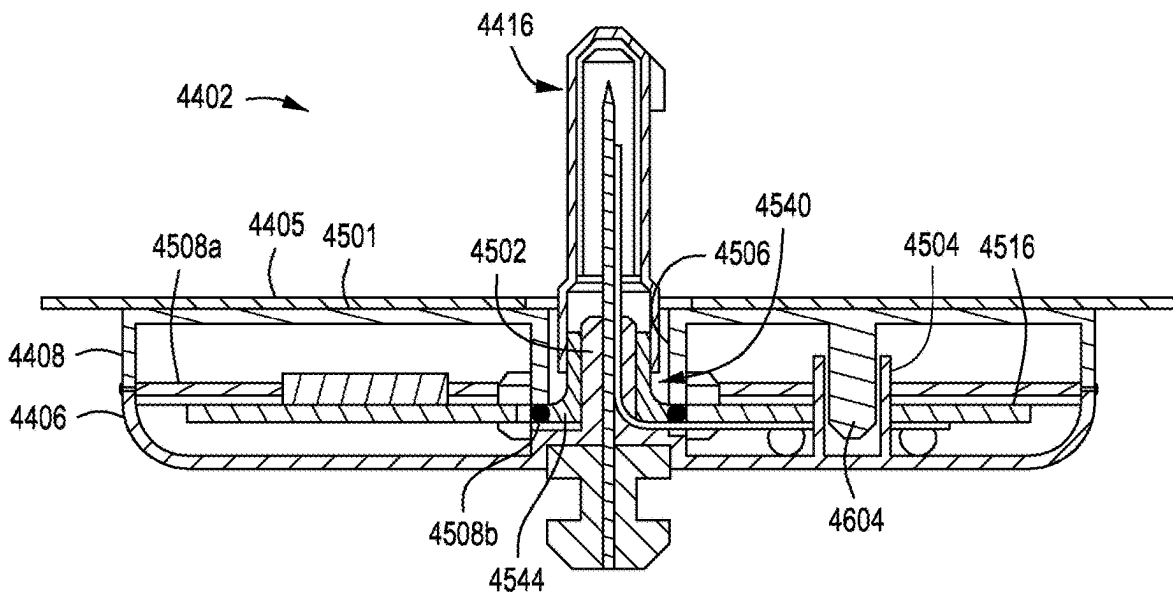


FIG. 46B

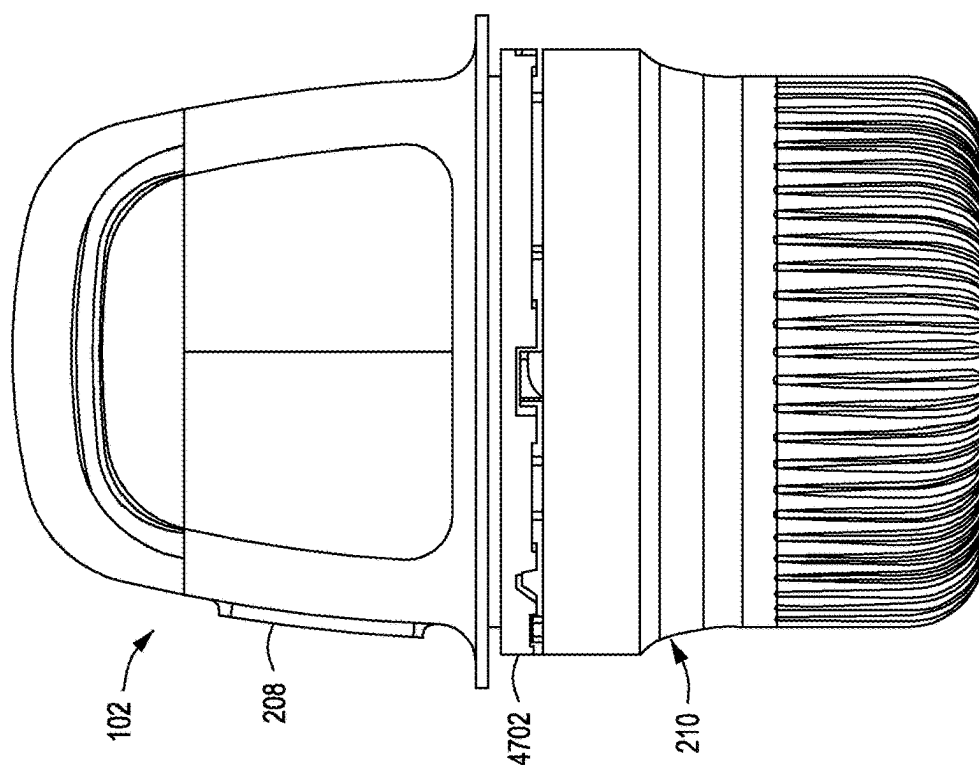


FIG. 47A

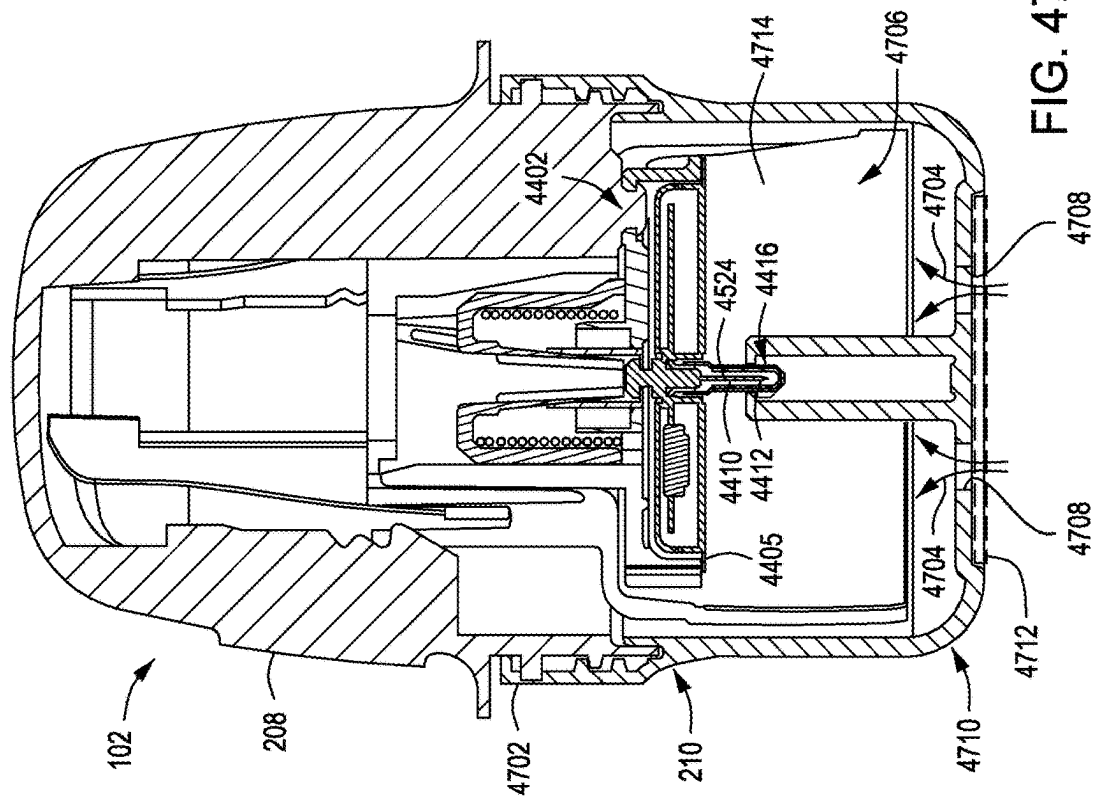


FIG. 47B

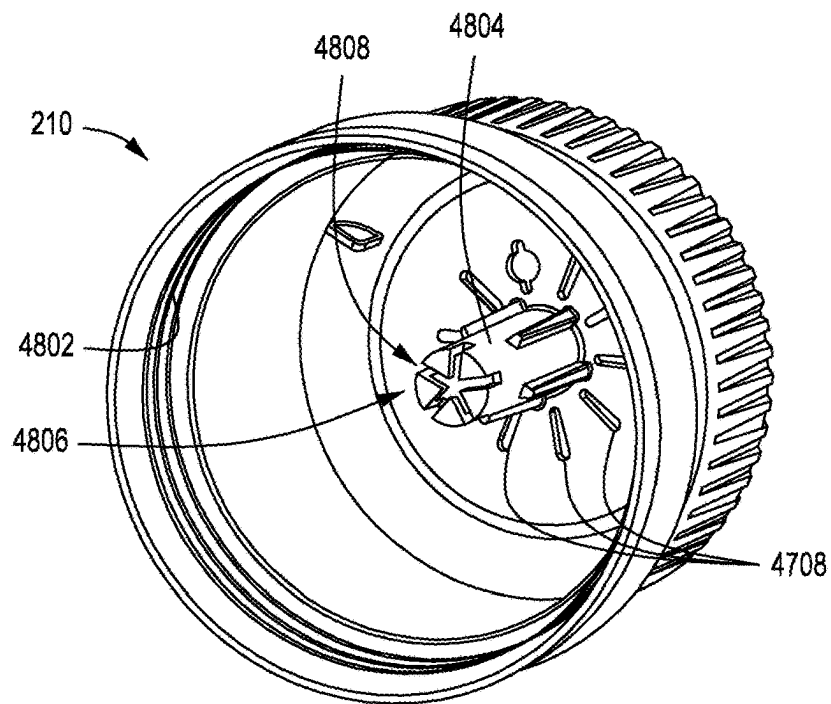


FIG. 48

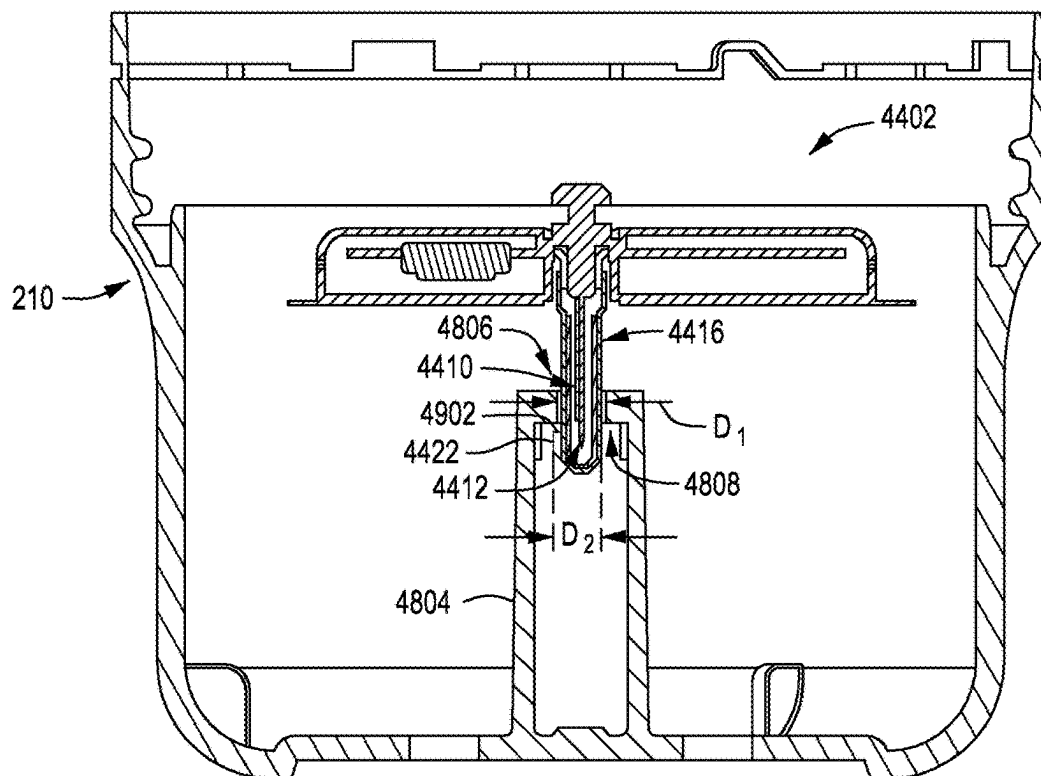


FIG. 49

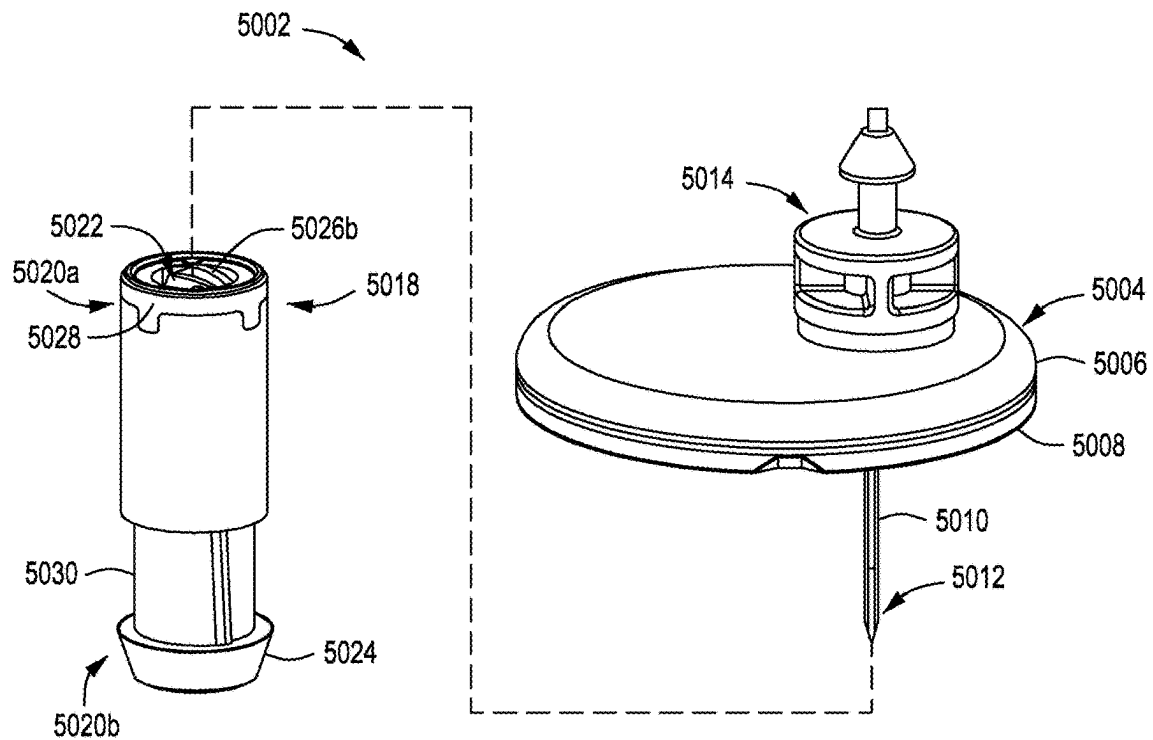


FIG. 50A

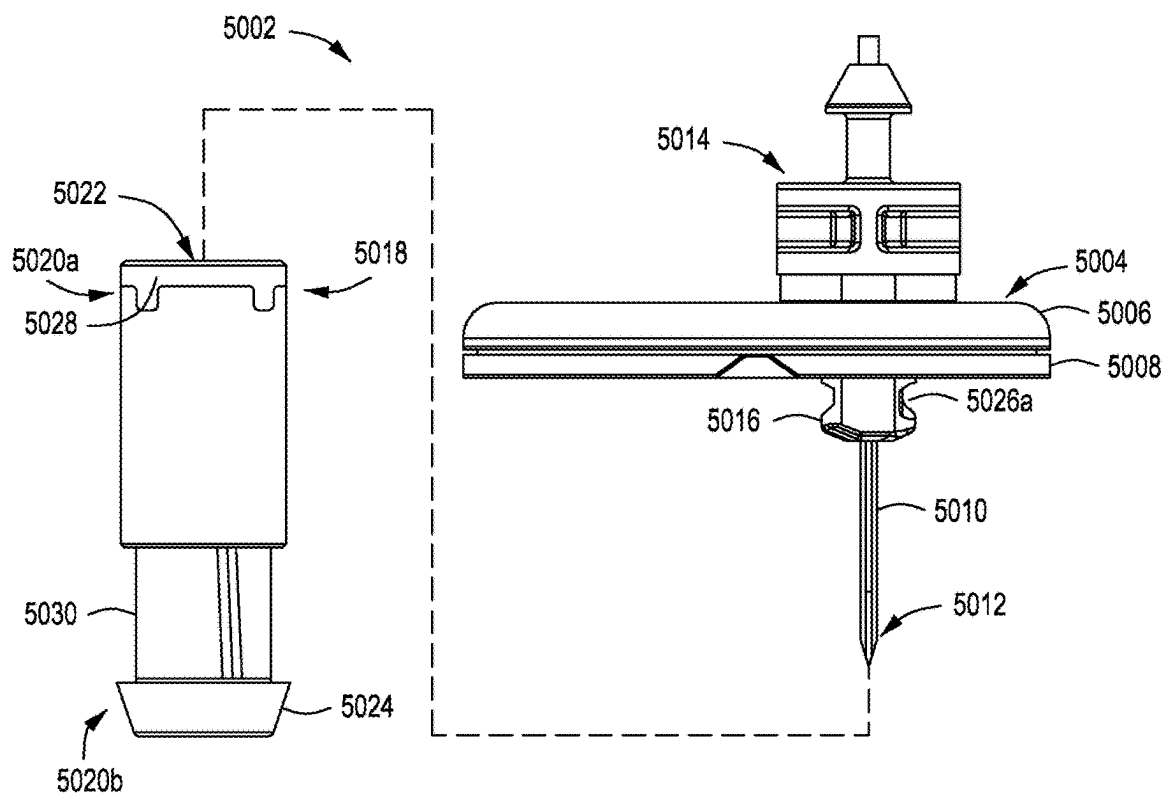


FIG. 50B

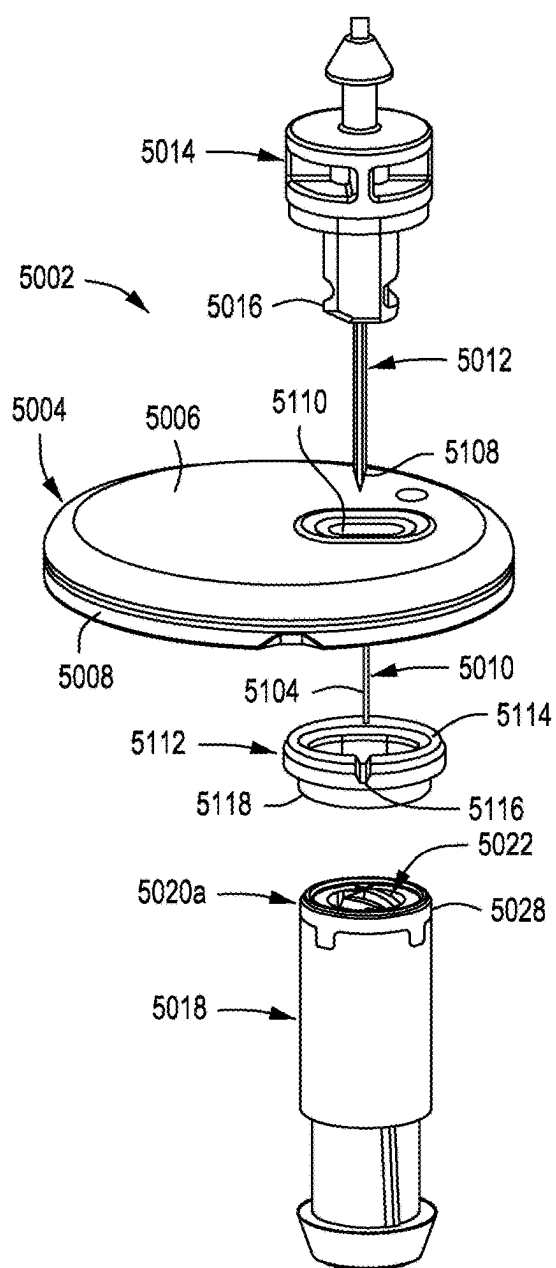


FIG. 51A

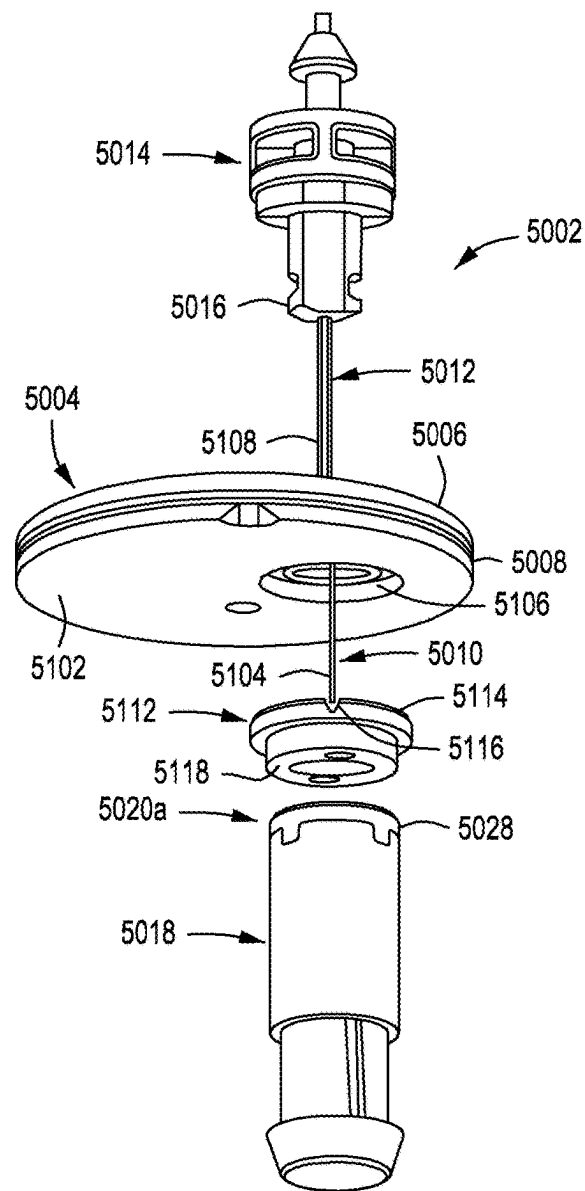


FIG. 51B

FIG. 52

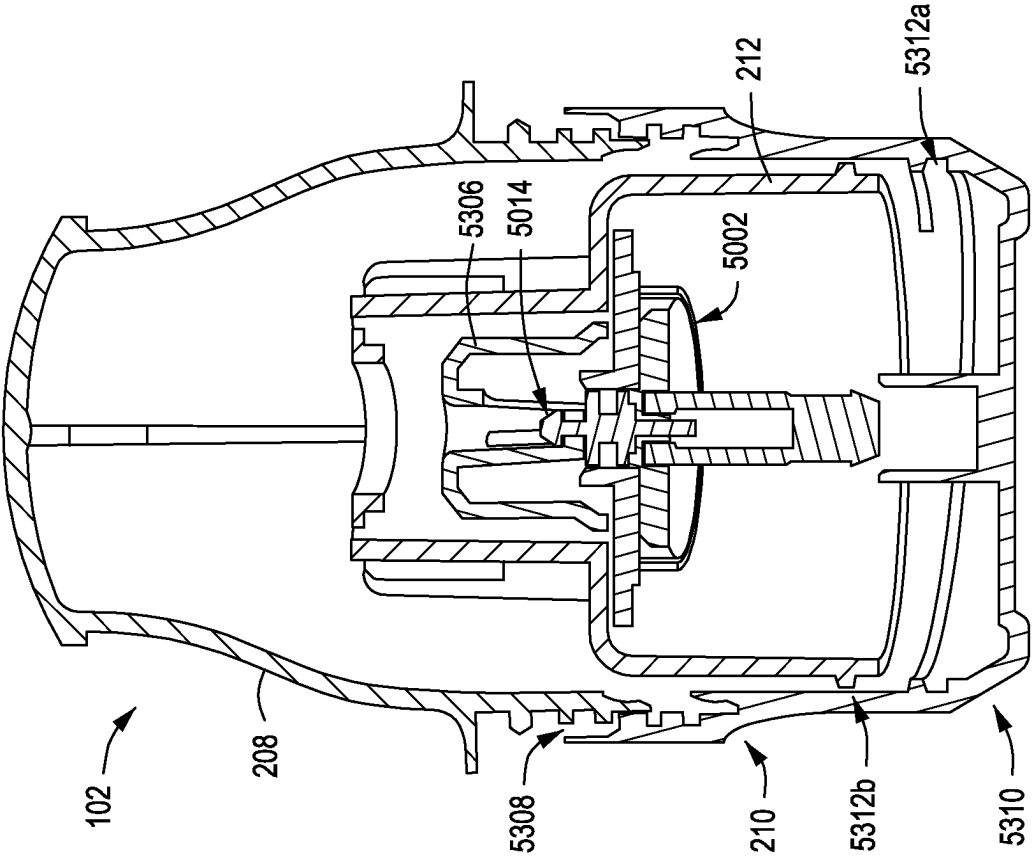


FIG. 53B

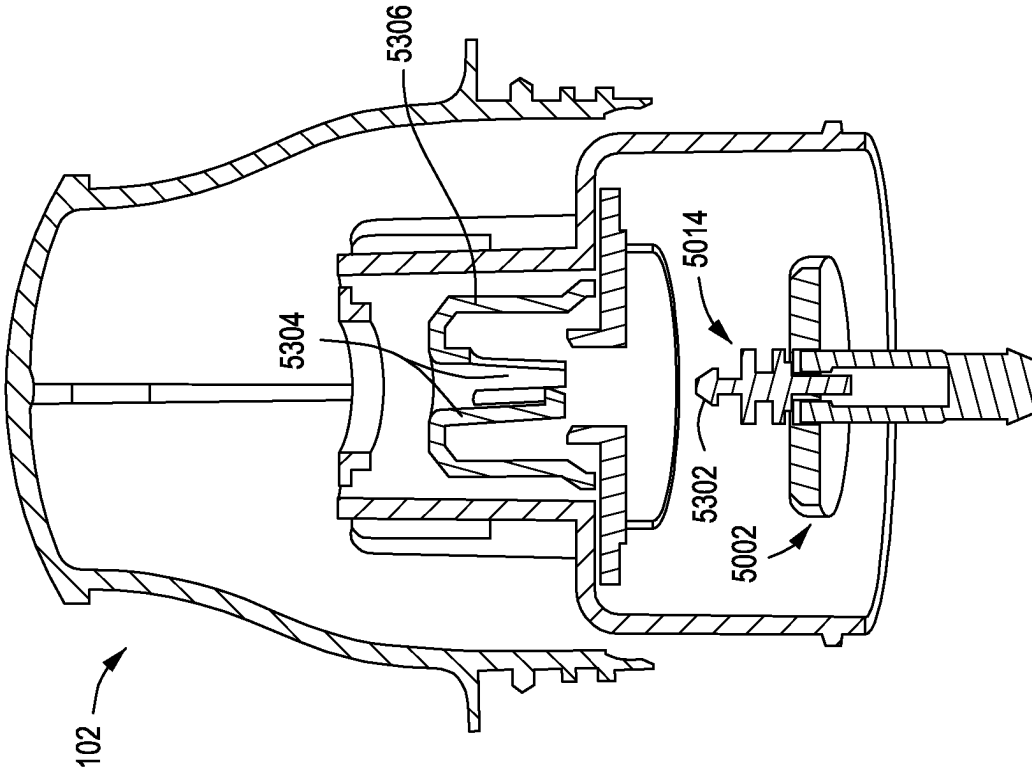


FIG. 53A

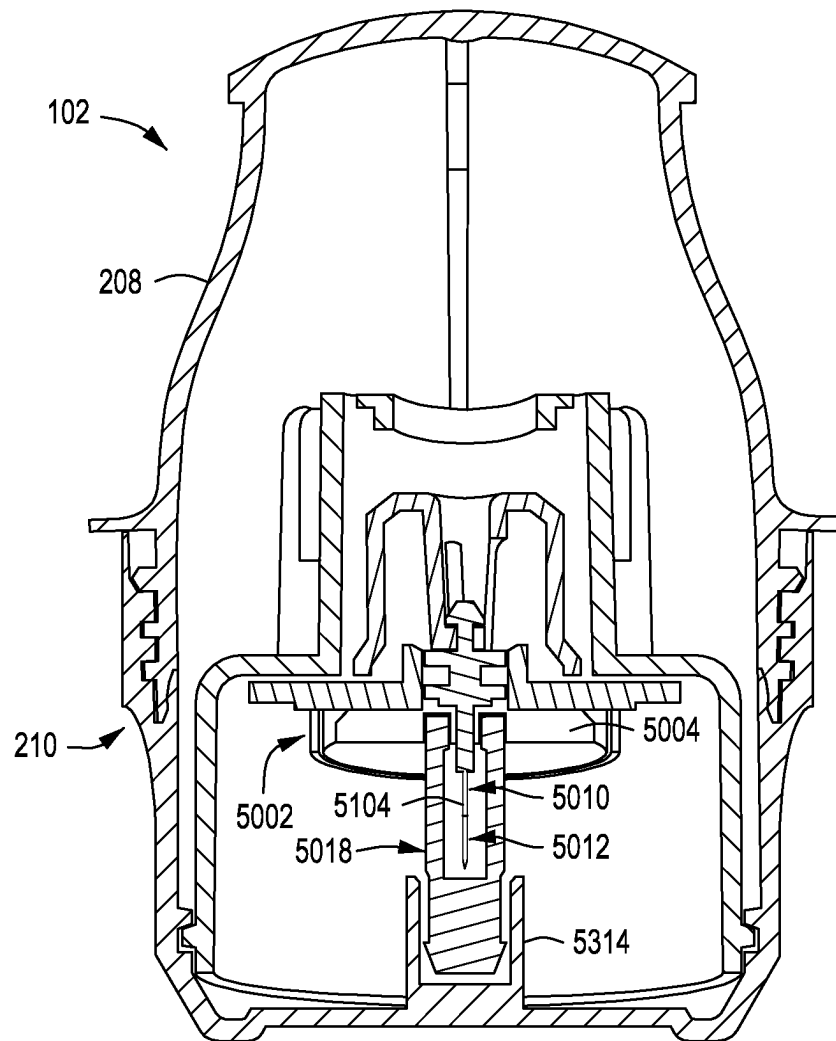


FIG. 53C

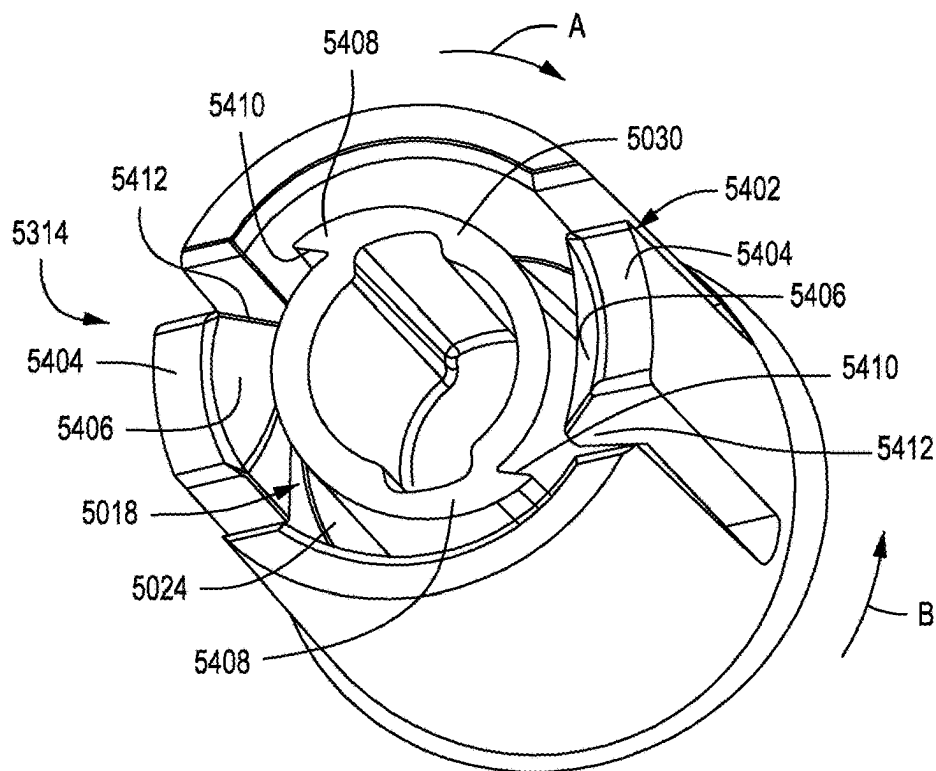


FIG. 54A

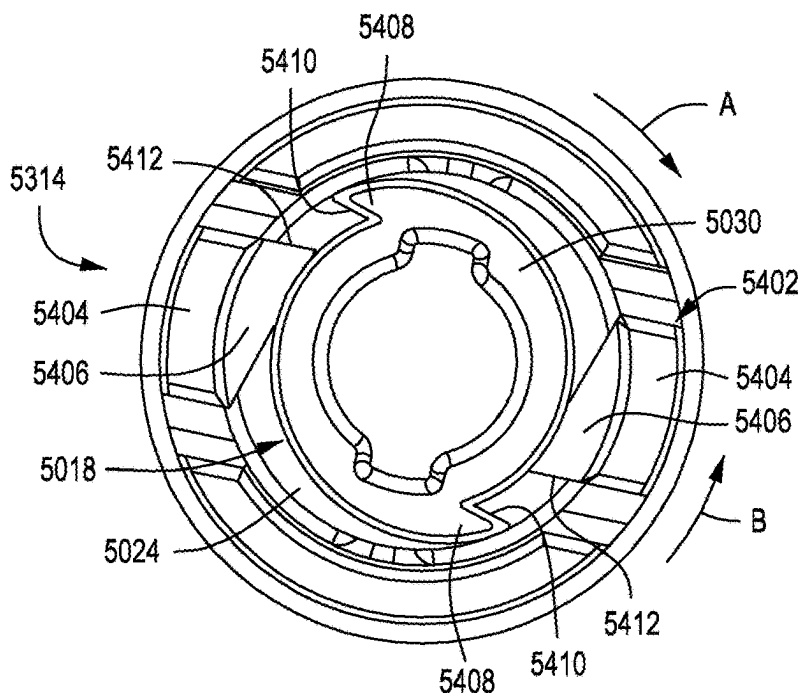


FIG. 54B

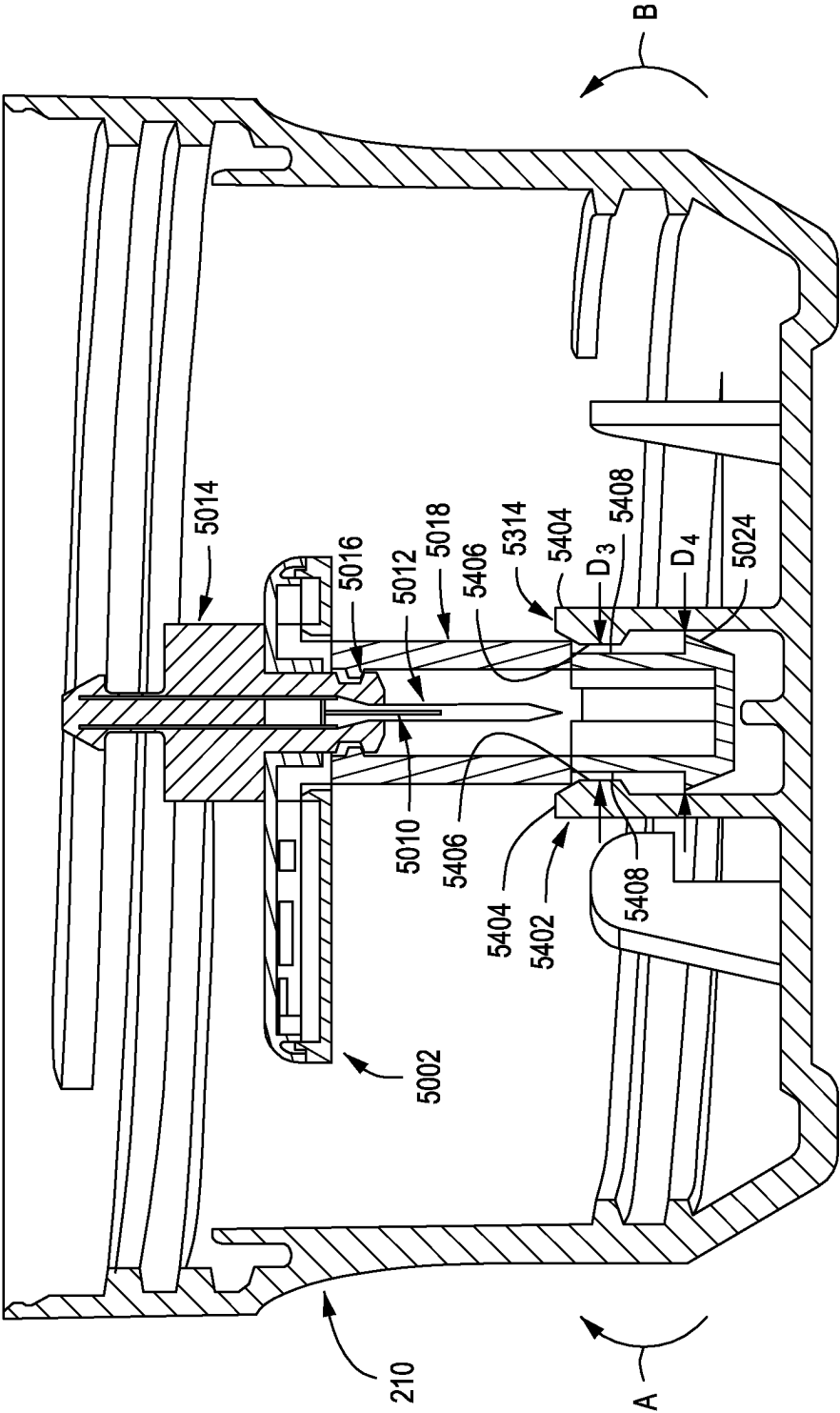


FIG. 55

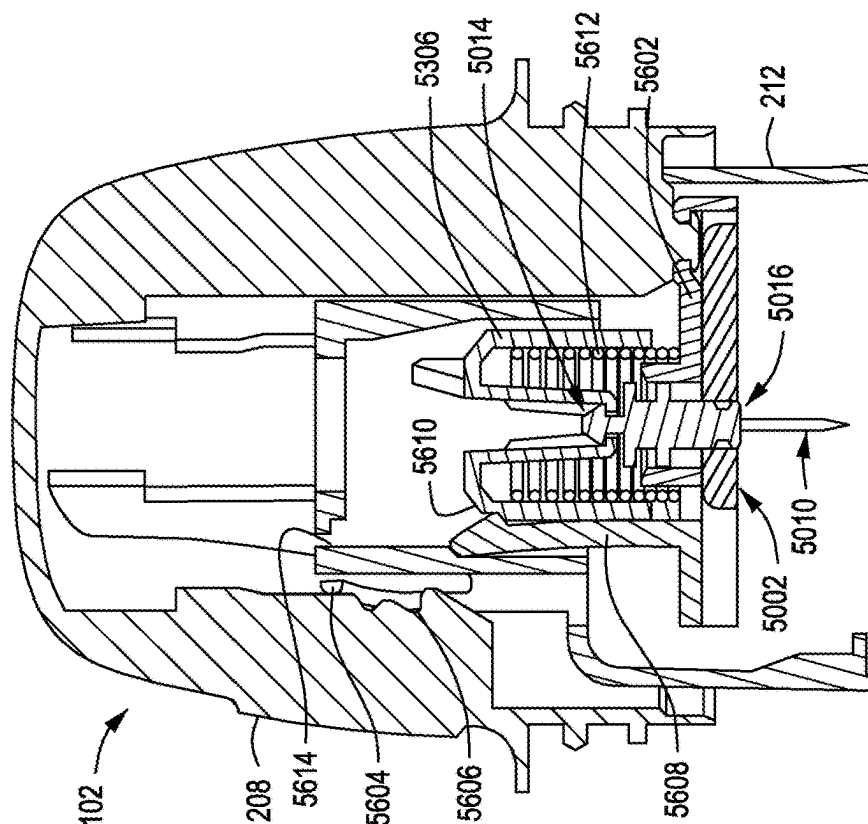


FIG. 56B

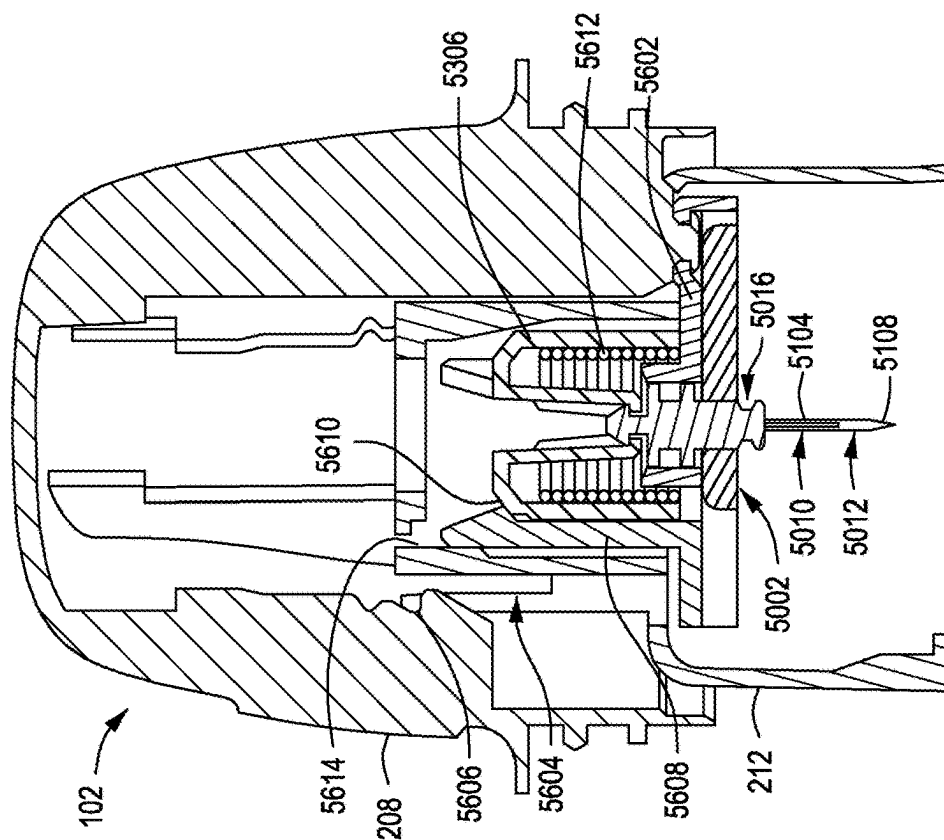


FIG. 56A

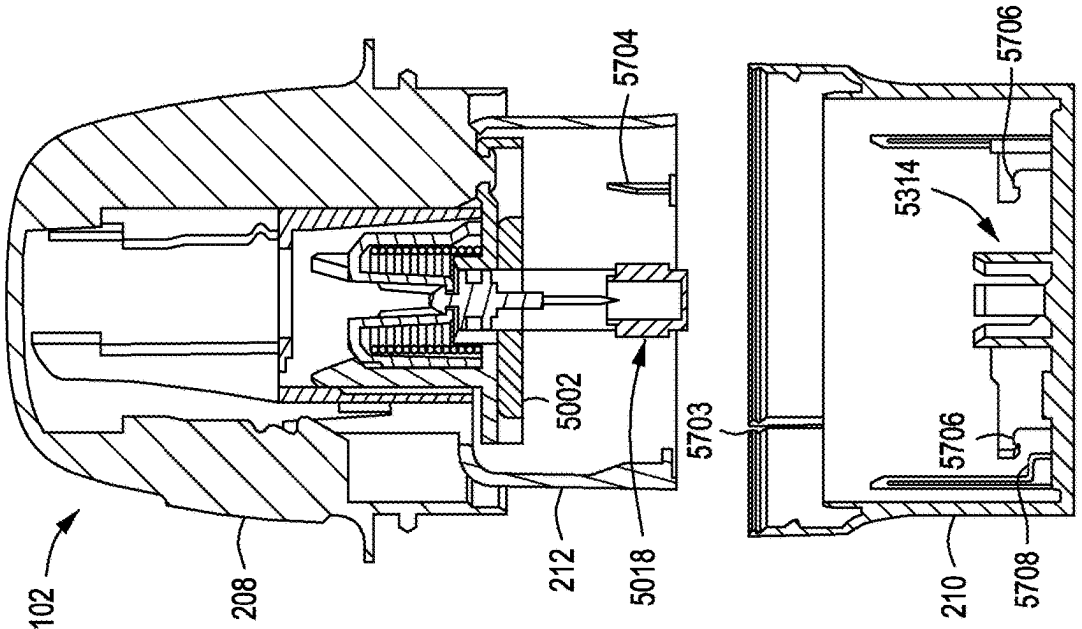


FIG. 57B

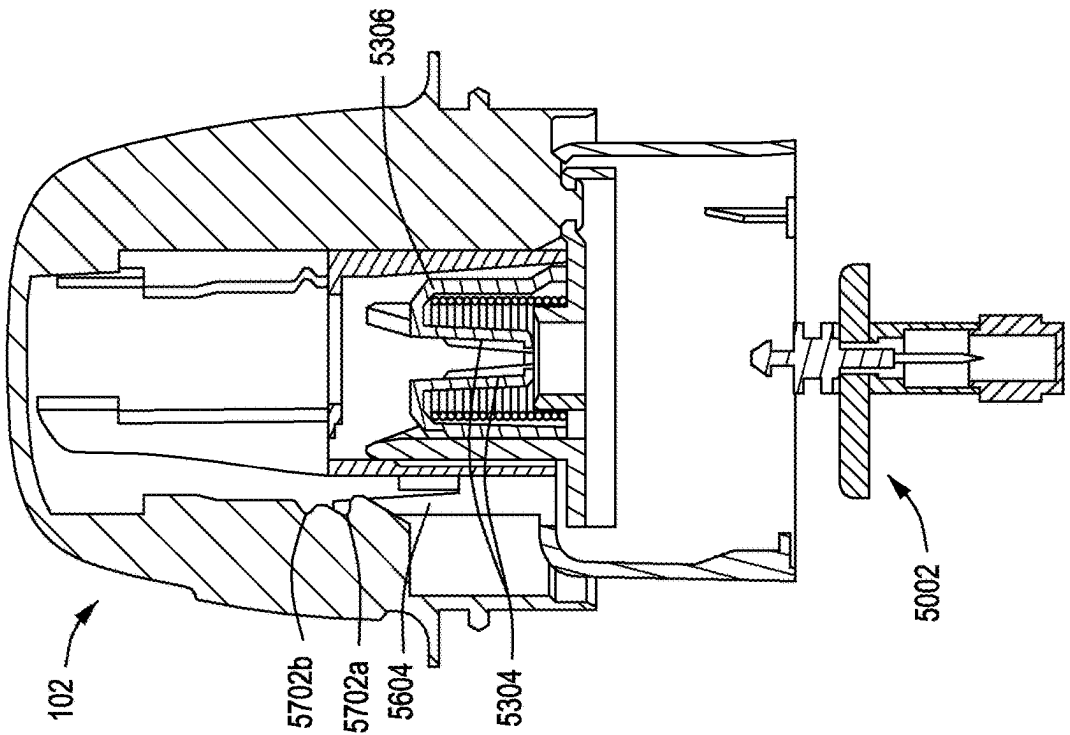


FIG. 57A

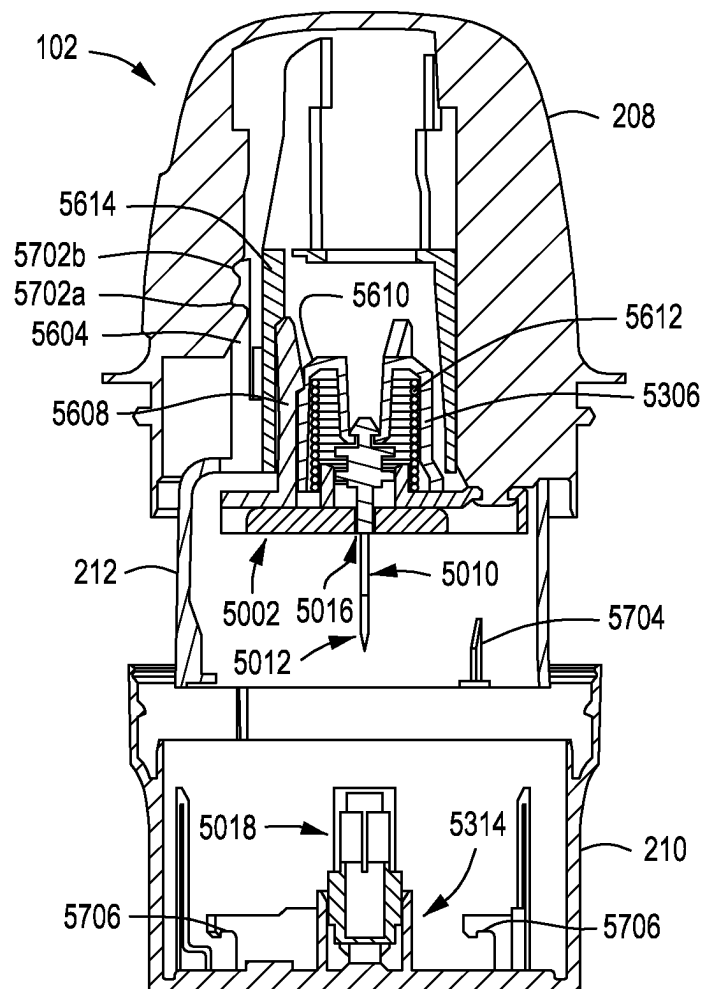


FIG. 57C

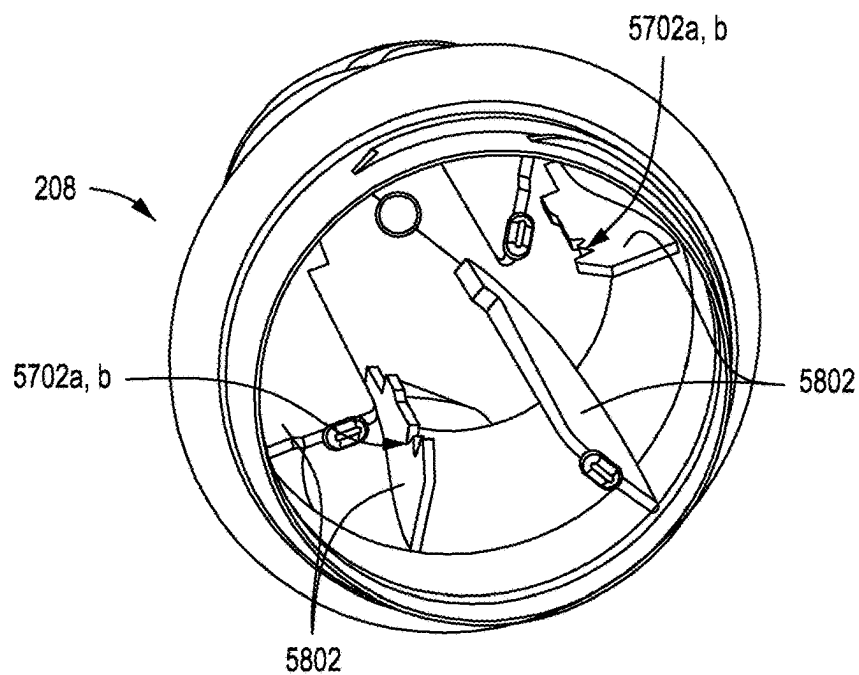


FIG. 58A

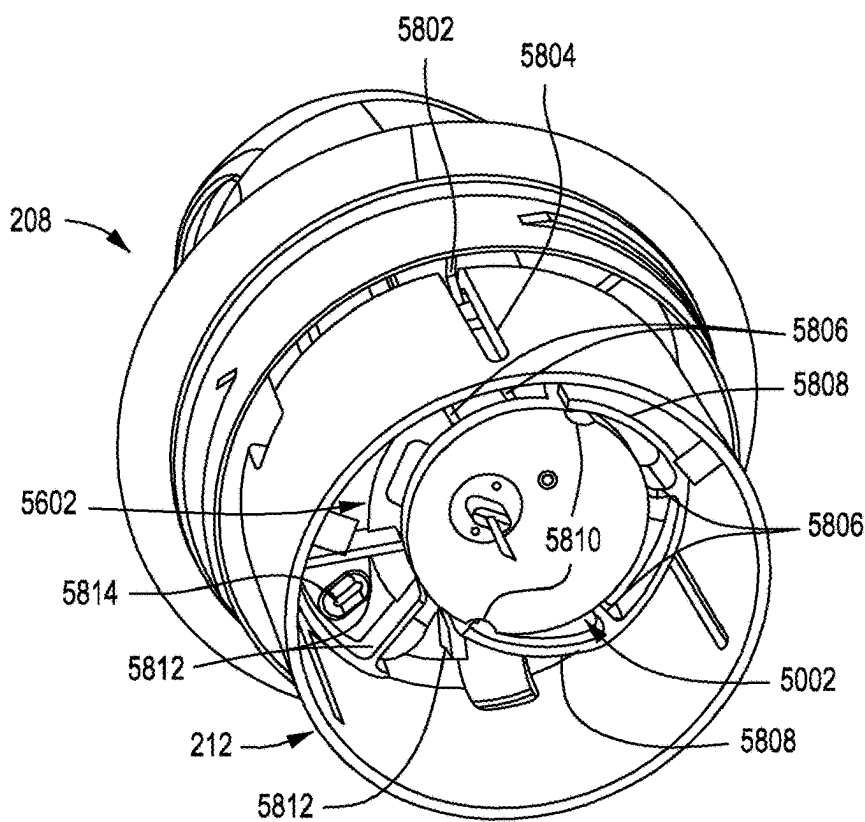


FIG. 58B

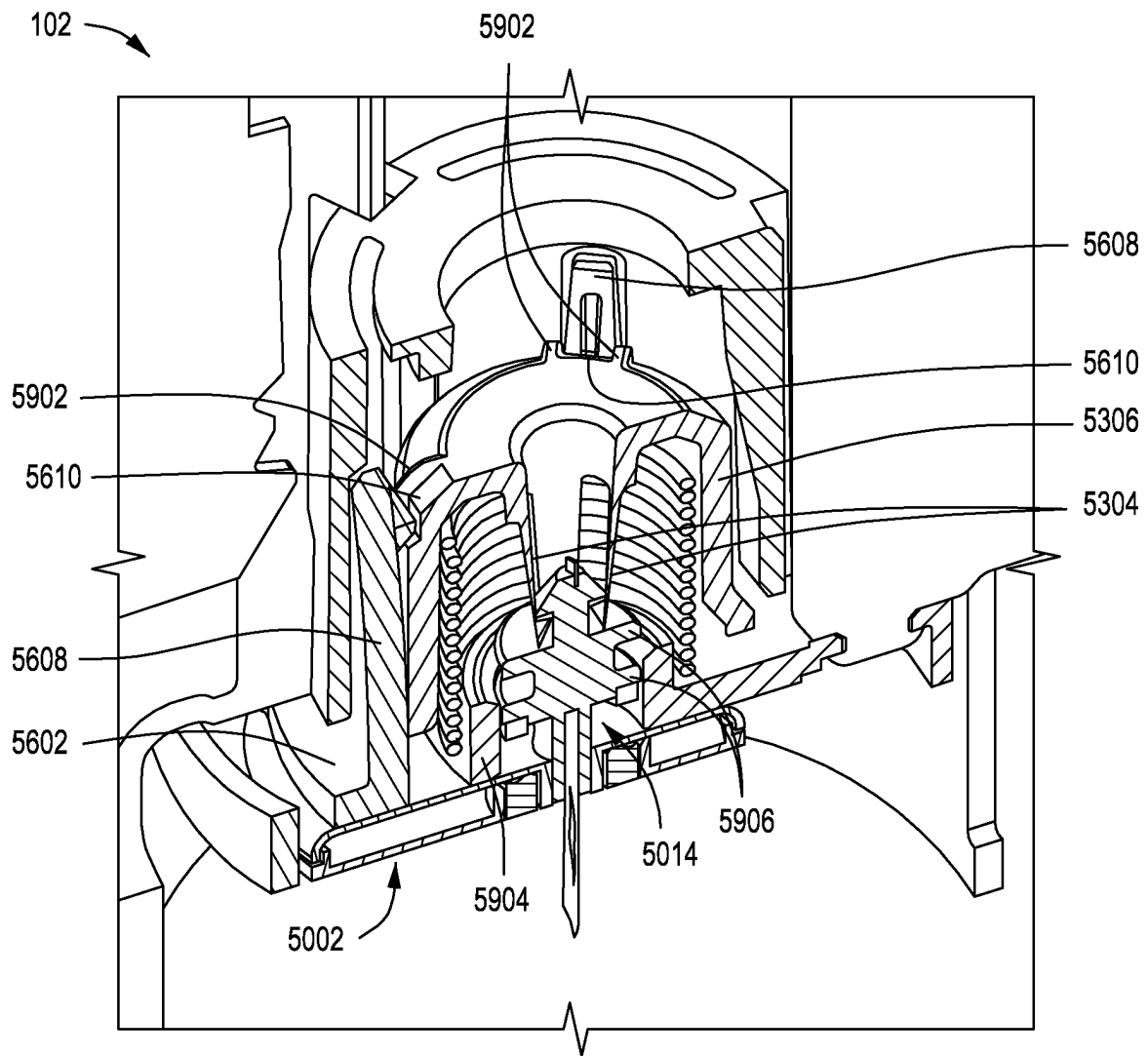


FIG. 59

FIG. 60A

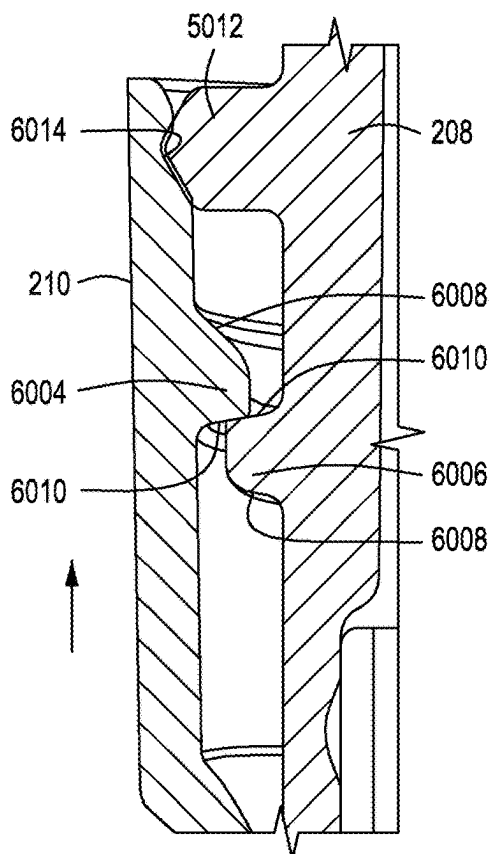
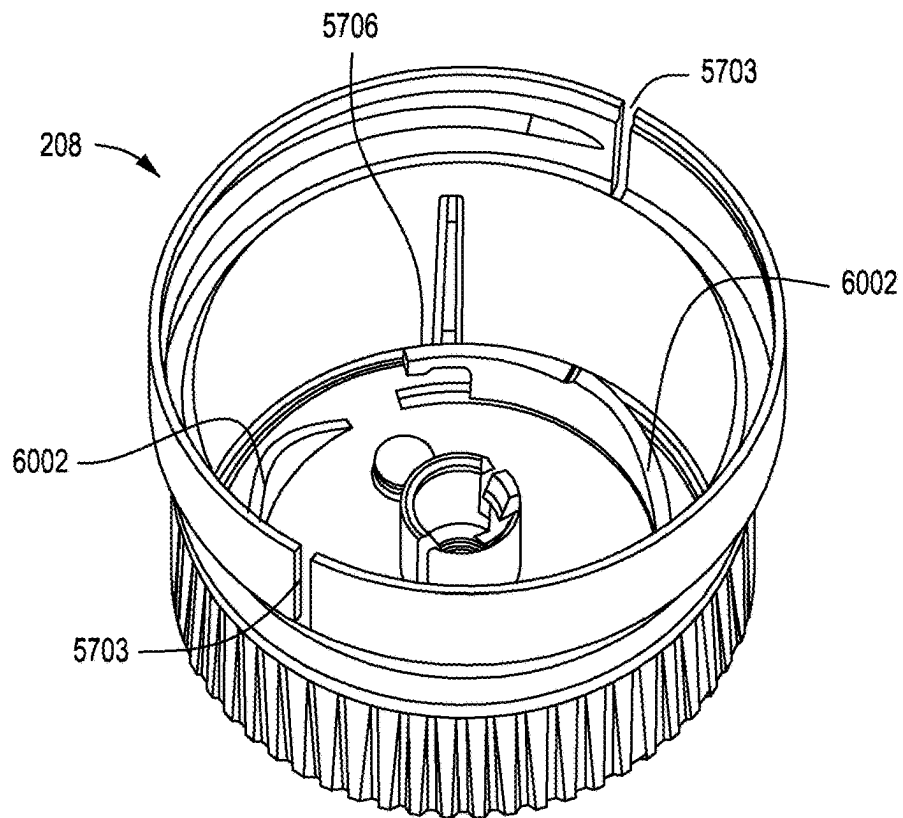


FIG. 60B

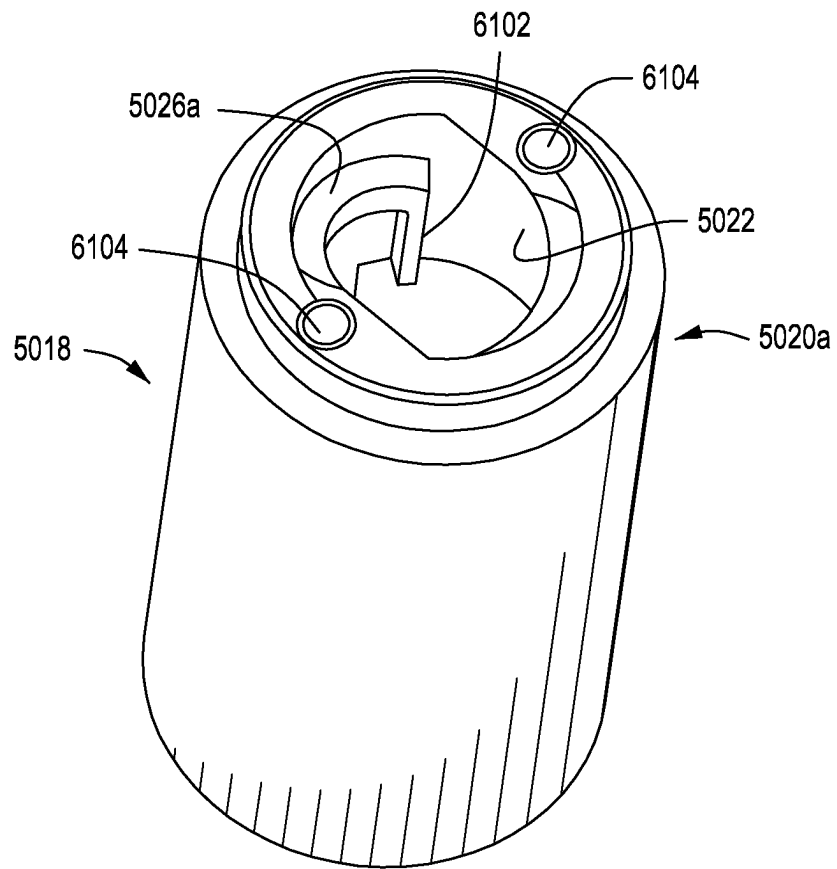


FIG. 61A

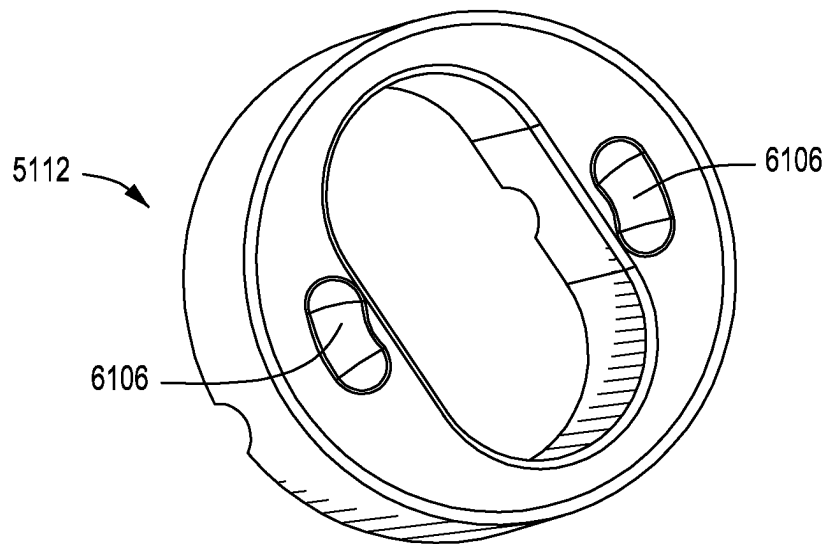


FIG. 61B

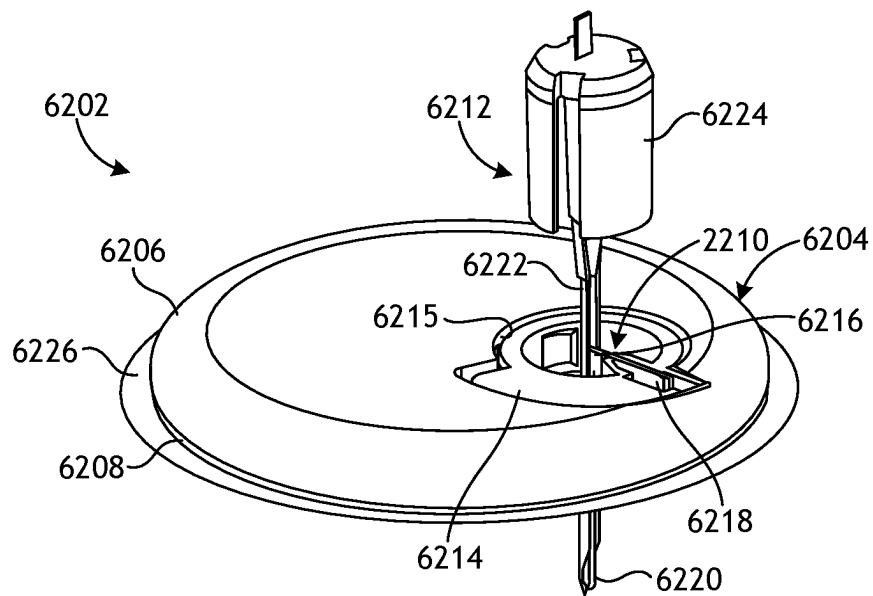


FIG. 62

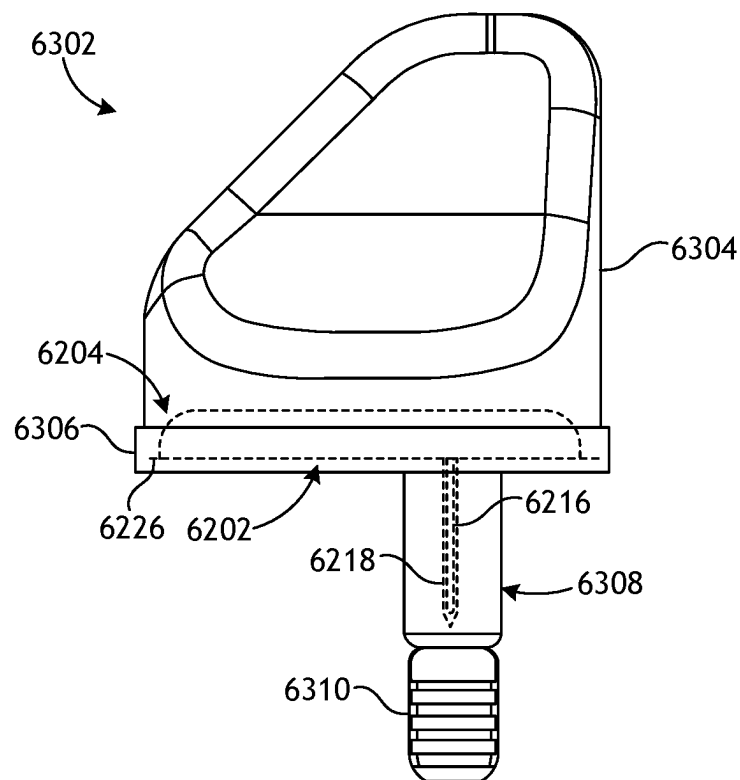


FIG. 63

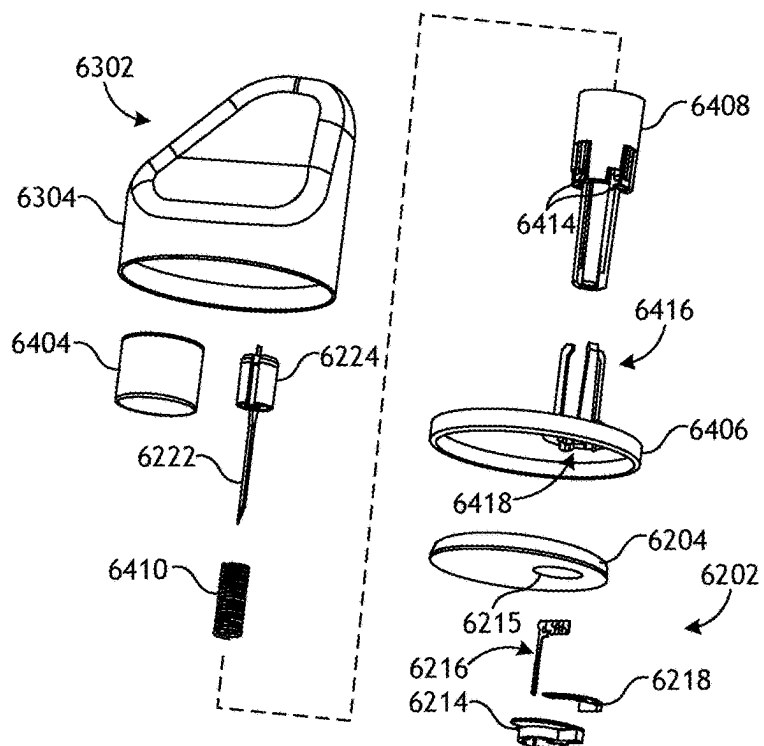


FIG. 64A

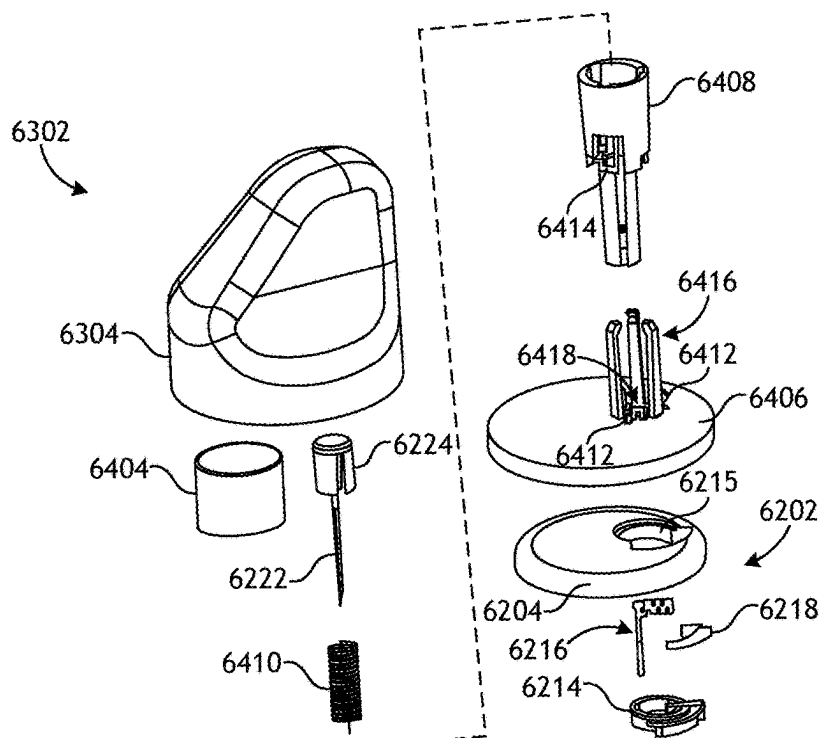


FIG. 64B

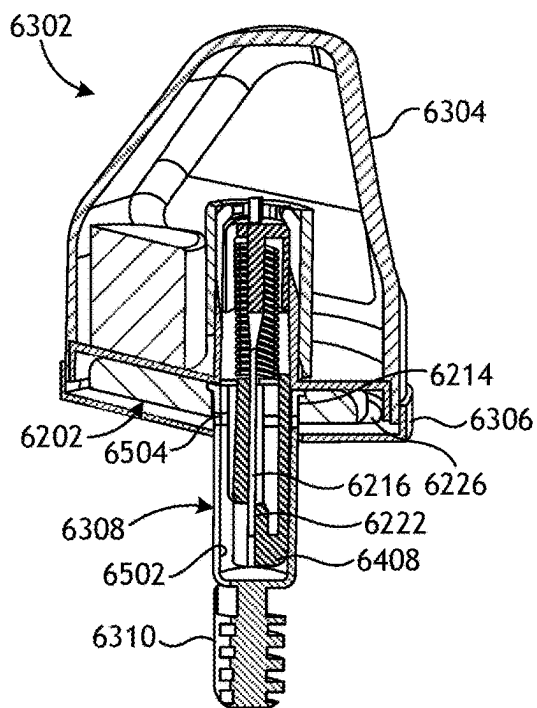


FIG. 65A

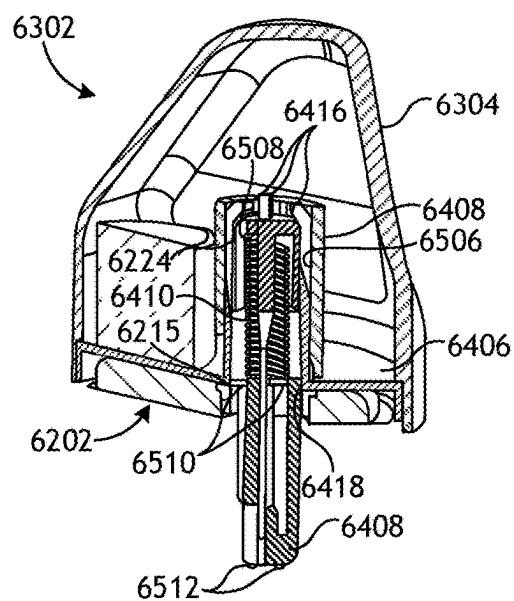


FIG. 65B

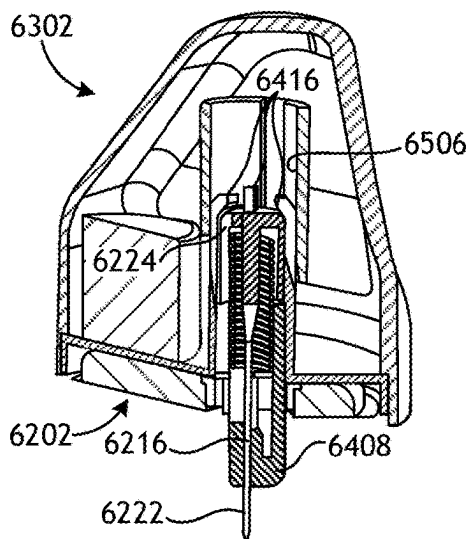


FIG. 65C

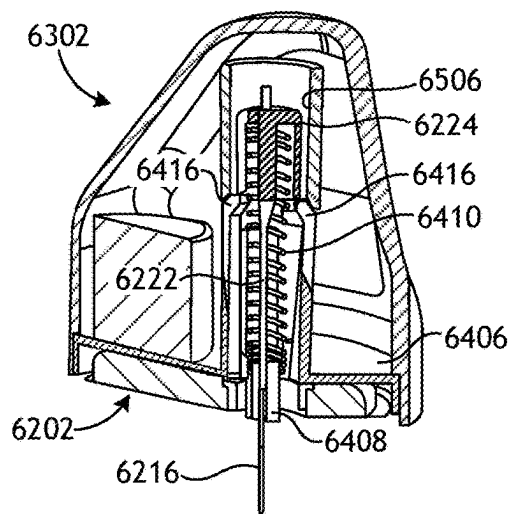


FIG. 65D

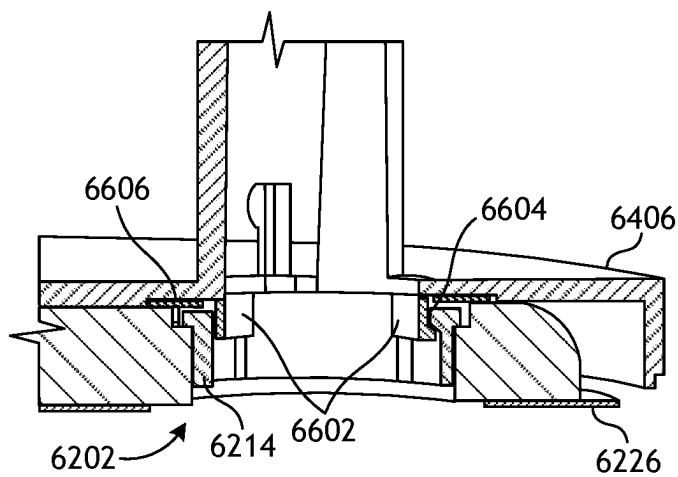


FIG. 66

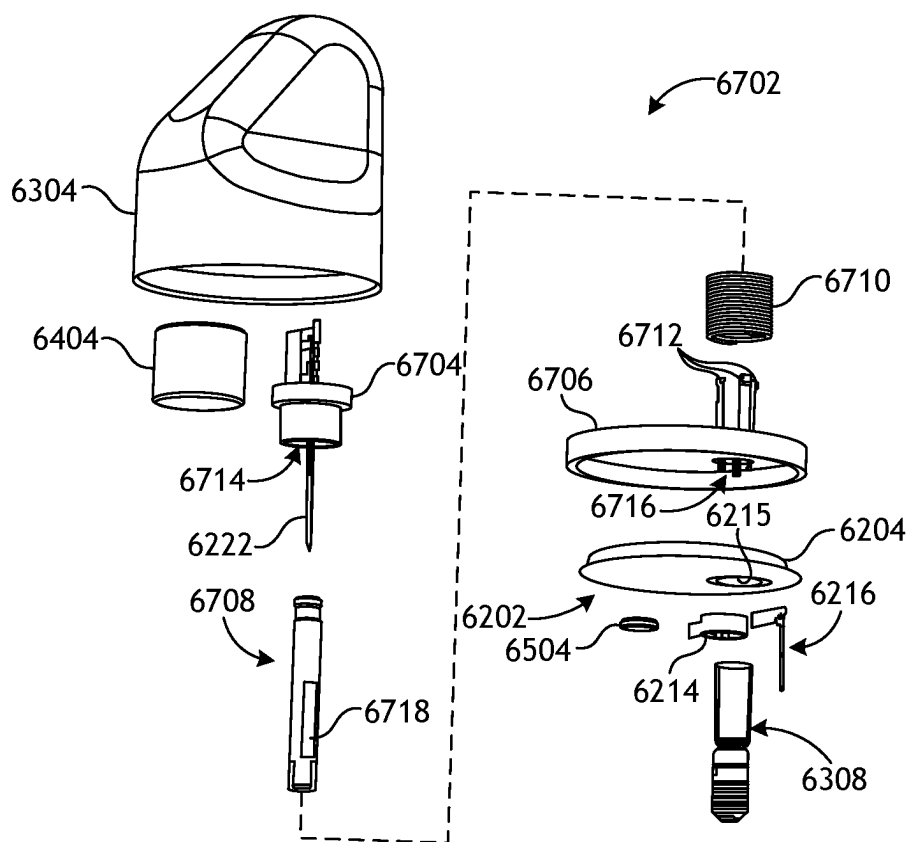


FIG. 67

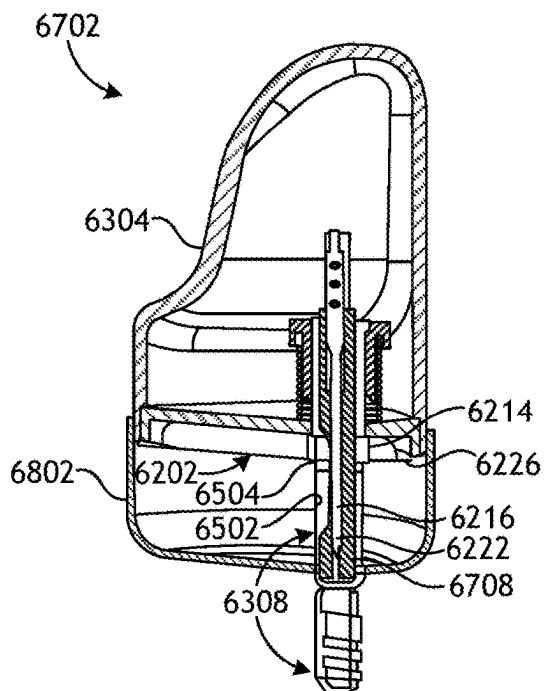


FIG. 68A

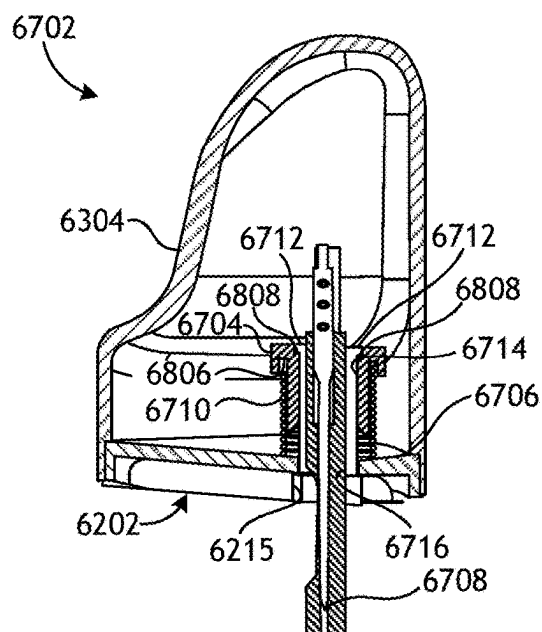


FIG. 68B

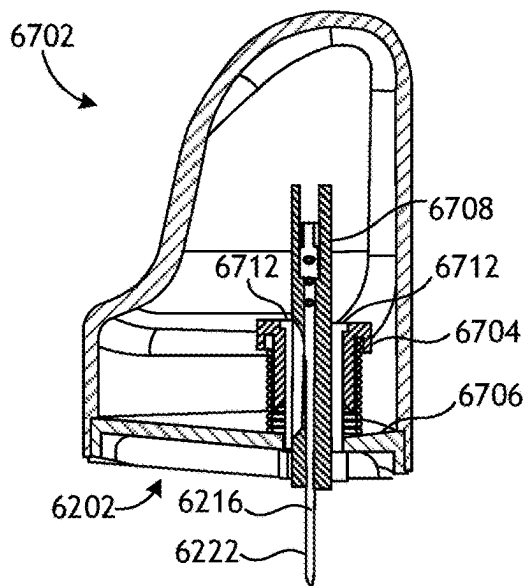


FIG. 68C

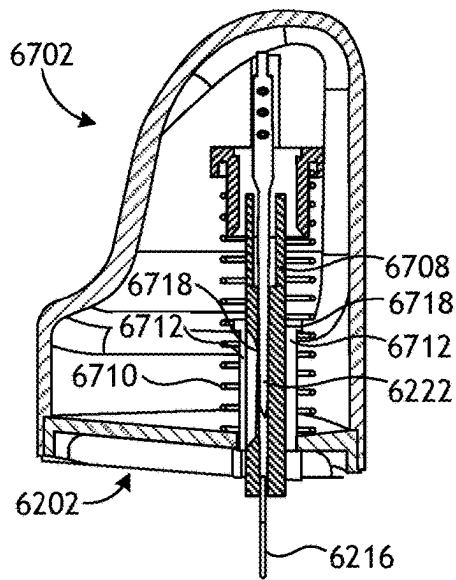


FIG. 68D

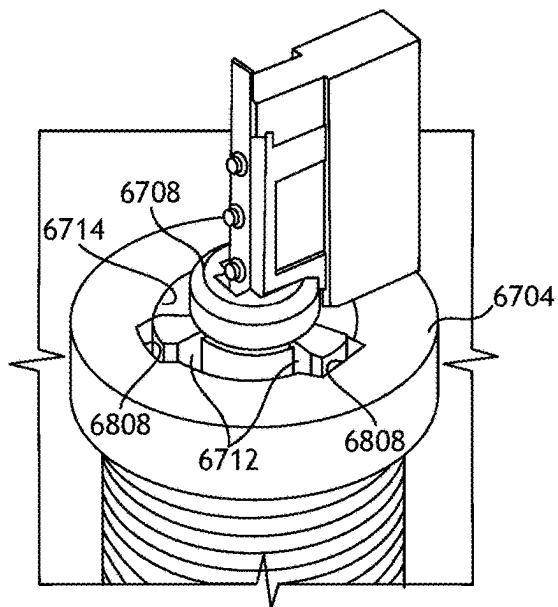


FIG. 69A

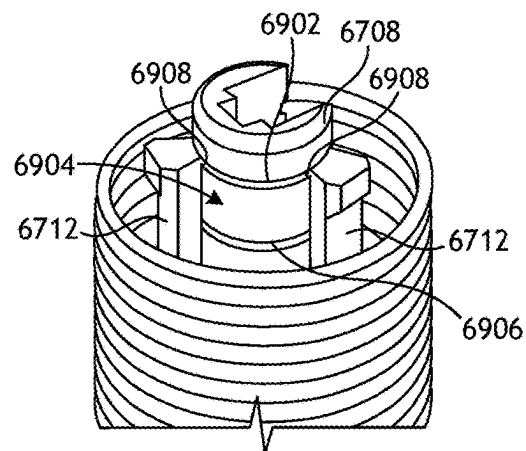


FIG. 69B

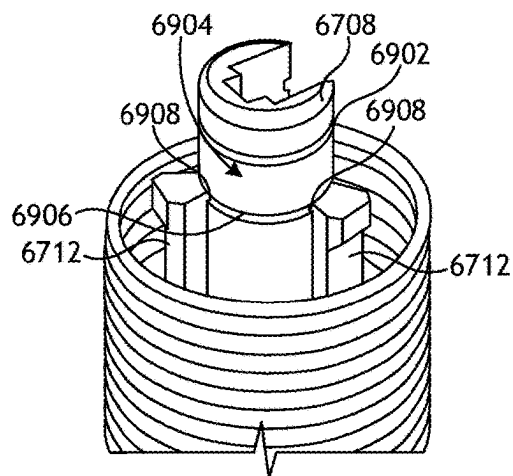


FIG. 69C

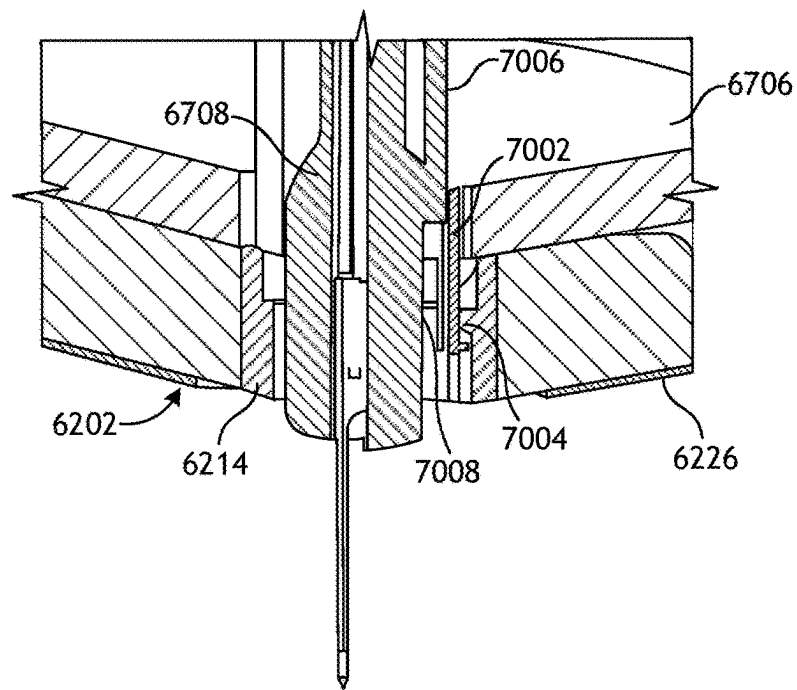


FIG. 70A

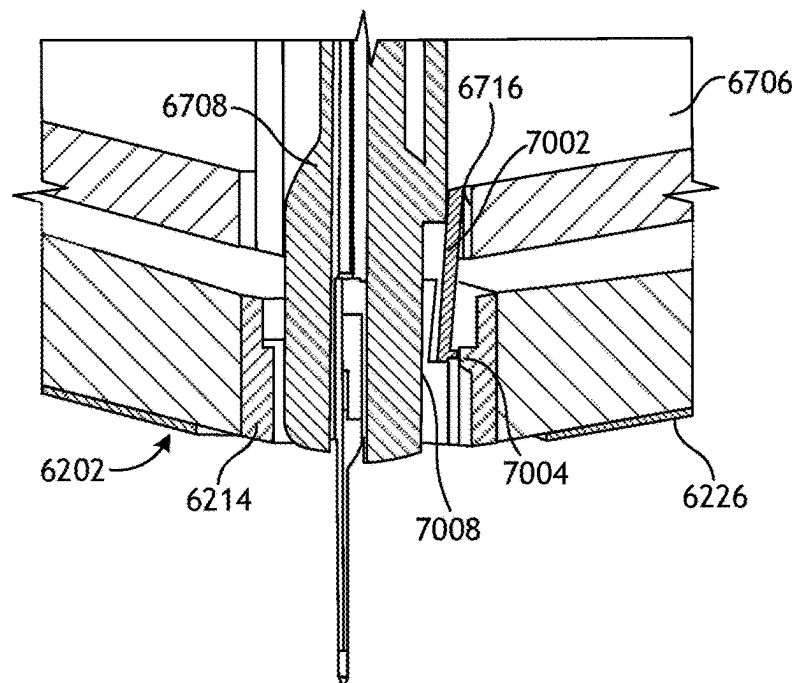


FIG. 70B

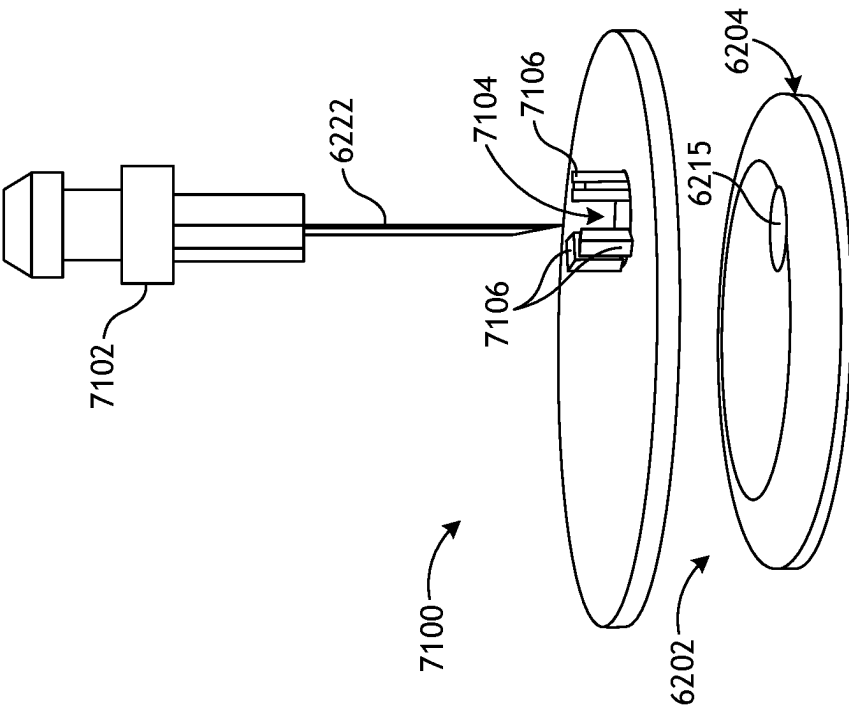


FIG. 71A

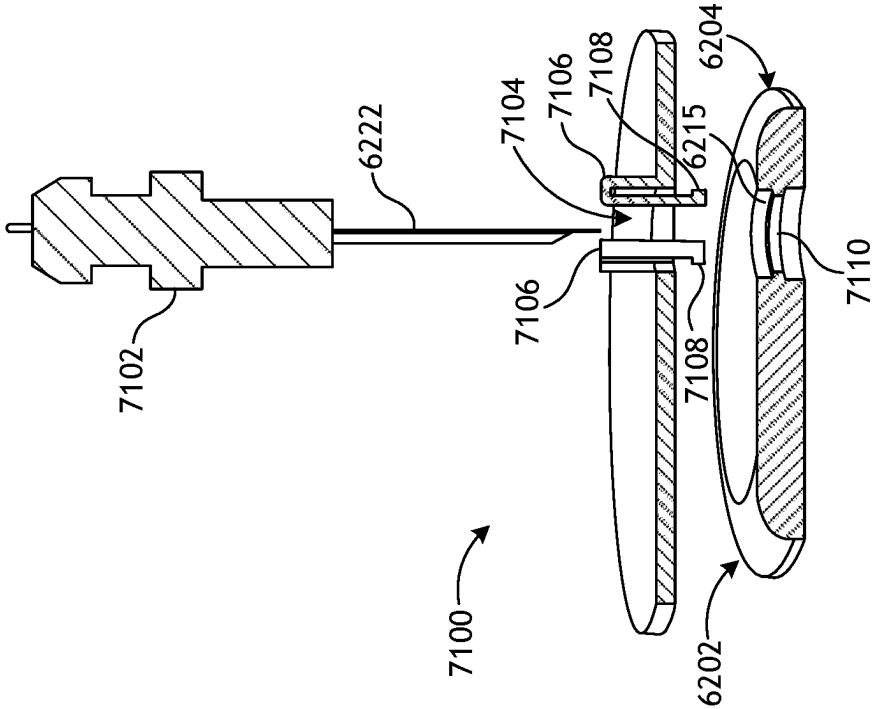
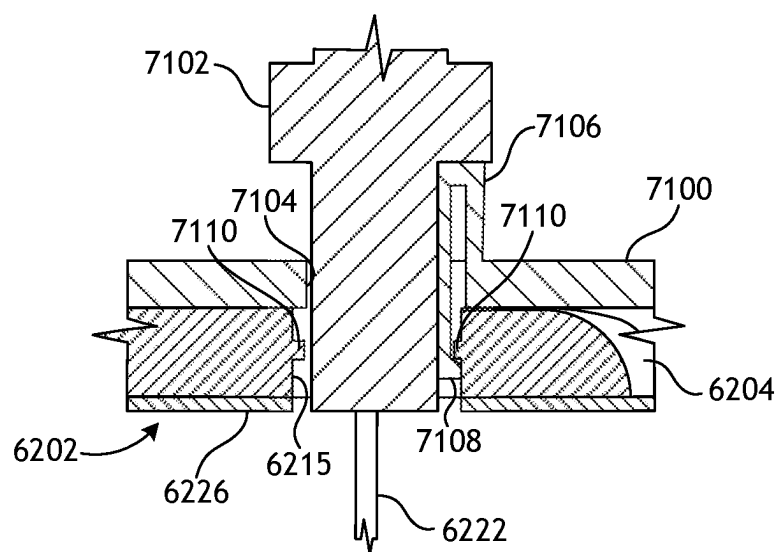
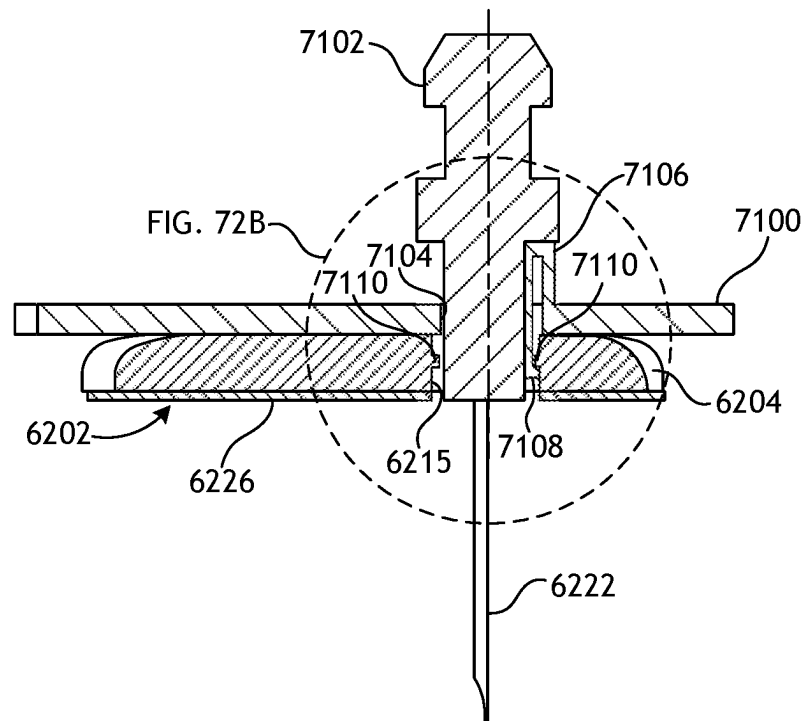


FIG. 71B



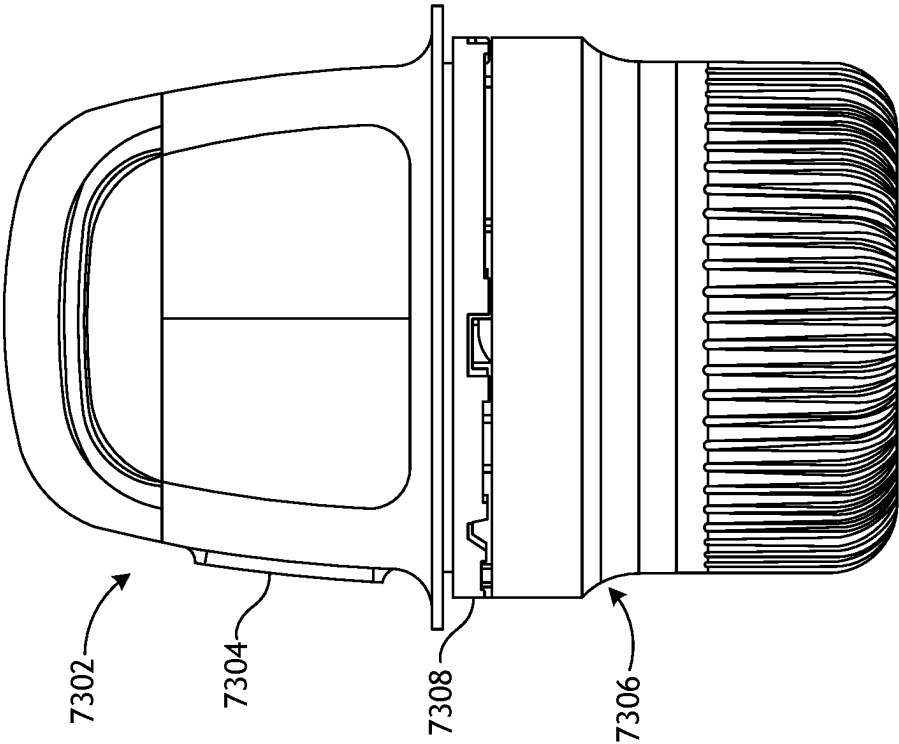


FIG. 73A

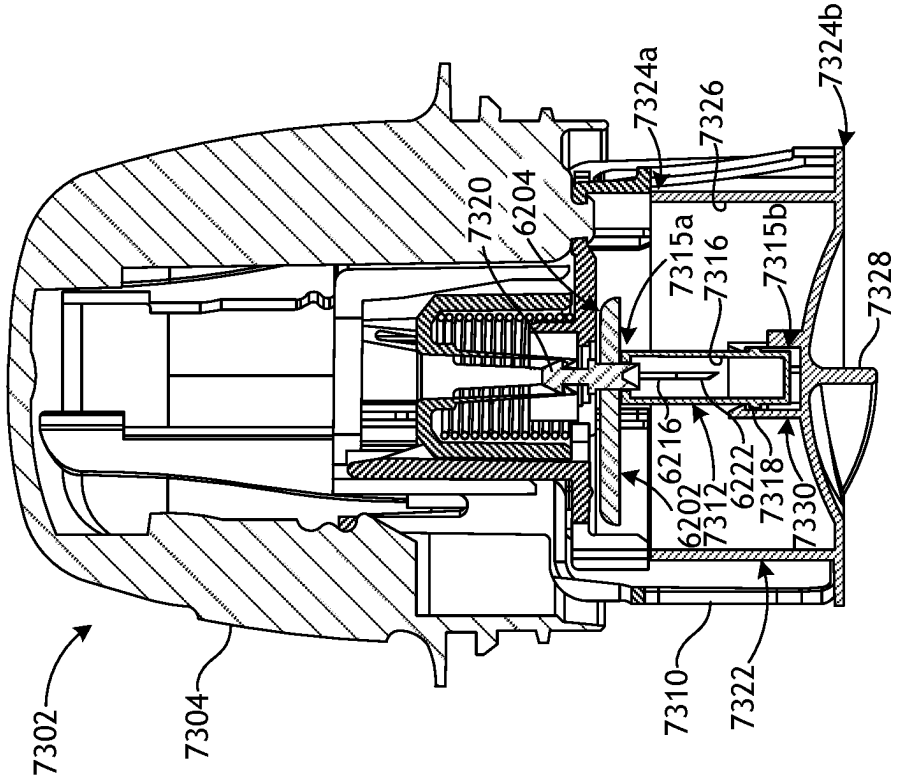


FIG. 73B

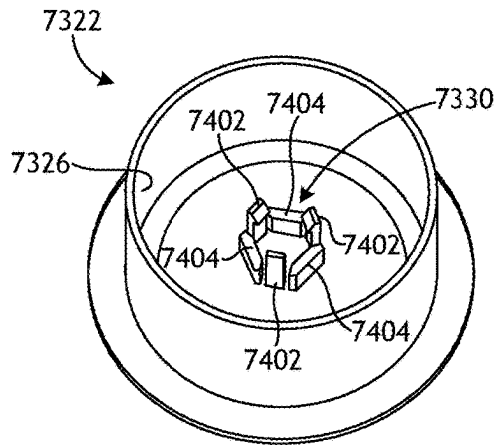


FIG. 74A

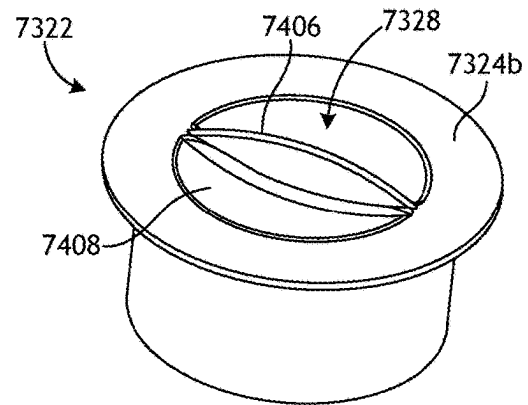


FIG. 74B

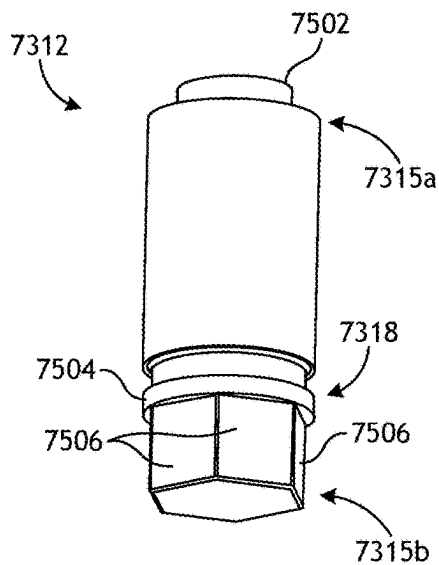


FIG. 75

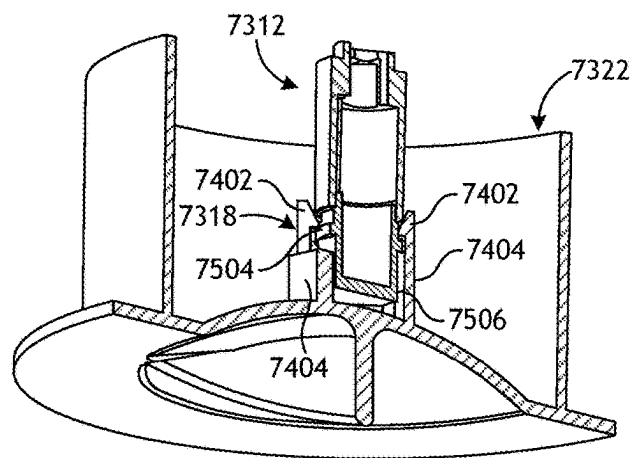


FIG. 76

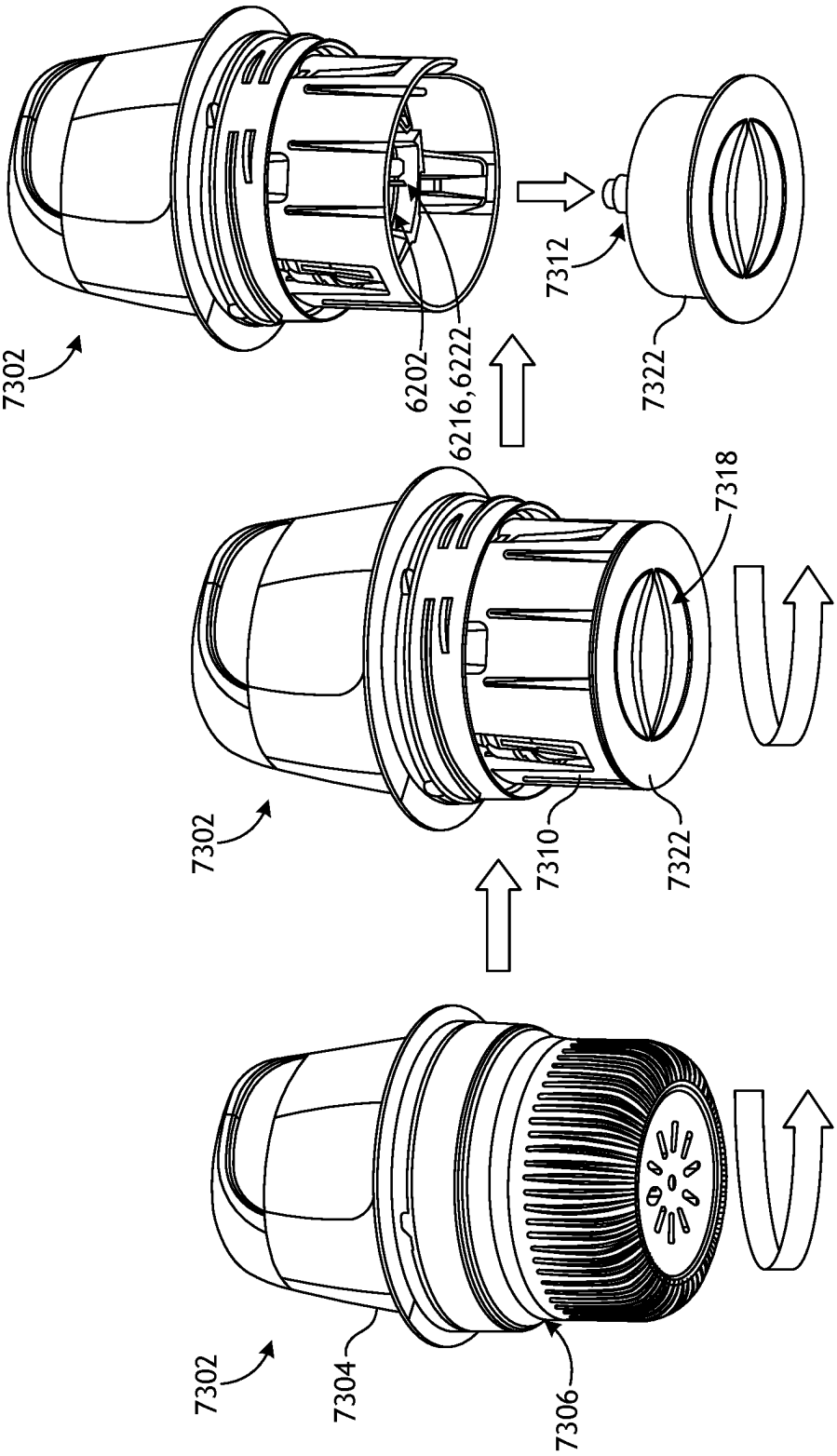


FIG. 77

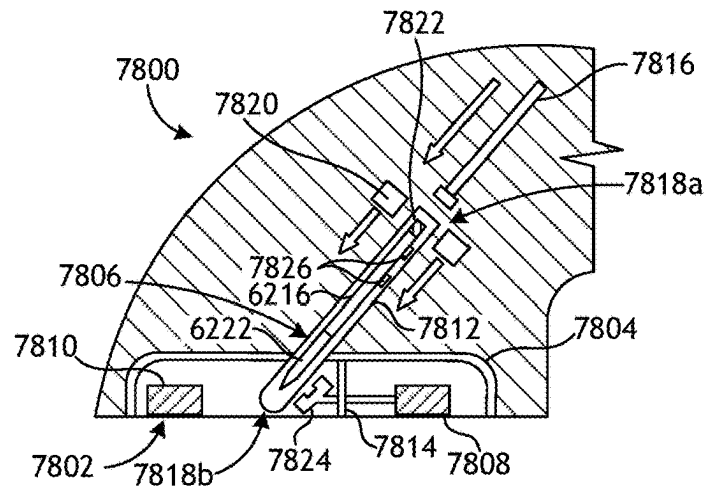


FIG. 78

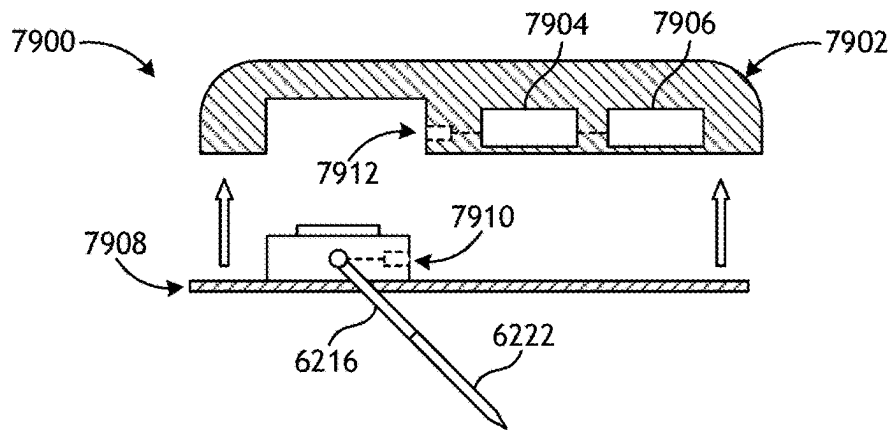


FIG. 79

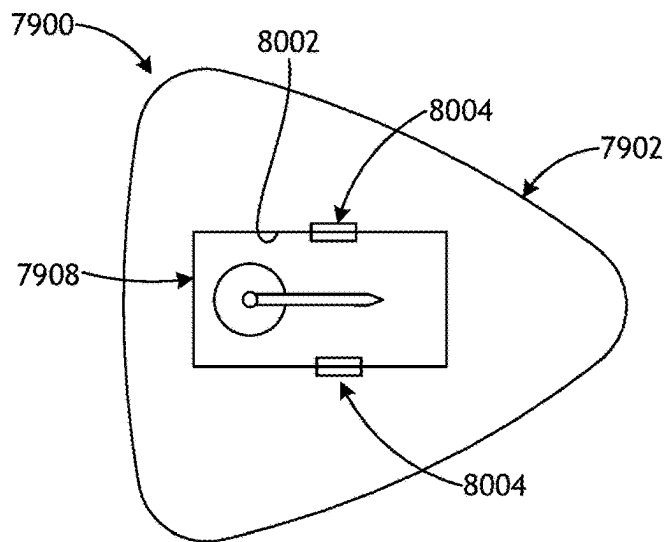


FIG. 80

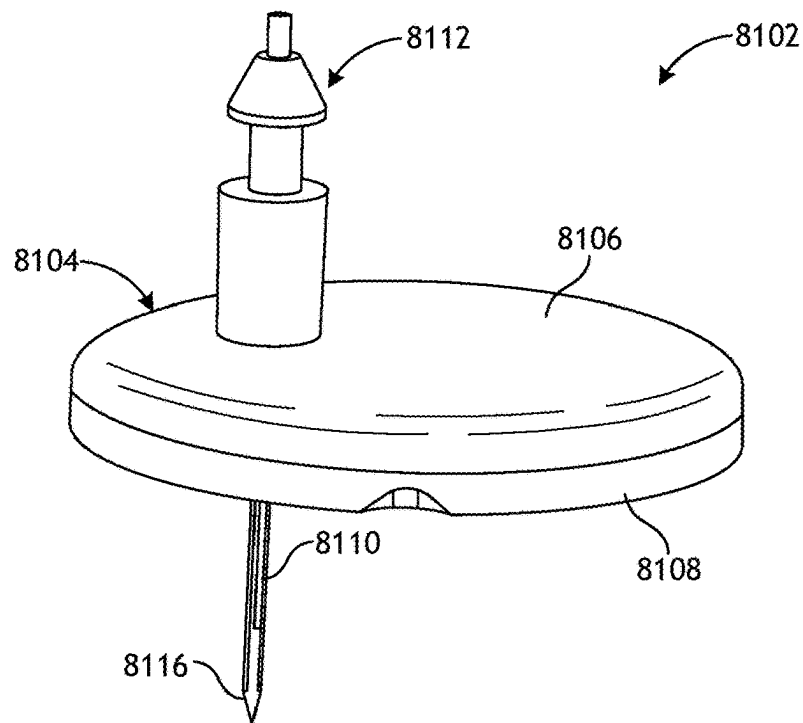


FIG. 81A

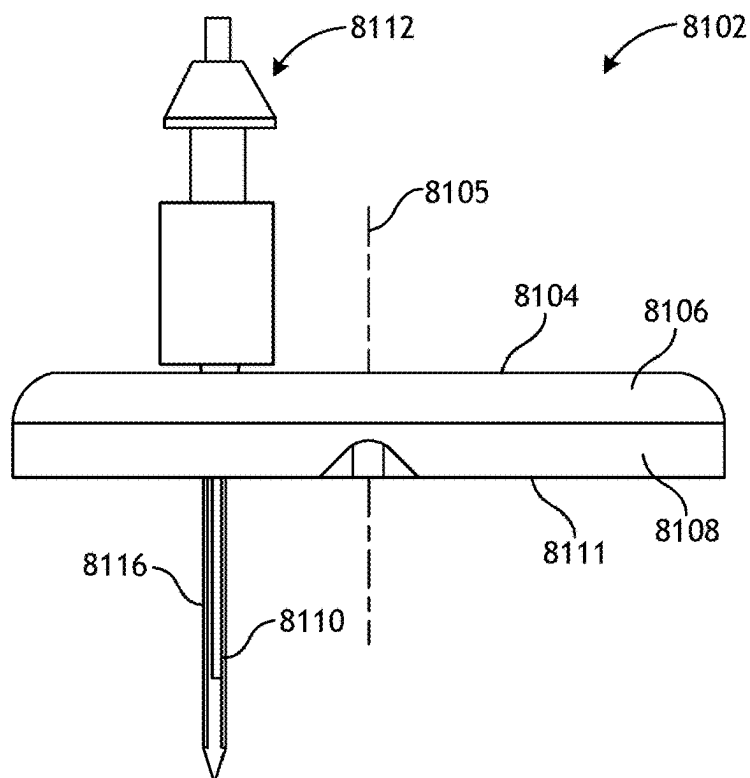


FIG. 81B

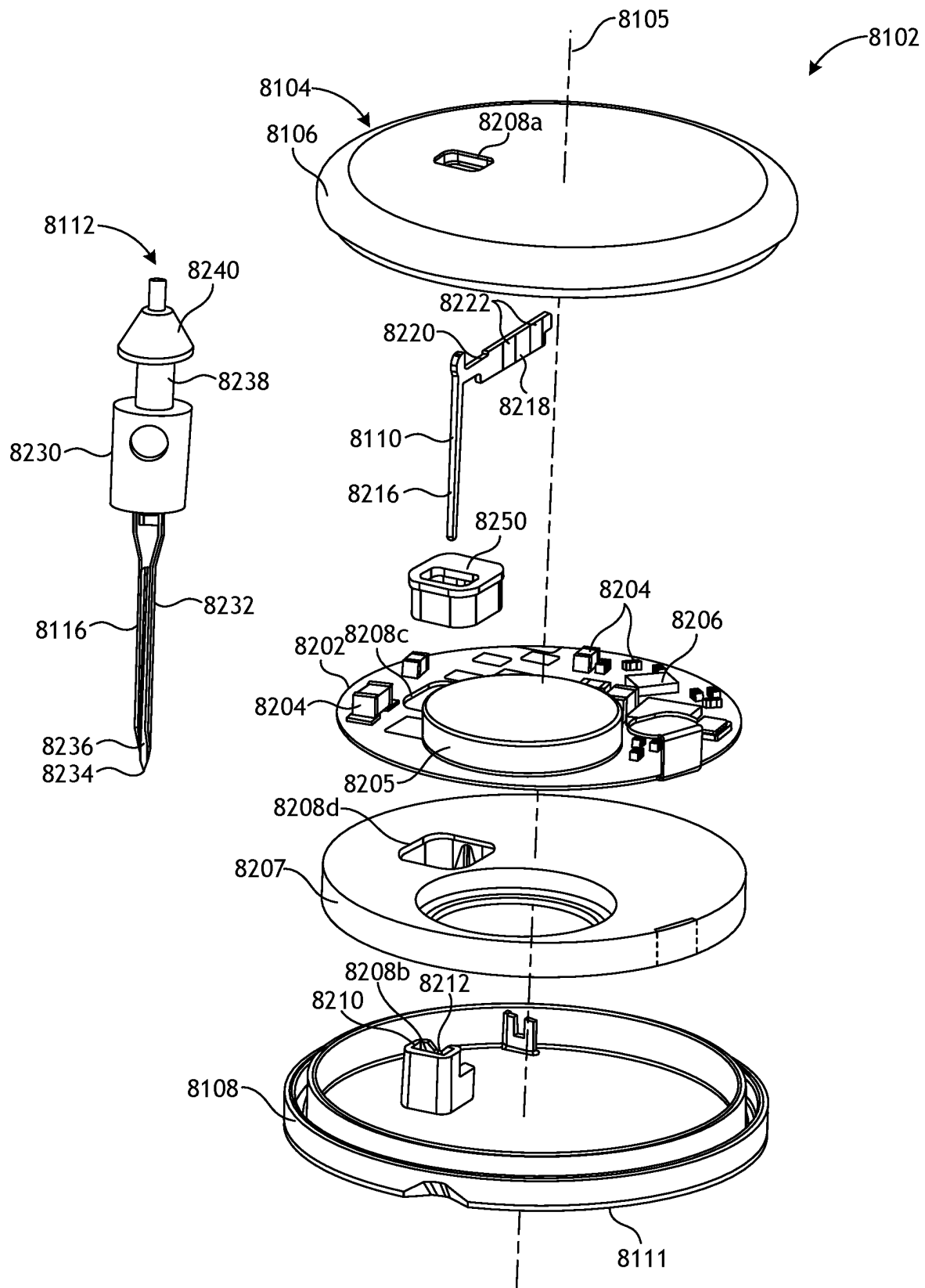


FIG. 82

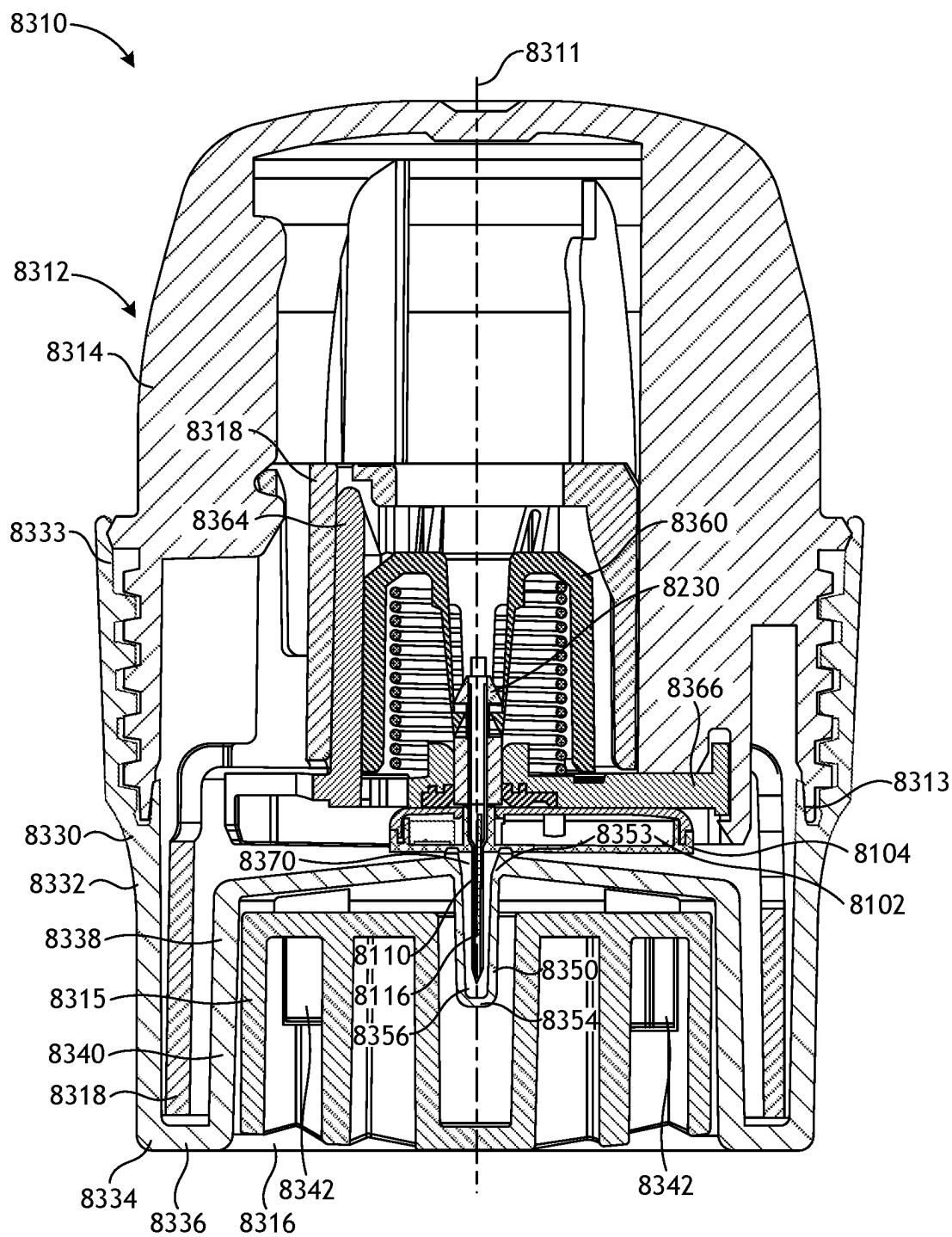


FIG. 83

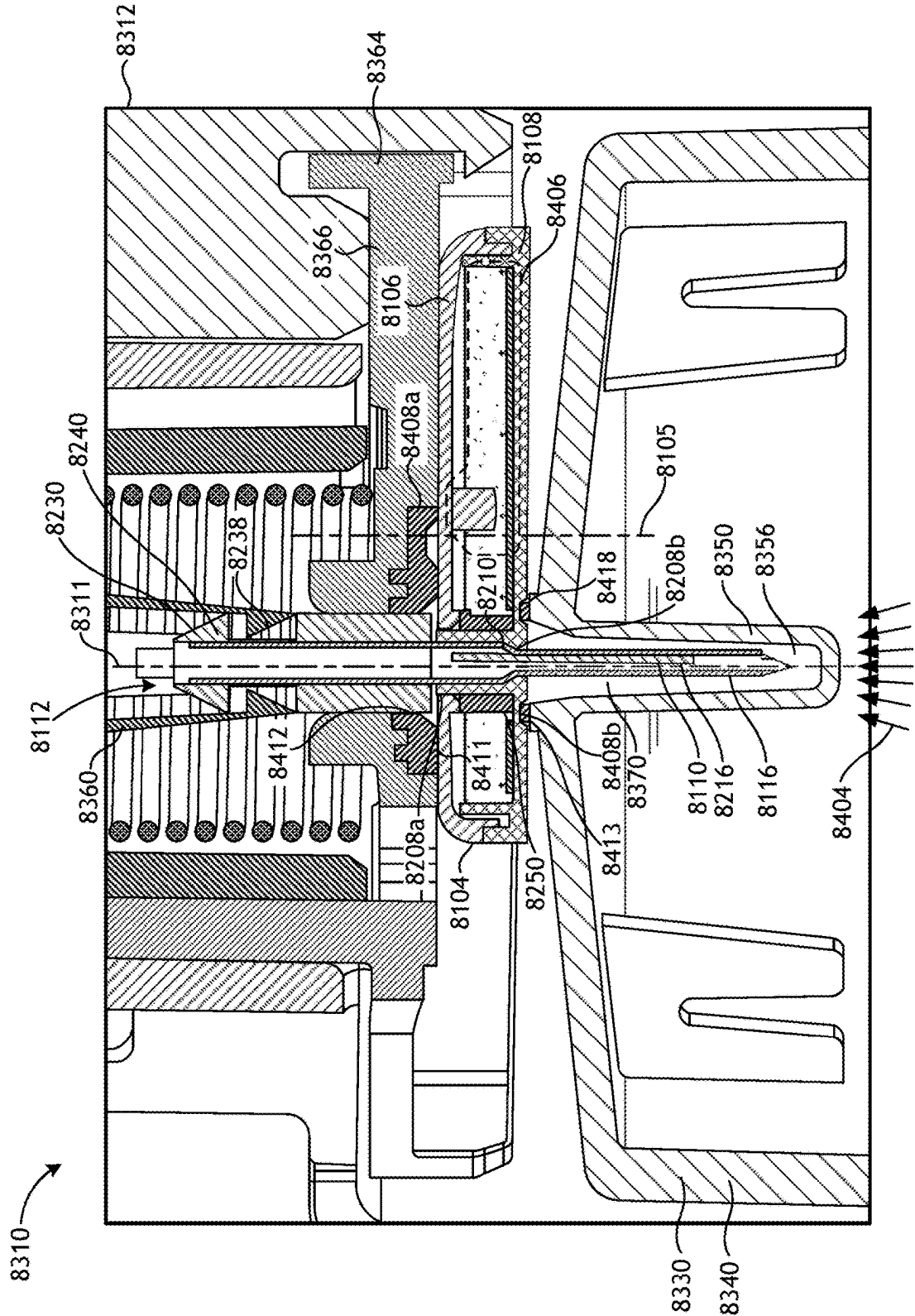
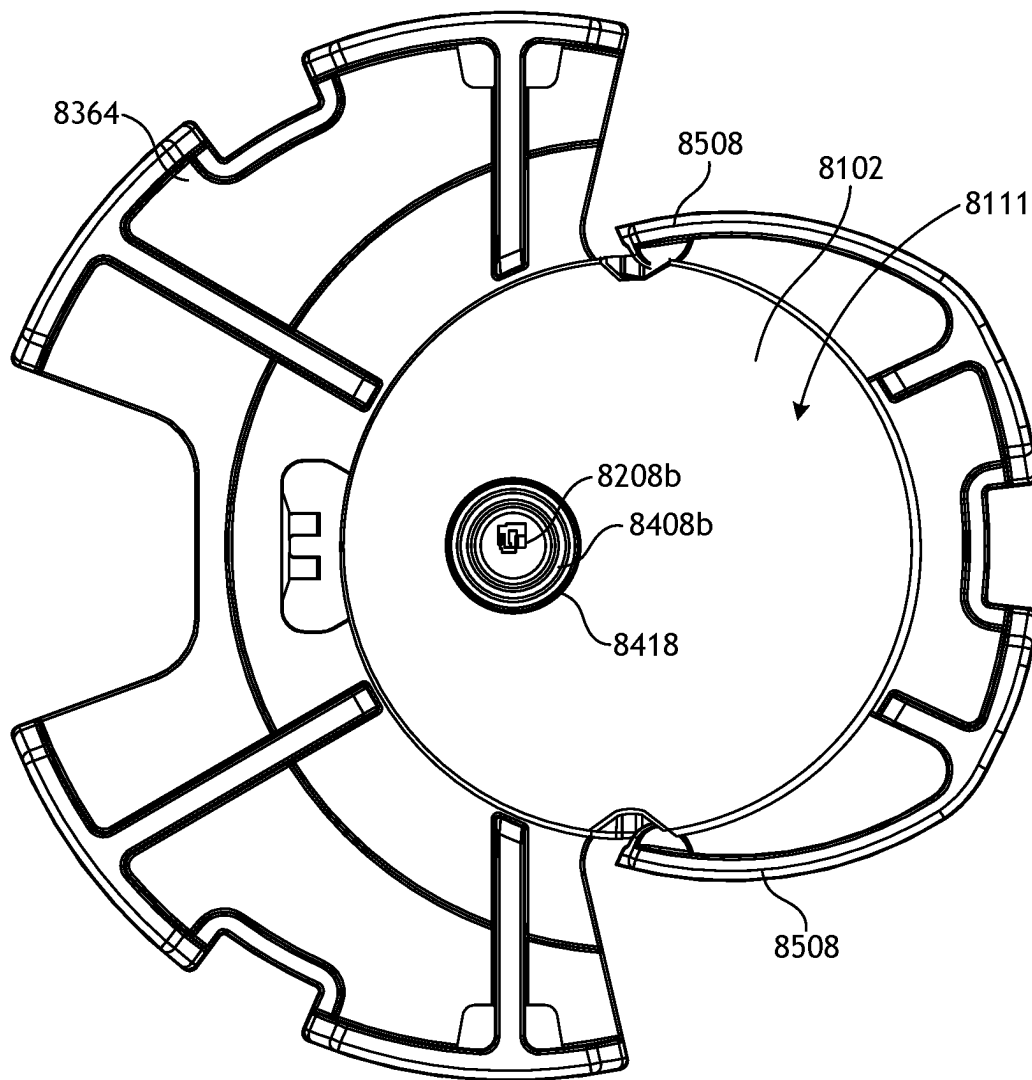


FIG. 84

*FIG. 85*

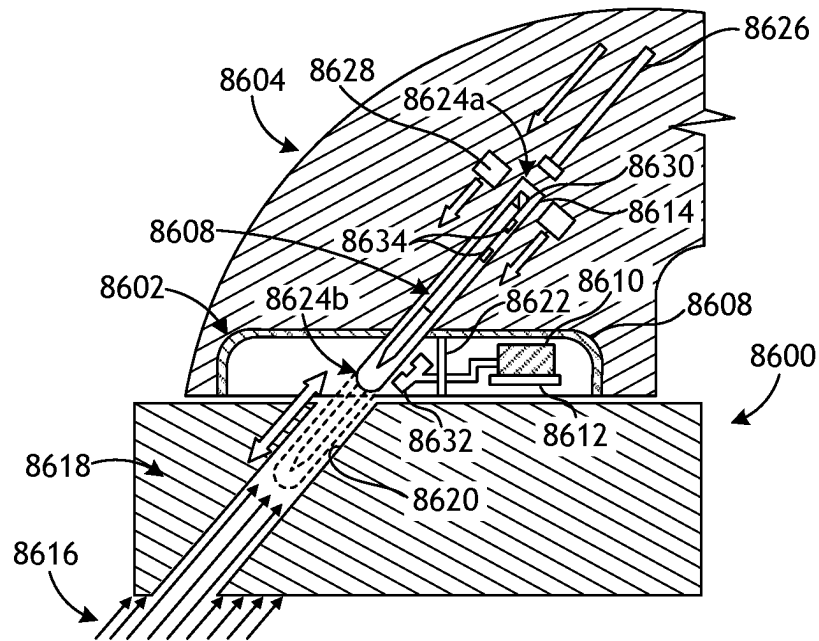


FIG. 86

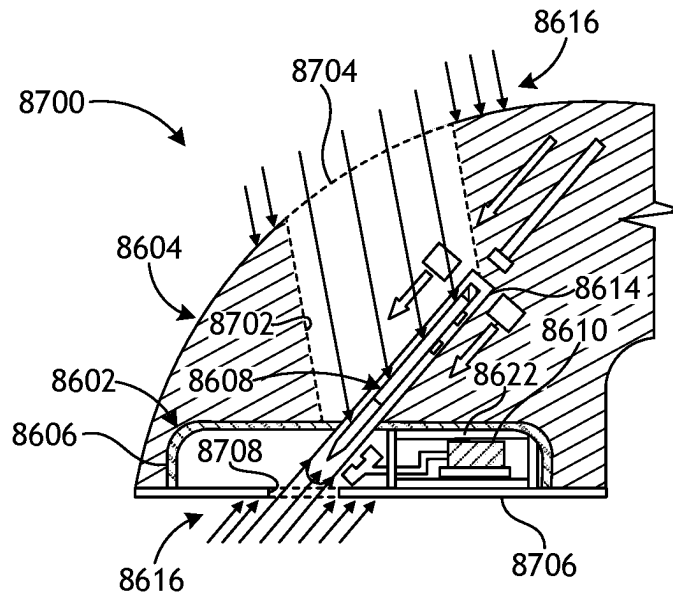


FIG. 87

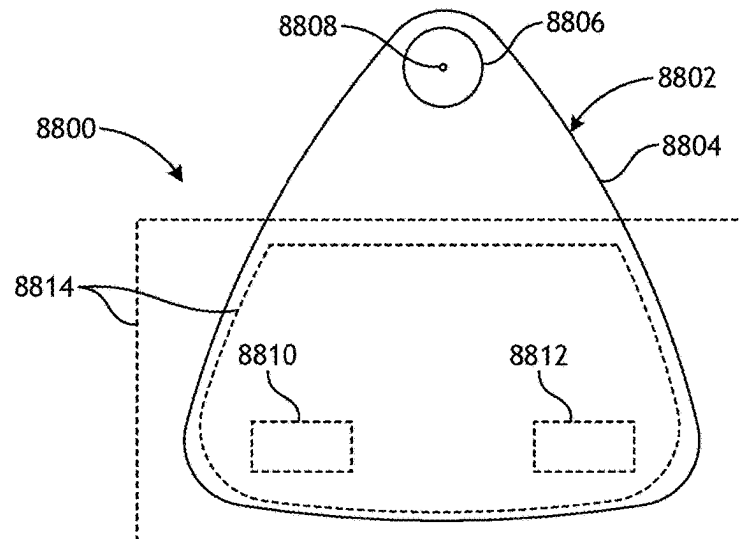


FIG. 88A

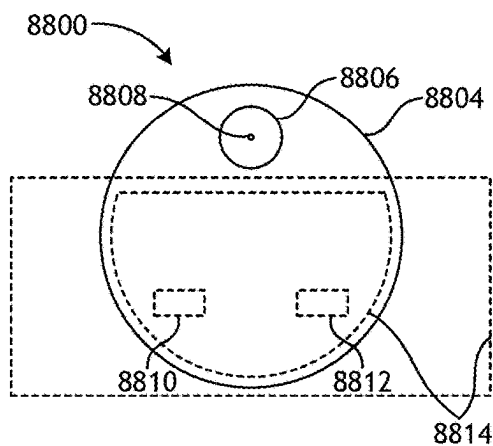


FIG. 88B

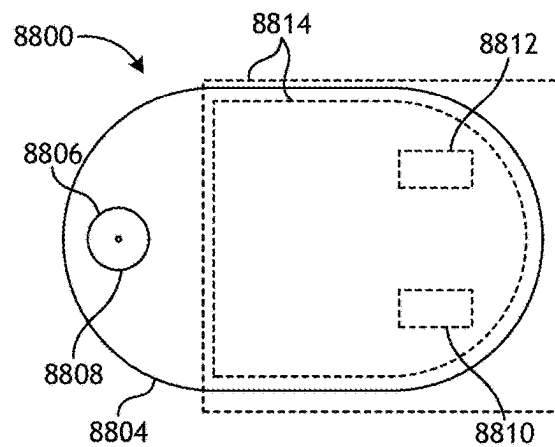


FIG. 88C

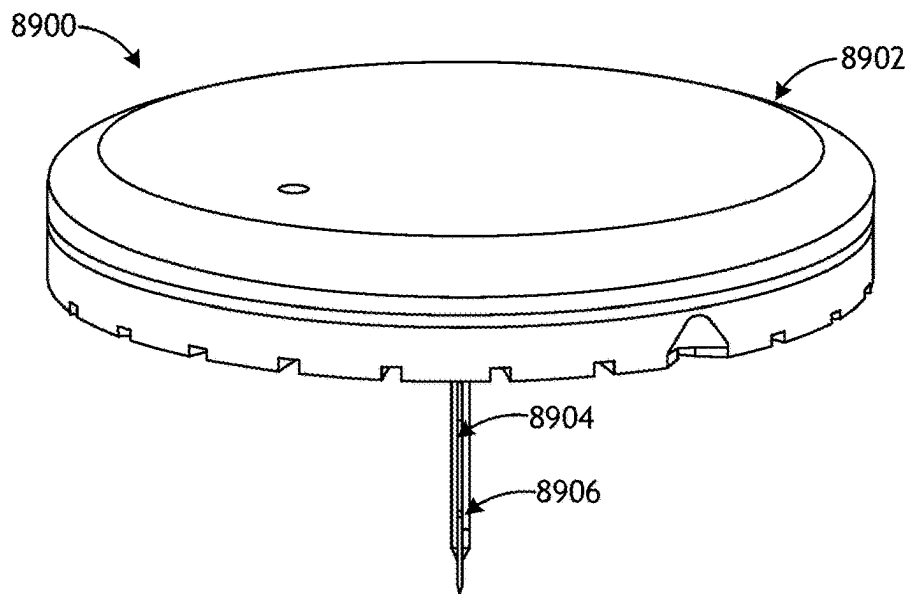


FIG. 89

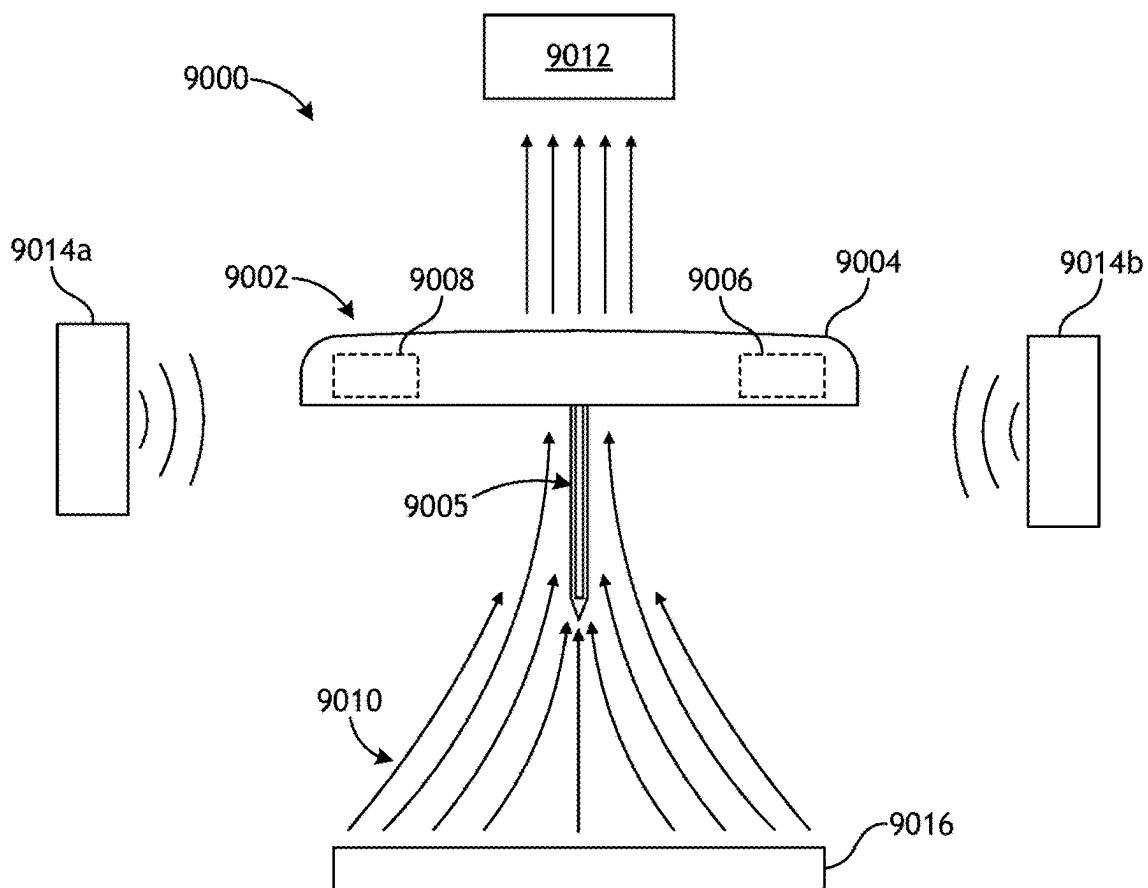


FIG. 90

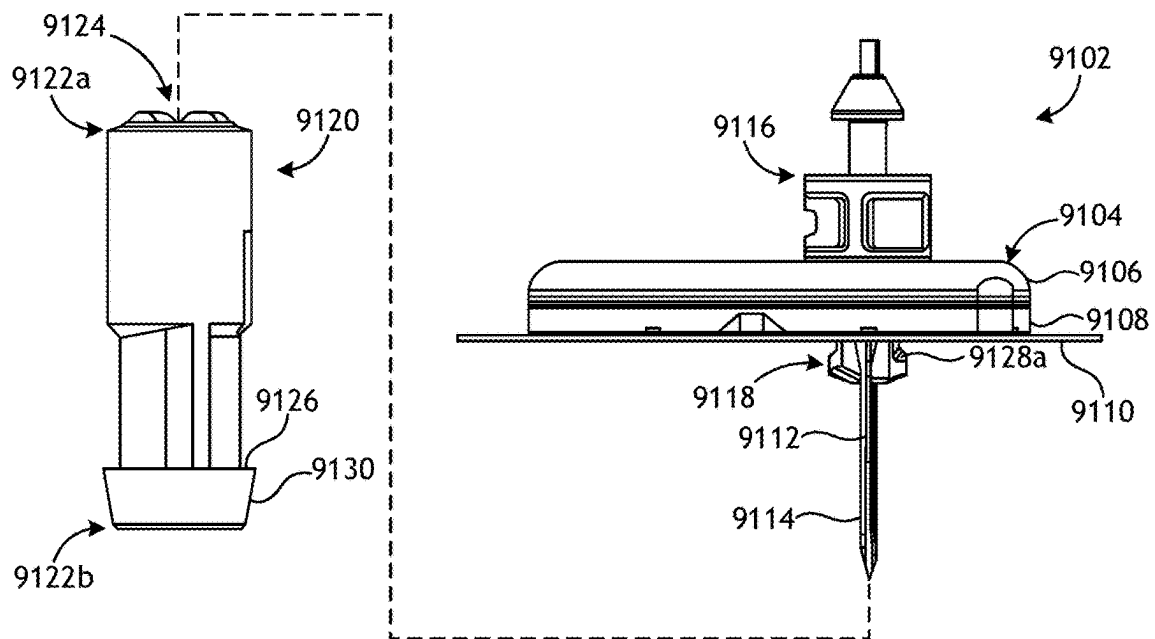


FIG. 91A

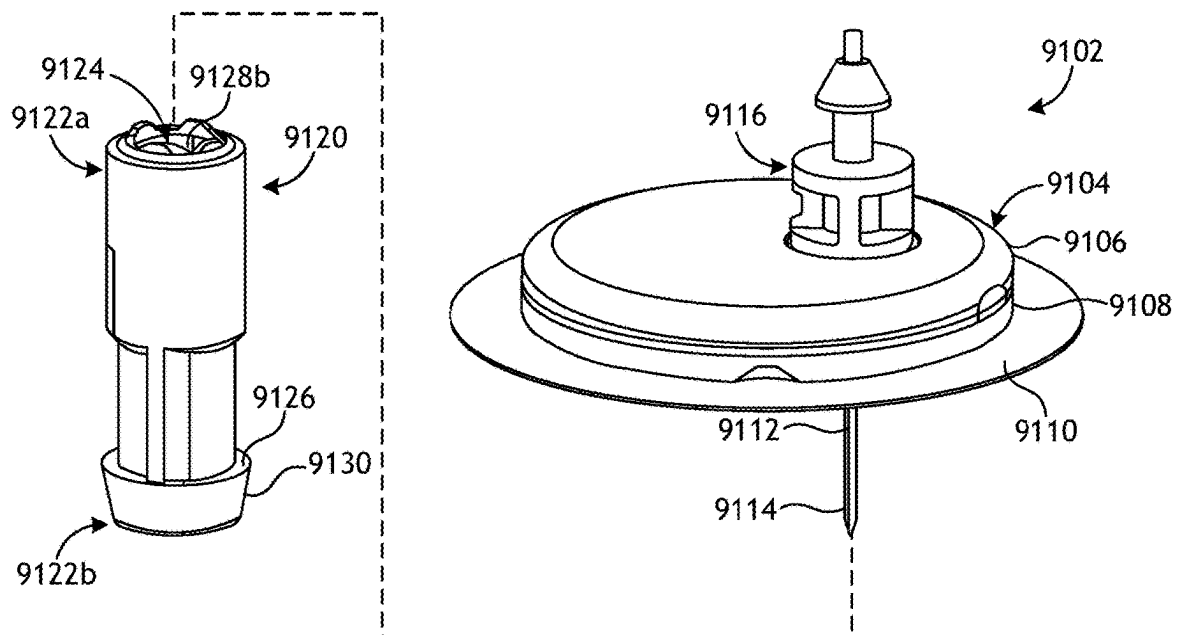


FIG. 91B

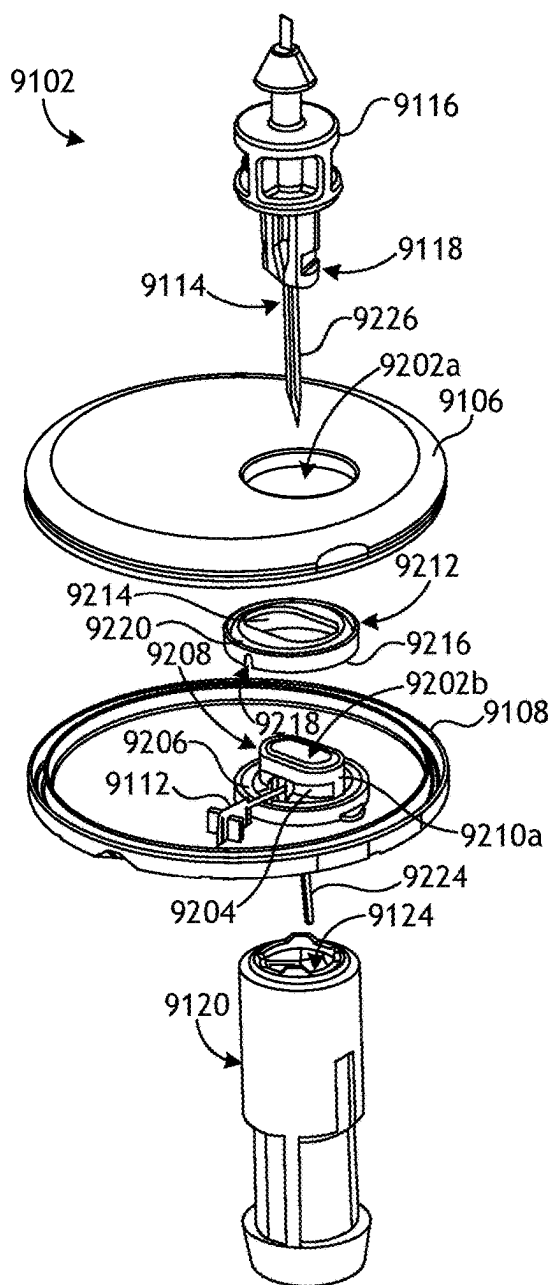


FIG. 92A

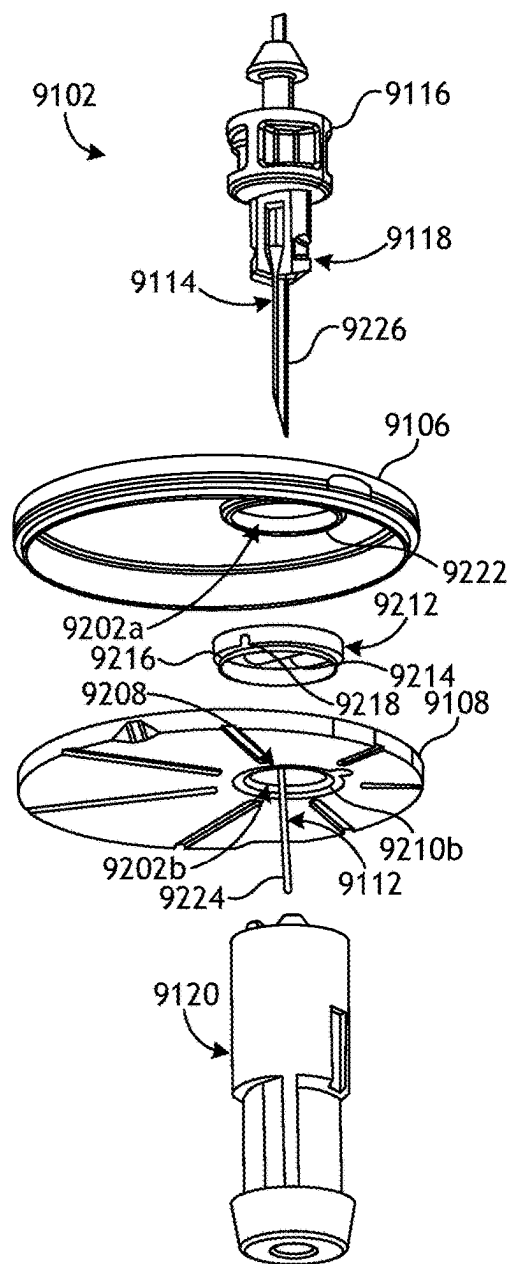


FIG. 92B

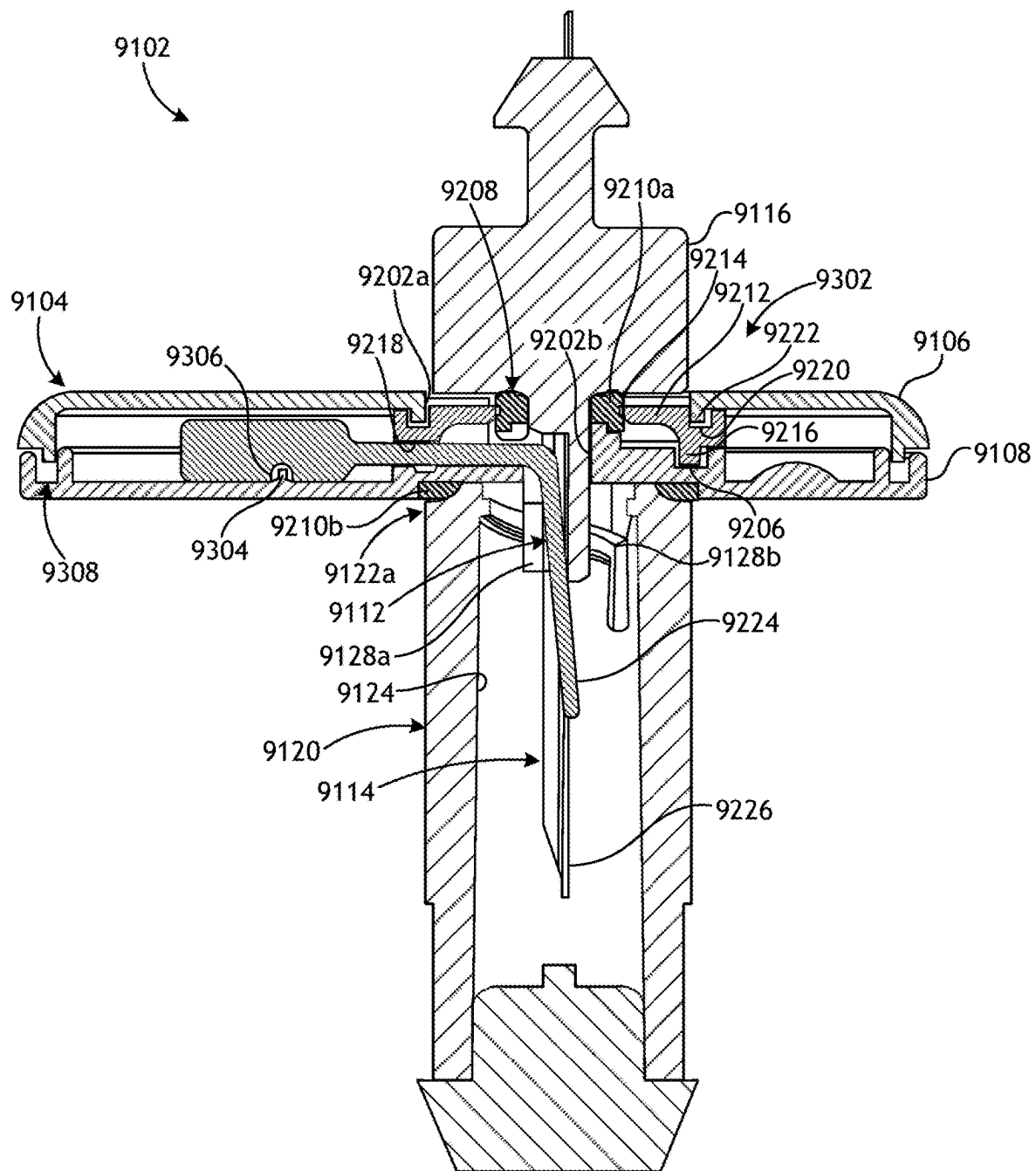


FIG. 93

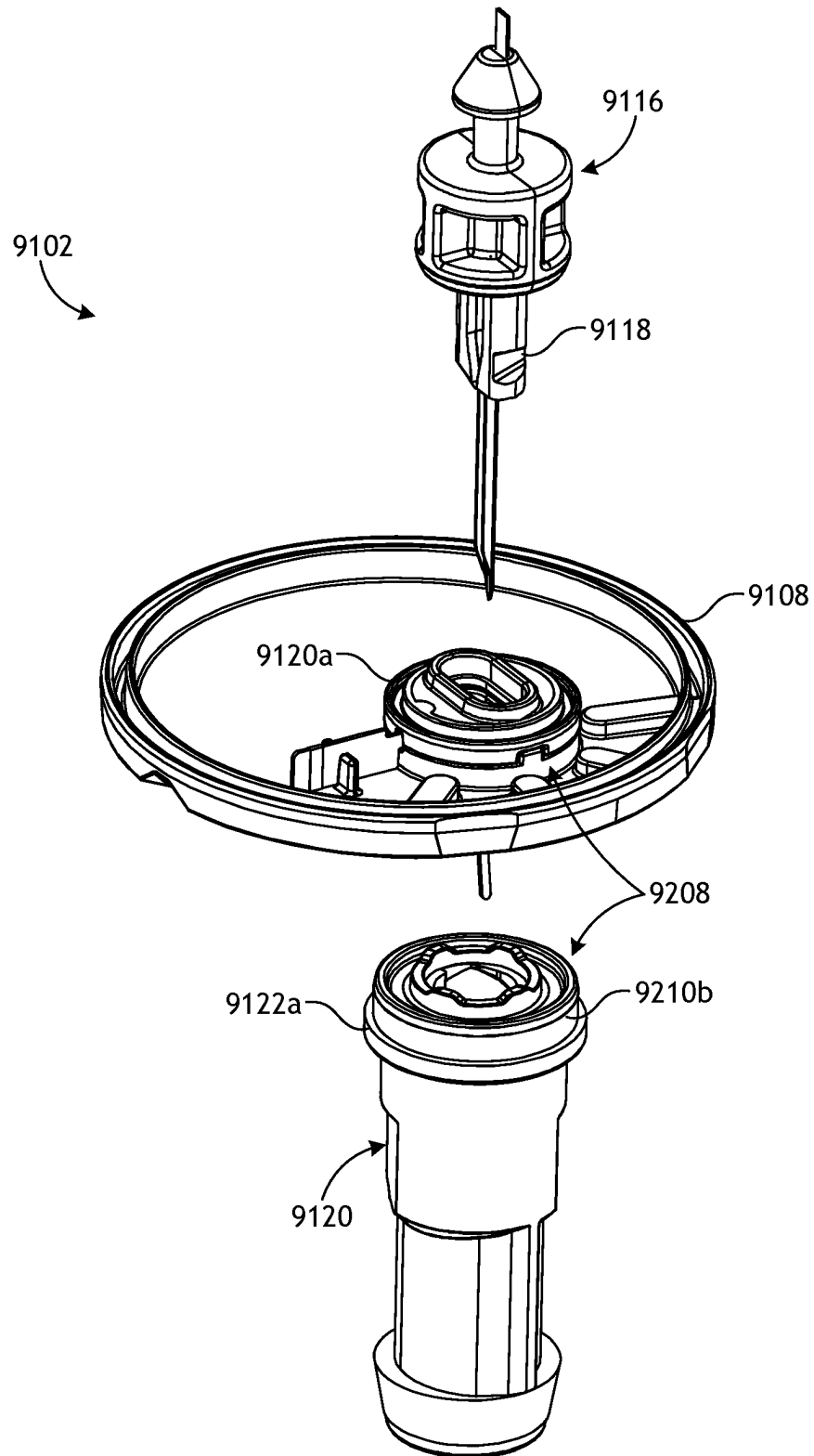


FIG. 93A

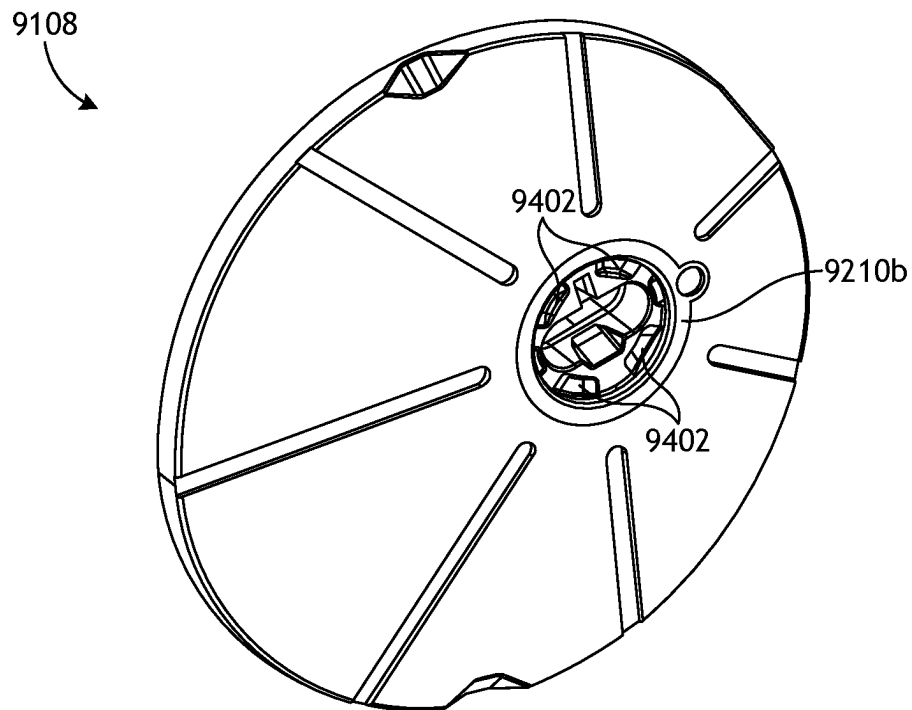


FIG. 94A

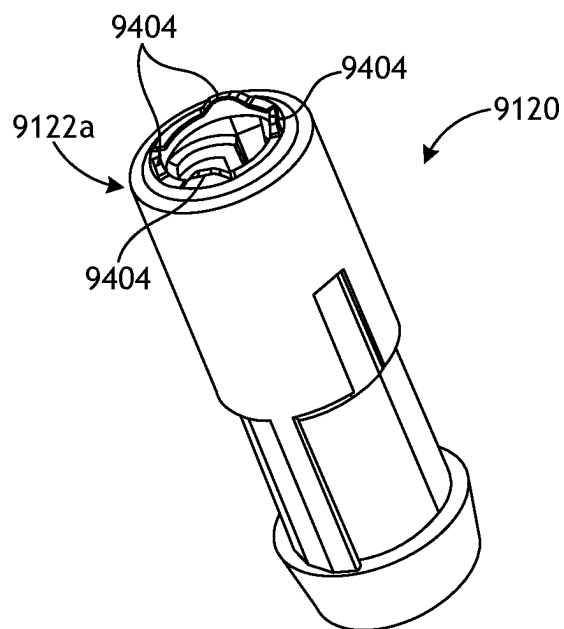


FIG. 94B

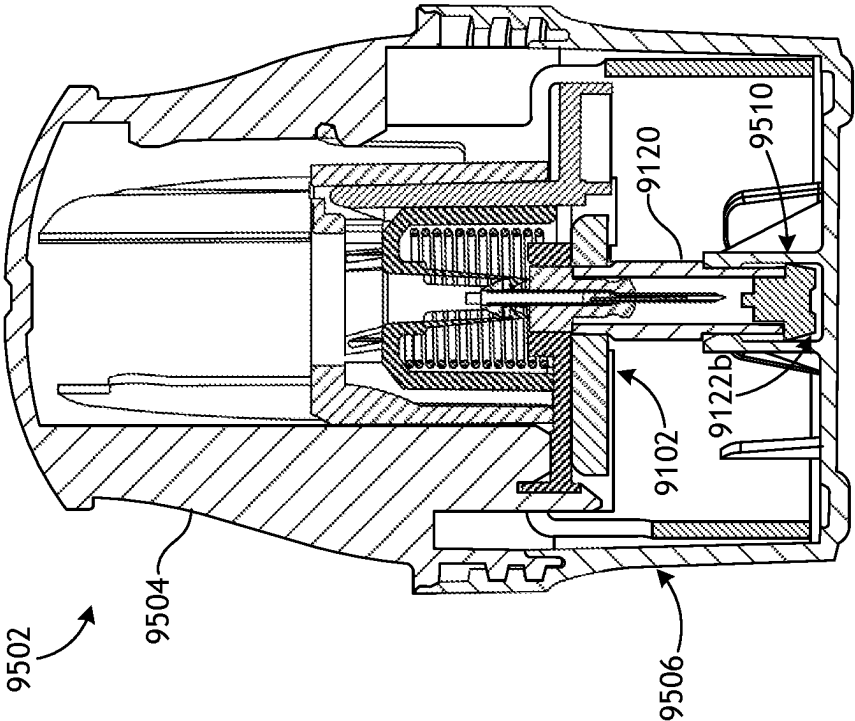


FIG. 95A

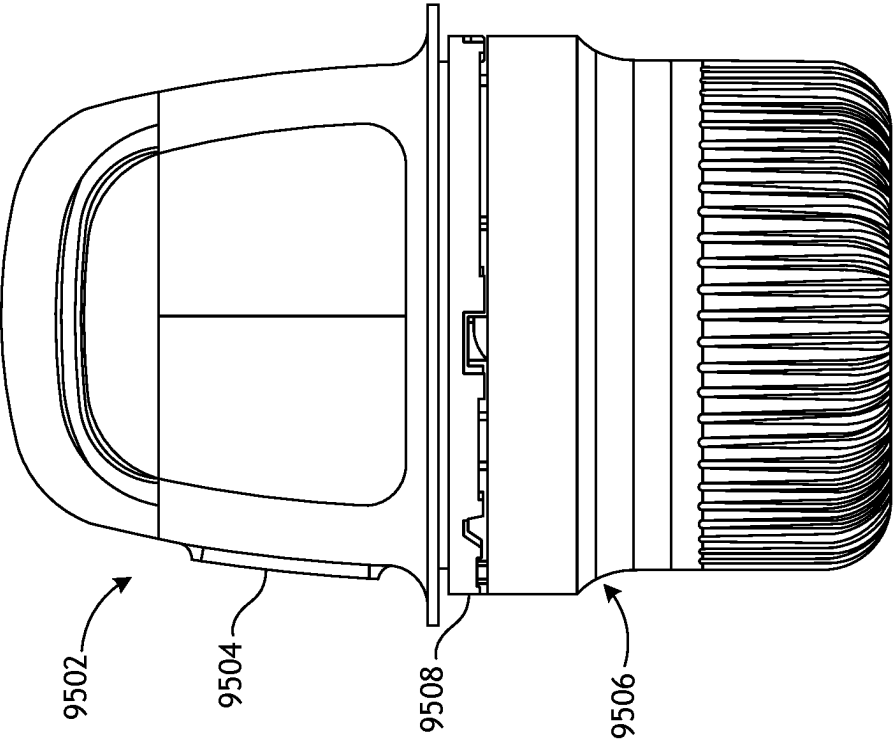


FIG. 95B

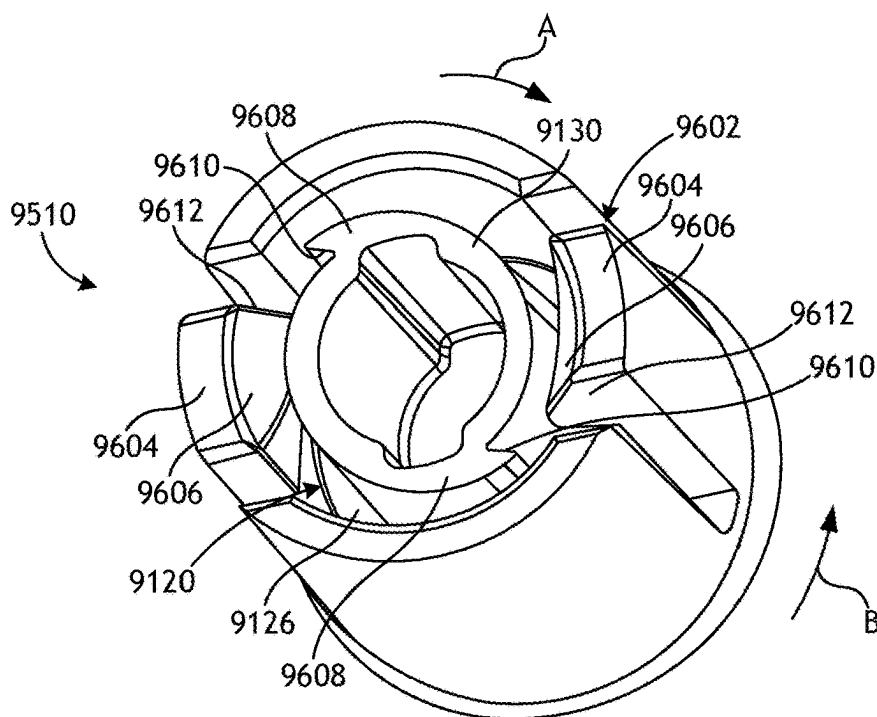


FIG. 96A

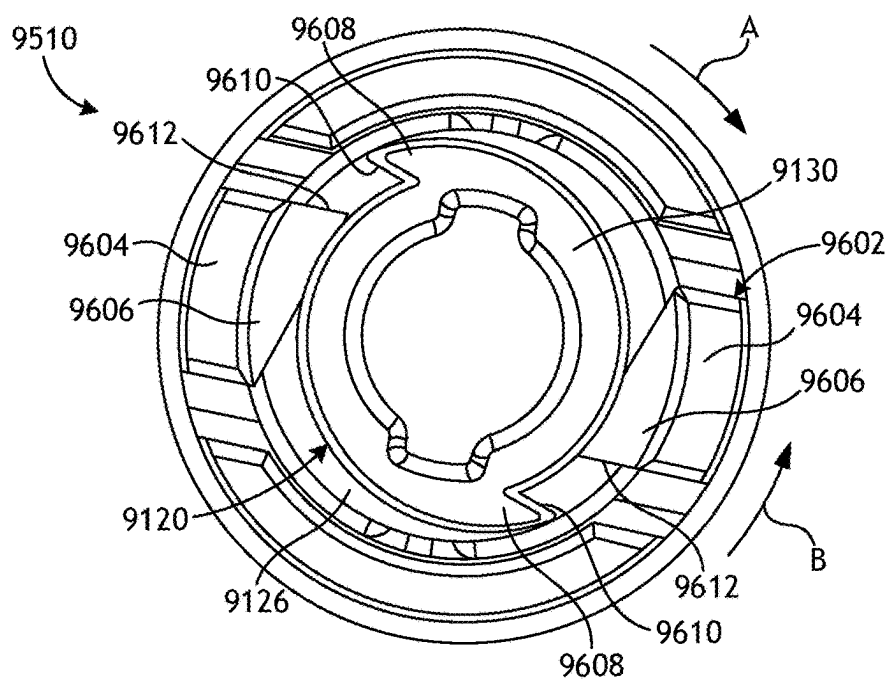


FIG. 96B

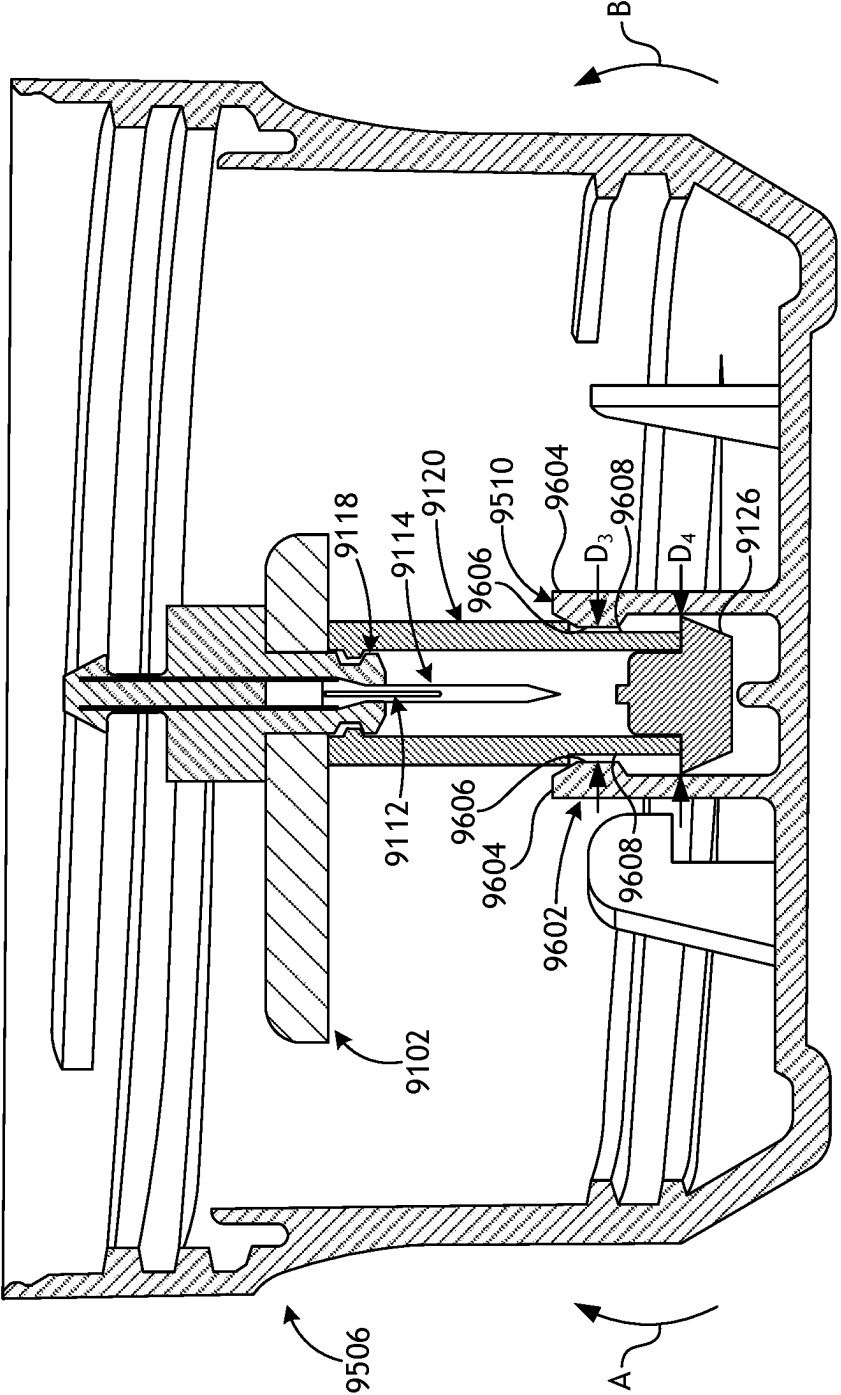


FIG. 97

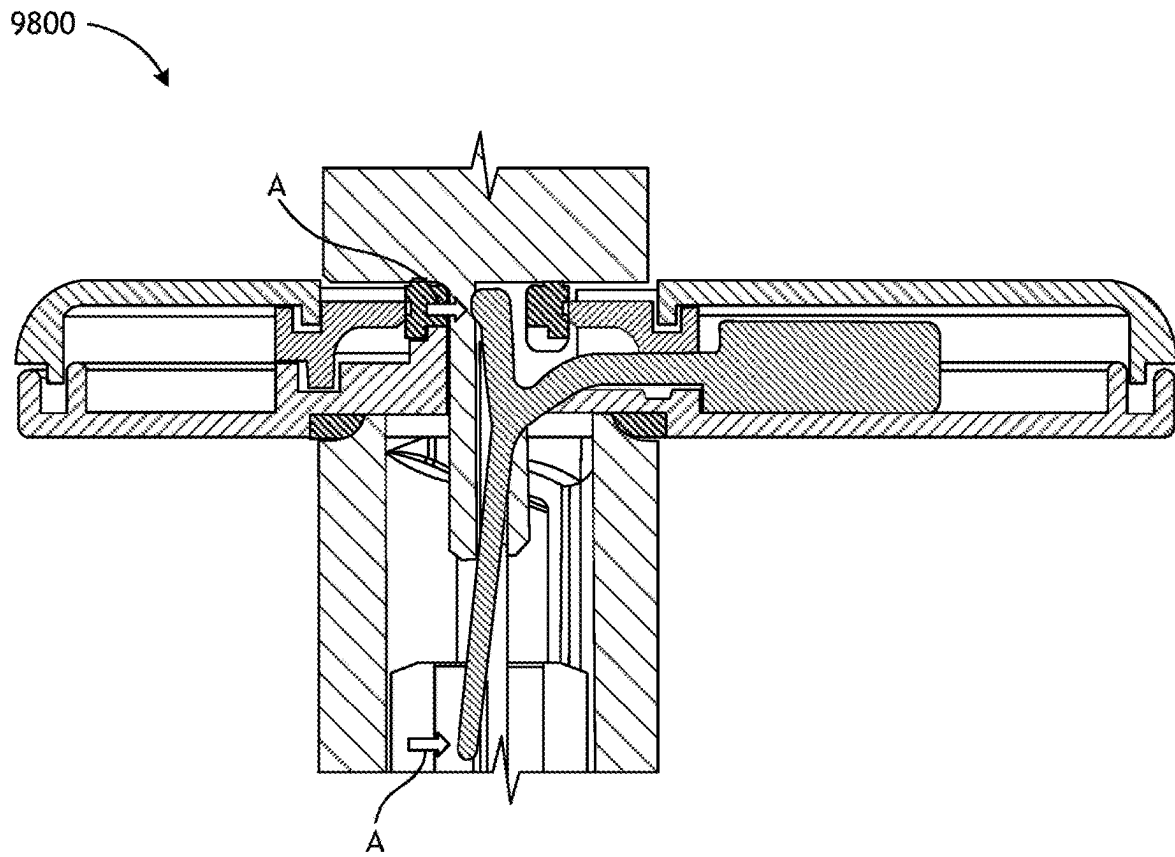
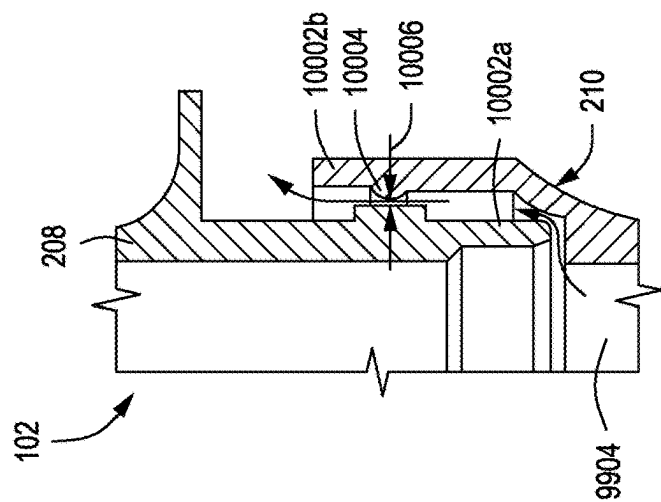
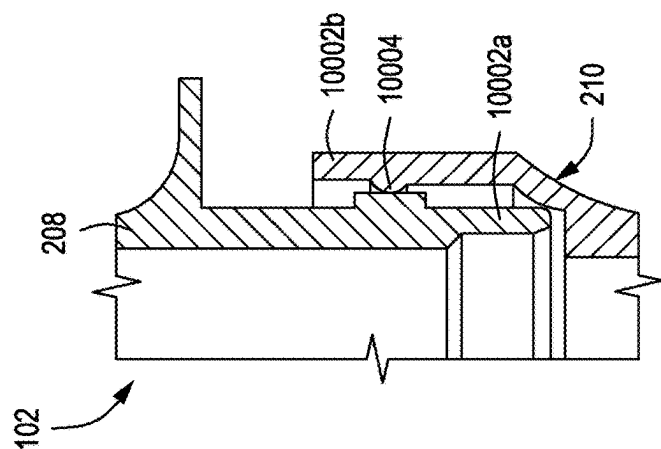
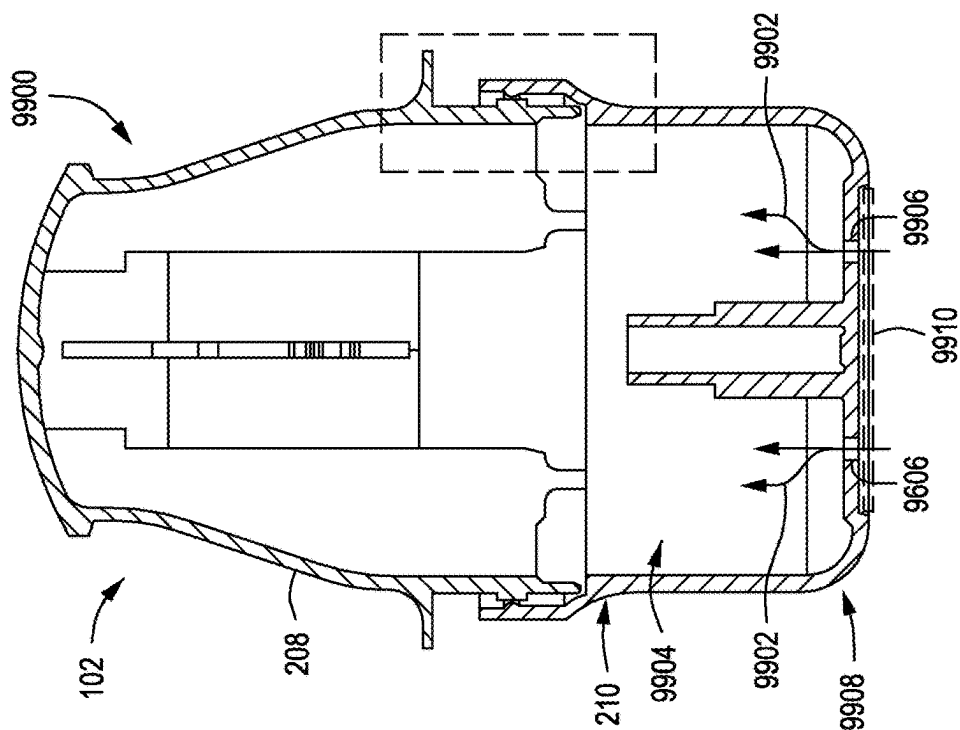


FIG. 98



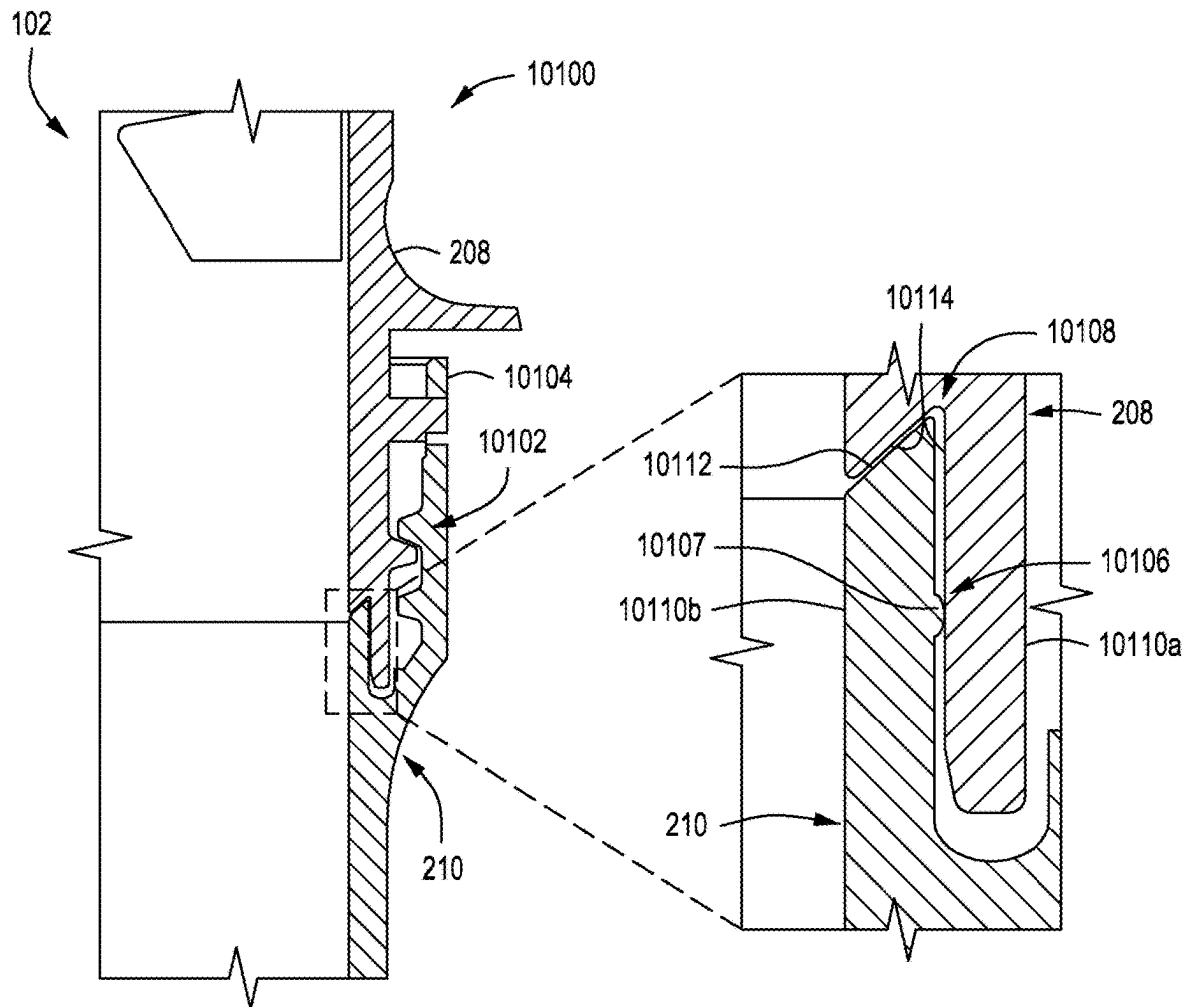


FIG. 101

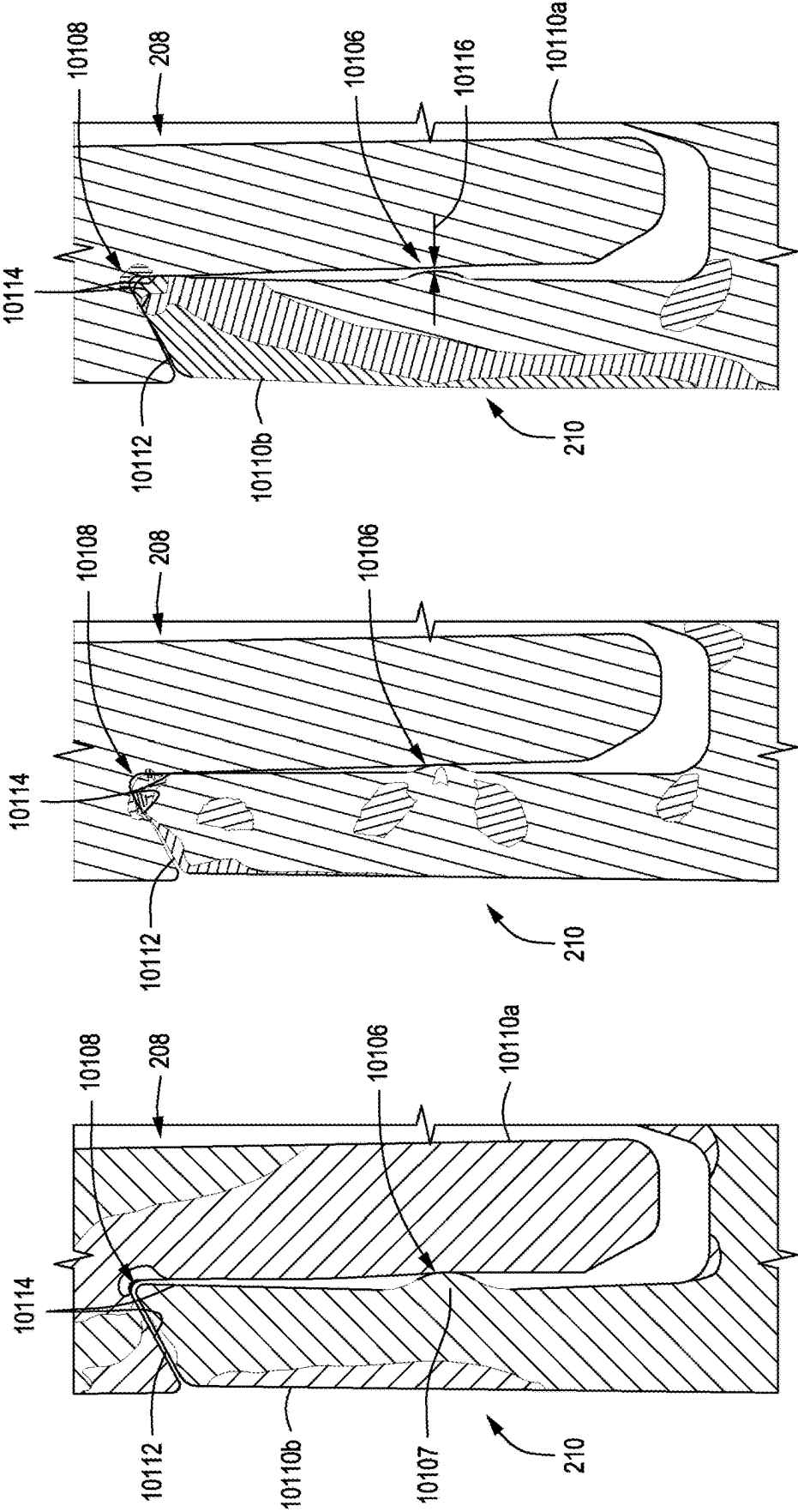


FIG. 102C

FIG. 102B

FIG. 102A

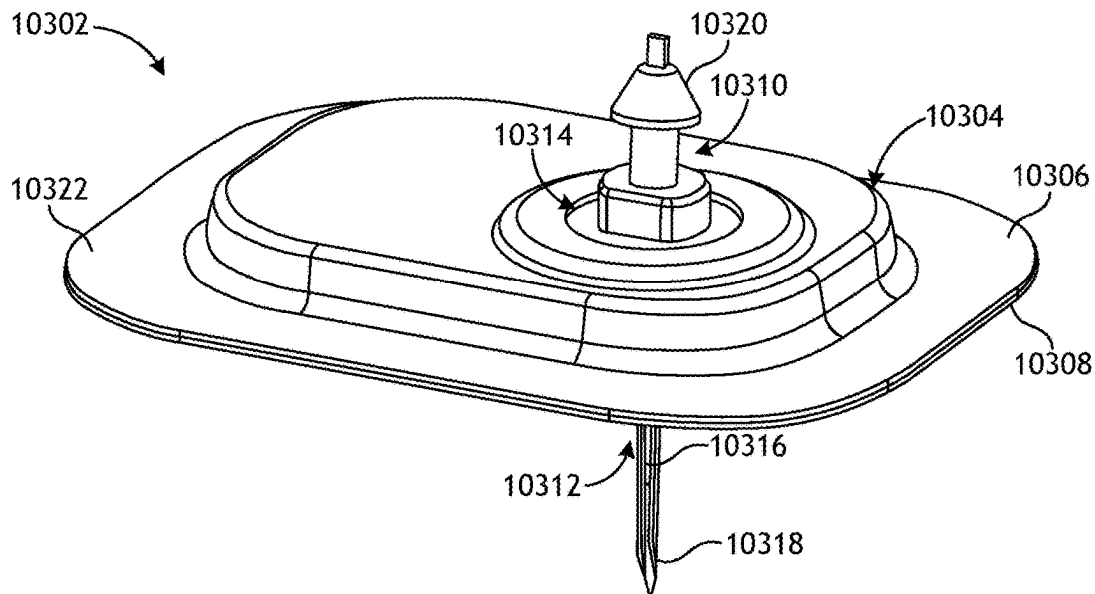


FIG. 103

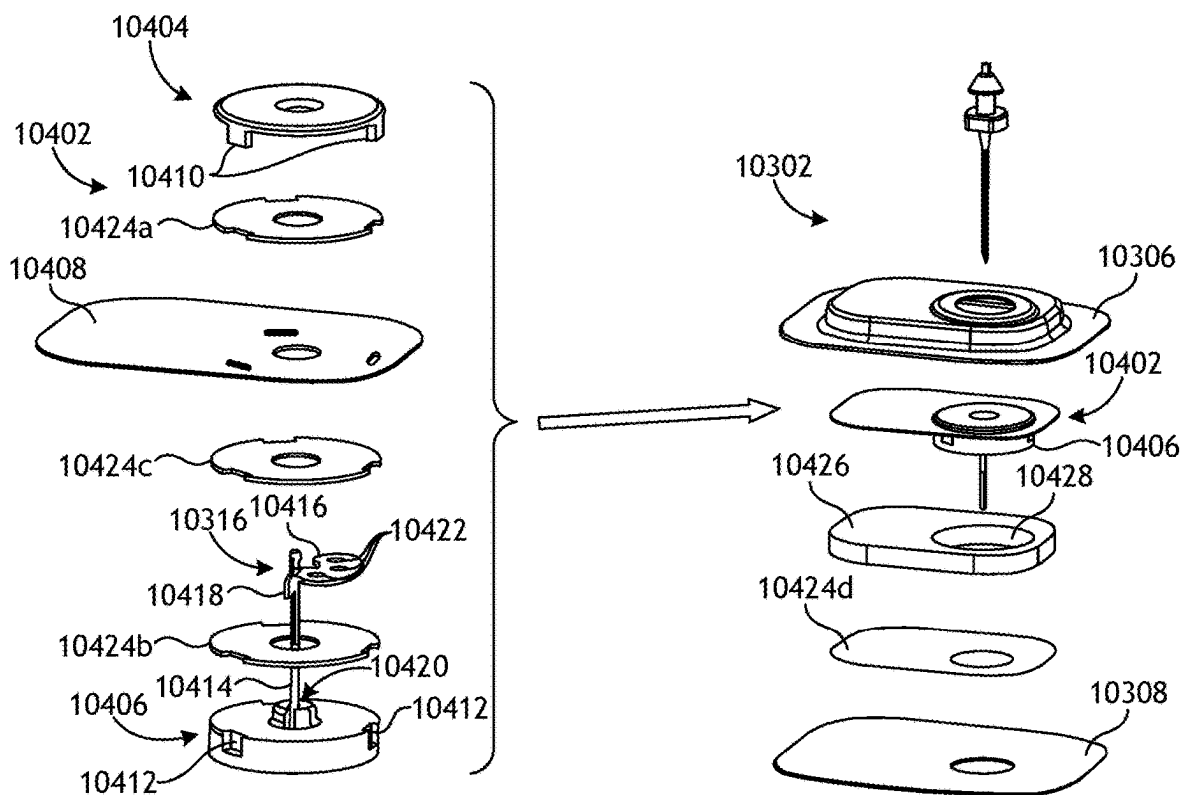


FIG. 104A

FIG. 104B

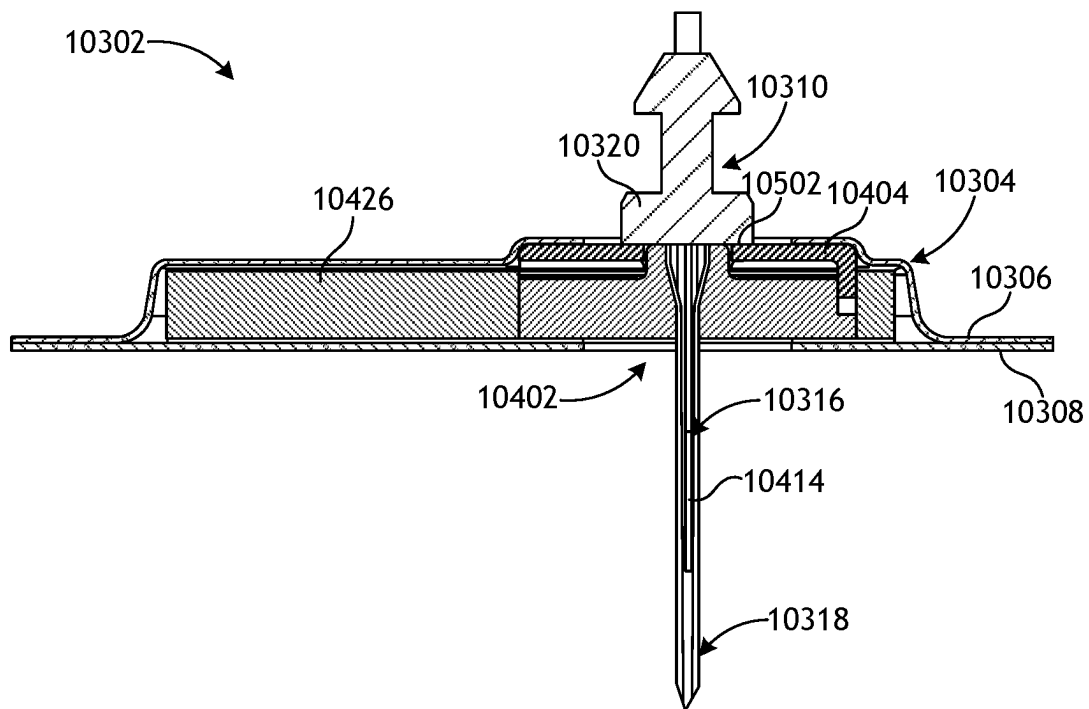


FIG. 105

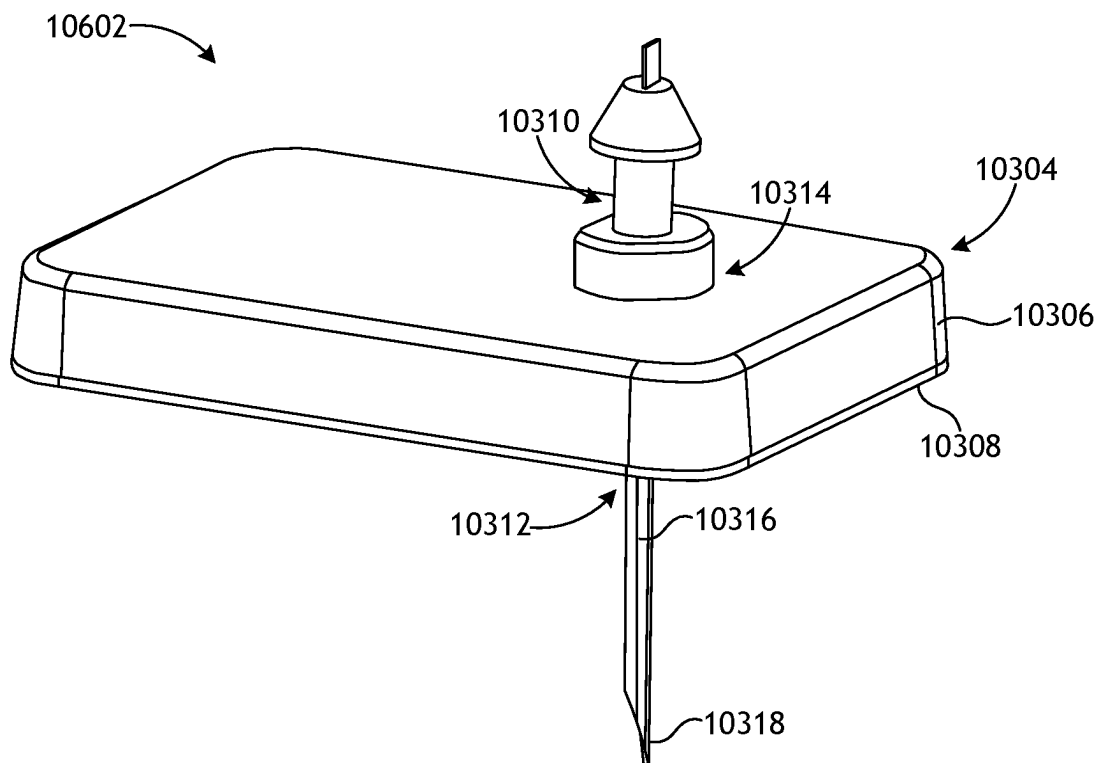


FIG. 106

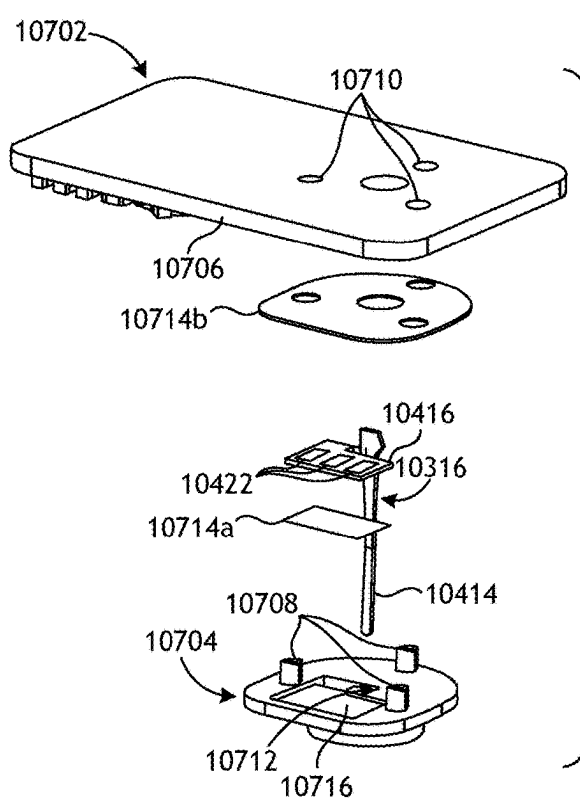


FIG. 107A

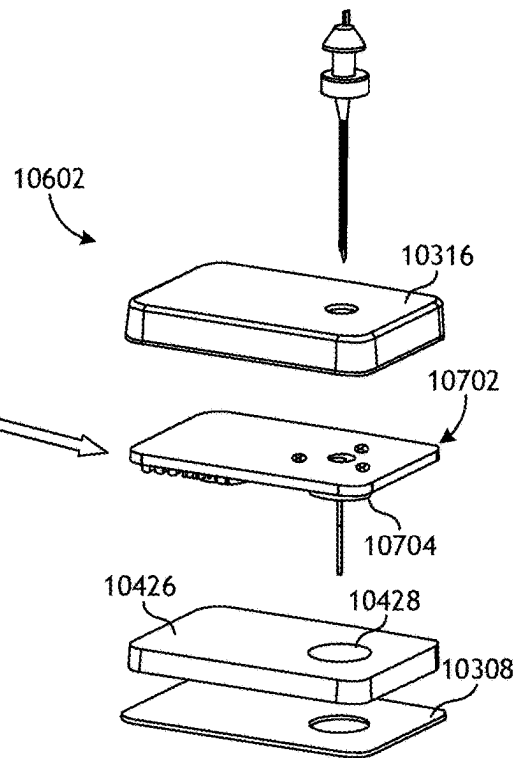


FIG. 107B

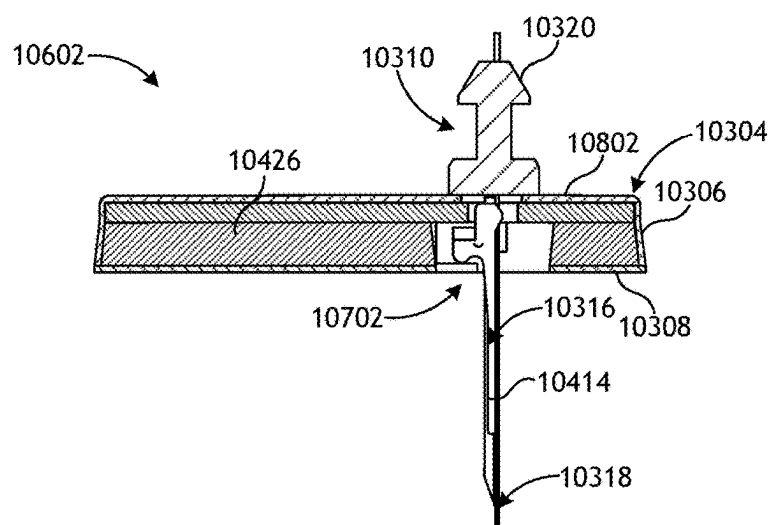
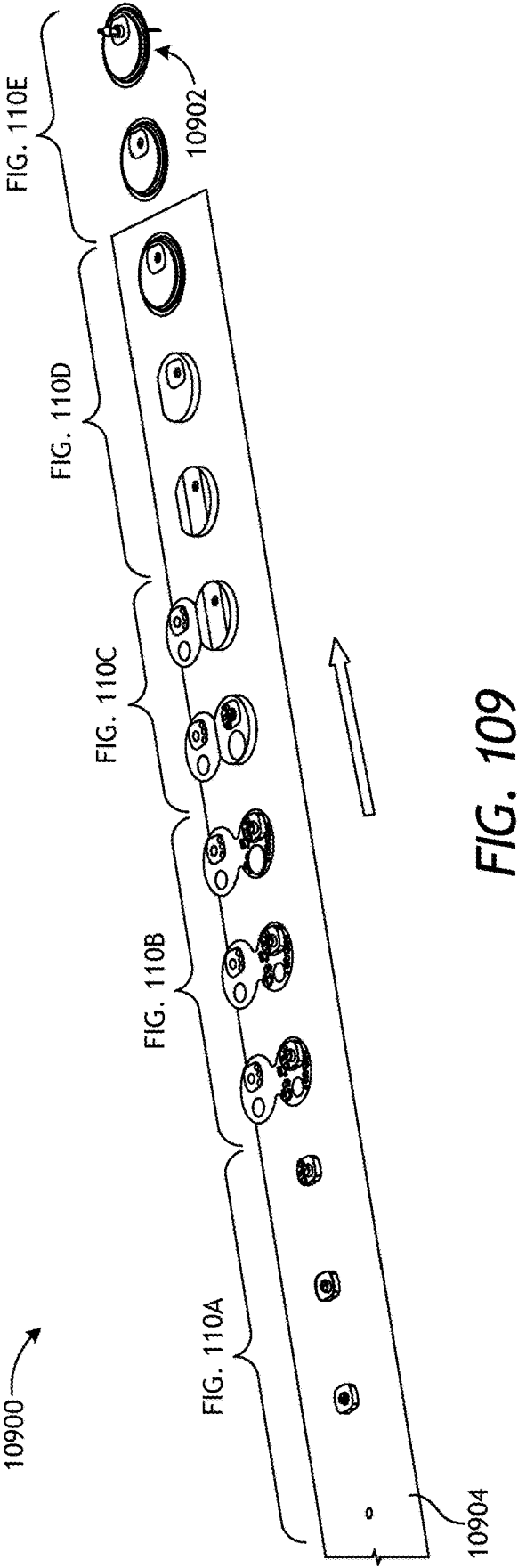
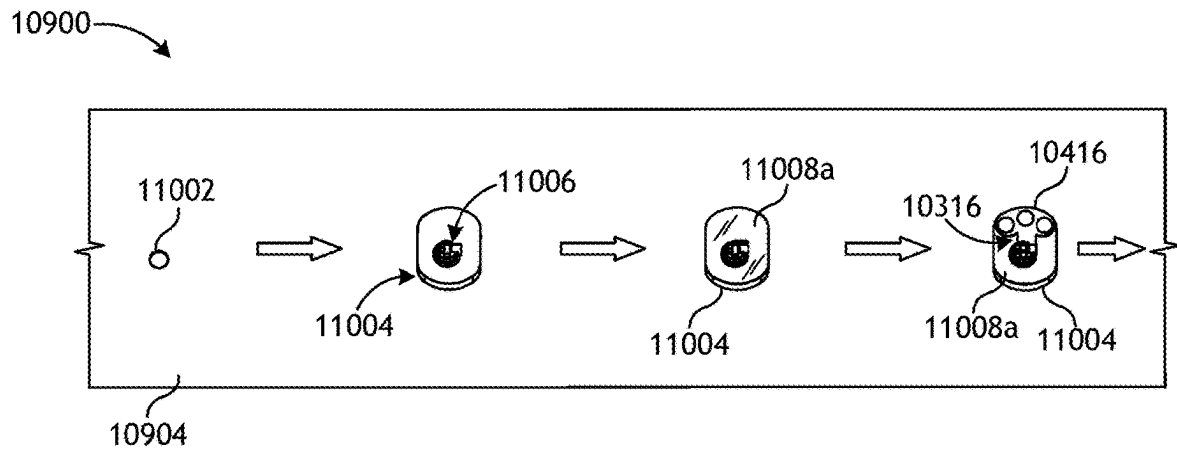
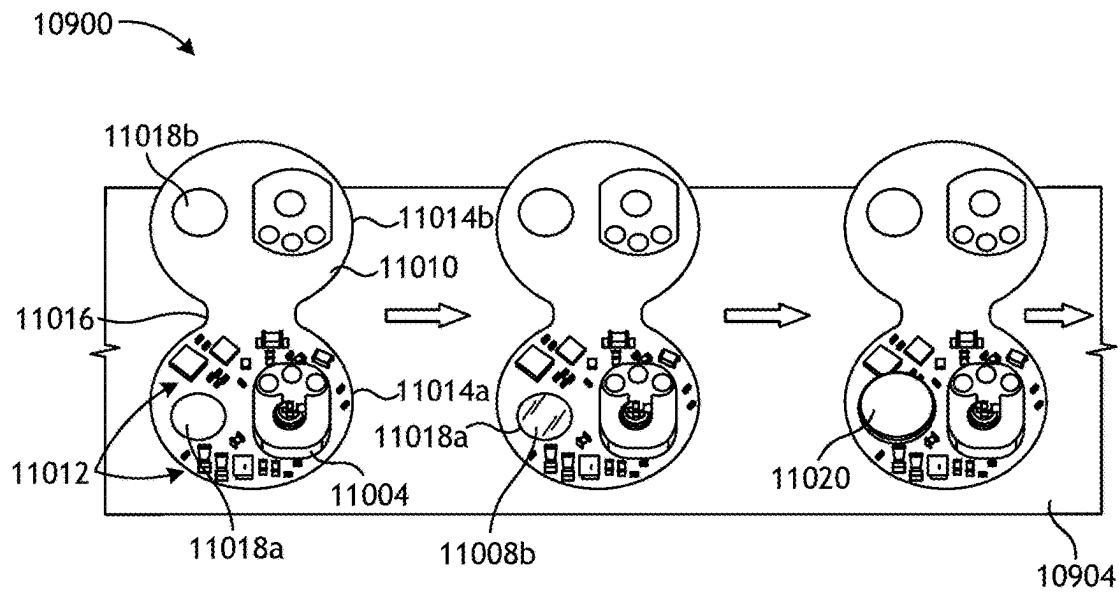


FIG. 108



*FIG. 110A**FIG. 110B*

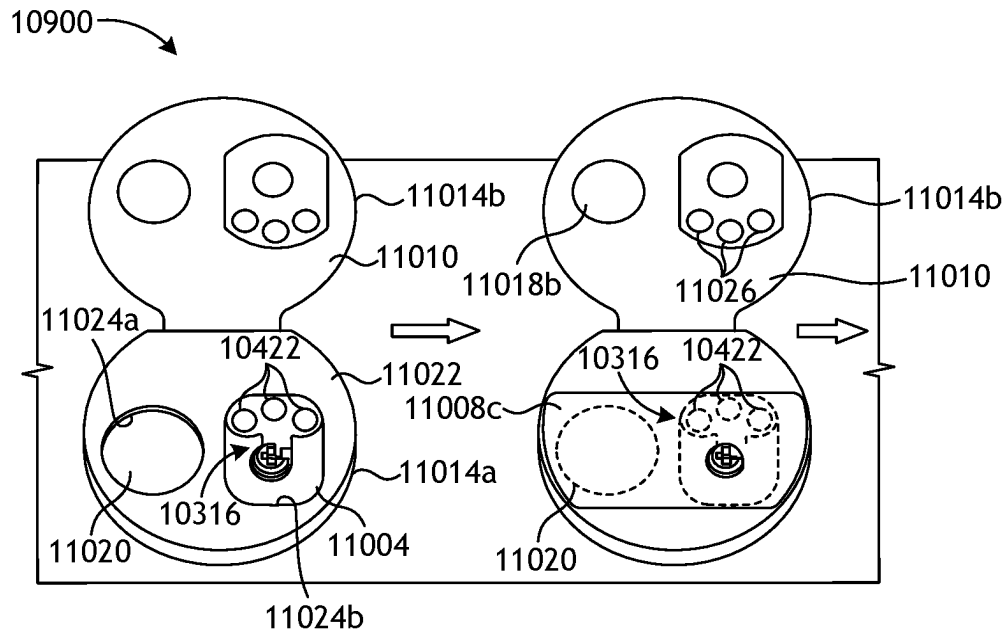


FIG. 110C

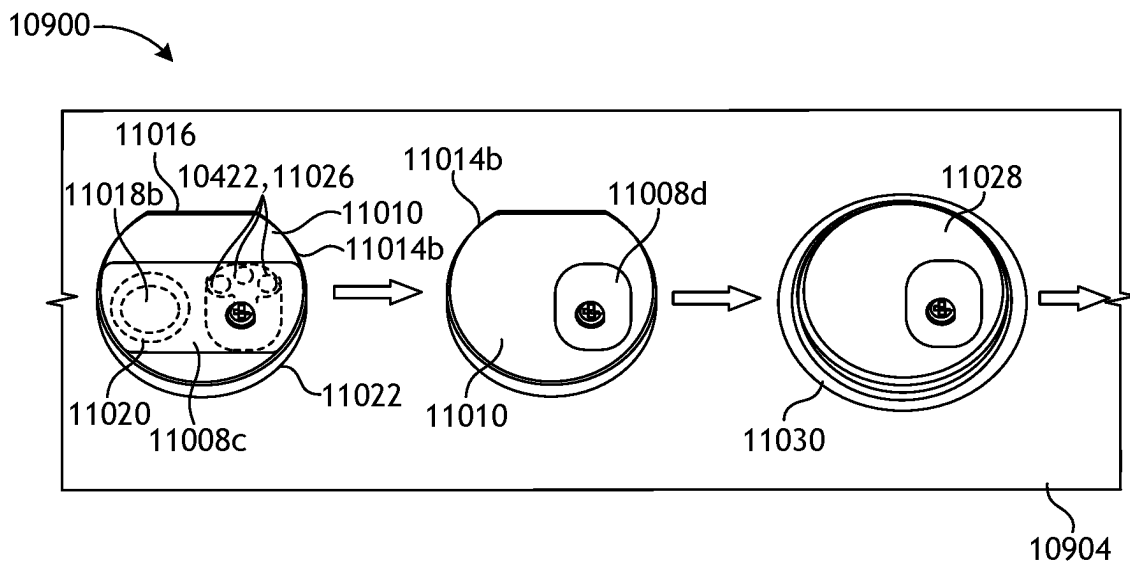


FIG. 110D

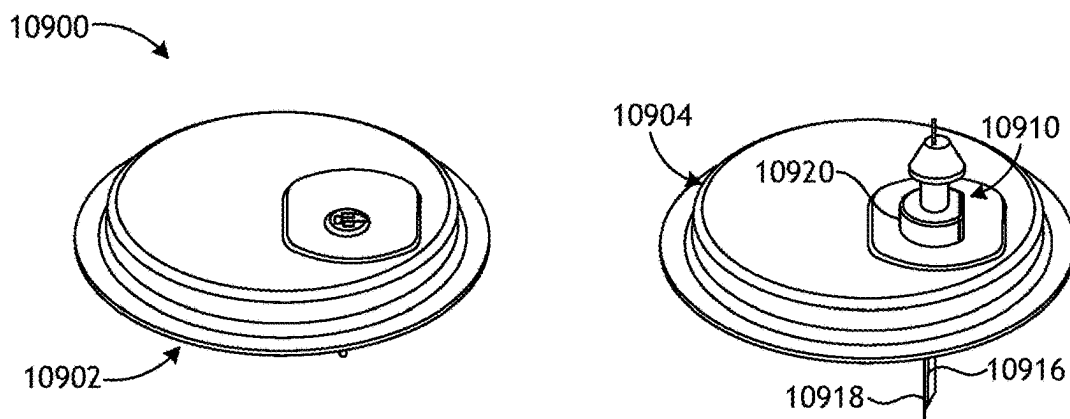


FIG. 110E

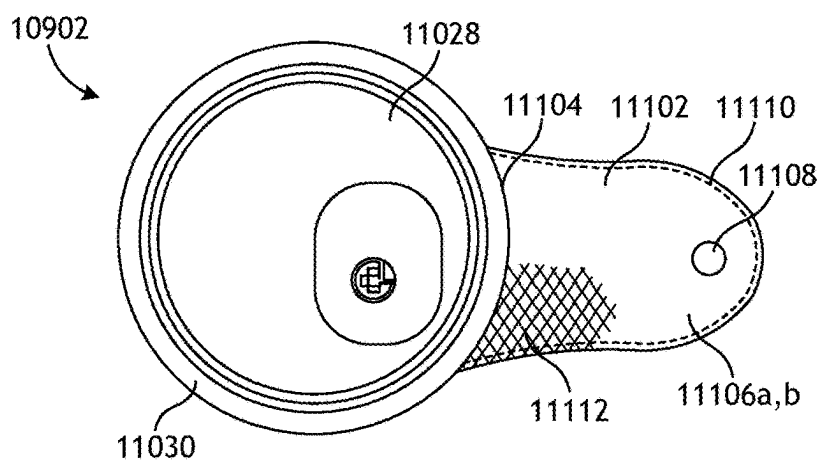


FIG. 111A

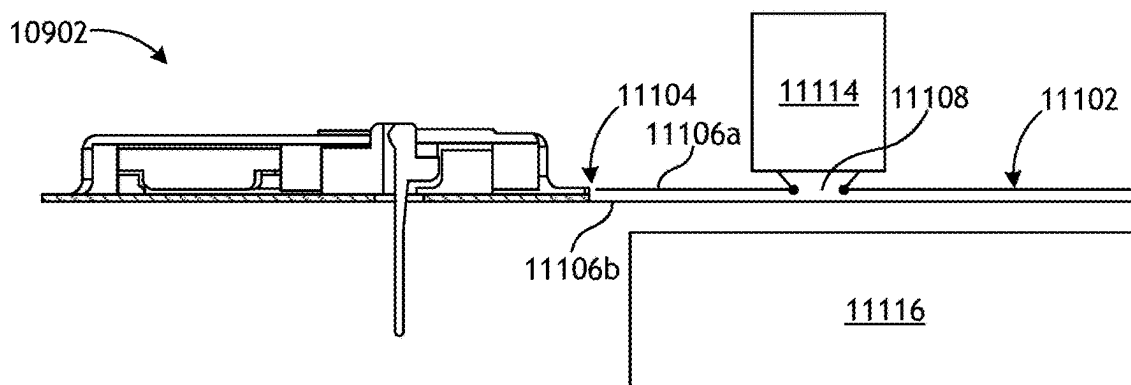


FIG. 111B

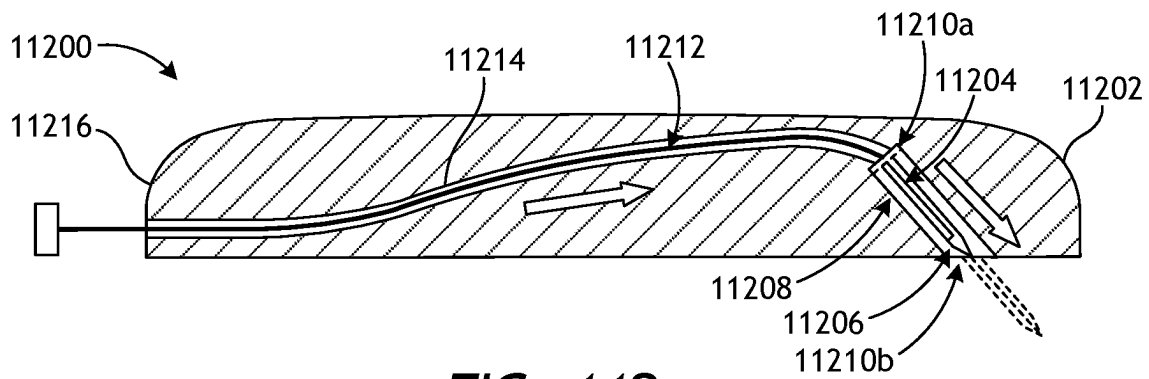


FIG. 112

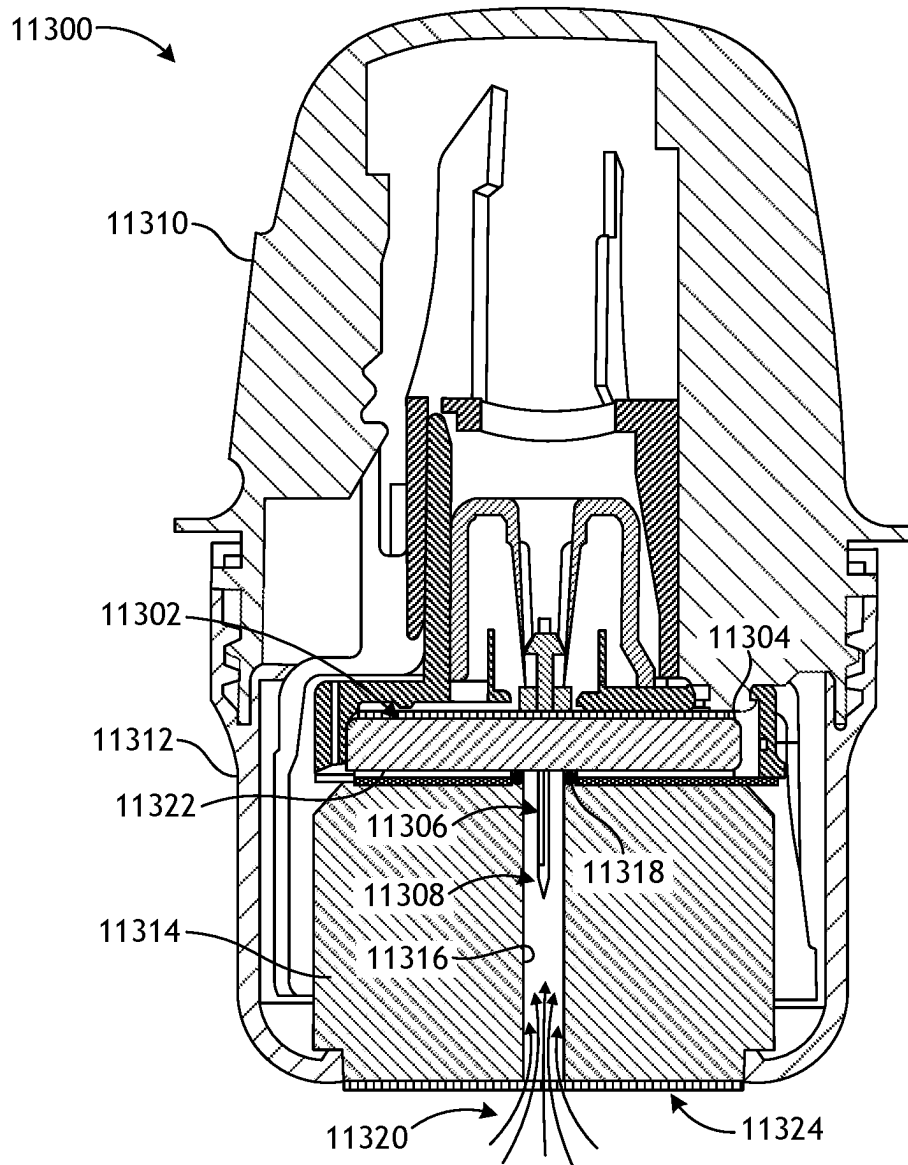


FIG. 113

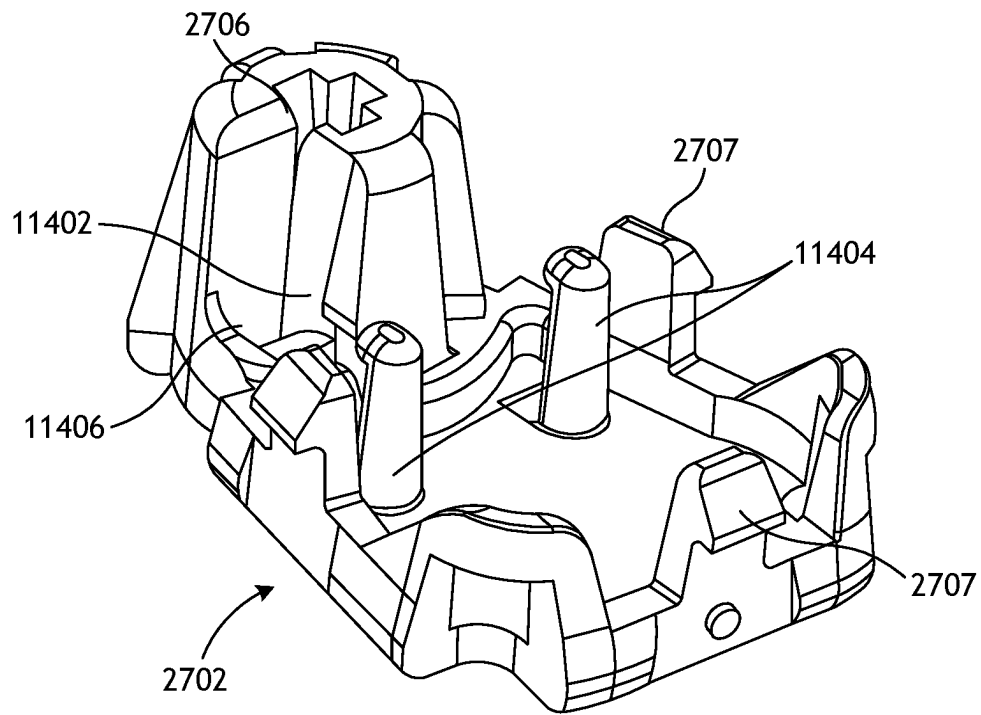


FIG. 114A

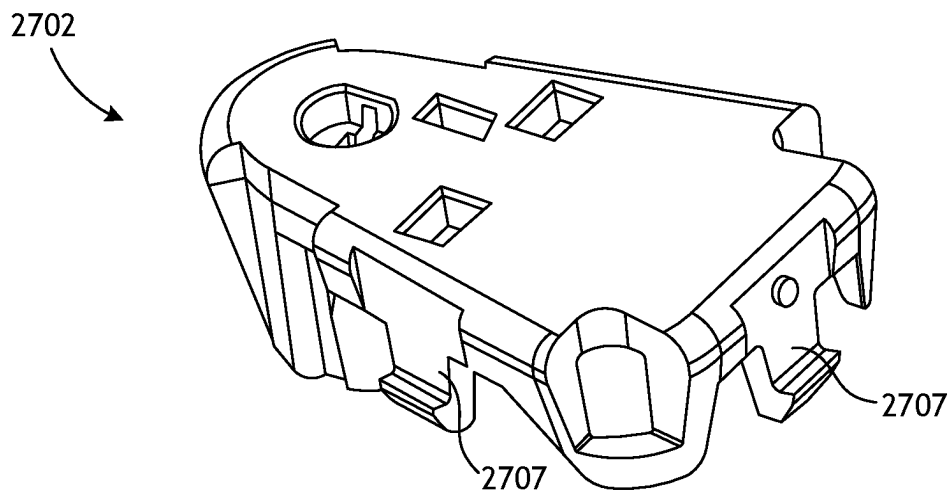


FIG. 114B

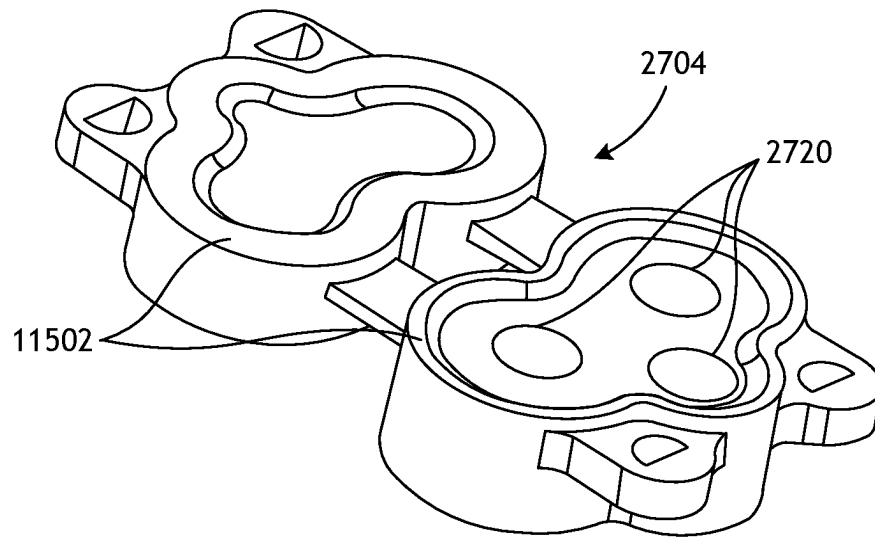


FIG. 115A

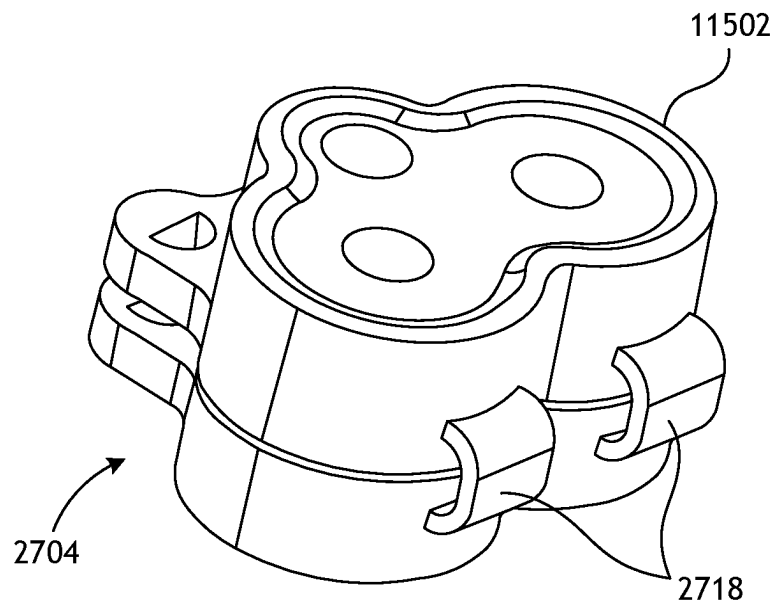


FIG. 115B

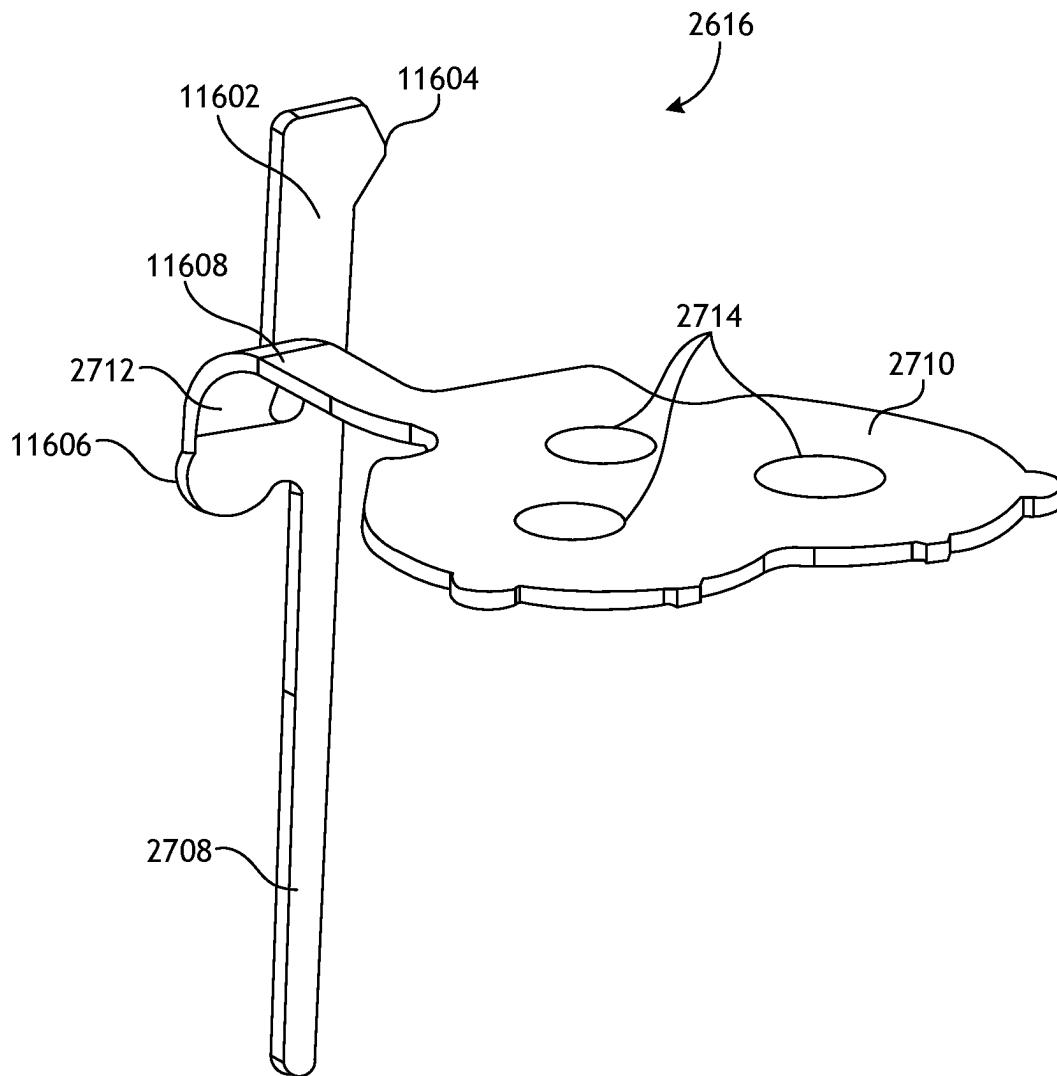


FIG. 116

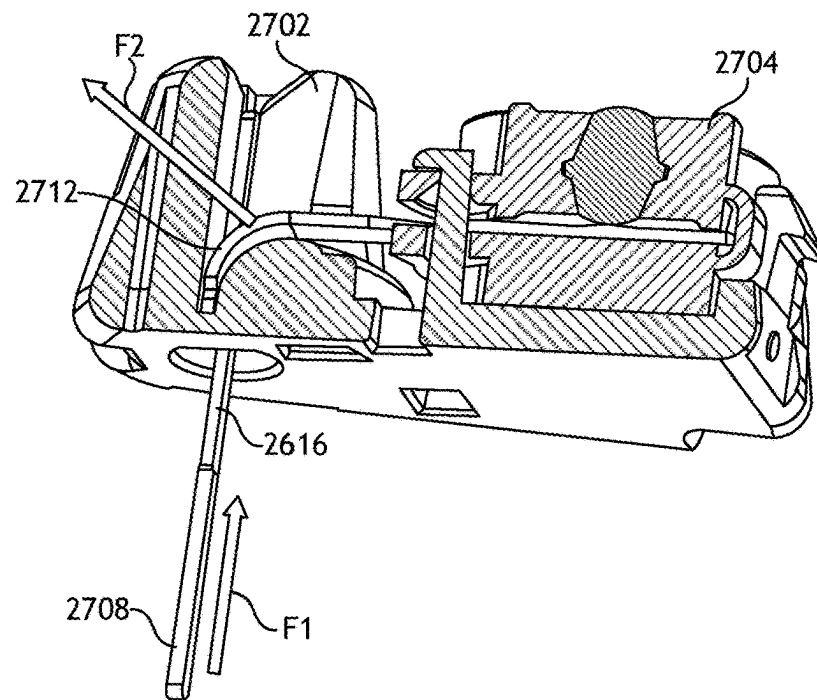


FIG. 117A

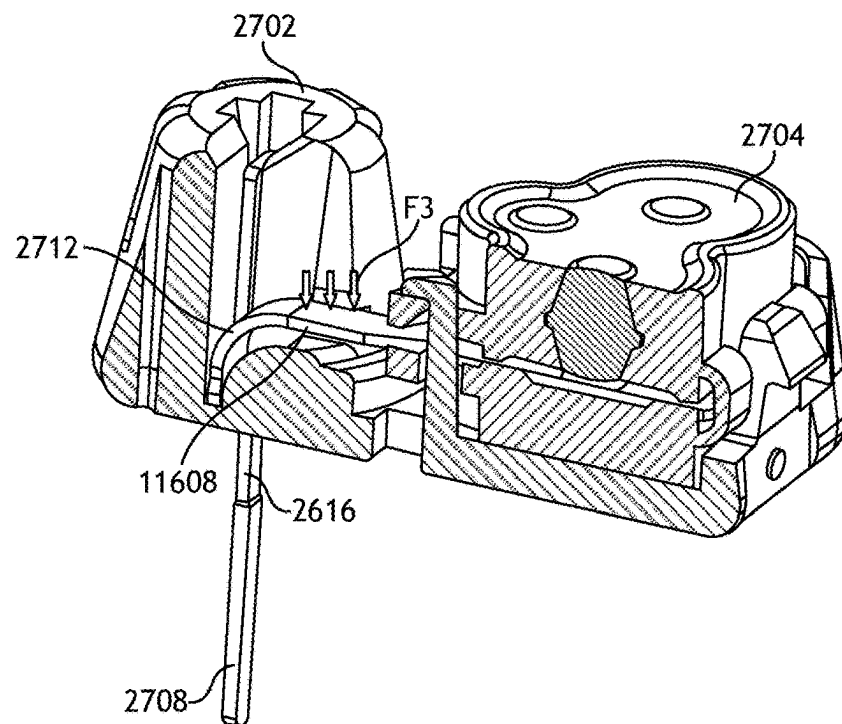


FIG. 117B

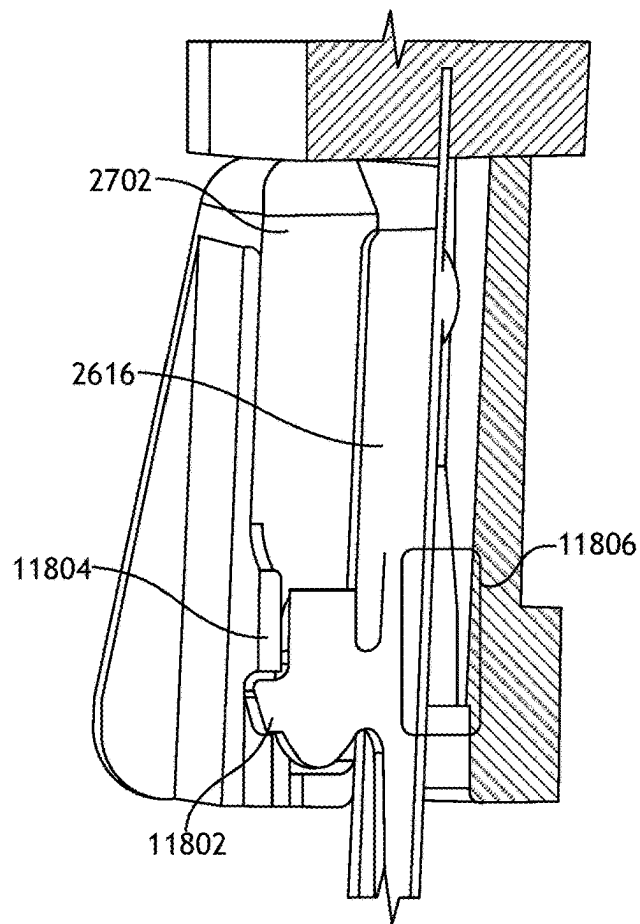


FIG. 118A

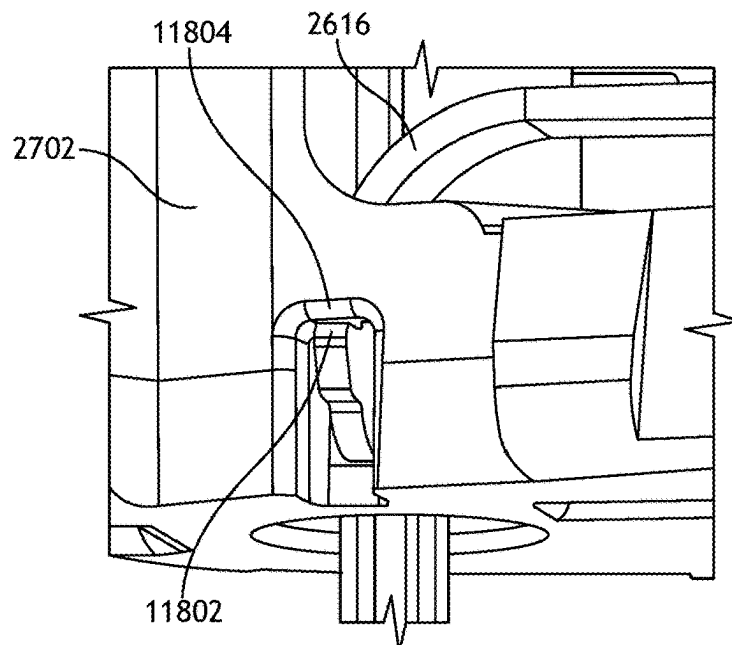


FIG. 118B

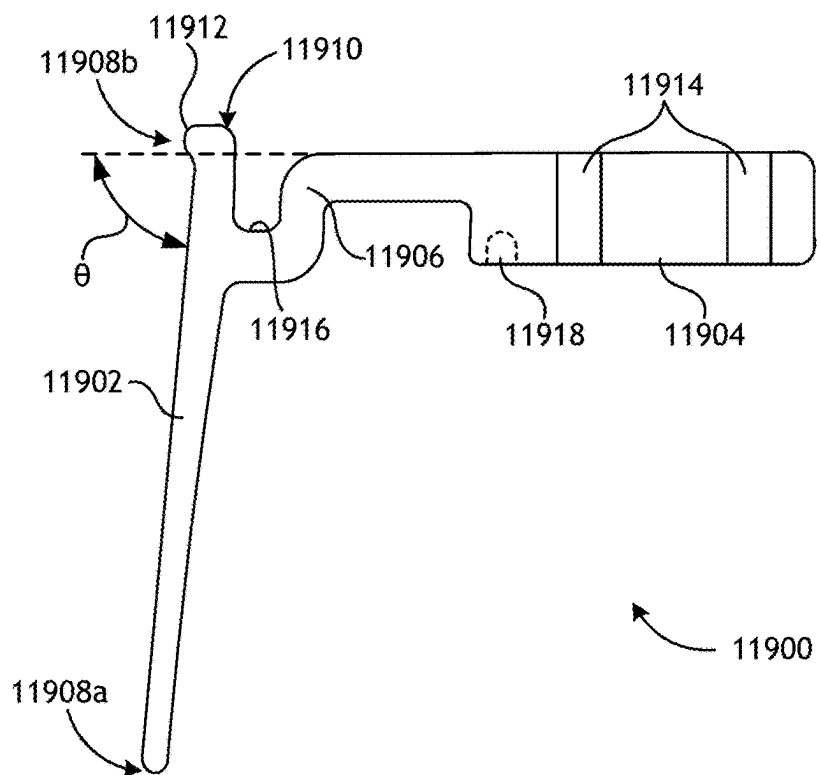


FIG. 119

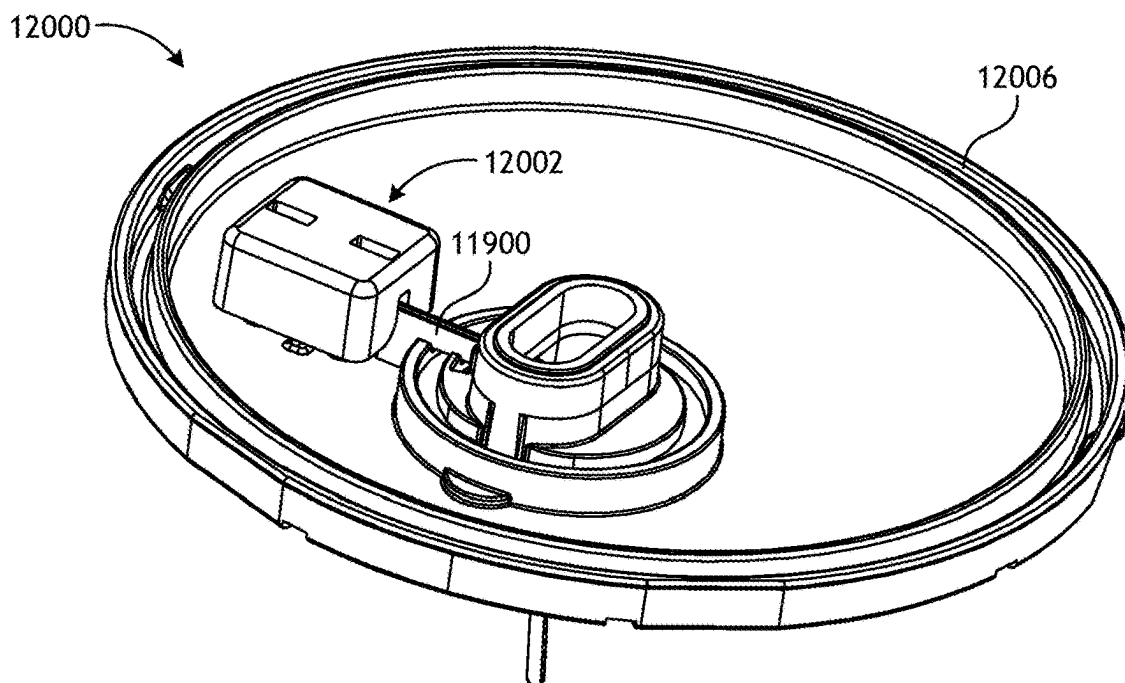


FIG. 120A

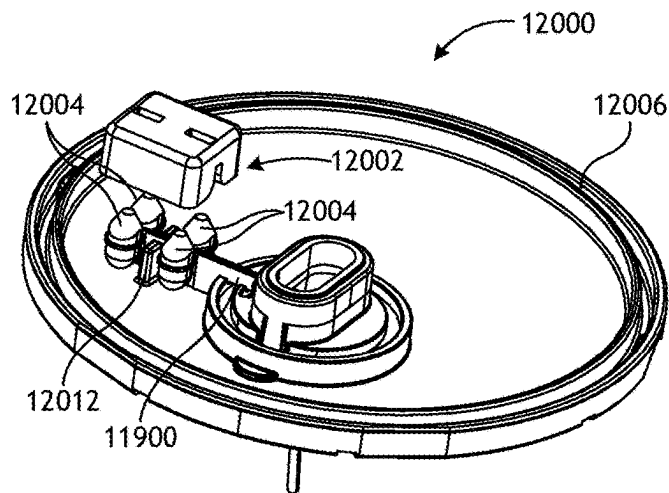


FIG. 120B

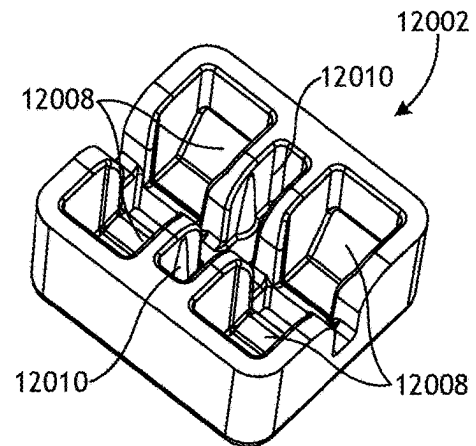


FIG. 120C

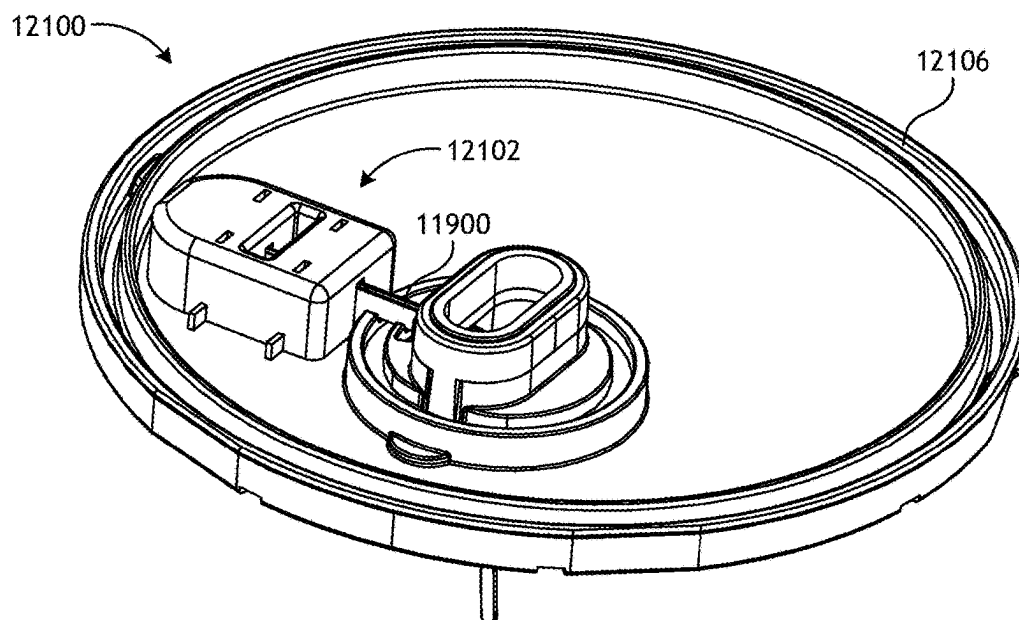


FIG. 121A

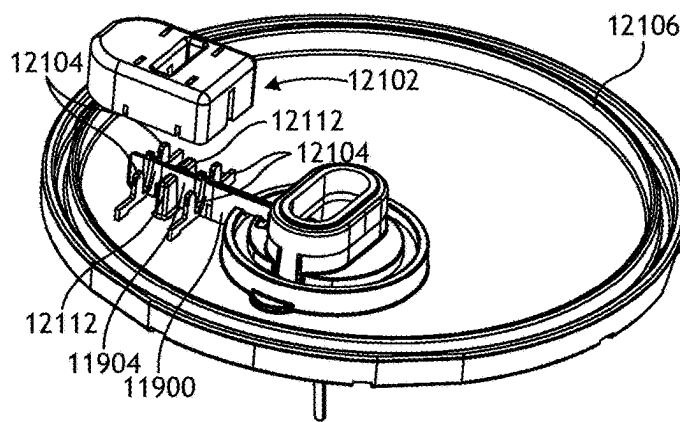


FIG. 121B

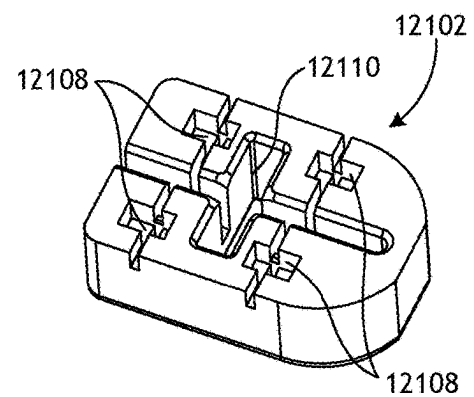


FIG. 121C

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FOCUSED STERILIZATION AND STERILIZED SUB-ASSEMBLIES FOR ANALYTE MONITORING SYSTEMS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Patent Application No. PCT/US2019/035797, filed Jun. 6, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/681,906 filed Jun. 7, 2018, U.S. Provisional Patent Application No. 62/681,908 filed Jun. 7, 2018, U.S. Provisional Patent Application No. 62/681,914 filed Jun. 7, 2018, U.S. Provisional Patent Application No. 62/776,536 filed Dec. 7, 2018, U.S. Provisional Patent Application No. 62/784,074 filed Dec. 21, 2018, U.S. Provisional Patent Application No. 62/788,475 filed Jan. 4, 2019, U.S. Provisional Patent Application No. 62/798,703 filed Jan. 30, 2019, U.S. Provisional Patent Application No. 62/798,700 filed Jan. 30, 2019, U.S. Provisional Patent Application No. 62/829,100 filed Apr. 4, 2019, U.S. Provisional Patent Application No. 62/836,198 filed Apr. 19, 2019, U.S. Provisional Patent Application No. 62/836,193 filed Apr. 19, 2019, U.S. Provisional Patent Application No. 62/836,203 filed Jun. Apr. 19, 2019, U.S. Provisional Patent Application No. 62/847,572 filed May 14, 2019, and U.S. Provisional Patent Application No. 62/849,442 filed May 17, 2019 which are hereby incorporated by reference in their entireties.

BACKGROUND

Diabetes is an incurable chronic disease in which the body does not produce or properly utilize insulin, a hormone produced by the pancreas that regulates blood glucose. When blood glucose levels rise, e.g., after a meal, insulin lowers the blood glucose levels by moving the blood glucose from the blood and into the body cells. When the pancreas does not produce sufficient insulin (a condition known as Type I Diabetes) or the body does not properly utilize insulin (a condition known as Type II Diabetes), the blood glucose remains in the blood, which could result in hyperglycemia or abnormally high blood sugar levels.

If symptoms of diabetes are not carefully monitored and treated, numerous complications can arise, including diabetic ketoacidosis, nonketotic hyperosmolar coma, cardiovascular disease, stroke, kidney failure, foot ulcers, eye damage, and nerve damage. Traditionally, monitoring has involved an individual pricking a finger to draw blood and testing the blood for glucose levels. Advancements that are more recent have allowed for continuous and long-term monitoring of blood glucose using biological sensors that are maintained in contact with bodily fluids for periods of days, weeks, or longer.

Analyte monitoring systems, for example, have been developed to facilitate long-term monitoring of bodily fluid analytes, such as glucose. Analyte monitoring systems typically include a sensor applicator configured to place a biological sensor into contact with a bodily fluid. More specifically, during delivery of the sensor to the skin of a user, at least a portion of the sensor is positioned below the skin surface, e.g., in the subcutaneous or dermal tissue.

It is important for devices implanted in the body or positioned below the skin to be sterile upon insertion. Sterilization can include any number of processes that effectively eliminate or kill transmissible agents, such as bacteria, fungi, and viruses. These transmittable agents, if

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not eliminated from the device, may be substantially detrimental to the health and safety of the user.

Some but not all analyte monitoring systems might require separate sterilization processes to sterilize the sensor and the electronic components. Electron beam sterilization, for example, is one example of radiation sterilization that can be used to terminally sterilize the sensor. Radiation sterilization, however, can harm the electronic components associated with the sensor. Consequently, the electronic components are commonly sterilized via gaseous chemical sterilization using, for example, ethylene oxide. Ethylene oxide, however, can damage the chemistry provided on the sensor. As such, integrating electronics and the sensor into one unit can complicate the sterilization process.

These issues can be worked around by separating the components into a sensor unit (e.g., a biological analyte sensor) and an adaptor unit (containing the data transmission electronics), so that each component can be packaged and sterilized separately using the appropriate sterilization method. This approach, however, requires additional components, additional packaging, additional process steps, and final user assembly of the two components, introducing a possibility of user error. Thus, a need exists for analyte monitoring systems that may be sterilized without separating the components.

BRIEF DESCRIPTION OF THE DRAWINGS

The following figures are included to illustrate certain aspects of the present disclosure, and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.

FIG. 1 is a conceptual diagram depicting an example analyte monitoring system that may incorporate one or more embodiments of the present disclosure.

FIGS. 2A-2G are progressive views of the assembly and application of the system of FIG. 1 incorporating a two-piece architecture.

FIGS. 3A and 3B are isometric and side views, respectively, of an example sensor control device.

FIGS. 4A and 4B are isometric and exploded views, respectively, of the plug assembly of FIGS. 3A-3B.

FIGS. 5A and 5B are exploded and bottom isometric views, respectively, of the electronics housing of FIGS. 3A-3B.

FIGS. 6A and 6B are side and cross-sectional side views, respectively, of the sensor applicator of FIG. 1 with the cap of FIG. 2B coupled thereto.

FIG. 7A is an enlarged cross-sectional side view of the sensor control device of FIG. 6B mounted within the cap of FIG. 6B.

FIG. 7B is an enlarged cross-sectional side view of another embodiment of the sensor control device of FIG. 6B mounted within the sensor applicator of FIG. 6B.

FIGS. 8-12 are schematic diagrams of example external sterilization assemblies, according to one or more embodiments of the present disclosure.

FIG. 13 is an isometric view of an example sensor control device.

FIG. 14A is a side view of the sensor applicator of FIG. 1.

FIG. 14B is a cross-sectional side view of the sensor applicator of FIG. 14A.

FIG. 15 is a cross-sectional side view of the sensor applicator of FIG. 14A and another example embodiment of

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the external sterilization assembly of FIG. 14B, according to one or additional more embodiments.

FIG. 16 is a cross-sectional side view of the sensor applicator of FIG. 14A and another example embodiment of the external sterilization assembly of FIG. 14B, according to one or more additional embodiments.

FIGS. 17A and 17B are isometric top and bottom views, respectively, of one example of the external sterilization assembly of FIG. 14B, according to one or more embodiments.

FIG. 18 is an isometric view of an example sensor control device.

FIG. 19A is a side view of the sensor applicator of FIG. 1.

FIG. 19B is a partial cross-sectional side view of the sensor applicator of FIG. 3A.

FIGS. 20A-20C are various views of the applicator insert of FIG. 19B, according to one or more embodiments of the disclosure.

FIG. 21 is another cross-sectional side view of the sensor applicator of FIG. 19A showing a hybrid sterilization assembly, according to one or more embodiments of the disclosure.

FIGS. 22A and 22B are isometric and cross-sectional side views, respectively, of another embodiment of the applicator insert of FIGS. 20A-20C.

FIG. 23 is a diagram of an example analyte monitoring system that may incorporate one or more embodiments of the present disclosure.

FIG. 24 is a schematic diagram of an example internal sterilization assembly, according to one or more additional embodiments of the present disclosure.

FIG. 25 is a schematic diagram of another example internal sterilization assembly, according to one or more additional embodiments of the present disclosure.

FIGS. 26A and 26B are isometric and side views, respectively, of an example sensor control device.

FIGS. 27A and 27B are isometric and exploded views, respectively, of the plug assembly of FIGS. 26A-26B.

FIG. 27C is an exploded isometric bottom view of the plug and the preservation vial.

FIGS. 28A and 28B are exploded and bottom isometric views, respectively, of the electronics housing of FIGS. 26A-26B.

FIGS. 29A and 29B are side and cross-sectional side views, respectively, of the sensor applicator of FIG. 1 with the cap of FIG. 2B coupled thereto.

FIG. 30 is a perspective view of an example embodiment of the cap of FIGS. 29A-29B.

FIG. 31 is a cross-sectional side view of the sensor control device positioned within the cap.

FIGS. 32A and 32B are isometric and side views, respectively, of an example sensor control device.

FIGS. 33A and 33B are exploded perspective top and bottom views, respectively, of the sensor control device of FIGS. 32A-32B.

FIGS. 34A and 34B are side and cross-sectional side views, respectively, of the sensor applicator of FIG. 1 with the cap of FIG. 2B coupled thereto.

FIG. 35 is an enlarged cross-sectional side view of the sensor control device mounted within the sensor applicator.

FIG. 36 is an enlarged cross-sectional bottom view of the sensor control device mounted atop the cap post.

FIGS. 37A-37C are isometric, side, and bottom views, respectively, of an example sensor control device.

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FIGS. 38A and 38B are isometric exploded top and bottom views, respectively, of the sensor control device of FIGS. 37A-37C.

FIGS. 39A-39D show example assembly of the sensor control device of FIGS. 37A-37C.

FIGS. 40A and 40B are side and cross-sectional side views, respectively, of a sensor applicator with the pre-assembled sensor control device of FIGS. 37A-37C arranged therein.

FIGS. 41A and 41B are enlarged cross-sectional views of the sensor control device during example radiation sterilization.

FIG. 42 is a plot that graphically depicts approximate penetration depth as a function of e-beam energy level for a one-sided e-beam sterilization (or irradiation) process.

FIG. 43 is a cross-sectional side view of a sensor applicator with the pre-assembled sensor control device of FIGS. 37A-37C arranged therein, according to one or more additional embodiments.

FIG. 44 is a side view of an example sensor control device.

FIG. 45 is an exploded view of the sensor control device of FIG. 44.

FIG. 46A is a cross-sectional side view of the assembled sealed subassembly of FIG. 45, according to one or more embodiments.

FIG. 46B is a cross-sectional side view of the fully assembled sensor control device of FIG. 44.

FIGS. 47A and 47B are side and cross-sectional side views, respectively, of an example embodiment of the sensor applicator of FIG. 1 with the cap of FIG. 2B coupled thereto.

FIG. 48 is a perspective view of an example embodiment of the cap of FIGS. 47A-47B.

FIG. 49 is a cross-sectional side view of the sensor control device positioned within the cap of FIGS. 47A-47B.

FIGS. 50A and 50B are isometric and side views, respectively, of another example sensor control device.

FIGS. 51A and 51B are exploded isometric top and bottom views, respectively, of the sensor control device of FIGS. 50A-50B.

FIG. 52 is a cross-sectional side view of an assembled sealed subassembly, according to one or more embodiments.

FIGS. 53A-53C are progressive cross-sectional side views showing assembly of the sensor applicator with the sensor control device of FIGS. 50A-50B.

FIGS. 54A and 54B are perspective and top views, respectively, of the cap post of FIG. 53C, according to one or more additional embodiments.

FIG. 55 is a cross-sectional side view of the sensor control device of FIGS. 50A-50B positioned within the cap of FIGS. 12B-12C.

FIGS. 56A and 56B are cross-sectional side views of the sensor applicator ready to deploy the sensor control device to a target monitoring location.

FIGS. 57A-57C are progressive cross-sectional side views showing assembly and disassembly of an example embodiment of the sensor applicator with the sensor control device of FIGS. 50A-50B.

FIG. 58A is an isometric bottom view of the housing, according to one or more embodiments.

FIG. 58B is an isometric bottom view of the housing with the sheath and other components at least partially positioned therein.

FIG. 59 is an enlarged cross-sectional side view of the sensor applicator with the sensor control device installed therein, according to one or more embodiments.

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FIG. 60A is an isometric top view of the cap, according to one or more embodiments.

FIG. 60B is an enlarged cross-sectional view of the engagement between the cap and the housing, according to one or more embodiments.

FIGS. 61A and 61B are isometric views of the sensor cap and the collar, respectively, according to one or more embodiments.

FIG. 62 is an isometric top view of an example sensor control device, according to one or more embodiments of the present disclosure.

FIG. 63 is a schematic side view of an example sensor applicator, according to one or more embodiments of the present disclosure.

FIGS. 64A and 64B are exploded isometric views of the sensor applicator and the sensor control device of FIGS. 62 and 63.

FIGS. 65A-65D are progressive cross-sectional side views of the sensor applicator of FIGS. 63 and 64A-64B depicting example deployment of a sensor control device, according to one or more embodiments.

FIG. 66 is an enlarged cross-sectional side view of an engagement between the sensor retainer and the sensor control device of FIGS. 65A-65D, according to one or more embodiments.

FIG. 67 is an exploded isometric view of another sensor applicator with the sensor control device of FIG. 62, according to one or more additional embodiments.

FIGS. 68A-68D are progressive cross-sectional side views of the sensor applicator of FIG. 67 depicting example deployment of the sensor control device, according to one or more embodiments.

FIG. 69A is an enlarged schematic view of the sharp hub and the fingers of the sensor retainer.

FIGS. 69B and 69C are enlarged schematic views of the fingers interacting with the upper portion of the needle shroud.

FIGS. 70A and 70B are enlarged cross-sectional side views of example engagement between the sensor retainer and the sensor control device, according to one or more embodiments.

FIGS. 71A and 71B are isometric and cross-sectional side views, respectively, of an example sensor retainer, according to one or more embodiments of the present disclosure.

FIGS. 72A and 72B are enlarged cross-sectional side views of the sensor retainer of FIGS. 71A-71B retaining the sensor control device, according to one or more embodiments.

FIGS. 73A and 73B are side and cross-sectional side views, respectively, of an example sensor applicator, according to one or more embodiments.

FIGS. 74A and 74B are isometric top and bottom views, respectively, of the internal applicator cover of FIG. 73B.

FIG. 75 is an isometric view of an example embodiment of the sensor cap of FIG. 73B, according to one or more embodiments.

FIG. 76 is an isometric, cross-sectional side view of the sensor cap of FIG. 75 received by the internal applicator cover of FIGS. 74A-74B, according to one or more embodiments.

FIG. 77 shows progressive removal of the applicator cap of FIG. 73A and the internal applicator cover of FIGS. 74A-74B from the sensor applicator of FIGS. 73A-73B, according to one or more embodiments.

FIG. 78 is a schematic diagram of an example sensor applicator, according to one or more additional embodiments of the present disclosure.

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FIG. 79 is an exploded view of an example sensor control device, according to one or more additional embodiments.

FIG. 80 is a bottom view of one embodiment of the sensor control device of FIG. 79.

FIGS. 81A and 81B are isometric and side views, respectively, of a sensor control device in accordance with one or more embodiments of the present disclosure.

FIG. 82 is an exploded perspective top view of the sensor control device of FIG. 81A.

FIG. 83 is a cross-sectional side view in perspective of an example sensor control device assembly including a sensor control device of FIG. 81A mounted within the sensor applicator, the sensor control device being compatible with the analyte monitoring system of FIG. 1.

FIG. 84 is an enlarged cross-sectional side view of the sensor control device assembly of FIG. 83.

FIG. 85 is a bottom view of a few members of the sensor control device assembly of FIG. 83, the members including the sensor control device held in a sensor carrier of the sensor applicator.

FIG. 86 is a schematic diagram of an example sterilization assembly, according to one or more embodiments of the present disclosure.

FIG. 87 is a schematic diagram of another example sterilization assembly, according to one or more embodiments of the present disclosure.

FIG. 88A is a schematic bottom view of another example sterilization assembly, according to one or more embodiments of the present disclosure.

FIGS. 88B and 88C are schematic bottom views of alternative embodiments of the sterilization assembly of FIG. 88A, according to one or more additional embodiments of the present disclosure.

FIG. 89 is an isometric schematic view of an example sensor control device, according to one or more embodiments.

FIG. 90 is a schematic diagram of another example sterilization assembly, according to one or more embodiments.

FIGS. 91A and 91B are side and isometric views, respectively, of an example sensor control device, according to one or more embodiments of the present disclosure.

FIGS. 92A and 92B are exploded, isometric top and bottom views, respectively, of the sensor control device of FIG. 2, according to one or more embodiments.

FIG. 93 is a cross-sectional side view of the sensor control device of FIGS. 91A-91B and 92A-92B, according to one or more embodiments.

FIG. 93A is an exploded isometric view of a portion of another embodiment of the sensor control device of FIGS. 91A-91B and 92A-92B.

FIG. 94A is an isometric bottom view of the mount of FIGS. 91A-91B and 92A-92B.

FIG. 94B is an isometric top view of the sensor cap of FIGS. 91A-91B and 92A-92B.

FIGS. 95A and 95B are side and cross-sectional side views, respectively, of an example sensor applicator, according to one or more embodiments.

FIGS. 96A and 96B are perspective and top views, respectively, of the cap post of FIG. 95B, according to one or more embodiments.

FIG. 97 is a cross-sectional side view of the sensor control device positioned within the applicator cap, according to one or more embodiments.

FIG. 98 is a cross-sectional view of a sensor control device showing example interaction between the sensor and the sharp.

FIG. 99 is a cross-sectional side view of an example analyte monitoring system enclosure used to house at least a portion of a sensor control device.

FIG. 100A is an enlarged cross-sectional side view of the interface between the sensor applicator and the cap as indicated by the dashed box of FIG. 99.

FIG. 100B is an enlarged cross-sectional side view of the interface between the sensor applicator and the cap as indicated by the dashed box of FIG. 99 during or after gaseous chemical sterilization.

FIG. 101 is a cross-sectional side view of another example analyte monitoring system enclosure used to house at least a portion of the sensor control device of FIG. 1.

FIGS. 102A-102C provide finite element analysis results corresponding to the interface between the housing and the cap during example gaseous chemical sterilization.

FIG. 103 is an isometric view of an example sensor control device.

FIGS. 104A and 104B are exploded, isometric views of the sensor control device of FIG. 103, according to one or more embodiments.

FIG. 105 is a cross-sectional side view of the assembled sensor control device of FIGS. 104A-104B, according to one or more embodiments.

FIG. 106 is an isometric view of another example sensor control device.

FIGS. 107A and 107B are exploded, isometric views of the sensor control device of FIG. 106, according to one or more embodiments.

FIG. 108 is a cross-sectional side view of the assembled sensor control device of FIGS. 107A-107B, according to one or more embodiments.

FIG. 109 is an isometric view of an example converting process for manufacturing a sensor control device in accordance with the principles of the present disclosure.

FIGS. 110A-110E depict progressive fabrication of the sensor control device of FIG. 109, according to one or more embodiments.

FIG. 111A is a top view of the sensor control device of FIG. 109 in preparation for pressure testing and/or vacuum sealing, according to one or more embodiments.

FIG. 111B is a cross-sectional side view of the sensor control device of FIG. 109 with a compressor.

FIG. 112 is a partial cross-sectional side view of an example sensor control device, according to one or more embodiments.

FIG. 113 is a cross-sectional side view of an example sensor applicator, according to one or more embodiments.

FIGS. 114A and 114B are top and bottom perspective views, respectively, of an example embodiment of the plug of FIGS. 27A-27B.

FIGS. 115A and 115B are perspective views depicting an example embodiment of the connector of FIGS. 27A-27B in open and closed states, respectively.

FIG. 116 is a perspective view of an example embodiment of the sensor of FIGS. 27A-27B.

FIGS. 117A and 117B are bottom and top perspective views, respectively, depicting an example embodiment of a sensor module assembly.

FIGS. 118A and 118B are close-up partial views of an example embodiment of the sensor plug of FIGS. 114A-114B having certain axial stiffening features.

FIG. 119 is a side view of an example sensor, according to one or more embodiments of the disclosure.

FIGS. 120A and 120B are isometric and partially exploded isometric views of an example connector assembly, according to one or more embodiments.

FIG. 120C is an isometric bottom view of the connector of FIGS. 120A-120B.

FIGS. 121A and 121B are isometric and partially exploded isometric views of another example connector assembly, according to one or more embodiments.

FIG. 121C is an isometric bottom view of the connector of FIGS. 121A-121B.

DETAILED DESCRIPTION

The present application is generally related to systems, devices, and methods for assembling an applicator and sensor control device for use in an in vivo analyte monitoring system.

FIG. 1 is a conceptual diagram depicting an example analyte monitoring system 100 that may incorporate one or more embodiments of the present disclosure. A variety of analytes can be detected and quantified using the system 100 (hereafter “the system 100”) including, but not limited to, acetyl choline, amylase, bilirubin, cholesterol, chorionic gonadotropin, creatine kinase (e.g., CK-MB), creatine, DNA, fructosamine, glucose, glutamine, growth hormones, hormones, ketones (e.g., ketone bodies), lactate, oxygen, peroxide, prostate-specific antigen, prothrombin, RNA, thyroid stimulating hormone, and troponin. The concentration of drugs, such as, but not limited to, antibiotics (e.g., gentamicin, vancomycin, and the like), digitoxin, digoxin, drugs of abuse, theophylline, and warfarin, may also be determined.

As illustrated, the system 100 includes a sensor applicator 102 (alternately referred to as an “insertor”), a sensor control device 104 (also referred to as an “in vivo analyte sensor control device”), and a reader device 106. The sensor applicator 102 is used to deliver the sensor control device 104 to a target monitoring location on a user’s skin (e.g., the arm of the user). Once delivered, the sensor control device 104 is maintained in position on the skin with an adhesive patch 108 coupled to the bottom of the sensor control device 104. A portion of a sensor 110 extends from the sensor control device 104 and is positioned such that it can be transcutaneously positioned and otherwise retained under the surface of the user’s skin during the monitoring period.

An introducer may be included to promote introduction of the sensor 110 into tissue. The introducer may comprise, for example, a needle often referred to as a “sharp.” Alternatively, the introducer may comprise other types of devices, such as a sheath or a blade. The introducer may transiently reside in proximity to the sensor 110 prior to tissue insertion and then be withdrawn afterward. While present, the introducer may facilitate insertion of the sensor 110 into tissue by opening an access pathway for the sensor 110 to follow. For example, the introducer may penetrate the epidermis to provide an access pathway to the dermis to allow subcutaneous implantation of the sensor 110. After opening the access pathway, the introducer may be withdrawn (retracted) so that it does not represent a hazard while the sensor 110 remains in place. In illustrative embodiments, the introducer may be solid or hollow, beveled or non-beveled, and/or circular or non-circular in cross-section. In more particular embodiments, suitable introducers may be comparable in cross-sectional diameter and/or tip design to an acupuncture needle, which may have a cross-sectional diameter of about 250 microns. It is to be recognized, however, that suitable introducers may have a larger or smaller cross-sectional diameter if needed for particular applications.

In some embodiments, a tip of the introducer (while present) may be angled over the terminus of the sensor 110,

such that the introducer penetrates a tissue first and opens an access pathway for the sensor **110**. In other illustrative embodiments, the sensor **110** may reside within a lumen or groove of the introducer, with the introducer similarly opening an access pathway for the sensor **110**. In either case, the introducer is subsequently withdrawn after facilitating sensor **110** insertion. Moreover, the introducer (sharp) can be made of a variety of materials, such as various types of metals and plastics.

When the sensor control device **104** is properly assembled, the sensor **110** is placed in communication (e.g., electrical, mechanical, etc.) with one or more electrical components or sensor electronics included within the sensor control device **104**. In some applications, for example, the sensor control device **104** may include a printed circuit board (PCB) having a data processor (e.g., an application specific integrated circuit or ASIC) mounted thereto, and the sensor **110** may be operatively coupled to the data processor which, in turn, may be coupled with an antenna and a power source.

The sensor control device **104** and the reader device **106** are configured to communicate with one another over a local communication path or link **112**, which may be wired or wireless, uni- or bi-directional, and encrypted or non-encrypted. The reader device **106** may constitute an output medium for viewing analyte concentrations and alerts or notifications determined by the sensor **110** or a processor associated therewith, as well as allowing for one or more user inputs, according to some embodiments. The reader device **106** may be a multi-purpose smartphone or a dedicated electronic reader instrument. While only one reader device **106** is shown, multiple reader devices **106** may be present in certain instances.

The reader device **106** may also be in communication with a remote terminal **114** and/or a trusted computer system **116** via communication path(s)/link(s) **118** and/or **120**, respectively, which also may be wired or wireless, uni- or bi-directional, and encrypted or non-encrypted. The reader device **106** may also or alternately be in communication with a network **122** (e.g., a mobile telephone network, the internet, or a cloud server) via communication path/link **124**. The network **122** may be further communicatively coupled to remote terminal **114** via communication path/link **126** and/or the trusted computer system **116** via communication path/link **128**.

Alternately, the sensor control device **104** may communicate directly with the remote terminal **114** and/or the trusted computer system **116** without an intervening reader device **106** being present. For example, the sensor **110** may communicate with the remote terminal **114** and/or the trusted computer system **116** through a direct communication link to the network **122**, according to some embodiments, as described in U.S. Pat. No. 10,136,816, incorporated herein by reference in its entirety.

Any suitable electronic communication protocol may be used for each of the communication paths or links, such as near field communication (NFC), radio frequency identification (RFID), BLUETOOTH® or BLUETOOTH® low energy protocols, WiFi, or the like. The remote terminal **114** and/or the trusted computer system **116** may be accessible, according to some embodiments, by individuals other than a primary user who have an interest in the user's analyte levels. The reader device **106** may include a display **130** and an optional input component **132**. The display **130** may comprise a touch-screen interface, according to some embodiments.

In some embodiments, the sensor control device **104** may automatically forward data to the reader device **106**. For example, analyte concentration data may be communicated automatically and periodically, such as at a certain frequency as data is obtained or after a certain time period has passed, with the data being stored in a memory until transmittal (e.g., every minute, five minutes, or other predetermined time period). In other embodiments, the sensor control device **104** may communicate with the reader device **106** in a non-automatic manner and not according to a set schedule. For example, data may be communicated from the sensor control device **104** using RFID technology when the sensor electronics are brought into communication range of the reader device **106**. Until communicated to the reader device **106**, data may remain stored in a memory of the sensor control device **104**. Thus, a patient does not have to maintain close proximity to the reader device **106** at all times, and can instead upload data when convenient. In yet other embodiments, a combination of automatic and non-automatic data transfer may be implemented. For example, data transfer may continue on an automatic basis until the reader device **106** is no longer in communication range of the sensor control device **104**.

The sensor control device **104** is often included with the sensor applicator **104** in what is known as a “two-piece” architecture that requires final assembly by a user before the sensor **110** can be properly delivered to the target monitoring location. More specifically, the sensor **110** and the associated electrical components included in the sensor control device **104** are provided to the user in multiple (two) packages, and the user must open the packaging and follow instructions to manually assemble the components before delivering the sensor **110** to the target monitoring location with the sensor applicator **102**.

More recently, however, advanced designs of sensor control devices and sensor applicators have resulted in a one-piece architecture that allows the system to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package and subsequently deliver the sensor control device to the target monitoring location. The one-piece system architecture may prove advantageous in eliminating component parts, various fabrication process steps, and user assembly steps. As a result, packaging and waste are reduced, and the potential for user error or contamination to the system is mitigated.

In the illustrated embodiment, the system **100** may comprise what is known as a “two-piece” architecture that requires final assembly by a user before the sensor **110** can be properly delivered to the target monitoring location. More specifically, the sensor **110** and the associated electrical components included in the sensor control device **104** are provided to the user in multiple (two) packages, where each may or may not be sealed with a sterile barrier but are at least enclosed in packaging. The user must open the packaging and follow instructions to manually assemble the components and subsequently deliver the sensor **110** to the target monitoring location with the sensor applicator **102**.

FIGS. 2A-2G are progressive views of the assembly and application of the system **100** incorporating a two-piece architecture. FIGS. 2A and 2B depict the first and second packages, respectively, provided to the user for final assembly. More specifically, FIG. 2A depicts a sensor container or tray **202** that has a removable lid **204**. The user prepares the sensor tray **202** by removing the lid **204**, which acts as a sterile barrier to protect the internal contents of the sensor tray **202** and otherwise maintain a sterile internal environ-

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ment. Removing the lid **204** exposes a platform **206** positioned within the sensor tray **202**, and a plug assembly **207** (partially visible) is arranged within and otherwise strategically embedded within the platform **206**. The plug assembly **207** includes a sensor module (not shown) and a sharp module (not shown). The sensor module carries the sensor **110** (FIG. 1), and the sharp module carries an associated sharp used to help deliver the sensor **110** transcutaneously under the user's skin during application of the sensor control device **104** (FIG. 1).

FIG. 2B depicts the sensor applicator **102** and the user preparing the sensor applicator **102** for final assembly. The sensor applicator **102** includes a housing **208** sealed at one end with an applicator cap **210**. In some embodiments, for example, an O-ring or another type of sealing gasket may seal an interface between the housing **208** and the applicator cap **210**. In at least one embodiment, the O-ring or sealing gasket may be molded onto one of the housing **208** and the applicator cap **210**. The applicator cap **210** provides a barrier that protects the internal contents of the sensor applicator **102**. In particular, the sensor applicator **102** contains an electronics housing (not shown) that retains the electrical components for the sensor control device **104** (FIG. 1), and the applicator cap **210** may or may not maintain a sterile environment for the electrical components. Preparation of the sensor applicator **102** includes uncoupling the housing **208** from the applicator cap **210**, which can be accomplished by unscrewing the applicator cap **210** from the housing **208**. The applicator cap **210** can then be discarded or otherwise placed aside.

FIG. 2C depicts the user inserting the sensor applicator **102** into the sensor tray **202**. The sensor applicator **102** includes a sheath **212** configured to be received by the platform **206** to temporarily unlock the sheath **212** relative to the housing **208**, and also temporarily unlock the platform **206** relative to the sensor tray **202**. Advancing the housing **208** into the sensor tray **202** results in the plug assembly **207** (FIG. 2A) arranged within the sensor tray **202**, including the sensor and sharp modules, being coupled to the electronics housing arranged within the sensor applicator **102**.

In FIG. 2D, the user removes the sensor applicator **102** from the sensor tray **202** by proximally retracting the housing **208** with respect to the sensor tray **202**.

FIG. 2E depicts the bottom or interior of the sensor applicator **102** following removal from the sensor tray **202** (FIG. 2). The sensor applicator **102** is removed from the sensor tray **202** with the sensor control device **104** fully assembled therein and positioned for delivery to the target monitoring location. As illustrated, a sharp **220** extends from the bottom of the sensor control device **104** and carries a portion of the sensor **110** within a hollow or recessed portion thereof. The sharp **220** is configured to penetrate the skin of a user and thereby place the sensor **110** into contact with bodily fluid.

FIGS. 2F and 2G depict example delivery of the sensor control device **104** to a target monitoring location **222**, such as the back of an arm of the user. FIG. 2F shows the user advancing the sensor applicator **102** toward the target monitoring location **222**. Upon engaging the skin at the target monitoring location **222**, the sheath **212** collapses into the housing **208**, which allows the sensor control device **104** (FIGS. 2E and 2G) to advance into engagement with the skin. With the help of the sharp **220** (FIG. 2E), the sensor **110** (FIG. 2E) is advanced transcutaneously into the patient's skin at the target monitoring location **222**.

FIG. 2G shows the user retracting the sensor applicator **102** from the target monitoring location, with the sensor

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control device **104** successfully attached to the user's skin. The adhesive patch **108** (FIG. 1) applied to the bottom of sensor control device **104** adheres to the skin to secure the sensor control device **104** in place. The sharp **220** (FIG. 2E) is automatically retracted when the housing **208** is fully advanced at the target monitoring location **222**, while the sensor **110** (FIG. 2E) is left in position to measure analyte levels.

For the two-piece architecture system, the sensor tray **202** (FIG. 2A) and the sensor applicator **102** (FIG. 2B) are provided to the user as separate packages, thus requiring the user to open each package and finally assemble the system. In some applications, the discrete, sealed packages allow the sensor tray **202** and the sensor applicator **102** to be sterilized in separate sterilization processes unique to the contents of each package and otherwise incompatible with the contents of the other.

More specifically, the sensor tray **202**, which includes the plug assembly **207** (FIG. 2A), including the sensor **110** (FIGS. 1 and 2E) and the sharp **220** (FIG. 2E), may be sterilized using radiation sterilization, such as electron beam (or "e-beam") irradiation. Radiation sterilization, however, can damage the electrical components arranged within the electronics housing of the sensor control device **104**. Consequently, if the sensor applicator **102**, which contains the electronics housing of the sensor control device **104**, needs to be sterilized, it may be sterilized via another method, such as gaseous chemical sterilization using, for example, ethylene oxide. Gaseous chemical sterilization, however, can damage the enzymes or other chemistry and biologics included on the sensor **110**. Because of this sterilization incompatibility, the sensor tray **202** and the sensor applicator **102** may be sterilized in separate sterilization processes and subsequently packaged separately, and thereby requiring the user to finally assemble the components upon receipt.

According to embodiments of the present disclosure, the system **100** (FIG. 1) may comprise a one-piece architecture that incorporates sterilization techniques specifically designed for a one-piece architecture. The one-piece architecture allows the system **100** to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package and subsequently deliver the sensor control device to the target monitoring location, as generally described above with reference to FIGS. 2E-2G. The one-piece system architecture described herein may prove advantageous in eliminating component parts, various fabrication process steps, and user assembly steps. As a result, packaging and waste are reduced, and the potential for user error or contamination to the system is mitigated.

Focused Electron Beam Sterilization with Collimator

FIGS. 3A and 3B are isometric and side views, respectively, of an example sensor control device **302**, according to one or more embodiments of the present disclosure. The sensor control device **302** (alternately referred to as a "puck") may be similar in some respects to the sensor control device **104** of FIG. 1 and therefore may be best understood with reference thereto. The sensor control device **302** may replace the sensor control device **104** of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator **102** (FIG. 1), which delivers the sensor control device **302** to a target monitoring location on a user's skin.

The sensor control device **302**, however, may be incorporated into a one-piece system architecture. Unlike the two-piece architecture system, for example, a user is not required to open multiple packages and finally assemble the sensor control device **302**. Rather, upon receipt by the user,

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the sensor control device 302 is already fully assembled and properly positioned within the sensor applicator 102. To use the sensor control device 302, the user need only break one barrier (e.g., the applicator cap 210 of FIG. 2B) before promptly delivering the sensor control device 302 to the target monitoring location.

As illustrated, the sensor control device 302 includes an electronics housing 304 that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing 304 may exhibit other cross-sectional shapes, such as ovoid (e.g., pill-shaped), a squircle, or polygonal, without departing from the scope of the disclosure. The electronics housing 304 may be configured to house or otherwise contain various electrical components used to operate the sensor control device 302.

The electronics housing 304 may include a shell 306 and a mount 308 that is matable with the shell 306. The shell 306 may be secured to the mount 308 via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, or one or more mechanical fasteners (e.g., screws). In some cases, the shell 306 may be secured to the mount 308 such that a sealed interface therebetween is generated. In such embodiments, a gasket or other type of seal material may be positioned at or near the outer diameter (periphery) of the shell 306 and the mount 308, and securing the two components together may compress the gasket and thereby generate a sealed interface. In other embodiments, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell 306 and the mount 308. The adhesive secures the shell 306 to the mount 308 and provides structural integrity, but may also seal the interface between the two components and thereby isolate the interior of the electronics housing 304 from outside contamination. If the sensor control device 302 is assembled in a controlled environment, there may be no need to terminally sterilize the internal electrical components. Rather, the adhesive coupling may provide a sufficient sterile barrier for the assembled electronics housing 304.

The sensor control device 302 may further include a plug assembly 310 that may be coupled to the electronics housing 304. The plug assembly 310 may be similar in some respects to the plug assembly 207 of FIG. 2A. For example, the plug assembly 310 may include a sensor module 312 (partially visible) interconnectable with a sharp module 314 (partially visible). The sensor module 312 may be configured to carry and otherwise include a sensor 316 (partially visible), and the sharp module 314 may be configured to carry and otherwise include a sharp 318 (partially visible) used to help deliver the sensor 316 transcutaneously under a user's skin during application of the sensor control device 302. As illustrated, corresponding portions of the sensor 316 and the sharp 318 extend from the electronics housing 304 and, more particularly, from the bottom of the mount 308. The exposed portion of the sensor 316 may be received within a hollow or recessed portion of the sharp 318. The remaining portion of the sensor 316 is positioned within the interior of the electronics housing 304.

FIGS. 4A and 4B are isometric and exploded views, respectively, of the plug assembly 310, according to one or more embodiments. The sensor module 312 may include the sensor 316, a plug 402, and a connector 404. The plug 402 may be designed to receive and support both the sensor 316 and the connector 404. As illustrated, a channel 406 may be defined through the plug 402 to receive a portion of the sensor 316. Moreover, the plug 402 may provide one or

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more deflectable arms 407 configured to snap into corresponding features provided on the bottom of the electronics housing 304 (FIGS. 3A-3B).

The sensor 316 includes a tail 408, a flag 410, and a neck 412 that interconnects the tail 408 and the flag 410. The tail 408 may be configured to extend at least partially through the channel 406 and extend distally from the plug 402. The tail 408 includes an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry. In use, the tail 408 is transcutaneously received beneath a user's skin, and the chemistry included thereon helps facilitate analyte monitoring in the presence of bodily fluids.

The flag 410 may comprise a generally planar surface having one or more sensor contacts 414 (three shown in FIG. 4B) arranged thereon. The sensor contact(s) 414 may be configured to align with a corresponding number of compliant carbon impregnated polymer modules (not shown) encapsulated within the connector 404.

The connector 404 includes one or more hinges 418 that enables the connector 404 to move between open and closed states. The connector 404 is depicted in FIGS. 4A-4B in the closed state, but can pivot to the open state to receive the flag 410 and the compliant carbon impregnated polymer module(s) therein. The compliant carbon impregnated polymer module(s) provide electrical contacts 420 (three shown) configured to provide conductive communication between the sensor 316 and corresponding circuitry contacts provided within the electronics housing 304 (FIGS. 3A-3B). The connector 404 can be made of silicone rubber and may serve as a moisture barrier for the sensor 316 when assembled in a compressed state and after application to a user's skin.

The sharp module 314 includes the sharp 318 and a sharp hub 422 that carries the sharp 318. The sharp 318 includes an elongate shaft 424 and a sharp tip 426 at the distal end of the shaft 424. The shaft 424 may be configured to extend through the channel 406 and extend distally from the plug 402. Moreover, the shaft 424 may include a hollow or recessed portion 428 that at least partially circumscribes the tail 408 of the sensor 316. The sharp tip 426 may be configured to penetrate the skin while carrying the tail 408 to put the active chemistry present on the tail 408 into contact with bodily fluids.

The sharp hub 422 may include a hub small cylinder 430 and a hub snap pawl 432, each of which may be configured to help couple the plug assembly 310 (and the entire sensor control device 302) to the sensor applicator 102 (FIG. 1).

FIGS. 5A and 5B are exploded and bottom isometric views, respectively, of the electronics housing 304, according to one or more embodiments. The shell 306 and the mount 308 operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device 302 (FIGS. 3A-3B).

A printed circuit board (PCB) 502 may be positioned within the electronics housing 304. A plurality of electronic modules (not shown) may be mounted to the PCB 502 including, but not limited to, a data processing unit, resistors, transistors, capacitors, inductors, diodes, and switches. The data processing unit may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device 302. More specifically, the data processing unit may be configured to perform data processing functions, where such functions may include but are not limited to, filtering and encoding of data signals,

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each of which corresponds to a sampled analyte level of the user. The data processing unit may also include or otherwise communicate with an antenna for communicating with the reader device 106 (FIG. 1).

As illustrated, the shell 306, the mount 308, and the PCB 502 each define corresponding central apertures 504, 506, and 508, respectively. When the electronics housing 304 is assembled, the central apertures 504, 506, and 508 coaxially align to receive the plug assembly 310 (FIGS. 4A-4B) therethrough. A battery 510 may also be housed within the electronics housing 304 and configured to power the sensor control device 302.

In FIG. 5B, a plug receptacle 512 may be defined in the bottom of the mount 308 and provide a location where the plug assembly 310 (FIGS. 4A-4B) may be received and coupled to the electronics housing 304, and thereby fully assemble the sensor control device 302 (FIG. 3A-3B). The profile of the plug 402 (FIGS. 4A-4B) may match or be shaped in complementary fashion to the plug receptacle 512, and the plug receptacle 512 may provide one or more snap ledges 514 (two shown) configured to interface with and receive the deflectable arms 407 (FIGS. 4A-4B) of the plug 402. The plug assembly 310 is coupled to the electronics housing 304 by advancing the plug 402 into the plug receptacle 512 and allowing the deflectable arms 407 to lock into the corresponding snap ledges 514. When the plug assembly 310 (FIGS. 4A-4B) is properly coupled to the electronics housing 304, one or more circuitry contacts 516 (three shown) defined on the underside of the PCB 502 may make conductive communication with the electrical contacts 420 (FIGS. 4A-4B) of the connector 404 (FIGS. 4A-4B).

FIGS. 6A and 6B are side and cross-sectional side views, respectively, of the sensor applicator 102 with the applicator cap 210 coupled thereto. More specifically, FIGS. 6A-6B depict how the sensor applicator 102 might be shipped to and received by a user, according to at least one embodiment. In some embodiments, however, the sensor applicator 102 might further be sealed within a bag (not shown) and delivered to the user within the bag. The bag may be made of a variety of materials that help prevent the ingress of humidity into the sensor applicator 102, which might adversely affect the sensor 316. In at least one embodiment, for example, the sealed back might be made of foil. Any and all of the sensor applicators described or discussed herein may be sealed within and delivered to the user within the bag.

According to the present disclosure, and as seen in FIG. 6B, the sensor control device 302 is already assembled and installed within the sensor applicator 102 prior to being delivered to the user. The applicator cap 210 may be threaded to the housing 208 and include a tamper ring 602. Upon rotating (e.g., unscrewing) the applicator cap 210 relative to the housing 208, the tamper ring 602 may shear and thereby free the applicator cap 210 from the sensor applicator 102. Following which, the user may deliver the sensor control device 302 to the target monitoring location, as generally described above with reference to FIGS. 2E-2G.

In some embodiments, as mentioned above, the applicator cap 210 may be secured to the housing 208 via a sealed engagement to protect the internal components of the sensor applicator 102. In at least one embodiment, for example, an O-ring or another type of sealing gasket may seal an interface between the housing 208 and the applicator cap 210. The O-ring or sealing gasket may be a separate component part or alternatively molded onto one of the housing 208 and the applicator cap 210.

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The housing 208 may be made of a variety of rigid materials. In some embodiments, for example, the housing 208 may be made of a thermoplastic polymer, such as polyketone. In other embodiments, the housing 208 may be made of cyclic olefin copolymer (COC), which can help prevent moisture ingress into the interior of the sensor applicator 102. As will be appreciated, any and all of the housings described or discussed herein may be made of polyketone or COC.

With specific reference to FIG. 6B, the sensor control device 302 may be loaded into the sensor applicator 102 by mating the sharp hub 422 with a sensor carrier 604 included within the sensor applicator 102. Once the sensor control device 302 is mated with the sensor carrier 604, the applicator cap 210 may then be secured to the sensor applicator 102.

In the illustrated embodiment, a collimator 606 is positioned within the applicator cap 210 and may generally help support the sensor control device 302 while contained within the sensor applicator 102. In some embodiments, the collimator 606 may form an integral part or extension of the applicator cap 210, such as being molded with or overmolded onto the applicator cap 210. In other embodiments, the collimator 606 may comprise a separate structure fitted within or attached to the applicator cap 210, without departing from the scope of the disclosure. In yet other embodiments, as discussed below, the collimator 606 may be omitted in the package received by the user, but otherwise used while sterilizing and preparing the sensor applicator 102 for delivery.

The collimator 606 may be designed to receive and help protect parts of the sensor control device 302 that need to be sterile, and isolate the sterile components of the sensor applicator 102 from microbial contamination from other locations within the sensor control device 302. To accomplish this, the collimator 606 may define or otherwise provide a sterilization zone 608 (alternately referred to as a "sterile barrier enclosure" or a "sterile sensor path") configured to receive the sensor 316 and the sharp 318 as extending from the bottom of the electronics housing 304. The sterilization zone 608 may generally comprise a hole or passageway extending at least partially through the body of the collimator 606. In the illustrated embodiment, the sterilization zone 608 extends through the entire body of the collimator 606, but may alternatively extend only partially therethrough, without departing from the scope of the disclosure.

When the sensor control device 302 is loaded into the sensor applicator 102 and the applicator cap 210 with the collimator 606 is secured thereto, the sensor 316 and the sharp 318 may be positioned within a sealed region 610 at least partially defined by the sterilization zone 608. The sealed region 610 is configured to isolate the sensor 316 and the sharp 318 from external contamination and may include (encompass) select portions of the interior of the electronics housing 304 and the sterilization zone 608 of the collimator 606.

While positioned within the sensor applicator 102, the fully assembled sensor control device 302 may be subjected to radiation sterilization 612. The radiation sterilization 612 may comprise, for example, e-beam irradiation, but other methods of sterilization may alternatively be used including, but not limited to, low energy X-ray irradiation. In some embodiments, the radiation sterilization 612 may be delivered either through continuous processing irradiation or through pulsed beam irradiation. In pulsed beam irradiation, the beam of radiation sterilization 612 is focused at a target

location and the component part or device to be sterilized is moved to the target location at which point the radiation sterilization **612** is activated to provide a directed pulse of radiation. The radiation sterilization **612** is then turned off, and another component part or device to be sterilized is moved to the target location and the process is repeated.

The collimator **606** may be configured to focus the radiation (e.g., beams, waves, energy, etc.) from the radiation sterilization **612** toward the components that are required to be sterile, such as the sensor **316** and the sharp **318**. More specifically, the hole or passageway of the sterilization zone **608** allows transmission of the radiation to impinge upon and sterilize the sensor **316** and the sharp **318**, while the remaining portions of the collimator **606** prevent (impede) the propagating radiation from disrupting or damaging the electronic components within the electronics housing **304**.

The sterilization zone **608** can exhibit any suitable cross-sectional shape necessary to properly focus the radiation on the sensor **316** and the sharp **318** for sterilization. In the illustrated embodiment, for example, the sterilization zone **608** is conical or frustoconical in shape. In other embodiments, however, the sterilization zone **608** may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure.

In the illustrated embodiment, the sterilization zone **608** provides a first aperture **614a** at a first end and a second aperture **614b** at a second end opposite the first end. The first aperture **614a** may be configured to receive the sensor **316** and the sharp **318** into the sterilization zone **608**, and the second aperture **614b** may allow the radiation (e.g., beams, waves, etc.) from the radiation sterilization **612** to enter the sterilization zone **608** and impinge upon the sensor **316** and the sharp **318**.

In embodiments where the sterilization zone **608** is conical or frustoconical in shape, the first aperture **614a** may have a diameter that is smaller than the diameter of the second aperture **614b**. In such embodiments, for example, the size of the first aperture **614a** may range between about 0.5 mm and about 3.0 mm, and the size of the second aperture **614b** may range between about 5.0 mm and about 16.0 mm. As will be appreciated, however, the respective diameters of the first and second apertures **614a,b** may be greater or less than the ranges provided herein, without departing from the scope of the disclosure, and depending on the application. Indeed, the diameters of the first and second apertures **614a,b** need only be large enough to allow a sufficient dose of radiation to impinge upon the sensor **316** and the sharp **318**. Moreover, in at least one embodiment, the sterilization zone **608** may be cylindrical in shape where the first and second apertures **614a,b** exhibit identical diameters.

The body of the collimator **606** reduces or eliminates the radiation sterilization **612** from penetrating through the body material and thereby damaging the electronic components within the electronics housing **304**. To accomplish this, in some embodiments, the collimator **606** may be made of a material that has a mass density greater than 0.9 grams per cubic centimeter (g/cc). One example material for the collimator **606** is polyethylene, but could alternatively comprise any material having a mass density similar to or greater than polyethylene. In some embodiments, for example, the material for the collimator **606** may comprise, but is not limited to, a metal (e.g., lead, stainless steel) or a high-density polymer.

In at least one embodiment, the design of the collimator **606** may be altered so that the collimator **606** may be made

of a material that has a mass density less than 0.9 grams per cubic centimeter (g/cc) but still operate to reduce or eliminate the radiation sterilization **612** from impinging upon the electronic components within the electronics housing **304**. To accomplish this, in some embodiments, the size (e.g., length) of the collimator **606** may be increased such that the propagating electrons from the radiation sterilization **612** are required to pass through a larger amount of material before potentially impinging upon sensitive electronics. The larger amount of material may help absorb or dissipate the dose strength of the radiation sterilization **612** such that it becomes harmless to the sensitive electronics. In other embodiments, however, the converse may equally be true. More specifically, the size (e.g., length) of the collimator **606** may be decreased as long as the material for the collimator **606** exhibits a large enough mass density.

In addition to the radiation blocking characteristics of the body of the collimator **606**, in some embodiments, one or more shields **616** (one shown) may be positioned within the sensor housing **304** to protect sensitive electronic components from radiation while the sensor control device **302** is subjected to the radiation sterilization **612**. The shield **616**, for example, may be positioned to interpose a data processing unit **618** and the radiation source (e.g., an e-beam electron accelerator). In such embodiments, the shield **616** may be positioned adjacent to and otherwise aligned with the data processing unit **618** and the radiation source to block or mitigate radiation exposure (e.g., e-beam radiation or energy) that might otherwise damage the sensitive electronic circuitry of the data processing unit **618**.

The shield **616** may be made of any material capable of blocking (or substantially blocking) the transmission of radiation. Suitable materials for the shield **616** include, but are not limited to, lead, tungsten, iron-based metals (e.g., stainless steel), copper, tantalum, tungsten, osmium, or any combination thereof. Suitable metals may be corrosion-resistant, austenitic, and any non-magnetic metal with a density ranging between about 5 grams per cubic centimeter (g/cc) and about 15 g/cc. The shield **616** may be fabricated via a variety of manufacturing techniques including, but not limited to, stamping, casting, injection molding, sintering, two-shot molding, or any combination thereof.

In other embodiments, however, the shield **616** may comprise a metal-filled thermoplastic polymer such as, but not limited to, polyamide, polycarbonate, or polystyrene. In such embodiments, the shield **616** may be fabricated by mixing the shielding material in an adhesive matrix and dispensing the combination onto shaped components or otherwise directly onto the data processing unit **618**. Moreover, in such embodiments, the shield **616** may comprise an enclosure that encapsulates (or substantially encapsulates) the data processing unit **618**.

In some embodiments, a collimator seal **620** may be applied to the end of the collimator **606** to seal off the sterilization zone **608** and, thus, the sealed region **610**. As illustrated, the collimator seal **620** may seal the second aperture **614b**. The collimator seal **620** may be applied before or after the radiation sterilization **612**. In embodiments where the collimator seal **620** is applied before undertaking the radiation sterilization **612**, the collimator seal **620** may be made of a radiation permeable microbial barrier material that allows radiation to propagate therethrough. With the collimator seal **620** in place, the sealed region **610** is able to maintain a sterile environment for the assembled sensor control device **302** until the user removes (unthreads) the applicator cap **210**.

In some embodiments, the collimator seal **620** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can be applied before or after the radiation sterilization **612**, and following the radiation sterilization **612**, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization zone **608** and the sealed region **610**. In other embodiments, the collimator seal **620** may comprise only a single protective layer applied to the end of the collimator **606**. In such embodiments, the single layer is gas permeable for the sterilization process, but is also capable of protection against moisture and other harmful elements once the sterilization process is complete. Accordingly, the collimator seal **620** may operate as a moisture and contaminant layer, without departing from the scope of the disclosure.

It is noted that, while the sensor **316** and the sharp **318** extend from the bottom of the electronics housing **304** and into the sterilization zone **608** generally concentric with a centerline of the sensor applicator **102** and the applicator cap **210**, it is contemplated herein to have an eccentric arrangement. More specifically, in at least one embodiment, the sensor **316** and the sharp **318** may extend from the bottom of the electronics housing **304** eccentric to the centerline of the sensor applicator **102** and the applicator cap **210**. In such embodiments, the collimator **606** may be re-designed and otherwise configured such that the sterilization zone **608** is also eccentrically positioned to receive the sensor **316** and the sharp **318**, without departing from the scope of the disclosure.

In some embodiments, the collimator **606** may comprise a first or “internal” collimator capable of being housed within the applicator cap **210** or otherwise within the sensor applicator **102**, as generally described above. A second or “external” collimator (not shown) may also be included or otherwise used in the assembly (manufacturing) process to help sterilize the sensor applicator **102**. In such embodiments, the external collimator may be positioned external to the sensor applicator **102** and the applicator cap **210** and used simultaneously with the internal collimator **606** to help focus the radiation sterilization **612** on the sensor **316** and the sharp **318**.

In one embodiment, for example, the external collimator may initially receive the radiation sterilization **612**. Similar to the internal collimator **606**, the external collimator may provide or define a hole or passageway extending through the external collimator. The beams of the radiation sterilization **612** passing through the passageway of the external collimator may be focused and received into the sterilization zone **608** of the internal collimator **606** via the second aperture **614b**. Accordingly, the external collimator may operate to pre-focus the radiation energy, and the internal collimator **606** may fully focus the radiation energy on the sensor **316** and the sharp **318**.

In some embodiments, the internal collimator **606** may be omitted if the external collimator is capable of properly and fully focusing the radiation sterilization **612** to properly sterilize the sensor **316** and the sharp **318**. In such embodiments, the sensor applicator may be positioned adjacent the external collimator and subsequently subjected to the radiation sterilization **612**, and the external collimator may prevent radiation energy from damaging the sensitive electronics within the electronics housing **304**. Moreover, in such

embodiments, the sensor applicator **102** may be delivered to the user without the internal collimator **606** positioned within the applicator cap **210**, thus eliminating complexity in manufacturing and use.

FIG. 7A is an enlarged cross-sectional side view of the sensor control device **302** mounted within the applicator cap **210**, according to one or more embodiments. As indicated above, portions of the sensor **316** and the sharp **318** may be arranged within the sealed region **610** and thereby isolated from external contamination. The sealed region **610** may include (encompass) select portions of the interior of the electronics housing **304** and the sterilization zone **608** of the collimator **606**. In one or more embodiments, the sealed region **610** may be defined and otherwise formed by at least a first seal **702a**, a second seal **702b**, and the collimator seal **620**.

The first seal **702a** may be arranged to seal the interface between the sharp hub **422** and the top of the electronics housing **304**. More particularly, the first seal **702a** may seal the interface between the sharp hub **422** and the shell **306**. Moreover, the first seal **702a** may circumscribe the first central aperture **504** defined in the shell **306** such that contaminants are prevented from migrating into the interior of the electronics housing **304** via the first central aperture **504**. In some embodiments, the first seal **702a** may form part of the sharp hub **422**. For example, the first seal **702a** may be overmolded onto the sharp hub **422**. In other embodiments, the first seal **702a** may be overmolded onto the top surface of the shell **306**. In yet other embodiments, the first seal **702a** may comprise a separate structure, such as an O-ring or the like, that interposes the sharp hub **422** and the top surface of the shell **306**, without departing from the scope of the disclosure.

The second seal **702b** may be arranged to seal the interface between the collimator **606** and the bottom of electronics housing **304**. More particularly, the second seal **702b** may be arranged to seal the interface between the mount **308** and the collimator **606** or, alternatively, between the collimator **606** and the bottom of the plug **402** as received within the bottom of the mount **308**. In applications including the plug **402**, as illustrated, the second seal **702b** may be configured to seal about and otherwise circumscribe the plug receptacle **512**. In embodiments that omit the plug **402**, the second seal **702b** may alternatively circumscribe the second central aperture **506** (FIG. 5A) defined in the mount **308**. Consequently, the second seal **702b** may prevent contaminants from migrating into the sterilization zone **608** of the collimator **606** and also from migrating into the interior of the electronics housing **304** via the plug receptacle **512** (or alternatively the second central aperture **506**).

In some embodiments, the second seal **702b** may form part of the collimator **606**. For example, the second seal **702b** may be overmolded onto the top of the collimator **606**. In other embodiments, the second seal **702b** may be overmolded onto the plug **402** or the bottom of the mount **308**. In yet other embodiments, the second seal **702b** may comprise a separate structure, such as an O-ring or the like, that interposes the collimator **606** and the plug **402** or the bottom of the mount **308**, without departing from the scope of the disclosure.

Upon loading the sensor control device **302** into the sensor applicator **102** (FIG. 6B) and securing the applicator cap **210** to the sensor applicator **102**, the first and second seals **702a,b** become compressed and generate corresponding sealed interfaces. The first and second seals **702a,b** may be made of a variety of materials capable of generating a sealed interface between opposing structures. Suitable mate-

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rials include, but are not limited to, silicone, a thermoplastic elastomer (TPE), polytetrafluoroethylene (PTFE or Teflon®), or any combination thereof.

As discussed above, the collimator seal **620** may be configured to seal off the bottom of the sterilization zone **608** and, thus, the bottom of the sealed region **610**. Accordingly, the first and second seals **702a,b** and the collimator seal **620** each create corresponding barriers at their respective sealing locations. The combination of these seals **702a,b** and **620** allows the sealed region **610** containing the sensor **316** and the sharp **318** to be terminally sterilized.

FIG. 7B is an enlarged cross-sectional side view of another embodiment of the sensor control device **302** mounted within the sensor applicator **102**, according to one or more embodiments. More specifically, FIG. 7B depicts alternative embodiments of the first and second seals **702a,b**. The first seal **702a** is again arranged to seal the interface between the sharp hub **422** and the top of the electronics housing **304** and, more particularly, seal off the first central aperture **504** defined in the shell **306**. In the illustrated embodiment, however, the first seal **702a** may be configured to seal both axially and radially. More particularly, when the sensor control device **302** is introduced into the sensor applicator **102**, the sharp hub **422** is received by the sensor carrier **604**. The first seal **702a** may be configured to simultaneously bias against one or more axially extending members **704** of the sensor carrier **604** and one or more radially extending members **706** of the sensor carrier **604**. Such dual biased engagement compresses the first seal **702a** both axially and radially and thereby allows the first seal **702a** to seal against the top of the electronics housing **304** in both the radial and axial directions.

The second seal **702b** is again arranged to seal the interface between the collimator **606** and the bottom of electronics housing **304** and, more particularly, between the mount **308** and the collimator **606** or, alternatively, between the collimator **606** and the bottom of the plug **402** as received within the bottom of the mount **308**. In the illustrated embodiment, however, the second seal **702b** may extend into the sterilization zone **608** and define or otherwise provide a cylindrical well **708** sized to receive the sensor **316** and the sharp **1408** as extending from the bottom of the mount **308**. In some embodiments, a desiccant **710** may be positioned within the cylindrical well to aid maintenance of a low humidity environment for biological components sensitive to moisture.

In some embodiments, the second seal **702b** may be omitted and the collimator **606** may be directly coupled to the electronics housing **304**. More specifically, in at least one embodiment, the collimator **606** may be threadably coupled to the underside of the mount **308**. In such embodiments, the collimator **606** may provide or otherwise define a threaded extension configured to mate with a threaded aperture defined in the bottom of the mount **308**. Threadably coupling the collimator **606** to the mount **308** may seal the interface between the collimator **606** and the bottom of electronics housing **304**, and thus operate to isolate sealed region **610**. Moreover, in such embodiments, the pitch and gauge of the threads defined on the collimator **606** and the mount **308** may match those of the threaded engagement between the applicator cap **210** and the sensor applicator **102**. As a result, as the applicator cap **210** is threaded to or unthreaded from the sensor applicator **102**, the collimator **606** may correspondingly be threaded to or unthreaded from the electronics housing **404**.

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Embodiments disclosed herein include:

A. An analyte monitoring system that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing, a sensor extending from a bottom of the electronics housing, a sharp hub positioned adjacent a top of the electronics housing, and a sharp carried by the sharp hub and extending through the electronics housing and from the bottom of the electronics housing. The analyte monitoring system further including a cap coupled to the sensor applicator, and a collimator positioned within the cap and defining a sterilization zone that receives the sensor and the sharp extending from the bottom of the electronics housing.

B. A method of preparing an analyte monitoring system includes loading a sensor control device into a sensor applicator, the sensor control device including an electronics housing, a sensor extending from a bottom of the electronics housing, a sharp hub positioned adjacent a top of the electronics housing, and a sharp carried by the sharp hub and extending through the electronics housing and from the bottom of the electronics housing. The method further including securing a cap to the sensor applicator, wherein a collimator is arranged within the cap and defines a sterilization zone that receives the sensor and the sharp extending from the bottom of the electronics housing, sterilizing the sensor and the sharp with radiation sterilization while positioned within the sterilization zone, and preventing radiation from the radiation sterilization from damaging electronic components within the electronics housing with the collimator.

C. A method of preparing an analyte monitoring system includes loading a sensor control device into a sensor applicator, the sensor control device including an electronics housing, a sensor extending from a bottom of the electronics housing, a sharp hub positioned adjacent a top of the electronics housing, and a sharp carried by the sharp hub and extending through the electronics housing and from the bottom of the electronics housing. The method further including positioning the sensor applicator adjacent a collimator, subjecting the sensor and the sharp to radiation sterilization, and preventing radiation from the radiation sterilization from damaging the electronic components within the electronics housing with the collimator.

Each of embodiments A, B, and C may have one or more of the following additional elements in any combination: Element 1: wherein the sterilization zone comprises a passageway extending at least partially through the collimator. Element 2: wherein the sterilization zone comprises a cross-sectional shape selected from the group consisting of conical, frustoconical, cubic, rectangular, pyramidal, and any combination thereof. Element 3: wherein the sterilization zone is frustoconical and defines a first aperture at a first end and a second aperture at a second end, and wherein the first aperture receives the sensor and the sharp extending from the bottom of the electronics housing and a seal is arranged at the second aperture. Element 4: further comprising a sealed region encompassing the sterilization zone and a portion of an interior of the electronics housing, wherein the sealed region is defined by a first seal that seals an interface between the sharp hub and the top of the electronics housing, a second seal that seals an interface between the collimator and the bottom of the electronics housing, and a third seal that seals an end of the sterilization zone. Element 5: wherein the first seal circumscribes a central aperture defined in the top of the electronics housing and prevents contaminants from migrating into the portion of the interior of the electronics housing via the central aperture, and

wherein the second seal circumscribes an aperture defined in the bottom of the electronics housing and prevents contaminants from migrating into the portion of the interior of the electronics housing via the aperture. Element 6: wherein the first seal provides one or both of an axial and a radial seal. Element 7: wherein the second seal extends into the sterilization zone and defines a cylindrical well that receives the sensor and the sharp. Element 8: further comprising a printed circuit board arranged within the electronics housing, a data processing unit mounted to the printed circuit board, and a shield positioned within the electronics housing to protect the data processing unit from radiation from a radiation sterilization process. Element 9: wherein the shield is made of a non-magnetic metal selected from the group consisting of lead, tungsten, iron, stainless steel, copper, tantalum, osmium, a thermoplastic polymer mixed with a non-magnetic metal, and any combination thereof.

Element 10: further comprising creating a sealed region as the cap is secured to the sensor applicator, the sealed region encompassing the sterilization zone and a portion of an interior of the electronics housing. Element 11: wherein creating the sealed region comprises sealing an interface between the sharp hub and the top of the electronics housing with a first seal, sealing an interface between the collimator and the bottom of the electronics housing with a second seal, and sealing an end of the sterilization zone with a third seal. Element 12: wherein sealing the interface between the sharp hub and the top of the electronics housing with the first seal comprises providing one or both of an axial seal and a radial seal with the first seal. Element 13: wherein the collimator comprises an internal collimator and sterilizing the sensor and the sharp with the radiation sterilization further comprises positioning the sensor applicator adjacent an external collimator arranged external to the sensor applicator, focusing the radiation with the external collimator to be received by the internal collimator, and preventing the radiation from damaging the electronic components within the electronics housing with the external and internal collimators. Element 14: wherein the sterilization zone defines a first aperture at a first end of the collimator and a second aperture at a second end of the collimator, and wherein sterilizing the sensor and the sharp comprises introducing radiation into the sterilization zone via the second aperture. Element 15: wherein preventing the radiation from the radiation sterilization from damaging the electronic components comprises blocking the radiation with the material of the collimator. Element 16: wherein a printed circuit board is arranged within the electronics housing and a data processing unit is mounted to the printed circuit board, the method further comprising protecting the data processing unit from radiation from the radiation sterilization process with a shield positioned within the electronics housing.

Element 17: wherein positioning the sensor applicator adjacent the collimator comprises arranging the collimator such that it resides external to the sensor applicator during the radiation sterilization.

By way of non-limiting example, exemplary combinations applicable to A, B, and C include: Element 2 with Element 3; Element 4 with Element 5; Element 4 with Element 6; Element 4 with Element 7; Element 8 with Element 9; Element 10 with Element 11; and Element 11 with Element 12.

External Sterilization Assemblies

Referring again briefly to FIG. 1, prior to being delivered to an end user, the sensor control device 104 must be sterilized to render the product free from viable microorganisms. The sensor 110 is commonly sterilized using

radiation sterilization, such as electron beam (“e-beam”) irradiation. Radiation sterilization, however, can damage the electronic components within the sensor control device 104, which are commonly sterilized via gaseous chemical sterilization (e.g., using ethylene oxide). Gaseous chemical sterilization, however, can damage the enzymes or other chemistry and biologics included on the sensor 110.

In the past, this sterilization incompatibility has been circumvented by separating the sensor 110 and the electronic components and sterilizing each individually. This approach, however, requires additional parts, packaging, process steps, and final assembly by the user, which introduces a possibility of user error. According to the present disclosure, the sensor control device 104, or any device requiring terminal sterilization, may be properly sterilized using an external sterilization assembly designed to focus sterilizing radiation (e.g., beams, waves, energy, etc.) toward component parts requiring sterilization, while simultaneously preventing the propagating radiation from disrupting or damaging sensitive electronic components.

FIG. 8 is a schematic diagram of an example external sterilization assembly 800, according to one or more embodiments of the present disclosure. The external sterilization assembly 800 (hereafter the “assembly 800”) may be designed and otherwise configured to help sterilize a medical device 802. The medical device 802 may comprise, for example, a sensor control device similar in some respects to the sensor control device 104 of FIG. 1, but could alternatively comprise other types of medical devices, health care products, or systems requiring terminal sterilization of specific component parts. Example medical devices or health care products that may incorporate the principles of the present disclosure include, but are not limited to, ingestible products, cardiac rhythm management (CRM) devices, under-skin sensing devices, externally mounted medical devices, or any combination thereof.

The medical device 802 may include a housing 804, a part 806 requiring sterilization, and one or more radiation sensitive components 808. In the illustrated embodiment, the radiation sensitive component 808 may be mounted to a printed circuit board (PCB) 810 positioned within the housing 804, and the housing 804 may comprise an electronics housing for a sensor control device. The radiation sensitive component 808 may include one or more electronic modules such as, but not limited to, a data processing unit (e.g., an application specific integrated circuit or ASIC), a resistor, a transistor, a capacitor, an inductor, a diode, and a switch. In other embodiments, however, the radiation sensitive component 808 may comprise a radiation sensitive chemical solution or analyte, as described herein with reference to FIG. 12.

In some embodiments, the part 806 may comprise a sensor (e.g., the sensor 110 of FIG. 1) that extends from the housing 804. As illustrated, the part 806 may extend at an angle from the bottom of the housing 804, but could alternatively extend perpendicular to the bottom or from another surface of the housing 804. In at least one embodiment, the part 806 may further include a sharp that may also require sterilization and may help implant the sensor beneath the skin of a user. In some embodiments, as illustrated, the part 806 may be encapsulated with a cap 812 that provides a sealed barrier that protects exposed portions of the part 806 (e.g., the sensor and associated sharp) until the part 806 is needed for use.

The medical device 802 may be subjected to radiation sterilization 814 to properly sterilize the part 806 for use. Suitable radiation sterilization 814 processes include, but are

not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. In embodiments that include the cap **812**, the cap **812** may be made of a material that permits propagation of the radiation **814** therethrough to facilitate radiation sterilization of the part **806**. Suitable materials for the cap **812** include, but are not limited to, a non-magnetic metal (e.g., aluminum, copper, gold, silver, etc.), a thermoplastic, ceramic, rubber (e.g., ebonite), a composite material (e.g., fiberglass, carbon fiber reinforced polymer, etc.), an epoxy, or any combination thereof. In some embodiments, the cap **812** may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

The assembly **800** may include a radiation shield **816** positioned external to the medical device **802** and configured to help sterilize the part **806** while preventing (impeding) propagating radiation **814** from disrupting or damaging the radiation sensitive component(s) **808**. To accomplish this, the radiation shield **816** may provide a collimator **818** that generally comprises a hole or passageway extending at least partially through the body of the radiation shield **816**. The collimator **818** defines a sterilization zone **820** configured to focus the radiation **814** toward the part **806**. In the illustrated embodiment, the part **806** may also be received within the sterilization zone **820** for sterilization.

While focusing the radiation **814** (e.g., beams, waves, energy, etc.) toward the part **806**, the radiation shield **816** may be made of a material that reduces or eliminates the radiation **814** from penetrating therethrough and thereby damaging the radiation sensitive component(s) **808** within the housing **804**. In other words, the radiation shield **816** may be made of a material having a density sufficient to absorb the dose of the beam energy being delivered. In some embodiments, for example, the radiation shield **816** may be made of any material that has a mass density greater than 0.9 grams per cubic centimeter (g/cc). In other embodiments, however, the mass density of a suitable material may be less than 0.9 g/cc, without departing from the scope of the disclosure. Suitable materials for the radiation shield **816** include, but are not limited to, a high-density polymer, (e.g., polyethylene, polypropylene, polystyrene, polytetrafluoroethylene, etc.), a metal (e.g., lead, stainless steel, aluminum, etc.), any combination thereof, or any material having a mass density greater than 0.9 g/cc.

The collimator **818** can exhibit any suitable cross-sectional shape necessary to focus the radiation on the part **806** for sterilization. In the illustrated embodiment, for example, the collimator **818** is conical or frustoconical in shape. In other embodiments, however, the collimator **818** may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure. In yet other embodiments, the collimator **818** may exhibit a circular cross-sectional shape with parallel sides.

In the illustrated embodiment, the collimator **818** provides a first aperture **822a** and a second aperture **822b** where the first and second apertures **822a,b** are defined at opposing ends of the sterilization zone **820**. The first aperture **822a** may allow the radiation **814** to enter the sterilization zone **820** and impinge upon the part **806**, and the second aperture **822b** may be configured to receive the part **806** into the sterilization zone **820**. In embodiments where the collimator **818** is conical or frustoconical in shape, the second aperture **822b** may have a diameter that is smaller than the diameter of the first aperture **822a**. In such embodiments, for example, the size of the second aperture **822b** may range between about 0.5 mm and about 3.0 mm, and the size of the

first aperture **822a** may range between about 5.0 mm and about 16.0 mm. As will be appreciated, however, the respective diameters of the first and second apertures **822a,b** may be greater or less than the ranges provided herein, without departing from the scope of the disclosure. Indeed, the diameters of the first and second apertures **822a,b** may be scaled to the device size and need only be large enough to allow a sufficient dose of radiation to impinge upon the part **806**. Moreover, in at least one embodiment, the collimator **818** may be cylindrical in shape where the first and second apertures **822a,b** exhibit identical diameters.

In some embodiments, the assembly **800** may further include a barrier shield **824** positioned within the housing **804**. The barrier shield **824** may be configured to help block radiation **814** (e.g., electrons) from propagating within the housing **804** toward the radiation sensitive component(s) **808**. The barrier shield **824** may be made of any of the materials mentioned above for the radiation shield **816**. In the illustrated embodiment, the barrier shield **824** is positioned vertically within the housing **804**, but may alternatively be positioned at any other angular configuration suitable for protecting the radiation sensitive component(s) **808**.

FIG. 9 is a schematic diagram of another example external sterilization assembly **900**, according to one or more additional embodiments of the present disclosure. The external sterilization assembly **900** (hereafter the “assembly **900**”) may be similar in some respects to the assembly **800** of FIG. 8 and therefore may be best understood with reference thereto, where like numerals will refer to similar components not described again. Similar to the assembly **800**, the assembly **900** may be designed and otherwise configured to help sterilize a medical device **902**. In the illustrated embodiment, the medical device **902** may comprise a two-piece sensor control device, but could alternatively comprise any of the medical devices mentioned herein with respect to the medical device **802**.

As illustrated, the medical device **902** includes a housing **904**, a part **906** requiring sterilization, and one or more radiation sensitive components **908** positioned within the housing **904**. The housing **904** may comprise packaging or an enclosure that contains the part **906** and the radiation sensitive component(s) **908**. The radiation sensitive component(s) **908** may comprise any of the electronic modules mentioned herein with respect to the radiation sensitive component(s) **808** of FIG. 8. The part **906** may comprise, for example a needle/sensor subassembly, and may be subjected to radiation sterilization **814** to properly sterilize the part **906** for use.

The assembly **900** may include a radiation shield **910** positioned external to the medical device **902** and configured to help sterilize the part **906** while preventing (impeding) propagating radiation **814** from damaging the radiation sensitive component(s) **908**. In the illustrated embodiment, the radiation shield **910** may define or otherwise provide an internal cavity **912** into which the medical device **902** may be positioned. Similar to the radiation shield **816** of FIG. 8, the radiation shield **910** may provide a collimator **914** that generally comprises a hole or passageway extending at least partially through the body of the radiation shield **910** and providing access into the cavity **912**. The collimator **914** may define a sterilization zone **916** that helps focus the radiation **814** toward the part **906**. The radiation shield **910** may be made of any of the materials mentioned above with respect to the radiation shield **816** to reduce or eliminate the radiation **814** from penetrating therethrough, except for at

the collimator **914**, and thereby damaging the radiation sensitive component(s) **908** within the housing **904**.

To properly sterilize the part **906**, the radiation sterilization **814** may be directed at the medical device **902**. The collimator **914** and sterilization zone **916** may be configured to concentrate and/or focus the radiation sterilization **814** toward the part **906**, while the remaining portions of the radiation shield **910** prevent (impede) the propagating radiation **814** from damaging the radiation sensitive component(s) **908** within the housing **904**. In the illustrated embodiment, the collimator **914** and sterilization zone **916** exhibit a circular cross-sectional shape with parallel sides, but could alternatively exhibit other cross-sectional shapes including, but not limited to, conical, frustoconical, pyramidal, polygonal, or any combination thereof.

In some embodiments, the assembly **900** may further include the barrier shield **824** positioned within the housing **904** to help block radiation **814** (e.g., electrons) from propagating within the housing **904** toward the radiation sensitive component(s) **908**.

FIG. **10** is a schematic diagram of another example external sterilization assembly **1000**, according to one or more additional embodiments of the present disclosure. The external sterilization assembly **1000** (hereafter the “assembly **1000**”) may be similar in some respects to the assembly **900** of FIG. **15** and therefore may be best understood with reference thereto, where like numerals will refer to similar components not described again. Similar to the assembly **900**, the assembly **1000** may be designed and otherwise configured to help sterilize a medical device **1002**. In the illustrated embodiment, the medical device **1002** may comprise a sensor control device similar to the sensor control device **104** of FIG. **1**, but could alternatively comprise any of the medical devices mentioned herein with respect to the medical device **802** of FIG. **8**.

As illustrated, the medical device **1002** includes a housing **1004**, a part **1006** requiring sterilization, and one or more radiation sensitive components **1008** positioned within the housing **1004**. In the illustrated embodiment, the housing **1004** may comprise an electronics housing for a sensor control device (e.g., the sensor control device **104** of FIG. **1**) and the radiation sensitive component(s) **1008** may comprise any of the electronic modules mentioned herein with respect to the radiation sensitive component(s) **808** of FIG. **8**. In some embodiments, the part **1006** may comprise a sensor (e.g., the sensor **110** of FIG. **1**) that extends from the housing **1004**, and may further include a sharp also requiring sterilization and used to help implant the sensor beneath the skin of a user.

The assembly **1000** may include a radiation shield **1010** positioned external to the medical device **1002** and configured to help sterilize the part **1006** while preventing (impeding) propagating radiation **814** from disrupting or damaging the radiation sensitive component(s) **1008**. The radiation shield **1010** may be made of any of the materials mentioned above with respect to the radiation shield **816** of FIG. **8** to reduce or eliminate the radiation **814** from penetrating therethrough and thereby damaging the radiation sensitive component(s) **1008** within the housing **1004**.

In the illustrated embodiment, the radiation shield **1010** may define or otherwise provide an internal cavity **1012** into which the medical device **1002** may be positioned for sterilization. In some embodiments, the radiation shield **1010** may comprise a box and the internal cavity **1012** may be formed within the interior of the box. The radiation shield **1010** may also provide a collimator **1014** that extends at least partially through the body of the radiation shield **1010**

and provides access into the cavity **1012**. The collimator **1014** may define a sterilization zone **1016** that focuses the radiation **814** toward the part **1006** for sterilization.

To properly sterilize the part **1006**, the radiation sterilization **814** may be directed at the medical device **1002**. The collimator **1014** and the sterilization zone **1016** may concentrate and/or focus the radiation sterilization **814** toward the part **1006**, while the remaining portions of the radiation shield **1010** prevent (impede) the propagating radiation **814** from damaging the radiation sensitive component(s) **1008** within the housing **1004**. In the illustrated embodiment, the collimator **1014** exhibits a circular cross-sectional shape with parallel sides, but could alternatively exhibit other cross-sectional shapes including, but not limited to, conical, frustoconical, pyramidal, polygonal, or any combination thereof.

FIG. **11** is a schematic diagram of another example external sterilization assembly **1100**, according to one or more additional embodiments of the present disclosure. The external sterilization assembly **1100** (hereafter the “assembly **1100**”) may be similar in some respects to the assemblies **800**, **900**, and **1000** of FIGS. **8**, **9**, and **10**, respectively, and therefore may be best understood with reference thereto. Similar to the assemblies **800-1000**, the assembly **1100** may be designed and otherwise configured to help sterilize a medical device **1102**. In the illustrated embodiment, the medical device **1102** may comprise a two piece sensor control device, but could alternatively comprise any of the medical devices mentioned herein with respect to the medical device **802**.

As illustrated, the medical device **1102** includes a housing **1104**, a part **1106** requiring sterilization, and one or more radiation sensitive components **1108** positioned within the housing **1104**. The radiation sensitive component(s) **1108** may comprise any of the electronic modules mentioned herein with respect to the radiation sensitive component(s) **808** of FIG. **8**. In the illustrated embodiment, the part **1106** may comprise, for example, a needle/sensor subassembly, and may be subjected to radiation sterilization **814** to properly sterilize the part **1106** for use.

The assembly **1100** may include a radiation shield **1110** positioned external to the medical device **1102** and configured to help sterilize the part **1106** while preventing (impeding) propagating radiation **814** from damaging the radiation sensitive component(s) **1108**. The radiation shield **1110** may be made of any of the materials mentioned above with respect to the radiation shield **816** of FIG. **8** to reduce or eliminate the radiation **814** from penetrating therethrough and thereby damaging the radiation sensitive component(s) **1108**.

In the illustrated embodiment, the radiation shield **1110** may comprise a clamshell structure including a first portion **1112a** and a second portion **1112b** matable (or engageable) with the first portion **1112a**. The radiation shield **1110** may also provide or otherwise define an internal cavity **1114** into which the medical device **1102** may be positioned for sterilization. In some embodiments, as illustrated, the first and second portions **1112a,b** may cooperatively define a portion of the internal cavity **1114** such that when the first and second portions **1112a,b** are properly mated, the internal cavity **1114** is formed. In other embodiments, however, the internal cavity **1114** may be defined wholly within the first portion **1112a** or wholly within the second portion **1112b**.

In some embodiments, the assembly **1100** may further include an absorber **1116** configured to protect the medical device **1102**. In at least one embodiment, as illustrated, portions of the absorber **1116** may be provided by or

otherwise form part of each of the first and second portions **1112a,b**. In such embodiments, the internal cavity **1114** may be defined, at least in part by the absorber **1116**. The absorber **1116** may be made of a material that absorbs stray radiation without causing Bremsstrahlung protons being generated. The material for the absorber **1116** may comprise, for example, any of the high-density polymers mentioned herein for the radiation shield **816** of FIG. 8.

Similar to the radiation shield **816** of FIG. 8, the radiation shield **1110** may provide a collimator. In the illustrated embodiment, however, the radiation shield **1110** provides or otherwise defines a first collimator **1118a** and a second collimator **1118b**, but could alternatively include only one of the collimators **1118a,b**, without departing from the scope of the disclosure. The first collimator **1118a** generally comprises a hole or passageway extending at least partially through the first portion **1112a** of the radiation shield **1110**, and the second collimator **1118b** generally comprises a hole or passageway extending at least partially through the second portion **1112b**. Each collimator **1118a,b** provides access into the internal cavity **1114** and the collimators **1118a,b** cooperatively define a sterilization zone **1120** that includes the internal cavity **1114** and helps focus the radiation **814** toward the part **1106** for sterilization.

To properly sterilize the part **1106**, the medical device **1102** may be positioned within the internal cavity **1114** and the opposing portions **1112a,b** may be mated to encapsulate the medical device **1102**. The medical device **1102** may be situated within the sterilization zone **1120** once properly positioned within the cavity **1114**. The radiation sterilization **814** may then be directed at the medical device **1102** on opposing sides of the radiation shield **1110**, and the collimators **1118a,b** may concentrate and/or focus the radiation sterilization **814** toward the part **1106** on opposing sides of the part **1106**. The remaining portions of the radiation shield **1110** prevent (impede) the propagating radiation **814** from damaging the radiation sensitive component(s) **1108** within the housing **1104**. In the illustrated embodiment, each collimator **1118a,b** exhibits a conical or frustoconical cross-sectional shape, but could alternatively exhibit other cross-sectional shapes including, but not limited to, circular, pyramidal, polygonal, or any combination thereof.

In some embodiments, the assembly **1100** may further include one or more barrier shields **824** (two shown) positioned within the housing **1104** to help block radiation **814** (e.g., electrons) from propagating within the housing **1104** toward the radiation sensitive component(s) **1108**.

FIG. 12 is a schematic diagram of another example external sterilization assembly **1200**, according to one or more additional embodiments of the present disclosure. The external sterilization assembly **1200** (hereafter the “assembly **1200**”) may be designed and otherwise configured to help sterilize a medical device **1202**, which, in the illustrated embodiment, comprises a hypodermic needle or syringe. As illustrated, the medical device **1202** includes a housing **1204** (e.g., a barrel or vial), a part **1206** requiring sterilization, and one or more radiation sensitive components **1208** positioned within the housing **1204**. In the illustrated embodiment, the radiation sensitive component **1208** may comprise a chemical solution or an analyte (e.g., an active agent, pharmaceutical, biologic, etc.) that may be sensitive to irradiation, and the part **1206** may comprise a needle designed to deliver the chemical solution.

In some embodiments, as illustrated, the part **1206** may be encased or otherwise surrounded by a cap **1210** (e.g., a needle cap) that encapsulates the part **1206**. Moreover, in at least one embodiment, the cap **1210** may be sealed against

the housing **1204** with a sealing element **1212**, such as an O-ring or the like. The cap **1210** and the sealing element **1212** may cooperatively provide a sterile barrier system that surrounds and protects exposed portions of the part **1206** until required to be used. The part **1206** may be subjected to radiation sterilization **814** to properly sterilize the part **1206** for use.

The assembly **1200** may include a radiation shield **1214** positioned external to the medical device **1202** and configured to help sterilize the part **1206** while preventing (impeding) propagating radiation **814** from damaging the radiation sensitive component **1208**. As illustrated, the radiation shield **1214** may provide a collimator **1216** that generally comprises a hole or passageway extending at least partially through the body of the radiation shield **1214** and defines a sterilization zone **1218** configured to focus the radiation **814** toward the part **1206** for sterilization. In the illustrated embodiment, the part **1206** may also be received within the sterilization zone **1218**. The collimator **1216** allows transmission of the radiation **814** to impinge upon and sterilize the part **1206**, while the remaining portions of the radiation shield **1214** prevent (impede) the propagating radiation **814** from damaging the radiation sensitive component(s) **1208** within the housing **1204**. In the illustrated embodiment, the collimator **1216** is conical or frustoconical in shape, but may alternatively exhibit other cross-sectional shapes, such as polygonal, pyramidal, circular, or any combination thereof.

In embodiments including the cap **1210**, the body of the cap **1210** may comprise a material that permits propagation of radiation **814** therethrough to facilitate radiation sterilization of the part **1206**. Suitable materials for the cap **1210** may be the same as mentioned herein for the cap **812** of FIG. 8.

In some embodiments, the assembly **1200** may further include the barrier shield **824** positioned to help block radiation **814** (e.g., electrons) from propagating within the housing **1204** toward the radiation sensitive component **1208** (e.g., the chemical solution). In the illustrated embodiment, the barrier shield **824** may define or otherwise provide a central aperture **1220** configured to allow the radiation sensitive component **1208** to exit the housing **1204** via the part **1206** (e.g., the needle). In other embodiments, the barrier shield **824** may provide a tortuous pathway that allows the radiation sensitive component **1208** to exit the housing **1204** via the part **1206**.

FIG. 13 is an isometric view of an example sensor control device **1302**, according to one or more additional embodiments of the present disclosure. The sensor control device **1302** may be the same as or similar to the sensor control device **104** of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator **102** (FIG. 1), which delivers the sensor control device **1302** to a target monitoring location on a user's skin. Moreover, the sensor control device **1302** may be alternately characterized as a medical device, similar to one or more of the medical devices **1402-1202** of FIGS. 8-12 described herein. Accordingly, the sensor control device **1302** may also require proper sterilization prior to being used.

As illustrated, the sensor control device **1302** includes an electronics housing **1304** that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing **1304** may exhibit other cross-sectional shapes, such as ovoid (e.g., pill-shaped), a squire, or polygonal, without departing from the scope of the disclosure. The electronics housing **1304** may be configured to house or otherwise contain various electronic components used to operate the sensor control device **1302**.

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The electronics housing **1304** may include a shell **1306** and a mount **1308** that is matable with the shell **1306**. The shell **1306** may be secured to the mount **1308** via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, one or more mechanical fasteners (e.g., screws), or any combination thereof. In some cases, the shell **1306** may be secured to the mount **1308** such that a sealed interface therebetween is generated. In such embodiments, a gasket or other type of seal material may be positioned at or near the outer diameter (periphery) of the shell **1306** and the mount **1308**, and securing the two components together may compress the gasket and thereby generate a sealed interface. In other embodiments, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell **1306** and the mount **1308**. The adhesive secures the shell **1306** to the mount **1308** and provides structural integrity, but may also seal the interface between the two components and thereby isolate the interior of the electronics housing **1304** from outside contamination.

In the illustrated embodiment, the sensor control device **1302** may further include a plug assembly **1310** that may be coupled to the electronics housing **1304**. The plug assembly **1310** may include a sensor module **1312** (partially visible) interconnectable with a sharp module **1314** (partially visible). The sensor module **1312** may be configured to carry and otherwise include a sensor **1316** (partially visible), and the sharp module **1314** may be configured to carry and otherwise include a sharp **1318** (partially visible) used to help deliver the sensor **1316** transcutaneously under a user's skin during application of the sensor control device **1302**. The sharp module **1314** may include a sharp hub **1320** that carries the sharp **1318**.

As illustrated, corresponding portions of the sensor **1316** and the sharp **1318** extend from the electronics housing **1304** and, more particularly, from the bottom of the mount **1308**. The exposed portion of the sensor **1316** (alternately referred to as the "tail") may be received within a hollow or recessed portion of the sharp **1318**. The remaining portions of the sensor **1316** are positioned within the interior of the electronics housing **1304**.

FIG. **14A** is a side view of the sensor applicator **102** of FIG. **1**. As illustrated, the sensor applicator **102** includes a housing **1402** and an applicator cap **1404** that may be removably coupled to the housing **1402**. In some embodiments, the applicator cap **1404** may be threaded to the housing **1402** and include a tamper ring **1406**. Upon rotating (e.g., unscrewing) the applicator cap **1404** relative to the housing **1402**, the tamper ring **1406** may shear and thereby free the applicator cap **1404** from the sensor applicator **102**. Once the applicator cap **1404** is removed, a user may then use the sensor applicator **102** to position the sensor control device **1302** (FIGS. **13** and **14B**) at a target monitoring location on the user's body.

In some embodiments, the applicator cap **1404** may be secured to the housing **1402** via a sealed engagement to protect the internal components of the sensor applicator **102**. In at least one embodiment, for example, an O-ring or another type of sealing gasket may seal an interface between the housing **1402** and the applicator cap **1404**. The O-ring or sealing gasket may be a separate component part or alternatively molded onto one of the housing **1402** and the applicator cap **1404**.

FIG. **14B** is a cross-sectional side view of the sensor applicator **102**. As illustrated, the sensor control device **1302** may be received within the sensor applicator **102** and the applicator cap **1404** may be coupled to the sensor applicator **102** to secure the sensor control device **1302** therein. The

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sensor control device **1302** may include one or more radiation sensitive components **1408** arranged within the electronics housing **1304**. The radiation sensitive component **1408** can include an electronic component or module such as, but not limited to, a data processing unit, a resistor, a transistor, a capacitor, an inductor, a diode, a switch, or any combination thereof. The data processing unit may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **1302**. In operation, the data processing unit may perform data processing functions, such as filtering and encoding of data signals corresponding to a sampled analyte level of the user. The data processing unit may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. **1**).

In the illustrated embodiment, a cap fill **1410** may be positioned within the applicator cap **1404** and may generally help support the sensor control device **1302** within the sensor applicator **102**. In one or more embodiments, the cap fill **1410** may comprise an integral part or extension of the applicator cap **1404**, such as being molded with or overmolded onto the applicator cap **1404**. In other embodiments, the cap fill **1410** may comprise a separate structure fitted within or otherwise attached to the applicator cap **1404**, without departing from the scope of the disclosure.

The sensor control device **1302** and, more particularly, the distal ends of the sensor **1316** and the sharp **1318** extending from the bottom of the electronics housing **1304**, may be sterilized while positioned within the sensor applicator **102**. More specifically, the fully assembled sensor control device **1302** may be subjected to radiation sterilization **1412**, which may be similar to the radiation sterilization **814** of FIGS. **8-12**. The radiation sterilization **1412** may be delivered either through continuous processing irradiation or through pulsed beam irradiation. In pulsed beam irradiation, the beam of radiation sterilization **1412** is focused at a target location and the component part or device to be sterilized is moved to the target location at which point the irradiation is activated to provide a directed pulse of radiation. The radiation sterilization **1412** is then turned off, and another component part or device to be sterilized is moved to the target location and the process is repeated.

According to the present disclosure, an external sterilization assembly **1414** may be used to help focus the radiation **1412** in sterilizing the distal ends of the sensor **1316** and the sharp **1318**, while simultaneously preventing (impeding) propagating radiation **1412** from damaging the radiation sensitive component **1408**. As illustrated, the external sterilization assembly **1414** (hereafter the "assembly **1414**") may include a radiation shield **1416** positioned at least partially external to the sensor applicator **102**. The radiation shield **1416** may provide or define an external collimator **1418** configured to help focus the radiation **1412** (e.g., beams, waves, energy, etc.) toward the components to be sterilized. More specifically, the external collimator **1418** allows transmission of the radiation **1412** to impinge upon and sterilize the sensor **1316** and the sharp **1318**, but prevent the radiation **1412** from damaging the radiation sensitive component **1408** within the electronics housing **1304**.

In the illustrated embodiment, the external collimator **1418** is designed to align with an internal collimator **1420** defined by the cap fill **1410**. Similar to the external collimator **1418**, the internal collimator **1420** may help focus the radiation **1412** toward the components to be sterilized. As illustrated, the cap fill **1410** may define a radial shoulder **1422** sized to receive and otherwise mate with an end of the

radiation shield **1416**, and the external collimator **1418** transitions to the internal collimator **1420** at the radial shoulder **1422**. In some embodiments, the transition between the external and internal collimators **1418**, **1420** may be continuous, flush, or smooth. In other embodiments, however, the transition may be discontinuous or stepped, without departing from the scope of the disclosure.

The external and internal collimators **1418**, **1420** may cooperatively define a sterilization zone **1424** that focuses the radiation **1412** and into which the distal ends of the sensor **1316** and the sharp **1318** may be positioned. The propagating radiation **1412** may traverse the sterilization zone **1424** to impinge upon and sterilize the sensor **1316** and the sharp **1318**. However, the cap fill **1410** and the radiation shield **1416** may each be made of materials that substantially prevent the radiation **1412** from penetrating the inner wall(s) of the sterilization zone **1424** and thereby damaging the radiation sensitive component **1408** within the housing **1304**. In other words, the cap fill **1410** and the radiation shield **1416** may each be made of materials having a density sufficient to absorb the dose of the beam energy being delivered. In some embodiments, for example, one or both of the cap fill **1410** and the radiation shield **1416** may be made of a material that has a mass density greater than 0.9 grams per cubic centimeter (g/cc). In other embodiments, however, the mass density of a suitable material may be less than 0.9 g/cc, without departing from the scope of the disclosure. Suitable materials for the cap fill **1410** and the radiation shield **1416** include, but are not limited to, a high-density polymer, (e.g., polyethylene, polypropylene, polystyrene, polytetrafluoroethylene, etc.), a metal (e.g., lead, stainless steel, aluminum, etc.), any combination thereof, or any material having a mass density greater than 0.9 g/cc. In at least one embodiment, the cap fill **1410** may be made of machined or 3D printed polypropylene and the radiation shield **1416** may be made of stainless steel.

In some embodiments, the design of the sterilization zone **1424** may be altered so that one or both of the cap fill **1410** and the radiation shield **1416** may be made of a material that has a mass density less than 0.9 g/cc but may still operate to prevent the radiation sterilization **1412** from damaging the radiation sensitive component **1408**. In such embodiments, the size (e.g., length) of the sterilization zone **1424** may be increased such that the propagating electrons from the radiation sterilization **1412** are required to pass through a larger amount of material before potentially impinging upon the radiation sensitive component **1408**. The larger amount of material may help absorb or dissipate the dose strength of the radiation **1412** such that it becomes harmless to the sensitive electronics. In other embodiments, however, the converse may equally be true. More specifically, the size (e.g., length) of the sterilization zone **1424** may be decreased as long as the material for the cap fill **1410** and/or the radiation shield **1416** exhibits a large enough mass density.

The sterilization zone **1424** defined by the external and internal collimators **1418**, **1420** can exhibit any suitable cross-sectional shape necessary to properly focus the radiation **1412** on the sensor **1316** and the sharp **1318** for sterilization. In the illustrated embodiment, for example, the external and internal collimators **1418**, **1420** are each conical or frustoconical in shape. In other embodiments, however, one or both of the external and internal collimators **1418**, **1420** may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure. In yet other embodiments, one or both of the external and

internal collimators **1418**, **1420** may exhibit a circular cross-sectional shape with parallel sides.

In the illustrated embodiment, the sterilization zone **1424** provides a first aperture **1426a** defined by the external collimator **1418** and a second aperture **1426b** defined by the internal collimator **1420**, where the first and second apertures **1426a,b** are located at opposing ends of the sterilization zone **1424**. The first aperture **1426a** permits the radiation **1412** to enter the sterilization zone **1424**, and the second aperture **1426b** provides a location where radiation **1412** can impact the sensor **1316** and the sharp **1318**. In the illustrated embodiment, the second aperture **1426b** also provides a location where the sensor **1316** and the sharp **1318** may be received into the sterilization zone **1424**.

In embodiments where the sterilization zone **1424** is conical or frustoconical in shape, the diameter of the first aperture **1426a** may be larger than the diameter of the second aperture **1426b**. In such embodiments, for example, the size of the first aperture **1426a** may range between about 5.0 mm and about 16.0 mm, and the size of the second aperture **1426b** may range between about 0.5 mm and about 3.0 mm. The respective diameters of the first and second apertures **1426a,b**, however, may be greater or less than the ranges provided herein, without departing from the scope of the disclosure, and depending on the application. Indeed, the diameters of the first and second apertures **1426a,b** need only be large enough to allow a sufficient dose of radiation to impinge upon the sensor **1316** and the sharp **1318**.

In the illustrated embodiment, the inner wall(s) of the sterilization zone **1424** (e.g., the external and internal collimators **1418**, **1420**) extend between the first and second apertures **1426a,b** at a substantially constant angle relative to the centerline of the sensor applicator **102**. The angle of the wall(s) may be any angle between 0° and 90° relative to the centerline of the sensor applicator **102**. The angle of the wall(s), however, may preferably be between 45° and 90° relative to the centerline of the sensor applicator **102**. In other embodiments, however, the angle of the wall(s) may vary between the first and second apertures **1426a,b**, without departing from the scope of the disclosure. In such embodiments, portions of the wall(s) may extend short distances at an angle dissimilar to adjacent portions, or the wall(s) may otherwise undulate between the first and second apertures **1426a,b**.

In some embodiments, the sterilization zone **1424** defined by the external and internal collimators **1418** may be substantially cylindrical and otherwise exhibit a circular or polygonal cross-section. In such embodiments, the first and second apertures **1426a,b** may exhibit identical diameters and the walls of the sterilization zone **1424** may be substantially parallel between the first and second ends of the sterilization zone **1424**.

In some embodiments, a cap seal **1428** (shown in dashed lines) may be arranged at the interface between the cap fill **1410** and the radiation shield **1416**. The cap seal **1428** may comprise a radiation permeable microbial barrier. In some embodiments, for example, the cap seal **1428** may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as TYVEK® available from DuPont®. The cap seal **1428** may seal off a portion of the sterilization zone **1424** to help form part of a sealed region **1430** configured to isolate the sensor **1316** and the sharp **1318** from external contamination.

The sealed region **1430** may include (encompass) select portions of the interior of the electronics housing **1304** and the sterilization zone **1424**. In one or more embodiments, the sealed region **1430** may be defined and otherwise formed by

at least the cap seal **1428**, a first or “top” seal **1432a**, and a second or “bottom” seal **1432b**. The cap seal **1428** and the top and bottom seals **1432a,b** may each create corresponding barriers at their respective sealing locations, thereby allowing the sterilization zone **1424** containing the sensor **1316** and the sharp **1318** to be terminally sterilized.

The top seal **1432a** may be arranged to seal the interface between the sharp hub **1320** and the top of the electronics housing **1304** (i.e., the shell **1306** of FIG. 13) and thereby prevent contaminants from migrating into the interior of the electronics housing **1304**. In some embodiments, the top seal **1432a** may form part of the sharp hub **1320**, such as being overmolded onto the sharp hub **1320**. In other embodiments, however, the top seal **1432a** may form part of or be overmolded onto the top surface of the shell **1306**. In yet other embodiments, the top seal **1432a** may comprise a separate structure, such as an O-ring or the like, that interposes the sharp hub **1320** and the top surface of the shell **1306**, without departing from the scope of the disclosure.

The bottom seal **1432b** may be arranged to seal the interface between the cap fill **1410** and the bottom of electronics housing **1304** (i.e., the mount **1308** of FIG. 13). The bottom seal **1432b** may prevent contaminants from migrating into the sterilization zone **1424** and from migrating into the interior of the electronics housing **1304**. In some embodiments, the bottom seal **1432b** may form part of the cap fill **1410**, such as being overmolded onto the top of the cap fill **1410**. In other embodiments, the bottom seal **1432b** may form part of or be overmolded onto the bottom of the mount **1308**. In yet other embodiments, the bottom seal **1432b** may comprise a separate structure, such as an O-ring or the like, that interposes the cap fill **1410** and the bottom of the mount **1308**, without departing from the scope of the disclosure.

Upon loading the sensor control device **1302** into the sensor applicator **102** and securing the applicator cap **1404** to the sensor applicator **102**, the top and bottom seals **1432a,b** may compress and generate corresponding sealed interfaces. The top and bottom seals **1432a,b** may be made of a variety of materials capable of generating a sealed interface between opposing structures. Suitable materials include, but are not limited to, silicone, a thermoplastic elastomer (TPE), polytetrafluoroethylene (e.g., TEFLON®), or any combination thereof.

It is noted that, while the sensor **1316** and the sharp **1318** extend from the bottom of the electronics housing **1304** and into the sterilization zone **1424** generally concentric with a centerline of the sensor applicator **102** and the applicator cap **1404**, it is contemplated herein to have an eccentric arrangement. More specifically, in at least one embodiment, the sensor **1316** and the sharp **1318** may extend from the bottom of the electronics housing **1304** eccentric to the centerline of the sensor applicator **102** and the applicator cap **1404**. In such embodiments, the external and internal collimators **1418**, **1420** may be re-designed and otherwise configured such that the sterilization zone **1424** is also eccentrically positioned to receive the sensor **1316** and the sharp **1318**, without departing from the scope of the disclosure.

In some embodiments, the external sterilization assembly **1414** may further include a sterilization housing or “pod” **1434** coupled to or forming part of the radiation shield **1416**. The sterilization pod **1434** provides and otherwise defines a chamber **1436** sized to receive all or a portion of the sensor applicator **102**. Once properly seated (received) within the sterilization pod **1434**, the sensor applicator **102** may be subjected to the radiation sterilization **1412** to sterilize the sensor **1316** and the sharp **1318**. The sterilization pod **1434**

may be made of any of the materials mentioned herein for the radiation shield **1416** to help prevent the radiation **1412** from propagating through the walls of the sterilization pod **1434**.

In some embodiments, the radiation shield **1416** may be removably coupled to the sterilization pod **1434** using one or more mechanical fasteners **1438** (one shown), but could alternatively be removably coupled via an interference fit, a snap fit engagement, etc. Removably coupling the radiation shield **1416** to the sterilization pod **1434** enables the radiation shield **1416** to be interchangeable with differently designed (sized) shields to fit particular sterilization applications for varying types and designs of the sensor applicator **102**. Accordingly, the sterilization pod **1434** may comprise a universal mount that allows the radiation shield **1416** to be interchanged with other shield designs having different parameters for the external collimator **1418**, as needed.

In some embodiments, the external sterilization assembly **1414** may further include a mounting tray **1440** coupled to or forming part of the sterilization pod **1434**. The sterilization pod **1434** may be removably coupled to the mounting tray **1440** using, for example, one or more mechanical fasteners **1442** (one shown). The mounting tray **1440** may provide or define a central aperture **1444** sized to receive the sensor applicator **102** and alignable with the chamber **1436** to enable the sensor applicator **102** to enter the chamber **1436**. As described below, in some embodiments, the mounting tray **1440** may define a plurality of central apertures **1444** for receiving a corresponding plurality of sensor applicators for sterilization.

FIG. 15 is a cross-sectional side view of the sensor applicator **102** and another example embodiment of the external sterilization assembly **1414**, according to one or more additional embodiments. As illustrated, the sensor control device **1302** is again received within the sensor applicator **102** and the applicator cap **1404** is coupled to the housing **1402** to secure the sensor control device **1302** therein.

In the illustrated embodiment, the applicator cap **1404** may be inverted and may define or otherwise provide a cap post **1502** sized to receive the distal ends of the sensor **1316** and the sharp **1318** extending from the bottom of the electronics housing **1304**. The cap post **1502** helps provide a portion of the sealed region **1430** configured to isolate the sensor **1316** and the sharp **1318** from external contamination. In the illustrated embodiment, the sealed region **1430** may be defined and otherwise formed by the cap post **1502** and the top and bottom seals **1432a,b**, which create corresponding barriers at their respective sealing locations. The top seal **1432a** may again be arranged to seal the interface between the sharp hub **1320** and the top of the electronics housing **1304** (i.e., the shell **1306** of FIG. 13), and the bottom seal **1432b** may be arranged to seal an interface between the applicator cap **1404** and the bottom of electronics housing **1304** (i.e., the mount **1308** of FIG. 13). In some embodiments, the bottom seal **1432b** may interpose the cap post **1502** and the bottom of electronics housing **1304**.

In the illustrated embodiment, the radiation shield **1416** may be positioned external to the sensor applicator **102** and may extend into the inverted portion of the applicator cap **1404**. The external collimator **1418** provided by the radiation shield **1416** defines a sterilization zone **1504** configured to focus the radiation **1412** toward the sensor **1316** and the sharp **1318**. In the illustrated embodiment, the cap post **1502** and portions of the sensor **1316** and the sharp **1318** positioned within the cap post **1502** extend into the sterilization

zone 1504. Propagating radiation 1412 may traverse the sterilization zone 1504 to sterilize the sensor 1316 and the sharp 1318 positioned within the cap post 1502. As indicated above, however, the radiation shield 1416 may be made of a material that substantially prevents the radiation 1412 from penetrating the wall(s) of the sterilization zone 1504 and thereby damaging the radiation sensitive component 1408 within the housing 1304.

In the illustrated embodiment, the external collimator 1418 defines a first aperture 1506a at a first end of the sterilization zone 1504 and a second aperture 1506b at the second end of the sterilization zone 1504. The first aperture 1506a permits the radiation 1412 to enter the sterilization zone 1504, and the second aperture 1506b provides a location where radiation 1412 is focused toward the sensor 1316 and the sharp 1318. The second aperture 1506b may also provide a location where the sensor 1316 and the sharp 1318 positioned within the cap post 1502 may be received into the sterilization zone 1504.

As illustrated, the external collimator 1418 and associated sterilization zone 1504 are conical or frustoconical in shape, and the diameter of the first aperture 1506a is larger than the diameter of the second aperture 1506b. The size of the first aperture 1506a may range between about 5.0 mm and about 16.0 mm, and the size of the second aperture 1506b may range between about 0.5 mm and about 3.0 mm, but could alternatively be greater or less than the provided ranges, without departing from the scope of the disclosure. Indeed, the sizes of the apertures 1506a,b may vary depending on the scale of the device. In other embodiments, however, the external collimator 1418 and associated sterilization zone 1504 may be substantially cylindrical and otherwise exhibit a circular or polygonal cross-section where the first and second apertures 1506a,b exhibit substantially identical diameters and the walls of the sterilization zone 1504 are substantially parallel.

FIG. 16 is a cross-sectional side view of the sensor applicator 102 and another example embodiment of the external sterilization assembly 1414, according to one or more additional embodiments. As illustrated, the sensor control device 1302 is again received within the sensor applicator 102 and the applicator cap 1404 is coupled to the housing 1402 to secure the sensor control device 1302 therein.

In the illustrated embodiment, the applicator cap 1404 may again be inverted and may define or otherwise provide a cap post 1602 sized to receive the distal ends of the sensor 1316 and the sharp 1318 extending from the bottom of the electronics housing 1304. Moreover, the radiation shield 1416 may be positioned external to the sensor applicator 102 and may extend into the inverted portion of the applicator cap 1404. More specifically, the radiation shield 1416 may extend into the inverted portion of the applicator cap 1404 and to the bottom of the cap post 1602. Unlike the cap post 1502 of FIG. 15, however, the bottom of the cap post 1602 may be open ended. In some embodiments, a cap seal 1604 may be arranged at the interface between the cap post 1602 and the radiation shield 1416 to seal off the open end of the cap post 1602. The cap seal 1604 may be similar to the cap seal 1428 of FIG. 14B, and therefore will not be described again.

In some embodiments, a cap fill 1606 may be positioned within the applicator cap 1404. In one or more embodiments, the cap fill 1606 may comprise an integral part or extension of the applicator cap 1404, such as being molded with or overmolded onto the applicator cap 1404. In other embodiments, the cap fill 1606 may comprise a separate

structure fitted within or otherwise attached to the applicator cap 1404, without departing from the scope of the disclosure. The cap fill 1606 may also provide or otherwise define an internal collimator 1608 that may help focus the radiation 1412 toward the components to be sterilized. In at least one embodiment, as illustrated, the cap post 1602 may be received within the internal collimator 1608.

The external and internal collimators 1418, 1608 may cooperatively define a sterilization zone 1610 that focuses the radiation 1412 toward the sensor 1316 and the sharp 1318. The propagating radiation 1412 may traverse the sterilization zone 1610 to impinge upon and sterilize the sensor 1316 and the sharp 1318. However, the cap fill 1606 and the radiation shield 1416 may each be made of any of the materials mentioned herein that substantially prevent the radiation 1412 from penetrating the inner wall(s) of the sterilization zone 1610 and thereby damaging the radiation sensitive component 1408 within the housing 1304. In at least one embodiment, the cap fill 1606 may be made of machined or 3D printed polypropylene and the radiation shield 1416 may be made of stainless steel.

The external and internal collimators 1418, 1608 can exhibit any suitable cross-sectional shape necessary to properly focus the radiation 1412 toward the sensor 1316 and the sharp 1318 for sterilization. In the illustrated embodiment, for example, the external collimator 1418 is conical or frustoconical in shape, and the internal collimator 1608 is substantially cylindrical with internal walls that are substantially parallel. In other embodiments, however, the external and internal collimators 1418, 1608 may exhibit other cross-sectional shapes, without departing from the scope of the disclosure.

In the illustrated embodiment, the external collimator 1418 defines a first aperture 1612a that permits the radiation 1412 to enter the sterilization zone 1610 and a second aperture 1612b positioned at or near the bottom opening to the cap post 1602 to focus the radiation 1412 at the sensor 1316 and the sharp 1318 positioned within the cap post 1602. The diameter of the first aperture 1612a is larger than the diameter of the second aperture 1612b and, as with prior embodiments, the size of the first aperture 1612a may range between about 5.0 mm and about 16.0 mm, and the size of the second aperture 1612b may range between about 0.5 mm and about 3.0 mm. In the illustrated embodiment, the external collimator 1418 funnels the electrons of the radiation 1412 toward the bottom opening to the cap post 1602 and amplifies the electrons at the sensor 1316 and the sharp 1318.

The cap seal 1604 may be arranged at the interface between the radiation shield 1416 and the cap post 1602 and/or the cap fill 1606. The cap seal 1604 may seal off a portion of the sterilization zone 1610 to help form part of the sealed region 1430 configured to isolate the sensor 1316 and the sharp 1318 from external contamination. The sealed region 1430 may include (encompass) select portions of the interior of the electronics housing 1304 and the sterilization zone 1610. In the illustrated embodiment, the sealed region 1430 may be defined and otherwise formed by the cap post 1602 and the top and bottom seals 1432a,b, which create corresponding barriers at their respective sealing locations. The bottom seal 1432b may be arranged to seal an interface between the applicator cap 1404 and the bottom of electronics housing 1304 (i.e., the mount 1308 of FIG. 13).

FIGS. 17A and 17B are partially exploded isometric top and bottom views, respectively, of one example of the external sterilization assembly 1414, according to one or more embodiments. In at least one embodiment, the assem-

bly **1414** may be designed and otherwise configured to accommodate and help sterilize a plurality of sensor applicators **102** (i.e., with the sensor control devices positioned therein). In the illustrated embodiment, the mounting tray **1440** defines a plurality of central apertures **1444** (FIG. **17A**), and a plurality of sterilization pods **1434** may be aligned with the central apertures **1444** and coupled to the mounting tray **1440**. The sensor applicators **102** may be received within the sterilization pods **1434** via the central apertures **1444**, and each sterilization pod **1434** may have a corresponding shield **1416** (FIG. **17B**) coupled thereto or otherwise forming part thereof.

In some embodiments, the assembly **1414** may further include a cover **1702** matable with the mounting tray **1440**. The cover **1702** may include or define a plurality of apertures **1106** (FIG. **17B**) sized to receive the tops of the sensor applicators **102** when the cover **1702** is placed on top of the mounting tray **1440**. In some embodiments, the cover **1702** may be made of any of the materials mentioned herein for the radiation shield **1416** to help prevent the radiation sterilization from propagating through the walls of the assembly **1414**. With the cover **1702** mated with the mounting tray **1414**, the sensor applicators **102** may be encapsulated or otherwise encased within the assembly **1414**.

Embodiments disclosed herein include:

D. An external sterilization assembly that includes a radiation shield positionable external to a medical device having a part requiring sterilization and a radiation sensitive component, and a collimator defined by the radiation shield and alignable with the part requiring sterilization, wherein the collimator focuses radiation from a radiation sterilization process toward the part requiring sterilization and the radiation shield prevents the radiation from damaging the radiation sensitive component.

E. An external sterilization assembly that includes a radiation shield positionable external to a sensor applicator that includes a housing, a cap coupled to the housing, and a sensor control device positioned within the housing, wherein the sensor control device includes an electronics housing, a radiation sensitive component arranged within the electronics housing, and a sensor and a sharp extending from the electronics housing. The external sterilization assembly further including an external collimator defined by the radiation shield and alignable with the sensor and the sharp, wherein the external collimator focuses radiation from a radiation sterilization process toward the sensor and the sharp and the radiation shield prevents the radiation from damaging the radiation sensitive component.

F. A method including arranging a radiation shield external to a sensor applicator having a housing, a cap coupled to the housing, and a sensor control device positioned within the housing, wherein the sensor control device includes an electronics housing, a radiation sensitive component arranged within the electronics housing, and a sensor and a sharp extending from the electronics housing. The method further including focusing radiation from a radiation sterilization process toward the sensor and the sharp with an external collimator defined by the radiation shield, and preventing the radiation from damaging the radiation sensitive component with the radiation shield.

Each of embodiments D, E, and F may have one or more of the following additional elements in any combination: Element 1: wherein the radiation shield is made of a material selected from the group consisting of a high-density polymer, a metal, and any combination thereof. Element 2: wherein the radiation sensitive component is selected from the group consisting of an electronic module, a chemical

solution, and any combination thereof. Element 3: wherein the collimator comprises a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof. Element 4: further comprising a cap that encapsulates the part requiring sterilization and provides a sealed barrier. Element 5: wherein the radiation shield defines an internal cavity that receives the medical device, and the collimator focuses the radiation into the internal cavity.

Element 6: wherein the radiation shield is made of a material selected from the group consisting of a high-density polymer, a metal, and any combination thereof. Element 7: wherein the external collimator comprises a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof. Element 8: further comprising a sterilization pod defining a chamber that receives at least a portion of the sensor applicator, wherein the radiation shield is removably coupled to the sterilization pod. Element 9: further comprising a mounting tray that defines a central aperture alignable with the chamber and sized to receive the sensor applicator, and a cover matable with the mounting tray to encase the sensor applicator. Element 10: wherein the external collimator is alignable with an internal collimator defined by a cap fill positioned within the cap, and wherein the external and internal collimators cooperatively define a sterilization zone into which the sensor and the sharp are received. Element 11: wherein the external and internal collimators each comprise a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof. Element 12: further comprising a cap seal arranged at an interface between the external and internal collimators. Element 13: wherein the cap is inverted and provides a cap post that receives the sensor and the sharp. Element 14: wherein the external collimator and the cap post cooperatively define a sterilization zone and the sensor and the sharp positioned within the cap post extend into the sterilization zone.

Element 15: wherein arranging the radiation shield external to the sensor applicator comprises positioning the sensor applicator within a chamber defined by a sterilization pod, the radiation shield being removably coupled to the sterilization pod. Element 16: wherein positioning the sensor applicator within the chamber defined by the sterilization pod further comprise extending the sensor applicator through a central aperture defined by a mounting tray and aligned with the chamber, positioning a cover on the mounting tray and thereby encasing the sensor applicator, and undertaking the radiation sterilization process while the sensor applicator is encased by the cover. Element 17: wherein the external collimator comprises a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof.

By way of non-limiting example, exemplary combinations applicable to D, E, and F include: Element 8 with Element 9; Element 10 with Element 11; Element 10 with Element 12; Element 13 with Element 14; and Element 15 with Element 16.

Hybrid Sterilization Assemblies

Referring again briefly, to FIG. **1**, prior to being delivered to an end user, the sensor control device **104** must be sterilized to render the product free from viable microorganisms. The sensor **110** is commonly sterilized using radiation sterilization, such as electron beam ("e-beam") irradiation. Radiation sterilization, however, can damage the

electronic components within the sensor control device **104**, which are commonly sterilized via gaseous chemical sterilization (e.g., using ethylene oxide). Gaseous chemical sterilization, however, can damage the enzymes or other chemistry and biologics included on the sensor **110**.

In the past, this sterilization incompatibility has been circumvented by separating the sensor **110** and the electronic components and sterilizing each individually. This approach, however, requires additional parts, packaging, process steps, and final assembly by the user, which introduces a possibility of user error. According to the present disclosure, the sensor control device **104**, or any device requiring terminal sterilization, may be properly sterilized using external sterilization assemblies designed to focus sterilizing radiation (e.g., beams, waves, energy, etc.) toward component parts

requiring sterilization, while simultaneously preventing the propagating radiation from disrupting or damaging sensitive electronic components.

FIG. **18** is an isometric view of an example sensor control device **1802**, according to one or more embodiments of the present disclosure. The sensor control device **1802** may be the same as or similar to the sensor control device **104** of FIG. **1** and, therefore, may be used in conjunction with the sensor applicator **102** (FIG. **1**), which delivers the sensor control device **1802** to a target monitoring location on a user's skin. Accordingly, the sensor control device **1802** also requires proper sterilization prior to being used.

As illustrated, the sensor control device **1802** includes an electronics housing **1804** that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing **1804** may exhibit other cross-sectional shapes, such as ovoid (e.g., pill- or egg-shaped), a squircle, polygonal, or any combination thereof, without departing from the scope of the disclosure. The electronics housing **1804** may be configured to house or otherwise contain various electronic components used to operate the sensor control device **1802**.

The electronics housing **1804** may include a shell **1806** and a mount **1808** that is matable with the shell **1806**. The shell **1806** may be secured to the mount **1808** via a variety of ways, such as a snap fit engagement, an interference fit, sonic or laser welding, one or more mechanical fasteners (e.g., screws), or any combination thereof. In some cases, the shell **1806** may be secured to the mount **1808** such that a sealed interface is generated therebetween. In such embodiments, a gasket or other type of seal material may be positioned at or near the outer diameter (periphery) of the shell **1806** and the mount **1808**, and securing the two components together may compress the gasket and thereby generate a sealed interface. In other embodiments, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell **1806** and the mount **1808**. The adhesive secures the shell **1806** to the mount **1808** and provides structural integrity, but may also seal the interface between the two components and thereby isolate the interior of the electronics housing **1804** from outside contamination.

In the illustrated embodiment, the sensor control device **1802** may optionally include a plug assembly **1810** that may be coupled to the electronics housing **1804**. The plug assembly **1810** may include a sensor module **1812** (partially visible) interconnectable with a sharp module **1814** (partially visible). The sensor module **1812** may be configured to carry and otherwise include a sensor **1816** (partially visible), and the sharp module **1814** may be configured to carry and otherwise include an introducer or sharp **1818** (partially visible) used to help deliver the sensor **1816** transcutaneously under a user's skin during application of the sensor

control device **1802**. In the illustrated embodiment, the sharp module **1814** includes a sharp hub **1820** that carries the sharp **1818**.

As illustrated, corresponding portions of the sensor **1816** and the sharp **1818** extend distally from the electronics housing **1804** and, more particularly, from the bottom of the mount **1808**. In at least one embodiment, the exposed portion of the sensor **1816** (alternately referred to as the "tail") may be received within a hollow or recessed portion of the sharp **1818**. The remaining portions of the sensor **1816** are positioned within the interior of the electronics housing **1804**.

FIG. **19A** is a side view of the sensor applicator **102** of FIG. **1**. As illustrated, the sensor applicator **102** includes a housing **1902** and an applicator cap **1904** that may be removably coupled to the housing **1902**. In some embodiments, the applicator cap **1904** may be threaded to the housing **1902** and include a tamper ring **1906**. Upon rotating (e.g., unscrewing) the applicator cap **1904** relative to the housing **1902**, the tamper ring **1906** may shear and thereby free the applicator cap **1904** from the sensor applicator **102**. Once the applicator cap **1904** is removed, a user may then use the sensor applicator **102** to position the sensor control device **1802** (FIG. **18**) at a target monitoring location on the user's body.

FIG. **19B** is a partial cross-sectional side view of the sensor applicator **102**. As illustrated, the sensor control device **1802** may be received within the sensor applicator **102** and the applicator cap **1904** may be coupled to the housing **1902** to secure the sensor control device **1802** within. The sensor control device **1802** may include one or more radiation sensitive components **1908** arranged within the electronics housing **1804**. The radiation sensitive component **1908** can include an electronic component or module such as, but not limited to, a data processing unit, a resistor, a transistor, a capacitor, an inductor, a diode, a switch, or any combination thereof. The data processing unit may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **1802**. In operation, the data processing unit may perform data processing functions, such as filtering and encoding of data signals corresponding to a sampled analyte level of the user. The data processing unit may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. **1**).

In the illustrated embodiment, an applicator insert **1910** may be positioned within the applicator cap **1904** and may generally help support the sensor control device **1802** within the sensor applicator **102**. In one embodiment, the applicator insert **1910** may comprise an integral part or extension of the applicator cap **1904**, such as being molded with or overmolded onto the applicator cap **1904**. In other embodiments, the applicator insert **1910** may comprise a separate structure fitted within or otherwise attached to the applicator cap **1904**, without departing from the scope of the disclosure. In such embodiments, for example, screwing the applicator cap **1904** onto the housing **1908** may progressively advance an inner surface **1912** of the applicator insert **1910** into axial and/or radial engagement with a bottom edge, surface or portion of the applicator insert **1910** to thereby axially secure the applicator insert **1910** within the applicator cap **1904**.

The sensor applicator **102** may further include a sheath **1914** and, in some embodiments, the applicator insert **1910** may engage the sheath **1914** to rotationally fix the applicator insert **1910** within the applicator cap **1904**. More specifi-

cally, the applicator insert **1910** may provide or otherwise define one or more radial alignment features **1916** (one shown) mateable with a corresponding groove or slot **1918** defined in the sheath **1914**. The radial alignment feature **1916** may comprise, for example, a rail, a flag, a tab, a protrusion, or the like extending from the main body of the applicator insert **1910** and may mate with the slot **1918** by sliding the radial alignment feature **1916** longitudinally into the slot **1918**, for example. Mating engagement between the radial alignment feature **1916** and the slot **1918** may also help angularly (rotationally) orient the applicator insert **1910** relative to the sensor control device **1802**. As will be appreciated, however, the mateable structures may alternatively be reversed, where the radial alignment feature **1916** is instead provided on the sheath **1914** and the slot **1918** is provided on the applicator insert **1910**.

The applicator insert **1910** may provide and otherwise define an internal collimator **1920a**, which forms part of a hybrid sterilization assembly described in more detail below. The internal collimator **1920a** may help define a portion of a sterilization zone **1922** and, more particularly, an upper portion **1924** of the sterilization zone **1922**. When the sensor control device **1802** is installed in the sensor applicator **102**, the distal ends of the sensor **1816** and the sharp **1818** may extend from the bottom of the electronics housing **1804** and reside within the upper portion **1924**.

In some embodiments, a microbial barrier **1926a** may be positioned at an opening to the upper portion **1924** of the sterilization zone **1922**. The microbial barrier **1926a** may help seal at least some of the upper portion **1924** of the sterilization zone **1922** to thereby isolate the distal ends of the sensor **1816** and the sharp **1818** from external contamination. The microbial barrier **1926a** may be made of a radiation permeable material, such as a synthetic material (e.g., a flash-spun high-density polyethylene fiber). One example synthetic material comprises TYVEK®, available from DuPont®. In other embodiments, however, the microbial barrier **1926a** may comprise, but is not limited to, tape, paper, film, foil, or any combination thereof. In at least one embodiment, the microbial barrier **1926a** may comprise or otherwise be formed by a thinned portion of the applicator insert **1910**, without departing from the scope of the disclosure.

In some embodiments, a moisture barrier **1926b** may be positioned or otherwise arranged at an opening **1928** to the applicator cap **1904**. Similar to the microbial barrier **1926a**, the moisture barrier **1926b** may be configured to help isolate portions of the sensor applicator **102** from external contamination. The moisture barrier **1926b** may be made of any of the materials mentioned above with reference to the microbial barrier **1926a**. In at least one embodiment, however, the moisture barrier **1926b** may comprise a thinned portion of the applicator cap **1904**, without departing from the scope of the disclosure. In such embodiments, the opening **1928** would not be necessary.

FIGS. 20A-20C are various views of the applicator insert **1910**, according to one or more embodiments of the disclosure. More specifically, FIG. 20A is an isometric top view, FIG. 20B is an isometric bottom view, and FIG. 20C is an isometric cross-sectional view of the applicator insert **1910**. As illustrated, the applicator insert **1910** includes a generally cylindrical body **2002** having a first or top end **2004a** and a second or bottom end **2004b** opposite the top end **2004a**. The top end **2004a** is generally closed except for an aperture **2005** sized to receive the sensor **1816** (FIG. 19B) and the sharp **1818** (FIG. 19B) therethrough, and the bottom end **2004b** is generally open.

The radial alignment feature **1916** described above is provided on a sidewall of the body **2002**. In some embodiments, additional radial alignment features **2006** (three shown) may be provided or otherwise defined on the sidewall of the body **2002**. In the illustrated embodiment, the additional radial alignment features **2006** each comprise a pair of longitudinally-extending tabs or projections **2008** angularly offset from each other on the sidewall to cooperatively define a slot **2010** therebetween. The slot **2010** may be sized to receive a projection or tab provided on the sheath **1914** (FIG. 19B) to help angularly (rotationally) orient the applicator insert **1910** relative to the sensor control device **1802** (FIG. 19B). Moreover, similar to the arrangement of the radial alignment feature **1916**, the mateable structures of the additional radial alignment features **2006** may alternatively be reversed, where the additional radial alignment features **2006** are instead provided on the sheath **1914** and the corresponding projection or tab is provided on the applicator insert **1910**.

As best seen in FIGS. 20A and 20C, the applicator insert **1910** may further include one or more sensor locating features **2012** that may be used to also help properly orient the applicator insert **1910** relative to the sensor control device **1802** (FIG. 19B) within the sensor applicator **102** (FIG. 19B). As illustrated, the sensor locating features **2012** may be defined on and extend axially from the top end **2004a** of the body **2002**. The sensor locating features **2012** may be sized to be received within corresponding apertures defined in the bottom of the sensor control device **1802**. In the illustrated embodiment, the sensor locating features **2012** comprise cylindrical projections, but could alternatively comprise other types of structural features suitable for mating with the corresponding features on the bottom of the sensor control device **1802**. The sensor locating features **2012**, in conjunction with the radial alignment feature **1916** and the additional radial alignment features **2006**, may prove especially advantageous in embodiments where the sensor control device **1802** comprises an eccentric orientation, where the sensor **1816** and the sharp **1818** are not concentric with the centerline of the sensor control device.

The internal collimator **1920a** may be formed or otherwise provided at the top end **2004a** of the applicator insert **1910**. As best seen in FIG. 20C, the internal collimator **1920a** may be defined by the applicator insert **1910** and may include a collimating insert **2014** and a gasket **2016**. The internal collimator **1920a** may be fabricated by first fabricating or otherwise producing the collimating insert **2014**. The applicator insert **1910** may then be overmolded onto the collimating insert **2014**. Also, the collimating insert **2014** could be insert molded into the applicator insert **1910**. Accordingly, the applicator insert **1910** may be made of a hard plastic. The gasket **2016** may then be molded onto the applicator insert **1910** in a second shot molding (overmolding) process.

The collimating insert **2014** may be made of a material that reduces or prevents sterilizing radiation from penetrating therethrough. Suitable materials for the collimating insert **2014** include, but are not limited to, a high-density polymer, (e.g., polyethylene, polypropylene, polystyrene, polytetrafluoroethylene, polyamide, etc.), a metal (e.g., lead, tungsten, stainless steel, aluminum, etc.), a composite material, or any combination thereof. In some embodiments, the collimating insert **2014** may be made of any material that has a mass density greater than 0.9 grams per cubic centimeter (g/cc).

The gasket **2016** may be made of any material that helps form a sealed interface with the bottom of the electronics

housing **1804** (FIG. **19B**) when the applicator insert **1910** is installed in the sensor applicator **102** (FIG. **19B**). Suitable materials for the gasket **2016** include, but are not limited to, silicone, a thermoplastic elastomer (TPE), polytetrafluoroethylene (e.g., TEFLON®), or any combination thereof. As illustrated, the gasket **2016** may fill a void **2018** defined by the applicator insert **1910** and may provide an annular projection **2020** that protrudes past and/or from the upper surface of the top end **2004a** of the body **2002**. The annular projection **2020** may prove advantageous in not only facilitating a sealed interface, but also in helping to take up tolerances as the applicator insert **1910** is installed in the sensor applicator **102**. Moreover, the mass of the gasket **2016** may also help absorb radiation during the sterilization processes described below, thus providing another layer of protection against radiation propagation. In at least one embodiment, the gasket **2016** may be large enough or of a material that absorbs sufficient radiation that the collimating insert **2014** may be omitted from the internal collimator **1920a**.

FIG. **21** is another cross-sectional side view of the sensor applicator **102** of FIG. **19A** showing a hybrid sterilization assembly **2102**, according to one or more embodiments of the disclosure. The hybrid sterilization assembly **2102**, alternately referred to as a “split collimation assembly” or “cooperative collimation assembly,” may be used to help sterilize the sensor control device **1802** and, more particularly, the distal ends of the sensor **1816** and the sharp **1818** extending from the bottom of the electronics housing **1804** while positioned within the sensor applicator **102**. More specifically, the fully assembled sensor control device **1802** may be subjected to radiation sterilization **2104** to sterilize the exposed portions of the sensor **1816** and the sharp **1818**. Suitable radiation sterilization **2104** processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof.

The radiation sterilization **2104** may be delivered either through continuous processing irradiation or through pulsed beam irradiation. In pulsed beam irradiation, the beam of radiation sterilization **2104** is focused at a target location and the component part or device to be sterilized is moved to the target location at which point the irradiation is activated to provide a directed pulse of radiation. The radiation sterilization **2104** is then turned off, and another component part or device to be sterilized is moved to the target location and the process is repeated.

According to the present disclosure, the hybrid sterilization assembly **2102** may be used to help focus the radiation **2104** in sterilizing the distal ends of the sensor **1816** and the sharp **1818**, while simultaneously preventing (impeding) propagating radiation **2104** from damaging the radiation sensitive component **1908**. As illustrated, the hybrid sterilization assembly **2102** (hereafter the “assembly **2102**”) may include the internal collimator **1920a** previously described above and an external collimator **1920b**. As illustrated, the internal collimator **1920a** may be arranged within the sensor applicator **102**, and the external collimator **1920b** may extend into the sensor applicator **102** (i.e., the applicator cap **1904**) by penetrating the opening **1928** to the applicator cap **1904**. The internal and external collimators **1920a,b** may cooperatively define the sterilization zone **1922** that focuses the radiation **2104** (e.g., beams, waves, energy, etc.) to impinge upon and sterilize the sensor **1816** and the sharp **1818**.

In the illustrated embodiment, the external collimator **1920b** is designed to align with the internal collimator **1920a**

and, more particularly, with the collimating insert **2014**. In at least one embodiment, for example, the collimating insert **2014**, may define a radial shoulder **2106** sized to receive and otherwise mate with an end of the external collimator **1920b** extended into the applicator cap **1904**. The external collimator **1920b** may transition to the internal collimator **1920a** at the radial shoulder **2106**. In some embodiments, the transition between the internal and external collimators **1920a,b** may be continuous, flush, or smooth. In other embodiments, however, the transition may be discontinuous or stepped, without departing from the scope of the disclosure.

Similar to the collimating insert **2014** of the internal collimator **1920a**, the external collimator **1920b** may be made of a material that substantially prevents the radiation **2104** from penetrating the inner wall(s) of the sterilization zone **1922** and thereby damaging the radiation sensitive component **1908** within the electronics housing **1804**. Accordingly, the external collimator **1920b** may be made of any of the materials mentioned herein as being suitable for the collimating insert **2014**. In at least one embodiment, the collimating insert **2014** and the external collimator **1920b** may each be made of stainless steel. Moreover, however, as mentioned above the gasket **2016** may also provide a degree of shielding or protection against the radiation from damaging the radiation sensitive component **1908**.

The sterilization zone **1922** defined by the internal and external collimators **1920a,b** can exhibit any suitable cross-sectional shape necessary to properly focus the radiation **2104** on the sensor **1816** and the sharp **1818** for sterilization. In the illustrated embodiment, for example, the internal and external collimators **1920a,b** are each conical or frustoconical in shape. In other embodiments, however, one or both of the internal and external collimators **1920a,b** may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure. In yet other embodiments, one or both of the internal and external collimators **1920a,b** may exhibit a circular cross-sectional shape with parallel sides.

In the illustrated embodiment, the sterilization zone **1922** provides a first aperture **2108a** defined by the external collimator **1920b** and a second aperture **2108b** defined by the internal collimator **1920a**, where the first and second apertures **2108a,b** are located at opposing ends of the sterilization zone **1922**. The first aperture **2108a** permits the radiation **2104** to enter the sterilization zone **1922**, and the second aperture **2108b** provides a location where the sensor **1816** and the sharp **1818** may be received into the sterilization zone **1922**.

In embodiments where the sterilization zone **1922** is conical or frustoconical in shape, the diameter of the first aperture **2108a** may be larger than the diameter of the second aperture **2108b**. In such embodiments, for example, the size of the first aperture **2108a** may range between about 5.0 mm and about 16.0 mm, and the size of the second aperture **2108b** may range between about 0.5 mm and about 5.0 mm. The respective diameters of the first and second apertures **2108a,b**, however, may be greater or less than the ranges provided herein, without departing from the scope of the disclosure, and depending on the application. Indeed, the diameters of the first and second apertures **2108a,b** need only be large enough to allow a sufficient dose of radiation to impinge upon the sensor **1816** and the sharp **1818**.

In embodiments where the sterilization zone **1922** is substantially cylindrical and otherwise exhibit a circular or polygonal cross-section, the first and second apertures

2108a,b may exhibit identical diameters. In such embodiments, the walls of the sterilization zone **1922** may or may not be substantially parallel between the first and second ends of the sterilization zone **1922**.

In the illustrated embodiment, the inner wall(s) of the sterilization zone **1922** (e.g., the internal and external collimators **1920a,b**) extend between the first and second apertures **2108a,b** at a substantially constant angle relative to the centerline of the sensor applicator **102**. The angle of the wall(s) may be any angle between 0° and 90° relative to the centerline of the sensor applicator **102**. The angle of the wall(s), however, may preferably be between 45° and 90° relative to the centerline. In other embodiments, however, the angle of the wall(s) may vary between the first and second apertures **2108a,b**, without departing from the scope of the disclosure. In such embodiments, portions of the wall(s) may extend short distances at an angle dissimilar to adjacent portions, or the wall(s) may otherwise undulate between the first and second apertures **2108a,b**.

The microbial barrier **1926a** may be installed at the interface between the internal and external collimators **1920a,b** and otherwise positioned at or near the radial shoulder **2106**. The microbial barrier **1926a** may be present during the radiation sterilization process. As indicated above, the microbial barrier **1926a** may help seal at least a portion of the sterilization zone **1922**. More particularly, the microbial barrier **1926a** may seal off a portion of the sterilization zone **1922** to help form part of a sealed region **2110** configured to isolate the sensor **1816** and the sharp **1818** from external contamination. The sealed region **2110** may include (encompass) select portions of the interior of the electronics housing **1804** and the sterilization zone **1922**. In one or more embodiments, the sealed region **2110** may be defined and otherwise formed by at least the microbial barrier **1926a**, a first or “top” seal **2112a**, and a second or “bottom” seal **2112b**. The microbial barrier **1926a** and the top and bottom seals **2112a,b** may each create corresponding barriers at their respective sealing locations, thereby allowing the sterilization zone **1922** containing the sensor **1816** and the sharp **1818** to be terminally sterilized.

The top seal **2112a** may be arranged to seal the interface between the sharp hub **1820** and the top of the electronics housing **1804** (i.e., the shell **1806** of FIG. **18**) and thereby prevent contaminants from migrating into the interior of the electronics housing **1804**. In some embodiments, the top seal **2112a** may form part of the sharp hub **1820**, such as being overmolded onto the sharp hub **1820**. In other embodiments, however, the top seal **2112a** may form part of or be overmolded onto the top surface of the shell **1806**. In yet other embodiments, the top seal **2112a** may comprise a separate structure, such as an O-ring or the like, that interposes the sharp hub **1820** and the top surface of the shell **1806**, without departing from the scope of the disclosure.

The bottom seal **2112b** may comprise the gasket **2016** (FIG. **20C**) and, more particularly, the annular projection **2020** (FIGS. **20A** and **20C**) overmolded onto the applicator insert **1910**. In operation, the bottom seal **2112b** may be arranged to seal the interface between the applicator insert **1910** and the bottom of electronics housing **1804** (i.e., the mount **1808** of FIG. **18**). The bottom seal **2112b** may prevent contaminants from migrating into the sterilization zone **1922** and from migrating into the interior of the electronics housing **1804**.

Upon loading the sensor control device **1802** into the sensor applicator **102** and securing the applicator cap **1904** to the sensor applicator **102**, the top and bottom seals **2112a,b** may become progressively compressed and thereby

generate corresponding sealed interfaces. The top and bottom seals **2112a,b** may be made of a variety of materials capable of generating a sealed interface between opposing structures. Suitable materials include, but are not limited to, silicone, a thermoplastic elastomer (TPE), polytetrafluoroethylene (e.g., TEFLON®), or any combination thereof.

Once the radiation sterilization process is finished, the external collimator **1920b** may be removed from the applicator cap **1904**, and the moisture barrier **1926b** may be placed to occlude the opening **1928** in the applicator cap **1904**. Upon delivery, a user may simply remove the applicator cap **1904** in preparation for delivering the sensor control device **1802**. In at least one embodiment, removing the applicator cap **1904** will simultaneously remove the applicator insert **1910**, which may be received into the applicator cap **1904** in a manner that allows the applicator insert **1910** to be secured to the applicator cap **1904** for disassembly. In such embodiments, for example, the applicator insert **1910** may be coupled to the applicator cap **1904** using a snap fit engagement or the like.

In some embodiments, the electronics housing **1804** may be filled with a potting material **2114** that fills in voids within the sensor control device **1802**. The potting material **2114** may comprise a biocompatible material that meets the requirements of ISO 10993. In some embodiments, for example, the potting material **2114** may comprise a urethane material, such as Resinaid® 3672, or silicone materials, such as SI 5055 or SI 5240 available from Henkel®. In other embodiments, the potting material **2114** may comprise an acrylate adhesive material, such as GE4949 available from Delo®.

The potting material **2114** may also serve as an additional safety barrier for absorbing or deflecting propagating radiation **2104**. In at least one embodiment, for example, the potting material **2114** may exhibit an e-beam resistance of at least 85 kGy. Accordingly, instead of passing through air typically present within the electronics housing **1804**, the radiation **2104** may be required to pass through the potting material **2114** before impinging upon the radiation sensitive component(s) **1908**. Although the potting material **2114** may not comprise a high density material, it may nonetheless serve as another level of radiation shielding. Moreover, the potting material **2114** may also increase the robustness of the sensor control device **1802** and the electronics housing **1804**. Consequently, using the potting material **2114** may allow the electronics housing **1804** to be made out of thinner materials, if desired.

It is noted that, while the sensor **1816** and the sharp **1818** extend from the bottom of the electronics housing **1804** and into the sterilization zone **1922** generally concentric with a centerline of the sensor applicator **102** and the applicator cap **1904**, it is contemplated herein to have an eccentric arrangement. More specifically, in at least one embodiment, the sensor **1816** and the sharp **1818** may extend from the bottom of the electronics housing **1804** eccentric to the centerline of the sensor applicator **102** and the applicator cap **1904**. In such embodiments, the internal and external collimators **1920a,b** may be re-designed and otherwise configured such that the sterilization zone **1922** is also eccentrically positioned to receive the sensor **1816** and the sharp **1818**, without departing from the scope of the disclosure.

FIGS. **22A** and **22B** are isometric and cross-sectional side views of another embodiment of the applicator insert **1910**. The applicator insert **1910** depicted in FIGS. **22A-22B** may be similar in most respects to the applicator insert **1910** of FIGS. **20A-20C**. Unlike the applicator insert **1910** of FIGS. **20A-20C**, however, the applicator insert **1910** of FIGS.

22A-22B exhibits an eccentric orientation where the internal collimator 1920a is located eccentric to a centerline 2202 (FIG. 22B) of the body 2002. In such embodiments, the sensor control device 1802 (FIGS. 19B and 21) may also exhibit an eccentric orientation such that the sensor 1816 (FIGS. 19B and 21) and the sharp 1818 (FIGS. 19B and 21) are able to extend into the aperture 2005 defined in the top end 2004a of the applicator insert 1910. Moreover, in such embodiments, the radial alignment feature 1916, the additional radial alignment features 2006, and the sensor locating features 2012 may prove particularly advantageous in helping to properly orient the applicator insert 1910 relative to the sensor control device 1802 within the sensor applicator 102 (FIGS. 19B and 21).

Embodiments disclosed herein include:

H. A sensor applicator that includes a housing having a sensor control device arranged therein, the sensor control device including a sensor, a sharp, and a radiation sensitive component, an applicator cap removably coupled to the housing, an applicator insert positionable within the applicator cap and defining an internal collimator that receives a distal end of the sensor and the sharp, and an external collimator extendable into the applicator cap, wherein the internal and external collimators cooperatively focus radiation from a radiation sterilization process toward the sensor and the sharp and simultaneously prevent the radiation from damaging the radiation sensitive component.

I. A method of sterilizing a sensor control device that includes positioning the sensor control device within a housing of a sensor applicator, the sensor control device including a sensor, a sharp, and a radiation sensitive component, receiving a distal end of the sensor and the sharp within an internal collimator defined by an applicator insert, removably coupling an applicator cap to the housing and thereby securing the applicator insert within the applicator cap, extending an external collimator into the applicator cap and aligning the external collimator with the internal collimator, and cooperatively focusing radiation from a radiation sterilization process toward the sensor and the sharp with the internal and external collimators while simultaneously preventing the radiation from damaging the radiation sensitive component.

J. A hybrid sterilization assembly that includes an applicator insert positionable within an applicator cap of a sensor applicator, an internal collimator defined by the applicator insert to receive a distal end of a sensor and a sharp of a sensor control device arranged within a housing of the sensor applicator, and an external collimator extendable into the applicator cap and alignable with the internal collimator, wherein the internal and external collimators cooperatively focus radiation from a radiation sterilization process toward the sensor and the sharp and simultaneously prevent the radiation from damaging the radiation sensitive component.

Each of embodiments H, I, and J may have one or more of the following additional elements in any combination: Element 1: wherein the applicator insert engages an inner surface of the applicator cap to axially secure the applicator insert within the applicator cap. Element 2: further comprising a sheath extending from the housing and into the applicator cap when the applicator cap is coupled to the housing, and one or more radial alignment features provided on the applicator insert and matable with one or more corresponding features provided on the sheath to rotationally orient the applicator insert relative to the sensor control device. Element 3: further comprising one or more sensor locating features provided on the applicator insert and matable with one or more corresponding features on the

sensor control device to rotationally orient the applicator insert relative to the sensor control device. Element 4: wherein the internal collimator includes a collimating insert and the external collimator is alignable with the collimating insert. Element 5: wherein the collimating insert and the external collimator are each made of a material selected from the group consisting of a high-density polymer, a metal, a composite material, and any combination thereof. Element 6: wherein the internal collimator further includes a gasket engageable with a bottom of the sensor control device to generate a sealed interface. Element 7: wherein the internal and external collimators cooperatively define a sterilization zone exhibiting a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof. Element 8: further comprising a potting material arranged within the sensor control device.

Element 9: further comprising engaging an inner surface of the applicator cap against the applicator insert and thereby axially securing the applicator insert within the applicator cap. Element 10: wherein the internal collimator includes a gasket, the method further comprising engaging the gasket against a bottom of the sensor control device as the applicator insert is axially secured within the applicator cap, and generating a sealed interface with the gasket against the bottom of the sensor control device. Element 11: wherein the internal and external collimators cooperatively define a sterilization zone that receives the sensor and the sharp, the method further comprising sealing at least a portion of the sterilization zone with a microbial barrier positioned at an interface between the internal and external collimators. Element 12: wherein the internal collimator includes a collimating insert and wherein aligning the external collimator with the internal collimator comprises aligning the external collimator with the collimating insert. Element 13: wherein the internal and external collimators cooperatively define a sterilization zone exhibiting a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof.

Element 14: further comprising a microbial barrier positioned at an interface between the internal and external collimators. Element 15: wherein the internal collimator includes a collimating insert and wherein the collimating insert and the external collimator are each made of a material selected from the group consisting of a high-density polymer, a metal, a composite material, and any combination thereof. Element 16: wherein the internal collimator further includes a gasket engageable with a bottom of the sensor control device to generate a sealed interface. Element 17: wherein the internal and external collimators cooperatively define a sterilization zone exhibiting a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof.

By way of non-limiting example, exemplary combinations applicable to H, I, and J include: Element 4 with Element 5; Element 4 with Element 6; Element 9 with Element 10; and Element 15 with Element 16.

Internal Sterilization Assemblies

Prior to being delivered to an end user, some medical devices must be sterilized to render the product free from viable microorganisms. Some medical devices, however, include under-skin sensing devices or sensors that must be sterilized using radiation sterilization, such as electron beam ("e-beam") irradiation. Radiation sterilization, however, can damage electronic components associated with the medical

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device, which are commonly sterilized via gaseous chemical sterilization (e.g., using ethylene oxide). Gaseous chemical sterilization, however, can damage the enzymes or other chemistry and biologics included on the under-skin sensing devices.

In the past, this sterilization incompatibility has been circumvented by separating the under-skin sensing devices and the electronic components and sterilizing each individually. This approach, however, requires additional parts, packaging, process steps, and final assembly by the user, which introduces a possibility of user error. According to the present disclosure, any device requiring terminal sterilization, may be properly sterilized using an internal sterilization assembly designed to focus sterilizing radiation (e.g., beams, waves, energy, etc.) toward component parts requiring sterilization, while simultaneously preventing the propagating radiation from disrupting or damaging sensitive electronic components.

FIG. 23 is a schematic diagram of an example internal sterilization assembly 2300, according to one or more embodiments of the present disclosure. The internal sterilization assembly 2300 (hereafter the “assembly 2300”) may be designed and otherwise configured to help sterilize a medical device 2302. The medical device 2302 may comprise a type of a health care product including any device, mechanism, assembly, or system requiring terminal sterilization of one or more component parts. Suitable examples of the medical device 2302 include, but are not limited to, ingestible products, cardiac rhythm management (CRM) devices, under-skin sensing devices, externally mounted medical devices, medication delivery devices, or any combination thereof.

In the illustrated embodiment, the medical device 2302 comprises an under-skin sensing device or “sensor control device,” also referred to as an “in vivo analyte sensor control device”. As illustrated, the medical device 2302 may be housed within a sensor applicator 2304 (alternately referred to as an “insertor”) and a cap 2306 may be removably coupled to the sensor applicator 2304. The medical device 2302 includes a housing 2308, a part 2310 requiring sterilization, and one or more radiation sensitive components 2312. In some embodiments, the part 2310 may comprise a sensor that extends from the housing 2308. In at least one embodiment, the part 2310 may further include a sharp that may also require sterilization and may help implant the sensor beneath the skin of a user. As illustrated, the part 2310 may extend at an angle from the bottom of the housing 2308, but could alternatively extend perpendicularly from the bottom or from another surface of the housing 2308. Moreover, as illustrated, the part 2310 may extend from one end of the housing 2308 or otherwise offset from a centerline of the housing 2308, but may alternatively extend concentric with the housing, without departing from the scope of the disclosure.

The sensor applicator 2304 is used to deliver the medical device 2302 to a target monitoring location on a user’s skin (e.g., the arm of the user). In some embodiments, the cap 2306 may be threaded to the sensor applicator 2304 and removed from the sensor applicator 2304 by unscrewing the cap 2306 from engagement with the sensor applicator 2304. Once the cap 2306 is removed, a user may then use the sensor applicator 2304 to position the medical device 2302 at a target monitoring location on the user’s body. The part 2310 is positioned such that it can be transcutaneously positioned and otherwise retained under the surface of the user’s skin. In some embodiments, the medical device 2302 may be spring loaded for ejection from the sensor applicator

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2304. Once delivered, the medical device 2302 may be maintained in position on the skin with an adhesive patch (not shown) coupled to the bottom of the medical device 2302.

In the illustrated embodiment, the radiation sensitive component 2312 may be mounted to a printed circuit board (PCB) 2314 positioned within the housing 2308. The radiation sensitive component 2312 may include one or more electronic modules such as, but not limited to, a data processing unit (e.g., an application specific integrated circuit or “ASIC”), a resistor, a transistor, a capacitor, an inductor, a diode, a switch, or any combination thereof. In other embodiments, however, the radiation sensitive component 2312 may comprise a radiation sensitive chemical solution or analyte (e.g., an active agent, pharmaceutical, biologic, etc.). In such embodiments, the medical device 2302 may alternatively comprise a hypodermic needle or syringe and the chemical solution or analyte may be positioned within an ampoule of the medical device 2302.

The medical device 2302 may be subjected to radiation sterilization 2316 to properly sterilize the part 2310 for use. Suitable radiation sterilization 2316 processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. The cap 2306 may define a collimator 2318 that allows the radiation 2316 to impinge upon and sterilize the part 2310. The cap 2306, however, may also act as a radiation shield that helps prevent (impede) propagating radiation 2316 from disrupting or damaging the radiation sensitive component(s) 2312. To accomplish this, the cap 2306 may be made of a material that reduces or prevents the radiation 2316 from penetrating therethrough.

More specifically, the cap 2306 may be made of a material having a density sufficient to absorb the dose of the radiation 2316 beam energy being delivered. In some embodiments, for example, the cap 2306 may be made of any material that has a mass density greater than 0.9 grams per cubic centimeter (g/cc). In other embodiments, however, the mass density of a suitable material may be less than 0.9 g/cc, without departing from the scope of the disclosure. Suitable materials for the cap 2306 include, but are not limited to, a high-density polymer, (e.g., polyethylene, polypropylene, polystyrene, polytetrafluoroethylene, etc.), a metal (e.g., lead, stainless steel, aluminum, etc.), any combination thereof, or any material having a mass density greater than 0.9 g/cc.

As illustrated, the collimator 2318 generally comprises a hole or passageway extending at least partially through the cap 2306. The collimator 2318 defines a sterilization zone 2320 configured to focus the radiation 2316 toward the part 2310. In the illustrated embodiment, the part 2310 may be received within the sterilization zone 2320 for sterilization. The collimator 2318 can exhibit any suitable cross-sectional shape necessary to focus the radiation 2316 on the part 2310 for sterilization. In the illustrated embodiment, for example, the collimator 2318 is conical or frustoconical in shape. In other embodiments, however, the collimator 2318 may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure. In yet other embodiments, the collimator 2318 may exhibit a circular cross-sectional shape with parallel sides.

In the illustrated embodiment, the collimator 2318 provides a first aperture 2322a and a second aperture 2322b where the first and second apertures 2322a,b are defined at opposing ends of the sterilization zone 2320. The first aperture 2322a may allow the radiation 2316 to enter the

sterilization zone **2320** and impinge upon the part **2310**, and the second aperture **2322b** may be configured to receive the part **2310** into the sterilization zone **2320**. In embodiments where the collimator **2318** is conical or frustoconical in shape, the second aperture **2322b** may have a diameter that is smaller than the diameter of the first aperture **2322a**. In such embodiments, for example, the size of the second aperture **2322b** may range between about 0.5 mm and about 3.0 mm, and the size of the first aperture **2322a** may range between about 5.0 mm and about 16.0 mm. As will be appreciated, however, the respective diameters of the first and second apertures **2322a,b** may be greater or less than the ranges provided herein, without departing from the scope of the disclosure. Indeed, the diameters of the first and second apertures **2322a,b** may be scaled to the device size and need only be large enough to allow a sufficient dose of radiation to impinge upon the part **2310**. Moreover, in at least one embodiment, the collimator **2318** may be cylindrical in shape where the first and second apertures **2322a,b** exhibit identical diameters.

In some embodiments, a cap seal **2324** (shown in dashed lines) may be positioned at the opening of the collimator **2318** and otherwise at the first aperture **2322a**. The cap seal **2324** may comprise a radiation permeable, microbial barrier. In some embodiments, for example, the cap seal **2324** may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as TYVEK® available from DuPont®. In other embodiments, however, the cap seal **2324** may comprise, but is not limited to, tape, paper, foil, or any combination thereof. In yet other embodiments, the cap seal **2324** may comprise a thinned portion of the cap **2306**, without departing from the scope of the disclosure. In such embodiments, the first aperture **2322a** would be omitted.

The cap seal **2324** may seal off a portion of the sterilization zone **2320** to isolate the part **2310** from external contamination, while simultaneously allowing the radiation **2316** to pass therethrough to sterilize the part **2310**. In some embodiments, a desiccant (not shown) may be arranged within the sterilization zone **2320**.

In some embodiments, the assembly **2300** may further include a barrier shield **2326** positioned within the housing **2308**. The barrier shield **2326** may be configured to help block radiation **2316** (e.g., electrons) from propagating within the housing **2308** toward the radiation sensitive component(s) **2312**. The barrier shield **2326** may be made of any of the materials mentioned above for the cap **2306**. In the illustrated embodiment, the barrier shield **2326** is positioned vertically within the housing **2308**, but may alternatively be positioned at any other angular configuration suitable for protecting the radiation sensitive component(s) **2312**.

FIG. **24** is a schematic diagram of another example internal sterilization assembly **2400**, according to one or more additional embodiments of the present disclosure. The internal sterilization assembly **2400** (hereafter the “assembly **2400**”) may be similar in some respects to the assembly **2300** of FIG. **23** and therefore may be best understood with reference thereto, where like numeral represent like components not described again in detail. Similar to the assembly **2300** of FIG. **23**, for example, the assembly **2400** may be designed and otherwise configured to help sterilize a medical device **2402**, which may be similar to the medical device **2302** of FIG. **23**. The medical device **2402** may comprise a sensor control device similar to the medical device **2302** of FIG. **23**, but may alternatively comprise any of the health care products mentioned herein.

As illustrated, the medical device **2402** may be housed within a sensor applicator **2404** and, more specifically, within a pocket **2406** defined in the sensor applicator **2404**. In some embodiments, a desiccant (not shown) may be arranged within the pocket **2406**. Similar to the medical device **2302** of FIG. **23**, the medical device **2402** may include the housing **2308**, the part **2310** requiring sterilization, and the radiation sensitive component(s) **2312**. In some embodiments, the assembly **2400** may further include the barrier shield **2326**, as generally described above. As illustrated, the part **2310** may extend perpendicularly from the bottom of the housing **2308**, but could alternatively extend at an angle or from another surface. Moreover, as illustrated, the part **2310** may extend along a centerline of the housing **2308**, but may alternatively extend eccentric to the centerline, without departing from the scope of the disclosure.

The sensor applicator **2404** is used to deliver the medical device **2402** to a target monitoring location on a user's skin (e.g., the arm of the user). As illustrated, the sensor applicator **2404** may include a spring-loaded button **2408** at least partially received within the sensor applicator **2404**. The button **2408** extends within a channel **2409** defined in the sensor applicator **2404** and is engageable with the top of the housing **2308** at its bottom end. In at least one embodiment, a sealed interface is created where the bottom of the button **2406** engages the housing **2308**. The medical device **2402** may be deployed for use from the pocket **2406** by pressing down on the button **2408**, which acts on the housing **2308** and thereby pushes the medical device **2402** distally and out of the pocket **2406** and away from the sensor applicator **2404**. The part **2310** is positioned such that it can be transcutaneously positioned and otherwise retained under the surface of the user's skin. Once delivered, the medical device **2402** may be maintained in position on the skin with an adhesive patch (not shown) coupled to the bottom of the medical device **2402**.

The medical device **2402** may be subjected to radiation sterilization **2316** to properly sterilize the part **2310** prior to use. In the illustrated embodiment, the radiation sterilization **2316** is directed to the top of the sensor applicator **2404** and the button **2408** defines a collimator **2410** that allows the radiation **2316** to impinge upon and sterilize the part **2310**. As illustrated, the collimator **2410** generally comprises a hole or passageway extending at least partially through the button **2408**. The collimator **2410** focuses the radiation **2316** toward the part **2310** and can exhibit any suitable cross-sectional shape necessary to focus the radiation **2316** on the part **2310** for sterilization. In the illustrated embodiment, for example, the collimator **2410** is at least partially conical or frustoconical in shape. In other embodiments, however, the collimator **2410** may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure. In yet other embodiments, the collimator **2410** may exhibit a circular cross-sectional shape with parallel sides.

Portions of the sensor applicator **2404** and the button **2408**, however, may also act as a radiation shield that helps prevent (impede) propagating radiation **2316** from disrupting or damaging the radiation sensitive component(s) **2312**, except through the collimator **2410**. To accomplish this, the sensor applicator **2404** and the button **2408** may be made of a material similar to the material of the cap **2306** of FIG. **23**. In at least one embodiment, the radiation sterilization **2316** may be emitted from a device or machine configured to focus and/or aim the radiation **2316** directly into the colli-

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mator **2410**, and thereby mitigating radiation **2316** exposure to adjacent portions of the sensor applicator **2404**.

In some embodiments, a first seal **2412a** (shown in dashed lines) may be positioned at the opening of the pocket **2406**, and a second seal **2412b** may be arranged at the opening to the collimator **2410** at the top of the button **2406**. The seals **2412a,b** may comprise radiation permeable, microbial barriers, similar to the cap seal **2324** of FIG. 23. The first seal **2412a** may seal off the pocket **2406** on the bottom of the sensor applicator **2404** to isolate the part **2310** from external contamination, and the second seal **2412b** may seal off the collimator **2410**, while simultaneously allowing the radiation **2316** to pass therethrough to sterilize the part **2310**.

FIG. 25 is a schematic diagram of another example internal sterilization assembly **2500**, according to one or more additional embodiments of the present disclosure. The internal sterilization assembly **2500** (hereafter the “assembly **2500**”) may be similar in some respects to the assemblies **2300** and **2400** of FIGS. 23 and 24 and therefore may be best understood with reference thereto, where like numeral represent like components not described again in detail. Similar to the assemblies **2300** and **2400** of FIGS. 23 and 24, for example, the assembly **2500** may be designed and otherwise configured to help sterilize a medical device **2502**, which may be similar to the medical devices **2302** and **2402** of FIGS. 23 and 24. The medical device **2502** may comprise a sensor control device similar to the medical devices **2302** and **2402** of FIGS. 23 and 24, but may alternatively comprise any of the health care products mentioned herein.

As illustrated, the medical device **2502** may be housed within a sensor applicator **2504**, which may include a spring-loaded sheath **2506**. The medical device **2502** may be positioned within a pocket **2508** defined at least partially by the sheath **2506**. In some embodiments, a desiccant (not shown) may be arranged within the pocket **2508**. Similar to the medical devices **2302** and **2402** of FIGS. 23 and 24, the medical device **2502** may include the housing **2308**, the part **2310** requiring sterilization, and the radiation sensitive component(s) **2312**. In some embodiments, the assembly **2500** may further include the barrier shield **2326**, as generally described above.

As illustrated, the part **2310** may extend perpendicularly from the bottom of the housing **2308**, but could alternatively extend at an angle or from another surface. Moreover, as illustrated, the part **2310** may extend along a centerline of the housing **2308**, but may alternatively extend eccentric to the centerline, without departing from the scope of the disclosure.

The sensor applicator **2504** is used to deliver the medical device **2502** to a target monitoring location on a user's skin (e.g., the arm of the user). The medical device **2502** may be deployed for use from the pocket **2508** by forcing the sheath **2506** against the user's skin and thereby causing the sheath **2506** to collapse into the body of the sensor applicator **2504**. Once the sheath **2506** collapses past the housing **2308**, the medical device **2502** may be discharged from the sensor applicator **2504**. The part **2310** is positioned such that it can be transcutaneously positioned and otherwise retained under the surface of the user's skin. Once delivered, the medical device **2502** may be maintained in position on the skin with an adhesive patch (not shown) coupled to the bottom of the medical device **2502**.

The medical device **2502** may be subjected to radiation sterilization **2316** to properly sterilize the part **2310** prior to use. In the illustrated embodiment, the radiation sterilization **2316** is directed to the top of the sensor applicator **2504**, which defines a collimator **2510** that allows the radiation

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2316 to impinge upon and sterilize the part **2310**. As illustrated, the collimator **2510** generally comprises a hole or passageway extending through the body of the sensor applicator **2504**. The collimator **2510** focuses the radiation **2316** toward the part **2310** and can exhibit any suitable cross-sectional shape necessary to focus the radiation **2316** on the part **2310** for sterilization. In the illustrated embodiment, for example, the collimator **2510** is conical or frustoconical in shape. In other embodiments, however, the collimator **2510** may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure. In yet other embodiments, the collimator **2510** may exhibit a circular cross-sectional shape with parallel sides.

The sensor applicator **2504**, however, may also act as a radiation shield that helps prevent (impede) propagating radiation **2316** from disrupting or damaging the radiation sensitive component(s) **2312**, except through the collimator **2510**. To accomplish this, the sensor applicator **2504** may be made of a material similar to the material of the cap **2306** of FIG. 23. In at least one embodiment, however, the radiation sterilization **2316** may be emitted from a device or machine configured to focus and/or aim the radiation **2316** directly into the collimator **2510**, and thereby mitigating radiation **2316** exposure to adjacent portions of the sensor applicator **2504**.

In some embodiments, a first seal **2512a** (shown in dashed lines) may be positioned at the opening of the pocket **2508**, and a second seal **2512b** may be arranged at the opening to the collimator **2510** at the top of the sensor applicator **2504**. The seals **2512a,b** may comprise radiation permeable, microbial barriers, similar to the cap seal **2324** of FIG. 23. The first seal **2512a** may seal off the pocket **2508** on the bottom of the sensor applicator **2504** to isolate the part **2310** from external contamination, and the second seal **2512b** may seal off the collimator **2510**, while simultaneously allowing the radiation **2316** to pass therethrough to sterilize the part **2310**.

Embodiments disclosed herein include:

K. An internal sterilization assembly that includes a sensor applicator, a medical device at least partially housed within the sensor applicator and having a part requiring sterilization and a radiation sensitive component, and a cap removably coupled to the sensor applicator and providing a collimator alignable with the part requiring sterilization, wherein the collimator focuses radiation from a radiation sterilization process toward the part requiring sterilization and the radiation is prevented from damaging the radiation sensitive component.

Embodiment K may have one or more of the following additional elements in any combination: Element 1: wherein the radiation sensitive component is selected from the group consisting of an electronic module, a chemical solution, and any combination thereof. Element 2: wherein the collimator comprises a cross-sectional shape selected from the group consisting of conical, frustoconical, pyramidal, circular, cubic, rectangular, and any combination thereof. Element 3: wherein the medical device comprises an in vivo analyte sensor control device and the part requiring sterilization comprises at least one of a sensor and a sharp extending from the housing of the in vivo analyte sensor control device. Element 4: wherein the at least one of the sensor and the sharp extends at an angle from the bottom of the housing. Element 5: wherein the at least one of the sensor and the sharp extends perpendicularly from the bottom of the housing. Element 6: wherein the at least one of the sensor and the sharp extends from the bottom of the housing along a

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centerline of the housing. Element 7: wherein the at least one of the sensor and the sharp extends from the bottom of the housing offset from a centerline of the housing. Element 8: wherein the cap is made of a material having a mass density greater than 0.9 g/cc. Element 9: wherein the cap is made of a material selected from the group consisting of a high-density polymer, a metal, and any combination thereof. Element 10: wherein the medical device comprises an in vivo analyte sensor control device having a housing that houses the radiation sensitive component, the internal sterilization assembly further comprising a barrier shield positioned within the housing to block the radiation from propagating within the housing toward the radiation sensitive component. Element 11: further comprising a spring-loaded button at least partially received within the sensor applicator and engageable with a top of the medical device, wherein the collimator is defined through the button. Element 12: further comprising a sealed interface at the intersection of the button and the medical device. Element 13: wherein at least one of the button and the sensor applicator is made of a material selected from the group consisting of a high-density polymer, a metal, and any combination thereof. Element 14: wherein the sensor applicator includes a spring-loaded sheath and the medical device is housed within a pocket at least partially defined by the sheath. Element 15: wherein the collimator is defined through the sensor applicator.

By way of non-limiting example, exemplary combinations applicable to A, B, and C include: Element 3 with Element 4; Element 3 with Element 5; Element 3 with Element 6; Element 3 with Element 7; Element 8 with Element 9; Element 11 with Element 12; Element 11 with Element 13; and Element 14 with Element 15.

One-Piece Bio-Sensor Design with Sensor Preservation Vial

FIGS. 26A and 26B are isometric and side views, respectively, of an example sensor control device 2602, according to one or more embodiments of the present disclosure. The sensor control device 2602 (alternately referred to as a “puck”) may be similar in some respects to the sensor control device 104 of FIG. 1 and therefore may be best understood with reference thereto. The sensor control device 2602 may replace the sensor control device 104 of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator 102 (FIG. 1), which delivers the sensor control device 2602 to a target monitoring location on a user’s skin.

The sensor control device 2602, however, may be incorporated into a one-piece system architecture in contrast to the sensor control device 104 of FIG. 1. Unlike the two-piece architecture, for example, a user is not required to open multiple packages and finally assemble the sensor control device 2602. Rather, upon receipt by the user, the sensor control device 2602 is already fully assembled and properly positioned within the sensor applicator 102 (FIG. 1). To use the sensor control device 2602, the user need only open one barrier (e.g., the applicator cap 210 of FIG. 2B) before promptly delivering the sensor control device 2602 to the target monitoring location.

As illustrated, the sensor control device 2602 includes an electronics housing 2604 that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing 2604 may exhibit other cross-sectional shapes, such as ovoid or polygonal, without departing from the scope of the disclosure. The electronics housing 2604 may be configured to house or otherwise contain various electrical components used to operate the sensor control device 2602.

The electronics housing 2604 may include a shell 2606 and a mount 2608 that is matable with the shell 2606. The

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shell 2606 may be secured to the mount 2608 via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, or one or more mechanical fasteners (e.g., screws). In some cases, the shell 2606 may be secured to the mount 2608 such that a sealed interface therebetween is generated. In such embodiments, a gasket or other type of seal material may be positioned at or near the outer diameter (periphery) of the shell 2606 and the mount 2608, and securing the two components together may compress the gasket and thereby generate a sealed interface. In other embodiments, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell 2606 and the mount 2608. The adhesive secures the shell 2606 to the mount 2608 and provides structural integrity, but may also seal the interface between the two components and thereby isolate the interior of the electronics housing 2604 from outside contamination. If the sensor control device 2602 is assembled in a controlled environment, there may be no need to terminally sterilize the internal electrical components. Rather, the adhesive coupling may provide a sufficient sterile barrier for the assembled electronics housing 2604.

The sensor control device 2602 may further include a plug assembly 2610 that may be coupled to the electronics housing 2604. The plug assembly 2610 may be similar in some respects to the plug assembly 207 of FIG. 2A. For example, the plug assembly 2610 may include a sensor module 2612 (partially visible) interconnectable with a sharp module 2614 (partially visible). The sensor module 2612 may be configured to carry and otherwise include a sensor 2616 (partially visible), and the sharp module 2614 may be configured to carry and otherwise include a sharp 2618 (partially visible) used to help deliver the sensor 2616 transcutaneously under a user’s skin during application of the sensor control device 2602. As illustrated, corresponding portions of the sensor 2616 and the sharp 2618 extend from the electronics housing 2604 and, more particularly, from the bottom of the mount 2608. The exposed portion of the sensor 2616 may be received within a hollow or recessed portion of the sharp 2618. The remaining portion of the sensor 2616 is positioned within the interior of the electronics housing 2604.

As discussed in more detail below, the sensor control device 2602 may further include a sensor preservation vial 2620 that provides a preservation barrier surrounding and protecting the exposed portions of the sensor 2616 and the sharp 2618 from gaseous chemical sterilization.

FIGS. 27A and 27B are isometric and exploded views, respectively, of the plug assembly 2610, according to one or more embodiments. The sensor module 2612 may include the sensor 2616, a plug 2702, and a connector 2704. The plug 2702 may be designed to receive and support both the sensor 2616 and the connector 2704. As illustrated, a channel 2706 may be defined through the plug 2702 to receive a portion of the sensor 2616. Moreover, the plug 2702 may provide one or more deflectable arms 2707 configured to snap into corresponding features provided on the bottom of the electronics housing 2604 (FIGS. 26A-26B).

The sensor 2616 includes a tail 2708, a flag 2710, and a neck 2712 that interconnects the tail 2708 and the flag 2710. The tail 2708 may be configured to extend at least partially through the channel 2706 and extend distally from the plug 2702. The tail 2708 includes an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry. In use, the tail 2708 is transcutaneously received beneath a user’s skin, and the chemistry included thereon helps facilitate analyte monitoring in the presence of bodily fluids.

The flag 2710 may comprise a generally planar surface having one or more sensor contacts 2714 (three shown in FIG. 27B) arranged thereon. The sensor contact(s) 2714 may be configured to align with a corresponding number of compliant carbon impregnated polymer modules (tops of which shown at 2720) encapsulated within the connector 2704.

The connector 2704 includes one or more hinges 2718 that enables the connector 2704 to move between open and closed states. The connector 2704 is depicted in FIGS. 27A-27B in the closed state, but can pivot to the open state to receive the flag 2710 and the compliant carbon impregnated polymer module(s) therein. The compliant carbon impregnated polymer module(s) provide electrical contacts 2720 (three shown) configured to provide conductive communication between the sensor 2616 and corresponding circuitry contacts provided within the electrical housing 2604 (FIGS. 26A-26B). The connector 2704 can be made of silicone rubber and may serve as a moisture barrier for the sensor 2616 when assembled in a compressed state and after application to a user's skin.

The sharp module 2614 includes the sharp 2618 and a sharp hub 2722 that carries the sharp 2618. The sharp 2618 includes an elongate shaft 2724 and a sharp tip 2726 at the distal end of the shaft 2724. The shaft 2724 may be configured to extend through the channel 2706 and extend distally from the plug 2702. Moreover, the shaft 2724 may include a hollow or recessed portion 2728 that at least partially circumscribes the tail 2708 of the sensor 2616. The sharp tip 2726 may be configured to penetrate the skin while carrying the tail 2708 to put the active chemistry present on the tail 2708 into contact with bodily fluids.

The sharp hub 2722 may include a hub small cylinder 2730 and a hub snap pawl 2732, each of which may be configured to help couple the plug assembly 2610 (and the entire sensor control device 2602) to the sensor applicator 102 (FIG. 1).

With specific reference to FIG. 27B, the preservation vial 2620 may comprise a generally cylindrical and elongate body 2734 having a first end 2736a and a second end 2736b opposite the first end 2736a. The first end 2736a may be open to provide access into an inner chamber 2738 defined within the body 2734. In contrast, the second end 2736b may be closed and may provide or otherwise define an enlarged head 2740. The enlarged head 2740 exhibits an outer diameter that is greater than the outer diameter of the remaining portions of the body 2734. In other embodiments, however, the enlarged head 2740 may be positioned at an intermediate location between the first and second ends 2736a,b.

FIG. 27C is an exploded isometric bottom view of the plug 2702 and the preservation vial 2620. As illustrated, the plug 2702 may define an aperture 2742 configured to receive the preservation vial 2620 and, more particularly, the first end 2736a of the body 2734. The channel 2706 may terminate at the aperture 2742 such that components extending out of and distally from the channel 2706 will be received into the inner chamber 2738 when the preservation vial 2620 is coupled to the plug 2702.

The preservation vial 2620 may be removably coupled to the plug 2702 at the aperture 2742. In some embodiments, for example, the preservation vial 2620 may be received into the aperture 2742 via an interference or friction fit. In other embodiments, the preservation vial 2620 may be secured within the aperture 2742 with a frangible member (e.g., a shear ring) or substance that may be broken with minimal separation force. In such embodiments, for example, the preservation vial 2620 may be secured within the aperture

2742 with a tag (spot) of glue, a dab of wax, or the preservation vial 2620 may include an easily peeled off glue. As described below, the preservation vial 2620 may be separated from the plug 2702 prior to delivering the sensor control device 2602 (FIGS. 26A-26B) to the target monitoring location on the user's skin.

Referring again to FIGS. 27A and 27B, the inner chamber 2738 may be sized and otherwise configured to receive the tail 2708, a distal section of the shaft 2724, and the sharp tip 2726, collectively referred to as the "distal portions of the sensor 2616 and the sharp 2618." The inner chamber 2738 may be sealed or otherwise isolated to prevent substances that might adversely interact with the chemistry of the sensor 2616 from migrating into the inner chamber 2738. More specifically, the inner chamber 2728 may be sealed to protect or isolate the distal portions of the sensor 2616 and the sharp 2618 during a gaseous chemical sterilization process since gases used during gaseous chemical sterilization can adversely affect the enzymes (and other sensor components, such as membrane coatings that regulate analyte influx) provided on the tail 2708.

In some embodiments, a seal 2744 (FIG. 27B) may provide a sealed barrier between the inner chamber 2738 and the exterior environment. In at least one embodiment, the seal 2744 may be arranged within the inner chamber 2738, but could alternatively be positioned external to the body 2734, without departing from the scope of the disclosure. The distal portions of the sensor 2616 and the sharp 2618 may penetrate the seal 2744 and extend into the inner chamber 2738, but the seal 2744 may maintain a sealed interface about the distal portions of the sensor 2616 and the sharp 2618 to prevent migration of contaminants into the inner chamber 2738. The seal 2744 may be made of, for example, a pliable elastomer or a wax.

In other embodiments (or in addition to the seal 2744), a sensor preservation fluid 2746 (FIG. 27B) may be present within the inner chamber 2738 and the distal portions of the sensor 2616 and the sharp 2618 may be immersed in or otherwise encapsulated by the preservation fluid 2746. The preservation fluid 2746 may generate a sealed interface that prevents sterilization gases from interacting with the enzymes provided on the tail 2708.

The plug assembly 2610 may be subjected to radiation sterilization to properly sterilize the sensor 2616 and the sharp 2618. Suitable radiation sterilization processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. In some embodiments, the plug assembly 2610 may be subjected to radiation sterilization prior to coupling the preservation vial 2620 to the plug 2702. In other embodiments, however, the plug assembly 2610 may be sterilized after coupling the preservation vial 2620 to the plug 2702. In such embodiments, the body 2734 of the preservation vial 2620 and the preservation fluid 2746 may comprise materials and/or substances that permit the propagation of radiation therethrough to facilitate radiation sterilization of the distal portions of the sensor 2616 and the sharp 2618.

Suitable materials for the body 2734 include, but are not limited to, a non-magnetic metal (e.g., aluminum, copper, gold, silver, etc.), a thermoplastic, ceramic, rubber (e.g., ebonite), a composite material (e.g., fiberglass, carbon fiber reinforced polymer, etc.), an epoxy, or any combination thereof. In some embodiments, the material for the body 2734 may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

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The preservation fluid **2746** may comprise any inert and biocompatible fluid (i.e., liquid, gas, gel, wax, or any combination thereof) capable of encapsulating the distal portions of the sensor **2616** and the sharp **2618**. In some embodiments, the preservation fluid **2746** may also permit the propagation of radiation therethrough. The preservation fluid **2746** may comprise a fluid that is insoluble with the chemicals involved in gaseous chemical sterilization. Suitable examples of the preservation fluid **2746** include, but are not limited to, silicone oil, mineral oil, a gel (e.g., petroleum jelly), a wax, fresh water, salt water, a synthetic fluid, glycerol, sorbitan esters, or any combination thereof. As will be appreciated, gels and fluids that are more viscous may be preferred so that the preservation fluid **2746** does not flow easily.

In some embodiments, the preservation fluid **2746** may include an anti-inflammatory agent, such as nitric oxide or another known anti-inflammatory agent. The anti-inflammatory agent may prove advantageous in minimizing local inflammatory response caused by penetration of the sharp **2618** and the sensor **2616** into the skin of the user. It has been observed that inflammation can affect the accuracy of glucose readings, and by including the anti-inflammatory agent the healing process may be accelerated, which may result in obtaining accurate readings more quickly.

FIGS. **28A** and **28B** are exploded and bottom isometric views, respectively, of the electronics housing **2604**, according to one or more embodiments. The shell **2606** and the mount **2608** operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device **2602** (FIGS. **26A-26B**).

A printed circuit board (PCB) **2802** may be positioned within the electronics housing **2604**. A plurality of electronic modules (not shown) may be mounted to the PCB **2802** including, but not limited to, a data processing unit, resistors, transistors, capacitors, inductors, diodes, and switches. The data processing unit may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **2602**. More specifically, the data processing unit may be configured to perform data processing functions, where such functions may include but are not limited to, filtering and encoding of data signals, each of which corresponds to a sampled analyte level of the user. The data processing unit may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. **1**).

As illustrated, the shell **2606**, the mount **2608**, and the PCB **2802** each define corresponding central apertures **2804**, **2806**, and **2808**, respectively. When the electronics housing **2604** is assembled, the central apertures **2804**, **2806**, **2808** coaxially align to receive the plug assembly **2610** (FIGS. **27A-27B**) therethrough. A battery **2810** may also be housed within the electronics housing **2604** and configured to power the sensor control device **2602**.

In FIG. **28B**, a plug receptacle **2812** may be defined in the bottom of the mount **2808** and provide a location where the plug assembly **2610** (FIGS. **27A-27B**) may be received and coupled to the electronics housing **2604**, and thereby fully assemble the sensor control device **2602** (FIG. **26A-3B**). The profile of the plug **2702** (FIGS. **27A-27C**) may match or be shaped in complementary fashion to the plug receptacle **2812**, and the plug receptacle **2812** may provide one or more snap ledges **2814** (two shown) configured to interface with and receive the deflectable arms **2707** (FIGS. **27A-27B**) of the plug **2702**. The plug assembly **2610** is coupled to the

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electronics housing **2604** by advancing the plug **2702** into the plug receptacle **2812** and allowing the deflectable arms **2707** to lock into the corresponding snap ledges **2814**. When the plug assembly **2610** (FIGS. **27A-27B**) is properly coupled to the electronics housing **2604**, one or more circuitry contacts **2816** (three shown) defined on the underside of the PCB **2802** may make conductive communication with the electrical contacts **2720** (FIGS. **27A-27B**) of the connector **2704** (FIGS. **27A-27B**).

FIGS. **29A** and **29B** are side and cross-sectional side views, respectively, of an example embodiment of the sensor applicator **102** with the applicator cap **210** coupled thereto. More specifically, FIGS. **29A-29B** depict how the sensor applicator **102** might be shipped to and received by a user. According to the present disclosure, and as seen in FIG. **29B**, the sensor control device **2602** is already assembled and installed within the sensor applicator **102** prior to being delivered to the user.

As indicated above, prior to coupling the plug assembly **2610** to the electronics housing **2604**, the plug assembly **2610** may be subjected to radiation sterilization to sterilize the distal portions of the sensor **2616** and the sharp **2618**. Once properly sterilized, the plug assembly **2610** may then be coupled to the electronics housing **2604**, as generally described above, and thereby form the fully assembled sensor control device **2602**. The sensor control device **2602** may then be loaded into the sensor applicator **102**, and the applicator cap **210** may be coupled to the sensor applicator **102**. The applicator cap **210** may be threaded to the housing **208** and include a tamper ring **2902**. Upon rotating (e.g., unscrewing) the applicator cap **210** relative to the housing **208**, the tamper ring **2902** may shear and thereby free the applicator cap **210** from the sensor applicator **102**.

According to the present disclosure, while loaded in the sensor applicator **102**, the sensor control device **2602** may be subjected to gaseous chemical sterilization **2904** configured to sterilize the electronics housing **2604** and any other exposed portions of the sensor control device **2602**. To accomplish this, a chemical may be injected into a sterilization chamber **2906** cooperatively defined by the sensor applicator **102** and the interconnected cap **210**. In some applications, the chemical may be injected into the sterilization chamber **2906** via one or more vents **2908** defined in the applicator cap **210** at its proximal end **2910**. Example chemicals that may be used for the gaseous chemical sterilization **2904** include, but are not limited to, ethylene oxide, vaporized hydrogen peroxide, and nitrogen oxide (e.g., nitrous oxide, nitrogen dioxide, etc.).

Since the distal portions of the sensor **2616** and the sharp **2618** are sealed within the preservation vial **2620**, the chemicals used during the gaseous chemical sterilization process do not interact with the enzymes, chemistry or biologics provided on the tail **2708**.

Once a desired sterility assurance level has been achieved within the sterilization chamber **2906**, the gaseous solution is removed and the sterilization chamber **2906** is aerated. Aeration may be achieved by a series of vacuums and subsequently circulating nitrogen gas or filtered air through the sterilization chamber **2906**. Once the sterilization chamber **2906** is properly aerated, the vents **2908** may be occluded with a seal **2912** (shown in dashed lines).

In some embodiments, the seal **2912** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can

be applied before the gaseous chemical sterilization process, and following the gaseous chemical sterilization process, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization chamber 2906. In other embodiments, the seal 2912 may comprise only a single protective layer applied to the applicator cap 210. In such embodiments, the single layer is gas permeable for the sterilization process, but is also capable of protection against moisture and other harmful elements once the sterilization process is complete.

With the seal 2912 in place, the applicator cap 210 provides a barrier against outside contamination, and thereby maintains a sterile environment for the assembled sensor control device 2602 until the user removes (un-threads) the applicator cap 210. The applicator cap 210 may also create a dust-free environment during shipping and storage that prevents an adhesive patch 2914 used to secure the sensor control device 2602 to the user's skin from becoming dirty.

FIG. 30 is a perspective view of an example embodiment of the applicator cap 210, according to the present disclosure. As illustrated, the applicator cap 210 has a generally circular cross-section and defines a series of threads 7302 used to couple the applicator cap 210 to the sensor applicator 102 (FIGS. 29A and 29B). The vents 2908 are also visible in the bottom of the applicator cap 210.

The applicator cap 210 may further provide and otherwise define a cap post 3004 centrally located within the interior of the applicator cap 210 and extending proximally from the bottom thereof. The cap post 3004 may be configured to help support the sensor control device 2602 while contained within the sensor applicator 102 (FIGS. 29A-29B). Moreover, the cap post 3004 may define an opening 3006 configured to receive the preservation vial 2620 as the applicator cap 210 is coupled to the sensor applicator 102.

In some embodiments, the opening 3006 to the cap post 3004 may include one or more compliant features 3008 that are expandable or flexible to enable the preservation vial 2620 to pass therethrough. In some embodiments, for example, the compliant feature(s) 3008 may comprise a collet-type device that includes a plurality of compliant fingers configured to flex radially outward to receive the preservation vial 2620. In other embodiments, however, the compliant feature(s) 3008 may comprise an elastomer or another type of compliant material configured to expand radially to receive the preservation vial 2620.

FIG. 31 is a cross-sectional side view of the sensor control device 2602 positioned within the applicator cap 210, according to one or more embodiments. As illustrated, the cap post 3004 defines a post chamber 3102 configured to receive the preservation vial 2620. The opening 3006 to the cap post 3004 provides access into the post chamber 3102 and exhibits a first diameter D_1 . In contrast, the enlarged head 2740 of the preservation vial 2620 exhibits a second diameter D_2 that is larger than the first diameter D_1 and greater than the outer diameter of the remaining portions of the preservation vial 2620. Accordingly, as the preservation vial 2620 is extended into the post chamber 3102, the compliant feature(s) 3008 of the opening 3006 may flex (expand) radially outward to receive the enlarged head 2740.

In some embodiments, the enlarged head 2740 may provide or otherwise define an angled outer surface that helps bias the compliant feature(s) 3008 radially outward. The enlarged head 2740, however, may also define an upper shoulder 3104 that prevents the preservation vial 2620 from reversing out of the post chamber 3102. More specifically,

the shoulder 3104 may comprise a sharp surface at the second diameter D_2 that will engage but not urge the compliant feature(s) 3008 to flex radially outward in the reverse direction.

Once the enlarged head 2740 bypasses the opening 3006, the compliant feature(s) 3008 flex back to (or towards) their natural state. In some embodiments, the compliant feature(s) 3008 may engage the outer surface of the preservation vial 2620, but may nonetheless allow the applicator cap 210 to rotate relative to the preservation vial 2620. Accordingly, when a user removes the applicator cap 210 by rotating the applicator cap 210 relative to the sensor applicator 102 (FIGS. 29A-29B), the preservation vial 2620 may remain stationary relative to the cap post 3004.

Upon removing the applicator cap 210 from the sensor applicator 102, and thereby also separating the sensor control device 2602 from the applicator cap 210, the shoulder 3104 defined on the enlarged head 2740 will engage the compliant feature(s) 3008 at the opening 3006. Because the diameter of the shoulder 3104 is greater than the diameter of the opening 3006, the shoulder 3104 will bind against the compliant feature(s) 3008 and thereby separate the preservation vial 2620 from the sensor control device 2602, which exposes the distal portions of the sensor 2616 and the sharp 2618. Accordingly, the compliant feature(s) 3008 may prevent the enlarged head 2740 from exiting the post chamber 3102 via the opening 3006 upon separating the applicator cap 210 from the sensor applicator 102 and the sensor control device 2602. The separated preservation vial 2620 will fall into and remain within the post chamber 3102.

In some embodiments, instead of the opening 3006 including the compliant feature(s) 3008, as generally described above, the opening 3006 may alternatively be threaded. In such embodiments, a small portion near the distal end of the preservation vial 2620 may also be threaded and configured to threadably engage the threads of the opening 3006. The preservation vial 2620 may be received within the post chamber 3102 via threaded rotation. Upon removing the applicator cap 210 from the sensor applicator 102, however, the opposing threads on the opening 3006 and the preservation vial 2620 bind and the preservation vial 2620 may be separated from the sensor control device 2602.

Accordingly, there are several advantages to incorporating the sensor control device 2602 into an analyte monitoring system (e.g., the analyte monitoring system 100 of FIG. 1). Since the sensor control device 2602 is finally assembled in a controlled environment, tolerances can be reduced or eliminated altogether, which allows the sensor control device 2602 to be thin and small. Moreover, since the sensor control device 2602 is finally assembled in a controlled environment, a thorough pre-test of the sensor control device 2602 can be undertaken at the factory, thus fully testing the sensor unit prior to packaging for final delivery.

Embodiments disclosed herein include:

L. A sensor control device that includes an electronics housing, a plug assembly matable with the electronics housing and including a sensor module that has a sensor and a sharp module that has a sharp, and a preservation vial coupled to the plug assembly and defining an inner chamber, wherein distal portions of the sensor and the sharp are receivable within the inner chamber and isolated within the inner chamber from gaseous chemical sterilization.

M. An analyte monitoring system that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing, a plug assembly coupled to the electronics housing and including a sensor module that has a sensor and a sharp

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module that has a sharp, and a preservation vial coupled to the plug assembly and defining an inner chamber. The analyte monitoring system further including a cap coupled to the sensor applicator to provide a barrier that seals the sensor control device within the sensor applicator, wherein distal portions of the sensor and the sharp are received within the inner chamber and isolated within the inner chamber from gaseous chemical sterilization.

N. A method of preparing an analyte monitoring system including loading a sensor control device into a sensor applicator, the sensor control device including an electronics housing, a plug assembly matable with the electronics housing and including a sensor module that has a sensor and a sharp module that has a sharp, and a preservation vial coupled to the plug assembly and defining an inner chamber. The method further including securing a cap to the sensor applicator and thereby providing a barrier that seals the sensor control device within the sensor applicator, sterilizing the sensor control device with gaseous chemical sterilization while the sensor control device is positioned within the sensor applicator, and isolating distal portions of the sensor and the sharp received within the inner chamber from the gaseous chemical sterilization.

Each of embodiments L, M, and N may have one or more of the following additional elements in any combination: Element 1: wherein the sensor module further includes a plug and the preservation vial is removably coupled to the plug. Element 2: wherein the preservation vial provides an enlarged head and a diameter of the enlarged head is greater than a diameter of remaining portions of the preservation vial. Element 3: further comprising a seal that provides a sealed barrier between the inner chamber and exterior to the inner chamber, wherein the distal portions of the sensor and the sharp penetrate the seal and extend into the inner chamber. Element 4: further comprising a preservation fluid within the inner chamber that isolates the distal portions of the sensor and the sharp from the gaseous chemical sterilization. Element 5: wherein the distal portions of the sensor and the sharp are at least partially immersed in the preservation fluid. Element 6: wherein the preservation fluid comprises an inert and biocompatible fluid selected from the group consisting of silicone oil, mineral oil, a gel, a wax, fresh water, salt water, a synthetic fluid, glycerol, sorbitan esters, and any combination thereof. Element 7: wherein the preservation fluid includes an anti-inflammatory agent.

Element 8: wherein the cap provides a cap post that defines a post chamber and an opening that receives an enlarged head of the preservation vial into the post chamber. Element 9: wherein the opening includes one or more compliant features that flex radially outward to receive the enlarged head. Element 10: wherein the one or more compliant features comprise a plurality of compliant fingers. Element 11: wherein the one or more compliant features prevent the enlarged head from exiting the post chamber through the opening upon separating the cap from the sensor applicator and the sensor control device. Element 12: wherein the cap is rotatable relative to the preservation vial when the preservation vial is received within the post chamber. Element 13: further comprising a preservation fluid within the inner chamber that isolates the distal portions of the sensor and the sharp from the gaseous chemical sterilization.

Element 14: wherein loading the sensor control device into a sensor applicator is preceded by assembling the plug assembly, coupling the preservation vial to the plug assembly such that the distal portions of the sensor and the sharp are received within the inner chamber, and coupling the plug

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assembly to an electronics housing and thereby providing the sensor control device. Element 15: wherein coupling the preservation vial to the plug assembly is preceded by sterilizing the plug assembly with radiation sterilization. Element 16: wherein isolating the distal portions of the sensor and the sharp from the gaseous chemical sterilization comprises at least partially immersing the distal portions of the sensor and the sharp within a preservation fluid present within the inner chamber. Element 17: wherein the cap provides a cap post that defines a post chamber having one or more compliant features arranged at an opening to the post chamber, and wherein securing the cap to the sensor applicator comprises receiving an enlarged head of the preservation vial into the post chamber via the opening, and flexing the one or more compliant features radially outward to receive the enlarged head.

By way of non-limiting example, exemplary combinations applicable to L, M, and N include: Element 4 with Element 5; Element 4 with Element 6; Element 4 with Element 7; Element 8 with Element 9; Element 9 with Element 10; Element 9 with Element 17; Element 8 with Element 12; Element 8 with Element 13; and Element 14 with Element 15.

Isolating One-Piece Sensor Design with Focused E-beam Sterilization

FIGS. 32A and 32B are isometric and side views, respectively, of an example sensor control device **3202**, according to one or more embodiments of the present disclosure. The sensor control device **3202** (alternately referred to as a “puck”) may be similar in some respects to the sensor control device **104** of FIG. 1 and therefore may be best understood with reference thereto. In some applications, the sensor control device **3202** may replace the sensor control device **104** of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator **102** (FIG. 1), which delivers the sensor control device **3202** to a target monitoring location on a user’s skin.

The sensor control device **3202**, however, may be incorporated into a one-piece system architecture in contrast to the sensor control device **104** of FIG. 1. Unlike the two-piece architecture, for example, a user is not required to open multiple packages and finally assemble the sensor control device **3202** before use. Rather, upon receipt by the user, the sensor control device **3202** is already fully assembled and properly positioned within the sensor applicator **102** (FIG. 1). To use the sensor control device **3202**, the user need only open one barrier (e.g., removing the applicator cap **210** of FIG. 2B) before promptly delivering the sensor control device **3202** to the target monitoring location.

As illustrated, the sensor control device **3202** includes an electronics housing **3204** that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing **3204** may exhibit other cross-sectional shapes, such as ovoid or polygonal, without departing from the scope of the disclosure. The electronics housing **3204** may be configured to house or otherwise contain various electrical components used to operate the sensor control device **3202**.

The electronics housing **3204** may include a shell **3206** and a mount **3208** that is matable with the shell **3206**. The shell **3206** may be secured to the mount **3208** via a variety of ways, such as a snap fit engagement, an interference fit, sonic (or ultrasonic) welding, using one or more mechanical fasteners (e.g., screws), or any combination thereof. In some embodiments, the interface between the shell **3206** and the mount **3208** may be sealed. In such embodiments, a gasket or other type of seal material may be positioned or applied

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at or near the outer diameter (periphery) of the shell **3206** and the mount **3208**. Securing the shell **3206** to the mount **3208** may compress the seal material and thereby generate a sealed interface. In at least one embodiment, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell **3206** and the mount **3208**, and the adhesive may not only secure the shell **3206** to the mount **3208** but may also seal the interface.

In embodiments where a sealed interface is created between the shell **3206** and the mount **3208**, the interior of the electronics housing **3204** may be effectively isolated from outside contamination between the two components. In such embodiments, if the sensor control device **3202** is assembled in a controlled and sterile environment, there may be no need to sterilize the internal electrical components (e.g., via gaseous chemical sterilization). Rather, the sealed engagement may provide a sufficient sterile barrier for the assembled electronics housing **3204**.

The sensor control device **3202** may further include a sensor module **3210** (partially visible in FIG. **32B**) and a sharp module **3212** (partially visible). The sensor and sharp modules **3210**, **3212** may be interconnectable and coupled to the electronics housing **3204**. The sensor module **3210** may be configured to carry and otherwise include a sensor **3214** (FIG. **32B**), and the sharp module **3212** may be configured to help deliver the sensor **3214** transcutaneously under a user's skin during application of the sensor control device **3202**.

As illustrated in FIG. **32B**, corresponding portions of the sensor **3214** and the sharp **3216** extend from the electronics housing **3204** and, more particularly, from the bottom of the mount **3208**. The exposed portion of the sensor **3214** may be received within a hollow or recessed portion of the sharp **3216**. The remaining portion(s) of the sensor **3214** is/are positioned within the interior of the electronics housing **3204**.

An adhesive patch **3218** may be positioned on and otherwise attached to the underside of the mount **3208**. Similar to the adhesive patch **108** of FIG. **1**, the adhesive patch **3218** may be configured to secure and maintain the sensor control device **3202** in position on the user's skin during operation. In some embodiments, a transfer adhesive **3220** may interpose the adhesive patch **3218** and the bottom of the mount **3208**. The transfer adhesive **3220** may help facilitate the assembly process of the sensor control device **3202**.

FIGS. **33A** and **33B** are exploded perspective top and bottom views, respectively, of the sensor control device **3202**, according to one or more embodiments. As illustrated, the shell **3206** and the mount **3208** of the electronics housing **3204** operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device **3202**.

A printed circuit board (PCB) **3302** may be positioned within the electronics housing **3204**. As shown in FIG. **33B**, a plurality of electronic modules **3304** may be mounted to the underside of the PCB **3302**. Example electronic modules **3304** include, but are not limited to, resistors, transistors, capacitors, inductors, diodes, and switches. A data processing unit **3306** (FIG. **33B**) may also be mounted to the PCB **3302** and may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **3202**. More specifically, the data processing unit **3306** may be configured to perform data processing functions, such as filtering and encoding of data signals, each of which corresponds to a sampled analyte

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level of the user. The data processing unit **3306** may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. **1**).

As illustrated, the shell **3206**, the mount **3208**, and the PCB **3302** each define corresponding central apertures **3308a**, **3308b**, **3308c**, respectively. When the sensor control device **3202** is assembled, the central apertures **3308a-c** coaxially align to receive portions of the sensor and sharp modules **3210**, **3212** therethrough.

A battery **3310** and a corresponding battery mount **3312** may also be housed within the electronics housing **3204**. The battery **3310** may be configured to power the sensor control device **3202**.

The sensor module **3210** may include the sensor **3214** and a connector **3314**. The sensor **3214** includes a tail **3316**, a flag **3318**, and a neck **3320** that interconnects the tail **3316** and the flag **3318**. The tail **3316** may be configured to extend through the central aperture **3308b** defined in the mount **3208** and extend distally from the underside thereof. The tail **3316** includes an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry. In use, the tail **3316** is transcutaneously received beneath a user's skin, and the chemistry included thereon helps facilitate analyte monitoring in the presence of bodily fluids.

The flag **3318** may comprise a generally planar surface having one or more sensor contacts **3322** (three shown in FIG. **33A**) disposed thereon. The flag **3318** may be configured to be received within the connector **3314** where the sensor contact(s) **3322** align with a corresponding number of compliant carbon impregnated polymer modules (not shown) encapsulated within the connector **3314**.

The connector **3314** includes one or more hinges **3324** that enables the connector **3314** to pivot between open and closed states. The connector **3314** is depicted in FIGS. **33A-33B** in the closed state, but can transition to the open state to receive the flag **3318** and the compliant carbon impregnated polymer module(s) therein. The compliant carbon impregnated polymer module(s) provide electrical contacts **3326** (three shown in FIG. **33A**) configured to provide conductive communication between the sensor **3214** and corresponding circuitry contacts **3328** provided on the PCB **3302**. When the sensor module **3210** is properly coupled to the electronics housing **3204**, the circuitry contacts **3328** make conductive communication with the electrical contacts **3326** of the connector **3314**. The connector **3314** can be made of silicone rubber and may serve as a moisture barrier for the sensor **3214**.

The sharp module **3212** includes the sharp **3216** and a sharp hub **3330** that carries the sharp **3216**. The sharp **3216** includes an elongate shaft **3332** and a sharp tip **3334** at the distal end of the shaft **3332**. The shaft **3332** may be configured to extend through each of the coaxially aligned central apertures **3308a-c** and extend distally from the bottom of the mount **3208**. Moreover, the shaft **3332** may include a hollow or recessed portion **3336** that at least partially circumscribes the tail **3316** of the sensor **3214**. The sharp tip **3334** may be configured to penetrate the skin while carrying the tail **3316** to put the active chemistry of the tail **3316** into contact with bodily fluids.

The sharp hub **3330** may include a hub small cylinder **3338** and a hub snap pawl **3340**, each of which may be configured to help couple the sensor control device **3202** to the sensor applicator **102** (FIG. **1**).

Referring specifically to FIG. **33A**, in some embodiments the sensor module **3210** may be at least partially received within a sensor mount pocket **3342** included within the electronics housing **3204**. In some embodiments, the sensor

mount pocket **3342** may comprise a separate structure, but may alternatively form an integral part or extension of the mount **3208**. The sensor mount pocket **3342** may be shaped and otherwise configured to receive and seat the sensor **3214** and the connector **3314**. As illustrated, the sensor mount pocket **3342** defines an outer periphery **3344** that generally circumscribes the region where the sensor **3214** and the connector **3314** are to be received. In at least one embodiment, the outer periphery **3344** may be sealed to the underside of the PCB **3302** when the electronics housing **3204** is fully assembled. In such embodiments, a gasket (e.g., an O-ring or the like), an adhesive, or another type of seal material may be applied (arranged) at the outer periphery **3344** and may operate to seal the interface between the sensor mount pocket **3342** and the PCB **3302**.

Sealing the interface between the sensor mount pocket **3342** and the underside of the PCB **3302** may help create or define a sealed zone or region within the electronics housing **3204**. The sealed region may prove advantageous in helping to isolate (protect) the tail **3316** of the sensor **3214** from potentially harmful sterilization gases used during gaseous chemical sterilization.

Referring specifically to FIG. **33B**, a plurality of channels or grooves **3346** may be provided or otherwise defined on the bottom of the mount **3208**. As illustrated, the grooves **3346** may form a plurality of concentric rings in combination with a plurality of radially extending channels. The adhesive patch **3218** (FIGS. **32A-32B**) may be attached to the underside of the mount **3208**, and, in some embodiments, the transfer adhesive **3220** (FIGS. **32A-32B**) may interpose the adhesive patch **3218** and the bottom of the mount **3208**. The grooves **3346** may prove advantageous in promoting the egress of moisture away from the center of the electronics housing **3204** beneath the adhesive patch **3218**.

In some embodiments, a cap post seal interface **3348** may be defined on the bottom of the mount **3208** at the center of the mount **3208**. As illustrated, the cap post seal interface **3348** may comprise a substantially flat portion of the bottom of the mount **3208**. The second central aperture **3308b** is defined at the center of the cap post seal interface **3348** and the grooves **3346** may circumscribe the cap post seal interface **3348**. The cap post seal interface **3348** may provide a sealing surface that may help isolate (protect) the tail **3316** of the sensor **3214** from potentially harmful sterilization gases used during gaseous chemical sterilization.

FIGS. **34A** and **34B** are side and cross-sectional side views, respectively, of the sensor applicator **102** with the applicator cap **210** coupled thereto. More specifically, FIGS. **34A-34B** depict how the sensor applicator **102** might be shipped to and received by a user. According to the present disclosure, and as seen in FIG. **34B**, the sensor control device **3202** is already assembled and installed within the sensor applicator **102** prior to being delivered to the user. The applicator cap **210** may be threaded to the housing **208** and include a tamper ring **3402**. Upon rotating (e.g., unscrewing) the applicator cap **210** relative to the housing **208**, the tamper ring **3402** may shear and thereby free the applicator cap **210** from the sensor applicator **102**. Following which, the user may deliver the sensor control device **3202** to the target monitoring location, as generally described above with reference to FIGS. **2E-2G**.

With specific reference to FIG. **34B**, the sensor control device **3202** may be loaded into the sensor applicator **102** by mating the sharp hub **3330** with a sensor carrier **3404** included within the sensor applicator **102**. More specifically,

the hub small cylinder **3338** and the hub snap pawl **3340** may be received by corresponding mating features of the sensor carrier **3404**.

Once the sensor control device **3202** is mated with the sensor carrier **3404**, the applicator cap **210** may then be secured to the sensor applicator **102**. As illustrated, the applicator cap **210** may provide and otherwise define a cap post **3406** centrally located within the interior of the applicator cap **210** and extending proximally from the bottom thereof. The cap post **3406** may be configured to help support the sensor control device **3202** while contained within the sensor applicator **102**. Moreover, the cap post **3406** may define a post chamber **3408** configured to receive the sensor **3214** and the sharp **3216** as extending from the bottom of the electronics housing **3204**. When the sensor control device **3202** is loaded into the sensor applicator **102**, the sensor **3214** and the sharp **3216** may be arranged within a sealed region **3410** at least partially defined by the post chamber **3408** and configured to isolate the sensor **3214** and the sharp **3216** during gaseous chemical sterilization.

In some embodiments, prior to assembling and loading the sensor control device **3202** into the sensor applicator **102**, the sensor and sharp modules **3210**, **3212** may be subjected to radiation sterilization to sterilize the distal portions of the sensor **3214** and the sharp **3216**. Once properly sterilized, the sensor and sharp modules **3210**, **3212** may then be coupled to the electronics housing **3204** and the fully assembled sensor control device **3202** may then be loaded into the sensor applicator **102** as described above.

In other embodiments, however, the fully assembled sensor control device **3202** may first be loaded into the sensor applicator **102** and the sensor and sharp modules **3210**, **3212** may then be subjected to radiation sterilization **3412** while positioned within the sensor applicator **102**. The radiation sterilization **3412** may comprise, for example, e-beam irradiation, but other methods of sterilization may alternatively be used including, but not limited to, gamma ray irradiation, X-ray irradiation, or any combination thereof.

In some embodiments, as illustrated, the sensor control device **3202** may be subjected to "focused" radiation sterilization **3412**, where the radiation (e.g., beams, waves, etc.) from the radiation sterilization **3412** is applied and otherwise directed only toward the sensor and sharp modules **3210**, **3212** (e.g., the sensor **3214** and the sharp **3216**). In such embodiments, the electrical components **3304** (FIG. **33B**) coupled to the PCB **3302** (FIGS. **33A-33B**), including the data processing unit **3306** (FIG. **33B**), may be positioned out of the range of the propagating radiation and, therefore, will not be affected by the radiation. The electrical components **3304** and the data processing unit **3306**, for example, may be positioned on the PCB **3302** near its outer periphery so as not to fall within the range (span) of the focused radiation sterilization **3412**. In other embodiments, this may be accomplished by shielding the sensitive electrical components **3304** with proper electromagnetic shields.

According to the present disclosure, while loaded in the sensor applicator **102**, the sensor control device **3202** may be subjected to gaseous chemical sterilization **3414** to sterilize the electronics housing **3204** and any other exposed portions of the sensor control device **3202**. To accomplish this, a chemical may be injected into a sterilization chamber **3416** cooperatively defined by the sensor applicator **102** and the interconnected cap **210**. In some applications, the chemical may be injected via one or more vents **3418** defined in the applicator cap **210** at its proximal end **3420**. Example chemicals that may be used for the gaseous chemical ster-

ilization **3414** include, but are not limited to, ethylene oxide, vaporized hydrogen peroxide, and nitrogen oxide (e.g., nitrous oxide, nitrogen dioxide, etc.).

Since the sensor **3214** and the sharp **3216** are sealed within the sealed region **3410**, the chemicals used during the gaseous chemical sterilization process do not interact with the enzymes, chemistry or biologics provided on the tail **3316**.

Once a desired sterility assurance level has been achieved within the sterilization chamber **3416**, the gaseous solution is removed and the sterilization chamber **3416** is aerated. Aeration may be achieved by a series of vacuums and subsequently circulating nitrogen gas or filtered air through the sterilization chamber **3416**. Once the sterilization chamber **3416** is properly aerated, the vents **3418** may be occluded with a seal **3422** (shown in dashed lines) applied to the proximal end **3420** of the applicator cap **210**.

In some embodiments, the seal **3422** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can be applied before the gaseous chemical sterilization **3414**, and following the gaseous chemical sterilization **3414**, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization chamber **3416**. In other embodiments, the seal **3422** may comprise only a single protective layer applied to the applicator cap **210**. In such embodiments, the single layer is gas permeable for the sterilization process, but is also capable of protection against moisture and other harmful elements once the sterilization process is complete.

With the seal **3422** in place, the applicator cap **210** provides a barrier against outside contamination, and thereby maintains a sterile environment for the assembled sensor control device **3202** until the user removes (un-threads) the applicator cap **210**. The applicator cap **210** may also create a dust-free environment during shipping and storage that prevents the adhesive patch **3218** used to secure the sensor control device **3202** to the user's skin from becoming dirty.

FIG. 35 is an enlarged cross-sectional side view of the sensor control device **3202** mounted within the sensor applicator **102** with the applicator cap **210** secured thereto, according to one or more embodiments. As indicated above, portions of the sensor **3214** and the sharp **3216** may be arranged within the sealed region **3410** and thereby protected from substances that might adversely interact with the chemistry of the sensor **3214**. More specifically, the gases used during the gaseous chemical sterilization **3414** (FIG. 34B) can adversely affect the enzymes provided on the tail **3316** of the sensor **3214**, and the sealed region **3410** protects the tail **3316** from the ingress of such chemicals.

As illustrated, the sealed region **3410** may include (encompass) select portions of the interior of the electronics housing **3204** and the post chamber **3408** of the cap post **3406**. In one or more embodiments, the sealed region **3410** may be defined and otherwise formed by at least a first seal **3502a**, a second seal **3502b**, and a third seal **3502c**. The first seal **3502a** may be arranged to seal the interface between the sharp hub **3330** and the shell **3206**. Moreover, the first seal **3502a** may circumscribe the first central aperture **3308a** defined in the shell **3206** such that fluids (e.g., gaseous

chemicals) are prevented from migrating into the interior of the electronics housing **3204** via the first central aperture **3308a**.

In some embodiments, the first seal **3502a** may form part of the sharp hub **3330**. For example, the first seal **3502a** may be overmolded onto the sharp hub **3330**. In other embodiments, the first seal **3502a** may be overmolded onto the top surface of the shell **3206**. In yet other embodiments, the first seal **3502a** may comprise a separate structure, such as an O-ring or the like, that interposes the sharp hub **3330** and the top surface of the shell **3206**, without departing from the scope of the disclosure.

The second seal **3502b** may be arranged to seal the interface between the cap post **3406** and the bottom of the mount **3208**, and the second seal **3502b** may circumscribe the second central aperture **3308b** defined in the mount **3208**. Consequently, the second seal **3502b** may prevent fluids (e.g., gaseous chemicals) from migrating into the post chamber **3408** of the cap post **3406** and also from migrating into the interior of the electronics housing **3204** via the second central aperture **3308b**.

In some embodiments, the second seal **3502b** may form part of the cap post **3406**. For example, the second seal **3502b** may be overmolded onto the top of the cap post **3406**. In other embodiments, the second seal **3502b** may be overmolded onto the cap post seal interface **3348** at the bottom of the mount **3208**. In yet other embodiments, the second seal **3502b** may comprise a separate structure, such as an O-ring or the like, that interposes the cap post **3406** and the bottom of the mount **3208**, without departing from the scope of the disclosure.

Upon loading the sensor control device **3202** into the sensor applicator **102** and securing the applicator cap **210** to the sensor applicator **102**, the first and second seals **3502a,b** become compressed and generate corresponding sealed interfaces. The first and second seals **3502a,b** may be made of a variety of materials capable of generating a sealed interface between opposing structures. Suitable materials include, but are not limited to, silicone, a thermoplastic elastomer (TPE), polytetrafluoroethylene (Teflon®), rubber, an elastomer, or any combination thereof.

The third seal **3502c** may be arranged to seal an interface between the sensor mount pocket **3342** and the PCB **3302** and, more particularly, between the outer periphery **3344** of the sensor mount pocket **3342** and the underside of the PCB **3302**. The third seal **3502c** may comprise a gasket (e.g., an O-ring or the like), an adhesive, or another type of seal material applied (arranged) at the outer periphery **3344**. In operation, the third seal **3502c** may prevent fluids (e.g., gaseous chemicals, liquids, etc.) from migrating into the interior of the sensor mount pocket **3342** and, therefore, into the post chamber **3408** to adversely react with the enzymes on the tail **3316**.

The applicator cap **210** may be secured to the sensor applicator **102** by threading the applicator cap **210** to the sensor applicator **102** via relative rotation. As the applicator cap **210** rotates relative to the sensor applicator **102**, the cap post **3406** advances until the second seal **3502b** engages the cap post seal interface **3348** at the bottom of the mount **3208**. Upon engaging the cap post seal interface **3348**, the second seal **3502b** may frictionally engage the mount **3208** and thereby urge corresponding rotation of the entire electronics housing **3204** in the same angular direction.

In prior art sensor control devices, such as the sensor control device **104** of FIG. 1, conical carrier grip features are commonly defined on the exterior of the electronics housing and configured to mate with corresponding conical features

provided on radially biased arms of the sensor mount pocket 3342. Mating engagement between these corresponding conical features helps prevent the electronics housing from rotating within the sensor applicator 102.

In contrast, the electronics housing 3204 of the presently disclosed sensor control device 3202 provides or otherwise defines an angled and otherwise continuously smooth exterior surface 3504 about its outer diameter (periphery). In some embodiments, as illustrated, the smooth exterior surface 3504 may be provided on the mount 3208, but may alternatively be provided on the shell 3206, without departing from the scope of the disclosure. One or more radially biased arms of the sensor mount pocket 3342 may be positioned to engage the exterior surface 3504 to help center the sensor control device 3202 within the sensor applicator 102. As the electronics housing 3204 is urged to rotate through frictional engagement between the second seal 3502b and the bottom of the mount 3208, the exterior surface 3504 slidably engages the radially biased arms, which do not inhibit rotation thereof.

FIG. 36 is an enlarged cross-sectional bottom view of the sensor control device 3202 positioned atop the cap post 3406, according to one or more embodiments. As illustrated, the adhesive patch 3218 is positioned on the underside of the mount 3208 and the transfer adhesive 3220 interposes the adhesive patch 3218 and the mount 3208.

The adhesive patch 3218 may occlude or otherwise cover most of the grooves 3346 defined on the bottom of the mount 3208. Moreover, as illustrated, the adhesive patch 3218 may extend a short distance into the cap post seal interface 3348. To enable the grooves 3346 to properly direct moisture away from the center of the electronics housing 3204 and from the cap post seal interface 3348, the adhesive patch 3218 (and the transfer adhesive 3220, if included) may provide or otherwise define one or more channels 3602 aligned with and otherwise arranged to fluidly communicate with the grooves 3346. In the illustrated embodiment, the channels 3602 extend radially outward from the center of the electronics housing 3204, but may alternatively be defined in other configurations and nonetheless interconnect with the grooves 3346 to facilitate fluid communication therebetween.

In operation, as moisture builds up around the center of the electronics housing 3204 and at the cap post seal interface 3348, the moisture is able to flow into the grooves 3346 via the channels 3602. Once in the grooves 3346, the moisture is able to flow radially outward beneath the adhesive patch 3218 and toward the outer periphery of the sensor control device 3202.

Embodiments disclosed herein include:

O. An analyte monitoring system that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing having a shell and a mount matable with the shell, a printed circuit board positioned within the electronics housing, a sensor extending from a bottom of the mount, a sharp hub positioned adjacent a top of the shell, and a sharp carried by the sharp hub and extending through the electronics housing and from the bottom of the mount. The analyte monitoring system further including a cap coupled to the sensor applicator and providing a cap post that defines a post chamber that receives the sensor and the sharp extending from the bottom of the mount, and a sealed region encompassing the post chamber and a portion of an interior of the electronics housing, wherein the sealed region is defined by a first seal that seals an interface between the sharp hub and the shell, a second seal that seals an interface between the cap post and

the bottom of the mount, and a third seal that seals an interface between the mount and the printed circuit board, and wherein portions of the sensor and the sharp reside within the sealed region and are thereby isolated from gaseous chemical sterilization.

P. A method of preparing an analyte monitoring system including loading a sensor control device into a sensor applicator, the sensor control device including an electronics housing having a shell and a mount matable with the shell, a printed circuit board positioned within the electronics housing, a sensor module having a sensor extending from a bottom of the mount, and a sharp module having a sharp hub and a sharp carried by the sharp hub, wherein the sharp extends through the electronics housing and from the bottom of the mount. The method further including securing a cap to the sensor applicator, wherein the cap provides a cap post that defines a post chamber that receives the sensor and the sharp extending from the bottom of the mount, creating a sealed region as the cap is secured to the sensor applicator, the sealed region encompassing the post chamber and a portion of an interior of the electronics housing, wherein portions of the sensor and the sharp reside within the sealed region, sterilizing the sensor control device with gaseous chemical sterilization while the sensor control device is positioned within the sensor applicator, and isolating the portions of the sensor and the sharp residing within the sealed region from the gaseous chemical sterilization.

Each of embodiments O and P may have one or more of the following additional elements in any combination: Element 1: wherein the first seal circumscribes a central aperture defined in the shell and prevents fluids from migrating into the portion of the interior of the electronics housing via the central aperture. Element 2: wherein the second seal circumscribes a central aperture defined in the mount and prevents fluids from migrating into the portion of the interior of the electronics housing via the central aperture and further prevents the fluids from migrating into the post chamber. Element 3: wherein the first seal is overmolded onto the sharp hub. Element 4: wherein the first seal interposes the sharp hub and a top surface of the shell. Element 5: wherein the second seal is overmolded onto the cap post. Element 6: wherein the second seal interposes the cap post and a bottom surface of the mount. Element 7: wherein the first and second seals are made of a material selected from the group consisting of silicone, a thermoplastic elastomer, polytetrafluoroethylene, and any combination thereof. Element 8: wherein the mount provides a sensor mount pocket that at least partially receives a sensor module within the electronics housing, and wherein the third seal is positioned at an outer periphery of the sensor mount pocket. Element 9: wherein the third seal comprises one of a gasket and an adhesive. Element 10: further comprising a plurality of grooves defined on the bottom of the mount, and a cap post seal interface defined on the bottom of the mount at a center of the mount, wherein the second seal seals against the cap post seal interface. Element 11: further comprising an adhesive patch coupled to the bottom of the mount and extending radially into the cap post seal interface, and one or more channels defined in the adhesive patch and interconnecting with the plurality of grooves to facilitate fluid communication between the cap post seal interface and the plurality of grooves. Element 12: wherein the electronics housing defines an angled and smooth exterior surface that allows the sensor control device to rotate unobstructed relative to the sensor applicator as the cap is coupled to the sensor applicator.

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Element 13: wherein creating the sealed region as the cap is secured to the sensor applicator comprises sealing an interface between the sharp hub and the shell with a first seal, sealing an interface between the cap post and the bottom of the mount with a second seal, and sealing an interface between the mount and the printed circuit board with a third seal. Element 14: wherein loading the sensor control device into a sensor applicator is preceded by sterilizing the sensor and the sharp with radiation sterilization, and assembling the sensor and sharp modules to the electronics housing. Element 15: wherein sterilizing the sensor control device with the gaseous chemical sterilization is preceded by sterilizing the sensor and the sharp with radiation sterilization while the sensor control device is positioned within the sensor applicator. Element 16: wherein the radiation sterilization is at least one of focused radiation sterilization and low-energy radiation sterilization. Element 17: wherein the electronics housing defines an angled and smooth exterior surface, the method further comprising allowing the sensor control device to rotate relative to the sensor applicator as the cap is secured to the sensor applicator.

By way of non-limiting example, exemplary combinations applicable to O and P include: Element 1 with Element 2; Element 1 with Element 3; Element 1 with Element 4; Element 1 with Element 5; Element 1 with Element 6; Element 1 with Element 7; Element 1 with Element 8; Element 3 with Element 4; Element 3 with Element 5; Element 3 with Element 6; Element 10 with Element 11; and Element 15 with Element 16.

One-Piece Puck Architecture with ASIC Shields, Use of Low and Medium Energy Radiation Sterilization, and Magnetic Deflection

FIGS. 37A-37C are isometric, side, and bottom views, respectively, of an example sensor control device 3702, according to one or more embodiments of the present disclosure. The sensor control device 3702 (alternately referred to as an on-body patch or unit) may be similar in some respects to the sensor control device 104 of FIG. 1 and therefore may be best understood with reference thereto. The sensor control device 3702 may replace the sensor control device 104 of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator 102 (FIG. 1), which delivers the sensor control device 3702 to a target monitoring location on a user's skin. However, in contrast to the sensor control device 104 of FIG. 1, various structural advantages and improvements allow the sensor control device 3702 to be incorporated into a one-piece system architecture.

Unlike the sensor control device 104 of FIG. 1, for example, a user is not required to open multiple packages and finally assemble the sensor control device 3702 prior to delivery to the target monitoring location. Rather, upon receipt by the user, the sensor control device 3702 may already be assembled and properly positioned within the sensor applicator 102. To use the sensor control device 3702, the user need only break one barrier (e.g., the applicator cap 210 of FIG. 2B) before promptly delivering the sensor control device 3702 to the target monitoring location.

Referring first to FIG. 37A, the sensor control device 3702 comprises an electronics housing 3704 that is generally disc-shaped and may have a generally circular cross-section. In other embodiments, however, the electronics housing 3704 may exhibit other cross-sectional shapes, such as ovoid or polygonal, without departing from the scope of the disclosure. The electronics housing 3704 may include a shell 3706 and a mount 3708 that is matable with the shell 3706.

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An adhesive patch 3710 may be positioned on and otherwise attached to the underside of the mount 3708. Similar to the adhesive patch 108 of FIG. 1, the adhesive patch 3710 may be configured to secure and maintain the sensor control device 3702 in position on the user's skin during operation.

In some embodiments, the shell 3706 may define a reference feature 3712. As illustrated, the reference feature 3712 may comprise a depression or blind pocket defined in the shell 3706 and extending a short distance into the interior of the electronics housing 3704. The reference feature 3712 may operate as a "datum c" feature configured to help facilitate control of the sensor control device 3702 in at least one degree of freedom during factory assembly. In contrast, prior sensor control devices (e.g., the sensor control device 104 of FIG. 1) typically include a tab extending radially from the side of the shell. The tab is used as an in-process clocking datum, but must be removed at the end of fabrication, and followed by an inspection of the shell where the tab once existed, which adds complexity to the prior fabrication process.

The shell 3706 may also define a central aperture 3714 sized to receive a sharp (not shown) that is extendable through the center of the electronics housing 3704.

FIG. 37B depicts a portion of a sensor 3716 extending from the electronics housing 3704. The remaining portion(s) of the sensor 3716 is/are positioned within the interior of the electronics housing 3704. Similar to the sensor 110 of FIG. 1, the exposed portion of the sensor 3716 is configured to be transcutaneously positioned under the user's skin during use. The exposed portion of the sensor 3716 can include an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry.

The sensor control device 3702 provides structural improvements that result in a height H and a diameter D that may be less than prior sensor control devices (e.g., the sensor control device 104 of FIG. 1). In at least one embodiment, for example, the height H may be about 1 mm or more less than the height of prior sensor control devices, and the diameter D may be about 2 mm or more less than the diameter of prior sensor control devices.

Moreover, the structural improvements of the sensor control device 3702 allows the shell 3706 to provide or otherwise define a chamfered or angled outer periphery 3718. In contrast, prior sensor control devices commonly require a rounded or outwardly arcuate outer periphery to accommodate internal components. The reduced height H, the reduced diameter D, and the angled outer periphery 3718 may each prove advantageous in providing a sensor control device 3702 that is thinner, smaller, and less prone to being prematurely detached by catching on sharp corners or the like while attached to the user's skin.

FIG. 37C depicts a central aperture 3720 defined in the underside of the mount 3708. The central aperture 3720 may be sized to receive a combination sharp (not shown) and sensor 3716, where the sensor 3716 is received within a hollow or recessed portion of the sharp. When the electronics housing 3704 is assembled, the central aperture 3720 coaxially aligns with the central aperture 3714 (FIG. 37A) of the shell 3706 (FIG. 37A) and the sharp penetrates the electronics housing by extending simultaneously through each central aperture 3714, 3720.

FIGS. 38A and 38B are exploded top and bottom views, respectively, of the sensor control device 3702, according to one or more embodiments. The shell 3706 and the mount 3708 operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device 3702. As illus-

trated, the sensor control device **3702** may include a printed circuit board assembly (PCBA) **3802** that includes a printed circuit board (PCB) **3804** having a plurality of electronic modules **3806** coupled thereto. Example electronic modules **3806** include, but are not limited to, resistors, transistors, capacitors, inductors, diodes, and switches. Prior sensor control devices commonly stack PCB components on only one side of the PCB. In contrast, the PCB components **3806** in the sensor control device **3702** can be dispersed about the surface area of both sides (i.e., top and bottom surfaces) of the PCB **3804**.

Besides the electronic modules **3806**, the PCBA **3802** may also include a data processing unit **3808** mounted to the PCB **3804**. The data processing unit **3808** may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **3702**. More specifically, the data processing unit **3808** may be configured to perform data processing functions, where such functions may include but are not limited to, filtering and encoding of data signals, each of which corresponds to a sampled analyte level of the user. The data processing unit **3808** may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. 1).

A battery aperture **3810** may be defined in the PCB **3804** and sized to receive and seat a battery **3812** configured to power the sensor control device **3702**. An axial battery contact **3814a** and a radial battery contact **3814b** may be coupled to the PCB **3804** and extend into the battery aperture **3810** to facilitate transmission of electrical power from the battery **3812** to the PCB **3804**. As their names suggest, the axial battery contact **3814a** may be configured to provide an axial contact for the battery **3812**, while the radial battery contact **3814b** may provide a radial contact for the battery **3812**. Locating the battery **3812** within the battery aperture **3810** with the battery contacts **3814a,b** helps reduce the height *H* (FIG. 37B) of the sensor control device **3702**, which allows the PCB **3804** to be located centrally and its components to be dispersed on both sides (i.e., top and bottom surfaces). This also helps facilitate the chamfer **3718** (FIG. 37B) provided on the electronics housing **3704**.

The sensor **3716** may be centrally located relative to the PCB **3804** and include a tail **3816**, a flag **3818**, and a neck **3820** that interconnects the tail **3816** and the flag **3818**. The tail **3816** may be configured to extend through the central aperture **3720** of the mount **3708** to be transcutaneously received beneath a user's skin. Moreover, the tail **3816** may have an enzyme or other chemistry included thereon to help facilitate analyte monitoring.

The flag **3818** may include a generally planar surface having one or more sensor contacts **3822** (three shown in FIG. 38B) arranged thereon. The sensor contact(s) **3822** may be configured to align with and engage a corresponding one or more circuitry contacts **3824** (three shown in FIG. 38A) provided on the PCB **3804**. In some embodiments, the sensor contact(s) **3822** may comprise a carbon impregnated polymer printed or otherwise digitally applied to the flag **3818**. Prior sensor control devices typically include a connector made of silicone rubber that encapsulates one or more compliant carbon impregnated polymer modules that serve as electrical conductive contacts between the sensor and the PCB. In contrast, the presently disclosed sensor contact(s) **3822** provide a direct connection between the sensor **3716** and the PCB **3804** connection, which eliminates the need for the prior art connector and advantageously reduces the height *H* (FIG. 37B). Moreover, eliminating the compliant

carbon impregnated polymer modules eliminates a significant circuit resistance and therefor improves circuit conductivity.

The sensor control device **3702** may further include a compliant member **3826**, which may be arranged to interpose the flag **3818** and the inner surface of the shell **3706**. More specifically, when the shell **3706** and the mount **3708** are assembled to one another, the compliant member **3826** may be configured to provide a passive biasing load against the flag **3818** that forces the sensor contact(s) **3822** into continuous engagement with the corresponding circuitry contact(s) **3824**. In the illustrated embodiment, the compliant member **3826** is an elastomeric O-ring, but could alternatively comprise any other type of biasing device or mechanism, such as a compression spring or the like, without departing from the scope of the disclosure.

The sensor control device **3702** may further include one or more electromagnetic shields, shown as a first shield **3828a** and a second shield **3828b**. The shields **3828a,b** may be arranged between the shell **3706** and the mount **3708**; i.e., within the electronics housing **3704** (FIGS. 37A-37B). In the illustrated embodiment, the first shield **3828a** is arranged above the PCB **3804** such that it faces the top surface of the PCB **3804**, and the second shield **3828b** is arranged below the PCB **3804** such that it faces the bottom surface of the PCB **3804**.

The shields **3828a,b** may be configured to protect sensitive electronic components from radiation while the sensor control device **3702** is subjected to radiation sterilization. More specifically, at least one of the shields **3828a,b** may be positioned to interpose the data processing unit **3808** and a radiation source, such as an e-beam electron accelerator. In some embodiments, for example, at least one of the shields **3828a,b** may be positioned adjacent to and otherwise aligned with the data processing unit **3808** and the radiation source to block or mitigate radiation absorbed dose that might otherwise damage the sensitive electronic circuitry of the data processing unit **3808**.

In the illustrated embodiment, the data processing unit **3808** interposes the first and second shields **3828a,b** such that the first and second shields **3828a,b** essentially bookend the data processing unit **3808** in the axial direction. In at least one embodiment, however, only one of the shields **3828a,b** may be necessary to properly protect the data processing unit **3808** during radiation sterilization. For example, if the sensor control device **3702** is subjected to radiation sterilization directed toward the bottom of the mount **3708**, only the second shield **3828b** may be needed to interpose the data processing unit **3808** and the radiation source, and the first shield **3828a** may be omitted. Alternatively, if the sensor control device **3702** is subjected to radiation sterilization directed toward the top of the shell **3706**, only the first shield **3828a** may be needed to interpose the data processing unit **3808** and the radiation source, and the second shield **3828b** may be omitted. In other embodiments, however, both shields **3828a,b** may be employed, without departing from the scope of the disclosure.

The shields **3828a,b** may be made of any material capable of attenuating (or substantially attenuating) the transmission of radiation. Suitable materials for the shields **3828a,b** include, but are not limited to, lead, tungsten, iron-based metals (e.g., stainless steel), copper, tantalum, tungsten, osmium, aluminum, carbon, or any combination thereof. Suitable materials for the shields **3828a,b** may be corrosion-resistant, austenitic, and any non-magnetic metal with a density ranging between about 2 grams per cubic centimeter (g/cc) and about 23 g/cc. The shields **3828a,b** may be

fabricated via a variety of manufacturing techniques including, but not limited to, stamping, casting, injection molding, sintering, two-shot molding, or any combination thereof.

In other embodiments, however, the shields **3828a,b** may comprise a metal-filled thermoplastic polymer such as, but not limited to, polyamide, polycarbonate, or polystyrene. In such embodiments, the shields **3828a,b** may be fabricated by mixing the shielding material in an adhesive matrix and dispensing the combination onto shaped components or otherwise directly onto the data processing unit **3808**. Moreover, in such embodiments, the shields **3828a,b** may comprise an enclosure that encapsulates (or substantially encapsulates) the data processing unit **3808**. In such embodiments, the shields **3828a,b** may comprise a metal-filled thermoplastic polymer, as mentioned above, or may alternatively be made of any of the materials mentioned herein that are capable of attenuating (or substantially attenuating) the transmission of radiation.

The shell **3706** may provide or otherwise define a first clocking receptacle **3830a** (FIG. **38B**) and a second clocking receptacle **3830b** (FIG. **38B**), and the mount **3708** may provide or otherwise define a first clocking post **3832a** (FIG. **38A**) and a second clocking post **3832b** (FIG. **38A**). Mating the first and second clocking receptacles **3830a,b** with the first and second clocking posts **3832a,b**, respectively, will properly align the shell **3706** to the mount **3708**.

Referring specifically to FIG. **38A**, the inner surface of the mount **3708** may provide or otherwise define a plurality of pockets or depressions configured to accommodate various component parts of the sensor control device **3702** when the shell **3706** is mated to the mount **3708**. For example, the inner surface of the mount **3708** may define a battery locator **3834** configured to accommodate a portion of the battery **3812** when the sensor control device **3702** is assembled. An adjacent contact pocket **3836** may be configured to accommodate a portion of the axial contact **3814a**.

Moreover, a plurality of module pockets **3838** may be defined in the inner surface of the mount **3708** to accommodate the various electronic modules **3806** arranged on the bottom of the PCB **3804**. Furthermore, a shield locator **3840** may be defined in the inner surface of the mount **3708** to accommodate at least a portion of the second shield **3828b** when the sensor control device **3702** is assembled. The battery locator **3834**, the contact pocket **3836**, the module pockets **3838**, and the shield locator **3840** all extend a short distance into the inner surface of the mount **3708** and, as a result, the overall height **H** (FIG. **37B**) of the sensor control device **3702** may be reduced as compared to prior sensor control devices. The module pockets **3838** may also help minimize the diameter of the PCB **3804** by allowing PCB components to be arranged on both sides (i.e., top and bottom surfaces).

Still referring to FIG. **38A**, the mount **3708** may further include a plurality of carrier grip features **3842** (two shown) defined about the outer periphery of the mount **3708**. The carrier grip features **3842** are axially offset from the bottom **3844** of the mount **3708**, where a transfer adhesive (not shown) may be applied during assembly. In contrast to prior sensor control devices, which commonly include conical carrier grip features that intersect with the bottom of the mount, the presently disclosed carrier grip features **3842** are offset from the plane (i.e., the bottom **3844**) where the transfer adhesive is applied. This may prove advantageous in helping ensure that the delivery system does not inadvertently stick to the transfer adhesive during assembly. Moreover, the presently disclosed carrier grip features **3842** eliminate the need for a scalloped transfer adhesive, which

simplifies the manufacture of the transfer adhesive and eliminates the need to accurately clock the transfer adhesive relative to the mount **3708**. This also increases the bond area and, therefore, the bond strength.

Referring to FIG. **38B**, the bottom **3844** of the mount **3708** may provide or otherwise define a plurality of grooves **3846**, which may be defined at or near the outer periphery of the mount **3708** and equidistantly spaced from each other. A transfer adhesive (not shown) may be coupled to the bottom **3844** and the grooves **3846** may be configured to help convey (transfer) moisture away from the sensor control device **3702** and toward the periphery of the mount **3708** during use. In some embodiments, the spacing of the grooves **3846** may interpose the module pockets **3838** (FIG. **38A**) defined on the opposing side (inner surface) of the mount **3708**. As will be appreciated, alternating the position of the grooves **3846** and the module pockets **3838** ensures that the opposing features on either side of the mount **3708** do not extend into each other. This may help maximize usage of the material for the mount **3708** and thereby help maintain a minimal height **H** (FIG. **37B**) of the sensor control device **3702**. The module pockets **3838** may also significantly reduce mold sink, and improve the flatness of the bottom **3844** that the transfer adhesive bonds to.

Still referring to FIG. **38B**, the inner surface of the shell **3706** may also provide or otherwise define a plurality of pockets or depressions configured to accommodate various component parts of the sensor control device **3702** when the shell **3706** is mated to the mount **3708**. For example, the inner surface of the shell **3706** may define an opposing battery locator **3848** arrangeable opposite the battery locator **3834** (FIG. **38A**) of the mount **3708** and configured to accommodate a portion of the battery **3812** when the sensor control device **3702** is assembled. Moreover, a shield locator **3850** may be defined in the inner surface of the shell **3706** to accommodate at least a portion of the first shield **3828a** when the sensor control device **3702** is assembled. The opposing battery locator **3848** and the shield locator **3850** extend a short distance into the inner surface of the shell **3706**, which helps reduce the overall height **H** (FIG. **37B**) of the sensor control device **3702**.

A sharp and sensor locator **3852** may also be provided by or otherwise defined on the inner surface of the shell **3706**. The sharp and sensor locator **3852** may be configured to receive both the sharp (not shown) and a portion of the sensor **3716**. Moreover, the sharp and sensor locator **3852** may be configured to align and/or mate with a corresponding sharp and sensor locator **2054** (FIG. **38A**) provided on the inner surface of the mount **3708**.

FIGS. **39A-39D** show progressive example assembly of the sensor control device **3702**, according to one or more embodiments. In FIG. **39A**, the battery **3812** has been loaded into the opposing battery locator **3848** and the first shield **3828a** has been loaded into the shield locator **3850** defined in the inner surface of the shell **3706**. The compliant member **3826** and the flag **3818** of the sensor **3716** may each be mounted to the first clocking receptacle **3830a**. The tail **3816** of the sensor **3716** may be inserted into the sharp and the sensor locator **3852**.

In FIG. **39B**, the PCB **3804** may be loaded into the shell **3706** to align the battery aperture **3810** with the battery **3812** and the axial and radial battery contacts **3814a,b** facilitate electrical communication.

In FIG. **39C**, the second shield **3828b** has been loaded into the shield locator **3840** defined in the inner surface of the mount **3708**. The mount **3708** is now ready to be coupled to the shell **3706** (FIGS. **39A** and **39B**). To accomplish this, the

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first and second clocking receptacles **3830a,b** (FIG. **39B**) of the shell **3706** may be coaxially aligned with the first and second clocking posts **3832a,b** of the mount **3708**, respectively. An adhesive may be applied to one or both of the shell **3706** and the mount **3708** to secure the two components together. In one embodiment, for example, the adhesive may be applied around the outer diameter (periphery) of the shell **3706**, and the shell **3706** may then be transferred to the mount **3708** and mated with the corresponding outer diameter (periphery) of the mount **3708**. In other embodiments, the adhesive may be applied around the outer diameter (periphery) of the mount **3708** or the outer diameter (periphery) of both the shell **3706** and the mount **3708**, without departing from the scope of the disclosure. In at least one embodiment, an adhesive may be used to secure the first and second clocking receptacles **3830a,b** to the first and second clocking posts **3832a,b**, respectively.

FIG. **39D** shows the assembled sensor control device **3702**, which may be tested to ensure the sensor **3716** and the corresponding electronics of the sensor control device **3702** function properly. The adhesive may not only secure the shell **3706** to the mount **3708** and provide structural integrity, but may also seal the interface between the two components and thereby isolate the interior of the electronics housing **3704** from outside contamination. Consequently, there may be no need to sterilize the internal electrical components of the sensor control device **3702** via gaseous chemical sterilization (e.g., ethylene oxide). Rather, the adhesive provides a sterile and moisture barrier to the interior of the assembled sensor control device **3702**.

The adhesive patch **3710** may be applied to the bottom **3844** of the mount **3708**. In some embodiments, the adhesive patch **3710** may have a removable release liner that is removed to enable the adhesive patch **3710** to be attached to the bottom **3844** of the mount **3708**.

Either before or after securing the adhesive patch **3710**, a sharp module **3904** may be coupled to the sensor control device **3702**. As illustrated, the sharp module **3904** may include a sharp hub **3906** and a sharp **3908** carried by the sharp hub **3906** and extending through the electronics housing **3704**. To couple the sharp module **3904** to the sensor control device **3702**, a sharp tip **3910** of the sharp **3908** may be extended through the coaxially aligned central apertures **3714**, **3720** (FIGS. **37A** and **37C**) of the shell **3706** and the mount **3708**, respectively. As the sharp tip **3910** penetrates the sensor control device **3702**, the tail **3816** may be received within a hollow or recessed portion of the sharp tip **3910**. The sharp tip **3910** may be configured to penetrate the skin while carrying the tail **3816** to put the active chemistry present on the tail **3816** into contact with bodily fluids.

The sharp tip **3910** may be advanced through the sensor control device **3702** until the sharp hub **3906** engages the upper surface of the shell **3706**. As illustrated, the sharp hub **3906** may include a hub small cylinder **3912** and a hub snap pawl **3914**, each of which may be configured to help couple the sensor control device **3702** to a sensor applicator (e.g., the sensor applicator **102** of FIG. **1**).

FIGS. **40A** and **40B** are side and cross-sectional side views, respectively, of the sensor applicator **102** sealed with the applicator cap **210**. According to the present disclosure, and as seen in FIG. **40B**, the sensor control device **3702** may already be assembled, as generally described above, and installed within the sensor applicator **102** prior to being delivered to a user. Accordingly, FIGS. **40A-40B** depict how the sensor applicator **102** might be shipped to and received by the user.

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The applicator cap **210** may be configured to provide a barrier against outside contamination, and thereby maintains a sterile environment for the assembled sensor control device **3702** positioned within the sensor applicator **102**. The applicator cap **210** may also create a dust-free environment during shipping and storage that prevents the adhesive patch **3710** (FIG. **40B**) from becoming dirty. The applicator cap **210** may be threaded to the housing **208** and include a tamper ring **4002**. Upon rotating (e.g., unscrewing) the applicator cap **210** relative to the housing **208**, the tamper ring **4002** may shear and thereby free the applicator cap **210** from the sensor applicator **102**.

As shown in FIG. **40B**, the sensor **3716** and the sharp **3908** are already incorporated into the assembled sensor control device **3702**. Consequently, there is no need for a two-piece architecture system that requires the sensor tray **202** (FIG. **2**) or a user to finally assemble the sensor control device **3702** as shown in and described with reference to FIGS. **2A-2D**. Rather, according to the present disclosure, the sensor control device **3702** may be fully sterilized while loaded in the sensor applicator **102** prior to being packaged for shipment to a user.

More specifically, the sensor control device **3702** may be subjected to radiation sterilization **4004** while loaded (positioned) within the sensor applicator **102** to sterilize the sensor **3716** and the sharp **3908**. The radiation sterilization **4004** may comprise, for example, e-beam irradiation, but other methods of sterilization may alternatively be used including, but not limited to, gamma ray irradiation, low energy X-ray irradiation, or any combination thereof.

In some embodiments, as illustrated, the radiation sterilization **4004** may be applied to the sensor control device **3702** through the applicator cap **210** and otherwise through a proximal end **4006** of the applicator cap **210**. The applicator cap **210** may be made of any material that allows radiation to pass therethrough. In at least one embodiment, for example, cap **210** may be made of a thermoplastic. The radiation sterilization **4004** may propagate through the applicator cap **210** and impinge upon the sensor control device **3702** to inactivate or kill microorganisms or other contaminants that may be present on the sensor **3716** and the sharp **3908**.

In some embodiments, the radiation sterilization **4004** may comprise electron beam (e-beam) irradiation. E-beam irradiation is a penetrating process that allows the sensor control device **3702** to be already mounted within the sensor applicator **102** before the irradiation process. By sterilizing the sensor control device **3702** after it has been packaged, the possibility of contamination during the time between sterilization and packaging is reduced.

FIGS. **41A** and **41B** are enlarged cross-sectional views of the sensor control device **3702** during example radiation sterilization **4004**, according to one or more embodiments of the present disclosure. In one aspect, one or more e-beam accelerators may be used to generate the radiation sterilization **4004** and, more particularly, to accelerate electrons into a concentrated highly charged electron stream. As materials pass through the stream of electrons, energy from the stream is absorbed and the absorption of this energy alters chemical and biological bonds. At certain levels of absorption, also known as the "absorbed dose," DNA chains and reproductive cells of microorganisms are destroyed, and thereby effectively sterilizing the target device or package. The irradiation dosage is important, as too low of a dosage may not result in complete sterilization, while too high of a dosage may result in adverse effects on the materials of the

sensor control device **3702** and the packaging (the applicator cap **210** of FIG. **40B**) being sterilized.

The electromagnetic shields **3828a,b** included within the sensor control device **3702** may prove advantageous in shielding and otherwise protecting sensitive electronic components, such as the data processing unit **3808**, while the sensor control device **3702** is subjected to the radiation sterilization **4004**.

In FIG. **41A**, one or both of the first and second shields **3828a,b** may help shield the data processing unit **3808** from the absorbed dose of radiation from the radiation sterilization **4004**. More specifically, the electromagnetic shields **3828a,b** may be aligned with and otherwise positioned to block or otherwise mitigate radiation exposure that might otherwise damage the data processing unit **3808**. In the illustrated embodiment, the radiation energy of the radiation sterilization **4004** propagates normal to the data processing unit **3808**, and at least the second shield **3828b** interposes the data processing unit **3808** and the source of the radiation sterilization **4004**.

In FIG. **41B**, the first shield **3828a** covers and otherwise encapsulates the data processing unit **3808** and thereby helps shield the data processing unit **3808** from the absorbed dose of radiation from the radiation sterilization **4004**. More specifically, by forming an enclosure around the data processing unit **3808**, the first shield **3828a** may be positioned to block or otherwise mitigate radiation exposure that might otherwise damage the data processing unit **3808**. In such embodiments, the second shield **3828b** may not be necessary.

The e-beam irradiation process of the radiation sterilization **4004** may include a continuous exposure or an intermittent exposure, and the e-beam accelerator may be of a continuous or a varying power, depending upon available machinery and determinations to achieve the desired internal and surface dosage limitations. The penetration power of e-beam irradiation correlates to the density of the underlying material being subjected to the radiation sterilization **4004** and the energy level of the e-beam accelerator. The larger and denser the material, the higher the energy the e-beam accelerator must output to achieve full penetration.

FIG. **42** is a plot **4200** that graphically depicts an approximation of penetration depth as a function of the energy level of e-beam radiation sterilization for unit density materials such as water. As indicated by the plot **4200**, the higher the energy level of the electrons of the e-beam radiation sterilization, the deeper the radiation will penetrate into a selected material. Most standard e-beam sterilization processes operate at a 10 mega electron-volt (MeV) energy level which, according to the plot **4200**, will penetrate into a given material about 3.8 cm for a unit density material such as water (density=1 g/cc).

According to embodiments of the present disclosure, e-beam sterilization (e.g., the radiation sterilization **4004** of FIGS. **40B** and **41A-41B**) may be undertaken at lower energy levels and nonetheless achieve comparable or commensurate sterilization dose achieved at high energy levels (e.g., 10 MeV or more). In some embodiments, for example, radiation sterilization may be undertaken at an energy level ranging between about 0.5 MeV and about 3.0 MeV and can achieve an equivalent dose to irradiating at higher energy levels. In yet other embodiments, the radiation sterilization may be undertaken at an energy level as low as 0.1 MeV, without departing from the scope of the disclosure.

According to the plot **4200**, dosing at an energy level ranging between about 0.5 MeV and about 3.0 MeV equates to a penetration depth ranging between about 0.2 cm and

about 1.0 cm for a material with density of 1 g/cc. Accordingly, at lower energy levels, it may be possible to shield sensitive electronic components with high density materials and small thicknesses such that little or no radiation penetrates the shield.

In view of the foregoing, the material and configuration of the shields **3828a,b** (FIGS. **41A-41B**) may be selected and optimized (tuned) in view of low energy radiation sterilization to protect the data processing unit **3808** (FIGS. **41A-41B**). The penetration depth for a given material may be determined for example in the range of 0.2 to 2.0 MeV, by Equation (1) below obtained from ISO/ASTM 51649: 2005 (E) "Standard Practice for Dosimetry in an Electron Beam Facility for Radiation Processing at Energies between 300 keV and 25 MeV."

$$Rp = \frac{(0.507E - 0.1243)}{\rho} \quad \text{Equation (1)}$$

where "E" is the energy level (MeV) of the e-beam accelerator and "p" is the density (g/cm³) of the given material. Equation 1 is derived from a Monte Carlo simulation for one-sided irradiation through polystyrene. As such, the computed penetration depth is an approximate value for polymeric and higher density materials. Based on the foregoing equation, Table 1 lists various materials that may be candidate materials for the shields **3828a,b**, their respective densities in g/cc, and their calculated penetration depth Rp at energy levels E of 1 MeV, 2 MeV, and 5 MeV:

TABLE 1

Element	Density (g/cc)	Penetration Depth (mm)		
		1 MeV	2 MeV	5 MeV
Carbon	2.3	1.69	3.94	10.67
Aluminum	2.7	1.42	3.30	8.93
Iron	7.9	0.49	1.13	3.06
Stainless Steel	8.1	0.47	1.10	2.99
Copper	8.9	0.43	1.00	2.71
Lead	11.4	0.34	0.78	2.12
Tantalum	16.7	0.23	0.53	1.45
Tungsten	19.4	0.20	0.46	1.25
Osmium	22.6	0.17	0.39	1.07

As indicated in Table 1, the higher the density of the material, the lower the penetration depth and, consequently, the thinner the material can be to adequately shield sensitive electronic components at lower energy levels. Moreover, the thinner the shield material, the thinner the product (e.g., the sensor control device **3702**) can be.

According to one or more embodiments of the present disclosure, the shields **3828a,b** that protect the data processing unit **3808** from radiation exposure may be any non-magnetic metal with a density of at least 2.0 g/cc. In other embodiments, the shields **3828a,b** may be a non-magnetic metal with a density of at least 5.0 g/cc. According to Table 1, suitable materials for the shields **3828a,b** can include, but are not limited to, iron, stainless steel, copper, lead, tantalum, tungsten, and osmium. Because of its low cost and availability, stainless steel may be a preferred material. In some embodiments, the material for the shields **3828a,b** may be any non-magnetic metal with a density ranging between about 2.0 g/cc and about 23.0 g/cc. In other embodiments, the material for the shields **3828a,b** may be a non-magnetic metal with a density ranging between about 5.0 g/cc and about 15.0 g/cc.

In other embodiments, the shields **3828a,b** that protect the data processing unit **3808** from radiation exposure may be a metal-filled thermoplastic polymer where the shielding metal exhibits a density of at least 2.0 g/cc. In such embodiments, the metal-filled thermoplastic polymer may be, but not limited to, polyamide, polycarbonate, or polystyrene. In such embodiments, the shields **3828a,b** may be fabricated by mixing the shielding material (metal) in an adhesive matrix and dispensing the combination onto shaped components or otherwise directly onto the data processing unit **3808**. Moreover, in such embodiments, the shield(s) **3828a,b** may comprise an enclosure that encapsulates (or substantially encapsulates) the data processing unit **3808**.

FIG. **43** is a cross-sectional view of the sensor control device **3702** mounted within the sensor applicator **102** with the applicator cap **210** secured thereto, according to one or more additional embodiments. Similar to the embodiments of FIGS. **41A-41B**, one or more shields may be used to protect sensitive electronic components of the sensor control device **3702**. Unlike the embodiments of FIGS. **41A-41B**, however, the shields of FIG. **43** are magnetic shields configured to divert propagating radiation from the radiation sterilization **4004** (FIGS. **40B** and **41A-41B**) away from or otherwise around the data processing unit **3808**.

More specifically, it is possible to locally deflect an electron beam away from a component of interest, such as the data processing unit **3808**, by generating a static magnetic field. Charged particles experience a force when travelling through a magnetic field, and the direction of this force is perpendicular to the direction of the field and the velocity of the charge. In equation form, a particle with mass m and charge q moving with velocity v in a magnetic field B experiences a force characterized by the following equation:

$$F = qv \times B \quad \text{Equation (2)}$$

This is a vector equation which indicates that the magnitude of the force F is:

$$F = (qvB) \sin \theta \quad \text{Equation (3)}$$

where θ is the angle between the velocity v and the magnetic field B , and the direction of the force is perpendicular to both the velocity v and the magnetic field B (in a sense given by the right hand rule). An electron (charge $-e$) injected into a uniform magnetic field B and moving perpendicular to the field B experiences a force:

$$F = -evB \quad \text{Equation (4)}$$

Now the force F remains perpendicular to the velocity v and the electron moves in a circular path of radius R . The radial (centripetal) acceleration is then:

$$a = -\frac{v^2}{R} \quad \text{Equation (5)}$$

Now apply Newton's second law of motion:

$$F = ma \quad \text{Equation (6)}$$

$$evB = m \frac{v^2}{R} \quad \text{Equation (7)}$$

Thus, the radius R of the electron's path is:

$$R = \frac{mv}{eB} \quad \text{Equation (8)}$$

Accordingly, an electron having a mass m with a charge e and traveling at a velocity v through a magnetic field B , perpendicular to the direction of the velocity v , will be deflected in a circle of radius R and at a tangent to this circle once outside the influence of the magnetic field B . The magnetic field may be placed (generated) anywhere along the path of the propagating radiation (e.g., the e-beam) before it can strike the component of interest (e.g., the data processing unit **3808**).

In one embodiment, a first magnet **4302a** may be arranged within the electronics housing **3704** adjacent the data processing unit **3808** to generate a static magnetic field. In the illustrated embodiment, the first magnet **4302a** is arranged where the second shield **3828b** of FIGS. **41A-41B** was placed. In such embodiments, a propagating radiation beam **4304** (e.g., e-beam) may pass through the first magnet **4302a** and the static magnetic field generated by the first magnet **4302a** will cause the radiation beam **4304** to be diverted away from the data processing unit **3808**.

In another embodiment, or in addition thereto, a second magnet **4302b** may be arranged within the applicator cap **210** to generate a static magnetic field. In the illustrated embodiment, the second magnet **4302b** is positioned to interpose the radiation source (e.g., an e-beam accelerator) and the data processing unit **3808**. A propagating radiation beam **4306** (e.g., e-beam) may pass through the second magnet **4302b** and the static magnetic field generated by the second magnet **4302b** will cause the radiation beam **4306** to be diverted away from the data processing unit **3808**.

In yet other embodiments, or in addition thereto, a third magnet **4302c** may be arranged external to the applicator cap **210** and the sensor applicator **102** to generate a static magnetic field. In the illustrated embodiment, the third magnet **4302c** is positioned outside of the applicator cap **210** and otherwise interposes the radiation source (e.g., an e-beam accelerator) and the data processing unit **3808**. A propagating radiation beam **4308** (e.g., e-beam) may pass through the third magnet **4302c** and the static magnetic field generated by the third magnet **4302c** will cause the radiation beam **4308** to be diverted away from the data processing unit **3808**.

As will be appreciated, precise alignment of the magnets **4302a-c** relative to sensor control device **3702** would need to be taken into consideration and sufficient margin be applied to the location and field strength accordingly.

Embodiments disclosed herein include:

Q. A sensor control device that includes an electronics housing, a printed circuit board positioned within the electronics housing and having a data processing unit mounted thereto, a sensor extending from a bottom of the electronics housing, a sharp module removably coupled to the electronics housing and having a sharp that extends through the electronics housing and receives a portion of the sensor extending from the bottom of the electronics housing, and at least one shield positioned within the electronics housing to protect the data processing unit from radiation from a radiation sterilization process.

R. An analyte monitoring system that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing, a

printed circuit board positioned within the electronics housing and having a data processing unit mounted thereto, a sensor extending from a bottom of the electronics housing, a sharp module removably coupled to the electronics housing and having a sharp that extends through the electronics housing and receives a portion of the sensor extending from the bottom of the electronics housing, and at least one shield positioned within the electronics housing to protect the data processing unit from radiation from a radiation sterilization process. The analyte monitoring system further including a cap coupled to the sensor applicator to provide a barrier that seals the sensor control device within the sensor applicator.

S. A method of preparing an analyte monitoring system including loading a sensor control device into a sensor applicator, the sensor control device including an electronics housing, a printed circuit board positioned within the electronics housing and having a data processing unit mounted thereto, a sensor extending from a bottom of the electronics housing, a sharp module removably coupled to the electronics housing and having a sharp that extends through the electronics housing and receives a portion of the sensor extending from the bottom of the electronics housing, and at least one shield positioned within the electronics housing. The method further including securing a cap to the sensor applicator and thereby providing a barrier that seals the sensor control device within the sensor applicator, sterilizing the sensor and the sharp with radiation sterilization while the sensor control device is positioned within the sensor applicator, and shielding the data processing unit with the at least one shield from radiation from the radiation sterilization.

T. A sensor control device that includes an electronics housing having a shell matable with a mount, a printed circuit board positioned within the electronics housing and defining a battery aperture sized to receive a battery, an axial battery contact extending into the battery aperture to provide electrical communication, and a radial battery contact extending into the battery aperture to provide electrical communication.

Each of embodiments Q, R, S, and T may have one or more of the following additional elements in any combination: Element 1: further comprising a battery aperture defined in the printed circuit board, a battery received within the battery aperture, an axial battery contact coupled to the printed circuit board and extending into the battery aperture to facilitate electrical communication, and a radial battery contact coupled to the printed circuit board and extending into the battery aperture to facilitate electrical communication. Element 2: further comprising one or more sensor contacts arranged on a flag of the sensor, and one or more circuitry contacts provided on the printed circuit board and engageable with the one or more sensor contacts to facilitate direct connection between the sensor and the printed circuit board. Element 3: wherein the at least one shield interposes the data processing unit and a radiation source that facilitates radiation sterilization. Element 4: wherein the at least one shield comprises a first shield facing a bottom of the printed circuit board and a second shield facing a top of the printed circuit board, and wherein the data processing unit interposes the first and second shields. Element 5: wherein the at least one shield comprises an enclosure that encapsulates the data processing unit. Element 6: wherein the at least one shield is made of a non-magnetic metal that exhibits a density ranging between about 2 g/cc and about 23 g/cc. Element 7: wherein the at least one shield is made of thermoplastic polymer mixed with a non-magnetic metal having a density of at least 2.0 g/cc. Element 8: further comprising a plurality of electronic modules coupled to top

and bottom surfaces of the printed circuit board. Element 9: wherein the electronics housing comprises a mount and the shell secured together and sealed with an adhesive. Element 10: wherein the at least one shield comprises a magnet arranged to divert the radiation away from the data processing unit.

Element 11: wherein the at least one shield interposes the data processing unit and a radiation source that facilitates radiation sterilization of the sensor and the sharp. Element 12: wherein the at least one shield is made with a non-magnetic metal having a density of at least 2.0 g/cc. Element 13: wherein the sensor control device is subjected to the radiation sterilization while positioned within the sensor applicator and at an energy level ranging between about 0.1 MeV and about 10.0 MeV. Element 14: wherein the at least one shield comprises a magnet arranged to divert the radiation away from the data processing unit.

Element 15: wherein the at least one shield interposes the data processing unit and a radiation source that facilitates the radiation sterilization, and wherein the at least one shield is made with a non-magnetic metal having a density of at least 2.0 g/cc, the method further comprising undertaking the radiation sterilization at an energy level ranging between about 0.1 MeV and about 10.0 MeV. Element 16: wherein the electronics housing comprises a shell matable with a mount, and wherein loading the sensor control device into the sensor applicator is preceded by sealing the shell to the mount with an adhesive and thereby generating a sterile barrier. Element 17: wherein the at least one shield comprises a magnet, and wherein shielding the data processing unit with the at least one shield comprises generating a static magnetic field with the magnet, and diverting the radiation away from the data processing unit with the static magnetic field.

Element 18: further comprising a plurality of electronic modules coupled to top and bottom surfaces of the printed circuit board. Element 19: wherein a plurality of module pockets are defined in an inner surface of the mount to accommodate the plurality of electronic modules. Element 20: wherein the mount and the shell are secured together and sealed with an adhesive. Element 21: wherein the shell defines a reference feature extending a short distance into an interior of the electronics housing. Element 22: further comprising an adhesive patch positioned on an underside of the mount. Element 23: wherein the shell defines an angled outer periphery. Element 24: further comprising a sensor partially arranged within the electronics housing and having a flag with one or more sensor contacts, and a compliant member arranged to interpose the flag and an inner surface of the shell and provide a passive biasing load against the flag to force the one or more sensor contacts into engagement with a corresponding one or more circuitry contacts provided on the printed circuit board. Element 25: wherein the compliant member comprises an elastomeric O-ring. Element 26: further comprising at least one shield positioned within the electronics housing, and a shield locator defined in an inner surface of the shell or the mount to accommodate at least a portion of the at least one shield. Element 27: wherein the at least one shield comprises a first shield and a second shield, and wherein the shield locator comprises a first shield locator defined in an inner surface of the shell to accommodate at least a portion of the first shield, and a second shield locator defined in an inner surface of the mount to accommodate at least a portion of the second shield. Element 28: further comprising one or more clocking receptacles defined on one of the mount or the shell, and one or more clocking posts defined on the other of the mount or

the shell and sized to be received within the one or more clocking receptacles to properly align the shell to the mount. Element 29: wherein a battery locator is defined in an inner surface of at least one of the shell and the mount and sized to accommodate a portion of the battery. Element 30: wherein the inner surface of the at least one of the shell and the mount further defines a contact pocket adjacent the battery locator and sized to accommodate a portion of the axial contact. Element 31: further comprising a plurality of carrier grip features defined about an outer periphery of the mount and axially offset from a bottom of the mount.

By way of non-limiting example, exemplary combinations applicable to Q, R, S, and T include: Element 3 with Element 4; Element 12 with Element 13; Element 18 and Element 19; Element 20 and Element 21; Element 24 and Element 25; Element 26 and Element 27; and Element 28 and Element 30.

One-Piece Analyte Monitoring Systems with Sensor Cap

Referring briefly again to FIGS. 1 and 2A-2G, for the two-piece architecture system, the sensor tray 202 and the sensor applicator 102 are provided to the user as separate packages, thus requiring the user to open each package and finally assemble the system. In some applications, the discrete, sealed packages allow the sensor tray 202 and the sensor applicator 102 to be sterilized in separate sterilization processes unique to the contents of each package and otherwise incompatible with the contents of the other. More specifically, the sensor tray 202, which includes the plug assembly 207, including the sensor 110 and the sharp 220, may be sterilized using radiation sterilization, such as electron beam (or "e-beam") irradiation. Radiation sterilization, however, can damage the electrical components arranged within the electronics housing of the sensor control device 104. Consequently, if the sensor applicator 102, which contains the electronics housing of the sensor control device 104, needs to be sterilized, it may be sterilized via another method, such as gaseous chemical sterilization using, for example, ethylene oxide. Gaseous chemical sterilization, however, can damage the enzymes or other chemistry and biologics included on the sensor 110. Because of this sterilization incompatibility, the sensor tray 202 and the sensor applicator 102 are commonly sterilized in separate sterilization processes and subsequently packaged separately, which requires the user to finally assemble the components for use.

According to embodiments of the present disclosure, the sensor control device 104 may be modified to provide a one-piece architecture that may be subjected to sterilization techniques specifically designed for a one-piece architecture sensor control device. A one-piece architecture allows the sensor applicator 102 and the sensor control device 104 to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package and subsequently deliver the sensor control device 104 to the target monitoring location. The one-piece system architecture described herein may prove advantageous in eliminating component parts, various fabrication process steps, and user assembly steps. As a result, packaging and waste are reduced, and the potential for user error or contamination to the system is mitigated.

FIG. 44 is a side view of an example sensor control device 4402, according to one or more embodiments of the present disclosure. The sensor control device 4402 may be similar in some respects to the sensor control device 104 of FIG. 1 and therefore may be best understood with reference thereto. Moreover, the sensor control device 4402 may replace the sensor control device 104 and, therefore, may be used in conjunction with the sensor applicator 102 of FIG. 1, which

may deliver the sensor control device 4402 to a target monitoring location on a user's skin.

Unlike the sensor control device 104 of FIG. 1, however, the sensor control device 4402 may comprise a one-piece system architecture not requiring a user to open multiple packages and finally assemble the sensor control device 4402 prior to application. Rather, upon receipt by the user, the sensor control device 4402 may already be fully assembled and properly positioned within the sensor applicator 102 (FIG. 1). To use the sensor control device 4402, the user need only open one barrier (e.g., the applicator cap 210 of FIG. 2B) before promptly delivering the sensor control device 4402 to the target monitoring location for use.

As illustrated, the sensor control device 4402 includes an electronics housing 4404 that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing 4404 may exhibit other cross-sectional shapes, such as ovoid or polygonal, without departing from the scope of the disclosure. The electronics housing 4404 may be configured to house or otherwise contain various electrical components used to operate the sensor control device 4402. In at least one embodiment, an adhesive patch 4405 may be arranged at the bottom of the electronics housing 4404. The adhesive patch 4405 may be similar to the adhesive patch 108 of FIG. 1, and may thus help adhere the sensor control device 4402 to the user's skin for use.

The electronics housing 4404 may include a shell 4406 and a mount 4408 that is matable with the shell 4406. The shell 4406 may be secured to the mount 4408 via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, one or more mechanical fasteners (e.g., screws), a gasket, an adhesive, or any combination thereof. In some cases, the shell 4406 may be secured to the mount 4408 such that a sealed interface therebetween is generated. In such embodiments, a seal member 4409, such as a gasket or an adhesive, may be positioned at or near the outer diameter (periphery) of the shell 4406 and the mount 4408, and securing the two components together may compress the seal member 4409 and thereby generate a sealed interface. The seal member 4409 secures the shell 4406 to the mount 4408 and provides structural integrity, but may also isolate the interior of the electronics housing 4404 from outside contamination. If the sensor control device 4402 is assembled in a controlled environment, there may be no need to terminally sterilize the internal electrical components. Rather, the sealed interface may provide a sufficient sterile barrier for the assembled electronics housing 4404.

The sensor control device 4402 may further include a sensor 4410 (partially visible) and a sharp 4412 (partially visible) used to help deliver the sensor 4410 transcutaneously under a user's skin during application of the sensor control device 4402. As illustrated, corresponding portions of the sensor 4410 and the sharp 4412 extend distally from the electronics housing 4404 and, more particularly, from the bottom of the mount 4408. The sharp 4412 may include a sharp hub 4414 configured to secure and carry the sharp 4412. To couple the sharp 4412 to the sensor control device 4402, the sharp 4412 may be advanced axially through the electronics housing 4404 until the sharp hub 4414 engages an upper portion of the shell 4406. As the sharp 4412 penetrates the electronics housing 4404, the exposed portion of the sensor 4410 may be received within a hollow or recessed (arcuate) portion of the sharp 4412. The remaining portion of the sensor 4410 is arranged within the interior of the electronics housing 4404.

The sensor control device **4402** may further include a sensor cap **4416**, as shown exploded (detached). The sensor cap **4416** may be removably coupled to the sensor control device **4402** (e.g., the electronics housing **4404**) at or near the bottom of the mount **4408**. As illustrated, the sensor cap **4416** may comprise a generally cylindrical and elongate body having a first end **4418a** and a second end **4418b** opposite the first end **4418a**. The first end **4418a** may be open to provide access into an inner chamber **4420** defined within the body. In contrast, the second end **4418b** may be closed and may provide or otherwise define an engagement feature **4422**. As described herein, the engagement feature **4422** may be configured to help the sensor cap **4416** mate with the cap (e.g., the applicator cap **210** of FIG. 2B) of a sensor applicator (e.g., the sensor applicator **102** of FIGS. 1 and 2A-2G) such that the sensor cap **4416** is removed from the sensor control device **4402** upon removing the cap from the sensor applicator. While the engagement feature **4422** is shown at or near the second end **4418b** of the sensor cap **4416**, the engagement feature **4422** may alternatively be positioned at an intermediate location between the first and second ends **4418a,b**.

As discussed in more detail below, the sensor cap **4416** may provide a sealed barrier surrounding and protecting the exposed portions of the sensor **4410** and the sharp **4412** from gaseous chemical sterilization. The sensor cap **4416** helps form a sealed sub-assembly that can first be sterilized using radiation sterilization, following which components of the sensor control device **4402** that are sensitive to radiation sterilization may be assembled to the sealed subassembly and then subjected to gaseous chemical sterilization.

FIG. 45 is an exploded view of the sensor control device **4402**, according to one or more embodiments. The shell **4406** and the mount **4408** operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device **4402**. The adhesive patch **4405** may be applied to a bottom **4501** of the mount **4408**.

As illustrated, the shell **4406** may provide or otherwise define a sharp and sensor locator **4502** and a clocking receptacle **4504**. The sharp and sensor locator **4502** may be configured to receive portions of both the sharp **4412** and the sensor **4410**. Moreover, the sharp and sensor locator **4502** may be configured to align with and be partially received within a central aperture **4506** defined in the mount **4408**. Similarly, the clocking receptacle **4504** may be configured to align with and be received within a clocking post (not shown) defined on the inner surface of the mount **4408**. Mating the sharp and sensor locator **4502** with the central aperture **4506**, and simultaneously mating the clocking receptacle **4504** with the clocking post may help axially and rotationally align the shell **4406** with the mount **4408**.

In some embodiments, a first seal member **4508a** (i.e., the seal member **4409** of FIG. 44) may be applied to one or both of the shell **4406** and the mount **4408** to secure the two components together. As illustrated, the first seal member **4508a** may be applied around the outer diameter (periphery) of the shell **4406**, the mount **4408**, or both. In another embodiment, or in addition thereto, a second seal member **4508b** may be used to seal the interface between the sharp and sensor locator **4502** and the central aperture **4506**. More specifically, the second seal member **4508b** may be configured to provide a sealed interface at an annular ridge **4510** that circumscribes the sharp and sensor locator **4502**. When the shell **4406** and the mount **4408** are mated, the annular ridge **4510** may juxtapose an opposing surface defined on the bottom of the mount **4408**, and the seal member **4508b**

may facilitate a seal between the opposing structures. The seal members **4508a,b** may comprise, for example, an adhesive or a gasket, and each may help secure the shell **4406** to the mount **4408** and seal the interface between the two components, and thereby isolate the interior of the electronics housing **4404** (FIG. 44) from outside contamination.

The sensor control device **4402** may include a printed circuit board (PCB) **4516** that may be arranged within the interior cavity formed by mating the shell **4406** and the mount **4408**. A data processing unit **4518** and a battery **4520** may be mounted to or otherwise interact with the PCB **4516**. The data processing unit **4518** may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **4402**. More specifically, the data processing unit **4518** may be configured to perform data processing functions, where such functions may include, but are not limited to, filtering and encoding of data signals, each of which corresponds to a sampled analyte level of the user. The data processing unit **4518** may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. 1).

The battery **4520** may provide power to the sensor control device **4402** and, more particularly, to the electronic components of the PCB **4516**. While not shown in FIG. 45, other electronic modules or components may be mounted to the PCB **4516** and may include, but are not limited to, one or more resistors, transistors, capacitors, inductors, diodes, and switches.

The sensor control device **4402** may provide or otherwise include a sealed subassembly **4522** (outlined in dashed lines), which includes (among other component parts) the shell **4406**, the sensor **4410**, the sharp **4412**, and the sensor cap **4416**. As discussed in more detail below, the sealed subassembly **4522** may help isolate the sensor **4410** and the sharp **4412** within the inner chamber **4420** of the sensor cap **4416** during a gaseous chemical sterilization process, which might otherwise adversely affect the chemistry provided on the sensor **4410**.

As illustrated, the sensor **4410** may include a tail **4524**, a flag **4526**, and a neck **4528** that interconnects the tail **4524** and the flag **4526**. The tail **4524** may be configured to extend through the central aperture **4506** of the mount **4408** to be transcutaneously received beneath a user's skin. Moreover, the tail **4524** may have an enzyme or other chemistry included thereon to help facilitate analyte monitoring. The flag **4526** may include a generally planar surface having one or more sensor contacts **4530** (three shown) configured to align with and engage a corresponding one or more circuitry contacts (not shown) provided on the PCB **4516**. In some embodiments, the sensor contacts **4530** may comprise a carbon impregnated polymer printed or otherwise digitally applied to the flag **4526**.

In assembling the sealed subassembly, the flag **4526** may be received at the clocking receptacle **4504** and the tail **4524** may be received within the sharp and sensor locator **4502**. In some embodiments, a groove **4532** may be defined in the annular ridge **4510** to receive and seat the neck **4528**, and may allow the neck **4528** to be sealed below and on top and thereby isolate the enzymes and other chemistry included on the tail **4524**.

The sensor control device **4402** may further include a compliant member **4534** receivable by the clocking receptacle **4504** and arranged to interpose the flag **4526** and the inner surface of the shell **4406**. The compliant member **4534** may be configured to provide a passive biasing load against

the flag 4526 that forces the sensor contacts 4530 into continuous engagement with the corresponding circuitry contacts on the PCB 4516. In the illustrated embodiment, the compliant member 4534 is an elastomeric O-ring, but could alternatively comprise any other type of biasing device or mechanism, such as a compression spring or the like. In other embodiments, however, the compliant member 4534 may form an integral part of the shell 4406, such as being an overmolded or co-molded portion of the shell 4406.

The sharp 4412 may include a sharp tip 4536 extendable through the coaxially aligned sharp and sensor locator 4502 and the central aperture 4506 of the shell 4406 and the mount 4408, respectively. In some embodiments, as the sharp tip 4536 extends through the sensor control device 4402, the tail 4524 of the sensor 4410 may be received within a hollow or recessed portion of the sharp tip 4536. The sharp tip 4536 may be configured to penetrate the skin while carrying the tail 4524 to put the active chemistry of the tail 4524 into contact with bodily fluids. The sharp tip 4536 may be advanced through the sensor control device 4402 until the sharp hub 4414 engages an upper surface of the shell 4406. In some embodiments, the sharp hub 4414 may form a sealed interface at the upper surface of the shell 4406.

In the illustrated embodiment, the sealed subassembly 4522 may further include a collar 4540 that provides or otherwise defines a column 4542 and an annular shoulder 4544 extending radially outward from the column 4542. In assembling the sealed subassembly 4522, at least a portion of the column 4542 may be received within the inner chamber 4420 of the sensor cap 4416 at the first end 4418a. The sensor cap 4416 may be removably coupled to the collar 4540 and separated from the collar 4540 prior to delivering the sensor control device 4402 to the target monitoring location on the user's skin. In some embodiments, the sensor cap 4416 may be removably coupled to the collar 4540 via an interference or friction fit. In other embodiments, the sensor cap 4416 may be threaded to the column 4542. In yet other embodiments, the sensor cap 4416 may be removably coupled to the collar 4540 with a frangible member (e.g., a shear ring) or substance that may be broken with minimal separation force (e.g., axial or rotational force). In such embodiments, for example, the sensor cap 4416 may be secured to the collar 4540 with a tag (spot) of glue or a dab of wax.

In some embodiments, a third seal member 4508c may interpose the annular shoulder 4544 and the annular ridge 4510 to form a sealed interface. In such embodiments, the third seal member 4508c may also extend (flow) into the groove 4532 defined in the annular ridge 4510 and thereby seal about the neck 4528 of the sensor 4410. Similar to the first and second seal members 4508a,b, the third seal member 4508c may comprise an adhesive or a gasket.

In some embodiments, however, the collar 4540 may be omitted from the sealed subassembly 4522 and the sensor cap 4416 may alternatively be removably coupled to the sharp and sensor locator 4502. In such embodiments, the sensor cap 4416 may be removably coupled to the sharp and sensor locator 4502 via an interference or friction fit, threading, with a frangible member or substance, or any combination thereof.

FIG. 46A is a cross-sectional side view of the assembled sealed subassembly 4522 of FIG. 45, according to one or more embodiments. To assemble the sealed subassembly 4522, the compliant member 4534 may first be received about the clocking receptacle 4504 and the flag 4526 of the sensor 4410 may subsequently be placed atop the compliant member 4534 and also about the clocking receptacle 4504.

Alternatively, the compliant member 4534 may form part of the shell 4406 (e.g., co-molded, overmolded, etc.) at the clocking receptacle 4504, and the flag 4526 may be arranged thereon. The tail 4524 of the sensor 4410 may be received within the sharp and sensor locator 4502, and the neck 4528 may be seated within the groove 4532 defined in the annular ridge 4510.

The collar 4540 may then be extended over the sharp and sensor locator 4502 until the annular shoulder 4544 rests against the annular ridge 4510. In some embodiments, the third seal member 4508c may interpose the annular shoulder 4544 and the annular ridge 4510 to form a sealed interface, and the third seal member 4508c may also extend (flow) into the groove 4532 to form a seal about the neck 4528. The sensor cap 4416 may then be removably coupled to the collar 4540, as generally described above, such that portions of one or both of the collar 4540 and the sharp and sensor locator 4502 are received within the inner chamber 4420. In some embodiments, however, the collar 4540 may be omitted and the sensor cap 4416 may instead be received on the sharp and sensor locator 4502 and the third seal member 4508c may seal the interface(s) between the sensor cap 4416 and the sharp and sensor locator 4502.

Before or after assembling the sensor cap 4416, the sharp 4412 may be coupled to the sensor control device 4402 by extending the sharp tip 4536 through an aperture 4602 defined in the top of the shell 4406 and advancing the sharp 4412 through the sharp and sensor locator 4502 until the sharp hub 4414 engages a top surface of the shell 4406. In the illustrated embodiment, the top surface where the sharp hub 4414 engages the shell 4406 comprises a recessed portion of the shell 4406, but could alternatively comprise an upper surface that is level with adjacent portions of the shell 4406.

The inner chamber 4420 may be sized and otherwise configured to receive the tail 4524 and the sharp tip 4536. Moreover, the inner chamber 4420 may be sealed to isolate the sensor 4410 from substances that might adversely interact with the chemistry of the tail 4524. More specifically, the inner chamber 4420 may be sealed at the interface between the hub 4414 and the shell 4406, at the interface between the annular shoulder 4544 and the annular ridge 4510 (e.g., with the third seal member 4508c), and at the interface between the sensor cap 4416 and the collar 4540 (e.g., via an interference fit or the like). In some embodiments, a desiccant 4603 may be present within the inner chamber 4420 to maintain preferred humidity levels.

Once properly assembled, the sealed subassembly 4522 may be subjected to radiation sterilization to properly sterilize the sensor 4410 and the sharp 4412. Advantageously, this sterilization step may be undertaken apart from the other component parts of the sensor control device 4402 (FIG. 45) since radiation sterilization can damage sensitive electrical components associated with the PCB 4516 (FIG. 45), such as the data processing unit 4518 (FIG. 45).

Suitable radiation sterilization processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. In some embodiments, the sealed subassembly 4522 may be subjected to radiation sterilization prior to coupling the sensor cap 4416 to the collar 4540 (or the sharp and sensor locator 4502). In other embodiments, however, the sealed subassembly 4522 may be sterilized after coupling the sensor cap 4416 to the collar 4540 (or the sharp and sensor locator 4502). In such embodiments, the body of the sensor cap 4416 may comprise a material that permits propagation of radiation therethrough to facilitate radiation

sterilization of the distal portions of the sensor **4410** and the sharp **4412**. Suitable materials include, but are not limited to, a non-magnetic metal (e.g., aluminum, copper, gold, silver, etc.), a thermoplastic, ceramic, rubber (e.g., ebonite), a composite material (e.g., fiberglass, carbon fiber reinforced polymer, etc.), an epoxy, or any combination thereof. In some embodiments, the sensor cap **4416** may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

FIG. **46B** is a cross-sectional side view of the fully assembled sensor control device **4402**, according to one or more embodiments. Once assembled and properly sterilized, as discussed above, the sealed subassembly **4522** of FIG. **46A** may be assembled to the remaining component parts of the sensor control device **4402**. The PCB **4516** may be positioned within the shell **4406**, and the mount **4408** may subsequently be secured to the shell **4406**. To axially and rotationally align the shell **4406** with the mount **4408**, the sensor cap **4416** may be aligned with and extended through the central aperture **4506** of the mount **4408**. The sharp and sensor locator **4502** may then be received within the central aperture **4506**, and the clocking receptacle **4504** may be mated with a clocking post **4604** defined by the mount **4408**.

As discussed above, the first and second seal members **4508a,b** may be used to secure the mount **4408** to the shell **4406** and also isolate the interior of the electronics housing **4404** from outside contamination. In the illustrated embodiment, the second seal member **4508b** may interpose the annular shoulder **4544** of the collar **4540** and a portion of the mount **4408** and, more particularly, the central aperture **4506**. The adhesive patch **4405** may then be applied to the bottom **4501** of the mount **4408**.

FIGS. **47A** and **47B** are side and cross-sectional side views, respectively, of an example embodiment of the sensor applicator **102** with the applicator cap **210** coupled thereto. More specifically, FIG. **47A** depicts how the sensor applicator **102** might be shipped to and received by a user, and FIG. **47B** depicts the sensor control device **4402** arranged within the sensor applicator **102**. Accordingly, the fully assembled sensor control device **4402** may already be assembled and installed within the sensor applicator **102** prior to being delivered to the user, thus removing any additional assembly steps that a user would otherwise have to perform.

The fully assembled sensor control device **4402** may be loaded into the sensor applicator **102**, and the applicator cap **210** may subsequently be coupled to the sensor applicator **102**. In some embodiments, the applicator cap **210** may be threaded to the housing **208** and include a tamper ring **4702**. Upon rotating (e.g., unscrewing) the applicator cap **210** relative to the housing **208**, the tamper ring **4702** may shear and thereby free the applicator cap **210** from the sensor applicator **102**.

According to the present disclosure, while loaded in the sensor applicator **102**, the sensor control device **4402** may be subjected to gaseous chemical sterilization **4704** configured to sterilize the electronics housing **4404** and any other exposed portions of the sensor control device **4402**. To accomplish this, a chemical may be injected into a sterilization chamber **4706** cooperatively defined by the sensor applicator **102** and the interconnected cap **210**. In some applications, the chemical may be injected into the sterilization chamber **4706** via one or more vents **4708** defined in the applicator cap **210** at its proximal end **610**. Example chemicals that may be used for the gaseous chemical sterilization **4704** include, but are not limited to, ethylene oxide,

vaporized hydrogen peroxide, nitrogen oxide (e.g., nitrous oxide, nitrogen dioxide, etc.), and steam.

Since the distal portions of the sensor **4410** and the sharp **4412** are sealed within the sensor cap **4416**, the chemicals used during the gaseous chemical sterilization process do not interact with the enzymes, chemistry, and biologics provided on the tail **4524** and other sensor components, such as membrane coatings that regulate analyte influx.

Once a desired sterility assurance level has been achieved within the sterilization chamber **4706**, the gaseous solution may be removed and the sterilization chamber **4706** may be aerated. Aeration may be achieved by a series of vacuums and subsequently circulating a gas (e.g., nitrogen) or filtered air through the sterilization chamber **4706**. Once the sterilization chamber **4706** is properly aerated, the vents **4708** may be occluded with a seal **4712** (shown in dashed lines).

In some embodiments, the seal **4712** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can be applied before the gaseous chemical sterilization process, and following the gaseous chemical sterilization process, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization chamber **4706**. In other embodiments, the seal **4712** may comprise only a single protective layer applied to the applicator cap **210**. In such embodiments, the single layer may be gas permeable for the sterilization process, but may also be capable of protection against moisture and other harmful elements once the sterilization process is complete.

With the seal **4712** in place, the applicator cap **210** provides a barrier against outside contamination, and thereby maintains a sterile environment for the assembled sensor control device **4402** until the user removes (unthreads) the applicator cap **210**. The applicator cap **210** may also create a dust-free environment during shipping and storage that prevents the adhesive patch **4714** from becoming dirty.

FIG. **48** is a perspective view of an example embodiment of the applicator cap **210**, according to the present disclosure. As illustrated, the applicator cap **210** is generally circular and defines a series of threads **4802** used to couple the applicator cap **210** to the sensor applicator **102** (FIGS. **47A** and **47B**). The vents **4708** discussed above are also visible in the bottom of the applicator cap **210**.

The applicator cap **210** may further provide and otherwise define a cap post **4804** centrally located within the interior of the applicator cap **210** and extending proximally from the bottom thereof. The cap post **4804** may be configured to receive the sensor cap **4416** (FIGS. **44**, **45**, **46A-46B**) upon coupling the applicator cap **210** to the sensor applicator **102**. More specifically, the cap post **4804** may define a receiver feature **4806** configured to interact with (e.g., receive) the engagement feature **4422** (FIG. **44**) of the sensor cap **4416**. Upon removing the applicator cap **210** from the sensor applicator **102**, however, the receiver feature **4806** may retain the engagement feature **4422** and thereby prevent the sensor cap **4416** from separating from the cap post **4804**. Consequently, removing the applicator cap **210** from the sensor applicator **102** will simultaneously detach the sensor cap **4416** from the sensor control device **4402** (FIG. **47B**), and thereby expose the distal portions of the sensor **4410** (FIG. **47B**) and the sharp **4412** (FIG. **47B**).

As will be appreciated, many design variations of the engagement and receiver features **4422**, **4806** may be employed, without departing from the scope of the disclosure. Any design may be used that allows the engagement feature **4422** to be received by the receiver feature **4806** upon coupling the applicator cap **210** to the sensor applicator **102**, and subsequently prevent the sensor cap **4416** from separating from the cap post **4804** upon removing the applicator cap **210**. In some embodiments, for example, the engagement and receiver features **4422**, **4806** may comprise a threaded interface or a keyed mating profile that allows initial engagement but prevents subsequent disengagement.

In the illustrated embodiment, the receiver feature **4806** includes one or more compliant members **4808** that are expandable or flexible to receive the engagement feature **4422** (FIG. **44**). The engagement feature **4422** may comprise, for example, an enlarged head or define one or more radial protrusions, and the compliant member(s) **4808** may comprise a collet-type device that includes a plurality of compliant fingers configured to flex radially outward to receive the enlarged head or radial protrusion(s). In other embodiments, however, the compliant member(s) **4808** may comprise an elastomer or another type of compliant material configured to expand radially to receive the enlarged head or radial protrusion(s).

FIG. **49** is a cross-sectional side view of the sensor control device **4402** positioned within the applicator cap **210**, according to one or more embodiments. In the illustrated depiction, the remaining portions of the sensor applicator **102** (FIGS. **47A-47B**) are omitted for simplicity. As illustrated, the opening to the receiver feature **4806** exhibits a first diameter D_1 , while the engagement feature **4422** of the sensor cap **4416** exhibits a second diameter D_2 that is larger than the first diameter D_1 and greater than the outer diameter of the remaining portions of the sensor cap **4416**. Accordingly, as the sensor cap **4416** is extended into the cap post **4804**, the compliant member(s) **4808** may flex (expand) radially outward to receive the engagement feature **4422**.

In some embodiments, the engagement feature **4422** may provide or otherwise define an angled outer surface that helps bias the compliant member(s) **4808** radially outward. The engagement feature **4422**, however, may also define an upper shoulder **4902** that prevents the sensor cap **4416** from reversing out of the cap post **4804**. More specifically, the shoulder **4902** may comprise a sharp surface at the second diameter D_2 that will engage but not urge the compliant member(s) **4808** to flex radially outward in the reverse direction.

Once the engagement feature **4422** bypasses the receiver feature **4806**, the compliant member(s) **4808** flex back to (or towards) their natural state. Upon removing the applicator cap **210** from the sensor applicator **102** (FIGS. **47A-47B**), the shoulder **4902** will engage and bind against the compliant member(s) **4808**, thereby separating the sensor cap **4416** from the sensor control device **4402** and exposing the distal portions of the sensor **4410** and the sharp **4412**.

In some embodiments, the receiver feature **4806** may alternatively be threaded and the engagement feature **4422** may also be threaded and configured to threadably engage the threads of the receiver feature **4806**. The sensor cap **4416** may be received within the cap post **4804** via threaded rotation. Upon removing the applicator cap **210** from the sensor applicator **102**, the opposing threads on the engagement and receiver features **4422**, **4806** bind and the sensor cap **4416** may be separated from the sensor control device **4402**.

FIGS. **50A** and **50B** are isometric and side views, respectively, of another example sensor control device **5002**, according to one or more embodiments of the present disclosure. The sensor control device **5002** may be similar in some respects to the sensor control device **4402** of FIG. **44** and therefore may be best understood with reference thereto. Moreover, the sensor control device **5002** may replace the sensor control device **104** of FIG. **1** and, therefore, may be used in conjunction with the sensor applicator **102** of FIG. **1**, which may deliver the sensor control device **5002** to a target monitoring location on a user's skin. Similar to the sensor control device **4402** of FIG. **44**, the sensor control device **5002** may comprise a one-piece architecture.

As illustrated, the sensor control device **5002** includes an electronics housing **5004** that includes a shell **5006** and a mount **5008** that is matable with the shell **5006**. The shell **5006** may be secured to the mount **5008** via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, one or more mechanical fasteners (e.g., screws), a gasket, an adhesive, or any combination thereof. In some cases, the shell **5006** may be secured to the mount **5008** such that a sealed interface is generated therebetween.

The sensor control device **5002** may further include a sensor **5010** (partially visible) and a sharp **5012** (partially visible), similar in function to the sensor **4410** and the sharp **4412** of FIG. **44**. Corresponding portions of the sensor **5010** and the sharp **5012** extend distally from the bottom of the electronics housing **5004** (e.g., the mount **5008**). The sharp **5012** may include a sharp hub **5014** configured to secure and carry the sharp **5012**. As best seen in FIG. **50B**, the sharp hub **5014** may include or otherwise define a mating member **5016**. To couple the sharp **5012** to the sensor control device **5002**, the sharp **5012** may be advanced axially through the electronics housing **5004** until the sharp hub **5014** engages an upper surface of the shell **5006** and the mating member **5016** extends distally from the bottom of the mount **5008**. As the sharp **5012** penetrates the electronics housing **5004**, the exposed portion of the sensor **5010** may be received within a hollow or recessed (arcuate) portion of the sharp **5012**. The remaining portion of the sensor **5010** is arranged within the interior of the electronics housing **5004**.

The sensor control device **5002** may further include a sensor cap **5018**, shown exploded or detached from the electronics housing **5004** in FIGS. **50A-50B**. Similar to the sensor cap **4416** of FIG. **44**, the sensor cap **5018** may help provide a sealed barrier that surrounds and protects the exposed portions of the sensor **5010** and the sharp **5012** from gaseous chemical sterilization. As illustrated, the sensor cap **5018** may comprise a generally cylindrical body having a first end **5020a** and a second end **5020b** opposite the first end **5020a**. The first end **5020a** may be open to provide access into an inner chamber **5022** defined within the body. In contrast, the second end **5020b** may be closed and may provide or otherwise define an engagement feature **5024**. Similar to the engagement feature **4422** of FIG. **44**, the engagement feature **5024** may help mate the sensor cap **5018** to the cap (e.g., the applicator cap **210** of FIG. **2B**) of a sensor applicator (e.g., the sensor applicator **102** of FIGS. **1** and **2A-2G**), and may help remove the sensor cap **5018** from the sensor control device **5002** upon removing the cap from the sensor applicator.

The sensor cap **5018** may be removably coupled to the electronics housing **5004** at or near the bottom of the mount **5008**. More specifically, the sensor cap **5018** may be removably coupled to the mating member **5016**, which extends distally from the bottom of the mount **5008**. In at least one embodiment, for example, the mating member **5016** may

define a set of external threads **5026a** (FIG. **50B**) matable with a set of internal threads **5026b** (FIG. **50A**) defined by the sensor cap **5018**. In some embodiments, the external and internal threads **5026a,b** may comprise a flat thread design (e.g., lack of helical curvature), which may prove advantageous in molding the parts. Alternatively, the external and internal threads **5026a,b** may comprise a helical threaded engagement. Accordingly, the sensor cap **5018** may be threadably coupled to the sensor control device **5002** at the mating member **5016** of the sharp hub **5014**. In other embodiments, the sensor cap **5018** may be removably coupled to the mating member **5016** via other types of engagements including, but not limited to, an interference or friction fit, or a frangible member or substance that may be broken with minimal separation force (e.g., axial or rotational force).

In some embodiments, the sensor cap **5018** may comprise a monolithic (singular) structure extending between the first and second ends **5020a,b**. In other embodiments, however, the sensor cap **5018** may comprise two or more component parts. In the illustrated embodiment, for example, the sensor cap **5018** may include a seal ring **5028** positioned at the first end **5020a** and a desiccant cap **5030** arranged at the second end **5020b**. The seal ring **5028** may be configured to help seal the inner chamber **5022**, as described in more detail below. In at least one embodiment, the seal ring **5028** may comprise an elastomeric O-ring. The desiccant cap **5030** may house or comprise a desiccant to help maintain preferred humidity levels within the inner chamber **5022**. The desiccant cap **5030** may also define or otherwise provide the engagement feature **5024** of the sensor cap **5018**.

FIGS. **51A** and **51B** are exploded isometric top and bottom views, respectively, of the sensor control device **5002**, according to one or more embodiments. The shell **5006** and the mount **5008** operate as opposing clamshell halves that enclose or otherwise substantially encapsulate various electronic components of the sensor control device **5002**. The electronic components housed within the electronics housing **5004** may be similar to the electronic components described with reference to FIG. **45** and, therefore, will not be described again. While not shown, the sensor control device **5002** may also include an adhesive patch that may be applied to the bottom **5102** (FIG. **51B**) of the mount **5008**, and may help adhere the sensor control device **5002** to the user's skin for use.

The sensor control device **5002** may provide or otherwise include a sealed subassembly that includes, among other component parts, the shell **5006**, the sensor **5010**, the sharp **5012**, and the sensor cap **5018**. Similar to the sealed subassembly **4522** of FIG. **45**, the sealed subassembly of the sensor control device **5002** may help isolate the sensor **5010** and the sharp **5012** within the inner chamber **5022** (FIG. **51A**) of the sensor cap **5018** during a gaseous chemical sterilization process, which might otherwise adversely affect the chemistry provided on the sensor **5010**.

The sensor **5010** may include a tail **5104** that extends out an aperture **5106** (FIG. **51B**) defined in the mount **5008** to be transcutaneously received beneath a user's skin. The tail **5104** may have an enzyme or other chemistry included thereon to help facilitate analyte monitoring. The sharp **5012** may include a sharp tip **5108** extendable through an aperture **5110** (FIG. **51A**) defined by the shell **5006**, and the aperture **5110** may be coaxially aligned with the aperture **5106** of the mount **5008**. As the sharp tip **5108** penetrates the electronics housing **5004**, the tail **5104** of the sensor **5010** may be received within a hollow or recessed portion of the sharp tip **5108**. The sharp tip **5108** may be configured to penetrate the

skin while carrying the tail **5104** to put the active chemistry of the tail **5104** into contact with bodily fluids.

The sharp tip **5108** may be advanced through the electronics housing **5004** until the sharp hub **5014** engages an upper surface of the shell **5006** and the mating member **5016** extends out the aperture **5106** in the bottom **5102** of the mount **5008**. In some embodiments, a seal member (not shown), such as an O-ring or seal ring, may interpose the sharp hub **5014** and the upper surface of the shell **5006** to help seal the interface between the two components. In some embodiments, the seal member may comprise a separate component part, but may alternatively form an integral part of the shell **5006**, such as being a co-molded or overmolded component part.

The sealed subassembly may further include a collar **5112** that is positioned within the electronics housing **5004** and extends at least partially into the aperture **5106**. The collar **5112** may be a generally annular structure that defines or otherwise provides an annular ridge **5114** on its top surface. In some embodiments, as illustrated, a groove **5116** may be defined in the annular ridge **5114** and may be configured to accommodate or otherwise receive a portion of the sensor **5010** extending laterally within the electronics housing **5004**.

In assembling the sealed subassembly, a bottom **5118** of the collar **5112** may be exposed at the aperture **5106** and may sealingly engage the first end **5020a** of the sensor cap **5018** and, more particularly, the seal ring **5028**. In contrast, the annular ridge **5114** at the top of the collar **5112** may sealingly engage an inner surface (not shown) of the shell **5006**. In at least one embodiment, a seal member (not shown) may interpose the annular ridge **5114** and the inner surface of the shell **5006** to form a sealed interface. In such embodiments, the seal member may also extend (flow) into the groove **5116** defined in the annular ridge **5114** and thereby seal about the sensor **5010** extending laterally within the electronics housing **5004**. The seal member may comprise, for example, an adhesive, a gasket, or an ultrasonic weld, and may help isolate the enzymes and other chemistry included on the tail **5104**.

FIG. **52** is a cross-sectional side view of an assembled sealed subassembly **5200**, according to one or more embodiments. The sealed subassembly **5200** may form part of the sensor control device **5002** of FIGS. **50A-50B** and **51A-51B** and may include portions of the shell **5006**, the sensor **5010**, the sharp **5012**, the sensor cap **5018**, and the collar **5112**. The sealed subassembly **5200** may be assembled in a variety of ways. In one assembly process, the sharp **5012** may be coupled to the sensor control device **5002** by extending the sharp tip **5108** through the aperture **5110** defined in the top of the shell **5006** and advancing the sharp **5012** through the shell **5006** until the sharp hub **5014** engages the top of the shell **5006** and the mating member **196** extends distally from the shell **5006**. In some embodiments, as mentioned above, a seal member **5202** (e.g., an O-ring or seal ring) may interpose the sharp hub **5014** and the upper surface of the shell **5006** to help seal the interface between the two components.

The collar **5112** may then be received over (about) the mating member **5016** and advanced toward an inner surface **5204** of the shell **5006** to enable the annular ridge **5114** to engage the inner surface **5204**. A seal member **5206** may interpose the annular ridge **5114** and the inner surface **5204** and thereby form a sealed interface. The seal member **5206** may also extend (flow) into the groove **5116** (FIGS. **51A-51B**) defined in the annular ridge **5114** and thereby seal about the sensor **5010** extending laterally within the elec-

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tronics housing **5004** (FIGS. **51A-51B**). In other embodiments, however, the collar **5112** may first be sealed to the inner surface **5204** of the shell **5006**, following which the sharp **5012** and the sharp hub **5014** may be extended through the aperture **5110**, as described above.

The sensor cap **5018** may be removably coupled to the sensor control device **5002** by threadably mating the internal threads **5026b** of the sensor cap **5018** with the external threads **5026a** of the mating member **5016**. Tightening (rotating) the mated engagement between the sensor cap **5018** and the mating member **5016** may urge the first end **5020a** of the sensor cap **5018** into sealed engagement with the bottom **5118** of the collar **5112**. Moreover, tightening the mated engagement between the sensor cap **5018** and the mating member **5016** may also enhance the sealed interface between the sharp hub **5014** and the top of the shell **5006**, and between the annular ridge **5114** and the inner surface **5204** of the shell **5006**.

The inner chamber **5022** may be sized and otherwise configured to receive the tail **5104** and the sharp tip **5108**. Moreover, the inner chamber **5022** may be sealed to isolate the tail **5104** and the sharp tip **5108** from substances that might adversely interact with the chemistry of the tail **5104**. In some embodiments, a desiccant **5208** (shown in dashed lines) may be present within the inner chamber **5022** to maintain proper humidity levels.

Once properly assembled, the sealed subassembly **5200** may be subjected to any of the radiation sterilization processes mentioned herein to properly sterilize the sensor **5010** and the sharp **5012**. This sterilization step may be undertaken apart from the remaining portions of the sensor control device (FIGS. **50A-50B** and **51A-51B**) to prevent damage to sensitive electrical components. The sealed subassembly **5200** may be subjected to radiation sterilization prior to or after coupling the sensor cap **5018** to the sharp hub **5014**. When sterilized after coupling the sensor cap **5018** to the sharp hub **5014**, the sensor cap **5018** may be made of a material that permits the propagation of radiation there-through. In some embodiments, the sensor cap **5018** may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

FIGS. **53A-53C** are progressive cross-sectional side views showing assembly of the sensor applicator **102** with the sensor control device **5002**, according to one or more embodiments. Once the sensor control device **5002** is fully assembled, it may then be loaded into the sensor applicator **102**. With reference to FIG. **53A**, the sharp hub **5014** may include or otherwise define a hub snap pawl **5302** configured to help couple the sensor control device **5002** to the sensor applicator **102**. More specifically, the sensor control device **5002** may be advanced into the interior of the sensor applicator **102** and the hub snap pawl **5302** may be received by corresponding arms **5304** of a sharp carrier **5306** positioned within the sensor applicator **102**.

In FIG. **53B**, the sensor control device **5002** is shown received by the sharp carrier **5306** and, therefore, secured within the sensor applicator **102**. Once the sensor control device **5002** is loaded into the sensor applicator **102**, the applicator cap **210** may be coupled to the sensor applicator **102**. In some embodiments, the applicator cap **210** and the housing **208** may have opposing, matable sets of threads **5308** that enable the applicator cap **210** to be screwed onto the housing **208** in a clockwise (or counter-clockwise) direction and thereby secure the applicator cap **210** to the sensor applicator **102**.

As illustrated, the sheath **212** is also positioned within the sensor applicator **102**, and the sensor applicator **102** may

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include a sheath locking mechanism **5310** configured to ensure that the sheath **212** does not prematurely collapse during a shock event. In the illustrated embodiment, the sheath locking mechanism **5310** may comprise a threaded engagement between the applicator cap **210** and the sheath **212**. More specifically, one or more internal threads **5312a** may be defined or otherwise provided on the inner surface of the applicator cap **210**, and one or more external threads **5312b** may be defined or otherwise provided on the sheath **212**. The internal and external threads **5312a,b** may be configured to threadably mate as the applicator cap **210** is threaded to the sensor applicator **102** at the threads **5308**. The internal and external threads **5312a,b** may have the same thread pitch as the threads **5308** that enable the applicator cap **210** to be screwed onto the housing **208**.

In FIG. **53C**, the applicator cap **210** is shown fully threaded (coupled) to the housing **208**. As illustrated, the applicator cap **210** may further provide and otherwise define a cap post **5314** centrally located within the interior of the applicator cap **210** and extending proximally from the bottom thereof. The cap post **5314** may be configured to receive at least a portion of the sensor cap **5018** as the applicator cap **210** is screwed onto the housing **208**.

With the sensor control device **5002** loaded within the sensor applicator **102** and the applicator cap **210** properly secured, the sensor control device **5002** may then be subjected to a gaseous chemical sterilization configured to sterilize the electronics housing **5004** and any other exposed portions of the sensor control device **5002**. The gaseous chemical sterilization process may be similar to the gaseous chemical sterilization **4704** of FIG. **47B** and, therefore, will not be described again in detail. Since the distal portions of the sensor **5010** and the sharp **5012** are sealed within the sensor cap **5018**, the chemicals used during the gaseous chemical sterilization process are unable to interact with the enzymes, chemistry, and biologics provided on the tail **5104**, and other sensor components, such as membrane coatings that regulate analyte influx.

FIGS. **54A** and **54B** are perspective and top views, respectively, of the cap post **5314**, according to one or more additional embodiments. In the illustrated depiction, a portion of the sensor cap **5018** is received within the cap post **5314** and, more specifically, the desiccant cap **5030** of the sensor cap **5018** is arranged within cap post **5314**.

As illustrated, the cap post **5314** may define a receiver feature **5402** configured to receive the engagement feature **5024** of the sensor cap **5018** upon coupling (e.g., threading) the applicator cap **210** (FIG. **53C**) to the sensor applicator **102** (FIGS. **53A-53C**). Upon removing the applicator cap **210** from the sensor applicator **102**, however, the receiver feature **5402** may prevent the engagement feature **914** from reversing direction and thus prevent the sensor cap **5018** from separating from the cap post **5314**. Instead, removing the applicator cap **210** from the sensor applicator **102** will simultaneously detach the sensor cap **5018** from the sensor control device **5002** (FIGS. **50A-50B** and **53A-53C**), and thereby expose the distal portions of the sensor **5010** (FIGS. **53A-53C**) and the sharp **5012** (FIGS. **53A-53C**).

Many design variations of the receiver feature **5402** may be employed, without departing from the scope of the disclosure. In the illustrated embodiment, the receiver feature **5402** includes one or more compliant members **5404** (two shown) that are expandable or flexible to receive the engagement feature **5024** (FIGS. **50A-50B**). The engagement feature **5024** may comprise, for example, an enlarged head and the compliant member(s) **5404** may comprise a

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collet-type device that includes a plurality of compliant fingers configured to flex radially outward to receive the enlarged head.

The compliant member(s) **5404** may further provide or otherwise define corresponding ramped surfaces **5406** configured to interact with one or more opposing camming surfaces **5408** provided on the outer wall of the engagement feature **5024**. The configuration and alignment of the ramped surface(s) **5406** and the opposing camming surface(s) **5408** is such that the applicator cap **210** is able to rotate relative to the sensor cap **5018** in a first direction A (e.g., clockwise), but the cap post **5314** binds against the sensor cap **5018** when the applicator cap **210** is rotated in a second direction B (e.g., counter clockwise). More particularly, as the applicator cap **210** (and thus the cap post **5314**) rotates in the first direction A, the camming surfaces **5408** engage the ramped surfaces **5406**, which urge the compliant members **5404** to flex or otherwise deflect radially outward and results in a ratcheting effect. Rotating the applicator cap **210** (and thus the cap post **5314**) in the second direction B, however, will drive angled surfaces **5410** of the camming surfaces **5408** into opposing angled surfaces **5412** of the ramped surfaces **5406**, which results in the sensor cap **5018** binding against the compliant member(s) **5404**.

FIG. **55** is a cross-sectional side view of the sensor control device **5002** positioned within the applicator cap **210**, according to one or more embodiments. As illustrated, the opening to the receiver feature **5402** exhibits a first diameter D_3 , while the engagement feature **5024** of the sensor cap **5018** exhibits a second diameter D_4 that is larger than the first diameter D_3 and greater than the outer diameter of the remaining portions of the sensor cap **5018**. As the sensor cap **5018** is extended into the cap post **5314**, the compliant member(s) **5404** of the receiver feature **5402** may flex (expand) radially outward to receive the engagement feature **5024**. In some embodiments, as illustrated, the engagement feature **5024** may provide or otherwise define an angled or frustoconical outer surface that helps bias the compliant member(s) **5404** radially outward. Once the engagement feature **5024** bypasses the receiver feature **5402**, the compliant member(s) **5404** are able to flex back to (or towards) their natural state and thus lock the sensor cap **5018** within the cap post **5314**.

As the applicator cap **210** is threaded to (screwed onto) the housing **208** (FIGS. **53A-53C**) in the first direction A, the cap post **5314** correspondingly rotates in the same direction and the sensor cap **5018** is progressively introduced into the cap post **5314**. As the cap post **5314** rotates, the ramped surfaces **5406** of the compliant members **5404** ratchet against the opposing camming surfaces **5408** of the sensor cap **5018**. This continues until the applicator cap **210** is fully threaded onto (screwed onto) the housing **208**. In some embodiments, the ratcheting action may occur over two full revolutions of the applicator cap **210** before the applicator cap **210** reaches its final position.

To remove the applicator cap **210**, the applicator cap **210** is rotated in the second direction B, which correspondingly rotates the cap post **5314** in the same direction and causes the camming surfaces **5408** (i.e., the angled surfaces **5410** of FIGS. **54A-54B**) to bind against the ramped surfaces **5406** (i.e., the angled surfaces **5412** of FIGS. **54A-54B**). Consequently, continued rotation of the applicator cap **210** in the second direction B causes the sensor cap **5018** to correspondingly rotate in the same direction and thereby unthread from the mating member **5016** to allow the sensor cap **5018** to detach from the sensor control device **5002**. Detaching the sensor cap **5018** from the sensor control device **5002**

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exposes the distal portions of the sensor **5010** and the sharp **5012**, and thus places the sensor control device **5002** in position for firing (use).

FIGS. **56A** and **56B** are cross-sectional side views of the sensor applicator **102** ready to deploy the sensor control device **5002** to a target monitoring location, according to one or more embodiments. More specifically, FIG. **56A** depicts the sensor applicator **102** ready to deploy (fire) the sensor control device **5002**, and FIG. **56B** depicts the sensor applicator **102** in the process of deploying (firing) the sensor control device **5002**. As illustrated, the applicator cap **210** (FIGS. **53A-53C** and **55**) has been removed, which correspondingly detaches (removes) the sensor cap **5018** (FIGS. **53A-53C** and **55**) and thereby exposes the tail **5104** of the sensor **5010** and the sharp tip **5108** of the sharp **5012**, as described above. In conjunction with the sheath **212** and the sharp carrier **5306**, the sensor applicator **102** also includes a sensor carrier **5602** (alternately referred to as a “puck” carrier) that helps position and secure the sensor control device **5002** within the sensor applicator **102**.

Referring first to FIG. **56A**, as illustrated, the sheath **212** includes one or more sheath arms **5604** (one shown) configured to interact with a corresponding one or more detents **5606** (one shown) defined within the interior of the housing **208**. The detent(s) **5606** are alternately referred to as “firing” detent(s). When the sensor control device **5002** is initially installed in the sensor applicator **102**, the sheath arms **5604** may be received within the detents **5606**, which places the sensor applicator **102** in firing position. In the firing position, the mating member **5016** extends distally beyond the bottom of the sensor control device **5002**. As discussed below, the process of firing the sensor applicator **102** causes the mating member **5016** to retract so that it does not contact the user’s skin.

The sensor carrier **5602** may also include one or more carrier arms **5608** (one shown) configured to interact with a corresponding one or more grooves **5610** (one shown) defined on the sharp carrier **5306**. A spring **5612** may be arranged within a cavity defined by the sharp carrier **5306** and may passively bias the sharp carrier **5306** upward within the housing **208**. When the carrier arm(s) **5608** are properly received within the groove(s) **5610**, however, the sharp carrier **5306** is maintained in position and prevented from moving upward. The carrier arm(s) **5608** interpose the sheath **212** and the sharp carrier **5306**, and a radial shoulder **5614** defined on the sheath **212** may be sized to maintain the carrier arm(s) **5608** engaged within the groove(s) **5610** and thereby maintain the sharp carrier **5306** in position.

In FIG. **56B**, the sensor applicator **102** is in the process of firing. As discussed herein with reference to FIGS. **2F-2G**, this may be accomplished by advancing the sensor applicator **102** toward a target monitoring location until the sheath **212** engages the skin of the user. Continued pressure on the sensor applicator **102** against the skin may cause the sheath arm(s) **5604** to disengage from the corresponding detent(s) **5606**, which allows the sheath **212** to collapse into the housing **208**. As the sheath **212** starts to collapse, the radial shoulder **5614** eventually moves out of radial engagement with the carrier arm(s) **5608**, which allows the carrier arm(s) **5608** to disengage from the groove(s) **5610**. The passive spring force of the spring **5612** is then free to push upward on the sharp carrier **5306** and thereby force the carrier arm(s) **5608** out of engagement with the groove(s) **5610**, which allows the sharp carrier **5306** to move slightly upward within the housing **208**. In some embodiments, fewer coils may be incorporated into the design of the spring **5612** to increase the spring force necessary to overcome the engagement

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between carrier arm(s) **5608** and the groove(s) **5610**. In at least one embodiment, one or both of the carrier arm(s) **5608** and the groove(s) **5610** may be angled to help ease disengagement.

As the sharp carrier **5306** moves upward within the housing **208**, the sharp hub **5014** may correspondingly move in the same direction, which may cause partial retraction of the mating member **5016** such that it becomes flush, substantially flush, or sub-flush with the bottom of the sensor control device **5002**. As will be appreciated, this ensures that the mating member **5016** does not come into contact with the user's skin, which might otherwise adversely impact sensor insertion, cause excessive pain, or prevent the adhesive patch (not shown) positioned on the bottom of the sensor control device **5002** from properly adhering to the skin.

FIGS. 57A-57C are progressive cross-sectional side views showing assembly and disassembly of an alternative embodiment of the sensor applicator **102** with the sensor control device **5002**, according to one or more additional embodiments. A fully assembled sensor control device **5002** may be loaded into the sensor applicator **102** by coupling the hub snap pawl **5302** into the arms **5304** of the sharp carrier **5306** positioned within the sensor applicator **102**, as generally described above.

In the illustrated embodiment, the sheath arms **5604** of the sheath **212** may be configured to interact with a first detent **5702a** and a second detent **5702b** defined within the interior of the housing **208**. The first detent **5702a** may alternately be referred to a "locking" detent, and the second detent **5702b** may alternately be referred to as a "firing" detent. When the sensor control device **5002** is initially installed in the sensor applicator **102**, the sheath arms **5604** may be received within the first detent **5702a**. As discussed below, the sheath **212** may be actuated to move the sheath arms **5604** to the second detent **5702b**, which places the sensor applicator **102** in firing position.

In FIG. 57B, the applicator cap **210** is aligned with the housing **208** and advanced toward the housing **208** so that the sheath **212** is received within the applicator cap **210**. Instead of rotating the applicator cap **210** relative to the housing **208**, the threads of the applicator cap **210** may be snapped onto the corresponding threads of the housing **208** to couple the applicator cap **210** to the housing **208**. Axial cuts or slots **5703** (one shown) defined in the applicator cap **210** may allow portions of the applicator cap **210** near its threading to flex outward to be snapped into engagement with the threading of the housing **208**. As the applicator cap **210** is snapped to the housing **208**, the sensor cap **5018** may correspondingly be snapped into the cap post **5314**.

Similar to the embodiment of FIGS. 53A-53C, the sensor applicator **102** may include a sheath locking mechanism configured to ensure that the sheath **212** does not prematurely collapse during a shock event. In the illustrated embodiment, the sheath locking mechanism includes one or more ribs **5704** (one shown) defined near the base of the sheath **212** and configured to interact with one or more ribs **5706** (two shown) and a shoulder **5708** defined near the base of the applicator cap **210**. The ribs **5704** may be configured to inter-lock between the ribs **5706** and the shoulder **5708** while attaching the applicator cap **210** to the housing **208**. More specifically, once the applicator cap **210** is snapped onto the housing **208**, the applicator cap **210** may be rotated (e.g., clockwise), which locates the ribs **5704** of the sheath **212** between the ribs **5706** and the shoulder **5708** of the applicator cap **210** and thereby "locks" the applicator cap **210** in place until the user reverse rotates the applicator cap **210** to remove the applicator cap **210** for use. Engagement

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of the ribs **5704** between the ribs **5706** and the shoulder **5708** of the applicator cap **210** may also prevent the sheath **212** from collapsing prematurely.

In FIG. 57C, the applicator cap **210** is removed from the housing **208**. As with the embodiment of FIGS. 53A-53C, the applicator cap **210** can be removed by reverse rotating the applicator cap **210**, which correspondingly rotates the cap post **5314** in the same direction and causes sensor cap **5018** to unthread from the mating member **5016**, as generally described above. Moreover, detaching the sensor cap **5018** from the sensor control device **5002** exposes the distal portions of the sensor **5010** and the sharp **5012**.

As the applicator cap **210** is unscrewed from the housing **208**, the ribs **5704** defined on the sheath **212** may slidably engage the tops of the ribs **5706** defined on the applicator cap **210**. The tops of the ribs **5706** may provide corresponding ramped surfaces that result in an upward displacement of the sheath **212** as the applicator cap **210** is rotated, and moving the sheath **212** upward causes the sheath arms **5604** to flex out of engagement with the first detent **5702a** to be received within the second detent **5702b**. As the sheath **212** moves to the second detent **5702b**, the radial shoulder **5614** moves out of radial engagement with the carrier arm(s) **5608**, which allows the passive spring force of the spring **5612** to push upward on the sharp carrier **5306** and force the carrier arm(s) **5608** out of engagement with the groove(s) **5610**. As the sharp carrier **5306** moves upward within the housing **208**, the mating member **5016** may correspondingly retract until it becomes flush, substantially flush, or sub-flush with the bottom of the sensor control device **5002**. At this point, the sensor applicator **102** is in firing position. Accordingly, in this embodiment, removing the applicator cap **210** correspondingly causes the mating member **5016** to retract.

FIG. 58A is an isometric bottom view of the housing **208**, according to one or more embodiments. As illustrated, one or more longitudinal ribs **5802** (four shown) may be defined within the interior of the housing **208**. The ribs **5802** may be equidistantly or non-equidistantly spaced from each other and extend substantially parallel to centerline of the housing **208**. The first and second detents **5702a,b** may be defined on one or more of the longitudinal ribs **5802**.

FIG. 58B is an isometric bottom view of the housing **208** with the sheath **212** and other components at least partially positioned within the housing **208**. As illustrated, the sheath **212** may provide or otherwise define one or more longitudinal slots **5804** configured to mate with the longitudinal ribs **5802** of the housing **208**. As the sheath **212** collapses into the housing **208**, as generally described above, the ribs **5802** may be received within the slots **5804** to help maintain the sheath **212** aligned with the housing during its movement. As will be appreciated, this may result in tighter circumferential and radial alignment within the same dimensional and tolerance restrictions of the housing **208**.

In the illustrated embodiment, the sensor carrier **5602** may be configured to hold the sensor control device **5002** in place both axially (e.g., once the sensor cap **5018** is removed) and circumferentially. To accomplish this, the sensor carrier **5602** may include or otherwise define one or more support ribs **5806** and one or more flexible arms **5808**. The support ribs **5806** extend radially inward to provide radial support to the sensor control device **5002**. The flexible arms **5808** extend partially about the circumference of the sensor control device **5002** and the ends of the flexible arms **5808** may be received within corresponding grooves **5810** defined in the side of the sensor control device **5002**. Accordingly, the flexible arms **5808** may be able to provide both axial and radial support to the sensor control device

5002. In at least one embodiment, the ends of the flexible arms **5808** may be biased into the grooves **5810** of the sensor control device **5002** and otherwise locked in place with corresponding sheath locking ribs **5812** provided by the sheath **212**.

In some embodiments, the sensor carrier **5602** may be ultrasonically welded to the housing **208** at one or more points **5814**. In other embodiments, however, the sensor carrier **5602** may alternatively be coupled to the housing **208** via a snap-fit engagement, without departing from the scope of the disclosure. This may help hold the sensor control device **5002** in place during transport and firing.

FIG. **59** is an enlarged cross-sectional side view of the sensor applicator **102** with the sensor control device **5002** installed therein, according to one or more embodiments. As discussed above, the sensor carrier **5602** may include one or more carrier arms **5608** (two shown) engageable with the sharp carrier **5306** at corresponding grooves **5610**. In at least one embodiment, the grooves **5610** may be defined by pairs of protrusions **5902** defined on the sharp carrier **5306**. Receiving the carrier arms **5608** within the grooves **5610** may help stabilize the sharp carrier **5306** from unwanted tilting during all stages of retraction (firing).

In the illustrated embodiment, the arms **5304** of the sharp carrier **5306** may be stiff enough to control, with greater refinement, radial and bi-axial motion of the sharp hub **5014**. In some embodiments, for example, clearances between the sharp hub **5014** and the arms **5304** may be more restrictive in both axial directions as the relative control of the height of the sharp hub **5014** may be more critical to the design.

In the illustrated embodiment, the sensor carrier **5602** defines or otherwise provides a central boss **5904** sized to receive the sharp hub **5014**. In some embodiments, as illustrated, the sharp hub **5014** may provide one or more radial ribs **5906** (two shown). In at least one embodiment, the inner diameter of the central boss **5904** helps provide radial and tilt support to the sharp hub **5014** during the life of sensor applicator **102** and through all phases of operation and assembly. Moreover, having multiple radial ribs **5906** increases the length-to-width ratio of the sharp hub **5014**, which also improves support against tilting.

FIG. **60A** is an isometric top view of the applicator cap **210**, according to one or more embodiments. In the illustrated embodiment, two axial slots **5703** are depicted that separate upper portions of the applicator cap **210** near its threading. As mentioned above, the slots **5703** may help the applicator cap **210** flex outward to be snapped into engagement with the housing **208** (FIG. **57B**). In contrast, the applicator cap **210** may be twisted (unthreaded) off the housing **208** by an end user.

FIG. **60A** also depicts the ribs **5706** (one visible) defined by the applicator cap **210**. By interlocking with the ribs **5704** (FIG. **57C**) defined on the sheath **212** (FIG. **57C**), the ribs **5706** may help lock the sheath **212** in all directions to prevent premature collapse during a shock or drop event. The sheath **212** may be unlocked when the user unscrews the applicator cap **210** from the housing (FIG. **59C**), as generally described above. As mentioned herein, the top of each rib **5706** may provide a corresponding ramped surface **6002**, and as the applicator cap **210** is rotated to unthread from the housing **208**, the ribs **5704** defined on the sheath **212** may slidably engage the ramped surfaces **6002**, which results in the upward displacement of the sheath **212** into the housing **208**.

In some embodiments, additional features may be provided within the interior of the applicator cap **210** to hold a desiccant component that maintains proper moisture levels

through shelf life. Such additional features may be snaps, posts for press-fitting, heat-staking, ultrasonic welding, etc.

FIG. **60B** is an enlarged cross-sectional view of the engagement between the applicator cap **210** and the housing **208**, according to one or more embodiments. As illustrated, the applicator cap **210** may define a set of inner threads **6004** and the housing **208** may define a set of outer threads **6006** engageable with the inner threads **6004**. As mentioned herein, the applicator cap **210** may be snapped onto the housing **208**, which may be accomplished by advancing the inner threads **6004** axially past the outer threads **6006** in the direction indicated by the arrow, which causes the applicator cap **210** to flex outward. To help ease this transition, as illustrated, corresponding surfaces **6008** of the inner and outer threads **6004**, **6006** may be curved, angled, or chamfered. Corresponding flat surfaces **6010** may be provided on each thread **6004**, **6006** and configured to matingly engage once the applicator cap **210** is properly snapped into place on the housing **208**. The flat surfaces **6010** may slidably engage one another as the user unthreads the applicator cap **210** from the housing **208**.

The threaded engagement between the applicator cap **210** and the housing **208** results in a sealed engagement that protects the inner components against moisture, dust, etc. In some embodiments, the housing **208** may define or otherwise provide a stabilizing feature **6012** configured to be received within a corresponding groove **1914** defined on the applicator cap **210**. The stabilizing feature **6012** may help stabilize and stiffen the applicator cap **210** once the applicator cap **210** is snapped onto the housing **208**. This may prove advantageous in providing additional drop robustness to the sensor applicator **102**. This may also help increase the removal torque of the applicator cap **210**.

FIGS. **61A** and **61B** are isometric views of the sensor cap **5018** and the collar **5112**, respectively, according to one or more embodiments. Referring to FIG. **61A**, in some embodiments, the sensor cap **5018** may comprise an injection molded part. This may prove advantageous in molding the internal threads **5026a** defined within the inner chamber **5022**, as opposed to installing a threaded core or threading the inner chamber **5022**. In some embodiments, one or more stop ribs **6102** (on visible) may be defined within the inner chamber **5022** to prevent over travel relative to mating member **5016** of the sharp hub **5014** (FIGS. **50A-50B**).

Referring to both FIGS. **61A** and **61B**, in some embodiments, one or more protrusions **6104** (two shown) may be defined on the first end **5020a** of the sensor cap **5018** and configured to mate with one or more corresponding indentations **6106** (two shown) defined on the collar **5112**. In other embodiments, however, the protrusions **6104** may instead be defined on the collar **5112** and the indentations **6106** may be defined on the sensor cap **5018**, without departing from the scope of the disclosure.

The matable protrusions **6104** and indentations **6106** may prove advantageous in rotationally locking the sensor cap **5018** to prevent unintended unscrewing of the sensor cap **5018** from the collar **5112** (and thus the sensor control device **5002**) during the life of the sensor applicator **102** and through all phases of operation/assembly. In some embodiments, as illustrated, the indentations **6106** may be formed or otherwise defined in the general shape of a kidney bean. This may prove advantageous in allowing for some over-rotation of the sensor cap **5018** relative to the collar **5112**. Alternatively, the same benefit may be achieved via a flat end threaded engagement between the two parts.

Embodiments disclosed herein include:

U. A sensor control device that includes an electronics housing, a sensor arranged within the electronics housing and having a tail extending from a bottom of the electronics housing, a sharp extending through the electronics housing and having a sharp tip extending from the bottom of the electronics housing, and a sensor cap removably coupled at the bottom of the electronics housing and defining a sealed inner chamber that receives the tail and the sharp.

V. An analyte monitoring system that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing, a sensor arranged within the electronics housing and having a tail extending from a bottom of the electronics housing, a sharp extending through the electronics housing and having a sharp tip extending from the bottom of the electronics housing, and a sensor cap removably coupled at the bottom of the electronics housing and defining an engagement feature and a sealed inner chamber that receives the tail and the sharp. The analyte monitoring system may further include a cap coupled to the sensor applicator and providing a cap post defining a receiver feature that receives the engagement feature upon coupling the cap to the sensor applicator, wherein removing the cap from the sensor applicator detaches the sensor cap from the electronics housing and thereby exposes the tail and the sharp tip.

W. A method of preparing an analyte monitoring system that includes loading a sensor control device into a sensor applicator, the sensor control device including an electronics housing, a sensor arranged within the electronics housing and having a tail extending from a bottom of the electronics housing, a sharp extending through the electronics housing and having a sharp tip extending from the bottom of the electronics housing, and a sensor cap removably coupled at the bottom of the electronics housing and defining a sealed inner chamber that receives the tail and the sharp. The method further including securing a cap to the sensor applicator, sterilizing the sensor control device with gaseous chemical sterilization while the sensor control device is positioned within the sensor applicator, and isolating the tail and the sharp tip within the inner chamber from the gaseous chemical sterilization.

Each of embodiments U, V, and W may have one or more of the following additional elements in any combination: Element 1: wherein the sensor cap comprises a cylindrical body having a first end that is open to access the inner chamber, and a second end opposite the first end and providing an engagement feature engageable with a cap of a sensor applicator, wherein removing the cap from the sensor applicator correspondingly removes the sensor cap from the electronics housing and thereby exposes the tail and the sharp tip. Element 2: wherein the electronics housing includes a shell matable with a mount, the sensor control device further comprising a sharp and sensor locator defined on an inner surface of the shell, and a collar received about the sharp and sensor locator, wherein the sensor cap is removably coupled to the collar. Element 3: wherein the sensor cap is removably coupled to the collar by one or more of an interference fit, a threaded engagement, a frangible member, and a frangible substance. Element 4: wherein an annular ridge circumscribes the sharp and sensor locator and the collar provides a column and an annular shoulder extending radially outward from the column, and wherein a seal member interposes the annular shoulder and the annular ridge to form a sealed interface. Element 5: wherein the annular ridge defines a groove and a portion of the sensor is seated within the groove, and wherein the seal member

extends into the groove to seal about the portion of the sensor. Element 6: wherein the seal member is a first seal member, the sensor control device further comprising a second seal member interposing the annular shoulder and a portion of the mount to form a sealed interface. Element 7: wherein the electronics housing includes a shell matable with a mount, the sensor control device further comprising a sharp hub that carries the sharp and is engageable with a top surface of the shell, and a mating member defined by the sharp hub and extending from the bottom of the electronics housing, wherein the sensor cap is removably coupled to the mating member. Element 8: further comprising a collar at least partially receivable within an aperture defined in the mount and sealingly engaging the sensor cap and an inner surface of the shell. Element 9: wherein a seal member interposes the collar and the inner surface of the shell to form a sealed interface. Element 10: wherein the collar defines a groove and a portion of the sensor is seated within the groove, and wherein the seal member extends into the groove to seal about the portion of the sensor.

Element 11: wherein the receiver feature comprises one or more compliant members that flex to receive the engagement feature, and wherein the one or more compliant members prevent the engagement feature from exiting the cap post upon removing the cap from the sensor applicator. Element 12: further comprising a ramped surface defined on at least one of the one or more compliant members, and one or more camming surfaces provided by the engagement feature and engageable with the ramped surface, wherein the ramped surface and the one or more camming surfaces allow the cap and the cap post to rotate relative to the sensor cap in a first direction, but prevent the cap and the cap post from rotating relative to the sensor cap in a second direction opposite the first direction. Element 13: wherein the electronics housing includes a shell matable with a mount, the sensor control device further comprising a sharp hub that carries the sharp and is engageable with a top surface of the shell, and a mating member defined by the sharp hub and extending from the bottom of the electronics housing, wherein the sensor cap is removably coupled to the mating member and rotating the cap in the second direction detaches the sensor cap from the mating member. Element 14: wherein the electronics housing includes a shell matable with a mount and the sensor control device further includes a sharp and sensor locator defined on an inner surface of the shell, and a collar received about the sharp and sensor locator, wherein the sensor cap is removably coupled to the collar.

Element 15: wherein the cap provides a cap post defining a receiver feature and the sensor cap defines an engagement feature, the method further comprising receiving the engagement feature with the receiver feature as the cap is secured to the sensor applicator. Element 16: further comprising removing the cap from the sensor applicator, and engaging the engagement feature on the receiver feature as the cap is being removed and thereby detaching the sensor cap from the electronics housing and exposing the tail and the sharp tip. Element 17: wherein loading the sensor control device into a sensor applicator is preceded by sterilizing the tail and the sharp tip with radiation sterilization, and sealing the tail and the sharp tip within the inner chamber.

By way of non-limiting example, exemplary combinations applicable to U, V, and W include: Element 2 with Element 3; Element 2 with Element 4; Element 4 with Element 5; Element 4 with Element 6; Element 7 with

Element 8; Element 8 with Element 9; Element 9 with Element 10; Element 11 with Element 12; and Element 15 with Element 16.

Sensor Applicator with Actuating Needle Shroud

Referring again briefly to FIG. 1, the sensor control device **104** is often included with the sensor applicator **104** in what is known as a “two-piece” architecture that requires final assembly by a user before the sensor **110** can be properly delivered to the target monitoring location. In such applications, the sensor **110** and the associated electrical components included in the sensor control device **104** are provided to the user in multiple (two) packages, and the user must open the packaging and follow instructions to manually assemble the components before delivering the sensor **110** to the target monitoring location with the sensor applicator **6302**. More recently, however, advanced designs of sensor control devices and associated sensor applicators have resulted in a one-piece architecture that allows the system to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package, remove an applicator cap, and subsequently deliver the sensor control device to the target monitoring location.

Notwithstanding these advances, conventional sensor applicators commonly include a shroud that surrounds the entire outer periphery of the sensor control device. To deploy the sensor control device, the shroud is forced against the skin and retracts into the sensor applicator, which causes the combination introducer and sensor to be delivered transcutaneously under the user's skin. Having the shroud positioned away from the insertion site near the introducer leaves the skin at the insertion site in a generally soft and uncompressed state. It can be difficult to insert a sensor in uncompressed soft tissue due to the skin depression that occurs as the introducer tip enters the skin, commonly referred to as skin “tenting”. Embodiments of the present disclosure include sensor applicators that incorporate a needle shroud to apply pressure to the skin at or near the insertion site.

FIG. 62 is an isometric top view of an example sensor control device **6202**, according to one or more embodiments of the present disclosure. The sensor control device **6202** may be the same as or similar to the sensor control device **104** of FIG. 1 and, therefore, may be designed to be delivered to a target monitoring location on a user's skin through operation of a sensor applicator (not shown). As illustrated, the sensor control device **6202** includes an electronics housing **6204** that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing **6204** may exhibit other cross-sectional shapes, such as oval, ovoid (e.g., pill- or egg-shaped), a squircle, polygonal, or any combination thereof, without departing from the scope of the disclosure. The electronics housing **6204** may house or otherwise contain various electronic components used to operate the sensor control device **6202**. For example, a printed circuit board (PCB) may be positioned within the electronics housing and may have thereto one or more of a battery, a data processing unit, and various resistors, transistors, capacitors, inductors, diodes, and switches.

The electronics housing **6204** may include a shell **6206** and a mount **6208** that is matable with the shell **6206**. The shell **6206** may be secured to the mount **6208** via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, one or more mechanical fasteners (e.g., screws), or any combination thereof. In some cases, the shell **6206** may be secured to the mount **6208** such that a sealed

interface is generated therebetween. In such embodiments, a gasket or other type of seal material may be positioned at or near the outer diameter (periphery) of the shell **6206** and the mount **6208**, and securing the two components together may compress the gasket and thereby generate a sealed interface. In other embodiments, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell **6206** and the mount **6208**. The adhesive secures the shell **6206** to the mount **6208** and provides structural integrity, but may also seal the interface between the two components and thereby isolate the interior of the electronics housing **6204** from outside contamination.

In the illustrated embodiment, the sensor control device **6202** also includes a sensor module **6210** interconnectable with a sharp module **6212**. The sensor module **6210** may be coupled to the electronics housing **6204** with a collar **6214**, and the collar **6214** may be mounted to the electronics housing **6204** within an aperture **6215** defined therethrough. The sensor module **6210** may include a sensor **6216** and a flexible connector **6218** used to help connect the sensor **6216** to the electronic components housed within the electronics housing **6204**. A tail **6220** of the sensor **6216** may extend distally from the electronics housing **6204** and, more particularly, from the bottom of the mount **6208**.

The sharp module **6212** may carry or otherwise include an introducer or sharp **6222** used to help deliver the sensor **6216** transcutaneously under a user's skin during deployment of the sensor control device **6202**. In the illustrated embodiment, the sharp module **6212** includes a sharp hub **6224** that carries the sharp **6222**. In one embodiment, the sharp hub **6224** may be overmolded onto the sharp **6222**, but could alternatively be fabricated from plastic, metal, or another suitable material as a separate component, and bonded, welded, or mechanically attached to the sharp **6222**. Similar to the tail **6220**, the distal end of the sharp **6222** may extend distally from the electronics housing **6204** and, more particularly, from the bottom of the mount **6208**. In at least one embodiment, the tail **6220** may be received within a hollow or recessed portion of the sharp **6222**.

While the sensor control device **6202** is depicted as an eccentric assembly, with the sensor **6216** and the sharp **6222** extending distally at a location offset from a central axis of the electronics housing **6204**, embodiments are contemplated herein where the sensor **6216** and the sharp **6222** are aligned with the central axis in a concentric design, without departing from the scope of the disclosure. Moreover, an adhesive patch **6226** may be positioned on and otherwise attached to the underside of the mount **6208**. Similar to the adhesive patch **108** of FIG. 1, the adhesive patch **6226** may be configured to secure and maintain the sensor control device **6202** in position on the user's skin during operation.

FIG. 63 is a schematic side view of an example sensor applicator **6302**, according to one or more embodiments of the present disclosure. The sensor applicator **6302** may be similar in some respects to the sensor applicator **102** of FIG. 1 and, therefore, may be configured to house and facilitate deployment of a sensor control device, such as the sensor control device **6202** (shown in dashed lines). As illustrated, the sensor applicator **6302** may include a housing **6304** sized to receive the sensor control device **6202** therein. In some embodiments, an applicator cap **6306** may be removably coupled to the housing **6304**. The applicator cap **6306** may be threaded to the housing **6304**, for example, but could alternatively be coupled thereto via a snap fit engagement, an interference fit, or the like, without departing from the scope of the disclosure. The applicator cap **6306** may help

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protect and shield the adhesive patch **6226** from contaminants or damage prior to deploying the sensor control device **6202**.

The sensor applicator **6302** may also include a sensor cap **6308** extending from the bottom of the sensor applicator **6302**. The sensor cap **6308** may be configured to receive and protect the distal ends of the sensor **6216** and the sharp **6222** extending from the bottom of the electronics housing **6204**. In some embodiments, the sensor cap **6308** may be coupled to or otherwise form an integral part or extension of the applicator cap **6306**. In other embodiments, however, the applicator and sensor caps **6306**, **6308** may constitute separate component parts that may be jointly or separately removable from the bottom of the housing **6304**.

In some embodiments, the sensor cap **6308** may extend from the sensor control device **6202** and form part of a sterile barrier with the collar **6214** (FIG. **62**) to protect the distal ends of the sensor **6216** and the sharp **6222**. In such embodiments, the sensor cap **6308** may be removably coupled to the collar **6214**, such as being threaded to the collar **6214** or coupled thereto using a bayonet coupling, an interference fit, a snap fit engagement, or any combination thereof. In other embodiments, however, the sensor cap **6308** may alternatively be removably coupled to another internal feature of the sensor applicator **6302**, without departing from the scope of the disclosure.

In one or more embodiments, the sensor cap **6308** may include a gripping interface **6310** that provides a location for a user to grasp onto and remove the sensor cap **6308** from the sensor applicator **6302**. The gripping interface **6310** may comprise, for example, a tab that can be grasped by the user with the thumb and forefinger. Once the applicator cap **6306** and the sensor cap **6308** are removed, a user may then use the sensor applicator **6302** to position the sensor control device **6202** (FIG. **62**) at a target monitoring location on the user's body, as will be described below.

FIGS. **64A** and **64B** are exploded isometric views of the sensor applicator **6302** and the sensor control device **6202**. The applicator cap **6306** and the sensor cap **6308** of FIG. **63** are not shown for simplicity. As illustrated, the collar **6214**, the sensor **6216**, and the flexible connector **6218** (collectively the sensor module **6210** of FIG. **62**) may each be mounted to the electronics housing **6204** at or within the aperture **6215** defined in the electronics housing **6204**.

The sensor applicator **6302** may include a desiccant **6404**, a sensor retainer **6406**, a needle shroud **6408**, and a driver spring **6410**. The desiccant **6404** may optionally contained within the housing **6304** to help maintain appropriate humidity levels. The housing **6304** may be matable with the sensor retainer **6406** (alternately referred to as a "puck retainer") to retain the needle shroud **6408**, the driver spring **6410**, the sharp hub **6224**, and the sharp **6222** within the housing **6304**. The sensor retainer **6406**, the needle shroud **6408**, the sharp hub **6224** with the sharp **6222**, and the driver spring **6410** may all be operatively coupled to help facilitate deployment of the sensor control device **6202**.

As described below, the needle shroud **6408** may be movable (actuatable) between an extended position and a retracted position to deploy the sensor control device **6202** from the sensor applicator **6302**. As best seen in FIG. **64B**, the sensor retainer **6406** may have one or more locking tabs **6412** engageable with a corresponding one or more locking members **6414** provided on the needle shroud **6408**. Coupling the locking members **6414** to the locking tabs **6412** helps secure the needle shroud **6408** in the extended posi-

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tion, whereas disengaging the locking members **6414** from the locking tabs **6412** allows the needle shroud **6408** to move to the retracted position.

Those skilled in the art will readily appreciate that the locking tabs and members **6412**, **6414** are merely one way to temporarily secure the needle shroud **6408** in the extended position. In other embodiments, for example, the locking tabs and members **6412**, **6414** may be replaced with corresponding detents and mating grooves or other common types of removable or releasable couplings, without departing from the scope of the disclosure.

The sensor retainer **6406** may further include a plurality of upwardly extending fingers **6414** (three shown) configured to extend partially into the needle shroud **6408** to help retain the sharp hub **6224** until the needle shroud **6408** moves to the retracted position. Once the needle shroud **6408** reaches the retracted position, the fingers **6414** may be able to flex radially outward to release the needle shroud **6408**, and the spring force of the driver spring **6410** may retract the sharp **6222** into the housing **6304**.

The sensor retainer **6406** may define an aperture **6418** through which the lower portion of the needle shroud **6408** can extend. The lower end of the needle shroud **6408** extends through the aperture **6418** (and the aperture **6215** provided in the electronics housing **6204**) when the needle shroud **6408** is in the extended position. Moving the needle shroud **6408** to the retracted position draws the lower end of the needle shroud **6408** upward through the aperture **6418** (and the aperture **6215** of the electronics housing **6204**).

FIGS. **65A-65D** are progressive cross-sectional side views of the sensor applicator **6302** depicting example deployment of the sensor control device **6202**, according to one or more embodiments. User operation (actuation) of the sensor applicator **6302** can cause the needle shroud **6408** to move from the extended position, as shown in FIGS. **65A** and **65B**, to the retracted position, as shown in FIG. **65D**. Once the needle shroud **6408** reaches the retracted position, the sensor control device **6202** may be able to be released (discharged) from the sensor retainer **6406**, as described below.

Referring first to FIG. **65A**, the applicator cap **6306** is removably coupled to the housing **6304**. In some embodiments, the interface between the applicator cap **6306** and the housing **6304** may be sealed to help protect and shield the adhesive patch **6226** from contamination or damage prior to deploying the sensor control device **6202**. The sensor cap **6308** is also depicted extending distally from the bottom of the sensor applicator **6302** and, more particularly, from the sensor control device **6202**.

The sensor cap **6308** may define an interior **6502** sized to receive a lower portion of the needle shroud **6408** in the extended position. Moreover, distal ends of the sensor **6216** and the sharp **6222** may also extend into the interior **6502** of the sensor cap **6308** and the needle shroud **6408** may generally cover the distal ends of the sensor **6216** and the sharp **6222** when the needle shroud **6408** is in the extended position. In some embodiments, a seal **6504** may be positioned at an interface between the top of the sensor cap **6308** and the collar **6214** and thereby help form a sterile barrier for the sensor **6216** and the sharp **6222**. In one embodiment, the seal **6504** may be co-molded or otherwise attached to the top of the sensor cap **6308**. In other embodiments, however, the seal **6504** may be co-molded or attached to the collar **6214**. In yet other embodiments, the seal **6504** may be a separate component part, such as an O-ring or the like placed between the top of the sensor cap **6308** and the collar **6214**.

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In one embodiment, as mentioned above, the sensor cap **6308** may be removably coupled to the collar **6214**, such as through a bayonet coupling, an interference fit, a snap fit engagement, or any combination thereof. In other embodiments, however, the sensor cap **6308** may be removably coupled to the needle shroud **6408**, without departing from the scope of the disclosure. Removably coupling the sensor cap **6308** to either the collar **6214** or the needle shroud **6408** may help maintain compression of the seal **6504**. To remove the sensor cap **6308** from the sensor applicator **6302**, a user may be able to grasp the gripping interface **6310** on the sensor cap **6308**. As indicated above, in some embodiments, both the applicator and sensor caps **6306**, **6308** may be removed simultaneously or separately.

In FIG. **65B**, the applicator cap **6306** and the sensor cap **6308** have been removed from the sensor applicator **6302**, thereby exposing the needle shroud **6408** and the bottom of the sensor control device **6202**. With the needle shroud **6408** in the extended position, as illustrated, the upper portion of the needle shroud **6408** resides within the housing **6304**, while the lower portion extends distally through the aperture **6418** defined in the sensor retainer **6406** and through the aperture **6215** defined through the sensor control device **6202**. The upwardly extending fingers **6414** of the sensor retainer **6406** may extend into or otherwise be positioned within an inner chamber **6506** defined by the upper portion of the needle shroud **6408**. Moreover, the sharp hub **6224** may be arranged within or between the fingers **6414**, and the driver spring **6410** may be arranged to interpose and engage the sharp hub **6224** and the sensor retainer **6406**.

More specifically, the top end of the driver spring **6410** may be received within a channel **6508** defined by the sharp hub **6224**, and the bottom end of the driver spring **6410** may engage one or more projections **6510** defined by the sensor retainer **6406** and extending radially into the aperture **6418**. Alternatively, the top end of the driver spring **6414** may engage an upper end of the sharp **6222**, thus eliminating the need for an overmolded sharp hub **6224**. The driver spring **6410** may be compressed between the sharp hub **6224** and the sensor retainer **6406** and prevented from releasing its spring force and expanding as long as the fingers **6414** are located within the inner chamber **6506**. More particularly, the top of one or more of the fingers **6414** may extend radially inward and over the sharp hub **6224**, thus preventing the sharp hub **6224** from moving upward until the fingers **6414** are no longer radially constrained by the inner chamber **6506**. Moving the needle shroud **6408** to the retracted position, however, correspondingly places the fingers **6414** outside of the inner chamber **6506**, which allows the driver spring **6410** force the sharp hub **6224** past the top of the fingers **6414**, as described below.

With the needle shroud **6408** in the extended position, the locking tabs **6412** (FIG. **64B**) of the sensor retainer **6406** may be engaged with the locking members **6414** (FIGS. **64A-64B**) provided on the needle shroud **6408**, which helps secure the needle shroud **6408** in the extended position. The locking members **6414** must be disengaged from the locking tabs **6412** to allow the needle shroud **6408** to move to the retracted position and thereby deploy the sensor control device **6202**. This can be accomplished by the user positioning the sensor applicator **6302** at the target monitoring location and forcing the needle shroud **6408** against the skin, which places an axial load on the bottom end of the needle shroud **6408**. The axial load will overcome the temporary engagement between the locking tabs **6412** and the locking

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members **6414**, thus freeing the needle shroud **6408** and enabling the needle shroud **6408** to start its transition to the retracted position.

In some embodiments, disengaging the locking members **6414** from the locking tabs **6412** may result in a tactile response, thus providing the user with haptic feedback. More particularly, upon disengaging the locking members **6414** from the locking tabs **6412**, a small vibration or tremor may result in the sensor applicator **6302**, thus indicating to a user that the deployment process has begun. This haptic feedback may encourage the user to continue to apply pressure to the needle shroud **6408**.

In some embodiments, one or more sensation features **6512** may be provided at the bottom end of the needle shroud **6408**. The sensation features **6512** may contact the underlying skin to stimulate the nerve endings on the skin at that location and thereby help to mask the sensation of the sharp **6222** penetrating the skin. In some embodiments, the sensation features **6512** may comprise nubs or small projections defined on the end of the needle shroud **6408**.

In FIG. **65C**, the needle shroud **6408** has moved a short distance from the extended position and toward the retracted position, thus exposing the sensor **6216** and the sharp **6222** as they extend out of the lower end of the needle shroud **6408**. More specifically, as the needle shroud **6408** is pressed against the skin, it compresses the skin, and moves relative to the sensor **6216** and the sharp **6222**, which causes the sensor **6216** and the sharp **6222** to extend out of the needle shroud **6408** to penetrate the skin. One advantage of the needle shroud **6408** is its proximity to the insertion site of the sensor **6216** and the sharp **6222**. More particularly, the needle shroud **6408** is able to provide local compression of the skin at the insertion site, which tightens the skin at the insertion site and thereby facilitates a more efficient insertion of the sensor **6216** and the sharp **6222**.

Moving the needle shroud **6408** to the retracted position also moves the upper portion of the needle shroud **6408** relative to the fingers **6414** and the sharp hub **6224** arranged within the inner chamber **6506** of the needle shroud **6408**. Friction between the fingers **6414** and the inner wall of the inner chamber **6506** provides a small amount of resistance while allowing motion of the housing towards the skin surface, which can be felt by the user during firing to help drive the sharp **6222** into the underlying skin by applying additional pressure to bypass the force bump.

In FIG. **65D**, the needle shroud **6408** has moved to the retracted position, and the bottom end of the needle shroud **6408** may be flush with or inset into the bottom of the sensor control device **6202**. Once the needle shroud **6408** has moved to the retracted position, the fingers **6414** of the sensor retainer **6406** may be positioned outside of the inner chamber **6506** and are therefore no longer radially constrained by the needle shroud **6408**. Consequently, the spring force built up in the driver spring **6410** may release and force the sharp hub **6224** against the tops of the fingers **6414**, which flexes the fingers **6414** radially outward and allows the sharp hub **6224** to move upward relative to the fingers **6414**. As the sharp hub **6224** moves upward, the sharp **6222** correspondingly retracts out of the underlying skin and into the sensor applicator **6302**, thus leaving only the sensor **6216** within the skin.

In some embodiments, the sensor applicator **6302** may provide haptic feedback to the user that provides an indication that the sensor deployment process is complete. More specifically, haptic or tactile feedback may be provided to the user when the needle shroud **6408** has moved to the retracted position and the sharp **6222** has fully retracted. In

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such embodiments, release of the driver spring 6410 may provide some degree of haptic feedback. However, springs, detents, or other elements may alternatively (or in addition) be included to also signal functionality and a completed firing process. In some applications, the forces generated by the experience may be tailored to be similar to taking a common retractable pen and pushing the thumb actuated “thruster” end against the skin.

FIG. 66 is an enlarged cross-sectional side view of an engagement between the sensor retainer 6406 and the sensor control device 6202, according to one or more embodiments. In some embodiments, the collar 6214 may be removably coupled to the sensor retainer 6406, which correspondingly retains the sensor control device 6202 to the sensor retainer 6406. In the illustrated embodiment, the sensor retainer 6406 may provide or otherwise define one or more first retention features 6602 operable to mate with one or more corresponding second retention features 6604 defined on the collar 6214. In the illustrated embodiment, the first and second retention features 6602, 6604 comprise tabs and corresponding lips or grooves that receive the tabs. However, the first and second retention features 6602, 6604 may comprise any type of removable coupling or engagement that temporarily couples the sensor control device 6202 to the sensor retainer 6406.

The sensor control device 6302 may be released from the sensor retainer 6406 by disengaging the first and second retention features 6602, 6604. This may be accomplished by attaching (sticking) the adhesive layer 6226 against the skin. The first and second retention features 6602, 6604 may be designed so that when the sensor control device 6202 is adhesively attached to the skin with the adhesive layer 6226, the engagement between the first and second retention features 6602, 6604 may be broken by retracting the sensor applicator 6302 away from the sensor control device 6202. This allows the sensor control device 6202 to separate from the sensor applicator 6302 and remain on the body.

In some embodiments, a seal 6606 may seal an interface between the top of the sensor control device 6202 and the bottom of the sensor retainer 6406, and thereby help form a sterile barrier for the sensor 6216 and the sharp 6222. In one embodiment, the seal 6606 may be co-molded or otherwise attached to the top of the sensor control device 6202 or the collar 6214. In other embodiments, however, the seal 6606 may be co-molded or attached to the bottom of the sensor retainer 6406. In yet other embodiments, the seal 6606 may be a separate component part, such as an O-ring or the like.

FIG. 67 is an exploded isometric view of another sensor applicator 6702 with the sensor control device 6202, according to one or more additional embodiments. The sensor applicator 6702 may be similar in some respects to the sensor applicator 6302 of FIGS. 63 and 64A-64B and may thus be best understood with reference thereto, where like numerals will correspond to like components not described again in detail. Similar to the sensor applicator 6302, for example, the sensor applicator 6702 may include the housing 6304 that may be sized to accommodate the desiccant 6404 and the sensor control device 6202 therein. The collar 6214 and the sensor 6216 of the sensor control device 6202 may each be mounted to the electronics housing 6204 at or within the aperture 6215 defined in the electronics housing 6204, as generally described above. Moreover, the sensor applicator 6702 may also include the sensor cap 6308 used to help form a sterile barrier with the collar 6214 and thereby protect the distal ends of the sensor 6216 and the sharp 6222. As described above, the seal 6504 may help form the sterile

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barrier by sealing the interface between the top of the sensor cap 6308 and the collar 6214 (or another portion of the sensor control device 6202).

A sharp hub 6704 carries the sharp 6222 and may be overmolded onto the sharp 6222, but could alternatively be fabricated from plastic, metal, or another suitable material as a separate component, and bonded, welded, or mechanically attached to the sharp 6222. The sensor applicator 6702 may also include a sensor retainer 6706, a needle shroud 6708, and a driver spring 6710. The sensor retainer 6706 (alternately referred to as a “puck retainer”) may be matable with the housing 6304 to help retain the needle shroud 6708, the driver spring 6710, and the sharp hub 6704 generally within or connected to the housing 6304. More specifically, the sensor retainer 6706, the needle shroud 6708, the sharp hub 6704, and the driver spring 6710 may all be operatively coupled to help facilitate deployment of the sensor control device 6202.

In the illustrated embodiment, the driver spring 6710 may be sized to be arranged about the sharp hub 6704, and the sensor retainer 6706 may provide a plurality of upwardly extending fingers 6712 (three shown) configured to extend into an inner chamber 6714 defined by the sharp hub 6704. The sharp 6222 and the needle shroud 6708 may be extendable through the inner chamber 6714, and further extendable through an aperture 6716 defined in the sensor retainer 6706 and the aperture 6215 provided in the electronics housing 6204. The needle shroud 6708 may be movable (actuatable) between an extended position and a retracted position to deploy the sensor control device 6202 from the sensor applicator 6702.

As described in more detail below, when the needle shroud 6708 is in the extended position, the fingers 6712 may be radially constrained between an outer surface of the needle shroud 6708 and an inner wall of the sharp hub 6704 within the inner chamber 6714, thus preventing the sharp hub 6704 (and the sharp 6222) from moving. Once the needle shroud 6708 moves to the extended position, however, the fingers 6712 may become aligned with one or more reliefs 6718 defined on the needle shroud 6708, which allow the fingers 6712 to flex radially inward and release the sharp hub 6704. In some embodiments, the driver spring 6710 may provide a spring force that urges the sharp hub 6704 upward and simultaneously flexes the fingers 6712 radially inward, which allows the sharp hub 6704 to move upward and retract the sharp 6222 into the housing 6304.

FIGS. 68A-68D are progressive cross-sectional side views of the sensor applicator 6702 depicting example deployment of the sensor control device 6202, according to one or more embodiments. User operation (actuation) of the sensor applicator 6702 can cause the needle shroud 6708 to move from the extended position, as shown in FIGS. 68A and 68B, to the retracted position, as shown in FIG. 68D. Once the needle shroud 6708 reaches the retracted position, the sensor control device 6202 may be able to be released (discharged) from the sensor retainer 6706.

Referring first to FIG. 68A, an applicator cap 6802 may be removably coupled to the housing 6304 and may be similar in some respects to the applicator cap 6306 of FIG. 63. In some embodiments, the interface between the applicator cap 6802 and the housing 6304 may be sealed to help protect and shield the adhesive patch 6226 from contamination or damage prior to deploying the sensor control device 6202. The sensor cap 6308 is also depicted extending distally from the bottom of the sensor applicator 6702 and, more particularly, from the sensor control device 6202. The interior 6502 of the sensor cap 6308 may accommodate the

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distal ends of the sensor 6216 and the sharp 6222 and the lower portion of the needle shroud 6708 in the extended position. Moreover, the seal 6504 may interpose the top of the sensor cap 6308 and the collar 6214 to help form a sterile barrier for the sensor 6216 and the sharp 6222.

In FIG. 68B, the applicator cap 6802 and the sensor cap 6308 have been removed from the sensor applicator 6702, thereby exposing the needle shroud 6708 and the bottom of the sensor control device 6202. With the needle shroud 6708 in the extended position, as illustrated, the upper portion of the needle shroud 6708 resides within the housing 6304, while the lower portion extends distally through the aperture 6716 defined in the sensor retainer 6706 and through the aperture 6215 defined through the sensor control device 6202. Moreover, the upper portion of the needle shroud 6708 extends into and through the inner chamber 6714 defined within the sharp hub 6704. The upwardly extending fingers 6712 of the sensor retainer 6706 extend into the inner chamber 6714 and interpose the needle shroud 6708 and the inner wall of the inner chamber 6714.

As indicated above, the driver spring 6710 may be positioned about an exterior portion of the sharp hub 6704 and may extend between the sharp hub 6704 and the sensor retainer 6706. More specifically, the top end of the driver spring 6710 may be received within a channel 6806 defined by the sharp hub 6704, and the bottom end of the driver spring 6710 may engage the sensor retainer 6706, such as a top surface of the sensor retainer 6706. The driver spring 6710 is compressed between the sharp hub 6704 and the sensor retainer 6706 when the needle shroud 6708 in the extended position. The driver spring 6710 is prevented from releasing its spring force and expanding as long as the fingers 6712 are radially constrained between the outer surface of the needle shroud 6708 and the inner wall of the inner chamber 6714. More particularly, the tops of the fingers 6712 may extend radially outward and received within a groove or notch 6808 defined on the sharp hub 6704. When the tops of the fingers 6712 are received within the notch(es) 6808, the sharp hub 6704 may be prevented from moving upward.

Referring briefly to FIG. 69A, depicted is an enlarged schematic view of the sharp hub 6704 and the fingers 6712 of the sensor retainer 6706 of FIG. 67. As illustrated, the tops of each finger 6712 may extend or protrude radially outward to be received within corresponding notches 6808 defined at an upper end of the sharp hub 6704. The fingers 6712 extend within the inner chamber 6714 and interpose the outer radial surface of the needle shroud 6708 and the inner wall of the inner chamber 6714. The sharp hub 6704 is prevented from moving upward as long as the tops of the fingers 6712 are constrained into engagement with the notches 6808.

Referring briefly to FIGS. 69B and 69C, depicted are enlarged schematic views of the fingers 6712 interacting with the upper portion of the needle shroud 6708. In some embodiments, as illustrated, the upper portion (end) of the needle shroud 6708 may define a groove 6902 and a detent profile 6904 that terminates in a force bump 6906. In such embodiments, the upper ends of the fingers 6712 may provide or otherwise define inwardly extending (protruding) lips or features 6908 configured to interact with the groove 6902, the detent profile 6904, and the force bump 6906. With the needle shroud 6708 in the extended position, the features 6908 provided on the fingers 6712 may be engaged with and otherwise received by the groove 6902 provided on the needle shroud 6708, which helps axially maintain the needle shroud 6708 in the extended position.

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The features 6908 must be disengaged from the groove 6902 to allow the needle shroud 6708 to move to the retracted position and thereby deploy the sensor control device 6202. This can be accomplished by the user positioning the sensor applicator 6702 (FIG. 68B) at the target monitoring location and forcing the bottom of the needle shroud 6708 against the skin, which places an axial load on the needle shroud 6708. The axial load will overcome the temporary engagement between the groove 6902 and the features 6908, thus freeing the needle shroud 6708 and enabling the needle shroud 6708 to start its upward transition to the retracted position.

As shown in FIG. 69C, the features 6908 have been disengaged from the groove 6902, and the features 6908 may slide along the detent profile 6904 as the needle shroud 6708 moves upward relative to the fingers 6712. When the features 6908 locate the force bump 6906, the user may apply additional pressure to overcome and otherwise bypass the force bump 6906. In some embodiments, disengaging the features 6908 from the groove 6902 or bypassing the force bump 6906 may result in a tactile response that may be felt by the user, thus providing the user with haptic feedback. More particularly, upon disengaging the features 6908 from the groove 6902 (or bypassing the force bump 6906), a small vibration or tremor may propagate through the sensor applicator 6702 (FIG. 68B), thus indicating to a user that the deployment process has begun. This haptic feedback may encourage the user to continue to apply pressure to the needle shroud 6708.

Referring again to FIGS. 68A-68D and, more particularly, to FIG. 68C, the needle shroud 6708 has moved from the extended position and toward the retracted position, thus exposing the sensor 6216 and the sharp 6222 as they extend out the lower end of the needle shroud 6708. More specifically, as the user presses the needle shroud 6708 against the skin, the needle shroud 6708 moves relative to the sensor 6216 and the sharp 6222, which causes the sensor 6216 and the sharp 6222 to extend out of the bottom of the needle shroud 6708 to penetrate the skin. One advantage of the needle shroud 6708 is its proximity to the insertion site of the sensor 6216 and the sharp 6222. More particularly, the needle shroud 6708 is able to provide local compression of the skin at the insertion site near the sharp 6222, which tightens the skin at the insertion site and thereby facilitates a more efficient insertion of the sharp 6222 and the sensor 6216.

Moving the needle shroud 6708 to the retracted position also moves the upper portion of the needle shroud 6708 relative to the fingers 6712 of the sensor retainer 6706 arranged within the inner chamber 6714 of the sharp hub 6704. Friction between the fingers 6712 and the outer surface of the needle shroud 6708 provides a small amount of resistance, which may be felt by the user during firing to help drive the sharp 6222 into the underlying skin without user hesitation.

In FIG. 68D, the needle shroud 6708 has moved to the retracted position, which aligns the fingers 6712 with the reliefs 6718 defined in the sidewall of the needle shroud 6708. Aligning the fingers 6712 with the reliefs 6718 allows the fingers 6712 to flex radially inward into the reliefs 6718 as the driver spring 6710 release and forces the sharp hub 6704 against the tops of the fingers 6712. Once the fingers 6712 enter the reliefs 6718, the sharp hub 6704 may be released and the spring force of the driver spring 6710 may move the sharp hub 6704 upward relative to the fingers

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6712, which correspondingly retracts the sharp 6222 into the sensor applicator 6702, thus leaving only the sensor 6216 within the skin.

In some embodiments, the sensor applicator 6702 may provide haptic feedback to the user that provides an indication that the sensor deployment process is complete. More specifically, haptic or tactile feedback may be provided to the user when the needle shroud 6708 moves to the retracted position and the sharp 6222 has fully retracted. In such embodiments, release of the driver spring 6710 may provide some degree of haptic feedback that propagates through the sensor applicator 6702 to be felt by the user. However, springs, detents, or other elements may alternatively (or in addition) be included to also signal functionality and a completed firing process. In some applications, the forces generated by the experience may be tailored to be similar to taking a common retractable pen and pushing the thumb actuated “thruster” end against the skin.

FIGS. 70A and 70B are enlarged cross-sectional side views of example engagement between the sensor retainer 6706 and the sensor control device 6202, according to one or more embodiments. In some embodiments, the collar 6214 may be removably coupled to the sensor retainer 6706, which correspondingly removably couples the sensor control device 6202 to the sensor retainer 6706. In the illustrated embodiment, the sensor retainer 6706 may provide or otherwise define one or more first retention features 7002 operable to mate with one or more corresponding second retention features 7004 defined on the collar 6214. In the illustrated embodiment, the first retention features 7002 comprise tabs that extend downwardly through the aperture 6716 of the sensor retainer 6706, and the second retention features 7004 comprise corresponding lips or grooves that receive the tabs. However, the first and second retention features 7002, 7004 may comprise any type of removable coupling or engagement that temporarily couples the sensor control device 6202 to the sensor retainer 6706.

As the needle shroud 6708 moves upward toward the retracted position, the first retention features 7002 may be radially constrained between an outer surface 7006 of the needle shroud 6708 and the collar 6214, which prevents the first retention features 7002 from disengaging from the second retention features 7004. Once the needle shroud 6708 reaches the retracted position, however, the first retention features 7002 may axially align with corresponding relief pockets 7008 defined in the sidewall of the needle shroud 6708. Once the first retention features 7002 axially align with the relief pockets 7008, the first retention features 7002 may be able to flex radially inward into the relief pockets 7008, which allows the sensor control device 6302 to be released from the sensor retainer 6706, as is shown in FIG. 70B. Flexing the first retention features 7002 radially inward may disengage the first and second retention features 7002, 7004, thus allowing the sensor control device to release from the sensor retainer 6706.

In some embodiments, the first and second retention features 7002, 7004 may be disengaged by attaching (sticking) the adhesive layer 6226 against the skin and pulling back on the sensor applicator 6702 (FIGS. 68A-68D). More specifically, the first and second retention features 7002, 7004 may be designed so that when the sensor control device 6202 is adhesively attached to the skin with the adhesive layer 6226, the engagement between the first and second retention features 7002, 7004 may be broken by retracting the sensor applicator 6702 away from the placed sensor

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control device 6202. This allows the sensor control device 6202 to separate from the sensor applicator 6702 and remain on the body.

FIGS. 71A and 71B are isometric and cross-sectional side views, respectively, of an example sensor retainer 7100, according to one or more embodiments. The sensor retainer 7100 may be similar in some respects to the sensor retainers 6406, 6706 of FIGS. 64A-64B and 67, respectively, and therefore may be best understood with reference thereto. Similar to the sensor retainers 6406, 6706, for example, the sensor retainer 7100 may be configured to retain the sensor control device 6202 prior to deployment within a sensor applicator, such as any of the sensor applicators 102, 6302, 6702 of FIGS. 1, 63, 67, respectively, described herein.

In contrast to the sensor retainers 6406, 6706 of FIGS. 64A-64B and 67, however, the sensor retainer 7100 may interact with a sharp hub 7102 that carries the sharp 6222 to releasably couple the sensor control device 6202 to the sensor retainer 7100. As illustrated, the sensor retainer 7100 may define an aperture 7104 through which a lower portion of the sharp hub 7102 (and the sharp 6222) may extend. The aperture 7104 may align with the aperture 6215 defined in the electronics housing 6204 of the sensor control device 6202, and the lower portion of the sharp hub 7102 may also extend into the aperture 6215 when the sensor control device 6202 is removably (releasably) coupled to the sensor retainer 7100.

As illustrated, the sensor retainer 7100 may define or otherwise provide one or more arms 7106 that extend downwardly into the aperture 7104 and past the bottom of the sensor retainer 7100. As best seen in FIG. 71B, each arm 7106 may provide or otherwise define one or more first retention features 7108 operable to mate with one or more corresponding second retention features 7110 defined on or otherwise provided by the sensor control device 6202. In some embodiments, the second retention features 7110 may be provided by the collar 6214 (FIGS. 62 and 67) positioned within the aperture 6215, but could alternatively be provided on another part of the sensor control device 6202, without departing from the scope of the disclosure.

In the illustrated embodiment, the first retention features 7108 may be provided at the bottom end of the arms 7106 and may comprise tabs or protrusions that extend (project) radially outward. The second retention feature 7110 may comprise a lip or annular shoulder extending radially inward at the aperture 6215 to receive and otherwise mate with the first retention features 7108. Those skilled in the art will readily appreciate, however, that the first and second retention features 7108, 7110 may comprise any type of removable coupling or engagement that temporarily couples the sensor control device 6202 to the sensor retainer 6706, without departing from the scope of the disclosure.

FIGS. 72A and 72B are enlarged cross-sectional side views of the sensor retainer 7100 retaining the sensor control device 6202. As illustrated, the lower portion of the sharp hub 7102 is received within the aperture 7104 of the sensor retainer 7100 and also extends at least partially through the aperture 6215 of the sensor control device 6202. The sharp hub 7102 is shown in FIGS. 72A-72B in an extended position, and may be movable to a retracted position where the sharp hub 7102 moves out of axial alignment with the apertures 6215, 7104. Moving the sharp hub 7102 to the retracted position may be accomplished through user intervention in firing the sensor applicator that houses the sensor control device 6202. Once the sensor applicator is fired, a spring or other biasing device (not shown) operatively

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coupled to the sharp hub **7102** may cause the sharp hub **7102** to quickly move upwardly relative to the sensor retainer **7100**.

With the sharp hub **7102** in the extended position, as depicted, the first retention features **7108** may be engaged with or otherwise mated to the second retention feature **7110**. Moreover, when the sharp hub **7102** is in the extended position, the arms **7106** may be radially constrained between the sidewall of the sharp hub **7102** and the second retention feature **7110**, which prevents the first retention features **7108** from disengaging from the second retention features **7110**. Once the sharp hub **7102** moves to the retracted position, however, the arms **7106** will no longer be backed by the sidewall of the sharp hub **7102**, thus enabling the arms **7106** to flex radially inward to disengage the first and second retention features **7108**, **7110** and thereby release the sensor control device **6302**.

In some embodiments, the arms **7106** may flex radially inward to disengage the first and second retention features **7108**, **7110** by attaching (sticking) the adhesive layer **6226** against the skin and pulling back on the sensor applicator that carries the sensor control device **6202**. More specifically, the first and second retention features **7108**, **7110** may be designed so that when the sensor control device **6202** is adhesively attached to the skin with the adhesive layer **6226**, the engagement between the first and second retention features **7108**, **7110** may be broken by retracting the sensor applicator away from the placed sensor control device **6202**. This allows the sensor control device **6202** to separate from the sensor applicator and remain on the body.

Electronics housings of prior sensor control devices are commonly manufactured of rigid plastic materials, and are retained within a sensor applicator by sensor retainers that have a plurality of flexible arms. Such electronics housings often define a plurality of semi-hemispherical notches or grooves on the outer periphery of the electronics housing that are sized to receive the ends of the flexible arms. According to embodiments of the present disclosure, however, the electronics housing **6204** of the sensor control device **6202** may be constructed of flexible or soft materials, such as a soft encapsulant, a foam, or small injection molded components. With flexible or soft materials, it can be a challenge to define features on the exterior of the electronics housing that can be used to retain the sensor control device **6202** to the sensor retainer **7100** during shipment and during the insertion process.

Accordingly, the sensor retainer **7100** includes the arms **7106** that help grasp and retain the sensor control device at the matable first and second retention features **7108**, **7110**. The arms **7106** are flexible and capable of deflecting away from the second retention feature **7110** when the sensor control device **6202** is pulled from the sensor applicator by adhesive attachment to the skin. Prior to insertion, however, the arms **7106** are prevented from deflecting and releasing the sensor control device **6202** by the presence of the sharp hub **7102** extended within (through) the apertures **6215**, **7104**. The sensor retainer **7100** may retain the sensor control device **6202** due to the arms **7106** not being able to deflect radially inwards. During the firing (insertion) process, however, and when the sharp **6222** and the sharp hub **7102** are retracted from the skin, the arms **7106** are no longer back supported and will be deflected as the sensor control device **6202** is pulled from the sensor applicator.

In addition to providing a method to retain the sensor control device **6202** in the sensor applicator, the features of the sensor retainer **7100** enable a more compact applicator design by replacing the flexible arms of conventional sensor

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retainers. By relocating the flexible retention arms to the apertures **6215**, **7104**, the overall size of the sensor applicator may be reduced.

FIGS. **73A** and **73B** are side and cross-sectional side views, respectively, of an example sensor applicator **7302**, according to one or more embodiments. The sensor applicator **7302** may be similar in some respects to the sensor applicator **102** of FIG. **1** and, therefore, may be designed to deliver (fire) a sensor control device, such as the sensor control device **6202**. FIG. **73A** depicts how the sensor applicator **7302** might be shipped to and received by a user, and FIG. **73B** depicts the sensor control device **6202** arranged within the interior of the sensor applicator **7302**.

As shown in FIG. **73A**, the sensor applicator **7302** includes a housing **7304** and an applicator cap **7306** removably coupled to the housing **7304**. In some embodiments, the applicator cap **7306** may be threaded to the housing **7304** and include a tamper ring **7308**. Upon rotating (e.g., unscrewing) the applicator cap **7306** relative to the housing **7304**, the tamper ring **7308** may shear and thereby free the applicator cap **7306** from the sensor applicator **7302**.

In FIG. **73B**, the applicator cap **7306** has been removed from the housing **7304**, thus exposing a sheath **7310** that generally surrounds the sensor control device **6202**. During firing of the sensor applicator **7302**, the sheath **7310** may be actuated (e.g. pushed or forced into the housing **7304**), which causes the sensor control device **6202** to be discharged from the sensor applicator **7302**.

In the illustrated embodiment, the sensor control device **6202** may include a sensor cap **7314** removably coupled to the sensor control device **6202** at or near the bottom of the electronics housing **6204**. The sensor cap **7314** may help provide or facilitate a sealed or sterile barrier surrounding and protecting the exposed portions of the sensor **6216** and the sharp **6222**. As illustrated, the sensor cap **7314** may comprise a generally cylindrical and elongate body having a first end **7315a** and a second end **7315b** opposite the first end **7315a**. The first end **7315a** may be open to provide access into an inner chamber **7316** defined within the body, and the second end **7315b** may be closed and may provide or otherwise define one or more engagement features **7318**.

In some embodiments, the sensor cap **7314** may be removably coupled to the sensor control device **6202** by being coupled to a sharp hub **7320** that carries the sharp **6222** and extends through the electronics housing **6204**. In such embodiments, the sharp hub **7320** may extend past the bottom of the electronics housing **6204** to provide a location where the sensor cap **7314** might engage the sharp hub **7320**. Consequently, at least a portion of the sharp hub **7320** may be extend into the inner chamber **7316** of the sensor cap **7314**. Prior to delivering the sensor control device **6202** to the target monitoring location on the user's skin, the sensor cap **7314** may be separated from the sharp hub **7320**. In some embodiments, the sensor cap **7314** may be removably coupled to the sharp hub **7320** via an interference or friction fit. In other embodiments, the sensor cap **7314** may be threaded to the sharp hub **7320**. In yet other embodiments, the sensor cap **7314** may be removably coupled to the sharp hub **7320** with a frangible member (e.g., a shear ring) or substance that may be broken with minimal separation force (e.g., axial or rotational force). In such embodiments, for example, the sensor cap **7314** may be secured to the sharp hub **7320** with a tag (spot) of glue or a dab of wax.

In some embodiments, however, the sharp hub **7320** may not extend past the bottom of the electronics housing **6204**. In such embodiments, the sensor cap **7314** may alternatively be removably coupled to another portion of the sensor

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control device 6202, such as the collar 6214 (FIGS. 62 and 67) or the mount 6208 (FIG. 62). In such embodiments, the sensor cap 7314 may be removably coupled to the collar 6214 or the mount 6208 (or both) via an interference or friction fit, threading, with a frangible member or substance, or any combination thereof.

The inner chamber 7316 may be sized and otherwise configured to receive the distal ends of the sensor 6216 and the sharp 6222. Moreover, the inner chamber 7316 may be sealed to isolate the sensor 6216 from substances that might adversely interact with the chemistry of the sensor 6216. More specifically, the inner chamber 7316 may be sealed at the interface between the first end 7315a of the sensor cap 7312 and the location where it is removably coupled to the sensor control device 6202. In some embodiments, a desiccant may be present within the inner chamber 7316 to help maintain preferred humidity levels.

As illustrated, the sensor applicator 7302 may further include an internal applicator cover 7322 that may extend at least partially into the sheath 7310. The internal applicator cover 7322 may comprise a generally cylindrical body having a first end 7324a and a second end 7324b opposite the first end 7324a. A sidewall of the internal applicator cover 7322 may extend between the first and second ends 7324a,b and into the interior of the sheath 7310 when the internal applicator cover 7322 is coupled to the sensor applicator 7302. The internal applicator cover 7322 may be open at the first end 7324a to provide access to a cover interior 7326. The second end 7324b may be closed and may provide or otherwise define a gripping interface 7328.

In some embodiments, the internal applicator cover 7322 may be removably coupled to the sheath 7310, such as via an interference fit or a threaded engagement. In other embodiments, the applicator cap 7306 (FIG. 73A) may be used to help retain the internal applicator cover 7322 within the sensor applicator 7302 while applicator cap 7306 is coupled (threaded) to the housing 7304. In yet other embodiments, the internal applicator cover 7322 may be coupled to the sensor cap 7312. More particularly, the internal applicator cover 7322 may provide or otherwise define receiving features 7330 within the cover interior 7326 at or near the second end 7324b. The receiving features 7330 may be configured to receive the second end 7315b of the sensor cap 7312 and, more particularly, mate with the engagement features 7318 of the sensor cap 7312.

The internal applicator cover 7322 may be removed from the sensor applicator 7302 by a user grasping the gripping interface 7328 and rotating and/or pulling on the internal applicator cover 7322 relative to the shroud 7310 and out of engagement with the sensor applicator 7302. As described below, as the internal applicator cover 7322 is removed, engagement between the receiving features 7330 and the engagement features 7318 causes the sensor cap 7312 to also be removed from the sensor control device 6202, thus exposing the sensor 6216 and the sharp 6222 and readying the sensor control device 6202 for firing.

FIGS. 74A and 74B are isometric top and bottom views, respectively, of the internal applicator cover 7322. As depicted, the receiving features 7330 may be provided within the cover interior 7326 at or near the bottom of the internal applicator cover 7322. As indicated above, the receiving features 7330 may be designed to receive the lower end 7315b (FIG. 73B) of the sensor cap 7312 (FIG. 73B) and mate with the engagement features 7318 (FIG. 73B). As will be appreciated, many design variations of the engagement features 7318 and the receiving features 7330 may be employed, without departing from the scope of the

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disclosure. Any design may be used that allows the engagement features 7318 to be received by the receiving features 7330, and subsequently prevent the sensor cap 7312 from separating from the receiving features 7330 upon removing the internal applicator cover 7322.

In some embodiments, for example, the engagement and receiving features 7318, 7330 may comprise a threaded interface or a keyed mating profile that allows initial engagement but prevents subsequent disengagement. In the illustrated embodiment, the receiving features 7330 include one or more compliant members 7402 that are expandable or flexible to receive the engagement features 7318. The receiving features 7330 may also include two or more planar members 7404 configured to receive the lower end 7315b (FIG. 73B) of the sensor cap 7312 (FIG. 73B) and prevent the sensor cap 7312 from rotating relative to the internal applicator cover 7322.

In FIG. 74B, the gripping interface 7328 may comprise an upright flange 7406 extending across a depression 7408 formed into the second end 7324b. A user may be able to grip the internal applicator cover 7322 with the thumb and forefinger at the upright flange 7406, and apply a rotational or axial load to the internal applicator cover 7322 via the gripping interface 7328.

FIG. 75 is an isometric view of an example embodiment of the sensor cap 7312, according to one or more embodiments. In some embodiments, as illustrated, the first end 7315a of the sensor cap 7312 may provide or define a reduced-diameter portion 7502 that may help facilitate removable coupling engagement to the sensor control device 6202 (FIG. 73B).

At the second end 7315b, the engagement features 7318 may comprise, for example, an enlarged head or annular ring 7504 that can interact with the compliant members 7402 (FIG. 74A) of the internal applicator cover 7322 (FIG. 74A). The annular ring 7504 may alternatively comprise one or more radial protrusions. In some embodiments, the engagement features 7318 may also provide or otherwise define two or more planar surfaces 7506 configured to interact with the planar members 7404 (FIG. 74A) of the internal applicator cover 7322. In at least one embodiment, the planar surfaces 7506 may provide a hexagonal shape to the second end 7315b and may mate with the planar members 7404.

FIG. 76 is an isometric, cross-sectional side view of the sensor cap 7312 received by the internal applicator cover 7322, according to one or more embodiments. As illustrated, the engagement features 7318 are received within the receiving features 7330 of the internal applicator cover 7322. More particularly, the annular ring 7504 is received by the compliant members 7402, and the compliant members 7402 may comprise, for example, a collet-type device that includes a plurality of compliant fingers configured to flex radially outward to receive the annular ring 7504. In other embodiments, however, the compliant members 7402 may comprise an elastomer or another type of compliant material configured to expand radially to receive the annular ring 7504. Accordingly, as the sensor cap 7312 is extended into the receiving features 7330, the compliant members 7402 may flex (expand) radially outward to receive the engagement features 7318. Once the annular ring 7504 bypasses the compliant members 7402, the compliant members 7402 flex back to their natural state and thereby prevent the sensor cap 7312 from disengaging from the internal applicator cover 7322.

Mating the engagement features 7318 to the receiving features 7330 may also include mating the planar surfaces 7502 of the sensor cap 7312 with the planar members 7404

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of the internal applicator cover **7322**. The opposing planar members and surfaces **7404**, **7502** may bind the sensor cap **7312** rotationally such that the sensor cap **7312** is unable to rotate relative to the internal applicator cover **7322**.

FIG. 77 shows progressive removal of the applicator cap **7306** and the internal applicator cover **7322** from the sensor applicator **7302**, according to one or more embodiments. Moving from left to right in FIG. 77, the applicator cap **7306** may be removed by unscrewing it from the housing **7304**. Removing the applicator cap **7306** exposes the sheath **7310** and the bottom of the internal applicator cover **7322**. At this point, the sensor cap **7312** remains removably coupled to the sensor control device **6202** within the sensor applicator **7302**. Consequently, the sterile barrier facilitated by the sensor cap **7312** is not broken by removal of the applicator cap **7306**, and the sensor **6216** and the sharp **6222** remain protected. This feature may prove advantageous in the event the user changes his/her mind about firing the sensor applicator **7302** (i.e., deploying the sensor control device **6202**) after removing the applicator cap **7306**. In the event of a decision change, the sensor **6216** and the sharp **6222** remain protected within the sensor cap **7312**, which is coupled to the internal applicator cover **7322**.

To be able to properly fire the sensor applicator **7302** and thereby deploy the sensor control device **6202**, the internal applicator cover **7322** must first be removed. As mentioned above, this can be done by the user gripping the internal applicator cover **7322** at the gripping interface **7328**. The user may then apply a rotational or axial load to the internal applicator cover **7322** via the gripping interface **7328** to remove the internal applicator cover **7322**. Upon removing the internal applicator cover **7322** from the sensor applicator **7302**, the receiving features **7330** (FIG. 74A) of the internal applicator cover **7322** may retain the engagement features **7318** of the sensor cap **7312** and thereby prevent the sensor cap **7312** from separating from the receiving features **7330**. Instead, removing the internal applicator cover **7322** from the sensor applicator **7302** will simultaneously detach the sensor cap **7312** from the sensor control device **6202**, and thereby expose the distal portions of the sensor **6216** and the sharp **6222**.

FIG. 78 is a schematic diagram of an example sensor applicator **7800**, according to one or more additional embodiments of the present disclosure. Similar to the other sensor applicators described herein, the sensor applicator **7800** may be configured to house and subsequently deploy a sensor control device **7802**, which may be similar in some respects to any of the sensor control devices described herein. Alternatively, the sensor control device **7802** may comprise a type of medical device, a health care product, or a system that might require terminal sterilization of specific component parts. Example medical devices or health care products that may incorporate the principles of the present disclosure include, but are not limited to, ingestible products, cardiac rhythm management (CRM) devices, under-skin sensing devices, externally mounted medical devices, or any combination thereof.

In the illustrated embodiment, the sensor control device **7802** includes a housing **7804**, a part **7806** requiring sterilization, one or more radiation sensitive components **7808**, and a battery **7810** that provides power to the sensor control device **7802**. In the illustrated embodiment, the radiation sensitive component **7808** may comprise one or more electronic modules such as, but not limited to, a data processing unit (e.g., an application specific integrated circuit or ASIC), a resistor, a transistor, a capacitor, an inductor, a diode, and a switch.

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In some embodiments, the part **7806** may comprise the sensor **6216** and the sharp **6222** described herein. As illustrated, the part **7806** may extend at an angle relative to the housing **7804**, but could alternatively extend perpendicular to the housing **7804**. In the illustrated embodiment, the part **7806** is arranged within a sterile chamber **7812** to protect the sensor **6216** and the sharp **6222** from external contamination. In some embodiments, the sterile chamber **7812** may have a desiccant arranged therein to help promote preferred humidity conditions.

The sensor **6216** and the sharp **6222** may be sterilized prior to being assembled in the sensor applicator **7800**, or alternatively while assembled in the sensor applicator **7800**. In at least one embodiment, the sensor **6216** and the sharp **6222** may be subjected to radiation sterilization to properly sterilize the part **7806** for use. Suitable radiation sterilization processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof.

In some embodiments, the sensor control device **7802** may include a barrier shield **7814** positioned within the housing **7804** to help block radiation (e.g., electrons) from propagating within the housing **7804** toward the radiation sensitive components **7808**. The barrier shield **7814** may be made of a material that reduces or eliminates radiation from penetrating therethrough and thereby damaging the radiation sensitive components **7808** within the housing **7804**. The barrier shield **7814** may be made of a material having a density sufficient to absorb the dose of the beam energy being delivered.

In some embodiments, the sterile chamber **7812** may be comprise a cap that encapsulates the sensor **6216** and the sharp **6222** to provide a sealed barrier that protects exposed portions of the part **7806** until the part **7806** is placed in use. In such embodiments, the sterile chamber **7812** may be removable or detachable to expose the sensor **6216** and the sharp **6222**, as described below. Moreover, in such embodiments, the cap may be made of a material that permits propagation of radiation therethrough to facilitate radiation sterilization of the part **7806**. Suitable materials for the sterile chamber **7812** include, but are not limited to, a non-magnetic metal (e.g., aluminum, copper, gold, silver, etc.), a thermoplastic, a ceramic, rubber (e.g., ebonite), a composite material (e.g., fiberglass, carbon fiber reinforced polymer, etc.), an epoxy, or any combination thereof. In some embodiments, the sterile chamber **7812** may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

In other embodiments, the sterile chamber **7812** may comprise a chamber or compartment defined within one or both of the sensor applicator **7800** and the sensor control device **7802**. In such embodiments, the sterile chamber **7812** may include a microbial barrier positioned at one or both ends of the sterile chamber **7812**. More specifically, the sterile chamber **7812** may provide or include an upper microbial barrier **7818a** and a lower microbial barrier **7818b** opposite the upper microbial barrier **7818a**. The upper and lower microbial barriers **7818a,b** may help seal the sterile chamber **7812** and thereby isolate the sensor **6216** and the sharp **6222** from external contamination. The microbial barriers **7818a,b** may be made of a radiation permeable material, such as a synthetic material (e.g., a flash-spun high-density polyethylene fiber). One example synthetic material comprises TYVEK®, available from DuPont®. In other embodiments, however, the microbial barriers **7818a,b** may comprise, but are not limited to, tape, paper, film, foil, or any combination thereof.

In some embodiments, the part **7806** may be deployable and otherwise movable relative to the sensor applicator **7800**. In such embodiments, the sensor **6216** and the sharp **6222** may be advanced distally out of the sterile chamber **7812** and past the bottom of the electronics housing **7804** to allow the sensor **6216** and the sharp **6222** to be transcutaneously received beneath a user's skin. Distally advancing the part **7806** may be accomplished via a variety of mechanical or electromechanical means. In some embodiments, for example, the sensor applicator **7800** may include a plunger **7816** configured to advance distally to push the sensor **6216** and the sharp **6222** out of the sterile chamber **7812**. In such embodiments, the plunger **7816** may also be configured to attach to the sharp **6222** and subsequently retract the sharp **6222** while leaving the sensor **6216** extended. During operation, the plunger **7816** may penetrate the upper microbial barrier **7818a** and force the sensor **6216** and the sharp **6222** distally through the lower microbial barrier **7818b**.

In other embodiments, the part **7806** may be advanced distally out of the sterile chamber **7812** using a magnetic coupling. More specifically, the sensor applicator **7800** may include a driver magnet **7820** movable within the sensor applicator **7800** and magnetically coupled to a driven magnet **7822** disposed on the part **7806**, such as on an upper end of the sharp **6222**. The driver magnet **7820** may be configured to advance distally and simultaneously push the sensor **6216** and the sharp **6222** out of the sterile chamber **7812** as magnetically coupled to the driven magnet **7822**. Once the sensor **6216** is properly placed, the driver magnet **7820** may be retracted proximally and simultaneously retract the sharp **6222** in the same direction while leaving the sensor **6216** extended. During operation, the driver magnet **7820** may cause the sensor **6216** and the sharp **6222** to penetrate distally through the lower microbial barrier **7818b**.

In embodiments where the sterile chamber **7812** comprises a cap, the plunger **7816** may also be operable to discharge or push the cap out of the sensor applicator **7800**. In such embodiments, a user may commence the firing process by priming the sensor applicator **7800**, which may cause the cap to be discharged from the sensor applicator **7800**. Further actuation of the sensor applicator **7800** by the user may cause the sensor **6216** and the sharp **6222** to be fully extended for subcutaneous implantation. In other embodiments, the cap may be removed either autonomously (e.g., it falls off or breaks away during firing) or the user may manually remove it by hand.

In some embodiments, the sensor applicator **7800** may further include an electrical connector **7824** in electrical communication with the electronics of the sensor control device **7802**, such as the radiation sensitive components **7808**. In at least one embodiment, the electrical connector **7824** may comprise one or more elastic pins made of a conductive polymer (e.g., a carbon impregnated polymer) and configured to facilitate electrical communication between the sensor **6216** and the radiation sensitive component **7808**. In such embodiments, the sensor **6216** may include one or more connectors **7826** alignable with the electrical connector **7824** when the part **7806** is advanced distally, as described above. Moreover, in embodiments where the sterile chamber **7812** comprises a cap, the electrical connector **7824** may be flexible to allow the cap to pass by the electrical connector **7824** until the connectors **7826** align with the electrical connector **7824**.

FIG. **79** is an exploded view of an example sensor control device **7900**, according to one or more additional embodiments. The sensor control device **7900** may be similar in some respects to any of the sensor control devices described

herein. For example, the sensor control device **7900** may include a housing **7902** that contains or otherwise houses a battery **7904** that powers the sensor control device **7900** and one or more radiation sensitive components **7906**. The radiation sensitive component **7906** may be similar to the radiation sensitive component **7808** of FIG. **78**, and therefore will not be described again. In some embodiments, the housing **7902** may be made of a flexible or deformable material.

The sensor control device **7900** may further include a sensor module **7908** that may be coupled to the housing **7902** to form the assembled sensor control device **7900**. As illustrated, the sensor module **7908** may include the sensor **6216** and the sharp **6222** extending distally therefrom. In the illustrated embodiment, the sensor **6216** and the sharp **6222** extend at an angle relative to the housing **7902**, but could alternatively extend perpendicular to the housing **7902**.

The sensor module **7908** may be sterilized separate from the housing **7902** to prevent damage to the radiation sensitive components **7906**. Following sterilization, the sensor module **7908** may be paired or coupled to the housing **7902** via a variety of permanent or removable attachment means. In some embodiments, for example, the sensor module **7908** may be coupled to the housing **7902** via a snap-fit engagement, an interference fit, or using one or more mechanical fasteners. In other embodiments, however, the sensor module **7908** may be coupled to the housing **7902** using an adhesive, sonic welding, or laser welding. Pairing the sensor module **7908** to the housing **7902** may be done during manufacturing or may be accomplished by a user prior to deploying the sensor control device.

Coupling the sensor module **7908** to the housing **7902** may also facilitate communication between the sensor **6216** and the radiation sensitive components **7906**. More particularly, in some embodiments, the sensor module **7908** may include one or more sensor contacts **7910** alignable with one or more electrical connectors **1912** provided on the housing **7902** when the sensor module **7908** is coupled to the housing **7902**. The sensor contacts **7910** and the electrical connectors **1912** may comprise one or more elastic pins made of a conductive polymer (e.g., a carbon impregnated polymer) and configured to facilitate electrical communication between the sensor **6216** and the radiation sensitive component **7906**.

FIG. **80** is a bottom view of one embodiment of the sensor control device **7900** of FIG. **79**. As illustrated, the housing **7902** exhibits a generally polygonal cross-sectional shape and, more particularly, a triangular shape with rounded corners. In other embodiments, however, the housing **7902** may exhibit other cross-sectional shapes including, but not limited to, circular, oval, ovoid, or other polygonal shapes (e.g., square, rectangular, pentagonal, etc.), without departing from the scope of the disclosure.

In the illustrated embodiment, the sensor module **7908** may be coupled to the housing **7902** via a snap-in or snap-fit engagement. More specifically, the housing **7902** may define a cavity **8002** sized to receive the sensor module **7908**, and one or both of the housing **7902** and the sensor module **7908** may define or otherwise provide tabs **8004** configured to matingly engage when the sensor module **7908** is received within the cavity **8002**. The tabs **8004** may mate to secure the sensor module **7908** within the cavity **8002**. As will be appreciated, the tabs **8004** may be replaced with any other type of device or mechanism that facilitates a snap-in or snap-fit engagement, without departing from the scope of the disclosure. As indicated above, coupling the sensor module **7908** to the housing **7902** may be done during

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manufacturing or may be accomplished by a user prior to deploying the sensor control device.

Embodiments disclosed herein include:

X. A sensor applicator that includes a housing and a sensor retainer arranged within the housing, a sensor control device removably coupled to the sensor retainer and including an electronics housing, a sensor arranged within the electronics housing and extending from a bottom of the electronics housing, and a sharp hub that carries a sharp extending through the electronics housing and from the bottom of the electronics housing. The sensor application further includes a needle shroud extendable through the sensor retainer and the electronics housing and movable between an extended position, where the needle shroud extends past the bottom of the electronics housing and covers distal ends of the sensor and the sharp, and a retracted position, where the needle shroud retracts into the housing and thereby exposes the distal ends of the sensor and the sharp.

Y. A method of deploying a sensor control device from a sensor applicator that includes positioning the sensor applicator adjacent a target monitoring location, the sensor applicator including a housing and a sensor retainer arranged within the housing, wherein the sensor control device is removably coupled to the sensor retainer and includes an electronics housing, a sensor arranged within the electronics housing and extending from a bottom of the electronics housing, and a sharp hub that carries a sharp extending through the electronics housing and from the bottom of the electronics housing. The method further includes aligning a needle shroud with the target monitoring location, the needle shroud extending through the sensor retainer and the electronics housing, engaging the needle shroud against the target monitoring location to move the needle shroud from an extended position, where the needle shroud extends past the bottom of the electronics housing and covers distal ends of the sensor and the sharp, and pushing on the sensor applicator to move the needle shroud to a retracted position, where the needle shroud retracts into the housing and exposes the distal ends of the sensor and the sharp to transcutaneously receive the sensor at the target monitoring location.

Each of embodiments X and Y may have one or more of the following additional elements in any combination: Element 1: further comprising a sensor cap defining an inner chamber that receives the distal ends of the tail and the sharp and forms a sterile barrier that protects the distal ends of the sensor and the sharp. Element 2: further comprising an applicator cap removably coupled to the housing, wherein the applicator cap and the sensor cap are simultaneously removable from the housing. Element 3: wherein the sensor cap extends from the sensor control device. Element 4: wherein the sensor control device further includes a collar coupled to the electronics housing, and wherein the sensor cap is removably coupled to the collar. Element 5: wherein the sensor cap provides a gripping interface for a user to grasp onto and remove the sensor cap from the sensor applicator. Element 6: wherein the needle shroud is received within the sensor cap when the needle shroud is in the extended position. Element 7: further comprising one or more first retention features provided on the sensor retainer, one or more second retention features provided on the sensor control device and matable with the one or more first features, wherein disengaging the one or more second retention features from the one or more first features deploys the sensor control device for use. Element 8: wherein the sensor retainer provides a plurality of upwardly extending fingers

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engageable with the sharp hub to prevent the sharp hub from moving relative to the sensor retainer when the needle shroud is in the extended position. Element 9: wherein the plurality of fingers are extendable into an upper portion of the needle shroud and interpose the sharp hub and an inner wall of the upper portion of the needle shroud when the needle shroud is in the extended position. Element 10: further comprising a driver spring compressed between the sharp hub and the sensor retainer when the needle shroud is in the extended position, wherein moving the needle shroud to the retracted position allows the driver spring to expand and move the sharp hub to retract the sharp into the housing. Element 11: wherein the plurality of fingers are extendable into the sharp hub and interpose the needle shroud and an inner wall of the sharp hub when the needle shroud is in the extended position. Element 12: further comprising a driver spring compressed between the sharp hub and the sensor retainer when the needle shroud is in the extended position, wherein moving the needle shroud to the retracted position allows the driver spring to expand and move the sharp hub to retract the sharp into the housing. Element 13: wherein the needle shroud defines a groove at an upper end and the plurality of fingers provide inwardly extending features engageable with the groove to help maintain the needle shroud in the extended position. Element 14: wherein the sensor retainer includes one or more locking tabs matable with one or more locking members provided on the needle shroud to secure the needle shroud in the extended position.

Element 15: further comprising forming a sterile barrier with a sensor cap that receives the distal ends of the tail and the sharp, wherein the needle shroud is received within the sensor cap when the needle shroud is in the extended position, and removing the sensor cap prior to engaging the needle shroud against the target monitoring location. Element 16: wherein one or more first retention features provided on the sensor retainer are matable with one or more second retention features provided on the sensor control device to couple the sensor control device to the sensor retainer, the method further comprising adhesively attaching the sensor control device to the target monitoring location, and pulling the sensor applicator away from the target monitoring location to disengage the one or more second retention features from the one or more first retention features and thereby detach the sensor control device from the sensor retainer. Element 17: wherein the sensor retainer provides a plurality of upwardly extending fingers engageable with the sharp hub, the method further comprising preventing the sharp hub from moving relative to the sensor retainer with the plurality of fingers when the needle shroud is in the extended position. Element 18: wherein the plurality of fingers are extendable into an upper portion of the needle shroud and interpose the sharp hub and an inner wall of the upper portion of the needle shroud when the needle shroud is in the extended position, the method further comprising moving the sharp hub to retract the sharp into the housing when the needle shroud moves to the retracted position with a driver spring extending between the sharp hub and the sensor retainer. Element 19: wherein the plurality of fingers are extendable into the sharp hub and interpose the needle shroud and an inner wall of the sharp hub when the needle shroud is in the extended position, the method further comprising moving the sharp hub to retract the sharp into the housing when the needle shroud moves to the retracted position with a driver spring extending between the sharp hub and the sensor retainer.

By way of non-limiting example, exemplary combinations applicable to X and Y include: Element 1 with Element

2; Element 1 with Element 3; Element 3 with Element 4; Element 1 with Element 5; Element 1 with Element 6; Element 8 with Element 9; Element 9 with Element 10; Element 8 with Element 11; Element 11 with Element 12; Element 11 with Element 13; Element 15 with Element 16; Element 17 with Element 19; and Element 17 with Element 19.

Localized Axial-Radial Sensor Seal for Analyte Monitoring

Referring briefly again to FIG. 1, the system 100 may comprise what is known as a “two-piece” architecture that requires final assembly by a user before the sensor 110 can be properly delivered to the target monitoring location. According to embodiments of the present disclosure, the sensor control device assembly of FIG. 1 may instead comprise a one-piece architecture that incorporates sterilization techniques specifically designed for a one-piece architecture. The one-piece architecture allows the sensor control device assembly to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package and subsequently deliver the sensor control device to the target monitoring location. The one-piece system architecture described herein may prove advantageous in eliminating component parts, various fabrication process steps, and user assembly steps. As a result, packaging and waste are reduced, and the potential for user error or contamination to the system is mitigated.

FIGS. 81A and 81B are isometric and side views, respectively, of an example sensor control device 8102. The sensor control device 8102 may be similar in some respects to the sensor control device 104 of FIG. 1 and therefore may be best understood with reference thereto. In some applications, the sensor control device 8102 may replace the sensor control device 104 of FIG. 1 and, therefore, may be used in conjunction with the analyte monitoring system 100 (FIG. 1) or the sensor applicator 102, which delivers the sensor control device 8102 to a target monitoring location on a user's skin.

The sensor control device 8102 includes an electronics housing 8104 that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing 8104 may exhibit other cross-sectional shapes, such as ovoid or polygonal and may be non-symmetrical. The electronics housing 8104 may include a shell 8106 and a mount 8108 configured to engage or couple with the shell 8106. The shell 8106 may be secured to the mount 8108 via a variety of ways, such as a snap fit engagement, an interference fit, sonic (or ultrasonic) welding, using one or more mechanical fasteners (e.g., screws), or any combination thereof. In some embodiments, the interface between the shell 8106 and the mount 8108 may be sealed. In such embodiments, a gasket or other type of seal material may be positioned or applied at or near the outer diameter (periphery) of the shell 8106 and the mount 8108. Securing the shell 8106 to the mount 8108 may compress the seal material and thereby generate a sealed interface. In at least one embodiment, an adhesive may be applied to the outer diameter (periphery) of one or both of the shell 8106 and the mount 8108, and the adhesive may not only secure the shell 8106 to the mount 8108 but may also seal the interface.

In embodiments where a sealed interface is created between the shell 8106 and the mount 8108, the interior of the electronics housing 8104 may be effectively isolated from outside contamination between the two components. In such embodiments, if the sensor control device 8102 is assembled in a controlled and sterile environment, there may

be no need to sterilize the internal electrical components (e.g., via gaseous chemical sterilization). Rather, the sealed engagement may provide a sufficient sterile barrier for the assembled electronics housing 8104.

The sensor control device 8102 may further include a sensor 8110, a sharp module 8112 engaged with the sensor 8110. The sensor 8110 and the sharp module 8112 may be interconnectable and may be coupled to the electronics housing 8104. The sharp module 8112 may be configured to carry and otherwise include a sharp 8116 used to help deliver the sensor 8110 transcutaneously under a user's skin during application of the sensor control device 8102.

As best seen in FIG. 81B, corresponding portions of the sensor 8110 and the sharp 8116 extend from the electronics housing 8104 and, more particularly, from the bottom of the mount 8108. The exposed portion of the sensor 8110 may be received within a hollow or recessed portion of the sharp 8116. The remaining portion(s) of the sensor 8110 is/are positioned within the interior of the electronics housing 8104.

FIG. 82 is an exploded perspective top view of the sensor control device 8102, according to one or more embodiments. As illustrated, the shell 8106 and the mount 8108 of the electronics housing 8104 may operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device 8102. Various electrical components may be positioned within the electronics housing 8104, including a printed circuit board (PCB) 8202 having a plurality of electronic modules 8204 and a battery 8205 mounted to the PCB 8202. The battery 8205 may be configured to power the sensor control device 8102. Example electronic modules 8204 include, but are not limited to, resistors, transistors, capacitors, inductors, diodes, integrated circuits, and switches. A data processing unit 8206 (FIG. 82) may also be mounted to the PCB 8202 and may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device 8102. More specifically, the data processing unit 8206 may be configured to perform data processing functions, such as filtering and encoding of data signals, each of which corresponds to a sampled analyte level of the user. The data processing unit 8206 may also include or otherwise communicate with an antenna for communicating with the reader device 106 (FIG. 1). As shown in FIG. 82, the PCB 8202 and various components mounted to it may be encapsulated or otherwise contained within an encapsulating material 8207.

As illustrated in FIG. 82, the shell 8106, the mount 8108, and the PCB 8202, and encapsulating material 8207 each define corresponding channels or apertures 8208a, 8208b, 8208c, 8208d, respectively. Due to their placement with respect to the outer surface of electronics housing 8104, aperture 8208a in the shell 8106 may be referred to as a top aperture, and aperture 8208b in the mount 8108 may be referred to as a bottom aperture. The mount 8108 further includes a channel 8210 that extends upward from aperture 8208b and a slot 8212 that extends through a side wall of channel 8210. When the sensor control device 8102 is assembled, the apertures 8208a-c align and the channel 8210 extends through apertures 8208a, 8208c, 8208d to receive portions of the sensor 8110 and the sharp module 8112 therethrough. The centers or central regions of apertures 8208a, 8208b, 8208c, 8208d and channel 8210 are arranged in an eccentric manner with respect to electronics housing 8104, being spaced apart from the sensor central axis 8105. The sharp 8110 and sensor 8110, which may extend through

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at least one of these apertures and channel **8210**, are likewise spaced apart from the sensor central axis **8105** and are arranged in an eccentric manner.

The sensor control device **8102** may further include a housing support **8250** to be located in electronics housing **8104** in the vicinity of apertures **8208a**, **8208b**, **8208c**, **8208d** to provide support between shell **8106** and mount **8108**. The illustrated embodiment, housing support **8250** for electronics housing **8104** is a collar **8250**. The collar **8250** may exhibit a variety of shapes, such as cylindrical, tubular, annular, polygonal, or any combination thereof.

The sensor **8110** includes a tail **8216**, a flag **8218**, and a neck **8220** that interconnects the tail **8216** and the flag **8218**. The central aperture **8208b** and channel **8210** defined in the mount **8108** may be configured to receive the tail **8216**, which may extend therethrough and extend distally from the underside thereof. The slot **8212** in the mount **8108** may be configured to receive the sensor neck **8220**, allowing the flag **8218** to extend to or toward the PCB **8202**. The tail **8216** includes an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry. In use, the tail **8216** is transcutaneously received beneath a user's skin, and the chemistry included thereon helps facilitate analyte monitoring in the presence of bodily fluids.

The flag **8218** may comprise a generally planar surface having one or more sensor contacts **8222** (two shown in FIG. **82**) disposed thereon. The flag **8218** or the contacts **8222** are configured to couple electrically to the PCB **8202** or modules on PCB **8202**, which may include a corresponding number of contacts (not shown), such as contacts on compliant carbon impregnated polymer modules for example.

The sharp module **8112** includes the sharp **8116** and a sharp hub **8230** that carries the sharp **8116**. The sharp **8116** includes an elongate shaft **8232** and a sharp tip **8234** at the distal end of the shaft **8232**. The shaft **8232** may be configured to extend through each of the coaxially aligned central apertures **8208a-c** and extend distally from the bottom of the mount **8108**. Moreover, the shaft **8232** may include a hollow or recessed portion **8236** that at least partially circumscribes the tail **8216** of the sensor **8110**. The sharp tip **8234** may be configured to penetrate the skin while carrying the tail **8216** to put the active chemistry of the tail **8216** into contact with bodily fluids.

The sharp hub **8230** may include a hub small cylinder **8238** and a hub snap pawl **8240**, each of which may be configured to help couple the sensor control device **8102** to the sensor applicator **102** (FIG. **1**).

An adhesive or adhesive patch (not shown), similar to the adhesive patch **108** of FIG. **1**, may be positioned on and otherwise attached to the bottom **8111** of the mount **8108**. As discussed above, the adhesive patch may be configured to secure and maintain the sensor control device **8102** in position on the user's skin during operation.

FIG. **83** is a cross-sectional side view of a sensor control device assembly **8310** having a central or longitudinal assembly axis **8311** and including a sensor applicator **8312** with a cap **8330** coupled thereto and the sensor control device **8102** installed inside. In some applications, the sensor control device assembly **8310** with its sensor control device **8102** and applicator **8312** may replace the sensor control device **104** and the applicator **102** of FIG. **1** and, therefore, may be used in conjunction with the analyte monitoring system **100** (FIG. **1**).

The cap **8330** may be threaded to the sensor applicator **8312** and may include a tamper-evident ring or wrap (not shown) to evidence or inhibit premature unthreading. Moreover, the cap **8330** may define an undercut **8313** at the base

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of the threaded interface that provides additional stiffness in tilting at the interface between the cap **8330** and the housing **8314** and a detent force that may need to be overcome for the cap **8330** to unscrew. Upon rotating (e.g., unscrewing) the cap **8330** relative to sensor applicator **8312**, the tamper ring or wrap may shear and thereby free the cap **8330** and desiccant **8315** from the sensor applicator **8312**. Following which, the user may deliver the sensor control device **8102** to the target monitoring location.

The sensor applicator **8312** includes a housing **8314** that is disposed around and slidably coupled to a sheath **8318** and is configured to move a prescribed axial distance relative to the sheath **8318**. Sheath **8318** defines a bottom for sensor applicator **8312**, the bottom that rests against a user's skin, for example, when sensor control device assembly **8310** is used to place a sensor control device **8102** on the user. Sensor applicator **8312** also includes a sharp carrier **8360** and a sensor carrier **8364** interposed between the sheath **8318** and sharp carrier **8360**. Sensor carrier **8364** includes a radially extending platform **8366** located below sharp carrier **8360**, which may rest on the platform **8366**. Platform **8366** is coupled to housing **8314** to move when housing **8314** moves axially relative to sheath **8318**.

The cap **8330** may include an outer shell **8332** that extends from a threaded first end **8333** to a bottom or second end **8334**. A base **8336** may be located at the second end **8334**, a support structure **8338** may extend from the base **8336** upward toward the first end **8333**, and a post **8350** extending from the support structure **8338**. Likewise, when installed, support structure **8338** may extend upward from the bottom of the sheath **8318** of the sensor applicator **8312**. The support structure **8338** is located within the outer shell **8332** and includes an inner shell **8340** supported by a plurality of ribs **8342**. Viewed from base **8336**, inner shell **8340** is concave. The post **8350** is centrally located within the interior of the cap **8330** and may be aligned with assembly axis **8311**. The post **8350** extends downward from a first end **8353** at the top of inner shell **8340** to a second end **8354** closer to cap base **8336**. The post **8350** defines a post chamber **8356**, which is open at first end **8353** and closed at second end **8354**.

The support structure **8338** or the post **8350** may be configured to help support the sensor control device **8102** while contained within the sensor applicator **8312**. Moreover, the post chamber **8356** is configured to receive the sensor **8110** and the sharp **8116** when extending from the bottom of the electronics housing **8104**. When the sensor control device **8102** is loaded into the sensor applicator **8312**, the sensor **8110** and the sharp **8116** may be arranged within a sealed region **8370** at least partially defined by the post chamber **8356** and configured to isolate the sensor **8110** and the sharp **8116** from various other regions in sensor control device assembly **8310**, which may contain various fluids or contaminants at various times.

The cap **8330** provides a barrier against outside contamination, and thereby maintains a sterile environment for the sensor control device assembly **8310**, including the sensor control device **8102** contained therein, until the user removes (unthreads) the cap **8330**. The cap **8330** may also create a dust-free environment during shipping and storage.

A desiccant **8315** may be included in cap **8330**, being located within the outer volume of the inner shell **8340**, and a cover member or seal **8316**, which in this example includes foil, may be applied to base **8336** to contain and seal the desiccant **8315** against the intrusion of moisture and other contamination, and may also provide evidence of tampering.

In some embodiments, the seal **8316** may comprise only a single protective layer applied to the cap **8330**, such as foil. In some embodiments, the seal **8316** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can be applied before a gaseous chemical sterilization is performed, and following the gaseous chemical sterilization, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture.

Referring now to FIG. **84**, illustrated is an enlarged cross-sectional side view of the sensor control device assembly **8310** having sensor control device **8102** mounted within the sensor applicator **8312** and the cap **8330** secured thereto, according to one or more embodiments. The sensor control device **8102** may be loaded into the sensor applicator **8312** by mating the sharp hub **8230** with the sharp carrier **8360** and by mating the electronics housing **8104** of the sensor control device **8102** with the sensor carrier **8364** (alternately referred to as a “puck carrier”). More specifically, the hub small cylinder **8238** and the hub snap pawl **8240** of sharp hub **8230** may be received by corresponding mating features of the sharp carrier **8360**.

After installation in sensor control device assembly **8310**, the sensor control device **8102** may be subjected to “focused” radiation sterilization **8404**, where the radiation is applied and otherwise directed toward the sensor **8110** and the sharp **8116**. In such embodiments, some or all of the electrical components **8204** (FIG. **82**), such as components group **8406** indicated with a dashed enclosure in FIG. **84**, may be positioned out of the range (span) of the propagating radiation **8404** and, therefore, will not be affected by the radiation. For this purpose, apertures **8208a**, **8208b**, **8208c**, **8208d**, sensor **8110**, and sharp module **8112** are spaced apart from the sensor central axis **8105** to increase the distance between these features that receive radiation **8404** and the components group **8406** of PCB **8202** that may contain various of the components **8204**, **8206** that are to be protected from radiation **8404**. For example, some or all of the electrical components **8204** and the data processing unit **8206**, as examples, may be positioned on the PCB **8202** near its outer periphery so as not to fall within the range (span) of the focused radiation sterilization **8404**. In other embodiments, this protection from radiation may be accomplished by shielding some or all of the electrical components **8204** and the data processing unit **8206**, as examples, with proper electromagnetic shields.

As indicated above, portions of the sensor **8110** and the sharp **8116** may be arranged within the sealed region **8370** and thereby protected from substances that might adversely interact with the chemistry of the sensor **8110**. More specifically, the sealed region **8370** protects the tail **8216**. The sealed region **8370** may include (encompass) select portions of the interior of the electronics housing **8104** and the post chamber **8356** of the post **8350**. In one or more embodiments, the sealed region **8370** may be defined and otherwise formed by at least a first seal **8408a** and a second seal **8408b**. Coupling the shell **8106** to the mount **8108** may create a sealed interface therebetween that may also participate in defining the extent of sealed region **8370**.

The first seal **8408a** may be arranged to seal an interface between the sharp hub **8230** and the shell **8106**. In the present example, the first seal **8408a** may be arranged to seal a first interface **8411** between the sensor carrier **8364** and the

top of the electronics housing **8104**, e.g., the shell **8106**. The first seal **8408a** may also be arranged to seal a second interface **8412** between the sensor carrier **8364** and sharp hub **8230** of the sharp module **8112**. Moreover, at first interface **8411** the first seal **8408a** may circumscribe the first central aperture **8208a** defined in the shell **8106** such that contaminants are prevented from migrating in a radial direction (relative to sensor axis **8105**) into the interior of the electronics housing **8104** via the first central aperture **8208a** or channel **8210**. At second interface **8412** the first seal **8408a** may prevent fluid from migrating in an axial direction relative to assembly axis **8311** (or, alternatively, relative to sensor axis **8105**) into the interior of the electronics housing **8104** via the first central aperture **8208a** or channel **8210**. Therefore, the first seal **8408a** interposes the sensor carrier **8364** and the electronics housing **8104** and interposes sensor carrier **8364** and the sharp hub **8230** and is configured to provide axial and radial sealing. In this example, first seal **8408a** is interposed between sensor applicator **8312** (e.g., the sensor carrier **8364**) and sensor control device **8104** and is also interposed between sensor applicator **8312** and sharp module **8112**.

In at least one embodiment, the first seal **8408a** may be overmolded on to the sensor carrier **8364**, thus forming a part of sensor carrier **8364**. In other embodiments, however, the first seal **8408a** may form part of the sharp hub **8230**, such as by being overmolded onto the sharp hub **8230**. In yet other embodiments, the first seal **8408a** may be overmolded onto the top surface of the shell **8106**. In even further embodiments, the first seal **8408a** may comprise a separate structure, such as an O-ring or the like, that interposes the sharp hub **8230** and the top surface of the shell **8106**, without departing from the scope of the disclosure.

The second seal **8408b** may be arranged to seal an interface **8413** between the post **8350** and the bottom of the mount **8108**, and the second seal **8408b** may circumscribe the second central aperture **8208b** defined in the mount **8108**. The second seal **8408b** may also circumscribe the post chamber **8356**. Consequently, the second seal **8408b** may prevent contaminants from migrating into the post chamber **8356** of the post **8350** and also from migrating into the interior of the electronics housing **8104** via the second central aperture **8208b**. For clarity, interface **8413** may also be referred to as a third interface. At third interface **8413**, the second seal **8408b** may prevent fluid from migrating in the radial direction.

As illustrated in FIG. **84**, the housing support **8250**, which in this example is a collar **8250**, may be located in electronics housing **8104** in the vicinity of apertures **8208a**, **8208b**, **8208c**, **8208d** and around collar **8250** of mount **8108** to provide support between shell **8106** and mount **8108** when an axial force is applied to engage seals **8408a**, **8408b** with electronics housing **8104**. The collar **8250** extends between the top and bottom of the electronics housing **8104** (e.g. shell **8106** and mount **8108**, respectively) and is positioned about the sensor **8110** to support the top of the electronics housing **8104** against flexing toward the bottom of the electronics housing and to support the bottom of the electronics housing against flexing toward the top of the electronics housing. Thus, collar **8250** is configured to provide a reaction force between top and bottom of the electronics housing **8104** when seals **8408a**, **8408b** engage electronics housing **8104**. Some embodiments include a housing support **8250** that is formed or bonded as a portion of electronics housing **8104** and may be, as examples, an extension of shell **8106** or an extension of mount **8108**.

Upon loading the sensor control device **8102** into the sensor applicator **8312** and securing the cap **8330** to the sensor applicator **8312**, the first and second seals **8408a,b** become compressed and generate corresponding sealed interfaces. The first and second seals **8408a,b** may be made of a variety of materials capable of generating a sealed interface between opposing structures. Suitable materials include, but are not limited to, silicone, a thermoplastic elastomer (TPE), polytetrafluoroethylene (Teflon®), rubber, an elastomer, or any combination thereof.

The cap **8330** may be secured to the sensor applicator **8312** by threading the cap **8330** to the sensor applicator **8312** via relative rotation. As the cap **8330** rotates relative to the sensor applicator **8312**, the post **8350** advances axially until post **8350** or the inner shell **8340** of cap **8330** engages the second seal **8408b** on the sealable surface **8418** at the bottom of the mount **8108**, creating a sealed interface **8413** therebetween. As the electronics housing **8104** of sensor control device **8102** is urged to rotate through frictional engagement between the second seal **8408b** and post **8350** or the inner shell **8340** of cap **8330**, sensor carrier **8364** inhibits rotation of the sensor control device **8102**.

FIG. **85** shows a bottom view of sensor control device **8102** and sensor carrier **8364**. Sensor carrier **8364** includes a pair of arms **8506** that extend around sensor control device **8102**. Arms **8506** may grasp notches formed in electronic housing **8106**. As illustrated, a sealable surface **8418** that extends around second central aperture **8208b** may be defined on the bottom of the mount **8108**. The sealable surface **8418** may comprise a groove. The sealable surface **8418** may receive second seal **8408b** to isolate (protect) the tail **8216** of the sensor **8110** from environmental contamination or from potentially harmful sterilization gases when gaseous chemical sterilization is used. In the illustrated embodiment, the second seal **8408b** is overmolded onto the bottom of the mount **8108** within a groove of sealable surface **8418**. Thus, second seal **8408b** forms a part of the electronics housing **8104**. In other embodiments, however, the second seal **8408b** may form part of the post **8350** (FIG. **84**). For example, the second seal **8408b** may be overmolded onto the top of the post **8350**. In yet other embodiments, the second seal **8408b** may comprise a separate structure, such as an O-ring or the like, that interposes the post **8350** and the bottom of the mount **8108**, without departing from the scope of the disclosure.

FIG. **86** is a schematic diagram of an example sterilization assembly **8600**, according to one or more embodiments of the present disclosure. The sterilization assembly **8600** (hereafter the “assembly **8600**”) may be designed and otherwise configured to help sterilize a medical device **8602** that may be deployed for use from a sensor applicator **8604**. The medical device **8602** may comprise, for example, a sensor control device similar in some respects to any of the sensor control devices described herein. In such embodiments, the sensor applicator **8604** may be similar in some respects to any of the sensor applicators described herein. Alternatively, the medical device **8602** may comprise other types of medical devices, health care products, or systems requiring terminal sterilization of specific component parts. Example medical devices or health care products that may incorporate the principles of the present disclosure include, but are not limited to, ingestible products, cardiac rhythm management (CRM) devices, under-skin sensing devices, externally mounted medical devices, or any combination thereof.

As illustrated, the medical device **8602** may include a housing **8606**, a part **8608** requiring sterilization, and one or

more radiation sensitive components **8610**. In the illustrated embodiment, the radiation sensitive component **8610** may be mounted to a printed circuit board (PCB) **8612** positioned within the housing **8606** and may include one or more electronic modules such as, but not limited to, a data processing unit (e.g., an application specific integrated circuit or ASIC), a resistor, a transistor, a capacitor, an inductor, a diode, and a switch.

As illustrated, the part **8608** may extend at an angle relative to the housing **8606**, but could alternatively extend perpendicular to the housing **8606**. In some embodiments, the part **8608** may comprise a sensor (e.g., the sensor **8110** of FIGS. **81A-81B**) and a sharp (e.g., the sharp **8116** of FIGS. **81A-81B**) used to help implant the sensor beneath the skin of a user. In some embodiments, as illustrated, the part **8608** may be temporarily encapsulated within a sterile chamber **8614** that provides a sealed barrier to protect exposed portions of the part **8608** (e.g., the sensor and associated sharp) until the part **8608** is needed for use.

The medical device **8602** may be subjected to radiation sterilization **8616** to properly sterilize the part **8608** for use. Suitable radiation sterilization **8616** processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. As illustrated, the assembly **8600** may include a radiation shield **8618** positioned external to the medical device **8602** and configured to help sterilize the part **8608** while preventing (impeding) propagating radiation **8616** from disrupting or damaging the radiation sensitive components **8610**. To accomplish this, the radiation shield **8618** may provide a collimator **8620** that generally comprises a hole or passageway extending at least partially through the body of the radiation shield **8618**. The collimator **8620** provides a sterilization zone designed to direct (focus) the radiation **8616** toward the part **8608**.

While the collimator **8610** focuses the radiation **8616** (e.g., beams, waves, energy, etc.) toward the part **8608**, the remaining portions of the radiation shield **8618** may be made of a material that reduces or eliminates the radiation **8616** from penetrating therethrough and thereby damaging the radiation sensitive components **8610** within the housing **8606**. In other words, the radiation shield **8618** may be made of a material having a density sufficient to absorb the dose of the beam energy being delivered. In some embodiments, for example, the radiation shield **8618** may be made of any material that has a mass density greater than 0.9 grams per cubic centimeter (g/cc). In other embodiments, however, the mass density of a suitable material may be less than 0.9 g/cc, without departing from the scope of the disclosure. Suitable materials for the radiation shield **8618** include, but are not limited to, a high-density polymer, (e.g., polyethylene, polypropylene, polystyrene, polytetrafluoroethylene, etc.), a metal (e.g., lead, stainless steel, aluminum, etc.), any combination thereof, or any material having a mass density greater than 0.9 g/cc.

The collimator **8620** can exhibit any suitable cross-sectional shape necessary to focus the radiation on the part **8608** for sterilization. In the illustrated embodiment, for example, the collimator **8620** has a circular cross-section with parallel sides. In other embodiments, however, the collimator **8620** may be conical or frustoconical in shape, or may have a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure.

In some embodiments, the assembly **8600** may further include a barrier shield **8622** positioned within the housing **8606**. The barrier shield **8622** may be configured to help

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block radiation **8616** (e.g., electrons) from propagating within the housing **8606** toward the radiation sensitive components **8610**. The barrier shield **8622** may be made of any of the materials mentioned above for the radiation shield **8618**. In the illustrated embodiment, the barrier shield **8622** is positioned vertically within the housing **8606**, but may alternatively be positioned at any other angular configuration suitable for protecting the radiation sensitive components **8610**.

In some embodiments, the sterile chamber **8614** may comprise a cap that encapsulates the part **8608** to provide a sealed barrier that protects exposed portions of the part **8608** until the part **8608** is placed in use. In such embodiments, the sterile chamber **8614** may be removable or detachable to expose the part **8608**, as described below. Moreover, in such embodiments, the cap may be made of a material that allows radiation to propagate therethrough to allow sterilization of the part **8608**. Suitable materials for the sterile chamber **8614** include, but are not limited to, a non-magnetic metal (e.g., aluminum, copper, gold, silver, etc.), a thermoplastic, ceramic, rubber (e.g., ebonite), a composite material (e.g., fiberglass, carbon fiber reinforced polymer, etc.), an epoxy, or any combination thereof. In some embodiments, the sterile chamber **8614** may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

In other embodiments, the sterile chamber **8614** may comprise a chamber or compartment defined within one or both of the sensor applicator **8604** and the sensor control device **8602**. In such embodiments, the sterile chamber **8614** may include a microbial barrier positioned at one or both ends of the sterile chamber **8614**. More specifically, the sterile chamber **8614** may provide or include an upper microbial barrier **8624a** and a lower microbial barrier **8624b** opposite the upper microbial barrier **8624a**. The upper and lower microbial barriers **8624a,b** may help seal the sterile chamber **8614** to thereby isolate the part **8608** from external contamination. The microbial barriers **8624a,b** may be made of a radiation permeable material, such as a synthetic material (e.g., a flash-spun high-density polyethylene fiber). One example synthetic material comprises TYVEK®, available from DuPont®. In other embodiments, however, the microbial barriers **8624a,b** may comprise, but are not limited to, tape, paper, film, foil, or any combination thereof.

In embodiments where the sterile chamber **8614** comprises a cap, the sterile chamber **8614** may be movable distally to help facilitate the sterilization process. More specifically, the sterile chamber **8614** may be movable at least partially into the sterilization zone formed by the collimator **8620**. Once positioned within the sterilization zone, the part **8608** may be subjected to the radiation **8616** to sterilize the part **8608** for use. Once sterilization is done, the sterile chamber **8614** may be retracted proximally in preparation for firing the sensor control device **8602**. Distally advancing the sterile chamber **8614** may be accomplished via a variety of mechanical or electromechanical means. In some embodiments, for example, the sensor applicator **8604** may include a plunger **8626** configured to advance distally to push the sterile chamber **8614** distally, and subsequently retract the sterile chamber **8614** once the sterilization process is complete.

The part **8608** itself may also be deployable and otherwise movable relative to the sensor applicator **8604**. More particularly, the part **8608** may be advanced distally past the bottom of the electronics housing **8606** to allow the part **8608** to be transcutaneously received beneath a user's skin. In some embodiments, the plunger **8626** may be used to

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push the part **8608** out of the sterile chamber **8614**. In such embodiments, the plunger **8626** may also be configured to attach to a portion of the part **8608** (e.g., the sharp) and subsequently retract that portion of the part **8608** while leaving another portion of the part **8608** (e.g., the sensor) extended. Moreover, in such embodiments, the plunger **8626** may be configured to penetrate the upper microbial barrier **8624a** and force the part **8608** distally through the lower microbial barrier **8624b**.

In other embodiments, the part **8608** may be advanced distally out of the sterile chamber **8614** using a magnetic coupling. More specifically, the sensor applicator **8604** may include a driver magnet **8628** movable within the sensor applicator **8604** and magnetically coupled to a driven magnet **8630** disposed on the part **8608**, such as on an upper end of the sharp. The driver magnet **8628** may be configured to advance distally and simultaneously push the part **8608** out of the sterile chamber **8614** as magnetically coupled to the driven magnet **8630**. In such embodiments, actuation of the magnetic coupling may force the part **8608** distally through the lower microbial barrier **8624b**. Once the sensor is properly placed, the driver magnet **8628** may be retracted proximally and simultaneously retract the sharp in the same direction while leaving the sensor extended.

In embodiments where the sterile chamber **8614** comprises a cap, the plunger **8626** may also be operable to discharge or push the cap out of the sensor applicator **8604** to enable the part **8608** to be properly received by the user. In such embodiments, a user may commence the firing process by priming the sensor applicator **8604**, which may cause the cap to be discharged or ejected from the sensor applicator **8604**. Further actuation of the sensor applicator **8604** by the user may cause the part **8608** to be fully extended for subcutaneous implantation. In other embodiments, however, the cap may be removed either autonomously (e.g., it falls off or breaks away) or the user may manually remove it by hand.

In some embodiments, the sensor applicator **8604** may further include an electrical connector **8632** in electrical communication with the electronics of the sensor control device **8602**, such as the radiation sensitive component **8610**. In at least one embodiment, the electrical connector **8632** may comprise one or more elastic pins made of a conductive polymer (e.g., a carbon impregnated polymer) and configured to facilitate electrical communication between the sensor and the radiation sensitive component **8610**. In such embodiments, the sensor may include one or more connectors **8634** alignable with the electrical connector **8632** when the part **8608** is advanced distally, as described above. Moreover, in embodiments where the sterile chamber **8614** comprises a cap, the electrical connector **8632** may be flexible to allow the cap to pass by the electrical connector **8632** until the connectors **8634** align with the electrical connector **8632**.

FIG. **87** is a schematic diagram of another example sterilization assembly **8700**, according to one or more embodiments of the present disclosure. The sterilization assembly **8700** (hereafter the "assembly **8700**") may be similar in some respects to the assembly **8600** of FIG. **86** and therefore may be best understood with reference thereto, where like numerals will represent like components not described again in detail. Similar to the assembly **8600**, for example, the medical device **8602** may be arranged for deployment within the sensor applicator **8604**, and the part **8608** requiring sterilization may be temporarily encapsulated within the sterile chamber **8614**. Unlike the assembly

8600, however, the part **8608** may be subjected to the radiation sterilization **8616** through the body of the sensor applicator **8604**.

More specifically, the radiation sterilization **8616** may be directed to the top of the sensor applicator **8604**, which defines a collimator **8702** that allows the radiation **8616** to impinge upon and sterilize the part **8608**. As illustrated, the collimator **8702** generally comprises a hole or passageway extending through the body of the sensor applicator **8604**. The collimator **8702** focuses (guides) the radiation **8616** toward the part **8608** and can exhibit any suitable cross-sectional shape necessary to focus the radiation **8616** on the part **8608** for sterilization. In the illustrated embodiment, for example, the collimator **8702** has a circular cross-section with parallel sides, but may alternatively, be conical or frustoconical in shape, or may exhibit a polygonal cross-sectional shape, such as cubic, rectangular (e.g., including parallelogram), or pyramidal, without departing from the scope of the disclosure.

The sensor applicator **8604** may also act as a radiation shield that helps prevent (impede) propagating radiation **8616** from disrupting or damaging the radiation sensitive components **8610**, except through the collimator **8702**. To accomplish this, the sensor applicator **8604** may be made of a material similar to the material of the radiation shield **8618** of FIG. **86**. In at least one embodiment, however, the radiation sterilization **8616** may be emitted from a device or machine configured to focus and/or aim the radiation **8616** directly into the collimator **8702**, and thereby mitigating radiation **8616** exposure to adjacent portions of the sensor applicator **8604**.

In some embodiments, a seal **8704** may be arranged at the opening to the collimator **8702** at the top of the sensor applicator **8604**. The seal **8704** may comprise a radiation permeable, microbial barrier, similar to the microbial barriers **8624a,b** of FIG. **86**. The seal **8704** may seal off the collimator **8702**, while simultaneously allowing the radiation **8616** to pass therethrough to sterilize the part **8608**.

In at least one embodiment, the position of the radiation sensitive components **8610** may be moved away from the line of fire of the radiation **8616**. In other embodiments, the barrier shield **8622** may extend about at least two sides of the radiation sensitive components **8610** to ensure sufficient blockage of the radiation **8616**. In at least one embodiment, however, the barrier shield **8622** may fully encapsulate the radiation sensitive components **8610**.

In one embodiment, the radiation sterilization **8616** may be directed toward the part **8608** from the bottom of the sensor control device **8602** and the bottom of the sensor applicator **8604**. In such embodiments, a shield **8706** may be positioned at the bottom of one or both of the sensor control device **8602** and the bottom of the sensor applicator **8604**. The shield **8706** may be made of any of the materials mentioned above for the radiation shield **8618** of FIG. **86**. Consequently, the shield **8706** may be configured to help block the radiation **8616** (e.g., electrons) from propagating toward the radiation sensitive components **8610**. The shield **8706**, however, may define or otherwise provide an aperture **8708** aligned with the part **8608** to allow the radiation **8616** to impinge upon the part **8608** for proper sterilization.

In at least one embodiment, the shield **8706** may form part of the sensor control device **8602** and may be deployed simultaneously with the sensor control device **8602** from the sensor applicator **8604**. In some embodiments, the shield **8706** may be removable from the sensor control device **8602** and otherwise only used during the sterilization process. In other embodiments, the shield **8706** may be arranged within

the housing **8606** and otherwise form an integral part thereof, without departing from the scope of the disclosure.

FIG. **88A** is a schematic bottom view of another example sterilization assembly **8800**, according to one or more embodiments of the present disclosure. The sterilization assembly **8800** (hereafter the “assembly **8800**”) may be used to sterilize a medical device **8802**, which may comprise a sensor control device or any of the other types of medical devices mentioned herein. In the illustrated embodiment, the medical device **8802** comprises a sensor control device having a housing **8804** that defines an aperture **8806** through which a part **8808** requiring sterilization may extend. In the view of FIG. **88A**, the part **8808** extends through the aperture **8806** and out of the page. Moreover, the part **8808** may comprise one or both of a sensor and a sharp, as generally described herein. The medical device **8802** may also include a battery **8810** and a radiation sensitive component **8812** arranged within the housing **8804**. The battery **8810** may power the medical device **8802** and the radiation sensitive component **8812** may be similar to the radiation sensitive component **8610** of FIGS. **86** and **87**.

As illustrated, the housing **8804** may exhibit a generally polygonal cross-sectional shape. More specifically, the housing **8804** is generally triangular with rounded corners. The position of the radiation sensitive component **8812** relative to the part **8808** is effectively as far away as possible within the confines of the housing **8804**. As will be appreciated, this may help reduce the chances of the radiation sensitive component **8812** being damaged during a radiation sterilization process to sterilize the part **8808**.

The assembly **8800** may also include a shield **8814** (shown in dashed lines), which may be made of the materials mentioned above for the radiation shield **8618** of FIG. **86**. Consequently, the shield **8814** may be configured to help protect the radiation sensitive component **8812** from damaging radiation during a sterilization process. In one embodiment, the shield **8814** may be arranged external to the housing **8804** and otherwise arranged to interpose the radiation sensitive component **8812** and the propagating electrons from the radiation treatment. In other embodiments, however, the shield **8814** may be arranged within the housing **8804** and otherwise form part of the medical device **8802**, without departing from the scope of the disclosure.

FIGS. **88B** and **88C** are schematic bottom views of alternative embodiments of the sterilization assembly **8800** of FIG. **88A**, according to one or more additional embodiments of the present disclosure. In FIG. **88B**, the housing **8804** exhibits a generally circular shape, and in FIG. **88C**, the housing **8804** exhibits a generally oval or ovoid shape. As will be appreciated, the housing **8804** may alternatively exhibit other cross-sectional shapes, including additional polygonal shapes (e.g., square, rectangular, pentagonal, etc.), without departing from the scope of the disclosure.

In FIGS. **88B** and **88C**, the part **8808** extends through the aperture **8806** and out of the page. Moreover, the battery **8810** and the radiation sensitive component **8812** may be arranged within the housing **8804** and the radiation sensitive component **8812** may be positioned relative to the part **8808** as far away as possible within the confines of the housing **8804**. Again, this may help reduce the chances of the radiation sensitive component **8812** being damaged during a radiation sterilization process to sterilize the part **8808**. The shield **8814** (shown in dashed lines) may again be included and configured to help protect the radiation sensitive component **8812** from damaging radiation during a sterilization process. As illustrated, the shield **8814** may be arranged external to the housing **8804**, or alternatively within the

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housing **8804** and otherwise form part of the medical device **8802**, without departing from the scope of the disclosure.

FIG. **89** is an isometric schematic view of an example sensor control device **8900**, according to one or more embodiments. The sensor control device **8900** may be similar in some respects to the sensor control devices described herein and, therefore, may be used as an on-body monitoring device used to monitor blood glucose levels. As illustrated, the sensor control device **8900** includes a housing **8902** that may contain and otherwise housing electronics used to operate the sensor control device **8900**. In the illustrated embodiment, the housing **8902** is generally disc-shaped and with a circular cross-section, but could alternatively exhibit other cross-sectional shapes, such as ovoid or polygonal and may be non-symmetrical. While not shown, an adhesive patch may be attached to the bottom of the housing **8902** to help attach the sensor control device **8900** to the skin of a user at a target monitoring location.

The sensor control device **8900** may further include a sensor **8904** and a sharp **8906** extending distally from the bottom of the housing **8902**. The sensor **8904** and the sharp **8906** may be similar in some respects to the sensor **8110** and the sharp **8116** of FIGS. **81A-81B**. Accordingly, in some embodiments, the sharp **8906** may be used to help deliver the sensor **8904** transcutaneously under a user's skin during application of the sensor control device **8900**. The exposed portion of the sensor **8904** may be received within a hollow or recessed portion of the sharp **8906**, and the remaining portion(s) of the sensor **8904** is/are positioned within the interior of the electronics housing **8902**.

In some embodiments, the sharp **8906** may be made of a dermal-dissolving material. In such embodiments, the sharp **8906** may be used to help introduce the sensor **8904** into the user's skin, but may dissolve after a predetermined time period upon exposure to chemicals and/or substances commonly found in the human body. Consequently, in such embodiments, there is no need to retract the sharp **8906**. Rather, the sharp **8906** may remain embedded within the user's dermal layer until it safely dissolves. A dermal-dissolving sharp **8906** may also make sterilization applications much easier, since low-energy surface sterilization may only be needed.

In other embodiments, the sharp **8906** may be omitted from the sensor control device **8900**. In such embodiments, the sensor **8904** may be made of materials that are rigid enough to allow the sensor **8904** to be transcutaneously received beneath a user's skin for monitoring without the assistance of the sharp **8906**. Accordingly, the sensor **8906** may operate as both a sensor and a sharp or introducer. Such embodiments may prove advantageous in eliminating the mechanisms and assemblies typically required to retract the sharp **8906**.

As will be appreciated, any of the embodiments mentioned herein may incorporate a dermal dissolving sharp or introducer, or may alternatively include a sharp that operates as both a sensor and a sharp, without departing from the scope of the disclosure.

FIG. **90** is a schematic diagram of another example sterilization assembly **9000**, according to one or more embodiments. Similar to the other sterilization assemblies described herein, the sterilization assembly **9000** (hereafter the "assembly **9000**") may be used to help sterilize a medical device, such as a sensor control device **9002**. The sensor control device **9002** may be similar in some respects to some or all of the sensor control devices described herein. For example, the sensor control device **9002** includes a housing **9004** that may contain and otherwise house the electronics

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used to operate the sensor control device **9002**. The sensor control device **9002** may further include a part **9005** requiring sterilization, one or more radiation sensitive components **9006**, and a battery **9008** that powers the sensor control device **9002**. The radiation sensitive component **9006** may be arranged within the housing **9004** and may include one or more electronic modules such as, but not limited to, a data processing unit (e.g., an application specific integrated circuit or ASIC), a resistor, a transistor, a capacitor, an inductor, a diode, and a switch.

As illustrated, the part **9005** may extend perpendicularly from the bottom of the housing **9004**, but could alternatively extend at an angle relative to the housing **9004**. Moreover, while the part **9005** extends generally concentric with a centerline of the housing **9004**, the part **9005** could alternatively extend from the housing **9004** at a location eccentric to the centerline, without departing from the scope of the disclosure. In some embodiments, the part **9005** may comprise a sensor (e.g., the sensor **8110** of FIGS. **81A-81B**) and a sharp (e.g., the sharp **8116** of FIGS. **81A-81B**) used to help implant the sensor beneath the skin of a user.

The medical device **8602** may be subjected to radiation sterilization **9010** to properly sterilize the part **9005** for use. Suitable radiation sterilization **9010** processes include, but are not limited to, electron beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. To help guide and otherwise focus the radiation **9010** toward the part **9005** and simultaneously away from the radiation sensitive component **9006**, the assembly **9000** may include or otherwise employ one or more magnets configured to direct the electrons of the radiation **9010** in a predetermined sterilization path.

More particularly, as illustrated, the assembly **9000** may include a central magnet **9012** and opposing lateral magnets **9014a** and **9014b**. The central magnet **9012** may be arranged opposite a radiation source **9016** such that the part **9005** to be sterilized interposes the central magnet **9012** and the radiation source **9016**. The central magnet **9012** may be tuned and otherwise configured to draw the electrons of the radiation **9010** toward the central magnet **9012**, which generally urges the radiation **9010** toward the center of the sensor control device **9002** and otherwise to where the part **9005** is located. In addition, the lateral magnets **9014a,b** may be arranged on opposite sides of the sensor control device **9002** and tuned or otherwise configured to generate a magnetic field that pushes the electrons of the radiation **9010** toward the center of the sensor control device **9002** or otherwise to where the part **9005** is located. Accordingly, the central and lateral magnets **9012**, **9014a,b** may cooperatively urge the radiation **9010** away from the radiation sensitive components **9006** and instead toward the part **9005** to sterilize the part **9005**.

Embodiments disclosed herein include:

Z. A sensor control device assembly that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing, a sensor extending from a bottom of the electronics housing, a sharp hub positioned adjacent a top of the electronics housing, and a sharp carried by the sharp hub and extending through the electronics housing and from the bottom of the electronics housing, a cap removably coupled to the sensor applicator and providing a support structure that defines a post chamber that receives the sensor and the sharp extending from the bottom of the electronics housing, a first seal that provides a radial seal against the sharp hub and an axial

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seal against the top of the electronics housing, and a second seal that seals an interface between the post and the bottom of the electronics housing.

AA. A method including positioning a sensor control device within a sensor applicator, the sensor control device including an electronics housing, a sensor extending from a bottom of the electronics housing, a sharp hub positioned adjacent a top of the electronics housing, and a sharp carried by the sharp hub and extending through the electronics housing and from the bottom of the electronics housing, removably coupling a cap to the sensor applicator, the cap providing a support structure that defines a post chamber that receives the sensor and the sharp extending from the bottom of the electronics housing, providing a radial seal against the sharp hub with a first seal, providing an axial seal against the top of the electronics housing with the first seal, and sealing an interface between the post and the bottom of the electronics housing with a second seal.

BB. A sensor control device assembly includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing having a top and a bottom, a sensor coupled to the electronics housing, and a sharp module engageable with the electronics housing and having a sharp. The sensor control device assembly further includes a post having a first end positioned proximal the bottom of the electronics housing, a second end opposite the first end, and a post chamber extending between the first and second ends, wherein distal portions of the sensor and the sharp are receivable within the post chamber, a first seal interposing the sensor applicator and the electronics housing to seal an interface therebetween and interposing the sensor applicator and the sharp module to seal an interface therebetween, and a second seal interposing the first end of the post and the bottom of the electronics housing.

Each of embodiments Z, AA, and BB may have one or more of the following additional elements in any combination: Element 1: further comprising a sensor carrier arranged within the sensor applicator to secure the sensor control device, wherein the first seal is overmolded onto the sensor carrier. Element 2: wherein the cap comprises a first end threaded to the sensor applicator, and a second end opposite the first end, and wherein the support structure extends from the second end into the sensor applicator and toward the sensor control device. Element 3: wherein the first seal circumscribes a top aperture defined in the electronics housing and prevents contaminants from migrating into an interior of the electronics housing via the top aperture. Element 4: wherein the second seal circumscribes a bottom aperture defined on the bottom of the electronics housing and prevents contaminants from migrating into an interior of the electronics housing via the bottom aperture and into the post chamber. Element 5: wherein the sensor control device includes a housing support positioned within the electronics housing and extending between the top and bottom of the electronics housing and positioned about the sensor to support the top of the electronics housing against flexing toward the bottom of the electronics housing and to support the bottom of the electronics housing against flexing toward the top of the electronics housing. Element 7: wherein the sensor and the sharp are positioned eccentric from a central axis of the electronics housing. Element 8: wherein the first seal is overmolded onto the top of the electronics housing.

Element 9: further creating a sealed region as the cap is coupled to the sensor applicator, the sealed region encompassing the post chamber and a portion of an interior of the electronics housing, wherein portions of the sensor and the

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sharp reside within the sealed region. Element 10: further comprising sterilizing the sensor and the sharp with radiation sterilization while positioned within the sensor applicator. Element 11: wherein the radiation sterilization is at least one of focused radiation sterilization and low-energy radiation sterilization. Element 12: wherein the first seal is overmolded onto a sensor carrier arranged within the sensor applicator to secure the sensor control device. Element 13: wherein removably coupling the cap to the sensor applicator comprises advancing the support structure into the sensor applicator and thereby causing the second seal to seal the interface between the post and the bottom of the electronics housing. Element 14: wherein the sensor control device includes a housing support positioned within the electronics housing and extending between the top and bottom of the electronics housing, the method further comprising supporting the top of the electronics housing against flexing toward the bottom of the electronics housing with the housing support, and supporting the bottom of the electronics housing against flexing toward the top of the electronics housing with the housing support. Element 15: further comprising preventing contaminants from migrating into an interior of the electronics housing via a top aperture defined in the electronics housing with the first seal. Element 16: further comprising preventing contaminants from migrating into the post chamber and an interior of the electronics housing via a bottom aperture defined on the bottom of the electronics housing with the second seal.

Element 17: further comprising a sensor carrier positioned within the sensor applicator to secure the sensor control device, wherein the first seal seals a first interface between the sensor carrier and the electronics housing and a second interface between the sensor carrier and the sharp module. Element 18: further comprising a cap removably coupled to the sensor applicator and providing a support structure that extends from the bottom of the sensor applicator toward the sensor control device, wherein the post extends from the support structure.

By way of non-limiting example, exemplary combinations applicable to Z, AA, and BB include: Element 10 with Element 11; and Element 13 with Element 14.

Seal Arrangement for Analyte Monitoring Systems

FIGS. 91A and 91B are side and isometric views, respectively, of an example sensor control device **9102**, according to one or more embodiments of the present disclosure. The sensor control device **9102** may be similar in some respects to the sensor control device **104** of FIG. 1 and therefore may be best understood with reference thereto. Moreover, the sensor control device **9102** may replace the sensor control device **104** of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator **102** of FIG. 1, which may deliver the sensor control device **9102** to a target monitoring location on a user's skin.

As illustrated, the sensor control device **9102** includes an electronics housing **9104**, which may be generally disc-shaped and have a circular cross-section. In other embodiments, however, the electronics housing **9104** may exhibit other cross-sectional shapes, such as ovoid, oval, or polygonal, without departing from the scope of the disclosure. The electronics housing **9104** includes a shell **9106** and a mount **9108** that is matable with the shell **9106**. The shell **9106** may be secured to the mount **9108** via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, laser welding, one or more mechanical fasteners (e.g., screws), a gasket, an adhesive, or any combination thereof. In some cases, the shell **9106** may be secured to the mount **9108** such that a sealed interface is generated therebetween.

An adhesive patch **9110** may be positioned on and otherwise attached to the underside of the mount **9108**. Similar to the adhesive patch **108** of FIG. 1, the adhesive patch **9110** may be configured to secure and maintain the sensor control device **9102** in position on the user's skin during operation.

The sensor control device **9102** may further include a sensor **9112** and a sharp **9114** used to help deliver the sensor **9112** transcutaneously under a user's skin during application of the sensor control device **9102**. Corresponding portions of the sensor **9112** and the sharp **9114** extend distally from the bottom of the electronics housing **9104** (e.g., the mount **9108**). A sharp hub **9116** may be overmolded onto the sharp **9114** and configured to secure and carry the sharp **9114**. As best seen in FIG. 91A, the sharp hub **9116** may include or otherwise define a mating member **9118**. In assembling the sharp **9114** to the sensor control device **9102**, the sharp **9114** may be advanced axially through the electronics housing **9104** until the sharp hub **9116** engages an upper surface of the electronics housing **9104** or an internal component thereof and the mating member **9118** extends distally from the bottom of the mount **9108**. As described herein below, in at least one embodiment, the sharp hub **9116** may sealingly engage an upper portion of a seal overmolded onto the mount **9108**. As the sharp **9114** penetrates the electronics housing **9104**, the exposed portion of the sensor **9112** may be received within a hollow or recessed (arcuate) portion of the sharp **9114**. The remaining portion of the sensor **9112** is arranged within the interior of the electronics housing **9104**.

The sensor control device **9102** may further include a sensor cap **9120**, shown detached from the electronics housing **9104** in FIGS. 91A-91B. The sensor cap **9120** may help provide a sealed barrier that surrounds and protects exposed portions of the sensor **9112** and the sharp **9114**. As illustrated, the sensor cap **9120** may comprise a generally cylindrical body having a first end **9122a** and a second end **9122b** opposite the first end **9122a**. The first end **9122a** may be open to provide access into an inner chamber **9124** defined within the body. In contrast, the second end **9122b** may be closed and may provide or otherwise define an engagement feature **9126**. As described in more detail below, the engagement feature **9126** may help mate the sensor cap **9120** to an applicator cap of a sensor applicator (e.g., the sensor applicator **102** of FIG. 1), and may help remove the sensor cap **9120** from the sensor control device **9102** upon removing the sensor cap from the sensor applicator.

The sensor cap **9120** may be removably coupled to the electronics housing **9104** at or near the bottom of the mount **9108**. More specifically, the sensor cap **9120** may be removably coupled to the mating member **9118**, which extends distally from the bottom of the mount **9108**. In at least one embodiment, for example, the mating member **9118** may define a set of external threads **9128a** (FIG. 91A) matable with a set of internal threads **9128b** (FIG. 91B) defined within the inner chamber **9124** of the sensor cap **9120**. In some embodiments, the external and internal threads **9128a,b** may comprise a flat thread design (e.g., lack of helical curvature), but may alternatively comprise a helical threaded engagement. Accordingly, in at least one embodiment, the sensor cap **9120** may be threadably coupled to the sensor control device **9102** at the mating member **9118** of the sharp hub **9116**. In other embodiments, the sensor cap **9120** may be removably coupled to the mating member **9118** via other types of engagements including, but not limited to, an interference or friction fit, or a frangible member or substance (e.g., wax, an adhesive, etc.) that may be broken with minimal separation force (e.g., axial or rotational force).

In some embodiments, the sensor cap **9120** may comprise a monolithic (singular) structure extending between the first and second ends **9122a,b**. In other embodiments, however, the sensor cap **9120** may comprise two or more component parts. In the illustrated embodiment, for example, the body of the sensor cap **9120** may include a desiccant cap **9130** arranged at the second end **9122b**. The desiccant cap **9130** may house or comprise a desiccant to help maintain preferred humidity levels within the inner chamber **9124**. Moreover, the desiccant cap **9130** may also define or otherwise provide the engagement feature **9126** of the sensor cap **9120**. In at least one embodiment, the desiccant cap **9130** may comprise an elastomeric plug inserted into the bottom end of the sensor cap **9120**.

FIGS. 92A and 92B are exploded, isometric top and bottom views, respectively, of the sensor control device **9102**, according to one or more embodiments. The shell **9106** and the mount **9108** operate as opposing clamshell halves that enclose or otherwise substantially encapsulate various electronic components (not shown) of the sensor control device **9102**. Example electronic components that may be arranged between the shell **9106** and the mount **9108** include, but are not limited to, a battery, resistors, transistors, capacitors, inductors, diodes, and switches.

The shell **9106** may define a first aperture **9202a** and the mount **9108** may define a second aperture **9202b**, and the apertures **9202a,b** may align when the shell **9106** is properly mounted to the mount **9108**. As best seen in FIG. 92A, the mount **9108** may provide or otherwise define a pedestal **9204** that protrudes from the inner surface of the mount **9108** at the second aperture **9202b**. The pedestal **9204** may define at least a portion of the second aperture **9202b**. Moreover, a channel **9206** may be defined on the inner surface of the mount **9108** and may circumscribe the pedestal **9202**. In the illustrated embodiment, the channel **9206** is circular in shape, but could alternatively be another shape, such as oval, ovoid, or polygonal.

The mount **9108** may comprise a molded part made of a rigid material, such as plastic or metal. In some embodiments, a seal **9208** may be overmolded onto the mount **9108** and may be made of an elastomer, rubber, a polymer, or another pliable material suitable for facilitating a sealed interface. In embodiments where the mount **9108** is made of a plastic, the mount **9108** may be molded in a first "shot" of injection molding, and the seal **9208** may be overmolded onto the mount **9108** in a second "shot" of injection molding. Accordingly, the mount **9108** may be referred to or otherwise characterized as a "two-shot mount."

In the illustrated embodiment, the seal **9208** may be overmolded onto the mount **9108** at the pedestal **9204** and also on the bottom of the mount **9108**. More specifically, the seal **9208** may define or otherwise provide a first seal element **9210a** overmolded onto the pedestal **9204**, and a second seal element **9210b** (FIG. 92B) interconnected to (with) the first seal element **9210a** and overmolded onto the mount **9108** at the bottom of the mount **9108**. In some embodiments, one or both of the seal elements **9210a,b** may help form corresponding portions (sections) of the second aperture **9202b**. While the seal **9208** is described herein as being overmolded onto the mount **9108**, it is also contemplated herein that one or both of the seal elements **9210a,b** may comprise an elastomeric component part independent of the mount **9208**, such as an O-ring or a gasket.

The sensor control device **9102** may further include a collar **9212**, which may be a generally annular structure that defines a central aperture **9214**. The central aperture **9214** may be sized to receive the first seal element **9210a** and may

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align with both the first and second apertures **9202a,b** when the sensor control device **9102** is properly assembled. The shape of the central aperture **9214** may generally match the shape of the second aperture **9202b** and the first seal element **9210a**.

In some embodiments, the collar **9212** may define or otherwise provide an annular lip **9216** on its bottom surface. The annular lip **9216** may be sized and otherwise configured to mate with or be received into the channel **9206** defined on the inner surface of the mount **9108**. In some embodiments, a groove **9218** may be defined on the annular lip **9216** and may be configured to accommodate or otherwise receive a portion of the sensor **9112** extending laterally within the mount **9108**. In some embodiments, the collar **9212** may further define or otherwise provide a collar channel **9220** (FIG. 92A) on its upper surface sized to receive and otherwise mate with an annular ridge **9222** (FIG. 92B) defined on the inner surface of the shell **9106** when the sensor control device **9102** is properly assembled.

The sensor **9112** may include a tail **9224** that extends through the second aperture **9202b** defined in the mount **9108** to be transcutaneously received beneath a user's skin. The tail **9224** may have an enzyme or other chemistry included thereon to help facilitate analyte monitoring. The sharp **9114** may include a sharp tip **9226** extendable through the first aperture **9202a** defined by the shell **9106**. As the sharp tip **9226** penetrates the electronics housing **9104**, the tail **9224** of the sensor **9112** may be received within a hollow or recessed portion of the sharp tip **9226**. The sharp tip **9226** may be configured to penetrate the skin while carrying the tail **9224** to put the active chemistry of the tail **9224** into contact with bodily fluids.

The sensor control device **9102** may provide a sealed subassembly that includes, among other component parts, portions of the shell **9106**, the sensor **9112**, the sharp **9114**, the seal **9208**, the collar **9212**, and the sensor cap **9120**. The sealed subassembly may help isolate the sensor **9112** and the sharp **9114** within the inner chamber **9124** (FIG. 92A) of the sensor cap **9120**. In assembling the sealed subassembly, the sharp tip **9226** is advanced through the electronics housing **9104** until the sharp hub **9116** engages the seal **9208** and, more particularly, the first seal element **9210a**. The mating member **9118** provided at the bottom of the sharp hub **9116** may extend out the second aperture **9202b** in the bottom of the mount **9108**, and the sensor cap **9120** may be coupled to the sharp hub **9116** at the mating member **9118**. Coupling the sensor cap **9120** to the sharp hub **9116** at the mating member **9118** may urge the first end **9122a** of the sensor cap **9120** into sealed engagement with the seal **9208** and, more particularly, into sealed engagement with the second seal element **9210b** on the bottom of the mount **9108**. In some embodiments, as the sensor cap **9120** is coupled to the sharp hub **9116**, a portion of the first end **9122a** of the sensor cap **9120** may bottom out (engage) against the bottom of the mount **9108**, and the sealed engagement between the sensor hub **9116** and the first seal element **9210a** may be able to assume any tolerance variation between features.

FIG. 93 is a cross-sectional side view of the sensor control device **9102**, according to one or more embodiments. As indicated above, the sensor control device **9102** may include or otherwise incorporate a sealed subassembly **9302**, which may be useful in isolating the sensor **9112** and the sharp **9114** within the inner chamber **9124** of the sensor cap **9120**. To assemble the sealed subassembly **9302**, the sensor **9112** may be located within the mount **9108** such that the tail **9224** extends through the second aperture **9202b** at the bottom of the mount **9108**. In at least one embodiment, a locating

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feature **9304** may be defined on the inner surface of the mount **9108**, and the sensor **9112** may define a groove **9306** that is matable with the locating feature **9304** to properly locate the sensor **9112** within the mount **9108**.

Once the sensor **9112** is properly located, the collar **9212** may be installed on the mount **9108**. More specifically, the collar **9212** may be positioned such that the first seal element **9210a** of the seal **9208** is received within the central aperture **9214** defined by the collar **9212** and the first seal element **9210a** generates a radial seal against the collar **9212** at the central aperture **9214**. Moreover, the annular lip **9216** defined on the collar **9212** may be received within the channel **9206** defined on the mount **9108**, and the groove **9218** defined through the annular lip **9216** may be aligned to receive the portion of the sensor **9112** that traverses the channel **9206** laterally within the mount **9108**. In some embodiments, an adhesive may be injected into the channel **9206** to secure the collar **9212** to the mount **9108**. The adhesive may also facilitate a sealed interface between the two components and generate a seal around the sensor **9112** at the groove **9218**, which may isolate the tail **9224** from the interior of the electronics housing **9104**.

The shell **9106** may then be mated with or otherwise coupled to the mount **9108**. In some embodiments, as illustrated, the shell **9106** may mate with the mount **9108** via a tongue-and-groove engagement **9308** at the outer periphery of the electronics housing **9104**. An adhesive may be injected (applied) into the groove portion of the engagement **9308** to secure the shell **9106** to the mount **9108**, and also to create a sealed engagement interface. Mating the shell **9106** to the mount **9108** may also cause the annular ridge **9222** defined on the inner surface of the shell **9106** to be received within the collar channel **9220** defined on the upper surface of the collar **9212**. In some embodiments, an adhesive may be injected into the collar channel **9220** to secure the shell **9106** to the collar **9212**, and also to facilitate a sealed interface between the two components at that location. When the shell **9106** mates with the mount **9108**, the first seal element **9210a** may extend at least partially through (into) the first aperture **9202a** defined in the shell **9106**.

The sharp **9114** may then be coupled to the sensor control device **9102** by extending the sharp tip **9226** through the aligned first and second apertures **9202a,b** defined in the shell **9106** and the mount **9108**, respectively. The sharp **9114** may be advanced until the sharp hub **9116** engages the seal **9208** and, more particularly, engages the first seal element **9210a**. The mating member **9118** may extend (protrude) out the second aperture **9202b** at the bottom of the mount **9108** when the sharp hub **9116** engages the first seal element **9210a**.

The sensor cap **9120** may then be removably coupled to the sensor control device **9102** by threadably mating the internal threads **9128b** of the sensor cap **9120** with the external threads **9128a** of the mating member **9118**. The inner chamber **9124** may be sized and otherwise configured to receive the tail **9224** and the sharp tip **9226** extending from the bottom of the mount **9108**. Moreover, the inner chamber **9124** may be sealed to isolate the tail **9224** and the sharp tip **9226** from substances that might adversely interact with the chemistry of the tail **9224**. In some embodiments, a desiccant (not shown) may be present within the inner chamber **9124** to maintain proper humidity levels.

Tightening (rotating) the mated engagement between the sensor cap **9120** and the mating member **9118** may urge the first end **9122a** of the sensor cap **9120** into sealed engagement with the second seal element **9210b** in an axial direction (e.g., along the centerline of the apertures **9202a**,

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b), and may further enhance the sealed interface between the sharp hub 9116 and the first seal element 9210a in the axial direction. Moreover, tightening the mated engagement between the sensor cap 9120 and the mating member 9118 may compress the first seal element 9210a, which may result in an enhanced radial sealed engagement between the first seal element 9210a and the collar 9212 at the central aperture 9214. Accordingly, in at least one embodiment, the first seal element 9210a may help facilitate axial and radial sealed engagements.

As mentioned above, the first and second seal elements 9210a,b may be overmolded onto the mount 9108 and may be physically linked or otherwise interconnected. Consequently, a single injection molding shot may flow through the second aperture 9202b of the mount 9108 to create both ends of the seal 9208. This may prove advantageous in being able to generate multiple sealed interfaces with only a single injection molded shot. An additional advantage of a two-shot molded design, as opposed to using separate elastomeric components (e.g., O-rings, gaskets, etc.), is that the interface between the first and second shots is a reliable bond rather than a mechanical seal. Hence, the effective number of mechanical sealing barriers is effectively cut in half. Moreover, a two-shot component with a single elastomeric shot also has implications to minimizing the number of two-shot components needed to achieve all the necessary sterile barriers.

Once properly assembled, the sealed subassembly 9302 may be subjected to a radiation sterilization process to sterilize the sensor 9112 and the sharp 9114. The sealed subassembly 9302 may be subjected to the radiation sterilization prior to or after coupling the sensor cap 9120 to the sharp hub 9116. When sterilized after coupling the sensor cap 9120 to the sharp hub 9116, the sensor cap 9120 may be made of a material that permits the propagation of radiation therethrough. In some embodiments, the sensor cap 9120 may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

FIG. 93A is an exploded isometric view of a portion of another embodiment of the sensor control device 9102 of FIGS. 91A-91B and 92A-92B. Embodiments included above describe the mount 9108 and the seal 9208 being manufactured via a two-shot injection molding process. In other embodiments, however, as briefly mentioned above, one or both of the seal elements 9210a,b of the seal 9208 may comprise an elastomeric component part independent of the mount 9208. In the illustrated embodiment, for example, the first seal element 9210a may be overmolded onto the collar 9212 and the second seal element 9210b may be overmolded onto the sensor cap 9120. Alternatively, the first and second seal elements 9210a,b may comprise a separate component part, such as a gasket or O-ring positioned on the collar 9212 and the sensor cap 9120, respectively. Tightening (rotating) the mated engagement between the sensor cap 9120 and the mating member 9118 may urge the second seal element 9210b into sealed engagement with the bottom of the mount 9108 in an axial direction, and may enhance a sealed interface between the sharp hub 9116 and the first seal element 9210a in the axial direction.

FIG. 94A is an isometric bottom view of the mount 9108, and FIG. 94B is an isometric top view of the sensor cap 9120, according to one or more embodiments. As shown in FIG. 94A, the mount 9108 may provide or otherwise define one or more indentations or pockets 9402 at or near the opening to the second aperture 9202b. As shown in FIG. 94B, the sensor cap 9120 may provide or otherwise define one or more projections 9404 at or near the first end 9122a

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of the sensor cap 9120. The projections 9404 may be received within the pockets 9402 when the sensor cap 9120 is coupled to the sharp hub 9116 (FIGS. 92A-92B and 93). More specifically, as described above, as the sensor cap 9120 is coupled to the mating member 9118 (FIGS. 92A-92B and 93) of the sensor hub 9116, the first end 9122a of the sensor cap 9120 is brought into sealed engagement with the second seal element 9210b. In this process, the projections 9404 may also be received within the pockets 9402, which may help prevent premature unthreading of the sensor cap 9120 from the sharp hub 9116.

FIGS. 95A and 95B are side and cross-sectional side views, respectively, of an example sensor applicator 9502, according to one or more embodiments. The sensor applicator 9502 may be similar in some respects to the sensor applicator 102 of FIG. 1 and, therefore, may be designed to deliver (fire) a sensor control device, such as the sensor control device 9102. FIG. 95A depicts how the sensor applicator 9502 might be shipped to and received by a user, and FIG. 95B depicts the sensor control device 9102 arranged within the interior of the sensor applicator 9502.

As shown in FIG. 95A, the sensor applicator 9502 includes a housing 9504 and an applicator cap 9506 removably coupled to the housing 9504. In some embodiments, the applicator cap 9506 may be threaded to the housing 9504 and include a tamper ring 9508. Upon rotating (e.g., unscrewing) the applicator cap 9506 relative to the housing 9504, the tamper ring 9508 may shear and thereby free the applicator cap 9506 from the sensor applicator 9502.

In FIG. 95B, the sensor control device 9102 is positioned within the sensor applicator 9502. Once the sensor control device 9102 is fully assembled, it may then be loaded into the sensor applicator 9502 and the applicator cap 9506 may be coupled to the sensor applicator 9502. In some embodiments, the applicator cap 9506 and the housing 9504 may have opposing, matable sets of threads that enable the applicator cap 9506 to be screwed onto the housing 9504 in a clockwise (or counter-clockwise) direction and thereby secure the applicator cap 9506 to the sensor applicator 9502.

Securing the applicator cap 9506 to the housing 9504 may also cause the second end 9122b of the sensor cap 9120 to be received within a cap post 9510 located within the interior of the applicator cap 9506 and extending proximally from the bottom thereof. The cap post 9510 may be configured to receive at least a portion of the sensor cap 9120 as the applicator cap 9506 is coupled to the housing 9504.

FIGS. 96A and 96B are perspective and top views, respectively, of the cap post 9510, according to one or more additional embodiments. In the illustrated depiction, a portion of the sensor cap 9120 is received within the cap post 9510 and, more specifically, the desiccant cap 9130 of the sensor cap 9120 is arranged within cap post 9510.

The cap post 9510 may define a receiver feature 9602 configured to receive the engagement feature 9126 of the sensor cap 9120 upon coupling (e.g., threading) the applicator cap 9506 (FIG. 95B) to the sensor applicator 9502 (FIGS. 95A-95B). Upon removing the applicator cap 9506 from the sensor applicator 9502, however, the receiver feature 9602 may prevent the engagement feature 9126 from reversing direction and thus prevent the sensor cap 9120 from separating from the cap post 9510. Instead, removing the applicator cap 9506 from the sensor applicator 9502 will simultaneously detach the sensor cap 9120 from the sensor control device 9102 (FIGS. 91A-91B and 92A-92B), and thereby expose the distal portions of the sensor 9112 (FIGS. 92A-92B) and the sharp 9114 (FIGS. 92A-92B).

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Many design variations of the receiver feature **9602** may be employed, without departing from the scope of the disclosure. In the illustrated embodiment, the receiver feature **9602** includes one or more compliant members **9604** (two shown) that are expandable or flexible to receive the engagement feature **9126**. The engagement feature **9126** may comprise, for example, an enlarged head and the compliant member(s) **9604** may comprise a collet-type device that includes a plurality of compliant fingers configured to flex radially outward to receive the enlarged head.

The compliant member(s) **9604** may further provide or otherwise define corresponding ramped surfaces **9606** configured to interact with one or more opposing camming surfaces **9608** provided on the outer wall of the engagement feature **9126**. The configuration and alignment of the ramped surface(s) **9606** and the opposing camming surface(s) **9608** is such that the applicator cap **9506** is able to rotate relative to the sensor cap **9120** in a first direction A (e.g., clockwise), but the cap post **9510** binds against the sensor cap **9120** when the applicator cap **9506** is rotated in a second direction B (e.g., counter clockwise). More particularly, as the applicator cap **9506** (and thus the cap post **9510**) rotates in the first direction A, the camming surfaces **9608** engage the ramped surfaces **9606**, which urge the compliant members **9604** to flex or otherwise deflect radially outward and results in a ratcheting effect. Rotating the applicator cap **9506** (and thus the cap post **9510**) in the second direction B, however, will drive angled surfaces **9610** of the camming surfaces **9608** into opposing angled surfaces **9612** of the ramped surfaces **9606**, which results in the sensor cap **9120** binding against the compliant member(s) **9604**.

FIG. **97** is a cross-sectional side view of the sensor control device **9102** positioned within the applicator cap **9506**, according to one or more embodiments. As illustrated, the opening to the receiver feature **9602** exhibits a first diameter D_3 , while the engagement feature **9126** of the sensor cap **9120** exhibits a second diameter D_4 that is larger than the first diameter D_3 and greater than the outer diameter of the remaining portions of the sensor cap **9120**. As the sensor cap **9120** is extended into the cap post **9510**, the compliant member(s) **9604** of the receiver feature **9602** may flex (expand) radially outward to receive the engagement feature **9126**. In some embodiments, as illustrated, the engagement feature **9126** may provide or otherwise define an angled outer surface that helps bias the compliant member(s) **9604** radially outward. Once the engagement feature **9126** bypasses the receiver feature **9602**, the compliant member(s) **9604** are able to flex back to (or towards) their natural state and thus lock the sensor cap **9120** within the cap post **9510**.

As the applicator cap **9506** is threaded to (screwed onto) the housing **9504** (FIGS. **95A-95B**) in the first direction A, the cap post **9510** correspondingly rotates in the same direction and the sensor cap **9120** is progressively introduced into the cap post **9510**. As the cap post **9510** rotates, the ramped surfaces **9606** of the compliant members **9604** ratchet against the opposing camming surfaces **9608** of the sensor cap **9120**. This continues until the applicator cap **9506** is fully threaded onto (screwed onto) the housing **9504**. In some embodiments, the ratcheting action may occur over two full revolutions of the applicator cap **9506** before the applicator cap **9506** reaches its final position.

To remove the applicator cap **9506**, the applicator cap **9506** is rotated in the second direction B, which correspondingly rotates the cap post **9510** in the same direction and causes the camming surfaces **9608** (i.e., the angled surfaces **9610** of FIGS. **96A-96B**) to bind against the ramped surfaces **9606** (i.e., the angled surfaces **9612** of FIGS. **96A-96B**).

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Consequently, continued rotation of the applicator cap **9506** in the second direction B causes the sensor cap **9120** to correspondingly rotate in the same direction and thereby unthread from the mating member **9118** to allow the sensor cap **9120** to detach from the sensor control device **9102**. Detaching the sensor cap **9120** from the sensor control device **9102** exposes the distal portions of the sensor **9112** and the sharp **9114**, and thus places the sensor control device **9102** in position for firing (use).

FIG. **98** is a cross-sectional view of a sensor control device **9800** showing example interaction between the sensor and the sharp. After assembly of the sharp, the sensor should sit in a channel defined by the sharp. The sensor control device in FIG. **9** does not show the sensor deflected inwards and otherwise aligned fully with the sharp, but such may be the case upon full assembly as slight bias forces may be assumed by the sensor at the locations indicated by the two arrows A. Biasing the sensor against the sharp may be advantageous so that any relative motion between the sensor and the sharp during subcutaneous insertion does not expose the sensor tip (i.e., the tail) outside the sharp channel, which could potentially cause an insertion failure.

Embodiments disclosed herein include:

CC. A sensor control device that includes an electronics housing including a shell that defines a first aperture and a mount that defines a second aperture alignable with the first aperture when the shell is coupled to the mount, a seal overmolded onto the mount at the second aperture and comprising a first seal element overmolded onto a pedestal protruding from an inner surface of the mount, and a second seal element interconnected with the first seal element and overmolded onto a bottom of the mount, a sensor arranged within the electronics housing and having a tail extending through the second aperture and past the bottom of the mount, and a sharp that extends through the first and second apertures and past the bottom of the electronics housing.

DD. An assembly that includes a sensor applicator, a sensor control device positioned within the sensor applicator and including an electronics housing including a shell that defines a first aperture and a mount that defines a second aperture alignable with the first aperture when the shell is mated to the mount, a seal overmolded onto the mount at the second aperture and comprising a first seal element overmolded onto a pedestal protruding from an inner surface of the mount, and a second seal element interconnected with the first seal element and overmolded onto a bottom of the mount, a sensor arranged within the electronics housing and having a tail extending through the second aperture and past the bottom of the mount, and a sharp that extends through the first and second apertures and past the bottom of the electronics housing. The assembly further including a sensor cap removably coupled to the sensor control device at the bottom of the mount and defining a sealed inner chamber that receives the tail and the sharp, and an applicator cap coupled to the sensor applicator.

Each of embodiments CC and DD may have one or more of the following additional elements in any combination: Element 1: wherein the mount comprises a first injection molded part molded in a first shot, and the seal comprises a second injection molded part overmolded onto the first injection molded part in a second shot. Element 2: further comprising a sharp hub that carries the sharp and sealingly engages the first seal element, and a sensor cap removably coupled to the sharp hub at the bottom of the mount and sealingly engaging the second seal element, wherein the sensor cap defines an inner chamber that receives the tail and the sharp. Element 3: wherein the sharp hub provides a

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mating member that extends past the bottom of the mount and the sensor cap is removably coupled to the mating member. Element 4: further comprising one or more pockets defined on the bottom of the mount at the second aperture, and one or more projections defined on an end of the sensor cap and receivable within the one or more pockets when the sensor cap is coupled to the sharp hub. Element 5: further comprising a collar positioned within the electronics housing and defining a central aperture that receives and sealingly engages the first seal element in a radial direction. Element 6: further comprising a channel defined on the inner surface of the mount and circumscribing the pedestal, an annular lip defined on an underside of the collar and matable with the channel, and an adhesive provided in the channel to secure and seal the collar to the mount at the channel. Element 7: further comprising a groove defined through the annular lip to accommodate a portion of the sensor extending laterally within the mount, wherein the adhesive seals about the sensor at the groove. Element 8: further comprising a collar channel defined on an upper surface of the collar, an annular ridge defined on an inner surface of the shell and matable with the collar channel, and an adhesive provided in the collar channel to secure and seal the shell to the collar. Element 9: wherein one or both of the first and second seal elements define at least a portion of the second aperture. Element 10: wherein the first seal element extends at least partially through the first aperture when the shell is coupled to the mount.

Element 11: wherein the sensor control device further includes a sharp hub that carries the sharp and sealingly engages the first seal element, and wherein the sensor cap is removably coupled to the sharp hub at the bottom of the mount and sealingly engages the second seal element. Element 12: wherein the sensor control device further includes one or more pockets defined on the bottom of the mount at the second aperture, and one or more projections defined on an end of the sensor cap and receivable within the one or more pockets when the sensor cap is coupled to the sharp hub. Element 13: wherein the sensor control device further includes a collar positioned within the electronics housing and defining a central aperture that receives and sealingly engages the first seal element in a radial direction. Element 14: wherein the sensor control device further includes a channel defined on the inner surface of the mount and circumscribing the pedestal, an annular lip defined on an underside of the collar and matable with the channel, and an adhesive provided in the channel to secure and seal the collar to the mount at the channel. Element 15: wherein the sensor control device further includes a groove defined through the annular lip to accommodate a portion of the sensor extending laterally within the mount, and wherein the adhesive seals about the sensor at the groove. Element 16: wherein the sensor control device further includes a collar channel defined on an upper surface of the collar, an annular ridge defined on an inner surface of the shell and matable with the collar channel, and an adhesive provided in the collar channel to secure and seal the shell to the collar. Element 17: wherein one or both of the first and second seal elements define at least a portion of the second aperture. Element 18: wherein the first seal element extends at least partially through the first aperture.

By way of non-limiting example, exemplary combinations applicable to CC and DD include: Element 2 with Element 3; Element 2 with Element 4; Element 5 with Element 6; Element 6 with Element 7; Element 5 with

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Element 8; Element 11 with Element 12; Element 13 with Element 14; Element 14 with Element 15; and Element 13 with Element 16.

Axial-Radial Thermal Cycle Resistant Cap Seal

FIG. 99 is a cross-sectional side view of an example analyte monitoring system enclosure 9900 used to house at least a portion of the sensor control device 104 of FIG. 1, according to one or more embodiments. As illustrated, the analyte monitoring system enclosure 9900 includes the sensor applicator 102 and the applicator cap 210 matable with the sensor applicator 102. The applicator cap 210 provides a barrier that protects the internal contents of the sensor applicator 102. In some embodiments, the applicator cap 210 may be secured to the housing 208 by a threaded engagement and, upon rotating (e.g., unscrewing) the applicator cap 210 relative to the housing 208, the applicator cap 210 can be freed from the sensor applicator 102. In other embodiments, however, the applicator cap 210 may be secured to the housing 208 via an interference or shrink fit engagement.

As described herein below, the coupled engagement between the sensor applicator 102 and the applicator cap 210 may prove vital in properly sterilizing the components positioned within the sensor applicator 102 and maintaining a sterile environment as sealed with the applicator cap 210. The embodiments described herein below may be applicable to analyte monitoring systems that incorporate a two-piece or a one-piece architecture. More particularly, in embodiments employing a two-piece architecture, the electronics housing (not shown) that retains the electrical components for the sensor control device 104 (FIG. 1) may be positioned within the sensor applicator 102 and the applicator cap 210 maintains the sterile environment. In contrast, in embodiments employing a one-piece architecture, the sensor applicator 102 may contain the fully assembled sensor control device 104 (not shown), and the applicator cap 210 maintains the sterile environment for the fully assembled sensor control device.

The components arranged within the sensor applicator 102 and sealed with the applicator cap 210 may be subjected to gaseous chemical sterilization 9902 configured to sterilize exposed portions of such components. To accomplish this, a chemical may be injected into a sterilization chamber 9904 cooperatively defined by the housing 208 and the interconnected cap 210. In some applications, the chemical may be injected into the sterilization chamber 9904 via one or more vents 9906 (two shown) defined in the applicator cap 210 at its proximal end 9908. Example chemicals that may be used for the gaseous chemical sterilization 9902 include, but are not limited to, ethylene oxide, vaporized hydrogen peroxide, and nitrogen oxide (e.g., nitrous oxide, nitrogen dioxide, etc.).

Once a desired sterility assurance level has been achieved within the sterilization chamber 9904, the gaseous solution may be evacuated via the vents 9906 and the sterilization chamber 9904 is aerated. Aeration may be achieved by a series of vacuums and subsequently circulating nitrogen gas or filtered air through the sterilization chamber 9904. Once the sterilization chamber 9904 is properly aerated, the vents 9906 may be occluded with a seal 9910 (shown in dashed lines).

In some embodiments, the seal 9910 may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can

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be applied before the gaseous chemical sterilization process, and following the gaseous chemical sterilization process, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization chamber 9904. In other embodiments, the seal 9910 may comprise only a single protective layer applied to the applicator cap 210. In such embodiments, the single layer is gas permeable for the sterilization process, but is also capable of protection against moisture and other harmful elements once the sterilization process is complete.

With the seal 9910 in place, the applicator cap 210 provides a barrier against outside contamination, and thereby maintains a sterile environment for the components arranged within the sensor applicator 102 until the user removes (unthreads) the applicator cap 210 from the housing 208.

FIG. 100A is an enlarged cross-sectional side view of the interface between the sensor applicator 102 and the applicator cap 210, as indicated by the dashed box of FIG. 99. As illustrated the housing 208 provides a first axial extension 10002a and the applicator cap 210 provides a second axial extension 10002b mateable with the first axial extension 10002a. In the illustrated embodiment, the diameter of the second axial extension 10002b of the applicator cap 210 is sized to receive the diameter of the first axial extension 10002a of the housing 208. In other embodiments, however, the reverse may be employed, where the diameter of the first axial extension 10002a may be sized to receive the diameter of the second axial extension 10002b, without departing from the scope of the disclosure.

In either scenario, a radial seal 10004 may be defined or otherwise provided at the interface between the first and second axial extensions 10002a,b and the radial seal 10004 may help prevent migration of fluids or contaminants across the interface in either axial direction. In the illustrated embodiment, the radial seal 10004 comprises a radial protrusion formed on the inner radial surface of the second axial extension 10002b. In other embodiments, however, the radial seal 10004 may alternatively be formed on the outer radial surface of the first axial extension 10002a, without departing from the scope of the disclosure. In embodiments where the second axial extension 10002b is received within the first axial extension 10002a, the radial seal 10004 may be formed on the inner radial surface of the first axial extension 10002a or alternatively on the outer radial surface of the second axial extension 10002b.

Gaseous chemical sterilization 9902 (FIG. 99) is commonly undertaken at elevated temperatures reaching 60° C. (140° F.) or more. At such elevated temperatures, the housing 208 and the applicator cap 210 may be subjected to thermal expansion that may affect the integrity of the radial seal 10004. The housing 208 and the applicator cap 210 may be made of dissimilar materials that have dissimilar coefficients of thermal expansion. In some embodiments, for example, the housing 208 may be made of polycarbonate and the applicator cap 210 may be made of polypropylene. Polypropylene exhibits a coefficient of thermal expansion of about $100\text{--}180 \times 10^{-6} \text{K}^{-1}$ and polycarbonate exhibits a coefficient of thermal expansion of about $66\text{--}70 \times 10^{-6} \text{K}^{-1}$. Since polypropylene has a thermal coefficient that is higher than polycarbonate, the applicator cap 210 will tend to expand at a greater rate than the polycarbonate housing 208 during gaseous chemical sterilization 9902. Moreover, the increased expansion of the applicator cap 210 can affect the seal integrity (capability) of the radial seal 10004.

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FIG. 100B is an enlarged cross-sectional side view of the interface between the sensor applicator 102 and the applicator cap 210, as indicated by the dashed box of FIG. 99 during and/or after gaseous chemical sterilization. Since the applicator cap 210 exhibits a thermal coefficient greater than the thermal coefficient of the housing 208, the applicator cap 210 expands at a greater rate than the housing 208 upon being subjected to the elevated temperatures required for gaseous chemical sterilization 9902 (FIG. 99). Consequently, a gap 10006 may be created between the opposing radial surfaces of the first and second axial extensions 10002a,b as the radial seal 10004 separates from opposed radial engagement. As shown by the arrows, the gap 10006 may provide a flow path for the outflow of toxic gases used for gaseous chemical sterilization 9902.

Following gaseous chemical sterilization 9902, and as the temperature is lowered to ambient, the applicator cap 210 may radially contract and the gap 10006 may close, thereby sealing the interface at the radial seal 10004 once again. Such embodiments may prove advantageous in simplifying the design of the applicator cap 210. More specifically, and according to one or more embodiments of the present disclosure, the gaseous chemical sterilization 9902 process may be carried out entirely through the gap 10006 formed between the opposing radial surfaces of the first and second axial extensions 10002a,b. In such embodiments, the temperature of the housing 208 and the applicator cap 210 may be elevated until the gap 10006 is created. Once the gap 10006 is created, the gaseous chemicals (e.g., ethylene oxide) used during the gaseous chemical sterilization 9902 may be injected into the sterilization chamber 9904 through the gap 10006 and otherwise by bypassing the radial seal 10004. The sterilization chamber 9904 may be subsequently aerated by drawing out the gaseous chemicals through the gap 10006 and circulating another fluid, such as nitrogen, into and out of the sterilization chamber 9904 via the gap 10006.

In such embodiments, the vents 9906 (FIG. 99) defined in the applicator cap 210 and the seal 9910 (FIG. 99) attached to the bottom of the applicator cap 210 may be omitted and otherwise unnecessary. Accordingly, in such embodiments, the bottom of the applicator cap 210 may be solid. Moreover, in such embodiments, a desiccant may be positioned within the applicator cap 210 or the sterilization chamber 9904 to aid maintenance of a low humidity environment for biological components sensitive to moisture.

In other embodiments, however, the applicator cap 210 may undergo stress relaxation at the enlarged diameter during gaseous chemical sterilization 9902. This may occur in embodiments where the material of the applicator cap 210 exhibits a thermal coefficient greater than the material of the housing 208 and the gaseous chemical sterilization 9902 spans a long period of time (e.g., one hour, five hours, ten hours, fifteen hours, or more). As the temperature is lowered to ambient, the applicator cap 210 may remain substantially at the enlarged diameter and the gap 10006 may correspondingly remain, which jeopardizes the integrity of the radial seal 10004.

Stress relaxation of the applicator cap 210 may also occur in embodiments where the housing 208 is made of a material that has a higher thermal coefficient than the applicator cap 210. In such embodiments, the housing 208 will expand at a greater rate than the applicator cap 210 and thereby radially expand against the applicator cap 210. The gap 10006 will not be generated as the housing 208 continuously biases against the applicator cap 210 during thermal expansion. The material of the applicator cap 210, however, will

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undergo stress relaxation at an enlarged diameter, and upon cooling the system to ambient, the gap **10006** may be generated as the housing **208** radially contracts but the applicator cap **210** remains near the enlarged diameter. The resulting gap **10006** compromises the sealed interface at the radial seal **10004**, and thereby prevents the applicator cap **210** from providing a barrier.

FIG. **101** is an enlarged cross-sectional side view of another example analyte monitoring system enclosure **10100** used to house at least a portion of the sensor control device **104** of FIG. **1**, according to one or more embodiments. Similar to the analyte monitoring system enclosure **9900** of FIGS. **99** and **1007A-100B**, the analyte monitoring system enclosure **10100** includes the sensor applicator **102** and the applicator cap **210** matable with the sensor applicator **102**. In the illustrated embodiment, the applicator cap **210** is secured to the housing **208** by complimentary mating threads **10102**, and may include a tamper ring **10104**. Upon rotating (e.g., unscrewing) the applicator cap **210** relative to the housing **208**, the tamper ring **10104** may shear and thereby free the applicator cap **210** from the sensor applicator **102**.

As best seen in the enlarged view, the interface between the housing **208** and the applicator cap **210** may provide or otherwise define a radial seal **10106** and an axial-radial seal **10108**. More specifically, the housing **208** may provide a first axial extension **10110a** and the applicator cap **210** may provide a second axial extension **10110b** extending in the opposite direction. In the illustrated embodiment, the diameter of the first axial extension **10110a** may be sized to receive the smaller diameter second axial extension **10110b** of the applicator cap **210**. In other embodiments, however, the diameter of the second axial extension **10110b** may be sized to receive a smaller diameter first axial extension **10110a** of the housing **208**, without departing from the scope of the disclosure.

In either scenario, the radial seal **10106** may be defined or otherwise provided at an interface between the first and second axial extensions **10110a,b** and configured to help prevent the migration of fluids or contaminants across the interface in either axial direction. In the illustrated embodiment, the radial seal **10106** comprises a radial protrusion **10107** formed on the outer radial surface of the second axial extension **10110b**, but the radial protrusion **10107** may alternatively be formed on the inner radial surface of the first axial extension **10110a**, without departing from the scope of the disclosure. In embodiments where the first axial extension **10110a** is received within the second axial extension **10110b**, the radial seal **10106** may be formed on the outer radial surface of the first axial extension **10110a** or alternatively on the inner radial surface of the second axial extension **10110b**.

As its name suggests, the axial-radial seal **10108** may be configured to provide a sealed interface between the housing **208** and the applicator cap **210** in both axial and radial directions, and thereby prevent the migration of fluids or contaminants across the interface in both axial and radial directions. To accomplish this, the axial-radial seal **10108** may comprise a beveled or chamfered surface **10112** configured to mate with a fillet **10114**, where the fillet **10114** comprises angularly offset surfaces angled to substantially mate with the angled profile of the chamfered surface **10112** in both axial and radial directions. In the illustrated embodiment, the chamfered surface **10112** is defined on the end of the second axial extension **10110b** and the fillet **10114** is defined by the first axial extension **10110a**. In other embodiments, however, the chamfered surface **10112** may alterna-

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tively be defined on the end of the first axial extension **10110a** and the fillet **10114** may be defined by the second axial extension **10110b**, without departing from the scope of the disclosure.

The radial seal **10106** and the axial-radial seal **10108** may be configured to cooperatively help maintain fluid tight interfaces between the housing **208** and the applicator cap **210**. During gaseous chemical sterilization **9902** (FIG. **99**), however, and since the housing **208** and the applicator cap **210** may be made of dissimilar materials having dissimilar coefficients of thermal expansion, the elevated temperatures may result in loss of a fluid tight seal at the radial seal **10106**. Nonetheless, the axial-radial seal **10108** may be designed and otherwise configured to maintain a fluid tight interface between the housing **208** and the applicator cap **210** while withstanding the elevated temperatures of gaseous chemical sterilization **9902**. Regardless of the materials of either of the housing **208** or the applicator cap **210**, and regardless of the respective coefficients of thermal expansion, the axial-radial seal **10108** may prove advantageous in maintaining a fluid tight interface. In some embodiments, the applicator cap **210** may provide a sterile barrier.

FIGS. **102A-102C** depict finite element analysis (FEA) results corresponding to the interface between the housing **208** and the applicator cap **210** during example gaseous chemical sterilization, according to one or more embodiments. FIG. **102A** depicts FEA analysis results as the applicator cap **210** is secured to the housing **208**, such as by screwing the applicator cap **210** onto the housing **208** via the threads **10102** (FIG. **101**). As illustrated, a radial preload may be generated at the radial seal **10106** as the radial protrusion **10107** provided on the second axial extension **10110b** is urged into radial contact with the inner radial surface of the first axial extension **10110a**. Moreover, a combination axial and radial preload may be generated at the axial-radial seal **10108** as the chamfered surface **10112** is urged into both axial and radial engagement with the fillet **10114**.

FIG. **102B** depicts FEA analysis results during an increase in temperature resulting from gaseous chemical sterilization. The temperature increase results in differential expansion between the materials of the housing **208** and cap **210**. Depending on the materials chosen, the applicator cap **210** may expand radially more or less than the housing **208**. During this temperature increase and the radial expansion of the housing **208** and the applicator cap **210**, the axial-radial seal **10108** remains intact as the chamfered surface **10112** is wedged into both axial and radial engagement with the fillet **10114**. Hence, the expansion of the fillet **10114** may dictate the final position of the axial-radial seal **10108** at elevated temperature. Depending upon whether the housing **208** material has a higher coefficient of thermal expansion than the applicator cap **210** material, or vice-versa, this result may or may not apply to the radial seal **10106**.

The elevated temperatures during gaseous chemical sterilization are typically maintained for long periods of time. During this time, stress relaxation may occur in all the stressed zones of the applicator cap **210** and insignificant residual stress is expected at the end of the temperature cycle. This implies that most of the preload (and hence sealing) is lost at elevated temperature.

FIG. **102C** depicts FEA analysis results after decreasing the temperature following gaseous chemical sterilization. In embodiments where the applicator cap **210** is made of a material having a higher coefficient of thermal expansion than the housing **208**, the radial seal **10106** is likely lost upon decreasing the temperature to ambient due to stress relax-

ation at the elevated temperature. As a result, separation of the first and second axial extensions **10110a,b** occurs and a gap **2816** is formed between the two surfaces after cooling. In contrast, in embodiments where the housing **208** is made of a material having a higher coefficient of thermal expansion than the applicator cap **210**, the radial seal **10106** may be re-activated following cooling. In either scenario, however, the axial-radial seal **10108** may remain intact throughout the temperature cycle as the chamfered surface **10112** is continuously wedged into both axial and radial engagement with the fillet **10114**. Accordingly, the axial-radial seal **10108** may prove advantageous in maintaining sealed engagement between the housing **208** and the applicator cap **210** regardless of the materials used.

Embodiments disclosed herein include:

EE. An analyte monitoring system enclosure including a sensor applicator including a housing that provides a first axial extension, a cap matable with the housing and providing a second axial extension, and an axial-radial seal that seals an interface between the housing and the cap in both axial and radial directions, wherein the axial-radial seal includes a fillet defined by one of the first and second axial extensions, and a chamfered surface matable with the fillet and defined on an end of the other of the first and second axial extensions.

FF. A method of sterilizing contents within an analyte monitoring system enclosure including injecting a chemical gas into the analyte monitoring system enclosure, the analyte monitoring system enclosure comprising a sensor applicator including a housing that provides a first axial extension, and a cap matable with the housing and providing a second axial extension. The method further including sealing an interface between the housing and the cap in both axial and radial directions with an axial-radial seal, wherein the axial-radial seal includes a fillet defined by one of the first and second axial extensions, and a chamfered surface matable with the fillet and defined on an end of the other of the first and second axial extensions, increasing and decreasing a temperature of the analyte monitoring system enclosure, and maintaining the axial-radial seal as the temperature is increased and decreased.

GG. A method of sterilizing contents within an analyte monitoring system enclosure including providing the analyte monitoring system enclosure, the analyte monitoring system enclosure comprising a sensor applicator including a housing that provides a first axial extension, and a cap matable with the housing and providing a second axial extension. The method further including increasing a temperature of the analyte monitoring system enclosure until a gap forms between the first and second axial extensions, injecting a chemical gas into the analyte monitoring system enclosure through the gap, evacuating the chemical gas from the analyte monitoring system enclosure through the gap, and decreasing the temperature of the analyte monitoring system and sealing an interface between the first and second axial extensions with a radial seal.

Each of embodiments EE, FF, and GG may have one or more of the following additional elements in any combination: Element 1: wherein the housing and the cap are made of dissimilar materials having dissimilar coefficients of thermal expansion. Element 2: wherein the fillet comprises angularly offset surfaces angled to mate with an angled profile of the chamfered surface in both the axial and radial directions. Element 3: further comprising a radial seal provided between the first and second axial extensions. Element 4: wherein the radial seal comprises a radial protrusion formed on an inner or outer surface of one of the first and

second axial extensions. Element 5: wherein the first axial extension is received within the second axial extension and the radial protrusion is formed on the outer surface of the first axial extension or the inner surface of the second axial extension. Element 6: wherein the second axial extension is received within the first axial extension and the radial protrusion is formed on the inner surface of the first axial extension or the outer surface of the second axial extension. Element 7: wherein the cap is secured to the housing via a threaded engagement.

Element 8: wherein maintaining the axial-radial seal comprises wedging the chamfered surface into one or both of axial and radial engagement with the fillet as the temperature is increased and decreased. Element 9: wherein the housing and the cap are made of dissimilar materials having dissimilar coefficients of thermal expansion. Element 10: further comprising radially sealing an interface between the housing and the cap with a radial seal. Element 11: wherein the radial seal comprises a radial protrusion formed on an inner radial surface or an outer radial surface of one of the first and second axial extensions, and wherein radially sealing the interface comprises urging the radial protrusion into engagement with an opposing surface of the other of the first and second axial extensions. Element 12: wherein the cap is secured to the housing via a threaded engagement.

Element 13: wherein the housing and the cap are made of dissimilar materials having dissimilar coefficients of thermal expansion. Element 14: wherein the radial seal comprises a radial protrusion formed on an inner radial surface or an outer radial surface of one of the first and second axial extensions, and wherein radially sealing the interface comprises urging the radial protrusion into engagement with an opposing surface of the other of the first and second axial extensions. Element 15: wherein the bottom of the cap is solid without vents formed therein. Element 16: further maintaining a low humidity environment within the cap with a desiccant.

By way of non-limiting example, exemplary combinations applicable to EE, FF, and GG include: Element 3 with Element 4; Element 4 with Element 5; Element 4 with Element 6; and Element 10 with Element 11.

Conversion Process for Sensor Control Devices

Referring again briefly to FIG. 1, the sensor control device **104** is often included with the sensor applicator **104** in what is known as a “two-piece” architecture that requires final assembly by a user before the sensor **110** can be properly delivered to the target monitoring location. More specifically, the sensor **110** and the associated electrical components included in the sensor control device **104** are provided to the user in multiple (two) packages, and the user must open the packaging and follow instructions to manually assemble the components before delivering the sensor **110** to the target monitoring location with the sensor applicator **102**. More recently, advanced designs of sensor control devices and sensor applicators have resulted in a one-piece architecture that allows the system to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package and subsequently deliver the sensor control device to the target monitoring location. Notwithstanding these advancements, however, sensor control devices are still frequently made of hard plastic materials that contain several component parts.

According to the present disclosure, sensor control devices (e.g., the sensor control device **104**) may alternatively be manufactured through a converting process that incorporate large rolls of process material that are progres-

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sively modified to form or otherwise assemble flexible sensor control devices in step-wise fashion. The converting processes described herein may use pressure sensitive adhesives (PSAs) or tapes, thermoformed films, die-cut or layered components, and other materials that readily lend themselves to roll-to-roll or other high volume manufacturing processes. These high-volume manufacturing processes have the potential to greatly decrease the cost of manufacturing sensor control devices and increase the rate of assembly.

FIG. 103 is an isometric view of an example sensor control device 10302, according to one or more embodiments of the present disclosure. The sensor control device 10302 may be the same as or similar to the sensor control device 104 of FIG. 1 and, therefore, may be used in conjunction with the sensor applicator 102 (FIG. 1), which delivers the sensor control device 10302 to a target monitoring location on a user's skin.

As illustrated, the sensor control device 10302 includes an electronics housing 10304 that is generally planar in shape and can exhibit a variety of cross-sectional shapes. In the illustrated embodiment, the electronics housing 10304 is rectangular with rounded corners, but could exhibit other cross-sectional shapes, such as circular, oval, ovoid (e.g., pill- or egg-shaped), a squircle, another polygonal shape (e.g., square, pentagonal, etc.), or any combination thereof, without departing from the scope of the disclosure. The electronics housing 10304 may be configured to house or otherwise contain various electronic components used to operate the sensor control device 10302.

The electronics housing 10304 may include an upper cover 10306 and a lower cover 10308 that is matable with the upper cover 10306. In some embodiments, the upper and lower covers 10306, 10308 may comprise a film, a foil, a foam, a laminated material (e.g., a laminated metal or foil), a coextruded material, a cast film, a comolded material, or any combination thereof. Accordingly, the upper and lower covers 10306, 10308 may be made of a variety of semi-rigid or flexible materials including, but not limited to, a plastic or thermoplastic, a metal, a composite material (e.g., fiberglass, etc.), or any combination thereof. Moreover, the upper and lower covers 10306, 10308 may be formed via a variety of manufacturing processes including, but not limited to, thermoforming, vacuum forming, injection molding, die-cutting, stamping, compression molding, transfer molding, or any combination thereof.

The upper cover 10306 may be secured to the lower cover 10308 via a variety of mating techniques, such as sonic welding, ultrasonic welding, laser welding, heat sealing, an adhesive substrate (e.g., a pressure sensitive adhesive or tape), or any combination thereof. In some cases, the upper cover 10306 may be secured to the lower cover 10308 such that a sealed interface is generated therebetween. The sealed interface may provide structural integrity, but may also isolate the interior of the electronics housing 10304 from outside contamination. In the illustrated embodiment, securing the upper cover 10306 to the lower cover 10308 may result in the formation of a flange 10322 extending about the periphery of the electronics housing 10304. In other embodiments, however, the upper and lower covers 10306, 10308 may be secured without forming the flange 10322.

In the illustrated embodiment, the sensor control device 10302 may optionally include a plug assembly 10310 that may be coupled to the electronics housing 10304. The plug assembly 10310 may include a sensor module 10312 (partially visible) interconnectable with a sharp module 10314 (partially visible). The sensor module 10312 may be con-

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figured to carry and otherwise include a sensor 10316 (partially visible), and the sharp module 10314 may be configured to carry and otherwise include an introducer or sharp 10318 (partially visible) used to help deliver the sensor 10316 transcutaneously under a user's skin during application of the sensor control device 10302. In the illustrated embodiment, the sharp module 10314 includes a sharp hub 10320 that carries the sharp 10318.

As illustrated, corresponding portions of the sensor 10316 and the sharp 10318 extend distally from the electronics housing 10304 and, more particularly, from the bottom of the lower cover 10308. In at least one embodiment, the exposed portion of the sensor 10316 (alternately referred to as the "tail") may be received within a hollow or recessed portion of the sharp 10318. The remaining portions of the sensor 10316 are positioned within the interior of the electronics housing 10304.

FIGS. 104A and 104B are exploded, isometric views of the sensor control device 10302 of FIG. 103, according to one or more embodiments. More specifically, FIG. 104A is an exploded, isometric view of a sensor electronics module 10402 included in the sensor control device 10302, and FIG. 104B is an exploded, isometric view of the sensor control device 10302 with the sensor electronics module 10402.

Referring first to FIG. 104A, the sensor electronics module 10402 may include a cap 10404, a sensor holder 10406, the sensor 10316, and a printed circuit board (PCB) 10408. The cap 10404 and the sensor holder 10406 may be made of injection molded plastic, for example, and may be configured to secure the sensor 10316 within the sensor electronics module 10402. To accomplish this, the cap 10404 and the sensor holder 10406 may be engageable and matable. In the illustrated embodiment, for example, the cap 10404 includes or defines one or more castellations or projections 10410 sized to be received within or mate with one or more corresponding grooves or pockets 10412 defined on the sensor holder 10406. Mating the projections 10410 with the pockets 10412 may help secure the sensor 10316 within the sensor electronics module 10402 and may also clamp down on the PCB 10408 and the other component parts of the sensor electronics module 10402, thus resulting in a solid structural component. In other embodiments, however, the projections 10410 may alternatively be provided on the sensor holder 10406, and the cap 10404 may instead define the pockets 10412, without departing from the scope of the disclosure.

As illustrated, the sensor 10316 includes a tail 10314, a flag 10416, and a neck 10418 that interconnects the tail 10314 and the flag 10416. The tail 10314 may be configured to extend at least partially through a channel 10420 defined in the sensor holder 10406 and extend distally from the sensor electronics module 10402. The tail 10314 includes an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry. In use, the tail 10314 is transcutaneously received beneath a user's skin, and the chemistry included thereon helps facilitate analyte monitoring in the presence of bodily fluids. The flag 10416 may comprise a generally planar surface having one or more sensor contacts 10422 (three shown) arranged thereon. The sensor contacts 10422 may be configured to align with a corresponding number of circuitry contacts (not shown) included on the PCB 10408 that provide conductive communication between the sensor 10316 and the electronic components provided on the PCB 10408.

In some embodiments, the PCB 10408 may be flexible, and may be sized to be positioned within the electronics housing 10304 (FIG. 103). A plurality of electronic modules

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(not shown) may be mounted to the PCB **10408** including, but not limited to, a data processing unit, resistors, transistors, capacitors, inductors, diodes, and switches. The data processing unit may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **10302** (FIGS. **103** and **104B**). More specifically, the data processing unit may be configured to perform data processing functions, where such functions may include but are not limited to, filtering and encoding of data signals, each of which corresponds to a sampled analyte level of the user. The data processing unit may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. **1**). One or more batteries (not shown) may also be mounted to the PCB **10408** and used to power the sensor control device **10302**.

The sensor electronics module **10402** may further include one or more adhesive substrates, shown as a first adhesive substrate **10424a**, a second adhesive substrate **10424b**, and a third adhesive substrate **10424c**. In some embodiments, each adhesive substrate **10424a-c** may comprise a pressure-adhesive tape that forms a bond when pressure is applied. The first adhesive substrate **10424a** may interpose the cap **10404** and the PCB **10408** and may operate to secure the cap **10404** to the PCB **10408**. The second adhesive substrate **10424b** may interpose the sensor holder **10406** and the sensor **10316** (i.e., the flag **10416**) and may operate to secure the sensor **10316** to the sensor holder **10406**.

The third adhesive substrate **10424c** may interpose the sensor **10316** (i.e., the flag **10416**) and the flexible PCB **10408** to couple the sensor **10316** to the PCB **10408**. In some embodiments, the third adhesive substrate **10424c** may also comprise a Z-axis anisotropic (or conductive) pressure-adhesive tape. In such embodiments, the third adhesive substrate **10424c** may also facilitate electrical communication between the sensor contacts **10422** provided on the flag **10416** and the corresponding circuitry contacts included on the PCB **10408**. Coupling the cap **10404** and the sensor holder **10406** may help maintain sufficient pressure on the third adhesive substrate **10424c** to ensure reliable electrical connection between the sensor **10316** and the PCB **10408**. Each of the adhesive substrates **320a-c** may also seal against liquid and moisture, thus helping to mitigate the chances of shorting the sensor **10316** and the PCB **10408**.

Referring now to FIG. **104B**, the sensor electronics module **10402** may be sized to be received between the upper and lower covers **10306**, **10308**. In the illustrated embodiment, the upper cover **10306** provides or otherwise defines a cavity that may receive the sensor electronics module **10402**. In other embodiments, however, the lower cover **10308**, or both the upper and lower covers **10306**, **10308**, could alternatively define the cavity, without departing from the scope of the disclosure.

The sensor control device **10302** may also include a filler **10426** that may be arranged between the upper and lower covers **10306**, **10308**. In some embodiments, the filler **10426** may comprise foam made of a low-density polyethylene, polyolefin, or polyurethane. Moreover, the filler **10426** may be die cut and/or molded to mate with the sensor electronics module **10402**. As illustrated, for instance, the filler **10426** may define an aperture **328** sized to receive a portion of the sensor electronics module **10402** and, more particularly, the sensor holder **10406**. In some embodiments, the filler **10426** may operate similar to a potting material by taking up space within the electronics housing **10304** (FIG. **103**) that would otherwise be occupied by air. Moreover, the material of the

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filler **10426** may expand less than air at elevated altitudes, such as would be experienced during shipping. The filler **10426** may also help to stabilize the electrical components of the PCB **10408** (FIG. **104B**) and mitigate vibration.

The sensor control device **10302** may further include a fourth adhesive substrate **10424d**, which may also comprise a pressure-adhesive tape that forms a bond when pressure is applied. The fourth adhesive substrate **10424b** may interpose the lower cover **10308** and the filler **10426**, and may operate to secure the filler **10426** to the lower cover **10308**. The adhesive substrates **10424a-d** may each be die-cut, thermoformed, or stamped pieces of material.

FIG. **105** is a cross-sectional side view of the assembled sensor control device **10302**, according to one or more embodiments. Securing the upper and lower covers **10306**, **10308** to one another, as described above, secures the sensor electronics module **10402** and the filler **10426** within the electronics housing **10304**. Once the upper and lower covers **10306**, **10308** are secured, the plug assembly **10310** may be received by the sensor control device **10302** by extending the sharp **10318** through the electronics housing **10304** until the sharp hub **10320** engages a top surface **10502** of the sensor control device **10302**, such as a top surface of the cap **10404**. As the sharp **10318** extends through the electronics housing **10304**, the sensor **10316** (e.g., the tail **10314**) may be received within a hollow or recessed portion of the sharp **10318**.

As described in more detail below, the sensor control device **10302** may be manufactured via a converting process, where some parts of the sensor control device **10302** are assembled or otherwise formed in a step-wise fashion from large rolls of material. As a result, the sensor control device **10302** may be entirely made at a factory, thus eliminating user assembly. Moreover, whereas current sensor control devices commonly use glues, potting, or casting and encapsulating compounds to seal and enclose (encapsulate) the sensor **10316** and the PCB **10408**, fabricating the sensor control device **10302** using the presently disclosed converting processes eliminates the need for glues or “wet chemistry,” thus making the fabrication process not dependent on curing methods or time.

FIG. **106** is an isometric view of another example sensor control device **10602**, according to one or more embodiments of the present disclosure. The sensor control device **10602** may be the same as or similar to the sensor control device **104** of FIG. **1** and, therefore, may be used in conjunction with the sensor applicator **102** (FIG. **1**), which delivers the sensor control device **10602** to a target monitoring location on a user's skin. Moreover, the sensor control device **10602** may be similar in some respects to the sensor control device **10302** of FIGS. **103**, **104A-104B** and **105** and therefore may be best understood with reference thereto, where like numerals will represent like components not described again in detail.

Similar to the sensor control device **10302** of FIGS. **103**, **104A-104B** and **105**, the sensor control device **10602** includes the electronics housing **10304** made of the upper and lower covers **10306**, **10308**. The sensor control device **10602** may further include the plug assembly **10310**, the sensor module **10312** with the sensor **10316**, and the sharp module **10314** with the sharp **10318**. Corresponding portions of the sensor **10316** and the sharp **10318** extend distally from the electronics housing **10304** and, more particularly, from the bottom of the lower cover **10308**. Unlike the sensor control device **10302**, however, one or both of the upper and lower covers **10306**, **10308** may be made of a rigid material such as, but not limited to, a plastic, a metal, a composite

material, a ceramic, or any combination thereof. Alternatively, one or both of the upper and lower covers **10306**, **10308** can be made of a semi rigid or flexible materials, such as an elastomer.

FIGS. **107A** and **107B** are exploded, isometric views of the sensor control device **10602** of FIG. **106**, according to one or more embodiments. More specifically, FIG. **107A** is an exploded, isometric view of a sensor electronics module **10702** included in the sensor control device **10602**, and FIG. **107B** is an exploded, isometric view of the sensor control device **10602** with the sensor electronics module **10702**.

Referring first to FIG. **107A**, the sensor electronics module **10702** includes a sensor holder **10704**, the sensor **10316**, and a printed circuit board (PCB) **10706**, which may be similar in some respects to the PCB **10408** of FIG. **104A**. The sensor holder **10704** may be made of injection molded plastic, for example, and may be configured to secure the sensor **10316** to the sensor electronics module **10702**. To accomplish this, the sensor holder **10704** may be engageable and matable with the PCB **10706**. In the illustrated embodiment, for example, the sensor holder **10704** includes or defines one or more projections **107608** (three shown) sized to be received within or mate with one or more corresponding holes **10710** (three shown) defined on the PCB **10706**. Mating the projections **107608** with the holes **10710** may secure the sensor **10316** to the sensor electronics module **10702**, thus resulting in a solid structural component. In other embodiments, however, the projections **107608** may alternatively be provided on the PCB **10706**, and the sensor holder **10704** may instead define the holes **10710**, without departing from the scope of the disclosure.

The tail **10314** of the sensor **10316** may be configured to extend through a channel **10712** defined in the sensor holder **10704** and extend distally from the sensor electronics module **10702**. The sensor contacts **10422** of the flag **10416** may be configured to align with a corresponding number of circuitry contacts (not shown) included on the PCB **10706** that provide conductive communication between the sensor **10316** and corresponding electronic components provided on the PCB **10706**.

The sensor electronics module **10702** may further include one or more adhesive substrates, shown as a first adhesive substrate **10714a** and a second adhesive substrate **10714b**. Similar to the adhesive substrates **10424a-d** of FIGS. **104A-104B**, each adhesive substrate **10714a,b** may comprise a pressure-adhesive tape that forms a bond when pressure is applied, and may each be die-cut, thermoformed, or stamped pieces of material. The first adhesive substrate **10714a** may interpose the sensor holder **10704** and the sensor **10316** (i.e., the flag **10416**) and may operate to secure the sensor **10316** to the sensor holder **10704**. In some embodiments, the sensor holder **10704** may define a depression **10716** sized to receive one or both of the first adhesive substrate **10714a** and the flag **10416**.

The second adhesive substrate **10714b** may be configured to help attach the sensor **10316** and the sensor holder **10704** to the PCB **10706**. Moreover, the second adhesive substrate **10714b** may comprise a Z-axis anisotropic (or conductive) pressure-adhesive tape and may therefore also facilitate electrical communication between the sensor contacts **10422** provided on the flag **10416** with the corresponding circuitry contacts included on the PCB **10706**. Coupling the sensor holder **10704** to the PCB **10706** may help maintain sufficient pressure on the second adhesive substrate **10714b** to ensure reliable electrical contact between the sensor **10316** and the PCB **10706**. The adhesive substrates **10714a,b** may also seal

against liquid and moisture, thus helping to mitigate the chances of shorting the sensor **10316** and the PCB **10706**.

Referring now to FIG. **107B**, the sensor electronics module **10702** may be sized to be received between the upper and lower covers **10306**, **10308**. In the illustrated embodiment, the upper cover **10306** provides or otherwise defines a cavity that can receive the sensor electronics module **10702**. In other embodiments, however, the lower cover **10308**, or a combination of the upper and lower covers **10306**, **10308**, could alternatively define the cavity, without departing from the scope of the disclosure. The sensor control device **10602** may also include the filler **10426** arranged between the upper and lower covers **10306**, **10308** and defining the aperture **10428** sized to receive a portion of the sensor electronics module **10702** and, more particularly, the sensor holder **10704**.

FIG. **108** is a cross-sectional side view of the assembled sensor control device **10602**, according to one or more embodiments. Securing the upper and lower covers **10306**, **10308** to one another, as described herein, secures the sensor electronics module **10702** and the filler **10426** within the electronics housing **10304**. Once the upper and lower covers **10306**, **10308** are secured and otherwise sealed, the plug assembly **10310** may be received by the sensor control device **10602** by extending the sharp **10318** through the electronics housing **10304** until the sharp hub **10320** engages a top surface **10802** of the sensor control device **10602**, such as a top surface of the upper cover **10306**. As the sharp **10318** extends through the electronics housing **10304**, the sensor **10316** (e.g., the tail **10314**) may be received within a hollow or recessed portion of the sharp **10318**.

FIG. **109** is an isometric view of an example converting process **10900** for manufacturing a sensor control device **10902** in accordance with the principles of the present disclosure. More specifically, the converting process **10900** is depicted showing progressive, step-wise building of a web-based assembly that results in the fabrication of the sensor control device **10902**. The sensor control device **10902** may be the same as or similar to any of the sensor control devices **104**, **10302**, **10602** described herein with reference to FIGS. **1**, **103**, and **106**, respectively. Accordingly, any of the sensor control devices **104**, **10302**, **10602** may be fabricated using the presently described converting process **10900**.

Whereas current sensor control devices are commonly made of hard plastics and require use assembly, the sensor control device **10902** made by the converting process **10900** may be made of flexible materials that do not require user assembly. Alternatively, rigid materials may instead be incorporated, without departing from the scope of the disclosure. The converting process **10900** may incorporate the use of one or more continuous rolls of process materials, such as a base substrate **10904** that may eventually form the lower cover **10308** (FIGS. **103** and **106**) of the electronics housing **10304** (FIGS. **103** and **106**). The base substrate **10904** may be continuously unrolled (unwound) from an adjacent roll (not shown) of material. This web-based process may include or exclude the incorporate of injection molded parts, such as for the upper or lower covers **10306**, **10308**. Consequently, fabrication of sensor control devices (e.g., the sensor control device **10902**) using the converting process **10900** may proceed in a continuous process that progressively modifies and/or arranges the materials and component parts to form the sensor control devices **10902**.

FIGS. **110A-110E** are referenced in FIG. **109** and depict progressive fabrication of the sensor control device **10902**,

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according to one or more embodiments. FIGS. 110A-110E will be described below to detail the various steps of the example converting process 10900.

Referring first to FIG. 110A, in a first step of the process 10900, a hole 11002 may be punched or otherwise formed in the base substrate 10904, which may comprise a sheet of material that may eventually form the base or lower cover 10308 (FIGS. 103 and 106) of the sensor control device 10902 (FIG. 109). The base substrate 10904 may comprise a belt or thin film made of a variety of different materials including, but not limited to, a plastic, a metal, a composite material, or any combination thereof. In at least one embodiment, the base substrate 10904 may comprise a laminated aluminum foil having a polyester film on one side (e.g., the bottom side), and a polyolefin heat seal layer on the opposing side (e.g., the top side).

In a second step of the process 10900, a sensor holder 11004 may be coupled to the base substrate 10904. The sensor holder 11004 may be the same as or similar to either of the sensor holders 10406, 10704 of FIGS. 104A and 107A, respectively. Accordingly, the sensor holder 11004 may define a channel 11006 sized to receive the tail 10314 (FIGS. 104A and 107A) of the sensor 10316 (FIGS. 104A and 107A). In some embodiments, the sensor holder 11004 may be ultrasonically welded or heat-sealed to the base substrate 10904, thus resulting in a sealed and watertight engagement. In at least one embodiment, however, the base substrate 10904 may comprise or otherwise include an adhesive substrate on the top side to secure and seal the sensor holder in place.

In a third step of the process 10900, a first adhesive substrate 11008a may be attached to the top of the sensor holder 11004. The first adhesive substrate 11008a may be similar to any of the adhesive substrates 10424a-d (FIGS. 104A-104B), 10714a,b (FIGS. 107A-107B) described herein, and may thus comprise a pressure-adhesive tape that forms a bond when pressure is applied. In at least one embodiment, the first adhesive substrate 11008a may comprise double-sided polyolefin foam tape and may be pressure sensitive on both sides.

In a fourth step of the process 10900, the sensor 10316 may be secured to the sensor holder 11004 using the first adhesive substrate 11008a. More specifically, the tail 10314 (FIGS. 104A and 107A) may be extended through the channel 11006 and the flag 10416 may be bent generally orthogonal to the tail 10314 and coupled to the underlying first adhesive substrate 11008a.

Referring now to FIG. 110B, in a fifth step of the process 10900, a printed circuit board (PCB) 11010 may be positioned on the base substrate 10904 and about the sensor holder 11004. The PCB 11010 may be similar in some respects to the PCB 10408 of FIGS. 104A and 107A, and may thus include a plurality of electronic modules 11012 mounted thereto. The electronic modules 11012 may include one or both of a Bluetooth antenna and a near field communication (NFC) antenna. As illustrated, the PCB 11010 may define two opposing lobes 11014a and 11014b interconnected by a neck portion 11016. Opposing battery contacts 11018a and 11018b may be provided on the opposing lobes 11014a,b to facilitate electrical communication with a battery 11020.

In a sixth step of the process 10900, a second adhesive substrate 11008b may be applied to the first battery contact 11018a in preparation for receiving the battery 11020 in an adjacent seventh step of the process 10900. The second adhesive substrate 11008b may comprise a pressure-adhesive tape used to couple the battery 11020 to the first battery

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contact 11018a. The second adhesive substrate 11008b, however, may also comprise a Z-axis anisotropic (or conductive) pressure-adhesive tape that also facilitates electrical communication (i.e., transfer of electrical power) between the battery 11020 and the first battery contact 11018a.

Referring now to FIG. 110C, in an eighth step of the process 10900, a filler 11022 may be positioned or arranged on the first lobe 11014a of the PCB 11010. The filler 11022 may be the same as or similar to the filler 10426 of FIG. 104B or 107B, and may thus comprise foam made of a low-density polyethylene or polyolefin. Moreover, the filler 11022 may be die cut and/or molded to fit around one or both of the battery 11020 and the sensor holder 11004. In the illustrated embodiment, the filler 11022 may define apertures 11024a and 11024b to receive the battery 11020 and/or the sensor holder 11004. The filler 11022 may also operate as a potting material that takes up space that would otherwise be occupied by air, and thus help to stabilize the electronic modules 11012 (FIG. 110B) of the PCB 11010 and mitigate damaging vibration.

In a ninth step of the process 10900, a third adhesive substrate 11008c may be applied to a top of the filler 11022 to help couple the second lobe 11014b of the PCB 11010 to the top of the filler 11022 in a subsequent step of the process 10900. The third adhesive substrate 11008c may comprise a pressure-adhesive tape, but may also comprise a Z-axis anisotropic (or conductive) pressure-adhesive tape that also facilitates electrical communication (i.e., transfer of electrical power) between the battery 11020 and the second battery contact 11018b. The third adhesive substrate 11008c may also facilitate electrical communication between the sensor contacts 10422 provided on the sensor 10316 and corresponding circuitry contacts 11026 (three shown) included on the PCB 11010.

Referring now to FIG. 110D, in a tenth step of the process 10900, the second lobe 11014b of the PCB 11010 may be folded down at the neck 11016 to couple the PCB 11010 to the filler 11022. Coupling the PCB 11010 to the filler 11022 may also complete the conductive pathway via the third adhesive substrate 11008c between the battery 11020 and the second battery contact 11018b, and between the sensor contacts 10422 and the corresponding circuitry contacts 11026.

In an eleventh step of the process 10900, a fourth adhesive substrate 11008d may be applied to a portion of the top of the second lobe 11014b of the PCB 11010. The fourth adhesive substrate 11008d may also comprise a pressure-adhesive tape, and may be used to couple an upper cover 11028 to the PCB 11010, as provided in a twelfth step of the process 10900. The upper cover 11028 may be the same as or similar to the upper cover 10306 of FIGS. 103 and 106, and the fourth adhesive substrate 11008d may help secure the upper cover 10306 to the PCB 11010.

In some embodiments, the upper cover 11028 may be provided by another roll of material continuously provided to the web-based assembly in the process 10900. In some embodiments, the upper cover 11028 may be vacuum-formed, but could alternatively, be cold formed or injection molded, without departing from the scope of the disclosure. Accordingly, as indicated above, this web-based process 10900 may include or exclude injection molded parts, such as for the upper or lower covers 10306, 10308. In some embodiments, the upper cover 11028 may be formed or defined to provide a flange 11030 about its periphery, and the flange 11030 may provide a location to seal the upper cover 11028 to the base substrate 10904 (i.e., the "lower cover"). The upper cover 11028 may be secured to the base substrate

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10904 via one or more of sonic welding, ultrasonic welding, laser welding, photonic flash soldering, heat sealing, an adhesive substrate (e.g., a pressure sensitive adhesive or tape), or any combination thereof. Alternatively, the fourth adhesive substrate **11008d** may sufficiently couple the upper cover **11028** to the base substrate **10904**, or an additional adhesive substrate (not shown) may be applied at the flange **11030** to secure the upper cover **11028** to the base substrate **10904**, without departing from the scope of the disclosure.

Referring now to FIG. **110E**, in a thirteenth step of the process **10900**, the outer diameter of the sensor control device **10902** may be trimmed to remove the excess portions of the base substrate **10904** (FIGS. **110A** and **110D**). In some embodiments, as illustrated, the sensor control device **10902** may have a substantially circular cross-section, but could alternatively comprise any other cross-sectional shape, such as polygonal, oval, ovoid (e.g., pill- or egg-shaped), a squircle, or any combination thereof, without departing from the scope of the disclosure.

In a fourteenth and final step of the process **10900**, the plug assembly **10310** as described herein may be received by the sensor control device **10902** by extending the sharp **10318** through the sensor control device **10902** until the sharp hub **10320** engages a top surface of the sensor control device **10902**. As the sharp **10318** extends through the sensor control device **10902**, the sensor **10316** may be received within a hollow or recessed portion of the sharp **10318**.

FIG. **111A** is a top view of the sensor control device **10902** in preparation for pressure testing and/or vacuum sealing, according to one or more embodiments. In the illustrated embodiment, a web **11102** may form part of or otherwise extend from the sensor control device **10902** across a tab section **11104**. The tab section **11104** may form part of the flange **11030** or may otherwise extend therefrom. The web **11102** may comprise two layers of film **11106a** and **11106b**. In some embodiments, for instance, the upper layer **11106a** may be connected to or form part of the material that forms the upper cover **11028**, as described above with reference to FIGS. **110D** and **110E**, and the lower layer **11106b** may be connected to or form part of the base material **10904**, as described above with reference to FIGS. **109**, **110A** and **110D**.

An aperture **11108** may be defined through the upper layer **11106a** (or the lower layer **11106b**) to facilitate fluid communication between the two layers **11106a,b** and the interior of the sensor control device **10902**. A seal **11110** may be made about the periphery of the web **11102** to seal the upper and lower layers **11106a,b** together. Moreover, the flange **11030** may be sealed about the periphery of the sensor control device **10902** except across the tab section **11104**, thus facilitating fluid communication into and/or out of the sensor control device via the web **11102**. In some embodiments, one or both of the upper and lower layers **11106a,b** may provide or otherwise define a pattern or web of interconnected channels **11112** that help facilitate fluid communication between the aperture **11108** and the interior of the sensor control device **10902** via the tab section **11104**.

By injecting air (or another fluid) into the sensor control device **10902** via the aperture **11108** and the web **11102**, the sensor control device **10902** may be pressure tested to determine if the outer periphery (e.g., the flange **11030**) or other portions of the sensor control device **10902** are properly sealed. This is often referred to as “pressure decay testing,” and helps verify seal integrity of medical devices made of layers of film. Alternatively, air may be evacuated from the sensor control device **10902** via the aperture **11108**

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and the web **11102** to place the interior of the sensor control device **10902** under vacuum conditions. The channels **11112** may prove advantageous in helping to draw the vacuum without entirely collapsing the upper and lower layers **11106a,b**.

FIG. **111B** is a cross-sectional side view of the sensor control device **10902** with a compressor **11114**. The compressor **11114** may have proper fittings to fluidly couple to the web **11102** via the aperture **11108**. In some embodiments, the compressor **11114** may be arranged on a back support **11116** to help support the pressure fitting at the aperture **11108**.

To pressure test the sensor control device **10902** to determine if it meets pressure requirements, the compressor **11114** may inject air into the web **11102** via the aperture **11108**, and the air may circulate to the interior of the sensor control device **10902** between the opposing layers **11106a,b** and via the tab section **11104**. This allows seal integrity testing to be performed during the manufacturing process of the sensor control device **10902**. Once the seal integrity is verified, the periphery of the sensor control device **10902** at the tab section **11104** may be sealed and the web **11102** may be trimmed from the sensor control device **10902**.

In some embodiments, after the sensor control device **10902** has been pressure tested, operation of the compressor **11114** may be reversed to pull a vacuum on the sensor control device **10902**, as indicated above. Once the vacuum is drawn, the periphery of the sensor control device **10902** at the tab section **11104** may be sealed, thus leaving the sensor control device **10902** under vacuum conditions. As will be appreciated, vacuum conditions may prove advantageous since the sensor control device **10902** may be transported through high altitudes, where a non-vacuum sealed device would have a tendency to expand or “pillow” out. Moreover, the vacuum may be drawn during the manufacturing process, following which the web **11102** may be trimmed from the sensor control device **10902**.

FIG. **112** is a partial cross-sectional side view of an example sensor control device **11200**, according to one or more embodiments. The sensor control device **11200** may be similar in some respects to any of the sensor control devices described herein. As illustrated, the sensor control device **11200** may include a housing **11202** configured to house electronic modules or components used to operate the sensor control device. Example electronic modules include, but are not limited to a battery, a data processing unit (e.g., an application specific integrated circuit or ASIC), a resistor, a transistor, a capacitor, an inductor, a diode, and a switch.

The sensor control device **11200** may further include a sensor **11204** and a sharp **11206**, which may be similar to any of the sensors and sharps described herein. Consequently, the sharp **11206** may be used to help transcutaneously implant the sensor **11204** beneath a user’s skin for monitoring blood glucose levels. In the illustrated embodiment, the sensor **11204** and the sharp **11206** are arranged within a sterile chamber **11208** to protect the sensor **11204** and the sharp **11206** from external contamination. In some embodiments, the sterile chamber **11208** may have a desiccant arranged therein to help promote preferred humidity conditions.

In some embodiments, the sensor **11204** and the sharp **11206** may be sterilized while assembled within the sensor control device **11200**. In at least one embodiment, the sensor **11204** and the sharp **11206** may be subjected to radiation sterilization to properly sterilize the sensor **11204** and the sharp **11206** for use. Suitable radiation sterilization processes include, but are not limited to, electron beam

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(e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof.

In some embodiments, the sterile chamber **11208** may comprise a cap that provides a sealed barrier that protects exposed portions of the sensor **11204** and the sharp **11206** until placed in use. In such embodiments, the sterile chamber **11208** may be removable or detachable to expose the sensor **11204** and the sharp **11206**, as described below. Moreover, in such embodiments, the cap may be made of a material that permits propagation of radiation therethrough to facilitate radiation sterilization of the sensor **11204** and the sharp **11206**. Suitable materials for the sterile chamber **11208** include, but are not limited to, a non-magnetic metal (e.g., aluminum, copper, gold, silver, etc.), a thermoplastic, ceramic, rubber (e.g., ebonite), a composite material (e.g., fiberglass, carbon fiber reinforced polymer, etc.), an epoxy, or any combination thereof. In some embodiments, the sterile chamber **11208** may be transparent or translucent, but can otherwise be opaque, without departing from the scope of the disclosure.

In other embodiments, the sterile chamber **11208** may comprise a chamber or compartment defined within the sensor control device **11200**. In such embodiments, the sterile chamber **11208** may include a microbial barrier positioned at one or both ends of the sterile chamber **11208**. More specifically, the sterile chamber **11208** may provide or include an upper microbial barrier **11210a** and a lower microbial barrier **11210b** opposite the upper microbial barrier **11210a**. The upper and lower microbial barriers **11210a,b** may help seal the sterile chamber **11208** to thereby isolate the sensor **11204** and the sharp **11206** from external contamination. The microbial barriers **11210a,b** may be made of a radiation permeable material, such as a synthetic material (e.g., a flash-spun high-density polyethylene fiber). One example synthetic material comprises TYVEK®, available from DuPont®. In other embodiments, however, the microbial barriers **11210a,b** may comprise, but are not limited to, tape, paper, film, foil, or any combination thereof.

In some embodiments, the sensor **11204** and the sharp **11206** may be deployable and otherwise movable relative to the sensor control device **11200**. In such embodiments, the sensor **11204** and the sharp **11206** may be advanced distally out of the sterile chamber **11208** and past the bottom of the housing **11202** to allow the sensor **11204** and the sharp **11206** to be transcutaneously received beneath a user's skin. Distally advancing the sensor **11204** and the sharp **11206** may be accomplished via a variety of mechanical or electromechanical means. In some embodiments, for example, the sensor control device **11200** may include a pusher **11212** configured to advance to push the sensor **11204** and the sharp **11206** out of the sterile chamber **11208**. In such embodiments, the pusher **11212** may also be configured to attach to the sharp **11206** and subsequently retract the sharp **11206** while leaving the sensor **11204** extended. During operation, the pusher **11212** may penetrate the upper microbial barrier **11210a** and force the sensor **11204** and the sharp **11206** distally through the lower microbial barrier **11210b**.

As illustrated, the pusher **11212** may comprise a flexible shaft that extends within a curved pathway **11214** defined laterally through the housing **11202** and does not penetrate the top of the housing **11202**. The pathway **11214** may terminate at or near an upper end of the sterile chamber **11208**. In at least one embodiment, as illustrated, the pusher **11212** may extend out of the housing **11202** at a sidewall **11216** thereof. In such embodiments, actuation of the pusher **11212** may originate at the location of the sidewall **11216** to advance or retract the pusher **11212** within the pathway

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11214 and thereby act on the sterile chamber **11208** and/or the sensor **11204**, and the sharp **11206**.

In embodiments where the sterile chamber **11208** comprises a cap, the pusher **11212** may be operable to discharge or push the cap out of the sensor control device **11200**. In such embodiments, a user may commence the firing process by priming the sensor control device **11200**, which may cause the cap to be discharged from the sensor control device **11200**. Further actuation of the sensor control device **11200** by the user may cause the sensor **11204** and the sharp **11206** to be fully extended for subcutaneous implantation. In other embodiments, the cap may be removed either autonomously (e.g., it falls off or breaks away during firing) or the user may manually remove it by hand.

FIG. **113** is a cross-sectional side view of an example sensor applicator **11300**, according to one or more embodiments. The sensor applicator **11300** may be similar in some respects to any of the sensor applicators described herein. Accordingly, the sensor applicator **11300** may be configured to house a sensor control device **11302** and may be operable to deploy the sensor control device **11302** to a target monitoring location. The sensor control device **11302** may be similar in some respects to any of the sensor control devices described herein. As illustrated, the sensor control device **11302** may include an electronics housing **11304** configured to house electronic modules or components used to operate the sensor control device **11302**. The sensor control device **11302** may further include a sensor **11306** and a sharp **11308**, which may be similar to any of the sensors and sharps described herein. Consequently, the sharp **11308** may be used to help transcutaneously implant the sensor **11306** beneath a user's skin for monitoring blood glucose levels.

In the illustrated embodiment, the sensor applicator includes a housing **11310** and an applicator cap **11312** removably coupled to the housing **11310**. The applicator cap **11312** may be threaded to the housing **11310** and may be removed by rotating (e.g., unscrewing) the applicator cap **11312** relative to the housing **11310**.

In the illustrated embodiment, the sensor applicator **11300** may include a filler **11314** arranged at least partially within the applicator cap **11312**. In some embodiments, the filler **11314** may form an integral part or extension of the applicator cap **11312**, such as being molded with or overmolded onto the applicator cap **11312**. In other embodiments, the filler **11314** may comprise a separate structure fitted within or attached to the applicator cap **11312**, without departing from the scope of the disclosure. In some embodiments, the filler **11314** may generally help support the sensor control device **11302** while contained within the sensor applicator **11302**.

The filler **11314** may define or otherwise provide a sterilization zone **11316** configured to receive the sensor **11306** and the sharp **11308** as extending from the bottom of the electronics housing **11304**. The sterilization zone **11316** may generally comprise a hole or passageway extending at least partially through the body of the filler **11314**. When the sensor control device **11302** is loaded into the sensor applicator **11302** and the applicator cap **11312** is secured thereto, the sensor **11306** and the sharp **11308** may be positioned within the sterilization zone **11316** of the filler **11314**, which may be sealed to isolate the sensor **11306** and the sharp **11308** from external contamination.

The applicator cap **11312** and the filler **11314** may each be made of a gas impermeable material, such as a plastic or polycarbonate. Moreover, a gasket **11318** may be located at an interface between the filler **11314** and the bottom of the

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electronics housing **11304** to generate a gas-tight seal. In some embodiments, the gasket **11318** may be overmolded onto the filler **11314** or alternatively onto the bottom of the electronics housing **11304**. In other embodiments, however, the gasket **11318** may comprise a separate component part or seal, such as an O-ring or the like.

While the sensor control device **11302** is positioned within the sensor applicator **11302**, the sensor **11306** and the sharp **11308** may be sterilized. According to the present embodiment, sterilizing the sensor **11306** and the sharp **11308** may be accomplished by introducing a sterilizing gas **11320** into the sterilization zone **11316**. The sterilizing gas **11320** may comprise, for example, nitrogen dioxide (NO₂), which operates to sterilize the sensor **11306** and the sharp **11308** without adversely affecting the chemistry on the sensor **11306**. Moreover, the gasket **11318** may prevent the sterilizing gas **11320** from migrating laterally out of the sterilization zone **11316** and impinging upon and damaging an adhesive layer **11322** attached to the bottom of the electronics housing **11304**. Accordingly, the sterilization zone **11316** allows transmission of the sterilizing gas **11320** to impinge upon and sterilize the sensor **11306** and the sharp **11308**, while the remaining portions of the filler **11314** and the gasket **11318** prevent (impede) the sterilizing gas **11320** from damaging the integrity of the adhesive layer **11322**.

In some embodiments, a microbial barrier **11324** may be applied to the end of the filler **11314** and/or the applicator cap **11312** to seal off the sterilization zone **11316**. In some embodiments, the microbial barrier **11324** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors and gases. The Tyvek® layer can be applied before or after application of the sterilizing gas **11320**, and following the sterilizing process, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization zone **11316**. In other embodiments, the microbial barrier **11324** may comprise only a single protective layer applied to the end of the filler **11314**. In such embodiments, the single layer is gas permeable for the sterilization process, but is also capable of protection against moisture and other harmful elements once the sterilization process is complete. Accordingly, the microbial barrier **11324** may operate as a moisture and contaminant layer, without departing from the scope of the disclosure.

It is noted that, while the sensor **11306** and the sharp **11308** extend from the bottom of the electronics housing **11304** and into the sterilization zone **11316** generally concentric with a centerline of the sensor applicator **11302** and the applicator cap **11312**, it is contemplated herein to have an eccentric arrangement. More specifically, in at least one embodiment, the sensor **11306** and the sharp **11308** may extend from the bottom of the electronics housing **11304** eccentric to the centerline of the sensor applicator **11302** and the applicator cap **11312**. In such embodiments, the filler **11314** may be re-designed and otherwise configured such that the sterilization zone **11316** is also eccentrically positioned to receive the sensor **11306** and the sharp **11308**, without departing from the scope of the disclosure.

Embodiments disclosed herein include:

HH. A sensor control device that includes an electronics housing including an upper cover securable to a lower cover, a sensor electronics module positionable between the upper and lower covers and including a sensor holder defining a

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channel, a sensor including a tail extendable through the channel and a flag that includes one or more sensor contacts, a printed circuit board (PCB) having one or more circuitry contacts alignable with the one or more sensor contacts, a first adhesive substrate interposing the flag and the sensor holder to secure the sensor to the sensor holder, and a second adhesive substrate interposing the flag and the PCB to secure the sensor to the PCB and facilitate electrical communication between the one or more sensor contacts and the one or more circuitry contacts. The sensor control device further includes a sharp extendable through the electronics housing, wherein the sharp and the tail extend from a bottom of the electronics housing.

II. A converting process of fabricating a sensor control device that includes positioning a sensor holder defining a channel on a base substrate, extending a tail of a sensor through the channel and securing a flag of the sensor to the sensor holder with a first adhesive substrate applied to a top of the sensor holder, wherein the flag includes one or more sensor contacts, positioning a printed circuit board (PCB) on the base substrate and about the sensor holder, the PCB providing one or more circuitry contacts alignable with the one or more sensor contacts, attaching the PCB to the flag with a second adhesive substrate applied to a top of the flag, facilitating electrical communication between the one or more sensor contacts and the one or more circuitry contacts with the second adhesive substrate, positioning an upper cover over the PCB and securing the upper cover to the base substrate to form an electronics housing, trimming the base substrate about an outer periphery of the electronics housing, and extending a sharp through the electronics housing, wherein the sharp and the tail extend from a bottom of the electronics housing.

Each of embodiments HH and II may have one or more of the following additional elements in any combination: Element 1: further comprising a filler positionable between the upper and lower covers with the sensor electronics module. Element 2: further comprising a third adhesive substrate interposing the lower cover and the filler to secure the filler to the lower cover. Element 3: wherein the sensor electronics module further includes a cap matable with the sensor holder to help secure the sensor within the sensor electronics module. Element 4: wherein the sensor electronics module further includes a third adhesive substrate interposing the cap and the PCB to secure the cap to the PCB. Element 5: wherein the sensor holder is matable with the PCB. Element 6: wherein one or both of the upper and lower covers are made of a material selected from the group consisting of a film, a foil, a foam, a laminated material, and any combination thereof. Element 7: wherein one or both of the upper and lower covers are formed by a manufacturing process selected from the group consisting of thermoforming, vacuum forming, injection molding, die-cutting, stamping, compression molding, transfer molding, and any combination thereof. Element 8: wherein the upper cover is secured to the lower cover via at least one of sonic welding, ultrasonic welding, laser welding, heat sealing, an adhesive substrate, and any combination thereof.

Element 9: wherein the base substrate comprises a film of material disposed on a roll, and attaching the sensor holder to the base substrate is preceded by unrolling the base substrate from the roll, and forming a hole in the base substrate. Element 10: wherein positioning the sensor holder on the base substrate comprises securing the sensor holder to the base substrate using at least one of ultrasonic welding, heat sealing, an adhesive substrate, and any combination thereof. Element 11: wherein the PCB defines first and

second lobes interconnected by a neck portion and the one or more circuitry contacts are provided on the second lobe, and wherein attaching the PCB to the flag comprises folding the second lobe onto the first lobe at the neck portion, and aligning the one or more circuitry contacts with the one or more sensor contacts. Element 12: wherein each lobe provides a battery contact, and the method further comprises applying a third adhesive substrate to the battery contact on the first lobe, attaching a battery to the third adhesive substrate, wherein the second adhesive substrate is further applied to a top of the battery, and folding the second lobe onto the first lobe to align the battery contact on the second lobe with the top of the battery, wherein the second and third adhesive substrates comprise Z-axis anisotropic pressure-adhesive tapes that facilitate electrical communication between the battery and the battery contacts. Element 13: further comprising positioning a filler on the PCB and about the sensor holder, and mitigating vibration and stabilizing electronic modules of the PCB with the filler. Element 14: further comprising applying a third adhesive substrate between the PCB and the upper cover to secure the upper cover to the PCB. Element 15: wherein positioning the upper cover over the PCB comprises forming the upper cover using a process selected from the group consisting of thermoforming, cold forming, vacuum forming, injection molding, die-cutting, stamping, and any combination thereof. Element 16: wherein securing the upper cover to the base substrate comprises sealing the upper cover to the base substrate using a process selected from the group consisting of sonic welding, ultrasonic welding, laser welding, heat sealing, using an adhesive substrate, and any combination thereof. Element 17: further comprising forming a web extending from the outer periphery of the electronics housing and across a tab section, the web providing upper and lower layers sealed at a periphery, facilitating fluid communication into an interior of the electronics housing via the web and an aperture defined in the upper layer, and pressure testing the electronics housing by injecting air into the electronics housing via the aperture and the web. Element 18: further comprising extracting air from the interior of the electronics housing via the web and the aperture, and sealing the outer periphery of the electronics housing under vacuum conditions.

By way of non-limiting example, exemplary combinations applicable to HH and II include: Element 1 with Element 2; Element 3 with Element 4; Element 11 with Element 12; and Element 17 with Element 18.

Example Embodiments of Sensor Module and Plug

FIGS. 114A and 114B are top and bottom perspective views, respectively, of an example embodiment of the plug 2702 of FIGS. 27A-27B, according to one or more embodiments. As described above, the plug 2702 may be designed to hold the connector 2704 (FIGS. 27A-27B and 115A-115B) and the sensor 2616 (FIGS. 27B and 116). The plug 2702 is capable of being securely coupled with the electronics housing 2604 (FIGS. 26A-26B), and the deflectable arms 2707 are configured to snap into corresponding features provided on the bottom of the electronics housing 2604. The sharp slot 2706 can provide a location for the sharp tip 2726 (FIG. 27B) to pass through and the sharp shaft 2724 (FIGS. 27A-27B) to temporarily reside. As illustrated, a sensor ledge 11402 can define a sensor position in a horizontal plane, prevent a sensor from lifting the connector 2704 off of connector posts 11404 and maintain the sensor 2616 parallel to a plane of connector seals. It can also define sensor bend geometry and minimum bend radius. It can limit sensor travel in a vertical direction and prevent a tower from

protruding above an electronics housing surface and define a sensor tail length below a patch surface. A sensor wall 11406 can constrain the sensor 2616 and define a sensor bend geometry and minimum bend radius.

FIGS. 115A and 115B are perspective views depicting an example embodiment of the connector 2704 in open and closed states, respectively. The connector 2704 can be made of silicone rubber that encapsulates compliant carbon impregnated polymer modules that serve as the electrical conductive contacts 2720 between the sensor 2616 (FIGS. 27B and 116) and electrical circuitry contacts for the electronics within housing 2604. The connector 2704 can also serve as a moisture barrier for the sensor 2616 when assembled in a compressed state after transfer from a container to an applicator and after application to a user's skin. A plurality of seal surfaces 11502 can provide a watertight seal for electrical contacts and sensor contacts. The hinges 2718 connect two distal and proximal portions of the connector 2704.

FIG. 116 is a perspective view of an example embodiment of the sensor 2616. The neck 2712 can be a zone which allows folding of the sensor 2616, for example ninety degrees. A membrane on the tail 2708 can cover an active analyte sensing element of the sensor 2616. The tail 2708 can be the portion of the sensor 2616 that resides under a user's skin after insertion. The flag 2710 includes the contacts 2714 and also provides a sealing surface. A biasing tower 11602 can be a tab that biases the tail 2708 into the sharp slot 2706 (FIGS. 114A-114B). A bias fulcrum 11604 can be an offshoot of the biasing tower 11602 that contacts an inner surface of a needle to bias the tail 2708 into a slot defined by the sharp. A bias adjuster 11606 can reduce a localized bending of a tail connection and prevent sensor trace damage. The contacts 2714 can electrically couple the active portion of the sensor to the connector 2704, and a service loop 11608 can translate an electrical path from a vertical direction ninety degrees and engage with the sensor ledge 11402 (FIG. 114B).

FIGS. 117A and 117B are bottom and top perspective views, respectively, depicting an example embodiment of a sensor module assembly comprising the sensor plug 2702, the connector 2704, and the sensor 2616. According to one aspect of the aforementioned embodiments, during or after insertion, the sensor 2616 can be subject to axial forces pushing up in a proximal direction against the sensor 2616 and into the sensor module, as shown by force F1 of FIG. 15A. According to some embodiments, this can result in an adverse force F2 being applied to neck 2712 of the sensor 2616 and, consequently, result in adverse forces F3 being translated to service loop 11608 of the sensor 2616. In some embodiments, for example, axial forces F1 can occur as a result of a sensor insertion mechanism in which the sensor is designed to push itself through the tissue, a sharp retraction mechanism during insertion, or due to a physiological reaction created by tissue surrounding sensor 2616 (e.g., after insertion).

FIGS. 118A and 118B are close-up partial views of an example embodiment of the sensor plug 2702 having certain axial stiffening features. In a general sense, the embodiments described herein are directed to mitigating the effects of axial forces on the sensor 2616 as a result of insertion and/or retraction mechanisms, or from a physiological reaction to the sensor in the body. As illustrated, the sensor 2616 comprises a proximal portion having a hook feature 11802 configured to engage a catch feature 11804 of the plug 2702. In some embodiments, the plug 2702 can also include a clearance area 11806 to allow a distal portion of the sensor

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2616 to swing backwards during assembly to allow for the assembly of the hook feature 11802 of the sensor 2616 over and into the catch feature 11804 of the plug 2702.

According to another aspect of the embodiments, the hook and catch features 11802, 11084 operate in the following manner. The sensor 2616 includes a proximal sensor portion, coupled to the plug 2702, as described above, and a distal sensor portion that is positioned beneath a skin surface in contact with a bodily fluid. The proximal sensor portion may include the hook feature 11802 adjacent to the catch feature 11804 of the plug 2702. During or after sensor insertion, one or more forces are exerted in a proximal direction along a longitudinal axis of the sensor 2616. In response to the one or more forces, the hook feature 11802 engages the catch feature 11804 to prevent displacement of the sensor 2616 in a proximal direction along the longitudinal axis.

According to another aspect of the disclosure, the sensor 2616 can be assembled with the plug 2702 in the following manner. The sensor 2616 is loaded into the plug 2702 by displacing the proximal sensor portion in a lateral direction to bring the hook feature 11802 in proximity to the catch feature 11804 of the plug 2702. More specifically, displacing the proximal sensor portion in a lateral direction causes the proximal sensor portion to move into the clearance area 11806 of the plug 2702.

Although FIGS. 118A and 118B depict the hook feature 11802 as a part of the sensor 2616, and the catch feature 11804 as a part of the plug 2702, those of skill in the art will appreciate that the hook feature 11802 can instead be a part of the plug 2702, and, likewise, the catch feature 11804 can instead be a part of the sensor 3106. Similarly, those of skill in the art will also recognize that other mechanisms (e.g., detent, latch, fastener, screw, etc.) implemented on the sensor 2616 and the plug 2702 to prevent axial displacement of sensor 2616 are possible and within the scope of the present disclosure.

FIG. 119 is a side view of an example sensor 11900, according to one or more embodiments of the disclosure. The sensor 11900 may be similar in some respects to any of the sensors described herein and, therefore, may be used in an analyte monitoring system to detect specific analyte concentrations. As illustrated, the sensor 11900 includes a tail 11902, a flag 11904, and a neck 11906 that interconnects the tail 11902 and the flag 11904. The tail 11902 includes an enzyme or other chemistry or biologic and, in some embodiments, a membrane may cover the chemistry. In use, the tail 11902 is transcutaneously received beneath a user's skin, and the chemistry included thereon helps facilitate analyte monitoring in the presence of bodily fluids.

The tail 11902 may be received within a hollow or recessed portion (e.g., the recessed portion 2728 of FIG. 27B) of a sharp (not shown) to at least partially circumscribe the tail 11902 of the sensor 11900. As illustrated, the tail 11902 may extend at an angle Θ offset from horizontal. In some embodiments, the angle Θ may be about 85°. Accordingly, in contrast to other sensor tails, the tail 11902 may not extend perpendicularly from the flag 11904, but instead at an angle offset from perpendicular. This may prove advantageous in helping maintain the tail 11902 within the keep the recessed portion of the sharp.

The tail 11902 includes a first or bottom end 11908a and a second or top end 11908b opposite the top end 11908a. A tower 11910 may be provided at or near the top end 11908b and may extend vertically upward from the location where the neck 11906 interconnects the tail 11902 to the flag 11904. During operation, if the sharp moves laterally, the

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tower 11910 will help picot the tail 11902 toward the sharp and otherwise stay within the recessed portion (e.g., the recessed portion 2728 of FIG. 27B) of the sharp. Moreover, in some embodiments, the tower 11910 may provide or otherwise define a protrusion 11912 that extends laterally therefrom. When the sensor 11900 is mated with the sharp and the tail 11902 extends within the recessed portion of the sharp, the protrusion 11912 may engage the inner surface of the recessed portion. In operation, the protrusion 11912 may help keep the tail 11902 within the recessed portion.

The flag 11904 may comprise a generally planar surface having one or more sensor contacts 11914 arranged thereon. The sensor contact(s) 11914 may be configured to align with a corresponding number of compliant carbon impregnated polymer modules encapsulated within a connector.

In some embodiments, as illustrated, the neck 11906 may provide or otherwise define a dip or bend 11916 extending between the flag 11904 and the tail 11902. The bend 11916 may prove advantageous in adding flexibility to the sensor 11900 and helping prevent bending of the neck 11906.

In some embodiments, a notch 11918 (shown in dashed lines) may optionally be defined in the flag near the neck 11906. The notch 11918 may add flexibility and tolerance to the sensor 11900 as the sensor 11900 is mounted to the mount. More specifically, the notch 11918 may help take up interference forces that may occur as the sensor 11900 is mounted within the mount.

FIGS. 120A and 120B are isometric and partially exploded isometric views of an example connector assembly 12000, according to one or more embodiments. As illustrated, the connector assembly 12000 may include a connector 12002, and FIG. 120C is an isometric bottom view of the connector 12002. The connector 12002 may comprise an injection molded part used to help secure one or more compliant carbon impregnated polymer modules 12004 (four shown in FIG. 120B) to a mount 12006. More specifically, the connector 12002 may help secure the modules 12004 in place adjacent the sensor 11900 and in contact with the sensor contacts 11914 (FIG. 119) provided on the flag 11904 (FIG. 119). The modules 12004 may be made of a conductive material to provide conductive communication between the sensor 11900 and corresponding circuitry contacts (not shown) provided within the mount 12006.

As best seen in FIG. 120C, the connector 12002 may define pockets 12008 sized to receive the modules 12004. Moreover, in some embodiments, the connector 12002 may further define one or more depressions 12010 configured to mate with one or more corresponding flanges 12012 (FIG. 120B) on the mount 12006. Mating the depressions 12010 with the flanges 12012 may secure the connector 12002 to the mount 12006 via an interference fit or the like. In other embodiments, the connector 12002 may be secured to the mount 12006 using an adhesive or via sonic welding.

FIGS. 121A and 121B are isometric and partially exploded isometric views of another example connector assembly 12100, according to one or more embodiments. As illustrated, the connector assembly 12100 may include a connector 12102, and FIG. 121C is an isometric bottom view of the connector 12102. The connector 12102 may comprise an injection molded part used to help keep one or more compliant metal contacts 12104 (four shown in FIG. 121B) secured against the sensor 11900 on a mount 12106. More specifically, the connector 12102 may help secure the contacts 12104 in place adjacent the sensor 11900 and in contact with the sensor contacts 11914 (FIG. 119) provided on the flag 11904. The contacts 12104 may be made of a stamped conductive material that provides conductive com-

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munication between the sensor **11900** and corresponding circuitry contacts (not shown) provided within the mount **12106**. In some embodiments, for example, the contacts **12104** may be soldered to a PCB (not shown) arranged within the mount **12106**.

As best seen in FIG. **121C**, the connector **12102** may define pockets **12108** sized to receive the contacts **12104**. Moreover, in some embodiments, the connector **12102** may further define one or more depressions **12110** configured to mate with one or more corresponding flanges **12112** (FIG. **120B**) on the mount **12006**. Mating the depressions **12110** with the flanges **12112** may help secure the connector **12102** to the mount **12106** via an interference fit or the like. In other embodiments, the connector **12102** may be secured to the mount **12106** using an adhesive or via sonic welding.

Therefore, the disclosed systems and methods are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the teachings of the present disclosure may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above may be altered, combined, or modified and all such variations are considered within the scope of the present disclosure. The systems and methods illustratively disclosed herein may suitably be practiced in the absence of any element that is not specifically disclosed herein and/or any optional element disclosed herein. While compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods can also “consist essentially of” or “consist of” the various components and steps. All numbers and ranges disclosed above may vary by some amount. Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range is specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces. If there is any conflict in the usages of a word or term in this specification and one or more patent or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

The use of directional terms such as above, below, upper, lower, upward, downward, left, and right and the like are used in relation to the illustrative embodiments as they are

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depicted in the figures, the upward direction being toward the top of the corresponding figure and the downward direction being toward the bottom of the corresponding figure.

What is claimed is:

1. An assembly for delivery of an analyte sensor comprising:

a sensor control device including:

an electronics housing including:

a shell having a first aperture, and

a mount secured to the shell to define an interior space, the mount having

a second aperture aligned with the first aperture;

a circuit board disposed within the interior space of the electronics housing and including a plurality of electronics modules;

an analyte sensor including an in vivo portion extending through the second aperture and configured to measure an analyte level in a bodily fluid and an ex vivo portion within the interior space of the electronics housing and including a plurality of electrical contacts coupled with the circuit board; and

a sharp hub adapted for engagement with the sensor control device and including a sharp and a mating member, the mating member configured to extend through the first and second aperture and distally from an underside of the mount;

a sensor cap having a first end, a second end, and an inner chamber, wherein the inner chamber receives at least a portion of the in vivo portion of the analyte sensor and wherein the first end is removably coupled to the mating member to seal the inner chamber; and

an applicator for delivery of the analyte sensor including:

a housing including a sensor carrier configured to secure the sensor control device within an interior of the applicator; and

an applicator cap removably coupled to the housing to seal the interior of the applicator, wherein the applicator cap mates with an engagement feature on the second end of the sensor cap when applicator cap is coupled to the housing, and wherein the sensor cap detaches from the mating member and remains mated with the applicator cap when applicator cap is removed from the housing.

2. The assembly of claim 1, wherein the first end of the sensor cap includes one or more internal flat threads and the mating member includes one or more external flat threads corresponding to the one or more internal flat threads, the sensor cap being removably coupled to the mating member by engagement of the one or more external flat thread to the one or more internal flat thread.

3. The assembly of claim 1, wherein the sharp includes a hollow portion configured to receive at least a portion of the analyte sensor.

4. The assembly of claim 3, wherein the hollow portion includes a non-circular cross-section.

5. The assembly of claim 1, wherein the sensor cap is configured to isolate the analyte sensor for sterilization.

6. The assembly of claim 1, wherein the applicator cap seals the interior of the applicator when the applicator cap is coupled to the housing.

7. The assembly of claim 1, wherein the sensor cap includes a desiccant cap arranged at the second end of the sensor cap.

8. The assembly of claim 7, wherein the desiccant cap defines the engagement feature of the sensor cap.

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9. The assembly of claim 1, wherein the inner chamber of the sensor cap includes a desiccant to maintain a target humidity level.

10. The assembly of claim 1, wherein the applicator cap further comprising a cap post configured to receive the engagement feature of the sensor cap. 5

11. The assembly of claim 1, wherein the shell includes an upper surface with a first aperture and a depression configured to facilitate control of the electronics housing in at least one degree of freedom. 10

12. The assembly of claim 1, wherein the plurality of electronics modules are disposed on a top surface and a bottom surface of the circuit board.

13. The assembly of claim 1, wherein the circuit board includes a battery cutout portion sized to receive and seat a battery. 15

14. The assembly of claim 1, wherein the electronics housing further includes a plurality of carrier grip features along an outer periphery of the electronics housing. 20

15. The assembly of claim 14, wherein the plurality of carrier grip features are configured to prevent the electronics housing from rotating within the applicator.

16. The assembly of claim 1, further comprising an adhesive configured to secure the shell to the mount, wherein the adhesive isolates an interior of the electronics housing from outside contamination. 25

17. The assembly of claim 1, wherein the first end of the sensor cap is removably coupled to the mount to form a sterile barrier for the analyte sensor. 30

18. The assembly of claim 1, further comprising a collar positioned in the electronics housing, wherein the collar comprises a central aperture axially aligned with the first aperture and the second aperture, wherein the sharp hub is adapted for engagement with the collar. 35

19. The assembly of claim 1, further comprising a conical collimator positioned within the applicator cap and defining a sterilization zone configured to receive the in vivo portion of the analyte sensor extending through the aperture in the mount.

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20. A method of assembling an analyte sensor for delivery comprising:

providing a sensor control device including:

an electronics housing including a shell having a first aperture and a mount secured to the shell to define an interior space, the mount having a second aperture aligned with the first aperture, a circuit board disposed within the interior space of the electronics housing and including a plurality of electronics modules, an analyte sensor including an in vivo portion extending through the second aperture and configured to measure an analyte level in a bodily fluid, an ex vivo portion within the interior space of the electronics housing and including a plurality of electrical contacts coupled with the circuit board, and a neck portion interconnecting the in vivo portion and the ex vivo portion, a sharp hub adapted for engagement with the sensor control device and including a sharp and a mating member, the mating member configured to extend through the first and second aperture and distally from an underside of the mount, and a sensor cap having a first end, a second end, and an inner chamber, wherein the inner chamber receives at least a portion of the in vivo portion of the analyte sensor and wherein the first end is removably coupled to the mating member to seal the inner chamber; and

providing an applicator for delivery of the analyte sensor including:

a housing including a sensor carrier configured to secure the sensor control device within an interior of the applicator, and an applicator cap removably coupled to the housing to seal the interior of the applicator, wherein the applicator cap mates with an engagement feature on the second end of the sensor cap when applicator cap is coupled to the housing, and wherein the sensor cap detaches from the mating member and remains mated with the applicator cap when applicator cap is removed from the housing.

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